Table of Contents

Chapter 1: Orienting Yourself - The Use of Coordinates **4**

1.1 Complete Concept Cheat Sheet (read this before any question) 4

1.2 Question Types / Concept Map for Chapter 1 7

1.3 Practice Questions (Q1 - Q50) with Full Step-by-Step Solutions 8

1.4 Chapter MCQs (60 questions, answers inline) 32

1.5 Chapter Formula Sheet 53

1.6 Last-Minute Cheat Sheet 54

1.7 Mistake Buster - DON'T MAKE THESE MISTAKES 54

1.8 Exam Strategy for Chapter 1 55

1.9 Chapter Completion Checklist 56

Chapter 2: Introduction to Linear Polynomials **56**

2.1 Complete Concept Cheat Sheet (read this before any question) 56

2.2 Question Types / Concept Map for Chapter 2 59

2.3 Practice Questions (Q1 - Q50) with Full Step-by-Step Solutions 61

2.4 Chapter MCQs (60 questions, answers inline) 79

2.5 Chapter Formula Sheet 99

2.6 Last-Minute Cheat Sheet 100

2.7 Mistake Buster - DON'T MAKE THESE MISTAKES 100

2.8 Exam Strategy for Chapter 2 101

2.9 Chapter Completion Checklist 101

Chapter 3: The World of Numbers (Real Numbers) **102**

3.1 Complete Concept Cheat Sheet (read this before any question) 102

3.2 Question Types / Concept Map for Chapter 3 105

3.3 Practice Questions (Q1 - Q50) with Full Step-by-Step Solutions 106

3.4 Chapter MCQs (60 questions, answers inline) 126

3.5 Chapter Formula Sheet 146

3.6 Last-Minute Cheat Sheet 147

3.7 Mistake Buster - DON'T MAKE THESE MISTAKES 148

3.8 Exam Strategy for Chapter 3 149

3.9 Chapter Completion Checklist 149

Chapter 4: Algebraic Identities and Factorisation **150**

4.1 Complete Concept Cheat Sheet (read this before any question) . . . 150
4.2 Question Types / Concept Map for Chapter 4 . . . 152
4.3 Practice Questions (Q1 - Q70) with Full Step-by-Step Solutions . . . 153
4.4 Chapter MCQs (80 questions, answers inline) . . . 176
4.5 Chapter Formula Sheet . . . 202
4.6 Last-Minute Cheat Sheet . . . 203
4.7 Mistake Buster - DON'T MAKE THESE MISTAKES . . . 204
4.8 Exam Strategy for Chapter 4 . . . 205
4.9 Chapter Completion Checklist . . . 205

Chapter 6: Perimeter and Area 206

6.1 Complete Concept Cheat Sheet (read this before any question) . . . 206
6.2 Question Types / Concept Map for Chapter 6 . . . 208
6.3 Practice Questions (Q1 - Q70) with Full Step-by-Step Solutions . . . 209
6.4 Chapter MCQs (80 questions, answers inline) . . . 231
6.5 Chapter Formula Sheet . . . 258
6.6 Last-Minute Cheat Sheet . . . 258
6.7 Mistake Buster - DON'T MAKE THESE MISTAKES . . . 259
6.8 Exam Strategy for Chapter 6 . . . 259
6.9 Chapter Completion Checklist . . . 260

Chapter 7: Probability - Measuring Chance 261

7.1 Complete Concept Cheat Sheet (read this before any question) . . . 261
7.2 Question Types / Concept Map for Chapter 7 . . . 263
7.3 Practice Questions (Q1 - Q50) with Full Step-by-Step Solutions . . . 263
7.4 Chapter MCQs (60 questions, answers inline) . . . 278
7.5 Chapter Formula Sheet . . . 298
7.6 Last-Minute Cheat Sheet . . . 299
7.7 Mistake Buster - DON'T MAKE THESE MISTAKES . . . 299
7.8 Exam Strategy for Chapter 7 . . . 300
7.9 Chapter Completion Checklist . . . 300

Grand Revision Arena (All Chapters Combined) 301

R.1 Master Formula Table (all chapters in one glance) . . . 301
R.2 Thirty Mixed Questions (Q101 - Q130) with Full Solutions . . . 302
R.3 Hundred Mixed MCQs (Q501 - Q600, answers inline) . . . 309

R.4 Twenty-Five Challenge Questions (Q201 - Q225) 343

R.5 Exam-Day One-Pager (read this in the last 10 minutes before the bell) 349

Chapter 1: Orienting Yourself - The Use of Coordinates

This chapter teaches you how to describe the exact position of any point on a flat surface using nothing but two numbers. That sounds small, but it is the foundation of coordinate geometry, which you will use in every higher class. The uploaded PDF covers the 2-D Cartesian system (axes, origin, ordered pairs, quadrants, plotting), distances parallel to the axes, the distance formula built on the Baudhayana-Pythagoras Theorem, reflections in the axes, midpoint and trisection ideas, the collinearity test, and points on a circle centred at the origin. In this practice book we push every one of those ideas one level higher, with harder numbers, missing-value reasoning, and proof-style thinking.

1.1 Complete Concept Cheat Sheet (read this before any question)

1.1.1 The coordinate system and its parts

What is it? A coordinate system is a grid-based framework that locates every point in a plane using an ordered pair of numbers. Two perpendicular number lines are drawn: the horizontal one is the **x-axis** and the vertical one is the **y-axis**. They meet at the **origin O = (0, 0)**. Values to the right of O on the x-axis and above O on the y-axis are positive; values to the left and below are negative.

When do I use it? Whenever a question gives you points, positions on a map or a room layout, distances between locations, or asks you to 'plot', 'locate', 'identify the quadrant', or 'find the distance between points'.

How do I recognise it? Any mention of (x, y) pairs, axes, quadrants, graph paper, or a floor plan with walls along two directions. **What does each part mean?** In the pair (x, y), the first number x is the **perpendicular distance from the y-axis** (measured along the x-axis direction), and the second number y is the **perpendicular distance from the x-axis**. The pair is called an *ordered pair* because the order is fixed: (3, 5) and (5, 3) are different points. Logically, you can understand it as 'walk x units horizontally, then y units vertically from O'.

Tiny example. The point (2, 3) is 2 units right of O and 3 units up. The point (-2, 3) is 2 units LEFT and 3 up. Same distances, different signs, different points entirely.

1.1.2 Points on the axes and the quadrant sign rules

- A point on the x-axis always looks like (x, 0). Its y-coordinate is 0 because it has zero height above the x-axis. Positive x means right of O, negative x means left of O.
- A point on the y-axis always looks like (0, y). Its x-coordinate is 0. Positive y means above O, negative y means below O.
- The origin (0, 0) lies on BOTH axes; it belongs to no quadrant.
- Quadrant I: (+, +). Quadrant II: (-, +). Quadrant III: (-, -). Quadrant IV: (+, -). Memorise the cycle: starting from I and going anticlockwise the signs are ++, -+, --, +-. A quick memory trick: signs of y follow the C-shape of Roman numerals I, II, III, IV written anticlockwise; signs of x are + in I, - in II, - in III, + in IV.
- Any point with x = 0 and y = 0 must be the origin. Any point with both coordinates non-zero cannot lie on either axis.

> **Identify-the-method trick**
> See a question that says 'on the x-axis' or 'on the y-axis'? Instantly write the fixed coordinate: (something, 0) or (0, something). This single habit unlocks a whole family of missing-coordinate questions.

1.1.3 Distances parallel to the axes

What is it? If two points share the same y-coordinate, the segment joining them is horizontal, and the distance is simply the absolute gap between the x-values: distance = $|x_2 - x_1|$. If two points share the same x-coordinate, the segment is vertical and the distance is $|y_2 - y_1|$.

When do I use it? Whenever two given points have one identical coordinate - check this FIRST before reaching for the distance formula. **How do I recognise it?** Same second number means a horizontal move (like walking along a corridor east-west); same first number means a vertical move. **Each term:** the absolute value bars guarantee a positive (non-negative) length; order of subtraction does not matter.

Tiny example. Distance between (2, 5) and (9, 5) is $|9 - 2| = 7$. Distance between (4, -3) and (4, 6) is $|6 - (-3)| = 9$. Note the classic sign trap: $6 - (-3) = 9$, not 3.

1.1.4 The distance formula (the star of the chapter)

What is it? For any two points $A(x_1, y_1)$ and $D(x_2, y_2)$, the distance is $AD = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$. **Where does it come from?** Drop a right-angled helper triangle: the horizontal leg is $|x_2 - x_1|$ and the vertical leg is $|y_2 - y_1|$. The segment AD is the hypotenuse, so the Baudhayana-Pythagoras Theorem ($leg^2 + leg^2 = hypotenuse^2$) gives the formula. It is not a magic rule - it is Pythagoras wearing coordinates.

When do I use it? When the two points differ in BOTH coordinates. **How do I recognise it?** Words like 'find the length of AB', 'how far apart', 'show that the triangle is isosceles', 'find the perimeter' with slanted sides. **Each term:** subtract corresponding coordinates, square each difference (squares kill any minus sign, so the order of the points does not matter), add, then take the positive square root.

Can it be derived logically? Yes - see the helper-triangle picture in Figure 2. **Tiny example.** From (1, 2) to (4, 6): differences are 3 and 4, so distance = $\sqrt{(9 + 16)} = \sqrt{25} = 5$. Whenever the squared sum is a perfect square you get a clean integer - examiners love the triples 3-4-5, 5-12-13, 6-8-10, 8-15-17, 9-12-15, 7-24-25.

- **Method 1 (helper triangle):** sketch, mark legs |dx| and |dy|, apply Pythagoras. Best for learning and for verifying.
- **Method 2 (formula directly):** substitute into $\sqrt{(dx^2 + dy^2)}$. Best for speed once confident.
- **Method 3 (shift to origin):** subtract the first point from both - distance is unchanged - then compute from (0, 0). Great mental shortcut: distance of (x, y) from origin is $\sqrt{(x^2 + y^2)}$.

> **Common confusion**
> Students often compute $\sqrt{(dx)} + \sqrt{(dy)}$ or $dx + dy$ instead of $\sqrt{(dx^2 + dy^2)}$. Remember: square EACH difference first, THEN add, THEN take one single square root at the very end. Never take a root midway.

1.1.5 Reflections in the axes

What is it? Reflecting a point in the x-axis flips its height: (x, y) becomes (x, -y). Reflecting in the y-axis flips its horizontal position: (x, y) becomes (-x, y). Reflecting in BOTH axes (equivalent to a half-turn about O)

gives (-x, -y). **Why only one sign changes:** a mirror in the x-axis changes only how far up or down you are; your sideways position stays the same.

Key property: reflection preserves distances. If A reflects to A' and B reflects to B', then AB = A'B'. The PDF proves this by computing both distances - the squares of the coordinate differences come out identical. Use this to check your reflection answers. **Tiny example.** A(2, 3) reflects in the x-axis to A'(2, -3). Take another point B(6, 3); it reflects to B'(6, -3). Now AB = |6 - 2| = 4 and A'B' = |6 - 2| = 4: the distance is preserved because the reflection moves the WHOLE segment, not just one endpoint. (If you compare A' with the unreflected B you get √52 - but that was never a preserved pair; always compare each point with its own image.)

1.1.6 Midpoint and equal-distance ideas

What is it? The midpoint of a segment splits it into two equal halves. The PDF discovers the connection experimentally: if M(m_1, m_2) is the midpoint of S and T, then each coordinate of M is the average of the corresponding coordinates: $m_1 = (x_S + x_T)/2$ and $m_2 = (y_S + y_T)/2$. You can verify with the distance formula: distances from M to S and from M to T come out equal. **Reverse use:** if you know one endpoint A and the midpoint M, the other endpoint is B = ($2m_1 - x_A$, $2m_2 - y_A$) - just undo the averaging. **Trisection:** points P and Q that cut AB into three equal parts satisfy P = A + (B - A)/3 and Q = A + 2(B - A)/3; the midpoint of P and Q equals the midpoint of AB, which is a lovely self-check.

When do I use the average idea? Keywords: 'midpoint', 'bisects', 'equally spaced', 'middle point of the segment'. **Tiny example.** Midpoint of (-4, 1) and (6, 5) is ((-4 + 6)/2, (1 + 5)/2) = (1, 3).

1.1.7 Collinearity test without plotting

What is it? Three points are collinear (lie on one straight line) if one of the three distances equals the sum of the other two. Concretely, compute AB, BC, AC and check whether the largest equals the sum of the other two. **Why it works:** if B truly sits on the straight path from A to C, then walking A to B to C covers exactly the straight distance AC - no detour is possible on a straight line (triangle inequality with equality).

When do I use it? Questions like 'are these points on the same straight line?' or 'check without plotting'. **Trap:** the check must be done with EXACT values. If the distances involve roots, compare exact squared forms or keep radicals; do not round early. **Tiny example.** (0,0), (3,4), (6,8): AB = 5, BC = 5, AC = 10, and 5 + 5 = 10, so collinear.

1.1.8 Points on a circle centred at the origin

What is it? A circle with centre O(0, 0) and radius r is the set of points whose distance from O equals exactly r. So a point (x, y) lies ON the circle if $x^2 + y^2 = r^2$, INSIDE if $x^2 + y^2$ is less than r^2, and OUTSIDE if it is greater. This is just the distance-from-origin shortcut applied with an equality/inequality. **How do I recognise it?** 'Show that these points lie on a circle with centre the origin - find the radius' or 'is the point inside, on, or outside the circle?'. **Tiny example.** (1, -8), (-4, 7), (-7, -4): squares give 65, 65, 65 - all equal, so all three lie on the circle $x^2 + y^2 = 65$, radius √65.

1.1.9 Exam tips for this chapter

1. Before any distance computation, CHECK whether a coordinate is shared - the parallel-to-axes shortcut saves time and avoids errors.
2. Keep the ordered-pair order sacred. (x, y) and (y, x) coincide only when x = y.

3. Squaring removes minus signs, so $(x_2 - x_1)^2 = (x_1 - x_2)^2$. Do not waste time worrying about subtraction order.
4. Memorise Pythagorean triples; they turn ugly roots into integers.
5. For 'inside/on/outside circle' questions, compare SQUARES of distances - never square roots.
6. After finding a reflection, quickly verify one preserved distance.
7. In multi-part geometry questions, draw a rough sketch even if the question says 'without plotting' - your sketch guides the algebra; the algebra is what you present.

1.2 Question Types / Concept Map for Chapter 1

Almost every coordinate question in exams falls into one of the following types. Use this map to label what you are looking at before solving - half the battle is knowing WHICH type of fight it is.

Type	Name	How to recognise it in the question	Core method
T1	Direct identification	Given coordinates, find quadrant / axis / sign pattern	Sign rules of the four quadrants
T2	Plotting and reading	Mark points, read positions from a room/map figure	Ordered pair = walk x then y
T3	Axis-parallel distance	Two points share one coordinate	|dx| or |dy| shortcut
T4	Distance formula	Two points, both coordinates differ	$\sqrt{(dx^2 + dy^2)}$
T5	Reverse / missing value	Distance given, one coordinate unknown (find k)	Set up equation, solve, check BOTH roots
T6	Triangle / quadrilateral classification	Show isosceles / right / square / rhombus / parallelogram	Compute all sides (and diagonals), compare
T7	Reflection reasoning	Image of a point in x-axis / y-axis / both	Flip the relevant sign; verify preserved distance
T8	Midpoint / trisection	'M is the midpoint', 'points of trisection', 'bisects'	Average coordinates; reverse-average for missing endpoint
T9	Collinearity test	'On the same line?', 'without plotting'	Largest distance = sum of other two (exact values)
T10	Circle membership	'lie on circle with centre origin', inside/outside check	Compare $x^2 + y^2$ with r^2
T11	Applied grid problems	City streets, screen pixels, room maps	Model positions as coordinates, then use T3/T4
T12	Perimeter / area from coordinates	Vertices given, find perimeter or area	Distance formula on each side; 1/2 base x height for right triangles
T13	Conceptual trap questions	Order of pair, meaning of coordinates, statements to justify	Reason from definitions, not memory

Type	Name	How to recognise it in the question	Core method
T14	Multi-concept HOTS	Combines distance + midpoint + circle or reflection chains	Break into labelled mini-steps, verify at each junction

Distribution of the 50 practice questions below: T1-T3 feed Q1-Q10, T4-T5 feed Q11-Q20, T6-T9 feed Q21-Q35, T10-T13 feed Q31-Q40, and T14 dominates Q41-Q50.

1.3 Practice Questions (Q1 - Q50) with Full Step-by-Step Solutions

Difficulty climbs in five bands: Q1-Q10 strong basics, Q11-Q20 moderate, Q21-Q30 difficult, Q31-Q40 high-level, Q41-Q50 very challenging (Olympiad-lite). Every question carries the concept tag so you can trace it back to the cheat sheet.

Q1 · Basic · Concept: Quadrant sign rules (T1)

Q1. Name the quadrant (or axis) in which each point lies, and state the sign pattern: A(7, -9), B(-6, 2), C(-8, -11), D(9, 4), E(0, -3), F(-12, 0).

SOLUTION

Step 1: Given: six points with their coordinates. Needed: location of each point.

Step 2: Recall the sign pattern: Quadrant I (+, +), Quadrant II (-, +), Quadrant III (-, -), Quadrant IV (+, -). Points with x = 0 lie on the y-axis; points with y = 0 lie on the x-axis.

Step 3: A(7, -9): x is positive, y is negative, so A is in Quadrant IV with signs (+, -).

Step 4: B(-6, 2): x negative, y positive, so Quadrant II with signs (-, +).

Step 5: C(-8, -11): both negative, so Quadrant III with signs (-, -).

Step 6: D(9, 4): both positive, so Quadrant I with signs (+, +).

Step 7: E(0, -3): x-coordinate is 0, so E lies on the y-axis, 3 units below the origin, in no quadrant.

Step 8: F(-12, 0): y-coordinate is 0, so F lies on the x-axis, 12 units left of the origin.

Hinglish: *Beta, yahan sirf signs dekhne hain. Pehle number jo hai wo x ki duniya hai - positive matlab right side, negative matlab left. Doosra number y hai - positive matlab upar, negative matlab neeche. A mein x positive, y negative, matlab right aur neeche = Quadrant IV. Bas ye pattern yaad rakho: I mein dono plus, II mein x minus y plus, III mein dono minus, IV mein x plus y minus. Aur agar koi coordinate 0 hai toh point axis par baitha hai, quadrant mein nahi.*

Answer: A in Q IV (+, -); B in Q II (-, +); C in Q III (-, -); D in Q I (+, +); E on the y-axis below O; F on the x-axis left of O.

Q2 · Basic · Concept: Points on the axes (T1/T2)

Q2. Write the coordinates of (i) the point on the x-axis that is 6 units to the left of the origin, (ii) the point on the y-axis that is 4.5 units below the origin, (iii) the point where the two axes meet.

SOLUTION

Step 1: Given: descriptions along the axes. Needed: coordinates.

Step 2: A point on the x-axis has the form (x, 0); 'left of the origin' means the x-value is negative.

Step 3: So 6 units left of O gives (-6, 0).

Step 4: A point on the y-axis has the form (0, y); 'below the origin' means the y-value is negative.

Step 5: So 4.5 units below O gives (0, -4.5).

Step 6: The axes meet at the origin, whose coordinates are (0, 0).

» **Why this step?** We fix the zero coordinate FIRST because 'on the axis' forces it; then only the sign and the size of the other coordinate remain.

Hinglish: *X-axis par rehne wale har point ka second number hamesha 0 hota hai - ye pakka rule hai. Left jaana hai matlab negative. Isliye (-6, 0). Y-axis par rehne wale ka pehla number 0 hota hai, aur neeche matlab negative y, isliye (0, -4.5). Origin dono axis ka intersection hai, wahan dono zero - (0, 0). Sign ka khayal rakho, bas yahi galti hoti hai.*

Answer: (i) (-6, 0) (ii) (0, -4.5) (iii) O(0, 0)

Q3 · Basic · Concept: Meaning of coordinates (T13)

Q3. For the point P(5, -7), state (i) its perpendicular distance from the y-axis, (ii) its perpendicular distance from the x-axis, (iii) which side of each axis it lies on.

SOLUTION

Step 1: Given: P(5, -7). Needed: distances from the two axes and directions.

Step 2: By definition, the x-coordinate is the perpendicular distance from the y-axis, measured along the x-direction.

Step 3: So the distance from the y-axis is 5 units, and since 5 is positive, P lies to the right of the y-axis.

Step 4: The y-coordinate is the perpendicular distance from the x-axis.

Step 5: So the distance from the x-axis is 7 units, and since -7 is negative, P lies below the x-axis.

Step 6: Hence P sits in Quadrant IV, 5 units right and 7 units down from the origin.

» **Why this step?** Distances are always written positive; the SIGN of the coordinate only tells you the side. Never say the distance is -7.

Hinglish: *Bahut important concept: x-coordinate batata hai y-axis se doori, aur y-coordinate batata hai x-axis se doori. Students yahan ulta kar dete hain! P(5, -7) ka matlab: y-axis se 5 unit door (right side), x-axis se 7 unit door (neeche, kyunki minus hai). Doori kabhi negative nahi hoti - minus sirf direction batata hai.*

Answer: (i) 5 units, right of the y-axis (ii) 7 units, below the x-axis (iii) Quadrant IV

Q4 · Basic · Concept: Ordered pair order matters (T13)

Q4. Are the points (3, 5) and (5, 3) the same point? Justify. For what values of a and b does the point (a, b) coincide with the point (b, a)?

SOLUTION

Step 1: Given: two ordered pairs. Needed: whether they name the same location.

Step 2: The point (3, 5) means 3 units along the x-direction and 5 units along the y-direction; (5, 3) means 5 along x and 3 along y.

Step 3: These are different walks from the origin, so they land at different places: (3, 5) is above-left of (5, 3). They coincide only if the two coordinates are equal.

Step 4: Setting (a, b) = (b, a) gives a = b and b = a - the same single condition a = b.

Step 5: So for any value with a = b (for example (4, 4)), the two pairs name one and the same point on the line y = x.

» **Why this step?** The justification is definitional, not visual: ordered pairs are equal exactly when they agree coordinate by coordinate.

Hinglish: *(3, 5) aur (5, 3) bilkul alag points hain kyunki ordered pair mein ORDER matter karta hai - pehle x ki baat, phir y ki. Dono tabhi ek hi point banenge jab x aur y barabar hon, yaani a = b. Tab dono us line par baithte hain jahan har point ka x aur y same hota hai (line y = x). Exam mein ye 'true/false with reason' type bahut aata hai.*

Answer: No, (3, 5) and (5, 3) are different points; (a, b) = (b, a) exactly when a = b.

Q5 · Basic · Concept: Plotting and locating (T2)

Q5. Without actually drawing, describe the position of P(-3.5, 2.5) and Q(2.5, -3.5). Are P and Q reflections of each other in either axis? Give a reason.

SOLUTION

Step 1: Given: two points with fractional coordinates. Needed: rough location and reflection relation.

Step 2: P has x = -3.5 (left of the y-axis) and y = +2.5 (above the x-axis), so P sits in Quadrant II.

Step 3: Q has x = +2.5 (right) and y = -3.5 (below), so Q sits in Quadrant IV.

Step 4: Reflecting P in the x-axis would give (-3.5, -2.5); reflecting P in the y-axis would give (3.5, 2.5). Neither equals Q(2.5, -3.5).

Step 5: Also notice P and Q have swapped magnitudes: 3.5 with 2.5 exchanged places. A reflection in an axis flips ONE sign and keeps both magnitudes in place, so P and Q are not axis reflections of each other.

Hinglish: *P left-upar wale hisse (Quadrant II) mein hai, Q right-neeche (Quadrant IV) mein. Reflection ka rule yaad karo: x-axis mein reflect karne par sirf y ka sign ulta hota hai, y-axis mein sirf x ka. Yahan signs ka pattern milta nahi (P ko flip karo toh (−3.5, −2.5) ya (3.5, 2.5) banta hai, Q toh (2.5, −3.5) hai). Aur dhyan do - P aur Q mein numbers ne apas mein jagah bhi badal li hai (3.5 aur 2.5 swap). Reflection mein swapping kabhi nahi hoti. Isliye ye reflections nahi hain.*

Answer: P is in Quadrant II, Q is in Quadrant IV; they are NOT reflections of each other in either axis (a reflection flips exactly one sign and swaps no coordinates).

Q6 · Basic · Concept: Axis-parallel distances (T3)

Q6. Find the distance between (i) A(2, 0) and B(9, 0), (ii) C(0, -4) and D(0, 6), (iii) E(-5, 3) and F(4, 3).

SOLUTION

Step 1: Given: three pairs of points. Needed: distances.

Step 2: Pair (i): both points lie on the x-axis (y = 0 for both), so use the horizontal shortcut |x2 - x1| = |9 - 2| = 7.

Step 3: Pair (ii): both on the y-axis, so distance = |y2 - y1| = |6 - (-4)| = |6 + 4| = 10.

Step 4: Pair (iii): both have y = 3, a horizontal segment: distance = |4 - (-5)| = |4 + 5| = 9.

Step 5: In every case the shared coordinate confirms the segment is parallel to one axis, so the single-difference shortcut is valid - no square roots needed.

» Why this step? Checking the shared coordinate FIRST is the habit that separates fast, accurate students from slow ones.

Hinglish: *Jab do points mein se koi ek coordinate same ho, toh seedha difference lo - formula ki zaroorat hi nahi. (ii) mein dhyan do: 6 - (-4) = 6 + 4 = 10. Minus ke saath minus mila toh plus ban gaya. Ye sign wala jhatka sabse common galti hai, isliye maine absolute value ke andar rakha hai. (iii) bhi same khel hai: 4 - (-5) = 9.*

Answer: (i) 7 units (ii) 10 units (iii) 9 units

Q7 · Basic · Concept: Perimeter from coordinates (T12/T3)

Q7. The corners of a rectangular room are O(0, 0), A(8, 0), B(8, 5) and C(0, 5). Find the perimeter and state which walls are parallel to which axes.

SOLUTION

Step 1: Given: four corners in order. Needed: perimeter and axis directions of walls.

Step 2: OA joins (0, 0) to (8, 0): both y-values are 0, so OA is horizontal with length |8 - 0| = 8.

Step 3: AB joins (8, 0) to (8, 5): both x-values are 8, so AB is vertical with length |5 - 0| = 5.

Step 4: BC joins (8, 5) to (0, 5): horizontal, length |8 - 0| = 8. CO joins (0, 5) to (0, 0): vertical, length 5.

Step 5: Perimeter = 8 + 5 + 8 + 5 = 26 units.

Step 6: OA and BC run along the x-direction (parallel to the x-axis); AB and CO run along the y-direction (parallel to the y-axis).

Hinglish: *Rectangle ke opposite sides barabar hoti hain, isliye do sides hi kaafi hain: ek horizontal 8, ek vertical 5. Perimeter = 2(8 + 5) = 26. Dekho, coordinates same hone se hi pata chal gaya ki kaun si deewar kis axis ke parallel hai - y same hai toh horizontal wall, x same hai toh vertical wall. Ye floor-plan wale questions mein bahut kaam aata hai.*

Answer: Perimeter = 26 units; OA and BC are parallel to the x-axis, AB and CO are parallel to the y-axis.

Q8 · Basic · Concept: Distance from origin (T4)

Q8. Find the distance of the point (6, 8) from the origin. Which Pythagorean triple is hiding inside this question, and how can the triple help you answer faster next time?

SOLUTION

Step 1: Given: point (6, 8). Needed: distance from O(0, 0).

Step 2: Distance from origin = $\sqrt{(x^2 + y^2)} = \sqrt{(6^2 + 8^2)}$.

Step 3: Compute the squares: 36 + 64 = 100.

Step 4: Take the root: $\sqrt{100} = 10$.

Step 5: The hidden triple is 3-4-5 scaled by 2: 6-8-10. If you spot (6, 8), you can jump straight to 10 without squaring.

Step 6: So the distance is 10 units.

» **Why this step?** Spotting the triple is a safe shortcut only when BOTH numbers of a known triple appear exactly; otherwise use the formula.

Hinglish: *Origin se distance ka short formula: $\sqrt{(x^2 + y^2)}$. 6 aur 8 ko dekhte hi 3-4-5 wala triple yaad karo - dono numbers 2 se multiply ho rahe hain, isliye answer 5 ka 2 guna = 10. Triples ratt lo: 3-4-5, 5-12-13, 8-15-17, 7-24-25. Par dhyan rahe - triple tabhi use karo jab numbers EXACT match karein, warna seedha formula lagao.*

Answer: 10 units; the triple 6-8-10 (which is 2 x (3-4-5)).

Q9 · Basic · Concept: Sign reasoning (T1/T13)

Q9. A point has coordinates (a, -a) where a is a non-zero real number. (i) In which quadrant does it lie when a is positive? (ii) In which quadrant when a is negative? (iii) What happens at a = 0? (iv) What is always true about the distances of this point from the two axes?

SOLUTION

Step 1: Given: the family of points (a, -a). Needed: quadrant analysis by sign of a.

Step 2: Case a positive: x = a is positive, y = -a is negative, so the point is in Quadrant IV.

Step 3: Case a negative: writing a = -m with m positive gives x = -m (negative) and y = m (positive), so the point is in Quadrant II.

Step 4: Case a = 0: the point is (0, 0), the origin itself - on both axes, in no quadrant.

Step 5: For every a, the distance from the y-axis is |a| and the distance from the x-axis is |-a| = |a|. These are always equal.

Step 6: So every point of this family stays exactly equally far from both axes - the family slides along the two lines y = -x.

Hinglish: *Ye question sign-logic ka gym hai. a positive toh x positive, y negative = Quadrant IV. a negative karo toh dono ke sign flip ho jaate hain: x negative, y positive = Quadrant II. a = 0 par origin. Aur sundar observation: is point ki dono axis se doori hamesha barabar hai |a|, kyunki x aur y ka size same hai, sirf sign ulta hai. Aage Ch 2 mein y = -x line padhoge, ye usi ka preview hai.*

Answer: (i) Quadrant IV when a > 0; (ii) Quadrant II when a < 0; (iii) the origin when a = 0; (iv) distances from both axes are always equal (each equals |a|).

Q10 · Basic · Concept: Axis membership (T1/T13)

Q10. From the list (2, 0), (0, 2), (-5, 0), (0, -5), (7, 0), (0, 7), (-3, -3), pick out (i) all points on the x-axis, (ii) all points on the y-axis, (iii) any point equidistant from both axes, and (iv) state how many of the listed points lie in a quadrant.

SOLUTION

Step 1: Given: eight points. Needed: classify by axis membership and symmetry.

Step 2: x-axis means y = 0: the points (2, 0), (-5, 0), (7, 0) qualify.

Step 3: y-axis means x = 0: the points (0, 2), (0, -5), (0, 7) qualify.

Step 4: Equidistant from both axes means |x| = |y|. Only (-3, -3) satisfies this (both distances 3).

Step 5: Six points lie on the axes, and exactly one point, (-3, -3), lies strictly inside a quadrant (Quadrant III).

Hinglish: *Simple filter lagao: y = 0 hai toh x-axis ka banda, x = 0 hai toh y-axis ka. Dono conditions fail ho jaayein tabhi point kisi quadrant ke andar hai. Yahan sirf (-3, -3) quadrant mein girta hai. Aur |x| = |y| wala idea yaad rakho - dono axis se barabar doori ka matlab. Ye chhota sa observation aage bahut bade questions kholta hai.*

Answer: (i) (2, 0), (-5, 0), (7, 0) (ii) (0, 2), (0, -5), (0, 7) (iii) (-3, -3) (iv) exactly 1 point lies in a quadrant (Q III).

Q11 · Moderate · Concept: Distance formula (T4)

Q11. Find the distance between A(1, 2) and B(4, 6) using (i) the helper-triangle method and (ii) the direct formula. Confirm both give the same value.

SOLUTION

Step 1: Given: A(1, 2), B(4, 6). Needed: AB by two methods.

Step 2: Helper triangle: the horizontal leg is |4 - 1| = 3 and the vertical leg is |6 - 2| = 4.

Step 3: By Pythagoras, AB = $\sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 4: Direct formula: AB = $\sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{(3^2 + 4^2)} = 5$.

Step 5: Both methods agree: AB = 5 units. The helper triangle simply explains where the formula comes from.

» **Why this step?** Showing both methods builds the habit of verifying with geometry - on hard questions this catches sign slips instantly.

Hinglish: *Yahi chapter ka hero question hai. Helper triangle mein humne sirf differences liye - 3 aur 4 - aur wo 3-4-5 triple nikla, isliye answer 5. Formula bhi wahi karta hai, bas ek line mein. Dono ka jawab same aana hi chahiye kyunki formula Pythagoras ka hi shortcut hai. Aage se seedha formula use karo, par agar kabhi doubt ho, helper triangle bana lo.*

Answer: AB = 5 units by both methods.

Q12 · Moderate · Concept: Distance formula with negatives (T4)

Q12. Find the distance between P(-2, 3) and Q(4, -5). Then find the distance of the midpoint of PQ from the origin.

SOLUTION

Step 1: Given: P(-2, 3), Q(4, -5). Needed: PQ, then distance of midpoint M from O.

Step 2: Differences: x-difference = 4 - (-2) = 6; y-difference = -5 - 3 = -8.

Step 3: PQ = $\sqrt{(6^2 + (-8)^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$. (The minus vanished on squaring, as promised.)

Step 4: Midpoint M = $((-2 + 4)/2, (3 + (-5))/2) = (2/2, -2/2) = (1, -1)$.

Step 5: Distance of M from origin = $\sqrt{(1^2 + (-1)^2)} = \sqrt{2}$.

Step 6: So PQ = 10 units and OM = $\sqrt{2}$ units.

» **Why this step?** Squaring (-8) gives 64 - the sign never survives a square, so never 'pre-correct' it to 8.

Hinglish: *Negatives ke saath formula lagane mein sabse badi galti: log -8 ko 8 karke beech mein hi root le lete hain. Aisa kuch nahi karna - square laga do, sign khud khatam ho jayega. $6^2 + (-8)^2 = 36 + 64 = 100$, root 10. Midpoint = average of coordinates = (1, −1). Aur (1, −1) ka origin se distance $\sqrt{2}$. Chhota point: $\sqrt{2}$ ko decimal mein mat likho jab tak manga na ho.*

Answer: PQ = 10 units; OM = $\sqrt{2}$ units.

Q13 · Moderate · Concept: Collinearity test (T9)

Q13. Without plotting, check whether M(0, 0), A(3, 4) and G(6, 8) lie on the same straight line. State the reasoning principle you used.

SOLUTION

Step 1: Given: three points. Needed: collinearity verdict without plotting.

Step 2: Compute OM = $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$.

Step 3: Compute AG = $\sqrt{((6 - 3)^2 + (8 - 4)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 4: Compute MG = $\sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$.

Step 5: Test: the largest distance (10) equals the sum of the other two (5 + 5 = 10).

Step 6: Since MG = OM + AG exactly, the point A sits on the straight path from M to G; the three points are collinear.

Step 7: Principle used: on a straight line, the whole equals the sum of the parts (equality case of the triangle inequality). Any detour would make the total strictly larger.

Hinglish: *Ye 'triangle inequality equality' ka test hai - bina graph banaye line check karne ka tareeka. Teen distances nikaalo: 5, 5, 10. Agar sabse bada = baaki dono ka sum, toh teeno ek hi line par hain. Agar sum bada aaye (strictly), toh beech ka point line se hat ke baitha hai. Yahan 5 + 5 = 10 exact match - collinear. Ek baat yaad rakho: values EXACT rakho, round mat karo, warna test galat ho sakta hai.*

Answer: Yes - OM + AG = MG (5 + 5 = 10), so M, A, G are collinear.

Q14 · Moderate · Concept: Reflection chain (T7)

Q14. The point P(4, -3) is reflected in the x-axis to get P_1, and then P_1 is reflected in the y-axis to get P_2. Write the coordinates of P_1 and P_2, and find the length PP_2.

SOLUTION

Step 1: Given: P(4, -3) and a chain of two reflections. Needed: images and PP_2.

Step 2: Reflection in the x-axis keeps x and flips the sign of y: $P_1 = (4, 3)$.

Step 3: Reflection in the y-axis keeps y and flips the sign of x: $P_2 = (-4, 3)$.

Step 4: PP_2 joins (4, -3) to (-4, 3): x-difference = -4 - 4 = -8, y-difference = 3 - (-3) = 6.

Step 5: $PP_2 = \sqrt{((-8)^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 6: Check with preserved distances: $PP_1 = |3 - (-3)| = 6$ (vertical segment), $P_1P_2 = |(-4) - 4| = 8$ (horizontal segment), and indeed $6^2 + 8^2 = 10^2$, so the numbers are internally consistent.

Hinglish: *Reflection ke do simple rules: x-axis mein ungli x-axis par - sirf y ka sign palto. y-axis mein - sirf x ka sign palto. Chain mein dono lagane se (4, -3) pehle (4, 3) bana, phir (-4, 3). Ab distance: 8 aur 6 ka 8-6-10 triple! Aur verification sundar hai: $PP_1 = 6$ vertical, $P_1P_2 = 8$ horizontal, inka Pythagoras bhi 10 hi deta hai. Double confirmation mil gaya.*

Answer: $P_1 = (4, 3)$, $P_2 = (-4, 3)$, $PP_2 = 10$ units.

Q15 · Moderate · Concept: Triangle classification (T6)

Q15. The vertices of a triangle are A(1, 2), B(5, 2) and C(3, 6). Find all three side lengths and classify the triangle. Is it equilateral? Is it right-angled?

SOLUTION

Step 1: Given: three vertices. Needed: side lengths and classification.

Step 2: AB: same y-coordinate (2), so AB = |5 - 1| = 4.

Step 3: AC = $\sqrt{((3 - 1)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20} = 2\sqrt{5}$.

Step 4: BC = $\sqrt{((3 - 5)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20} = 2\sqrt{5}$.

Step 5: Two sides equal (AC = BC), one different: the triangle is isosceles, not equilateral.

Step 6: Right-angle check by the converse of Pythagoras: is the square of the longest side equal to the sum of squares of the other two? Longest is $2\sqrt{5}$, its square 20. Others: $4^2 = 16$ and $(2\sqrt{5})^2 = 20$, and 16 + 20 = 36 which is not 20.

Step 7: Not right-angled. In fact $4 + 2\sqrt{5} + 2\sqrt{5}$ is the perimeter = $4 + 4\sqrt{5}$.

Hinglish: *Side nikaalne ke baad classification ka recipe: (1) do barabar? isosceles. (2) teen barabar? equilateral. (3) Pythagoras satisfy? right-angled. Yahan AC aur BC dono $2\sqrt{5}$ nikle - isosceles pakka. Right angle ka test: sabse lambi side ka square = baaki dono ke squares ka sum? 20 vs 36 - fail. Toh na equilateral, na right. Simple hai, bas teeno tests order mein lagao.*

Answer: AB = 4, AC = BC = $2\sqrt{5}$; the triangle is isosceles; it is neither equilateral nor right-angled.

Q16 · Moderate · Concept: Square verification (T6)

Q16. Show that the points (2, 2), (6, 2), (6, 6) and (2, 6) are the corners of a square. Then find its area and the length of its diagonal.

SOLUTION

Step 1: Given: four points in cyclic order. Needed: prove square, then area and diagonal.

Step 2: Bottom side (2, 2) to (6, 2): horizontal, length 4.

Step 3: Right side (6, 2) to (6, 6): vertical, length 4.

Step 4: Top side (6, 6) to (2, 6): horizontal, length 4. Left side (2, 6) to (2, 2): vertical, length 4.

Step 5: All four sides equal to 4, and adjacent sides meet at the corners of an axis-aligned rectangle (horizontal meets vertical), so every corner angle is 90 degrees.

Step 6: Hence the figure is a square of side 4. Area = side^2 = 16 square units.

Step 7: Diagonal d joins (2, 2) to (6, 6): d = $\sqrt{((6 - 2)^2 + (6 - 2)^2)} = \sqrt{32} = 4\sqrt{2}$. This matches the square-diagonal relation d = side x $\sqrt{2}$.

Hinglish: *Square prove karne ka two-step formula: (1) chaaron sides barabar, (2) koi bhi ek corner ka angle 90 (yahan horizontal aur vertical lines mil rahe hain toh automatic 90). Side 4 hai, toh area 16 aur diagonal side x $\sqrt{2} = 4\sqrt{2}$. Distance formula se bhi verify kiya - $\sqrt{32} = 4\sqrt{2}$, dono match. Aisi cross-checking exam mein full marks dilati hai.*

Answer: It is a square of side 4; area = 16 sq units; diagonal = $4\sqrt{2}$ units.

Q17 · Moderate · Concept: Right-angle by converse Pythagoras (T6)

Q17. A(1, 1), B(4, 1) and C(4, 5) are three points. Use the distance formula and the converse of the Baudhayana-Pythagoras Theorem to show that angle B is a right angle, and find the area of triangle ABC.

SOLUTION

Step 1: Given: three vertices. Needed: prove right angle at B, then area.

Step 2: AB: same y = 1, so AB = |4 - 1| = 3.

Step 3: BC: same x = 4, so BC = |5 - 1| = 4.

Step 4: AC = $\sqrt{((4 - 1)^2 + (5 - 1)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 5: Check the converse: AC^2 = 25 and AB^2 + BC^2 = 9 + 16 = 25. They are equal, so the angle opposite AC - which is angle B - is a right angle.

Step 6: Area of a right triangle = (1/2) x leg x leg = (1/2) x 3 x 4 = 6 square units.

Step 7: The hidden triple is again 3-4-5.

Hinglish: *Converse Pythagoras ka use yaad karo: agar $c^2 = a^2 + b^2$, toh c ke saamne wala angle 90 hai. Yahan AC sabse lambi hai (5), aur $3^2 + 4^2 = 5^2$ nikla, toh B par right angle. Area ke liye right triangle ka bonus: do legs ko base-height maan lo, (1/2)(3)(4) = 6. Coordinate geometry mein area aise hi nikalte hain jab triangle axis-aligned legs rakhta hai.*

Answer: Angle B = 90 degrees (since $3^2 + 4^2 = 5^2 = AC^2$); area = 6 square units.

Q18 · Moderate · Concept: Missing value, two roots (T5)

Q18. The distance between the points (k, 3) and (4, 3) is 7. Find all possible values of k. How many points satisfy the condition?

SOLUTION

Step 1: Given: two points sharing y = 3, distance 7. Needed: k.

Step 2: Both points have y-coordinate 3, so the segment is horizontal and distance = |x2 - x1|.

Step 3: So |4 - k| = 7. An absolute-value equation splits into two cases.

Step 4: Case 1: 4 - k = 7, giving k = -3. Case 2: 4 - k = -7, giving k = 11.

Step 5: Verification: from (4, 3) to (-3, 3): |4 - (-3)| = 7. From (4, 3) to (11, 3): |11 - 4| = 7. Both work.

Step 6: So there are TWO points, (-3, 3) and (11, 3), one on each side of (4, 3).

» **Why this step?** The absolute value MUST open into two cases. Dropping one case is the single most common error in 'find k' questions.

Hinglish: *Yahi wo question hai jahan aadhe marks ud jaate hain! |4 - k| = 7 ke DO solutions hote hain: point (4, 3) ke left mein bhi ek aur right mein bhi ek. k = -3 aur k = 11. Agar tumne sirf ek nikala, toh samajh lo absolute value ke do case khulane bhool gaye. Verification bhi kar li - dono se distance 7 hi aati hai.*

Answer: k = -3 or k = 11; there are two such points.

Q19 · Moderate · Concept: Circle membership (T10)

Q19. A circle has centre the origin and radius 5. Classify each point as on, inside, or outside the circle: (3, 4), (4, 2), (-3, 4), (0, 5), (-4, -4).

SOLUTION

Step 1: Given: circle $x^2 + y^2 = 25$. Needed: classify five points.

Step 2: Method: compare $x^2 + y^2$ with $r^2 = 25$. Equal means on the circle; smaller means inside; larger means outside.

Step 3: (3, 4): 9 + 16 = 25, exactly equal, so ON the circle.

Step 4: (4, 2): 16 + 4 = 20, which is less than 25, so INSIDE (distance $\sqrt{20}$ slightly under 5).

Step 5: (-3, 4): 9 + 16 = 25, so ON the circle (signs never matter after squaring).

Step 6: (0, 5): 0 + 25 = 25, so ON the circle.

Step 7: (-4, -4): 16 + 16 = 32, greater than 25, so OUTSIDE (distance $\sqrt{32} = 4\sqrt{2}$ approx 5.66).

» **Why this step?** Comparing squares instead of square roots avoids every rounding issue - 20 vs 25 is cleaner than 4.47 vs 5.

Hinglish: *Circle membership ka super-fast test: $x^2 + y^2$ ko r^2 se compare karo. Barabar = line par, kam = andar, zyada = bahar. Root nikalne ki zaroorat hi nahi. (-3, 4) dekho - minus hone ke bawajood square ke baad 25 hi aaya, toh circle par hi hai. (-4, -4) dono square 16+16 = 32 > 25, bahar. Ye technique time bhi bachati hai aur rounding errors se bhi bachati hai.*

Answer: On the circle: (3, 4), (-3, 4), (0, 5). Inside: (4, 2). Outside: (-4, -4).

Q20 · Moderate · Concept: Missing y-value on a circle (T5/T10)

Q20. The point (3, y) lies at distance 5 from the origin. Find all possible values of y. Where do the two possible points sit relative to each other?

SOLUTION

Step 1: Given: distance from O equals 5. Needed: y.

Step 2: Distance from origin: $\sqrt{(3^2 + y^2)} = 5$.

Step 3: Square both sides (both sides are non-negative, so squaring is safe): $9 + y^2 = 25$.

Step 4: So $y^2 = 16$, and taking square roots gives y = 4 or y = -4.

Step 5: Verification: (3, 4) gives $\sqrt{(9 + 16)} = 5$, and (3, -4) also gives 5. Both are genuine solutions.

Step 6: The two points (3, 4) and (3, -4) are reflections of each other in the x-axis - same x, opposite y - and both lie on the circle of radius 5.

Hinglish: *Yahan bhi do answers aayenge, aur wajah samjho: circle ke ek vertical line par do points hote hain - ek upar, ek neeche. y = 4 aur y = -4. Dono verify kiye, dono sahi. Aur ye dono points x-axis ke reflection partners hain. Ye pattern (quadratic se do roots) Chapter 1 ka signature hai - jab bhi circle ya distance se equation bane, do*

answers ki ummeed rakho.

Answer: y = 4 or y = -4; the two points are mirror images in the x-axis, both on the circle of radius 5.

Q21 · Difficult · Concept: Perimeter + classification (T6/T12)

Q21. Find the perimeter of the triangle with vertices A(2, -2), B(-2, 1) and C(5, 2). Along the way, show that the triangle is isosceles.

SOLUTION

Step 1: Given: three vertices. Needed: all sides, isosceles check, perimeter.

Step 2: AB: differences are (-2 - 2) = -4 and (1 - (-2)) = 3. AB = √(16 + 9) = √25 = 5.

Step 3: BC: differences are (5 - (-2)) = 7 and (2 - 1) = 1. BC = √(49 + 1) = √50 = 5√2.

Step 4: CA: differences are (2 - 5) = -3 and (-2 - 2) = -4. CA = √(9 + 16) = √25 = 5.

Step 5: AB = CA = 5, so the triangle is isosceles with apex at A.

Step 6: Perimeter = AB + BC + CA = 5 + 5√2 + 5 = 10 + 5√2 units (approximately 17.07).

Step 7: Keep the exact form 10 + 5√2 in the answer; convert to decimals only if the question asks.

Hinglish: *Do sides 5 nikli aur ek 5√2 - isosceles confirm. Perimeter mein exact value rakho: 10 + 5√2. Exam mein log √50 ko 7.07 leke ghumaa dete hain aur final answer gandaa ho jata hai. Rule yaad rakho: jab tak decimal manga nahi, root form hi rakho. Aur ek observation: (4, 3) wala 3-4-5 triple do baar mila - coordinate questions mein triples baar-baar ghuumte hain, unhe pakadna speed deta hai.*

Answer: Perimeter = 10 + 5√2 units; the triangle is isosceles (AB = CA = 5).

Q22 · Difficult · Concept: Reverse midpoint (T8)

Q22. M(-1, 3) is the midpoint of the segment joining A(2, -4) and B(x, y). Find B, and verify your answer by checking that MA = MB.

SOLUTION

Step 1: Given: midpoint and one endpoint. Needed: the other endpoint.

Step 2: Midpoint equations: (-1) = (2 + x)/2 and 3 = (-4 + y)/2.

Step 3: Solve the first: multiply by 2: -2 = 2 + x, so x = -4.

Step 4: Solve the second: 6 = -4 + y, so y = 10.

Step 5: So B = (-4, 10).

Step 6: Verify MA: differences (2 - (-1)) = 3 and (-4 - 3) = -7, MA = √(9 + 49) = √58.

Step 7: Verify MB: differences (-4 - (-1)) = -3 and (10 - 3) = 7, MB = √(9 + 49) = √58. Equal, so B is correct.

» **Why this step?** Reverse midpoint = undo the average: multiply by 2, then subtract the known endpoint. Verification with equal distances is cheap and catches slips.

Hinglish: *Midpoint wala equation ulta chalana hai: average diya hai, original nikalna hai. 2 se multiply karo, phir known endpoint minus karo. x ke liye: (-1)(2) - 2 = -4. y ke liye: (3)(2) - (-4) = 10. Verification zaroor karo - MA aur MB dono √58 aaye, matlab M sach mein beech mein baitha hai. Bina verification ke answer likhna adhoora hai.*

Answer: B = (-4, 10), verified since MA = MB = √58.

Q23 · Difficult · Concept: Parallelogram check (T6)

Q23. Show that the points A(0, 0), B(4, 0), C(6, 3), D(2, 3) form a parallelogram, and then show that it is NOT a rectangle.

SOLUTION

Step 1: Given: four points in order. Needed: parallelogram yes, rectangle no.

Step 2: AB: from (0, 0) to (4, 0): horizontal, length 4.

Step 3: CD: from (6, 3) to (2, 3): horizontal, length |6 - 2| = 4. So AB = CD and both are horizontal - parallel and equal.

Step 4: BC: from (4, 0) to (6, 3): $\sqrt{(2^2 + 3^2)} = \sqrt{13}$. DA: from (2, 3) to (0, 0): $\sqrt{(2^2 + 3^2)} = \sqrt{13}$. So BC = DA, both with the same slant.

Step 5: Both pairs of opposite sides are equal (and clearly parallel: horizontal pairs, and BC, DA share the same step pattern 2-across, 3-up). So ABCD is a parallelogram.

Step 6: Rectangle test: a parallelogram is a rectangle exactly when its diagonals are equal.

Step 7: AC = $\sqrt{(6^2 + 3^2)} = \sqrt{45} = 3\sqrt{5}$. BD = $\sqrt{((2 - 4)^2 + (3 - 0)^2)} = \sqrt{(4 + 9)} = \sqrt{13}$.

Step 8: Since $3\sqrt{5}$ is not equal to $\sqrt{13}$, the diagonals are unequal, so the parallelogram is not a rectangle.

» **Why this step?** The diagonal test is the cleanest rectangle check using ONLY distance - no angle measurement needed.

Hinglish: *Parallelogram proof ke liye opposite sides barabar kafi hain (parallel toh step pattern se dikh hi raha hai). Rectangle check ka shortcut yaad rakho: parallelogram mein diagonals barabar hona = rectangle. Yahan AC = 3√5 aur BD = √13 - barabar nahi, toh rectangle nahi. Ye diagonal-test chapter ke andar hi aata hai kyunki wo sirf distance formula use karta hai. Sketch banaoge toh slanted box dikhega - jaise side se dhakela hua rectangle.*

Answer: AB = CD = 4 and BC = DA = √13, so ABCD is a parallelogram; its diagonals $3\sqrt{5}$ and √13 are unequal, so it is not a rectangle.

Q24 · Difficult · Concept: Equidistant point on an axis (T5/T8)

Q24. Find the point on the y-axis that is equidistant from P(-2, 0) and Q(6, 4). Verify your answer.

SOLUTION

Step 1: Given: unknown point R on the y-axis. Needed: R equidistant from P and Q.

Step 2: On the y-axis the point has the form R(0, y) - fix the zero first.

Step 3: $RP^2 = (0 - (-2))^2 + (y - 0)^2 = 4 + y^2$.

Step 4: $RQ^2 = (0 - 6)^2 + (y - 4)^2 = 36 + (y - 4)^2$.

Step 5: Set them equal: $4 + y^2 = 36 + y^2 - 8y + 16$.

Step 6: Cancel y^2 from both sides (this is why the equation stays linear and easy): $4 = 52 - 8y$.

Step 7: So $8y = 48$, giving $y = 6$. Hence R = (0, 6).

Step 8: Verify: RP = $\sqrt{(4 + 36)} = \sqrt{40} = 2\sqrt{10}$. RQ = $\sqrt{(36 + (6 - 4)^2)} = \sqrt{(36 + 4)} = \sqrt{40} = 2\sqrt{10}$. Equal, so R is correct.

» **Why this step?** Comparing squared distances avoids square roots entirely; and y^2 cancels, leaving a linear equation.

Hinglish: *Y-axis par point = (0, y) - pehle yeh lock karo. Phir dono distances ke SQUARES barabar rakho, root ka jhanjhat khatam. Sabse sundar cheez: y^2 dono taraf cancel ho gaya aur seedha linear equation mil gaya, y = 6. Verify karo: dono distances 2√10. Aisa equidistant point technically perpendicular bisector par baithta hai - yahan*

humne uska sirf y-axis wala intersection nikala. 2√10 ko 6.32 likhne ki zaroorat nahi.

Answer: R = (0, 6), verified since RP = RQ = 2√10.

Q25 · Difficult · Concept: Right isosceles triangle (T6/T14)

Q25. A right-angled isosceles triangle has its right angle at the origin and its two equal legs of length 6 lying along the positive x-axis and positive y-axis. Write the coordinates of all three vertices, find the hypotenuse, and find the area. Generalise: if the legs are a, what is the hypotenuse?

SOLUTION

Step 1: Given: right angle at O, legs 6 along positive axes. Needed: vertices, hypotenuse, area, general rule.

Step 2: One vertex is O(0, 0). Walking 6 along the positive x-axis gives (6, 0); walking 6 up the positive y-axis gives (0, 6).

Step 3: Hypotenuse joins (6, 0) to (0, 6): length = $\sqrt{(6^2 + 6^2)} = \sqrt{72} = 6\sqrt{2}$.

Step 4: Area = (1/2) x base x height = (1/2) x 6 x 6 = 18 square units.

Step 5: General case: legs a and a give hypotenuse $\sqrt{(a^2 + a^2)} = \sqrt{(2a^2)} = a\sqrt{2}$.

Step 6: This is exactly the PDF's special-case observation: putting a = b in the general Pythagoras relation forces hypotenuse = a√2 for isosceles right triangles.

Hinglish: *Right isosceles ka signature result: hypotenuse hamesha leg ka √2 guna hota hai. 6 ka leg toh 6√2. Yaad rakhne ka trick: square ka diagonal bhi side ka √2 guna hota hai - same baat hai, kyunki aadha square hi toh ye triangle hai! Area half-product = 18. Generalisation likhna (a√2) exam mein extra impression banata hai kyunki PDF mein special-case wala box isi baat ko highlight karta hai.*

Answer: Vertices (0, 0), (6, 0), (0, 6); hypotenuse = 6√2; area = 18 sq units; in general the hypotenuse is a√2.

Q26 · Difficult · Concept: City-grid application (T11)

Q26. In a city, streets run in two perpendicular directions, 200 m apart. Intersections are labelled (row, column). An auto rickshaw goes from intersection (1, 1) to intersection (4, 3). (i) What is the shortest distance it can travel moving only along streets? (ii) What is the straight-line (as-the-crow-flies) distance? (iii) How much shorter is the direct route?

SOLUTION

Step 1: Given: grid spacing 200 m, start (1, 1), end (4, 3). Needed: street distance, direct distance, saving.

Step 2: Moving along streets from row 1 to row 4 needs 3 steps; from column 1 to column 3 needs 2 steps.

Step 3: Shortest street path = 3 + 2 = 5 steps = 5 x 200 = 1000 m.

Step 4: Direct distance uses the distance formula with coordinate gap 3 rows and 2 columns: $\sqrt{(3^2 + 2^2)}$ x 200 = √13 x 200 approx 3.606 x 200 approx 721 m.

Step 5: Saving = 1000 - 200√13 approx 1000 - 721 = 279 m (about 28 percent).

Step 6: Note the concept: along-streets distance adds the gaps; direct distance applies Pythagoras to the same gaps.

Hinglish: *City grid hi bada coordinate plane hai - bas unit 200 m hai. Galiyon mein chalna hai toh horizontal + vertical steps JODO (3 + 2 = 5 blocks = 1000 m). Seedha udna hai toh Pythagoras lagao: √(9 + 4) = √13 blocks. Difference dekho - seedha rasta karib 280 m bachata hai. Yahi idea PDF ke athletics track aur map questions mein bhi chalta hai: grid distance vs straight distance.*

Answer: (i) 1000 m (ii) 200√13 approx 721 m (iii) approx 279 m shorter

Q27 · Difficult · Concept: Rhombus from coordinates (T6)

Q27. The points (0, 5), (-3, 0), (0, -5) and (3, 0) are the corners of a quadrilateral taken in order. Show that it is a rhombus and find the lengths of its diagonals and its area.

SOLUTION

Step 1: Given: four symmetric points. Needed: prove rhombus, diagonals, area.

Step 2: Side from (0, 5) to (-3, 0): $\sqrt{(3^2 + 5^2)} = \sqrt{34}$.

Step 3: Side from (-3, 0) to (0, -5): $\sqrt{(3^2 + 5^2)} = \sqrt{34}$. By the same computation each of the remaining two sides is $\sqrt{(3^2 + 5^2)} = \sqrt{34}$.

Step 4: All four sides are equal to $\sqrt{34}$, so the quadrilateral is a rhombus (a parallelogram with equal sides; symmetry about both axes guarantees the parallelogram behaviour, and can be double-checked: opposite sides are translations of each other).

Step 5: Diagonal 1 joins (0, 5) to (0, -5): vertical, length 10.

Step 6: Diagonal 2 joins (-3, 0) to (3, 0): horizontal, length 6.

Step 7: Area of a rhombus = (1/2) x product of diagonals = (1/2) x 10 x 6 = 30 square units.

Hinglish: *Chaaron sides √34 - rhombus pakka. Diagonals yahan axis par hi baithi hain, isliye unki length seedhi mili: 10 aur 6. Rhombus ka area = (1/2) d1 d2 = 30 - ye formula yaad rakho, square/rectangle ke bhai ki tarah. Ye PDF ke Ch 6 se connect hota hai jahan rhombus-area questions hain, par classification toh pure Chapter 1 ka khel hai (distance formula).*

Answer: All sides √34, so it is a rhombus; diagonals 10 and 6; area = 30 square units.

Q28 · Difficult · Concept: Algebraic distance (T4/T14)

Q28. Find the distance between the points (a + b, a - b) and (a - b, a + b), where a and b are real numbers with b not equal to 0. The answer must be in terms of b.

SOLUTION

Step 1: Given: two symbolic points. Needed: distance in terms of b.

Step 2: x-difference: (a + b) - (a - b) = 2b.

Step 3: y-difference: (a - b) - (a + b) = -2b.

Step 4: Distance = $\sqrt{((2b)^2 + (-2b)^2)} = \sqrt{(4b^2 + 4b^2)} = \sqrt{(8b^2)}$.

Step 5: Simplify: $\sqrt{(8b^2)} = \sqrt{8} \times \sqrt{(b^2)} = 2\sqrt{2} \times |b|$.

Step 6: Since b is not zero, for positive b this is $2\sqrt{2}$ b. The cleanest exact answer is $2\sqrt{2}\,|b|$; note that a cancels out completely - the distance does not depend on a at all.

» Why this step? Notice how the a-terms cancel inside both differences. Spotting cancellations before squaring is a mark of an algebra-fluent student.

Hinglish: *Symbolic distance question - numbers ki jagah letters, darne ki zaroorat nahi. Difference lo, a dono jagah cancel ho gaya, bacha 2b aur -2b. Square ke baad $8b^2$, root se $2\sqrt{2}\,|b|$. Bar (absolute value) isliye kyunki distance kabhi negative nahi hoti. Interesting observation: answer a par depend hi nahi karta - dono points a se same doori par slide karte hain, unke beech ka gap sirf b se banta hai.*

Answer: Distance = $\sqrt{(8b^2)} = 2\sqrt{2}\,|b|$ (independent of a).

Q29 · Difficult · Concept: Equidistant locus (T5/T14)

Q29. A point P(x, y) moves so that its distance from A(3, 4) always equals its distance from B(-1, 6). Show that x and y must satisfy the relation y = 2x + 3, and verify that the point (0, 3) obeys it.

SOLUTION

Step 1: Given: PA = PB for a moving point. Needed: the relation between x and y.

Step 2: Write squared distances (roots dropped on both sides - legal since both sides are non-negative).

Step 3: $PA^2 = (x - 3)^2 + (y - 4)^2 = x^2 - 6x + 9 + y^2 - 8y + 16$.

Step 4: $PB^2 = (x + 1)^2 + (y - 6)^2 = x^2 + 2x + 1 + y^2 - 12y + 36$.

Step 5: Set $PA^2 = PB^2$: $x^2 - 6x + 9 + y^2 - 8y + 16 = x^2 + 2x + 1 + y^2 - 12y + 36$.

Step 6: Cancel x^2 and y^2 from both sides: $-6x - 8y + 25 = 2x - 12y + 37$.

Step 7: Move everything to one side: $-8x + 4y - 12 = 0$, that is $4y = 8x + 12$, giving $y = 2x + 3$.

Step 8: Verify (0, 3): $PA = \sqrt{((0 - 3)^2 + (3 - 4)^2)} = \sqrt{(9 + 1)} = \sqrt{10}$. $PB = \sqrt{((0 + 1)^2 + (3 - 6)^2)} = \sqrt{(1 + 9)} = \sqrt{10}$. Equal - the point obeys the condition.

Hinglish: *Ye Chapter 1 ka sabse powerful idea hai: distance ki equality ek RELATION bana deti hai. Square karke x^2 aur y^2 cancel kar do - jo bacha wo straight line hai, $y = 2x + 3$. Is line ke har point dono A aur B se barabar door hai (isko perpendicular bisector kehte hain - Class 11 mein formally aayega). Verification: (0, 3) se dono distances $\sqrt{10}$. Jab bhi 'equidistant from two points' dikhe, yahi square-and-cancel recipe chalao.*

Answer: PA = PB forces y = 2x + 3; verified at (0, 3) where PA = PB = $\sqrt{10}$.

Q30 · Difficult · Concept: Circle membership with bigger radius (T10)

Q30. Consider the circle with centre the origin and radius 13. (i) Verify that (5, 12), (12, 5), (-12, 5) and (0, 13) all lie on the circle. (ii) Determine whether (6, 12) and (4, 12) lie inside, on, or outside it.

SOLUTION

Step 1: Given: circle $x^2 + y^2 = 169$. Needed: classify six points.

Step 2: (5, 12): 25 + 144 = 169, so on the circle (the 5-12-13 triple).

Step 3: (12, 5): 144 + 25 = 169, on the circle (same triple, swapped).

Step 4: (-12, 5): 144 + 25 = 169, on the circle (signs vanish on squaring).

Step 5: (0, 13): 0 + 169 = 169, on the circle (topmost point).

Step 6: (6, 12): 36 + 144 = 180 which is greater than 169, so OUTSIDE the circle.

Step 7: (4, 12): 16 + 144 = 160 which is less than 169, so INSIDE the circle.

» **Why this step?** The 5-12-13 triple is the second most useful triple after 3-4-5; radius 13 questions are built around it.

Hinglish: *Radius 13 = 5-12-13 triple ka ghar. Chaaron points check karo - sab 169 de rahe hain, sab circle par. Phir (6, 12) aur (4, 12) ka fun part: 180 > 168... nahi, 180 > 169 toh bahar; 160 < 169 toh andar. Dekho kamaal: (6, 12) aur (4, 12) dono lagbhag same jagah hain, par ek bahar aur ek andar! Circle ki seedhi borderline - yahi precision Chapter 1 sikhata hai.*

Answer: All four triple points lie on the circle; (6, 12) is outside; (4, 12) is inside.

Q31 · High · Concept: Trisection points (T8)

Q31. P and Q are the points of trisection of the segment AB, with P closer to A and Q closer to B, where A(4, 7) and B(16, -2). Find P and Q. Then verify that the midpoint of PQ coincides with the midpoint of AB.

SOLUTION

Step 1: Given: A(4, 7), B(16, -2), trisection needed. Needed: P, Q, and the midpoint check.

Step 2: Step size: one third of the way from A to B means adding one third of each gap. Gap in x: 16 - 4 = 12; gap in y: -2 - 7 = -9.

Step 3: One third of the gaps: 12/3 = 4 and -9/3 = -3.

Step 4: P (closer to A) = A + one step = (4 + 4, 7 - 3) = (8, 4).

Step 5: Q = A + two steps = (4 + 8, 7 - 6) = (12, 1). Equivalently Q = B - one step = (16 - 4, -2 + 3) = (12, 1). Both give the same.

Step 6: Midpoint of P and Q = ((8 + 12)/2, (4 + 1)/2) = (10, 2.5).

Step 7: Midpoint of A and B = ((4 + 16)/2, (7 - 2)/2) = (10, 2.5). They coincide - the check passes.

» Why this step? The middle of the two trisectors is always the midpoint of the whole segment - a beautiful structural fact worth remembering for instant verification.

Hinglish: *Trisection matlab segment ko 3 barabar tukdon mein kaatna. Fundaa simple: total gap ko 3 se divide karo, wo ek 'step' hai. P = A + ek step, Q = A + do step. Yahan steps (4, -3) the. Aur jo self-check diya na - PQ ka midpoint hamesha AB ke midpoint ke barabar hota hai - ye free verification hai, exam mein zaroor lagao. (10, 2.5) = (10, 2.5), perfect match.*

Answer: P(8, 4), Q(12, 1); both midpoints equal (10, 2.5), verification passes.

Q32 · High · Concept: Missing rectangle vertex (T6/T8)

Q32. Three vertices of a rectangle are A(1, 2), B(5, 2) and C(5, 7). Find the fourth vertex D, and verify the rectangle property that its diagonals are equal.

SOLUTION

Step 1: Given: three consecutive vertices. Needed: D and the diagonal check.

Step 2: AB is horizontal (y = 2 for both) with length 4; BC is vertical (x = 5 for both) with length 5.

Step 3: In a rectangle the fourth side DA must be vertical (above A) and CD horizontal - so D must sit directly above A at the same height as C.

Step 4: Directly above A(1, 2) means x = 1; at the same height as C means y = 7. So D = (1, 7).

Step 5: Verify with side lengths: DA from (1, 7) to (1, 2) has length 5, CD from (5, 7) to (1, 7) has length 4 - matching the opposite sides.

Step 6: Diagonal AC = $\sqrt{((5 - 1)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$.

Step 7: Diagonal BD = $\sqrt{((1 - 5)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$. Equal, so the rectangle property holds.

Hinglish: *Rectangle ka chautha vertex pattern se milta hai: A ke seedha upar, C ke level par. x = 1 (A jaisa), y = 7 (C jaisa). Phir property check - diagonals barabar? Dono √41. Yahan √41 ko simplify nahi kar sakte (41 prime hai), toh root form hi final hai. Notice karo: har rectangle mein diagonals barabar hote hain - ye test humne parallelogram question (Q23) mein bhi use kiya tha, wahan fail hua tha, yahan pass.*

Answer: D = (1, 7); diagonals AC = BD = √41, confirming the rectangle.

Q33 · High · Concept: Area of right triangle from coordinates (T12)

Q33. A triangle has vertices O(0, 0), A(8, 0) and B(0, 6). Show that it is right-angled, find its hypotenuse, and compute its area and perimeter.

SOLUTION

Step 1: Given: three vertices, two of them on the axes. Needed: right angle, hypotenuse, area, perimeter.

Step 2: OA lies along the x-axis: length 8. OB lies along the y-axis: length 6.

Step 3: The x-axis and y-axis are perpendicular, and OA, OB lie along them, so the angle at O is 90 degrees.

Step 4: Hypotenuse AB = $\sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$ (the 6-8-10 triple).

Step 5: Area = (1/2) x OA x OB = (1/2) x 8 x 6 = 24 square units.

Step 6: Perimeter = 8 + 6 + 10 = 24 units. A charming coincidence: area 24 and perimeter 24.

Hinglish: *Jab triangle ke do sides axes par baithe hon, right angle free mein milta hai - axis parallel hi perpendicular hote hain. Phir 8-6-10 triple se hypotenuse 10. Area = 24, perimeter bhi 24 - dono number same! Aisi coincidences questions mein jaan-bujh ke rakhi jaati hain, toh apne practice questions mein bhi aisi numbers chuno toh checking easy hoti hai.*

Answer: Right angle at O; hypotenuse AB = 10; area = 24 sq units; perimeter = 24 units.

Q34 · High · Concept: Square from its diagonal (T14)

Q34. Two opposite corners of a square are P(-3, 2) and R(5, -4). Find the length of the diagonal, the side of the square, its area, and the coordinates of the other two corners.

SOLUTION

Step 1: Given: diagonal endpoints of a square. Needed: diagonal, side, area, remaining vertices.

Step 2: Diagonal PR = $\sqrt{((5 - (-3))^2 + (-4 - 2)^2)} = \sqrt{(8^2 + (-6)^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 3: For a square of side s, the diagonal is $s\sqrt{2}$, so $s = 10/\sqrt{2} = 5\sqrt{2}$.

Step 4: Area = $s^2 = (5\sqrt{2})^2 = 50$ square units. (Shortcut: area = diagonal^2/2 = 100/2 = 50.)

Step 5: Centre of the square = midpoint of PR = ((-3 + 5)/2, (2 - 4)/2) = (1, -1).

Step 6: Half-diagonal vector from centre to R is (5 - 1, -4 - (-1)) = (4, -3). To build the other diagonal, swap the components and flip one sign: (3, 4).

Step 7: Other corners: centre + (3, 4) = (4, 3) and centre - (3, 4) = (-2, -5).

Step 8: Verify: Q(4, 3) to P(-3, 2): $\sqrt{(7^2 + 1)} = \sqrt{50} = 5\sqrt{2}$ (a side). Q to R(5, -4): $\sqrt{(1 + 49)} = \sqrt{50}$ (a side). Q to S(-2, -5): $\sqrt{(36 + 64)} = 10$ (the other diagonal). All consistent.

» Why this step? The swap-and-flip move rotates the half-diagonal by 90 degrees - it is just Pythagoras thinking in vector form, and it is fully verifiable with the distance formula.

Hinglish: *Square ka diagonal diya hai toh side = diagonal/$\sqrt{2}$ = $5\sqrt{2}$, area = 50. Baaki do corners nikalne ka tareeka: pehle centre (midpoint) lo - (1, -1). Half-diagonal (4, -3) hai. Uske components SWAP karo aur ek sign flip karo: (3, 4) ban jayega - ye 90 degree ghuma hua half-diagonal hai. Centre mein plus-minus lagao: (4, 3) aur (-2, -5). Phir verification: sab sides $5\sqrt{2}$, doosra diagonal 10. Ye wala question HOTS-zone ka gateway hai - rotation ka idea samajh gaye toh aage Olympiad-level coordinate proofs easy lagenge.*

Answer: Diagonal = 10; side = $5\sqrt{2}$; area = 50 sq units; other corners Q(4, 3) and S(-2, -5).

Q35 · High · Concept: Collinearity - negative coordinates (T9)

Q35. Check, without plotting, whether the points (-2, -1), (0, 0) and (4, 3) lie on one straight line. Show your exact arithmetic.

SOLUTION

Step 1: Given: three points with negative coordinates. Needed: collinearity verdict with exact values.

Step 2: d1 = distance from (-2, -1) to (0, 0) = $\sqrt{(2^2 + 1^2)} = \sqrt{5}$.

Step 3: d2 = distance from (0, 0) to (4, 3) = $\sqrt{(16 + 9)} = \sqrt{25} = 5$.

Step 4: d3 = distance from (-2, -1) to (4, 3) = $\sqrt{(6^2 + 4^2)} = \sqrt{(36 + 16)} = \sqrt{52} = 2\sqrt{13}$.

Step 5: Collinearity requires the largest to equal the sum of the other two: is $2\sqrt{13} = \sqrt{5} + 5$?

Step 6: Compare squares of both sides: left squared is 52. Right squared is $(\sqrt{5} + 5)^2 = 5 + 25 + 10\sqrt{5} = 30 + 10\sqrt{5}$ approx $30 + 22.36 = 52.36$.

Step 7: Since 52 differs from $30 + 10\sqrt{5}$ (equivalently $\sqrt{5} + 5$ approx 7.236 while $2\sqrt{13}$ approx 7.211), the equality FAILS.

Step 8: So the three points are NOT collinear - the middle point sits slightly off the line through the outer two.

» **Why this step?** When radicals are involved, compare by squaring (both sides positive) rather than by decimals - decimals can mislead near-miss cases.

Hinglish: *Negative coordinates aane par bhi process same hai. Yahan twist ye hai ki distances roots mein hain - √5, 5, 2√13. Comparison exact karo: dono side ko square karo. Left 52, right 30 + 10√5 approx 52.36. Thoda sa difference hai - matlab NOT collinear. Ye question isliye rakha hai ki tum consider karo ki values kabhi approximate judge na karo. Agar maine jaldi mein 7.24 vs 7.21 dekh ke 'barabar hai' bol diya hota, toh galat ho jaata!*

Answer: Not collinear: $\sqrt{5} + 5$ is not equal to $2\sqrt{13}$ (52 is not equal to $30 + 10\sqrt{5}$).

Q36 · High · Concept: Applied screen/circle problem (T11/T10)

Q36. A computer screen is a rectangle 800 pixels wide and 600 pixels high, with the origin at the bottom-left corner. A circular icon of radius 80 has centre A(100, 150); another of radius 100 has centre B(250, 230). (i) Does any part of either icon stick out of the screen? (ii) Do the two icons overlap each other?

SOLUTION

Step 1: Given: screen 800 x 600, two circles with centres and radii. Needed: boundary check and overlap check.

Step 2: Icon A: leftmost x = 100 - 80 = 20 (inside, positive), rightmost x = 100 + 80 = 180 (inside, below 800), lowest y = 150 - 80 = 70 (inside), highest y = 150 + 80 = 230 (inside, below 600). Icon A is fully on the screen.

Step 3: Icon B: leftmost x = 250 - 100 = 150, rightmost = 250 + 100 = 350, lowest y = 230 - 100 = 130, highest y = 330. All within 0 to 800 and 0 to 600. Fully on screen too.

Step 4: Overlap test for two circles: compare the centre distance d with r1 + r2 and |r1 - r2|.

Step 5: Centre distance AB = $\sqrt{((250 - 100)^2 + (230 - 150)^2)} = \sqrt{(150^2 + 80^2)} = \sqrt{(22500 + 6400)} = \sqrt{28900}$ = 170 (the 8-15-17 triple scaled by 10).

Step 6: Now r1 + r2 = 180 and |r1 - r2| = 20. Since 20 is less than 170 and 170 is less than 180, the circles intersect in two points - they overlap partially.

Step 7: So: no icon leaves the screen, and the icons DO overlap.

Hinglish: *Screen check: har icon ka extreme left/right/top/bottom compute karo aur 0-800, 0-600 range se compare karo. Dono andar hain. Overlap ka test yaad rakho: centres ki doori vs radii ka sum. Distance 170 nikla (8-15-17 triple x 10!). 170 < 180 isliye overlap hota hai, aur 170 > 20 isliye poora ek doosre ke andar nahi hai - partial overlap. Ye 2-line test high-school geometry ka swiss knife hai.*

Answer: (i) Both icons lie fully inside the screen. (ii) Yes, the icons overlap: AB = 170 lies strictly between |100 - 80| = 20 and 80 + 100 = 180.

Q37 · High · Concept: Locus: equidistant from both axes (T14)

Q37. Describe the set of all points that are equidistant from the x-axis and the y-axis. If such a point must also lie at distance 5 from the origin, list ALL points that satisfy both conditions (exact values).

SOLUTION

Step 1: Given: equidistant from both axes, then distance-5 filter. Needed: locus, then exact points.

Step 2: Distance from the x-axis is |y|; distance from the y-axis is |x|. Equality |x| = |y| means y = x or y = -x.

Step 3: So the locus is the pair of lines y = x and y = -x (the two diagonals through the origin).

Step 4: Now add: $\sqrt{(x^2 + y^2)} = 5$. On the line y = x this becomes $\sqrt{(2x^2)} = \sqrt{2}\,|x| = 5$, so $|x| = 5/\sqrt{2} = (5\sqrt{2})/2$.

Step 5: That gives two points: $((5\sqrt{2})/2, (5\sqrt{2})/2)$ and $(-(5\sqrt{2})/2, -(5\sqrt{2})/2)$.

Step 6: On the line y = -x the same computation gives $((5\sqrt{2})/2, -(5\sqrt{2})/2)$ and $(-(5\sqrt{2})/2, (5\sqrt{2})/2)$.

Step 7: Total: four points, one in each quadrant, all at distance 5 from O and equidistant from both axes. Verify one: $((5\sqrt{2})/2)^2 + ((5\sqrt{2})/2)^2 = 25/2 + 25/2 = 25$, distance 5. Correct.

Hinglish: *Dono axis se barabar door = |x| = |y| = do diagonal lines y = x aur y = -x. Ab distance 5 bhi chahiye: $\sqrt{(x^2 + y^2)} = 5$. y = x rakhne par $2x^2 = 25$, $x = 5/\sqrt{2} = 5\sqrt{2}/2$. Rationalise kar dena toh aur saaf dikhta hai. Chaar points bane - har quadrant mein ek, sab circle (radius 5) par aur dono diagonals par. Locus thinking + circle membership ka combo - yahi HOTS ka taste hai.*

Answer: Locus: the two lines y = x and y = -x. The four points are (plus/minus $5\sqrt{2}/2$, plus/minus $5\sqrt{2}/2$) with matching signs on y = x and opposite signs on y = -x.

Q38 · High · Concept: Triple spotting at scale (T4/T12)

Q38. A triangle has vertices (0, 0), (5, 0) and (0, 12). Identify the right angle, find all three sides, and compute the area and the perimeter. Then state the general rule for a right triangle with legs a and b.

SOLUTION

Step 1: Given: vertices on both axes. Needed: sides, area, perimeter, general rule.

Step 2: The side from (0, 0) to (5, 0) lies on the x-axis (length 5); the side from (0, 0) to (0, 12) lies on the y-axis (length 12).

Step 3: These two axis lines are perpendicular, so the right angle is at the origin.

Step 4: Hypotenuse = $\sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = \sqrt{169} = 13$ (the 5-12-13 triple).

Step 5: Area = (1/2) x 5 x 12 = 30 square units.

Step 6: Perimeter = 5 + 12 + 13 = 30 units. Again area equals perimeter - these triple-based triangles love that coincidence.

Step 7: General rule: for a right triangle with legs a and b, hypotenuse = $\sqrt{(a^2 + b^2)}$, area = ab/2.

Hinglish: *5-12-13 triple ka full Dabangg avatar: area bhi 30, perimeter bhi 30. Chhota triangle 3-4-5 ka bhi area 6 aur perimeter 12 tha - wahan match nahi tha, par 5-12-13 aur 6-8-10 mein area = perimeter nikalta hai (8-6-10: area 24, perimeter 24). Aisi patterns notice karna maths ko interesting banata hai. General rule likhna zaroori hai - examiner ko dikhta hai ki tum case se general tak soch rahe ho.*

Answer: Right angle at the origin; sides 5, 12, 13; area = 30 sq units; perimeter = 30 units; general: hypotenuse $\sqrt{(a^2 + b^2)}$, area ab/2.

Q39 · High · Concept: Rectangle verification kit (T6)

Q39. K(1, 1), L(5, 1), M(5, 4) and N(1, 4) are joined in order. Using distances only, prove the figure is a rectangle, and find its perimeter, area and the length of each diagonal.

SOLUTION

Step 1: Given: four points in cyclic order. Needed: prove rectangle; perimeter, area, diagonals.

Step 2: KL: y fixed at 1, length |5 - 1| = 4 (horizontal).

Step 3: LM: x fixed at 5, length |4 - 1| = 3 (vertical).

Step 4: MN: y fixed at 4, length |5 - 1| = 4 (horizontal). NK: x fixed at 1, length |4 - 1| = 3 (vertical).

Step 5: Opposite sides are equal, and every side is axis-parallel, so adjacent sides meet at right angles: the figure is a rectangle.

Step 6: Diagonals: KM = $\sqrt{((5 - 1)^2 + (4 - 1)^2)} = \sqrt{(16 + 9)} = \sqrt{25} = 5$. LN = $\sqrt{((1 - 5)^2 + (4 - 1)^2)} = \sqrt{(16 + 9)} = 5$. Equal diagonals confirm the rectangle property.

Step 7: Perimeter = 2(4 + 3) = 14 units. Area = 4 x 3 = 12 square units. Both diagonals are 5.

Hinglish: *Ye question rectangle-proof ka full kit hai: (1) opposite sides barabar, (2) sab sides axis-parallel toh right angles automatic, (3) diagonals barabar - teeno checks ho gaye. 3-4-5 triple yahan diagonals mein chhupa hai! Perimeter 14, area 12, diagonal 5. Aise structured answers examiner ko bahut pasand aate hain kyunki har property alag line par verified hai.*

Answer: It is a rectangle (equal opposite sides, axis-parallel sides, equal diagonals of length 5); perimeter 14 units, area 12 sq units.

Q40 · High · Concept: Quadrant-restricted distances (T1/T4)

Q40. A point P lies in Quadrant II. Its perpendicular distance from the x-axis is 5 and from the y-axis is 3. Write the coordinates of P and find its distance from the origin. If a twin point P' in Quadrant I has the same axis distances, what is the length PP'?

SOLUTION

Step 1: Given: Quadrant II point with axis distances 5 (from x-axis) and 3 (from y-axis). Needed: P, OP, and PP'.

Step 2: Distance from the x-axis is the size of the y-coordinate: |y| = 5. Distance from the y-axis is |x| = 3.

Step 3: Quadrant II means x negative, y positive. So x = -3 and y = +5, giving P(-3, 5).

Step 4: OP = $\sqrt{((-3)^2 + 5^2)} = \sqrt{(9 + 25)} = \sqrt{34}$.

Step 5: P' in Quadrant I with the same distances: x = +3, y = +5, so P'(3, 5).

Step 6: PP' joins (-3, 5) to (3, 5): same y, so PP' = |3 - (-3)| = 6 (horizontal segment).

Step 7: Note the traps avoided: the distance from the x-axis (5) became the y-coordinate, and the Quadrant II condition fixed both signs.

» Why this step? Translating 'distance from the x-axis' into 'absolute y-value' is the key step; students who mix this up put 3 and 5 in the wrong slots.

Hinglish: *Sabse pehle translation: x-axis se doori 5 matlab |y| = 5; y-axis se doori 3 matlab |x| = 3. Ab quadrant II ka rule lagao - x negative, y positive: P(-3, 5). OP = √34 (koi triple nahi bana, root form chal raha hai). Quadrant I wala twin (3, 5). Dono ka y same, toh distance sirf x ka gap = 6. Ye question pure sign-discipline ka test hai - ek sign ulta kiya toh poora answer flip.*

Answer: P(-3, 5); OP = √34; P'(3, 5); PP' = 6 units.

Q41 · HOTS · Concept: Two intersection points on an axis (T5)

Q41. Find ALL points on the x-axis that lie at distance 10 from the point C(3, -6). Explain geometrically why there must be two answers.

SOLUTION

Step 1: Given: points (x, 0) at distance 10 from C(3, -6). Needed: all such x.

Step 2: Distance equation: $\sqrt{((x - 3)^2 + (0 - (-6))^2)} = 10$.

Step 3: Square both sides: $(x - 3)^2 + 36 = 100$.

Step 4: So $(x - 3)^2 = 64$, giving $x - 3 = 8$ or $x - 3 = -8$.

Step 5: Thus $x = 11$ or $x = -5$. The points are (11, 0) and (-5, 0).

Step 6: Verify: from (11, 0) to (3, -6): $\sqrt{(64 + 36)} = \sqrt{100} = 10$. From (-5, 0) to (3, -6): $\sqrt{(64 + 36)} = 10$. Both valid.

Step 7: Geometry: a circle of radius 10 centred at C(3, -6) cuts the x-axis at exactly two points because the centre sits 6 units above the axis, which is less than the radius 10 - the circle dips below the axis on both sides of $x = 3$.

» **Why this step?** The equation $(x - 3)^2 = 64$ is a disguised two-case problem; the geometric picture (circle cutting a line) tells you in advance how many answers to expect.

Hinglish: *Circle-axis ka milaap socho: C se radius 10 ka circle banao, wo x-axis ko kata kyun? Kyunki C axis se sirf 6 door hai aur radius 10 hai - circle axis ke paar niche tak jaata hai, dono taraf se cut karta hai. Equation mein bhi wahi dikh raha hai: $(x-3)^2 = 64$ ke do roots: $x = 11$ aur $x = -5$. Jab bhi square = positive number dikhe, do answers ready rakho. Aur agar centre axis se doori radius se ZYADA hoti, toh zero answers aate - wo case bhi soch ke dekho!*

Answer: (11, 0) and (-5, 0); two answers because the circle of radius 10 around C(3, -6) crosses the x-axis twice.

Q42 · HOTS · Concept: Circumcentre discovery (T10/T14)

Q42. For the right triangle with vertices A(0, 0), B(8, 0) and C(8, 6), find the point that is equidistant from all three vertices. Where does it sit, and what pattern do you notice?

SOLUTION

Step 1: Given: right triangle with legs on the axes... (adapted: right angle at B since AB is horizontal and BC is vertical). Needed: the common-equidistance point.

Step 2: AB = 8 (horizontal), BC = 6 (vertical), so the right angle is at B(8, 0), and AC is the hypotenuse.

Step 3: $AC = \sqrt{((8 - 0)^2 + (6 - 0)^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 4: Claim: the midpoint of the hypotenuse is equidistant from all three vertices. Midpoint M of AC = ((0 + 8)/2, (0 + 6)/2) = (4, 3).

Step 5: Check $MA = \sqrt{(4^2 + 3^2)} = 5$. $MC = \sqrt{((8 - 4)^2 + (6 - 3)^2)} = \sqrt{(16 + 9)} = 5$.

Step 6: $MB = \sqrt{((8 - 4)^2 + (0 - 3)^2)} = \sqrt{(16 + 9)} = 5$. All three distances are 5 - verified.

Step 7: Pattern: in a right triangle, the midpoint of the hypotenuse is equidistant from all three vertices (it is the centre of the circle passing through the three corners). This is the coordinate-proof version of a famous theorem.

Hinglish: *Ye question ek theorem khud discover karwa raha hai! Right angle B par hai (horizontal + vertical legs), hypotenuse AC = 10. Uska midpoint (4, 3) lo - teeno vertices se distance 5, 5, 5. Matlab: right triangle mein hypotenuse ka midpoint hi wo magical point hai jo teeno corners se barabar door baithta hai (circumcentre). 3-4-5*

triple yahan bhi kaam kar gaya. Ye result yaad rakho - contest maths mein baar-baar kaam aayega.

Answer: M(4, 3), the midpoint of the hypotenuse, is equidistant (distance 5) from all three vertices.

Q43 · HOTS · Concept: Sum-of-distances locus (T14)

Q43. A(-4, 0) and B(4, 0) are two fixed points. P(0, y) lies on the y-axis such that PA + PB = 10. Find all possible y. Why is y = 0 NOT a solution?

SOLUTION

Step 1: Given: PA + PB = 10 with P on the y-axis. Needed: y values.

Step 2: PA = $\sqrt{((0 + 4)^2 + y^2)} = \sqrt{(16 + y^2)}$. By symmetry PB has the same value $\sqrt{(16 + y^2)}$.

Step 3: So the condition reads $2\sqrt{(16 + y^2)} = 10$, hence $\sqrt{(16 + y^2)} = 5$.

Step 4: Square: $16 + y^2 = 25$, so $y^2 = 9$, giving y = 3 or y = -3.

Step 5: Verify y = 3: PA = $\sqrt{(16 + 9)} = 5$, PB = 5, total 10. Same for y = -3. Both work.

Step 6: Why not y = 0: then PA = PB = 4 and the sum is 8, which is short of 10 - P at the origin sits strictly between A and B, so PA + PB just equals AB = 8, the minimum possible value. To reach 10 the point must climb off the line.

Hinglish: *Symmetry dekho: A aur B origin ke dono taraf barabar door hain, isliye PA = PB hoga hi. Toh 2 x distance = 10, distance = 5. 16 + y^2 = 25 se y = plus-minus 3. Aur y = 0 kyun nahi: origin par P ke paas PA + PB = AB hi hota hai = 8 - shortest possible. 10 chahiye toh P ko upar ya neeche jhukna padega (y = 3 ya -3). Ye question aage ke classes mein ellipse ka darwaza hai - PA + PB constant wala idea.*

Answer: y = 3 or y = -3; y = 0 fails because PA + PB = AB = 8 there, the minimum, not 10.

Q44 · HOTS · Concept: Box method for area (T12/T14)

Q44. A triangle has vertices A(0, 0), B(1, 2) and C(3, 1). Show that it is isosceles, and find its area using the box (enclosing rectangle) method.

SOLUTION

Step 1: Given: three lattice points. Needed: isosceles proof and area.

Step 2: AB = $\sqrt{(1^2 + 2^2)} = \sqrt{5}$. BC = $\sqrt{((3 - 1)^2 + (1 - 2)^2)} = \sqrt{(4 + 1)} = \sqrt{5}$. AC = $\sqrt{(3^2 + 1^2)} = \sqrt{10}$.

Step 3: AB = BC = $\sqrt{5}$, so the triangle is isosceles (apex at B).

Step 4: Box method: enclose the triangle in the smallest axis-aligned rectangle: corners (0, 0), (3, 0), (3, 2), (0, 2). Box area = 3 x 2 = 6.

Step 5: The box minus the triangle leaves three right triangles: T1 below AB with legs 1 and 2 (area 1), T2 right of BC with legs 2 and 1 (area 1), T3 above AC with legs 3 and 1 (area 1.5).

Step 6: Area of ABC = 6 - 1 - 1 - 1.5 = 2.5 square units.

Step 7: Cross-check by another triangle with same side data is not needed; instead sanity-check magnitude: the triangle fits inside a 3 x 2 box, so 2.5 is plausible.

» Why this step? When the triangle is not right-angled and the base-height pair is not obvious, the box method computes area using only right triangles - fully within Chapter 1 tools.

Hinglish: *Jab triangle tedha ho aur base-height na dikhe, box method use karo: triangle ko chhote rectangle mein band karo, rectangle ka area lo, aur charo taraf ke right triangles ghata do. 6 - 1 - 1 - 1.5 = 2.5. Ye technique squares ke root wale triangles ka area nikalne ka desi jugaad hai - bina kisi naye formula ke. Aur dekho, AB = BC = √5 se isosceles bhi proof ho gaya. Do kaam ek question mein.*

Answer: Isosceles (AB = BC = √5); area = 2.5 square units via the box method.

Q45 · HOTS · Concept: Chord comparison on a circle (T10/T4)

Q45. Three points A(3, 4), B(-4, 3) and C(0, -5) all lie on the circle of radius 5 centred at the origin (verify this). Find the lengths of the chords AB, BC and CA, and state which chord is the longest.

SOLUTION

Step 1: Given: three points, circle radius 5. Needed: verify membership, three chord lengths, longest chord.

Step 2: Membership: A: 9 + 16 = 25. B: 16 + 9 = 25. C: 0 + 25 = 25. All equal to $r^2 = 25$, so all three lie on the circle.

Step 3: AB = $\sqrt{((-4 - 3)^2 + (3 - 4)^2)} = \sqrt{(49 + 1)} = \sqrt{50} = 5\sqrt{2}$ (approximately 7.07).

Step 4: BC = $\sqrt{((0 + 4)^2 + (-5 - 3)^2)} = \sqrt{(16 + 64)} = \sqrt{80} = 4\sqrt{5}$ (approximately 8.94).

Step 5: CA = $\sqrt{((3 - 0)^2 + (4 + 5)^2)} = \sqrt{(9 + 81)} = \sqrt{90} = 3\sqrt{10}$ (approximately 9.49).

Step 6: Comparing exact squares: 90 > 80 > 50, so CA > BC > AB.

Step 7: The longest chord is CA with length $3\sqrt{10}$. (No chord can exceed the diameter 10, and indeed all three are below 10.)

Hinglish: *Pehle membership - teeno 25 de rahe hain, sab circle par. Phir chords: √50, √80, √90. Compare karne ka clean tareeka: squares compare karo (50 < 80 < 90), roots ki zaroorat nahi. Longest CA = 3√10 approx 9.49. Aur ek boundary-check: diameter 10 hai, koi chord 10 se bada nahi - hamara answer us limit ke andar hi hai. Yahi internal consistency marks dilati hai.*

Answer: All three lie on the circle; AB = $5\sqrt{2}$, BC = $4\sqrt{5}$, CA = $3\sqrt{10}$; the longest chord is CA.

Q46 · HOTS · Concept: Parallelogram via diagonal midpoints (T8/T6)

Q46. A(1, 2), B(4, 3) and C(6, 7) are three vertices of parallelogram ABCD taken in order. Find the fourth vertex D using the fact that the diagonals of a parallelogram bisect each other, and verify your answer.

SOLUTION

Step 1: Given: three consecutive vertices of a parallelogram. Needed: D.

Step 2: Diagonals of a parallelogram bisect each other, so midpoint of AC must equal midpoint of BD.

Step 3: Midpoint of AC = ((1 + 6)/2, (2 + 7)/2) = (3.5, 4.5).

Step 4: Let D = (x, y). Midpoint of BD = ((4 + x)/2, (3 + y)/2). Set equal to (3.5, 4.5).

Step 5: From x: (4 + x)/2 = 3.5 gives x = 3. From y: (3 + y)/2 = 4.5 gives y = 6.

Step 6: So D = (3, 6).

Step 7: Verify as a parallelogram by sides: AB = $\sqrt{(3^2 + 1^2)} = \sqrt{10}$, DC = $\sqrt{((6 - 3)^2 + (7 - 6)^2)} = \sqrt{(9 + 1)} = \sqrt{10}$; BC = $\sqrt{(2^2 + 4^2)} = \sqrt{20} = 2\sqrt{5}$, AD = $\sqrt{((3 - 1)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20}$. Opposite sides equal - consistent.

» Why this step? The diagonal-midpoint method converts a geometry condition into two tiny linear equations - far safer than guessing a translation vector.

Hinglish: *Parallelogram ka chautha vertex nikalne ka RULE: diagonals ka midpoint same hota hai. AC ka midpoint lo, phir BD ka midpoint uske barabar rakho - do chhote equations mil gaye: x = 3, y = 6. Verification mein opposite sides bhi barabar nikli (√10 aur 2√5 ke pairs). Ye midpoint-trick Q31 (trisection) ke saath milke midpoint-based toolkit complete karta hai: aage class 10 mein yehi idea hi midpoint formula ka base banega.*

Answer: D = (3, 6); verified since AB = DC = $\sqrt{10}$ and BC = AD = $2\sqrt{5}$.

Q47 · HOTS · Concept: Grid-walk vs direct distance (T11/T4)

Q47. A delivery robot moves on a grid of unit squares. It must travel from station S(1, 1) to station T(7, 4). (i) What is the length of the shortest path using only horizontal and vertical moves? (ii) What is the straight-line distance? (iii) A second robot flies directly. How much distance does it save compared to the ground robot?

SOLUTION

Step 1: Given: grid unit 1, S(1, 1) to T(7, 4). Needed: street length, direct length, saving.

Step 2: Horizontal gap = 7 - 1 = 6 units; vertical gap = 4 - 1 = 3 units.

Step 3: Ground robot (only axis moves): 6 + 3 = 9 units. Every such path has exactly this length - no order of moves can beat it.

Step 4: Flying robot (straight line): $\sqrt{(6^2 + 3^2)} = \sqrt{(36 + 9)} = \sqrt{45} = 3\sqrt{5}$ (approximately 6.71 units).

Step 5: Saving = $9 - 3\sqrt{5}$ (exactly), approximately 9 - 6.71 = 2.29 units.

Step 6: Keep the exact answer $9 - 3\sqrt{5}$ in the final statement; the decimal is only for intuition.

Hinglish: *Grid-walk ka funda: horizontal steps + vertical steps hamesha fix hote hain (yahan 6 + 3 = 9), chahe kisi bhi order mein chalo. Direct distance Pythagoras se $3\sqrt{5}$. Saving $9 - 3\sqrt{5}$ approx 2.3 units. Ye question Q26 (city auto) ka bhai hai - wahan 200 m blocks the, yahan 1 unit. Same skeleton, naye numbers - yahi 'question structure' samajhna original practice hai, ratna nahi.*

Answer: (i) 9 units (ii) $3\sqrt{5}$ units (approx 6.71) (iii) saving $9 - 3\sqrt{5}$ approx 2.29 units.

Q48 · HOTS · Concept: Equally spaced collinear chain (T9/T8)

Q48. P(1, 2), Q(4, 6) and R(7, 10) are three points. Show they are collinear and equally spaced. Then find the point T on the same line, beyond R, with QT = 10.

SOLUTION

Step 1: Given: three points. Needed: collinearity, equal spacing, then T.

Step 2: PQ = $\sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{(9 + 16)} = 5$.

Step 3: QR = $\sqrt{((7 - 4)^2 + (10 - 6)^2)} = \sqrt{(9 + 16)} = 5$.

Step 4: PR = $\sqrt{((7 - 1)^2 + (10 - 2)^2)} = \sqrt{(36 + 64)} = 10$.

Step 5: Since PR = PQ + QR = 10, the points are collinear, and PQ = QR = 5 shows Q is exactly the midpoint - equally spaced.

Step 6: The repeated step from P to Q is (+3, +4). The same step continues from Q to R, and one more step lands at T = (7 + 3, 10 + 4) = (10, 14).

Step 7: Check QT: from Q(4, 6) to T(10, 14) is two steps: 2 x 5 = 10. Verified.

» Why this step? A constant coordinate-step (+3, +4) between consecutive points is the signature of equal spacing on a line - recognise the pattern and distances become trivial.

Hinglish: *Pattern-spotting question! Har agle point mein (+3, +4) jud raha hai - 3-4-5 triple ka repeated use. PQ = QR = 5, PR = 10: collinear + equally spaced confirmed. Ab 'beyond R with QT = 10' ka matlab: Q se 2 step aage = R se 1 step aage = T(10, 14). Verify kiya QT = 10. Aise 'chain' questions mein step-pakadna hi asli skill hai, har baar distance formula chalana nahi.*

Answer: Collinear with PQ = QR = 5, PR = 10; T = (10, 14) with QT = 10 verified.

Q49 · HOTS · Concept: Chord and centre distance (T10/T8)

Q49. A circle of radius 10 is centred at the origin. A and B(−8, 6) are two points on it, with A(6, 8). Find the length of chord AB and the perpendicular distance of the chord from the centre. Verify the right-triangle relation between them.

SOLUTION

Step 1: Given: circle $x^2 + y^2 = 100$, chord joining A(6, 8) and B(-8, 6). Needed: chord length, distance from centre, consistency check.

Step 2: First confirm membership: A: 36 + 64 = 100. B: 64 + 36 = 100. Both on the circle.

Step 3: Chord AB = $\sqrt{((-8 - 6)^2 + (6 - 8)^2)} = \sqrt{(196 + 4)} = \sqrt{200} = 10\sqrt{2}$.

Step 4: Midpoint M of the chord = ((6 + (-8))/2, (8 + 6)/2) = (-1, 7).

Step 5: The line from the centre to the midpoint of a chord is perpendicular to the chord, so the required distance is OM = $\sqrt{((-1)^2 + 7^2)} = \sqrt{50} = 5\sqrt{2}$.

Step 6: Right-triangle check: half-chord = $5\sqrt{2}$, OM = $5\sqrt{2}$, radius = 10. Indeed $(5\sqrt{2})^2 + (5\sqrt{2})^2 = 50 + 50 = 100 = 10^2$.

Step 7: Everything is consistent - the chord, its distance from the centre and the radius obey the Baudhayana-Pythagoras relation.

Hinglish: *Do kaam ek saath: chord length (distance formula se $10\sqrt{2}$) aur centre se doori. Fundaa: chord ke midpoint se centre ko jodo - wo line chord par perpendicular hoti hai, toh wahi distance hai. OM = $\sqrt{50} = 5\sqrt{2}$. Phir master-check: (half-chord)² + (distance)² = radius². 50 + 50 = 100. Perfect. Ye relation har circle-chord question ka backbone hai - yahin se 'perpendicular from centre bisects chord' wali theory aati hai.*

Answer: AB = $10\sqrt{2}$; distance from centre = $5\sqrt{2}$; verified since $(5\sqrt{2})^2 + (5\sqrt{2})^2 = 100 = \text{radius}^2$.

Q50 · HOTS · Concept: Full square proof chain (T14)

Q50. Prove that the points P(1, 7), Q(4, 2), R(-1, -1) and S(-4, 4), taken in this order, are the vertices of a square. You must verify ALL necessary properties using the distance formula and midpoints.

SOLUTION

Step 1: Given: four points in cyclic order. Needed: full square proof.

Step 2: PQ = $\sqrt{((4 - 1)^2 + (2 - 7)^2)} = \sqrt{(9 + 25)} = \sqrt{34}$.

Step 3: QR = $\sqrt{((-1 - 4)^2 + (-1 - 2)^2)} = \sqrt{(25 + 9)} = \sqrt{34}$.

Step 4: RS = $\sqrt{((-4 + 1)^2 + (4 + 1)^2)} = \sqrt{(9 + 25)} = \sqrt{34}$.

Step 5: SP = $\sqrt{((1 + 4)^2 + (7 - 4)^2)} = \sqrt{(25 + 9)} = \sqrt{34}$. All four sides equal: it is at least a rhombus-shaped candidate.

Step 6: Diagonals: PR = $\sqrt{((-1 - 1)^2 + (-1 - 7)^2)} = \sqrt{(4 + 64)} = \sqrt{68} = 2\sqrt{17}$. QS = $\sqrt{((-4 - 4)^2 + (4 - 2)^2)} = \sqrt{(64 + 4)} = \sqrt{68}$. Diagonals are equal.

Step 7: Bisecting check: midpoint of PR = ((1 + (-1))/2, (7 + (-1))/2) = (0, 3). Midpoint of QS = ((4 + (-4))/2, (2 + 4)/2) = (0, 3). Diagonals bisect each other, so the figure is a parallelogram.

Step 8: A parallelogram with all sides equal is a rhombus; a rhombus with equal diagonals is a square. Hence PQRS is a square.

Step 9: Optional right-angle cross-check: $PR^2 = 68$ and $PQ^2 + QR^2 = 34 + 34 = 68$, so the angle at Q (opposite diagonal PR) is 90 degrees by the converse of Pythagoras.

» Why this step? The clean chain for proving a square with distances only: sides equal (rhombus candidate) + diagonals bisect (parallelogram) + diagonals equal (square). Each link uses only Chapter 1 tools.

Hinglish: *Ye Chapter 1 ka final boss hai. Strategy 3-layer ki hai: (1) chaaron sides √34 - rhombus material, (2) diagonals ke midpoints dono (0, 3) - parallelogram confirm, (3) diagonals dono √68 - square confirm. Bonus check: 68 = 34 + 34 se right angle bhi mil gaya. Notice karo kitna systematic hai - 'sides equal + diagonals bisect + diagonals equal' ye teen-step recipe kisi bhi square-proof question mein chalegi. Ye last question mein jaan-bujh ke negative coordinates aur non-axis orientation rakha hai taaki tum axes par dependent na raho.*

Answer: PQRS is a square: all sides √34, diagonals equal (2√17) and bisecting each other at (0, 3), with right angles confirmed by 68 = 34 + 34.

1.4 Chapter MCQs (60 questions, answers inline)

Each MCQ lists the options, the correct answer, a full explanation, the trap analysis for the tempting wrong options, and a Hinglish note. Options are engineered from real student mistakes, so think before you pick.

MCQ 1 · Normal · Concept: Quadrant signs

1. The point (-7, 4) lies in which quadrant?

A. Quadrant I
B. Quadrant II
C. Quadrant III
D. Quadrant IV

Correct Answer: (B)

x = -7 is negative and y = +4 is positive. The sign pattern (-, +) is exactly the definition of Quadrant II (above the x-axis, left of the y-axis).

Why (A) is wrong: (+, +) would need BOTH positive; here x is negative.

Why (C) is wrong: Quadrant III is (-, -) - y must also be negative.

Why (D) is wrong: Quadrant IV is (+, -) - both signs are wrong here.

Hinglish: *Pehla number minus (left), doosra plus (upar) - ye Quadrant II ka signature hai. Bas signs match karo, answer mil gaya.*

MCQ 2 · Normal · Concept: Points on axes

2. The point on the y-axis lying 6 units above the origin has coordinates:

A. (6, 0)
B. (0, 6)
C. (-6, 0)
D. (0, -6)

Correct Answer: (B)

On the y-axis the x-coordinate must be 0. Above the origin means positive y. Hence (0, 6).

Why (A) is wrong: (6, 0) is on the x-axis, not the y-axis.

Why (D) is wrong: (0, -6) is 6 units BELOW the origin.

Hinglish: *Y-axis = pehla coordinate zero. Upar = positive. (0, 6). Itni si baat, par (6, 0) likhne wale har saal milte hain!*

MCQ 3 · Normal · Concept: Coordinate meaning

3. The perpendicular distance of the point (-4, 9) from the y-axis is:

A. 9 units
B. 4 units
C. -4 units
D. √97 units

Correct Answer: (B)

Distance from the y-axis is given by the size of the x-coordinate: |x| = |-4| = 4. Distances are never negative.

Why (A) is wrong: 9 is the distance from the x-axis, not the y-axis.

Why (C) is wrong: Distances are always non-negative; -4 is a coordinate, not a distance.

Why (D) is wrong: √97 is the distance from the origin.

Hinglish: *y-axis se doori = |x|. Minus sirf side batata hai (left), distance toh 4 hi hai. Distance kabhi negative nahi hota - option C khatam. √97 toh origin se distance hai ($4^2 + 9^2$).*

MCQ 4 · Normal · Concept: Vertical distance

4. The distance between the points (3, -2) and (3, 7) is:

A. 5 units
B. 9 units
C. √85 units
D. 10 units

Correct Answer: (B)

Same x-coordinate means a vertical segment. Distance = |7 - (-2)| = |7 + 2| = 9.

Why (A) is wrong: 5 confuses the gap with the smaller coordinate.

Why (C) is wrong: √85 results from a wrong application of the distance formula.

Hinglish: *x same hai toh seedha y ka gap: 7 - (-2) = 9. Minus-minus plus bana - yahi sabse common slip hai. √85 aata hai agar log dono differences ghalat jod dein.*

MCQ 5 · Normal · Concept: Distance formula 6-8-10

5. The distance between (1, -2) and (7, 6) is:

A. 10 units
B. 12 units
C. √84 units
D. 14 units

Correct Answer: (A)

Differences: 7 - 1 = 6 and 6 - (-2) = 8. Distance = √(36 + 64) = √100 = 10, the 6-8-10 triple.

Why (B) is wrong: 12 would come from adding instead of applying Pythagoras.

Why (C) is wrong: √84 comes from miscomputing the y-difference as 4 or 8-(-2) as something else.

Hinglish: *6 aur 8 dikh rahe hain? 3-4-5 ka do guna - seedha 10. Triple yaad rakho toh square karne ki bhi zaroorat nahi.*

MCQ 6 · Normal · Concept: Conceptual: free coordinate

6. For how many values of k is the distance between (k, 3) and (k, 8) equal to 5?

A. Exactly one value of k
B. Exactly two values of k
C. All real values of k
D. No value of k

Correct Answer: (C)

Both points have the same x-coordinate k, so the distance is |8 - 3| = 5 regardless of k. Every real k works - there are infinitely many such pairs.

Why (A) is wrong: There is no equation restricting k.

Why (B) is wrong: The distance does not depend on k at all.

Hinglish: *Ye trick question hai! Distance dono points ke y-gap se banti hai jo hamesha 5 hai - k ka kuch lena dena hi nahi. Isliye saare real k chalega. Log sochte hain k = 5 hoga - nahi bhai, k toh bas dono ko same rakhta hai.*

MCQ 7 · Normal · Concept: Distance from origin

7. The distance of the point (-6, 8) from the origin is:

A. 2 units
B. 10 units
C. 14 units
D. 100 units

Correct Answer: (B)

$\sqrt{((-6)^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$. Signs die inside squares; the 6-8-10 triple does the rest.

Why (A) is wrong: 2 = |-6 - 8| is a meaningless subtraction of coordinates.

Why (C) is wrong: Adding coordinate sizes is not a distance.

Why (D) is wrong: 100 is the squared distance, not the distance.

Hinglish: *Minus square mein mar jaata hai: 36 + 64 = 100, root 10. 14 wo log chunte hain jo -6 + 8 ko distance samajh lete hain - aisa formula exist hi nahi karta.*

MCQ 8 · Normal · Concept: Half-turn symmetry

8. If P(a, b) lies in Quadrant II, then the point (-a, -b) lies in:

A. Quadrant I
B. Quadrant II
C. Quadrant III
D. Quadrant IV

Correct Answer: (D)

In Q II, a is negative and b is positive. So -a is positive and -b is negative: signs (+, -), which is Quadrant IV. Flipping both signs reflects the point through the origin (a half-turn).

Why (A) is wrong: Both signs must flip, not stay one-negative-one-positive.

Why (B) is wrong: Same quadrant is impossible after a half-turn.

Hinglish: *Q II mein a negative, b positive. Dono ke sign palte toh (+, -) = Q IV. Yaad rakho: (x, y) aur (-x, -y) hamesha opposite quadrants mein hote hain - origin ke through mirror.*

MCQ 9 · Normal · Concept: Reflection in y-axis

9. The reflection of the point (-3, -5) in the y-axis is:

A. (3, -5)
B. (3, 5)
C. (-3, 5)
D. (-3, -5)

Correct Answer: (A)

Reflection in the y-axis flips only the sign of x: (-3, -5) becomes (3, -5). The y-coordinate is untouched.

Why (B) is wrong: (3, 5) flips both signs - that is reflection in BOTH axes.

Why (C) is wrong: (-3, 5) flips y only - that is the x-axis reflection.

Hinglish: *y-axis mein mirror = sirf x ka sign ulta. (-3, -5) ka x minus tha, ab plus: (3, -5). y ko mat chhedo - wahi sabse common galti hai.*

MCQ 10 · Normal · Concept: Reflection in x-axis

10. The reflection of (2, 7) in the x-axis, followed by reflection of the image in the y-axis, gives:

A. (-2, -7)
B. (2, -7)
C. (-2, 7)
D. (7, -2)

Correct Answer: (A)

x-axis reflection: (2, 7) to (2, -7). Then y-axis reflection flips x: (2, -7) to (-2, -7). Two axis reflections = half-turn = both signs flip.

Why (B) is wrong: Only one reflection was performed.

Why (C) is wrong: Wrong axis was flipped.

Why (D) is wrong: Coordinates were swapped - reflections never swap coordinates.

Hinglish: *Do reflections ki chain = dono signs flip. (2, 7) se seedha (-2, -7). Shortcut yaad rakho: x-axis phir y-axis reflect karna = origin ke around 180 degree ghumaana.*

MCQ 11 · Normal · Concept: Midpoint

11. The midpoint of the segment joining (-3, 8) and (5, -2) is:

A. (1, 3)
B. (2, 6)
C. (1, 5)
D. (4, 3)

Correct Answer: (A)

Midpoint = ((-3 + 5)/2, (8 + (-2))/2) = (2/2, 6/2) = (1, 3).

Why (B) is wrong: (2, 6) is the sum of the coordinates without halving.

Why (C) is wrong: The y-average was computed as (8 - 2)/2 = 5 by dropping a sign.

Why (D) is wrong: Adds only the x-pair wrongly.

Hinglish: *Average lo dono coordinates ka: (-3 + 5)/2 = 1 aur (8 - 2)/2 = 3. (2, 6) wo galti hai jab divide karna bhool jaate ho.*

MCQ 12 · Normal · Concept: Reverse midpoint

12. If M(2, -1) is the midpoint of A(-4, 3) and B(x, y), then B is:

A. (8, -5)
B. (-10, 7)
C. (8, 5)
D. (0, 1)

Correct Answer: (A)

Undo the average: x = 2(2) - (-4) = 8 and y = 2(-1) - 3 = -5. So B = (8, -5).

Why (B) is wrong: Adds instead of subtracting the known endpoint.

Why (C) is wrong: y computed as -2 + 3.

Why (D) is wrong: Halves something unnecessarily.

Hinglish: *Midpoint se endpoint nikalna = 2 x midpoint minus known point. x: 4 + 4 = 8. y: -2 - 3 = -5. Sign ka khayal: 3 ko MINUS karna hai, plus nahi.*

MCQ 13 · Normal · Concept: Collinearity

13. Which set of three points is collinear?

A. (0, 0), (2, 1), (4, 2)
B. (0, 1), (2, 2), (5, 4)
C. (1, 1), (2, 3), (4, 6)
D. (0, 0), (1, 2), (3, 7)

Correct Answer: (A)

In option A each step from one point to the next is (+2, +1) - a constant step means equal slope, so the points lie on one line (y = x/2). The other sets change the step pattern, so they bend.

Why (B) is wrong: Steps (+2, +1) then (+3, +2) - slopes 1/2 vs 2/3 differ.

Why (C) is wrong: Slopes 2 vs 3/2 differ.

Why (D) is wrong: Slopes 2 vs 5/2 differ.

Hinglish: *Collinear ka fastest check: consecutive points ke beech ka STEP same hai ya nahi. Option A mein har baar (+2, +1) - perfect line. Baaki options mein step badal jaata hai, matlab line mudti hai.*

MCQ 14 · Normal · Concept: Circle membership

14. Which point lies ON the circle $x^2 + y^2 = 10$ with centre the origin?

A. (1, 3)
B. (2, 2)
C. (1, 2)
D. (3, 3)

Correct Answer: (A)

Check $x^2 + y^2$: (1, 3) gives 1 + 9 = 10 exactly. (2, 2) gives 8 (inside), (1, 2) gives 5 (inside), (3, 3) gives 18 (outside).

Why (B) is wrong: 8 < 10 means inside, not on.

Why (C) is wrong: 5 is far from 10.

Why (D) is wrong: 18 > 10 means outside.

Hinglish: *Equation mein plug karo, 10 aana chahiye. (1, 3): 1 + 9 = 10 perfect. Baaki sab 10 nahi de rahe - andar ya bahar.*

MCQ 15 · Normal · Concept: 5-12-13 triple

15. The distance of (5, -12) from the origin is:

A. 7 units
B. 13 units
C. 17 units
D. 169 units

Correct Answer: (B)

$\sqrt{(25 + 144)} = \sqrt{169} = 13$. The 5-12-13 triple, again with a harmless minus sign.

Why (A) is wrong: 5 + (-12) size subtraction - meaningless.

Why (C) is wrong: 17 = 5 + 12 (adding instead of Pythagoras).

Why (D) is wrong: 169 is 13 squared.

Hinglish: *5-12-13 triple ratt lo. Minus ko ignore karo - square ke baad koi matlab nahi. 169 square hai, root nahi - aisi chhoti galti bahut log karte hain.*

MCQ 16 · Normal · Concept: Symbolic distance

16. The distance between the points (a, 0) and (0, a) (a > 0) is:

A. 2a units
B. $a\sqrt{2}$ units
C. a units
D. 0 units

Correct Answer: (B)

Differences: a - 0 = a and 0 - a = -a. Distance = $\sqrt{(a^2 + a^2)} = \sqrt{(2a^2)} = a\sqrt{2}$.

Why (A) is wrong: 2a would be a + a, but legs add only through squares.

Why (C) is wrong: a is just one leg.

Why (D) is wrong: The points are distinct unless a = 0.

Hinglish: *Dono differences ka size a hai - isosceles right triangle ka hypotenuse = $a\sqrt{2}$. 2a karna Pythagoras ko addition samajhna hai - bada classic mistake.*

MCQ 17 · Normal · Concept: Perimeter of right triangle

17. The perimeter of the triangle with vertices (0, 0), (4, 0), (0, 3) is:

A. 7 units
B. 12 units
C. 10 units
D. 6 units

Correct Answer: (B)

Legs are 4 and 3; hypotenuse = 5 (3-4-5). Perimeter = 4 + 3 + 5 = 12.

Why (A) is wrong: 3 + 4 only - forgot the hypotenuse.

Why (C) is wrong: Confused with a different triple.

Why (D) is wrong: Half the perimeter.

Hinglish: *Axis par baithe legs 4 aur 3, hypotenuse 5. Sab JODO: 12. 7 sirf do legs ka sum hai - hypotenuse bhool gaye.*

MCQ 18 · Normal · Concept: Area of right triangle

18. The area of the triangle with vertices (0, 0), (8, 0) and (0, 6) is:

A. 48 sq units
B. 24 sq units
C. 14 sq units
D. 10 sq units

Correct Answer: (B)

Right angle at the origin with legs 8 and 6. Area = (1/2)(8)(6) = 24.

Why (A) is wrong: Missing the factor 1/2.

Why (C) is wrong: 8 + 6 - a sum, not an area.

Why (D) is wrong: Applied some ratio incorrectly.

Hinglish: *Right triangle ka area = aadha x base x height = 24. 48 wo galti hai jab 1/2 multiply karna bhool jaate ho.*

MCQ 19 · Normal · Concept: Assertion-reason

19. Assertion (A): The distance of a point (x, y) from the origin is $\sqrt{(x^2 + y^2)}$. Reason (R): The Baudhayana-Pythagoras Theorem gives hypotenuse = √(sum of squares of legs). Choose the correct option:

A. Both A and R are true and R explains A
B. Both A and R are true but R does not explain A
C. A is true, R is false
D. A is false, R is true

Correct Answer: (A)

The distance formula from the origin is exactly Pythagoras applied to the legs x and y. The assertion is true and the reason is the actual justification behind it.

Why (B) is wrong: R genuinely explains A here - do not dismiss the connection.

Why (C) is wrong: R is a correct statement of the theorem.

Why (D) is wrong: A is the standard distance-from-origin result.

Hinglish: *A sahi hai, aur wo sahi ISLIYE hai kyunki origin se jaane wala segment right triangle ka hypotenuse hai jiske legs x aur y hain. R wahi theorem hai - toh R, A ko explain karta hai. Option A.*

MCQ 20 · Normal · Concept: Horizontal segment

20. The line joining (-5, 2) and (5, 2) is parallel to:

A. the x-axis
B. the y-axis
C. both axes
D. neither axis

Correct Answer: (A)

Both points have y = 2, so the segment is horizontal, i.e. parallel to the x-axis (and perpendicular to the y-axis).

Why (B) is wrong: Same y means horizontal, not vertical.

Why (C) is wrong: A non-degenerate segment cannot be parallel to both perpendicular axes.

Hinglish: *y same = horizontal line = x-axis ke parallel. Same x hota toh y-axis ke parallel hota.*

MCQ 21 · Normal · Concept: Equidistant on x-axis

21. The point (x, 0) is equidistant from (2, 3) and (6, 3). Then x equals:

A. 4
B. 2
C. 8
D. 3

Correct Answer: (A)

$(x - 2)^2 + 9 = (x - 6)^2 + 9$. The 9s cancel: $(x - 2)^2 = (x - 6)^2$, so x - 2 = plus/minus (x - 6). The negative-sign branch gives a contradiction (0 = -8... no solution from that branch), the positive-sign branch gives x - 2 = x - 6? Also impossible - so take x - 2 = -(x - 6): 2x = 8, x = 4. Geometrically, x = 4 is the midpoint of 2 and 6, directly below the midpoint of the two given points.

Why (B) is wrong: That is the first point itself.

Why (C) is wrong: 8 is beyond the second point - not symmetric.

Why (D) is wrong: 3 is not the average of 2 and 6.

Hinglish: *Dono given points ka y same hai (3), isliye equidistant point unke beech ke THEEK beech mein hoga: (2 + 6)/2 = 4. Equation se bhi x = 4 hi aata hai. Ye symmetry ka use hai - jab dono points same height par hon toh answer unka midpoint hota hai.*

MCQ 22 · Normal · Concept: Rectangle perimeter

22. The points (0, 0), (3, 0), (3, 4) and (0, 4) form a rectangle. Its perimeter is:

A. 12 units
B. 14 units
C. 24 units
D. 7 units

Correct Answer: (B)

Sides are 3 and 4. Perimeter = 2(3 + 4) = 14. (Its diagonal is 5, area 12.)

Why (A) is wrong: 12 is the area, not the perimeter.

Why (C) is wrong: Doubled the area by mistake.

Why (D) is wrong: Added only two sides.

Hinglish: *Perimeter = 2(length + width) = 2 x 7 = 14. 12 toh AREA hai - dono ko mix karna classic confusion hai.*

MCQ 23 · Normal · Concept: Order independence of distance

23. Statement: If the two points in a distance calculation are interchanged, the computed distance changes. Is the statement TRUE or FALSE, and why?

A. TRUE, because x and y swap places
B. FALSE, because the differences are squared, which erases the sign change
C. TRUE, because the formula is order-dependent
D. FALSE, because distance is always zero

Correct Answer: (B)

Swapping the points changes (x2 - x1) into (x1 - x2) = negative of it. But squaring makes both equal. Hence the distance is unchanged.

Why (A) is wrong: Coordinates do not swap - only the subtraction order flips.

Why (C) is wrong: The formula is order-independent.

Why (D) is wrong: Distance is zero only for identical points.

Hinglish: *Points ulta karne se differences ka sign ulta hota hai, par square laga-te-hi sign khatam. Isliye distance same rehta hai. Ye freedom hai - jis point ko pehle maano, jawab same.*

MCQ 24 · Normal · Concept: Axis membership

24. A point whose x-coordinate is 0 must lie:

A. on the x-axis
B. on the y-axis
C. in Quadrant II
D. at the origin

Correct Answer: (B)

x = 0 means zero sideways distance from the y-axis, so the point sits on the y-axis - at (0, y) for some y. It is the origin only if additionally y = 0.

Why (A) is wrong: On the x-axis it is y that equals 0.

Why (C) is wrong: Quadrant points have both coordinates non-zero.

Why (D) is wrong: Origin needs y = 0 as well.

Hinglish: *x = 0 matlab left-right displacement zero = seedha y-axis par. Origin tabhi banega jab y bhi 0 ho. x-axis wale options mein y = 0 hota hai - ulta mat karo.*

MCQ 25 · Normal · Concept: Equidistant on y-axis

25. The point (0, y) is equidistant from A(-3, 0) and B(5, 2). Then y equals:

A. 5
B. 2
C. 10
D. 4

Correct Answer: (A)

$9 + y^2 = 25 + (y - 2)^2$. Expand: $9 + y^2 = 25 + y^2 - 4y + 4$, so $9 = 29 - 4y$, giving $4y = 20$, $y = 5$. The point is (0, 5).

Why (B) is wrong: That is B's y-coordinate, not the solution.

Why (C) is wrong: Result of a sign slip in $(y - 2)^2$.

Why (D) is wrong: Arithmetic slip in moving terms.

Hinglish: *(0, y) rakho, dono squared distances barabar karo, y^2 cancel - linear equation: y = 5. Verify: (0,5) se A doori √34, B se bhi √34. Perfect.*

MCQ 26 · HOTS · Concept: Hypotenuse from coordinates

26. For the triangle with vertices (1, 1), (4, 1) and (1, 5), the hypotenuse has length:

A. 5 units
B. √41 units
C. 7 units
D. 6 units

Correct Answer: (A)

Legs along the axes directions: 4 - 1 = 3 and 5 - 1 = 4, meeting at the right angle (1, 1). Hypotenuse = √(9 + 16) = 5. The whole 3-4-5 triangle is just shifted to start at (1, 1).

Why (B) is wrong: √41 = √(16 + 25) comes from squaring wrong gaps.

Why (C) is wrong: 3 + 4 again.

Why (D) is wrong: Wrong leg difference.

Hinglish: *Origin par nahi hai triangle, (1, 1) par hai - par legs toh wahi 3 aur 4 hain! Translation se triple badalta nahi. Hypotenuse 5. √41 aata hai agar coordinates se seedha formula lagao bina right-angle pehchane.*

MCQ 27 · HOTS · Concept: Missing value on a line

27. If the points (2, 3), (4, k) and (6, 5) are collinear, then k equals:

A. 4

B. 3.5

C. 5

D. 2

Correct Answer: (A)

From (2, 3) to (6, 5) the total step is (+4, +2); halfway (at x = 4) the y-value must be the average: k = (3 + 5)/2 = 4. Indeed (4, 4) is the midpoint.

Why (B) is wrong: Averaged wrongly.

Why (C) is wrong: Copied the endpoint value.

Why (D) is wrong: No basis.

Hinglish: *Collinear matlab beech wala point line par. (2,3) se (6,5) ke beech ka x = 4 hai toh y bhi average hoga: (3+5)/2 = 4. Slope-check bhi karo: dono taraf slope 1/2. Clean.*

MCQ 28 · HOTS · Concept: Equally spaced chain

28. If (3, 2), (6, y) and (9, 8) are equally spaced collinear points, then y equals:

A. 5

B. 6

C. 4

D. 7

Correct Answer: (A)

Equal spacing means a constant step. x-step is 3 twice, so the y-step must also repeat: y = 2 + (8 - 2)/2 = 2 + 3 = 5. Check: steps (+3, +3) then (+3, +3).

Why (B) is wrong: Copied the x-step into y.

Why (C) is wrong: Halved wrongly.

Why (D) is wrong: No pattern use.

Hinglish: *Equally spaced = same step dono baar. x mein +3, +3 dikh raha hai, toh y mein bhi same jump: 2 se 8 tak total 6, half = 3, y = 5. Pattern dekho, equation ki zaroorat nahi.*

MCQ 29 · HOTS · Concept: Two-circle intersection

29. How many points are at distance 5 from BOTH (0, 0) and (8, 0)?

A. 0

B. 1

C. 2

D. Infinitely many

Correct Answer: (C)

The two circles (radius 5, centres 8 apart) overlap since $8 < 5 + 5 = 10$ and $8 > 0$. Two overlapping circles meet in exactly 2 points. By symmetry they are (4, 3) and (4, -3): check $16 + 9 = 25$. Correct.

Why (A) is wrong: Would need the circles separate ($8 > 10$).

Why (B) is wrong: Would need exactly touching ($8 = 10$).

Why (D) is wrong: That happens only for identical circles.

Hinglish: *Do circles socho - centres 8 door, radius 5-5. Wo do jagah kaatenge. Symmetry se x = 4 par, y = plus-minus 3 (kyunki 3-4-5 triple). Answer 2. Ek point tab hota jab circles ko touch karte (centres ki doori = radius ka sum).*

MCQ 30 · HOTS · Concept: Square from centre and vertex

30. A square has centre (2, 3) and one vertex at (5, 7). The area of the square is:

A. 50 sq units

B. 25 sq units

C. 100 sq units

D. $10\sqrt{2}$ sq units

Correct Answer: (A)

Centre-to-vertex distance = half the diagonal = $\sqrt{(3^2 + 4^2)} = 5$. So the full diagonal is 10, and area = diagonal$^2/2 = 100/2 = 50$. (Side = $10/\sqrt{2} = 5\sqrt{2}$, area 50.)

Why (B) is wrong: Used 5 as the side instead of half-diagonal.

Why (C) is wrong: Used 10 as the side.

Why (D) is wrong: That is a length, not an area.

Hinglish: *Centre se vertex tak = aadha diagonal = 5 (3-4-5 triple!). Poora diagonal 10, area = $d^2/2 = 50$. Ye diagonal-area relation square ke liye yaad rakho - side nikal-ghumane se fast hai.*

MCQ 31 · Normal · Concept: Circle membership

31. Which of these points does NOT lie on the circle $x^2 + y^2 = 25$?

A. (-3, -4)

B. (0, -5)

C. (4, -3)

D. (-4, -4)

Correct Answer: (D)

Check each: $9 + 16 = 25$ (on), $0 + 25 = 25$ (on), $16 + 9 = 25$ (on), but $16 + 16 = 32$ which is not 25. So (-4, -4) is off the circle (distance $4\sqrt{2}$ approx 5.66).

Why (A) is wrong: Both negatives still square to 9 and 16.

Why (B) is wrong: Zero coordinate is fine: $0 + 25 = 25$.

Why (C) is wrong: Same triple as A with signs flipped.

Hinglish: *Teeno pehle options 25 de rahe hain - circle par. (-4, -4) dekho: 32 nikla - bahar. Aise 'NOT' questions mein har option check karna padta hai, shortcuts nahi chalte.*

MCQ 32 · Normal · Concept: Distance expression

32. The distance of the point (a, b) from the origin equals:

A. a + b

B. $\sqrt{(a^2 + b^2)}$

C. $\sqrt{(a^2 - b^2)}$

D. $\sqrt{a} + \sqrt{b}$

Correct Answer: (B)

By the distance formula with the second point (0, 0): $\sqrt{((a - 0)^2 + (b - 0)^2)} = \sqrt{(a^2 + b^2)}$.

Why (A) is wrong: Adding lengths of legs never gives the hypotenuse.

Why (C) is wrong: A minus in the formula is fictitious.

Why (D) is wrong: Roots distribute over multiplication, not addition.

Hinglish: *Formula yaad karo - squares JUDTE hain, minus ya addition nahi. a + b isliye galat kyunki 3-4 wale case mein 7 deta hai jabki asli distance 5 hai.*

MCQ 33 · HOTS · Concept: Two possible y values

33. If the distance between A(1, 2) and B(4, y) is 5, the sum of ALL possible values of y is:

A. 4

B. 6

C. 8

D. 2

Correct Answer: (A)

$9 + (y - 2)^2 = 25$ gives $(y - 2)^2 = 16$, so y = 6 or y = -2. Both points (4, 6) and (4, -2) are at distance 5 from A. Sum = 6 + (-2) = 4.

Why (B) is wrong: Took only the positive root y = 6.

Why (C) is wrong: Added 6 + 6 or wrong pairing.

Why (D) is wrong: Took only the negative root.

Hinglish: *Do roots aayenge - circle A se guzar raha hai, vertical line x = 4 par do milaap. y = 6 aur y = -2. Question ne SUM manga hai = 4. Sirf ek value likhne wale question dhyan se nahi padhte.*

MCQ 34 · Normal · Concept: Double reflection

34. Reflecting (-2, -3) in the x-axis and then the image in the y-axis gives:

A. (2, 3)

B. (-2, 3)

C. (2, -3)

D. (-3, -2)

Correct Answer: (A)

x-axis reflection flips y: (-2, 3). Then the y-axis reflection flips x: (2, 3). Both flips together = sign of both coordinates reversed.

Why (B) is wrong: Only y flipped - one reflection.

Why (C) is wrong: Only x flipped - one reflection.

Why (D) is wrong: Swapped coordinates - reflections do not do that.

Hinglish: *Do mirror = dono sign flip. Seedha result (2, 3). Order matter nahi karta yahan - x pehle ya y pehle, same end point.*

MCQ 35 · Normal · Concept: Reflection on axis point

35. The reflection of the point (0, -7) in the x-axis lies:

A. on the x-axis
B. on the y-axis
C. in Quadrant III
D. at the origin

Correct Answer: (B)

Reflection flips y: (0, -7) becomes (0, 7). The x-coordinate is still 0, so the image sits on the y-axis. Points on the reflecting axis stay fixed, but this point was on the y-axis, not the x-axis, so it moves - along the y-axis.

Why (A) is wrong: Only points ON the x-axis stay put under x-reflection.

Why (C) is wrong: Quadrant III needs x negative too.

Why (D) is wrong: Origin only if both coordinates are zero.

Hinglish: *(0, -7) y-axis par baitha hai. x-axis mein reflect karne se wo y-axis ke upar jump karta hai: (0, 7). Axis par baithe point ko reflect karne ka rule: jis axis par hai wahi rehta hai (agar usi axis mein reflect karo), warna apni axis par hi travel karta hai.*

MCQ 36 · HOTS · Concept: Quarter-division point

36. P(-1, 4) and Q(7, -4) are joined. The point one quarter of the way from P to Q is:

A. (1, 2)
B. (3, 0)
C. (2, 1)
D. (1, 3)

Correct Answer: (A)

Total step Q - P = (8, -8). One quarter of the step = (2, -2). Add to P: (-1 + 2, 4 - 2) = (1, 2).

Why (B) is wrong: That is the midpoint (half-way), not the quarter point.

Why (C) is wrong: Step subtracted from P instead of added.

Why (D) is wrong: Wrong fraction of the step.

Hinglish: *Trisection wala idea hi hai, bas 1/4 ka step. Total jump (8, -8), uska chauthai (2, -2), P mein jodo: (1, 2). Option C aata hai jab log step ko P se MINUS kar dein.*

MCQ 37 · Normal · Concept: Sign logic

37. If a > 0 and b > 0, then the point (-a, -b) lies in:

A. Quadrant I
B. Quadrant II
C. Quadrant III
D. Quadrant IV

Correct Answer: (C)

-a and -b are both negative, giving the sign pattern (-, -), which is Quadrant III.

Why (A) is wrong: Needs both positive.

Why (B) is wrong: Needs x negative, y positive.

Why (D) is wrong: Needs x positive, y negative.

Hinglish: *Positive dono the, minus laga-te-hi dono negative = Q III. Signs hi sab kuch hain is chapter mein.*

MCQ 38 · Normal · Concept: Statement-based

38. Consider: (i) (x, y) equals (y, x) only when x = y. (ii) The distance between two points can be negative. (iii) Every point on the x-axis has y-coordinate 0. Which statements are TRUE?

A. (i) and (iii) only
B. (i) and (ii) only
C. (ii) and (iii) only
D. all three

Correct Answer: (A)

(i) is true: equal ordered pairs need equal coordinates. (ii) is false: distance is a length, always at least 0. (iii) is true by definition of the x-axis.

Why (B) is wrong: (ii) is never true.

Why (C) is wrong: (ii) is never true.

Why (D) is wrong: (ii) spoils the set.

Hinglish: *(i) sahi - ordered pair ka matlab hi coordinate-by-coordinate match hai. (ii) galat - distance kabhi negative nahi. (iii) sahi - x-axis ka definition. Toh (i) aur (iii).*

MCQ 39 · Normal · Concept: Symbolic horizontal distance

39. The distance between (a + b, c) and (a - b, c) is:

A. 2b units
B. 2|b| units
C. 2a units
D. |c| units

Correct Answer: (B)

Same y-coordinate c, so the distance is |(a + b) - (a - b)| = |2b| = 2|b|. The a-terms cancel.

Why (A) is wrong: Correct only when b is known positive; the safe answer carries the bar.

Why (C) is wrong: a cancels out completely.

Why (D) is wrong: c is irrelevant to a horizontal segment.

Hinglish: *y same hai, horizontal distance: |a + b - a + b| = |2b|. Bar isliye - b negative ho sakta hai, distance nahi ho sakti. a cancel ho gaya - uska koi role nahi.*

MCQ 40 · HOTS · Concept: Diagonal-step distance

40. Q = (3 + k, 4 + k) lies at distance $5\sqrt{2}$ from P(3, 4). Then k equals:

A. 5 only
B. -5 only
C. 5 or -5
D. no real value

Correct Answer: (C)

Distance = $\sqrt{(k^2 + k^2)} = \sqrt{(2k^2)} = |k|\sqrt{2}$. Setting $|k|\sqrt{2} = 5\sqrt{2}$ gives |k| = 5, so k = 5 or k = -5. Both directions along the diagonal line y = x work.

Why (A) is wrong: Missed the negative direction along the line.

Why (B) is wrong: Missed the positive direction.

Why (D) is wrong: Real solutions exist.

Hinglish: *Step (k, k) diagonal direction mein hai, distance $|k|\sqrt{2}$. $5\sqrt{2}$ se match karne par $|k| = 5$ - dono sign possible. Plus-minus bhoolna is question ki asli trap hai.*

MCQ 41 · Normal · Concept: Square perimeter

41. The perimeter of the square with vertices (-1, 2), (3, 2), (3, 6) and (-1, 6) is:

A. 16 units

B. 8 units

C. 32 units

D. 14 units

Correct Answer: (A)

Side = 3 - (-1) = 4. Perimeter = 4 x 4 = 16.

Why (B) is wrong: Two sides only.

Why (C) is wrong: Doubled for no reason.

Why (D) is wrong: Arithmetic slip.

Hinglish: *Side = x-gap = 4. Perimeter = 16. 8 sirf do sides, 32 toh area ke jaisa lagega ($4^2 = 16$ area actually) - perimeter 4 x side formula use karo.*

MCQ 42 · Normal · Concept: Equidistant + quadrant

42. A point is equidistant from both axes, lies in Quadrant II, and is at distance $4\sqrt{2}$ from the origin. The point is:

A. (4, 4)

B. (-4, 4)

C. (-4, -4)

D. $(4\sqrt{2}, -4\sqrt{2})$

Correct Answer: (B)

Equidistant from the axes with distance d from the origin forces $|x| = |y| = d/\sqrt{2} = 4$. Quadrant II demands x negative, y positive: (-4, 4).

Why (A) is wrong: Wrong quadrant (I).

Why (C) is wrong: Wrong quadrant (III).

Why (D) is wrong: Distances from axes would be $4\sqrt{2}$, not equal-to-origin relation.

Hinglish: *Distance $4\sqrt{2} = |x|\sqrt{2}$ (diagonal line) isliye $|x| = 4$. Q II: x minus, y plus. (-4, 4). Signs hi toh asli question the!*

MCQ 43 · HOTS · Concept: Circumcentre of right triangle

43. For the right triangle with vertices (0, 0), (6, 0) and (0, 8), the point equidistant from all three vertices is:

A. (3, 4)

B. (3, 5)

C. (4, 3)

D. (2, 3)

Correct Answer: (A)

Hypotenuse joins (6, 0) to (0, 8); its midpoint is (3, 4). In a right triangle the midpoint of the hypotenuse is equidistant from all three vertices (distance 5 here: 3-4-5 again).

Why (B) is wrong: Not the midpoint of the hypotenuse.

Why (C) is wrong: Swapped coordinates.

Why (D) is wrong: No geometric basis.

Hinglish: *Q42 wala theorem yaad karo - right triangle mein hypotenuse ka MIDPOINT hi teeno se equidistant. (6,0) aur (0,8) ka midpoint = (3, 4). Distance check: origin se 5, dono vertices se 5. Bang on.*

MCQ 44 · Normal · Concept: Circle through origin

44. A circle is centred at (3, 4) with radius 5. Which statement is TRUE?

A. The origin lies on the circle

B. The origin lies inside the circle

C. The origin lies outside the circle

D. The circle does not reach the axes

Correct Answer: (A)

Distance from (3, 4) to (0, 0) = $\sqrt{(9 + 16)} = 5$ = radius exactly, so the origin lies ON the circle. The circle touches both axes at (0, 4)... wait, distance from centre to the y-axis is 3 (less than 5) and to the x-axis is 4 (less than 5), so it crosses both axes.

Why (B) is wrong: Would need distance less than 5.

Why (C) is wrong: Would need distance more than 5.

Why (D) is wrong: The circle crosses both axes.

Hinglish: *Centre (3, 4) se origin ki doori = 5 = radius. Isliye origin circle PAR hai. Aur kyunki centre axis se 3 aur 4 door hai (5 se kam), circle dono axes ko kaat bhi raha hai. Option D bhi galat hai.*

MCQ 45 · Normal · Concept: Grid vs direct

45. On a unit grid, the shortest path from (2, 3) to (6, 7) moving only along grid lines has length 8. The straight-line distance is:

A. $4\sqrt{2}$ units

B. 8 units

C. $2\sqrt{2}$ units

D. 16 units

Correct Answer: (A)

Gaps: 6 - 2 = 4 and 7 - 3 = 4. Direct distance = $\sqrt{(16 + 16)} = \sqrt{32} = 4\sqrt{2}$ (approximately 5.66), versus 8 along the grid.

Why (B) is wrong: That is the grid distance.

Why (C) is wrong: Underestimated the root.

Why (D) is wrong: That is the squared distance 32 misread as 16 x ... no, 16 is not even the square here.

Hinglish: *4 aur 4 ka square: 32, root $4\sqrt{2}$. Grid par 8, seedha 5.66 - seedha rasta jeet gaya. $\sqrt{32}$ ko simplify karna mat bhoolo: $4\sqrt{2}$.*

MCQ 46 · Normal · Concept: Reflection preserves midpoints

46. A and B are two points and M is the midpoint of AB. A and B are reflected in the x-axis to A' and B'. Which statement is TRUE?

A. M' (the image of M) is the midpoint of A'B'

B. A'B' is longer than AB

C. M' does not exist on segment A'B'

D. The midpoint of A'B' is M itself, unchanged

Correct Answer: (A)

Reflection preserves distances and maps the segment onto the segment, so it maps the midpoint to the midpoint. The image M' is the midpoint of A'B'. (M itself moves unless it lies on the x-axis.)

Why (B) is wrong: Reflections preserve all distances.

Why (C) is wrong: The midpoint image always exists.

Why (D) is wrong: M moves under reflection unless it sits on the x-axis.

Hinglish: *Reflection poore segment ko utha ke shift karta hai - beech ka point beech ka hi rahega (image mein). AB ki length bhi same rehti hai - option B galat.*

MCQ 47 · Normal · Concept: Right triangle area, shifted

47. A(-1, -5), B(3, -5) and C(3, -2) form a right triangle. Its area is:

A. 6 sq units

B. 12 sq units

C. 5 sq units

D. 7 sq units

Correct Answer: (A)

AB is horizontal with length 4; BC is vertical with length 3. Area = (1/2)(4)(3) = 6. Negative coordinates do not change lengths or areas.

Why (B) is wrong: Forgot the 1/2.

Why (C) is wrong: Used the hypotenuse 5 as a leg.

Why (D) is wrong: Sum of legs instead of half-product.

Hinglish: *Sab negative zone mein hai par length toh gap hai: 4 aur 3. Area 6. Negative coordinates se dar mat - absolute gap lo.*

MCQ 48 · HOTS · Concept: Counting locus points

48. How many points satisfy BOTH |x| = 5 and |y| = 3?

A. 4

B. 2

C. 8

D. 1

Correct Answer: (A)

|x| = 5 gives x = 5 or -5; |y| = 3 gives y = 3 or -3. Combinations: (5, 3), (5, -3), (-5, 3), (-5, -3) - four points, one in each quadrant.

Why (B) is wrong: Counted only one sign combination.

Why (C) is wrong: Over-counted.

Why (D) is wrong: Missed the sign freedom.

Hinglish: *Do signs, do choices har ek ka = 2 x 2 = 4 points. Har quadrant mein ek. Ye counting-sawaal HOTS mein easy marks hai agar system se lo.*

MCQ 49 · Normal · Concept: Product of roots

49. If the point (x, 4) is at distance 5 from the origin, the PRODUCT of all possible values of x is:

A. -9
B. 9
C. 0
D. 3

Correct Answer: (A)

$x^2 + 16 = 25$ gives x = 3 or x = -3. Product = 3 x (-3) = -9.

Why (B) is wrong: Missed the negative root.

Why (C) is wrong: Sum would be 0, not product.

Why (D) is wrong: Single root only.

Hinglish: *3-4-5 triple phir! x = plus-minus 3. Product manga hai: 3 x (-3) = -9. Dono values nikalna zaroori tha, warna product hi galat.*

MCQ 50 · Normal · Concept: Nearest axis point

50. The point on the x-axis NEAREST to (4, 3) is:

A. (4, 0)
B. (0, 3)
C. (0, 0)
D. (3, 0)

Correct Answer: (A)

The shortest route from a point to a line is along the perpendicular. Dropping a perpendicular from (4, 3) to the x-axis lands at (4, 0), with distance 3.

Why (B) is wrong: (0, 3) is on the y-axis.

Why (C) is wrong: Origin is distance 5 away.

Why (D) is wrong: (3, 0) is distance √10 away.

Hinglish: *Line ka sabse nazdeeki point = perpendicular wala pair. (4, 3) se seedha niche = (4, 0), doori 3. y-coordinate hi doori thi - 3.*

MCQ 51 · Normal · Concept: Coinciding ordered pairs

51. The points (5, k) and (k, 5) coincide when k equals:

A. 0
B. 5
C. 5 or -5
D. any real number

Correct Answer: (B)

Two ordered pairs coincide when both coordinates match: 5 = k and k = 5 - the single condition k = 5.

Why (A) is wrong: k = 0 gives (5, 0) vs (0, 5) - different.

Why (C) is wrong: No distance or equation here to create two roots.

Why (D) is wrong: Only the diagonal line y = x hosts such coincidences.

Hinglish: *Ek hi point tab banega jab dono coordinate same: k = 5. Plus-minus wala confusion Q40 jaisa hai par wahan distance thi, yahan equality hai - yahan sirf ek value.*

MCQ 52 · Normal · Concept: 13-triple membership

52. Which of the following CANNOT be the coordinates of a point at distance 13 from the origin?

A. (5, 12)
B. (12, 5)
C. (6, 12)
D. (13, 0)

Correct Answer: (C)

Check squares against 169: 25 + 144 = 169 (yes), 144 + 25 = 169 (yes), 36 + 144 = 180 (no), 169 + 0 = 169 (yes). (6, 12) is outside the circle of radius 13.

Why (A) is wrong: Exactly the 5-12-13 triple.

Why (B) is wrong: Same triple swapped.

Why (D) is wrong: Axis point at full radius.

Hinglish: *$6^2 + 12^2 = 180$, 169 nahi - bahar. Baaki sab 13-triple family ke hain. Aise 'which CANNOT' questions mein har option ko 169 se compare karo.*

MCQ 53 · Normal · Concept: Symbolic diagonal distance

53. The distance between (t, t) and (t + 3, t + 7) is:

A. √58 units
B. 10 units
C. √52 units
D. 58 units

Correct Answer: (A)

Differences: (t + 3) - t = 3 and (t + 7) - t = 7. Distance = √(9 + 49) = √58. The t cancels - the distance is the same wherever the pair sits.

Why (B) is wrong: Confused with a non-existent triple.

Why (C) is wrong: Arithmetic slip: 9 + 49 is not 52.

Why (D) is wrong: Forgot the root.

Hinglish: *t cancel ho gaya dono difference mein. Bacha 3 aur 7: 9 + 49 = 58, root √58. Simplify nahi hota (58 = 2 x 29, koi square factor nahi). 58 ko root ke bahar mat likho - wo squared distance hai.*

MCQ 54 · HOTS · Concept: Classify quadrilateral

54. The points (1, 1), (4, 1), (6, 3), (3, 3) joined in order form a:

A. rectangle
B. parallelogram but not a rectangle
C. square
D. rhombus

Correct Answer: (B)

AB = 3 and CD = 3 (both horizontal); BC = √(4 + 4) = 2√2 and DA = 2√2 (both slanted with the same step (+2, +2)). Opposite sides equal and parallel: parallelogram. Diagonals: AC = √(25 + 4) = √29, BD = √(1 + 4) = √5, unequal - so not a rectangle. Sides are unequal too, so not a rhombus or square.

Why (A) is wrong: Diagonals unequal - fails the rectangle test.

Why (C) is wrong: Sides unequal.

Why (D) is wrong: Rhombus needs all sides equal.

Hinglish: *Opposite sides barabar (3, 3) aur (2√2, 2√2) - parallelogram. Diagonals check: √29 vs √5 - barabar nahi, toh rectangle nahi. Sides bhi unequal - rhombus/square nahi. Ye Q23-style classification ka MCQ avatar hai.*

MCQ 55 · HOTS · Concept: Ratio division point

55. P divides the segment from A(0, 0) to B(12, 8) in the ratio AP : PB = 1 : 3. Then P is:

A. (3, 2)

B. (4, 8/3)

C. (6, 4)

D. (9, 6)

Correct Answer: (A)

AP is one part out of four total parts, so P = A + (1/4)(B - A) = (12/4, 8/4) = (3, 2). Check AP = √13 and PB = √(81 + 36) = √117 = 3√13, ratio 1 : 3. Correct.

Why (B) is wrong: Took 1/3 of the way instead of 1/4.

Why (C) is wrong: That is the midpoint (ratio 1 : 1).

Why (D) is wrong: That is the 3 : 1 point.

Hinglish: *Ratio 1 : 3 matlab total 4 parts, P pehla part leke baitha hai. Step = B - A ka 1/4 = (3, 2). Verify: AP = √13, PB = 3√13 - ratio sahi. (6, 4) toh MIDPOINT hai (1:1 wala), ratio dhyan se padho!*

MCQ 56 · Normal · Concept: Vertical distance, negatives

56. The distance between (-2, 4) and (-2, -9) is:

A. 13 units

B. 11 units

C. √193 units

D. 5 units

Correct Answer: (A)

Same x, so distance = |4 - (-9)| = |4 + 9| = 13.

Why (B) is wrong: Dropped the sign of -9.

Why (C) is wrong: Used the full distance formula unnecessarily.

Why (D) is wrong: Subtracted 4 from 9 without the negative.

Hinglish: *x same - vertical line. 4 - (-9) = 13. Minus ke saath minus ka khel phir. 11 kaise? 4 + (-9) karke -7 ka absolute... nahi, 9 - 4 = 5? Option D wahi hai. Dono galat - sahi formula |y2 - y1|.*

MCQ 57 · Normal · Concept: Square from diagonal

57. A square has diagonal of length √72. Its area is:

A. 36 sq units

B. 72 sq units

C. 18 sq units

D. 6√2 sq units

Correct Answer: (A)

Area = diagonal^2/2 = 72/2 = 36. (Side = √72/√2 = √36 = 6.)

Why (B) is wrong: Forgot to halve d^2.

Why (C) is wrong: Halved twice.

Why (D) is wrong: That is the side, not the area.

Hinglish: *Square area = $d^2/2$ - seedha 36. Side 6, side-square 36. 72 likhne wale 1/2 bhool gaye.*

MCQ 58 · HOTS · Concept: Compare three distances

58. For the point (6, 8), which distance is the SMALLEST: to the x-axis, to the y-axis, or to the origin?

A. to the x-axis (8 units)

B. to the y-axis (6 units)

C. to the origin (10 units)

D. all are equal

Correct Answer: (B)

Distance to the x-axis = |y| = 8. Distance to the y-axis = |x| = 6. Distance to the origin = √(36 + 64) = 10. The comparison is 6 < 8 < 10, so the smallest is the 6-unit distance to the y-axis.

Why (A) is wrong: x-axis distance is 8, not 6.

Why (C) is wrong: Origin distance is always the largest of the three here.

Why (D) is wrong: 6, 8, 10 are distinct.

Hinglish: *Ruko, ye question galti-pakadne wala hai! x-axis se doori |y| = 8 hai, y-axis se doori |x| = 6 hai. Sabse chhoti 6 hai = y-axis se doori. Asli seekh: x-axis se doori y-coordinate hoti hai, y-axis se doori x-coordinate - ULTA mat karna, warna 6 aur 8 ki jagah badal jaayegi.*

MCQ 59 · Normal · Concept: Assertion-reason axis

59. Assertion (A): The point (0, 4) lies on the y-axis. Reason (R): Every point whose x-coordinate is 0 lies on the y-axis.

A. Both A and R are true and R explains A

B. Both A and R are true but R does not explain A

C. A is true, R is false

D. A is false, R is true

Correct Answer: (A)

(0, 4) has x = 0, which is exactly the reason R states for lying on the y-axis. Both are true and R is the direct justification of A.

Why (B) is wrong: The connection is direct, not incidental.

Why (C) is wrong: R is the standard definition.

Why (D) is wrong: A is clearly true.

Hinglish: *(0, 4) ka x zero hai - aur y-axis ka definition bhi yahi hai. R hi A ka logic hai. Straightforward A-R: option A.*

MCQ 60 · HOTS · Concept: 9-12-15 triangle

60. The perimeter of the triangle with vertices (0, 0), (9, 0) and (0, 12) is:

A. 36 units

B. 30 units

C. 42 units

D. 39 units

Correct Answer: (A)

Legs 9 and 12; hypotenuse = √(81 + 144) = √225 = 15 (the 9-12-15 triple = 3 x (3-4-5)). Perimeter = 9 + 12 + 15 = 36.

Why (B) is wrong: Used 5-12-13 by mistake.

Why (C) is wrong: Overshoot on the hypotenuse.

Why (D) is wrong: Partial sum.

Hinglish: *9 aur 12 ko dekho: dono 3 se divide hote hain - 3-4-5 ka scaled version, hypotenuse 15. Perimeter 36. Triples ka scaling trick bade numbers ko chhota kar deta hai.*

1.5 Chapter Formula Sheet

Every working formula of Chapter 1 in one table. Read the conditions column carefully - most wrong answers in exams are correct formulas applied in illegal situations.

Formula	Meaning and when to use	Conditions	Tiny example	Common mistake
Distance from O = $\sqrt{(x^2 + y^2)}$	Length of the segment joining any point to the origin; use whenever 'how far from O' appears	Any real x, y	(6, 8): $\sqrt{(36 + 64)} = 10$	Writing x + y instead of the square-root sum
$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$	Distance between ANY two points; the master formula	Works for all real coordinates, any order of the points	(1, 2) to (4, 6): $\sqrt{(9 + 16)} = 5$	Taking the root separately on each difference
Horizontal distance = $\lvert x_2 - x_1 \rvert$	Two points with the same y-coordinate	$y_1 = y_2$	(2, 5) to (9, 5): 7	Forgetting the absolute value when $x_2 < x_1$
Vertical distance = $\lvert y_2 - y_1 \rvert$	Two points with the same x-coordinate	$x_1 = x_2$	(0, -4) to (0, 6): 10	Computing 6 - (-4) as 2 instead of 10
Midpoint M = $((x_1 + x_2)/2, (y_1 + y_2)/2)$	The point that bisects a segment; also proves diagonals bisect	Always valid for a segment	(-3, 8) and (5, -2) give (1, 3)	Subtracting instead of adding, or forgetting to halve
Reverse midpoint: B = $(2m_1 - x_1, 2m_2 - y_1)$	One endpoint known with the midpoint; find the other endpoint	Midpoint given	M(2, -1), A(-4, 3) give B(8, -5)	Adding the coordinates instead of subtracting
Trisection: P = A + (B - A)/3	Points cutting a segment into three equal parts	P closer to A; Q = A + 2(B - A)/3	A(4, 7), B(16, -2) give P(8, 4), Q(12, 1)	Using half the gap (that gives the midpoint)
Reflection in x-axis: $(x, y) \to (x, -y)$	Mirror image across the horizontal axis	Only y flips	(4, -3) gives (4, 3)	Flipping x as well (that is a double reflection)
Reflection in y-axis: $(x, y) \to (-x, y)$	Mirror image across the vertical axis	Only x flips	(-3, -5) gives (3, -5)	Flipping y instead of x

Formula	Meaning and when to use	Conditions	Tiny example	Common mistake
Circle test: on $x^2 + y^2 = r^2$; inside if $x^2 + y^2 < r^2$; outside if $x^2 + y^2 > r^2$	Classify a point against a circle centred at O	Centre must be the origin	(4, 2) with r = 5: 20 < 25, inside	Comparing roots instead of squares
Collinearity: largest distance = sum of other two	Three points on one line	Use EXACT values, no early rounding	5 + 5 = 10 for (0,0), (3,4), (6,8)	Comparing rounded decimals in near-miss cases
Perimeter of polygon = sum of side distances	Vertices given; add up the side lengths	Compute each side exactly	8 + 5 + 8 + 5 = 26 for a 8 x 5 rectangle	Using a wrong pair of coordinates for one side
Area of right triangle = (1/2) x leg x leg	Right angle visible (legs axis-parallel or via converse Pythagoras)	Legs must be the two sides forming the right angle	(0,0), (8,0), (0,6): area 24	Forgetting the 1/2, or using the hypotenuse as a leg

1.6 Last-Minute Cheat Sheet

> **30-second revision card - Chapter 1**
> SIGNS: Q I (+,+), Q II (-,+), Q III (-,-), Q IV (+,-). x = 0 means y-axis, y = 0 means x-axis. DISTANCES: same y, use $|x_2-x_1|$; same x, use $|y_2-y_1|$; otherwise $\sqrt{(dx^2+dy^2)}$. From origin: $\sqrt{(x^2+y^2)}$. TRIPLES: 3-4-5, 5-12-13, 8-15-17, 7-24-25 and their multiples. MIDPOINT = averages; reverse = double minus. Reflection: x-axis flips y, y-axis flips x. CIRCLE: compare x^2+y^2 with r^2. COLLINEAR: biggest = sum of other two. (a,b) = (b,a) only if a = b.

- **If you see two points sharing a coordinate** → single-difference shortcut, no square root.
- **If you see 'find k' with a distance** → absolute value opens BOTH ways; expect two answers.
- **If you see 'inside / on / outside the circle'** → compare $x^2 + y^2$ with r^2, never roots.
- **If you see 'midpoint of ... is M, find the other end'** → double the midpoint, subtract the known end.
- **If you see 'reflected in the axis'** → flip exactly ONE sign; two reflections flip both.
- **If you see 'without plotting, same line?'** → largest distance equals sum of the other two (exact values only).
- **If you see legs on the axes of a triangle** → right angle at the shared axis point; area = half product of legs.
- **If a question shifts a 3-4-5-style shape away from the origin** → the triple survives translation; just use gaps.

1.7 Mistake Buster - DON'T MAKE THESE MISTAKES

1. **Coordinate-slot swap.** Distance from the y-axis is the x-coordinate, and from the x-axis is the y-coordinate. Writing them the other way round silently ruins the answer. Fix: say aloud 'x talks about the y-axis' while practising.

2. **Sign collapse in subtraction.** 6 - (-3) = 9, not 3. Every vertical-distance error trace leads here. Fix: rewrite a - (negative) as a + positive BEFORE computing.
3. **Root-splitting the formula.** $\sqrt{(dx^2 + dy^2)}$ is NOT $\sqrt{dx} + \sqrt{dy}$ and NOT dx + dy. Fix: chant 'square, square, add, then root - once'.
4. **Losing a root in 'find k' problems.** |4 - k| = 7 has two solutions. Fix: always write the two cases explicitly.
5. **Comparing rounded roots for collinearity.** $\sqrt{5} + 5$ vs $2\sqrt{13}$ differ by less than 0.03; decimals mislead. Fix: compare squared values exactly.
6. **Mixing up (x, y) and (y, x).** They coincide only on the line y = x. Fix: never 'sort' coordinates alphabetically or by size.
7. **Distance written as negative.** Distances are never negative; a negative coordinate is a DIRECTION. Fix: absolute value before reporting any length.
8. **Midpoint halving forgotten.** (x_1+x_2, y_1+y_2) is the double midpoint, not the midpoint. Fix: divide by 2 as a separate written step.
9. **Reflection in the wrong variable.** x-axis reflection flips y. Students flip x because the axis is named x. Fix: 'the axis you reflect in stays fixed - the other one flips'.
10. **Using the hypotenuse as a leg in area.** Area = half product of the two LEGS (the sides touching the right angle). Fix: locate the right angle first, then pick its two arms.

1.8 Exam Strategy for Chapter 1

Order of attack. Begin with identification and axis-distance questions (they take under a minute each and settle your nerves). Then do standard distance-formula questions. Keep classification questions (prove square/parallelogram) and multi-part HOTS for the end, when you can give them the calm they need.

Where marks are lost. In this chapter, examiners report three leak-points: sign errors under time pressure (a 2-mark question gone), single-root answers in 'find k' questions (half marks gone), and unverified classifications (method marks gone even when the answer is right). The antidote for all three is the same: write the check line - verify one distance, verify both roots, verify the defining property.

Formulas most frequently tested. The distance formula (appears in almost every question), the parallel-to-axis shortcuts, the midpoint formula in both directions (find midpoint, find missing endpoint), reflection rules, and the $x^2 + y^2$ versus r^2 comparison. The 3-4-5 family of triples is effectively a free speed boost in half of all numerical questions.

Concepts that get mixed. Distance with midpoint (find endpoints from midpoints), distance with circle membership (inside/outside tests), reflections chained with distance (prove preservation), and collinearity with equal spacing (chain-step pattern). When two concepts combine, handle them in labelled mini-steps rather than one mega-computation.

Safe shortcuts. Using recognised triples (only when the numbers match exactly), shifting both points by a common vector before computing distance, and comparing squared distances instead of roots. **Unsafe shortcuts.** Rounding roots early to compare, guessing the quadrant from a sketch drawn freehand, dropping the absolute value bars 'because the answer looks positive', and applying the midpoint formula when the question actually describes a trisection point.

How to check an answer. Distance: recompute with the points interchanged - the result must be identical. Midpoint: re-average your answer with the other endpoint - you must recover the given midpoint. Reflection: check one preserved distance. Circle: recompute $x^2 + y^2$. Two minutes of checking routinely saves four to six marks.

1.9 Chapter Completion Checklist

Tick each box only when you can honestly say DONE:

- [ ] Every concept covered: axes, origin, ordered pairs, quadrants, axis points, distances, distance formula, reflections, midpoint, trisection, collinearity, circle membership
- [ ] Every formula covered: 13 formulas in Section 1.5, each with conditions and a micro-example
- [ ] 50 advanced questions completed (Q1-Q50), difficulty ladder respected
- [ ] Every question has a full step-by-step solution
- [ ] Every solution has a Hinglish explanation
- [ ] 60 MCQs completed with answers
- [ ] Every MCQ has an explanation plus trap analysis
- [ ] Formula sheet completed
- [ ] Last-minute cheat sheet completed
- [ ] Mistake buster completed (10 named mistakes)
- [ ] Exam strategy completed

Chapter 2: Introduction to Linear Polynomials

From locating points we now move to describing relationships with expressions. The uploaded PDF builds the idea of a polynomial from daily life (boxes of pens, garden fencing, chess-club fees), defines degree, focuses on linear polynomials, shows them as input-output machines (functions), develops linear patterns and growth/decay, teaches how to fit the relationship $y = ax + b$ from data, and finishes by plotting straight lines and reading the meaning of a (slope) and b (y-intercept). This book sharpens every one of those ideas with reverse questions, two-equation fittings, plan comparisons and pattern proofs.

2.1 Complete Concept Cheat Sheet (read this before any question)

2.1.1 Algebraic expressions: terms, coefficients, constants, variables

What is it? An algebraic expression combines numbers, variables (letter-numbers) and operation signs. In $4x + 5y + 3$, the parts $4x$, $5y$ and 3 are the **terms**; x and y are **variables** (also called letter-numbers); 4 and 5 are the **coefficients**; 3 is a **constant**. **Why coefficients matter:** the coefficient tells you how many copies of the variable you have, and it is the number that multiplies the variable - including its sign. In $-7x$ the coefficient is -7, not 7.

When do I use this? Every 'identify the terms / coefficient / constant' question, and every modelling question where a real situation must become an expression (cost of fencing = rate x length, and so on). **Tiny example.**

For the garden in the PDF: wire fence cost 100 x 2l = 200l, wooden fence 80 x 2w = 160w, seeds 50 x area = 50lw; total 200l + 160w + 50lw - a three-term expression in two variables.

> **Hidden coefficient trap**
> A term like x has coefficient 1 (not 0!), and -x has coefficient -1. A term like $5x^2y$ has coefficient 5. Missing numbers never mean missing coefficients.

2.1.2 Degree and the polynomial family

What is it? A polynomial in ONE variable is an expression made of that variable and its powers (whole-number powers). The **degree** is the highest power of the variable. Degree names: 1 = linear, 2 = quadratic, 3 = cubic, 0 = constant. Examples: 3z + 7 is linear (degree 1); $x^2 + 5x + 1$ is quadratic; $5y^3 + y^2 - 8$ is cubic; the number 8 alone is a constant polynomial (it can be written $8x^0$).

How do I find the degree quickly? Scan the powers of the variable and take the maximum. In $2x^4 - 3x^3 + 6x^2 - 2x + 7$ the powers are 4, 3, 2, 1, 0, so the degree is 4. **Sign-awareness:** the degree ignores the sign and size of coefficients - $-9x^5$ still has degree 5. **Tiny example.** -13 is a constant polynomial of degree 0; 6z + 1 is linear.

Common confusion: 'linear polynomial' (degree exactly 1, like 2x + 3) versus 'linear equation' (a statement like 2x + 3 = 64 that equates a linear polynomial to something). The polynomial is the expression; the equation is the claim that two expressions are equal.

2.1.3 Evaluation: the input-output machine

What is it? Substituting a value of the variable into a polynomial gives its value at that input. The PDF pictures this as a machine: feed x = 4 into the machine 2x + 3, out comes 11; feed x = -6, out comes -9. This machine-view is the seed of the 'function' idea. **Golden rule of substitution:** write the substitution with brackets: 2(-6) + 3, never 2 -6 + 3. Brackets prevent the sign disaster.

Tiny example. For p(x) = 7x - 4: p(0) = -4, p(2) = 10, p(-3) = -25. **Reverse use:** given the output, find the input - that is solving a linear equation: 7x - 4 = 24 gives x = 4. Evaluation forward, equation backward - the same machine run both ways.

2.1.4 Linear equations from words

What is it? Translate a sentence into an equation with ONE unknown, solve, and answer the original question. Standard moves: 'one number is 10 more than the other' means x and x + 10; 'three times as many' means 3x; 'their sum is 64' means x + (x + 10) = 64.

Method (always the same skeleton). Step 1: define the unknown precisely ('let the smaller number be x'). Step 2: express every other quantity in terms of x. Step 3: write the balancing condition (sum, difference, total money). Step 4: solve. Step 5: ANSWER THE QUESTION ASKED - if the question asked for both numbers, report both; a bare x earns no full marks. **Sanity check:** substitute back into the words, not just the equation.

2.1.5 Linear patterns and the nth term

What is it? A linear pattern is a sequence whose consecutive terms differ by a CONSTANT. The tile pattern 1, 3, 5, 7, ... adds 2 tiles each stage, and its nth-stage count is 2n - 1. The constant difference becomes the coefficient of n; the constant term adjusts the start. **Finding the rule from the table:** if the difference is d, the rule looks like an = dn + c; find c by matching one known term (e.g. for 1, 3, 5: d = 2 and term 1 = 1 gives

1 = 2(1) + c, so c = -1).

Why the difference is the slope: on the graph of the rule, moving one stage right moves the value up by exactly d - the line rises by d per step. **Tiny example.** Matchstick squares in a row: 4, 7, 10, ... difference 3, rule 3n + 1. Stage 10 needs 31 matches. **Reverse use:** 'can the pattern ever equal 100?' means solving 3n + 1 = 100; here n = 33 - a whole number, so yes, stage 33 uses exactly 100 matches. If n had come out fractional, the answer would be NO - patterns only exist at whole stages.

2.1.6 Linear growth and linear decay

What is it? Growth: a quantity increases by a fixed amount over equal intervals - C(d) = 100 + 60d (journey cost). Decay: decreases by a fixed amount - h(t) = 3 - 0.5t (water level). The fixed change is the slope: positive for growth, negative for decay. **Structure:** value = starting value + (change per step) x (number of steps). On a graph, growth tilts up, decay tilts down.

How do I recognise it? Words like 'every month adds', 'decreases by 800 each year', 'reduces by 15 per day'. **Two standard questions:** forward (value after n steps - substitute) and backward (when does the value reach V? - solve). Decay also has the 'run out' question: set the expression to 0 (or the floor value) and solve. **Tiny example.** Phone bought at 10000, losing 800 per year: v = 10000 - 800t; after 3 years 7600; it touches 3600 when t = 8.

2.1.7 Fitting y = ax + b from two observations

What is it? Many real bills have the shape y = ax + b: a fixed part b plus a per-unit part a. Given two observations (x_1, y_1) and (x_2, y_2), substitute both into y = ax + b to get TWO equations, then eliminate b by subtraction: $a = (y_2 - y_1)/(x_2 - x_1)$. Then recover b from either observation: $b = y_1 - a\,x_1$.

Worked skeleton (the telecom example). 10 GB cost 350 and 20 GB cost 550. Equations: 350 = 10a + b and 550 = 20a + b. Subtract: 200 = 10a, so a = 20 (each extra GB costs Rs 20). Back-substitute: b = 350 - 200 = 150 (the fixed monthly fee). Answer: y = 20x + 150. **Meaning check:** a is the RATE per unit, b is the ENTRY fee - interpreting them is as important as computing them.

> **Subtraction is the whole trick**
> Subtracting the two equations kills b because b is the SAME in both. What remains is a clean one-step equation for a. If your two equations accidentally have different b values, you have copied the data wrongly - stop and re-read.

2.1.8 Graphs of lines: slope a and y-intercept b

What is it? Every equation y = ax + b draws a straight line. Plotting needs just two points: pick any two x-values, compute the y-values, join with a ruler and extend. Three facts carry all the exam weight: (1) y = ax lines all pass through the ORIGIN (0, 0); (2) larger positive a makes a steeper line, a between 0 and 1 makes a flatter line, negative a tilts the line down-left to up-right reversed (decay); (3) b is where the line CUTS THE Y-AXIS, at the point (0, b) - the y-intercept.

Effect of changing a vs changing b. Fix b and vary a: the line pivots about (0, b), changing steepness. Fix a and vary b: the line SLIDES up or down, staying PARALLEL - lines with the same slope are parallel and never meet. Growth means positive slope, decay means negative slope; the slope numerically equals the constant difference of the underlying pattern. **Tiny example.** y = 2x - 1, y = 2x + 1, y = 2x + 5 are three parallel lines

cutting the y-axis at (0, -1), (0, 1) and (0, 5).

Converting to standard shape. Equations often arrive disguised: 2y = 4x + 7 means y = 2x + 3.5 after dividing by 2; 5y = 6x - 10 means y = 1.2x - 2. ALWAYS isolate y first; only then read a and b. Reading the coefficient of x before this conversion is a classic error.

2.1.9 Exam tips for this chapter

1. Before reading slope and intercept, rewrite the equation as y = ...
2. Substitute with brackets - especially for negative inputs.
3. In word problems, define the unknown in words first; the equation follows automatically.
4. For pattern rules, lock the constant difference first, then fit the constant term from ONE known term.
5. When fitting y = ax + b, subtract the two data equations; never solve them separately for b first (double work, double error chances).
6. Reverse questions (which stage gives 100 matches?) need INTEGER answers - a fractional n means 'no such stage', and saying so IS the answer.
7. On graphs, label the y-intercept point (0, b) explicitly - it is usually the mark-earning feature.

2.2 Question Types / Concept Map for Chapter 2

Type	Name	How to recognise it	Core method
T1	Anatomy of expressions	Identify terms, coefficients, variables, constants	Term-wise reading, sign-aware coefficients
T2	Degree and classification	Find degree; label linear/quadratic/cubic/constant	Highest power rule
T3	Evaluation	Find value of polynomial at given input	Bracket substitution
T4	Reverse evaluation	Given output, find input	Solve the linear equation
T5	Word-problem modelling	Numbers/ages/coins/fences in words	Define x, translate, solve, answer in words
T6	Linear patterns	Table of stages; find nth term or a target stage	Constant difference = coefficient of n
T7	Reachability question	Can the pattern equal a value?	Solve dn + c = target; n must be a whole number
T8	Growth and decay	Increases/decreases by fixed amount per interval	value = start + rate x steps
T9	Fit y = ax + b	Two observations given	Subtract equations to get a, back-substitute for b
T10	Interpret a and b	What do the numbers mean?	a = rate per unit, b = fixed part
T11	Slope and intercept reading	Find slope/y-intercept of disguised equations	Isolate y first

Type	Name	How to recognise it	Core method
T12	Parallel lines	Same slope, different intercepts	Compare a values after isolating y
T13	Graph plotting / reading	Draw the line; identify where it cuts axes	Two points; (0, b) on y-axis; y = 0 for x-axis
T14	Multi-concept HOTS	Composition of evaluations, plan comparisons, concurrence	Break into labelled steps, verify each

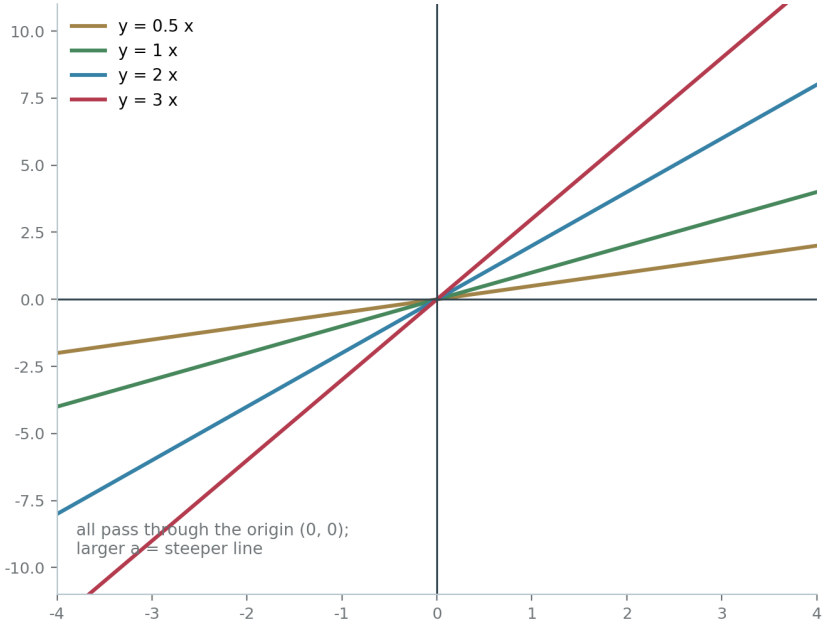


Figure 2.1: The family y = ax - all lines pass through the origin; larger a means steeper.

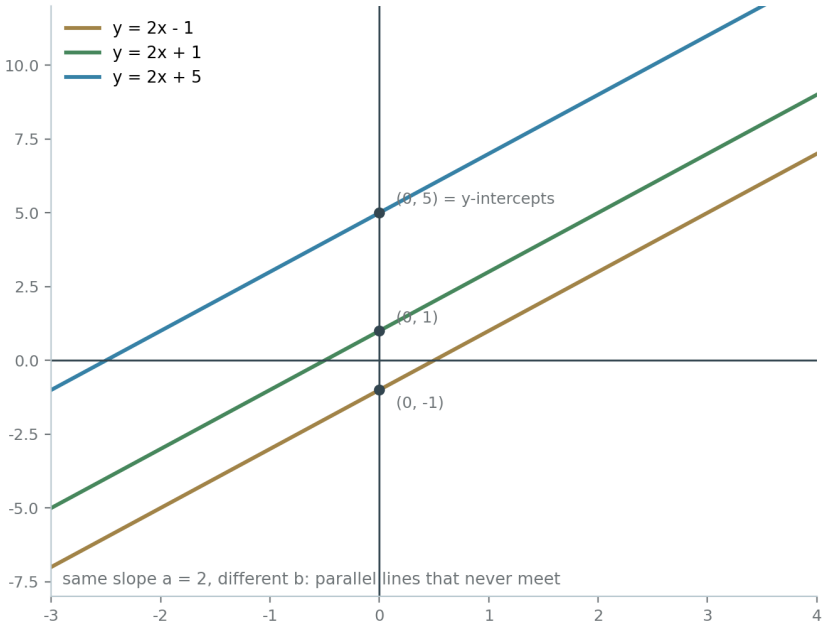


Figure 2.2: Same slope (a = 2), different intercepts b - parallel lines that never meet.

2.3 Practice Questions (Q1 - Q50) with Full Step-by-Step Solutions

Q1 · Basic · Concept: Anatomy of expressions (T1)

Q1. For the expression 6m + 4n - 9, list all the terms, the variables, the coefficient of each variable term, and the constant term. Then state the coefficient of m in the expression 6m + 4n - 9 - m after it is simplified.

SOLUTION

Step 1: Given: 6m + 4n - 9. Needed: terms, variables, coefficients, constant.

Step 2: Terms are the parts joined by plus or minus: 6m, 4n and -9. (The minus sign travels WITH its term.)

Step 3: Variables (letter-numbers): m and n.

Step 4: Coefficient of m is 6; coefficient of n is 4; the constant term is -9.

Step 5: Now simplify 6m + 4n - 9 - m: combine the m-terms, 6m - m = 5m.

Step 6: The simplified expression is 5m + 4n - 9, so the coefficient of m is 5.

» **Why this step?** Subtraction creates negative terms: -9 is a term, and combining like terms changes coefficients. Students who call the coefficient '6' after simplification have ignored the invisible -1.

Hinglish: *Terms ko minus ke saath padho - (-9) bhi ek term hai. Coefficient matlab 'kitne variable'. Aur dhyan do: simplified expression mein m ka coefficient 5 ban gaya kyunki -m ka coefficient -1 tha aur 6 - 1 = 5. Invisible coefficient dekhna hi asli skill hai.*

Answer: Terms: 6m, 4n, -9; variables: m, n; coefficients 6 and 4; constant -9. After simplification the coefficient of m is 5.

Q2 · Basic · Concept: Degree (T2)

Q2. Find the degree of each polynomial: (i) $3x^2 + 7x - 2$ (ii) $y^3 - y + 5$ (iii) -13 (iv) $6z + 1$ (v) $8 - t^4 + t^2$.

SOLUTION

Step 1: Degree = highest power of the variable present.

Step 2: (i) powers are 2, 1, 0, so degree 2 (quadratic).

Step 3: (ii) powers 3, 1, 0: degree 3 (cubic).

Step 4: (iii) the constant -13 can be written $-13x^0$, so degree 0 (constant polynomial).

Step 5: (iv) powers 1, 0: degree 1 (linear).

Step 6: (v) powers 0, 4, 2: the highest is 4, so degree 4 (a quartic - just report the number 4).

Hinglish: *Degree nikalne ka ek hi kaam: powers scan karo, sabse badi utha lo. Constant ka degree 0 hota hai - ye MCQ favourite hai. (v) mein dhyan do, powers beech mein ghumte hain (8, t^4, t^2) - order koi baat nahi, max chahiye.*

Answer: (i) 2 (ii) 3 (iii) 0 (iv) 1 (v) 4.

Q3 · Basic · Concept: Classification (T2)

Q3. Classify each as linear, quadratic, cubic or constant: $5 - 2x$, $x^2 + 3x$, $4t^3 - t$, 7, $0.5y + 3$.

SOLUTION

Step 1: Degree 1 means linear, 2 quadratic, 3 cubic, 0 constant.

Step 2: 5 - 2x has degree 1: linear.

Step 3: $x^2 + 3x$ has degree 2: quadratic.

Step 4: $4t^3 - t$ has degree 3: cubic.

Step 5: 7 has degree 0: constant.

Step 6: 0.5y + 3 has degree 1: linear (decimals in coefficients are perfectly fine).

Hinglish: *Naam yaad rakho: 1-linear, 2-quadratic, 3-cubic, 0-constant. Decimal coefficient wala bhi normal polynomial hi hai - 0.5y + 3 linear hi hai.*

Answer: Linear: 5 - 2x and 0.5y + 3. Quadratic: $x^2 + 3x$. Cubic: $4t^3 - t$. Constant: 7.

Q4 · Basic · Concept: Coefficients at high powers (T1)

Q4. In the polynomial $2x^4 - 5x^3 + x^2 - 9x + 1$, state the coefficients of x^4, x^3, x^2, x and the constant term.

SOLUTION

Step 1: Read term by term with signs attached.

Step 2: Coefficient of x^4 is 2.

Step 3: Coefficient of x^3 is -5 (the minus belongs to the coefficient).

Step 4: Coefficient of x^2 is 1 - the term x^2 is really $1x^2$.

Step 5: Coefficient of x is -9; the constant term is 1.

Hinglish: *Sign term ke saath chalta hai - (-5) hi coefficient hai. Aur x^2 ka coefficient 1 hai, zero nahi! Ye do hi cheezein is question mein marks katati hain.*

Answer: 2, -5, 1, -9 and constant 1.

Q5 · Basic · Concept: Evaluation (T3)

Q5. Find the value of the linear polynomial 7x - 4 at (i) x = 0, (ii) x = 2, (iii) x = -3.

SOLUTION

Step 1: Substitute with brackets every time.

Step 2: (i) 7(0) - 4 = 0 - 4 = -4.

Step 3: (ii) 7(2) - 4 = 14 - 4 = 10.

Step 4: (iii) 7(-3) - 4 = -21 - 4 = -25.

Step 5: Note how the outputs drop by exactly 7 each time x drops by 1 going 2, 0, -3: (10, -4, -25) - the constant difference at work.

Hinglish: *Bracket lagana mandatory hai: 7(-3) = -21. Bina bracket ke log 7 - 3 - 4 = 0 likh dete hain - jabardast galti. Aur observation: x ek kadam gira toh output 7 gira - linear ka dhadakta hua constant difference.*

Answer: -4, 10, -25.

Q6 · Basic · Concept: Evaluation of a quadratic (T3)

Q6. Find the value of the quadratic polynomial $2s^2 + 3s - 5$ at (i) s = 0, (ii) s = 2, (iii) s = -2.

SOLUTION

Step 1: (i) $2(0)^2 + 3(0) - 5 = -5$.

Step 2: (ii) $2(2)^2 + 3(2) - 5 = 2(4) + 6 - 5 = 8 + 6 - 5 = 9$.

Step 3: (iii) $2(-2)^2 + 3(-2) - 5 = 2(4) - 6 - 5 = 8 - 6 - 5 = -3$.

Step 4: Watch step (iii): $(-2)^2 = +4$ but $3(-2) = -6$. The square forgives the sign; the plain term does not.

Hinglish: *Do alag jagah sign ka risk hai: square wala term minus ko kha jaata hai (+4), par 3s wala term sign ko POORA follow karta hai (-6). Isi liye bracket - har value apne bracket mein.*

Answer: -5, 9, -3.

Q7 · Basic · Concept: Function machine (T3)

Q7. A machine computes y = 3x - 1 for every input x. Find the outputs for x = 5 and x = -2. Then find the input that produces the output 20.

SOLUTION

Step 1: Forward: x = 5 gives y = 3(5) - 1 = 15 - 1 = 14.

Step 2: Forward: x = -2 gives y = 3(-2) - 1 = -6 - 1 = -7.

Step 3: Reverse: we need x such that 3x - 1 = 20.

Step 4: Add 1 to both sides: 3x = 21. Divide by 3: x = 7.

Step 5: Verify: 3(7) - 1 = 20. Correct.

Hinglish: *Machine aage bhi chalti hai peeche bhi! Aage = substitute karo (14, -7). Peeche = equation solve karo: 3x - 1 = 20 se x = 7. Function ka yahi asli matlab hai - input-output ka judgementa rishta.*

Answer: Outputs 14 and -7; the input producing 20 is x = 7.

Q8 · Basic · Concept: Word problem: sum and difference (T5)

Q8. The sum of two numbers is 50, and one of them is 8 more than the other. Find the numbers.

SOLUTION

Step 1: Let the smaller number be x. Then the larger is x + 8.

Step 2: Sum condition: x + (x + 8) = 50.

Step 3: Simplify: 2x + 8 = 50, so 2x = 42, giving x = 21.

Step 4: The numbers are 21 and 29.

Step 5: Check in words: 21 + 29 = 50 and 29 - 21 = 8. Both conditions hold.

Hinglish: *Chota number x maan lo, bada apne aap x + 8 ban jaata hai. Sum equation se x = 21. Aur LAST STEP yaad rakho: dono numbers report karo - sirf 'x = 21' likha toh aadha answer hai. Words mein wapas check karo, equation mein hi nahi.*

Answer: The numbers are 21 and 29.

Q9 · Basic · Concept: Perimeter polynomial (T3/T5)

Q9. The perimeter of a square of side s is the linear polynomial 4s. Find the perimeter for s = 1, 2.5 and 7.5. If the side grows by 0.5 cm, by how much does the perimeter grow? Why?

SOLUTION

Step 1: p(1) = 4(1) = 4; p(2.5) = 10; p(7.5) = 30 (all in the side units).

Step 2: If the side grows by 0.5, the new perimeter is 4(s + 0.5) = 4s + 2.

Step 3: So the perimeter grows by exactly 2 units - for EVERY square, whatever its side.

Step 4: Why constant: 4 times a fixed growth is a fixed growth; this is the constant-difference signature of a linear pattern.

Hinglish: *PDF wala hi idea: side 0.5 badhe toh perimeter 2 badhta hai - hamesha, kisibhi square ke liye. Kyunki 4 x 0.5 = 2, aur multiplication by constant fixed growth ko fixed hi rakhta hai. Yehi 'linear pattern' ka matlab hai.*

Answer: Perimeters 4, 10, 30; growth per 0.5 cm of side is exactly 2 units (constant, because the rule is linear).

Q10 · Basic · Concept: Fee structure modelling (T5/T8)

Q10. A badminton academy charges a joining fee of Rs 300 plus Rs 60 for every session attended. (i) Write the total cost for n sessions. (ii) Find the cost for 12 sessions. (iii) A player paid Rs 780 in total. How many sessions did she attend?

SOLUTION

Step 1: Cost = fixed part + variable part = 300 + 60n.

Step 2: For 12 sessions: 300 + 60(12) = 300 + 720 = Rs 1020.

Step 3: Reverse: 300 + 60n = 780.

Step 4: Subtract 300: 60n = 480, so n = 8.

Step 5: Check: 300 + 60(8) = 300 + 480 = 780. Correct - she attended 8 sessions.

Hinglish: *Fixed + rate x count - yahi har bill ka formula hai. Aage jaane ke liye substitute, peeche aane ke liye solve. 780 mein se 300 hataya toh sessions ka khel bacha: 480/60 = 8.*

Answer: (i) 300 + 60n (ii) Rs 1020 (iii) 8 sessions.

Q11 · Moderate · Concept: Pattern rule (T6/T7)

Q11. A pattern of square tiles starts with 5 tiles in stage 1, and each new stage adds 3 tiles. (i) Write the number of tiles in stage n. (ii) How many tiles in stage 12? (iii) Which stage has exactly 65 tiles?

SOLUTION

Step 1: The sequence is 5, 8, 11, 14, ... with constant difference 3.

Step 2: Rule: tiles(n) = 3n + c. Match stage 1: 3(1) + c = 5 gives c = 2. So tiles(n) = 3n + 2.

Step 3: Stage 12: 3(12) + 2 = 38 tiles.

Step 4: Reverse: 3n + 2 = 65 gives 3n = 63, n = 21.

Step 5: n = 21 is a whole number, so stage 21 has exactly 65 tiles.

Hinglish: *Constant difference pehle pakdo (3), phir ek known term se c nikaalo (yahan 2). Aage ke liye substitute, peeche ke liye solve - aur n WHOLE NUMBER aana chahiye, warna stage exist nahi karti. Yahan 21 nikla, perfect.*

Answer: (i) 3n + 2 (ii) 38 tiles (iii) stage 21.

Q12 · Moderate · Concept: Linear decay money (T8)

Q12. Bela has Rs 500 and spends Rs 30 every day. (i) Write the amount left after n days. (ii) After how many days will she be left with Rs 200? (iii) What is the amount left after 15 days, and why does the model stay valid only up to 10 days of reaching zero? Actually find the exact day her money runs out.

SOLUTION

Step 1: Amount(n) = 500 - 30n.

Step 2: Set 500 - 30n = 200: 30n = 300, n = 10 days.

Step 3: After 15 days: 500 - 30(15) = 500 - 450 = Rs 50.

Step 4: Running out: 500 - 30n = 0 gives n = 500/30 = 16.67 days. Money reaches zero partway through the 17th day.

Step 5: So the linear model describes her cash only until about 16.67 days; beyond that the amount cannot go below 0.

» **Why this step?** Real decay models hit a floor. Reporting WHEN the floor is reached (and that the model stops there) is the mark of a careful modeller.

Hinglish: *Decay question mein teen sawaal hote hain: kitna bacha (substitute), kab itna bacha (solve), kab khatam (zero set karo). Yahan 16.67 din - matlab 17ve din ke beech mein khatam. Model ki limit batana bhi answer ka hissa hai - paisa negative nahi ho sakta!*

Answer: (i) 500 - 30n (ii) after 10 days (iii) Rs 50 after 15 days; money hits zero during the 17th day (n = 500/30 = 16.67), where the model ends.

Q13 · Moderate · Concept: Piecewise-style fare (T8)

Q13. A taxi charges Rs 40 for the first 3 km (and any part of it), then Rs 18 for every additional km. (i) Find the fare for a 12 km trip. (ii) Write the fare for a trip of n km where n is at least 3. (iii) For what distance is the fare Rs 148?

SOLUTION

Step 1: Beyond 3 km, chargeable extra distance = (n - 3) km.

Step 2: Fare for 12 km: 40 + 18 x (12 - 3) = 40 + 162 = Rs 202.

Step 3: General: F(n) = 40 + 18(n - 3) = 18n - 54 + 40 = 18n - 14, valid for n at least 3.

Step 4: Set 18n - 14 = 148: 18n = 162, n = 9 km.

Step 5: Check with the original description: 40 for the first 3 km plus 18 x 6 = 108 for the next 6 km: total 148. Correct.

Hinglish: *PDF ke auto-fare jaisa hi question, bas numbers naye. Formula banane ke liye (n - 3) ka idea zaroori hai - pehle 3 km ka paisa already fixed 40 mein hai. Simplify karke 18n - 14 milta hai aur reverse solve se 9 km. Hamesha original description se verify karo.*

Answer: (i) Rs 202 (ii) F(n) = 18n - 14 for n at least 3 (iii) 9 km.

Q14 · Moderate · Concept: Age problem (T5)

Q14. A mother is 4 times as old as her daughter. After 6 years, the sum of their ages will be 82 years. Find their present ages.

SOLUTION

Step 1: Let the daughter be x now; the mother is 4x.

Step 2: After 6 years: daughter x + 6, mother 4x + 6.

Step 3: Sum: (x + 6) + (4x + 6) = 82.

Step 4: Simplify: 5x + 12 = 82, so 5x = 70, x = 14.

Step 5: Present ages: daughter 14, mother 56.

Step 6: Check: after 6 years they are 20 and 62, sum 82. Correct.

Hinglish: *Age questions mein 'after' ka matlab dono ke age mein addition. 5x + 12 = 82 se x = 14. Check zaroori: 20 + 62 = 82. Mother 4x isliye 56 - equation se nahi, definition se.*

Answer: Daughter 14 years, mother 56 years.

Q15 · Moderate · Concept: Ratio and difference (T5)

Q15. Two positive integers are in the ratio 4 : 7 and their difference is 45. Find the numbers.

SOLUTION

Step 1: Let the numbers be 4x and 7x (ratio guarantees the common factor x).

Step 2: Difference: 7x - 4x = 3x = 45, so x = 15.

Step 3: Numbers: 4(15) = 60 and 7(15) = 105.

Step 4: Check: 105 - 60 = 45 and 60 : 105 simplifies to 4 : 7. Correct.

Hinglish: *Ratio diya hai toh numbers ko 4x, 7x maano - ye shortcut hi asli technique hai. Difference equation ekdum linear ban jaata hai: 3x = 45. Ratio wapas check karo - 60:105, dono ko 15 se divide karo, 4:7. Sahi.*

Answer: 60 and 105.

Q16 · Moderate · Concept: Coin problem (T5)

Q16. Rohit has five-rupee and ten-rupee coins. He has twice as many five-rupee coins as ten-rupee coins, and the total money is Rs 100. How many coins of each type does he have?

SOLUTION

Step 1: Let the number of ten-rupee coins be x. Then five-rupee coins = 2x.

Step 2: Total money: 10x + 5(2x) = 100.

Step 3: Simplify: 10x + 10x = 100, so 20x = 100, x = 5.

Step 4: Ten-rupee coins: 5 (Rs 50). Five-rupee coins: 10 (Rs 50).

Step 5: Check: 50 + 50 = 100 and 10 is twice 5. Correct.

Hinglish: *'Twice as many' ka matlab: jo KAM hai wo x, jo zyada hai wo 2x. Yahan 5-rupee zyada hain isliye 2x. Money equation mein VALUE se jodo - coin count nahi: 10x + 10x = 100. Sweet cancellation mili kyunki 5 x 2 = 10.*

Answer: 5 ten-rupee coins and 10 five-rupee coins.

Q17 · Moderate · Concept: Cutting a length (T5)

Q17. A 240 cm rope is cut into two pieces so that the longer piece is three times the shorter piece. Find the length of each piece.

SOLUTION

Step 1: Let the shorter piece be x. The longer is 3x.

Step 2: Total: x + 3x = 240, so 4x = 240, x = 60.

Step 3: Pieces: 60 cm and 180 cm.

Step 4: Check: 60 + 180 = 240 and 180 = 3 x 60. Correct.

Hinglish: *4x = 240 - chaar hisse jaise. Ek hissa 60, teen hisse 180. Simple, par 'both pieces report karo' wala discipline yaad rakho.*

Answer: 60 cm and 180 cm.

Q18 · Moderate · Concept: Rectangle dimensions (T5)

Q18. The length of a rectangle is 3 more than twice its width, and its perimeter is 36 cm. Find the dimensions.

SOLUTION

Step 1: Let the width be w; then the length is 2w + 3.

Step 2: Perimeter: 2(w + 2w + 3) = 36.

Step 3: Inside out: 3w + 3 = 18, so 3w = 15, w = 5.

Step 4: Width 5 cm, length 2(5) + 3 = 13 cm.

Step 5: Check: 2(5 + 13) = 36 and 13 is indeed 3 more than twice 5. Correct.

Hinglish: *Perimeter formula ke andar hi expression daal do, phir dheere-dheere simplify: pehle 2 se divide, phir 3 se. '3 more than twice' ka translation (2w + 3) hi question ka dil hai.*

Answer: Width 5 cm, length 13 cm.

Q19 · Moderate · Concept: Linear growth (T8)

Q19. A sapling is 1.75 feet tall and grows 0.5 feet each month. (i) Write its height after t months. (ii) Height after 7 months? (iii) After how many months will it be 6.75 feet tall?

SOLUTION

Step 1: h(t) = 1.75 + 0.5t.

Step 2: h(7) = 1.75 + 3.5 = 5.25 feet.

Step 3: Set 1.75 + 0.5t = 6.75: 0.5t = 5, t = 10 months.

Step 4: Check: 1.75 + 0.5(10) = 1.75 + 5 = 6.75. Correct.

Hinglish: *Growth = start + rate x time. Decimals se ghabrana nahi - 1.75 + 3.5 = 5.25 seedha hai. Reverse mein 5/0.5 = 10 - decimal divide karte waqt dono ko 100 se multiply karna comfortable ho jaata hai practice se.*

Answer: (i) h(t) = 1.75 + 0.5t (ii) 5.25 feet (iii) 10 months.

Q20 · Moderate · Concept: Linear decay depreciation (T8)

Q20. A phone bought for Rs 10,000 loses Rs 800 in value every year. (i) Value after t years? (ii) Value after 3 years? (iii) After how many years will the value be Rs 3,600?

SOLUTION

Step 1: v(t) = 10000 - 800t.

Step 2: v(3) = 10000 - 2400 = Rs 7,600.

Step 3: Set 10000 - 800t = 3600: 800t = 6400, t = 8 years.

Step 4: Check: 10000 - 800(8) = 10000 - 6400 = 3600. Correct.

Hinglish: *Depreciation = linear decay ka business avatar. 6400/800 = 8 - tables ka kaam. Model tab tak valid jab tak value non-negative: 10000/800 = 12.5 saal ke baad phone 'zero value' ka zone mein.*

Answer: (i) v(t) = 10000 - 800t (ii) Rs 7,600 (iii) 8 years.

Q21 · Difficult · Concept: Fit y = ax + b (T9)

Q21. A learning platform charges a fixed monthly fee plus a per-module charge. A student observed that accessing 10 modules cost Rs 400 and accessing 14 modules cost Rs 500. Assuming the bill is y = ax + b, find a and b, and interpret both numbers.

SOLUTION

Step 1: Substitute both observations into y = ax + b: 400 = 10a + b and 500 = 14a + b.

Step 2: Subtract the first from the second: 100 = 4a, so a = 25.

Step 3: Back-substitute into the first: 400 = 10(25) + b = 250 + b, so b = 150.

Step 4: The relationship is y = 25x + 150.

Step 5: Interpretation: a = 25 is the charge per module (Rs 25 per module); b = 150 is the fixed monthly fee (Rs 150 even with zero modules).

Step 6: Verify with the second point: 25(14) + 150 = 350 + 150 = 500. Correct.

» **Why this step?** Subtracting the equations kills b instantly - the entire technique rests on b being identical in both lines.

Hinglish: *Do observation, do equation, ek SUBTRACTION - b khatam, a seedha mil gaya (25). Phir wapas jao, b nikaalo (150). Interpretation zaroor likho: 25 = per-module charge, 150 = fixed fee. Ye interpretation wali line hi*

examiner ko dikhati hai ki tumne samjha hai, sirf calculation nahi kiya.

Answer: a = 25 (per-module rate), b = 150 (fixed fee); y = 25x + 150.

Q22 · Difficult · Concept: Fit y = ax + b gym version (T9)

Q22. A gym charges a fixed monthly fee plus an hourly rate for the badminton court. Using the court for 10 hours cost Rs 800; using it for 15 hours cost Rs 1100. Find the linear relationship y = ax + b and state what each number represents.

SOLUTION

Step 1: Equations: 800 = 10a + b and 1100 = 15a + b.

Step 2: Subtract: 300 = 5a, so a = 60.

Step 3: Back-substitute: 800 = 10(60) + b = 600 + b, so b = 200.

Step 4: Relationship: y = 60x + 200.

Step 5: Meaning: Rs 60 per hour of court time; Rs 200 fixed monthly membership.

Step 6: Verification: 60(15) + 200 = 900 + 200 = 1100. Correct.

Hinglish: *Same machine, naye numbers: subtract (300 = 5a), a = 60, wapas jao, b = 200. Practise se ye 30-second ka kaam ban jaata hai. Interpretation sentence banao hi - marks wahan hai.*

Answer: a = 60 (per hour), b = 200 (fixed fee); y = 60x + 200.

Q23 · Difficult · Concept: Temperature scales (T9)

Q23. The relation between Celsius C and Fahrenheit F has the linear form F = aC + b. Water melts at 0 degrees Celsius and 32 degrees Fahrenheit, and boils at 100 degrees Celsius and 212 degrees Fahrenheit. Find a and b.

SOLUTION

Step 1: First data point: C = 0, F = 32 gives 32 = a(0) + b, so b = 32 directly - a gift.

Step 2: Second data point: C = 100, F = 212 gives 212 = 100a + 32.

Step 3: So 100a = 180, giving a = 1.8 (which is 9/5).

Step 4: The relation is F = 1.8C + 32, the famous conversion formula.

Step 5: Verification: C = 100 gives 180 + 32 = 212. Correct.

Hinglish: *Yahan data ki kripa se b PEHLE mil gaya (zero input pe output hi b hai!). Phir sirf ek equation bachi, a = 1.8. Ye formula tum pehle se jaante ho - ab tumne use DERIVE kiya hai. Fitting ka asli maza yahi hai.*

Answer: a = 1.8 (= 9/5), b = 32; F = 1.8C + 32.

Q24 · Difficult · Concept: Kelvin-Fahrenheit relation (T9/T10)

Q24. The relation between Kelvin x and Fahrenheit y is y = (9/5)(x - 273) + 32. (i) Find y when x = 313. (ii) Find x when y = 158. (iii) Rewrite the relation in the form y = ax + b and state a and b.

SOLUTION

Step 1: (i) y = (9/5)(313 - 273) + 32 = (9/5)(40) + 32 = 72 + 32 = 104 degrees Fahrenheit.

Step 2: (ii) Start from 158 = (9/5)(x - 273) + 32. Subtract 32: 126 = (9/5)(x - 273).

Step 3: Multiply by 5/9: 126 x 5/9 = 70 = x - 273, so x = 343 K.

Step 4: (iii) Expand: y = (9/5)x - (9/5)(273) + 32 = 1.8x - 491.4 + 32 = 1.8x - 459.4.

Step 5: So a = 1.8 and b = -459.4. The graph is a straight line with slope 1.8 cutting the y-axis far below zero.

Step 6: Both checks agree with the formula's internal logic (313 - 273 = 40 is a clean number; 158 maps back to 343 K exactly).

Hinglish: *Bracket PEHLE kholo mental math mein: 313 - 273 = 40, phir 9/5 x 40 = 72. (ii) mein ulta chale: minus 32, multiply 5/9, plus 273 - teen saaf kadam. (iii) expand karne par b = -459.4 - bura lagta hai par bilkul theek hai, Kyunki Kelvin zero hi itna neeche Fahrenheit mein hota hai.*

Answer: (i) 104 degrees F (ii) 343 K (iii) y = 1.8x - 459.4, so a = 1.8 and b = -459.4.

Q25 · Difficult · Concept: Matchstick pattern (T6/T7)

Q25. Squares are made with matchsticks placed in a row, each new square sharing one side with the previous one. The counts are 4, 7, 10, ... (i) How many matchsticks for n squares? (ii) For 22 squares? (iii) Can exactly 100 matchsticks form such a row of squares? Justify.

SOLUTION

Step 1: Sequence 4, 7, 10 has constant difference 3: rule M(n) = 3n + c. From stage 1: 3 + c = 4, so c = 1. Rule: M(n) = 3n + 1.

Step 2: For 22 squares: M(22) = 66 + 1 = 67 matchsticks.

Step 3: Reachability: set 3n + 1 = 100, giving 3n = 99, n = 33.

Step 4: n = 33 is a whole number, so YES - a row of 33 squares uses exactly 100 matchsticks.

Step 5: The justification is the integrality of n: patterns exist only at whole-number stages.

Hinglish: *3n + 1 wala rule - ek square 4, har naya square sirf 3 extra le raha hai kyunki ek side share hoti hai. 'Can 100 form a stage?' ka matlab: solve karo aur INTEGRITY check karo. Yahan n = 33 whole number - YES. Agar 33.33 aata toh answer 'No, kabhi nahi' hota - aur wahi asli answer hota.*

Answer: (i) 3n + 1 (ii) 67 (iii) Yes - stage n = 33 uses exactly 100 matchsticks.

Q26 · Difficult · Concept: Hexagon matchstick pattern (T6/T7)

Q26. Hexagons are joined in a row, each new hexagon sharing one side with the previous one. The matchstick counts for 1, 2, 3 hexagons are 6, 11, 16. (i) Rule for n hexagons? (ii) Matchsticks for 15 hexagons? (iii) Can 200 matchsticks build such a row? Justify.

SOLUTION

Step 1: Sequence 6, 11, 16 has difference 5: rule M(n) = 5n + c; stage 1 gives 5 + c = 6, c = 1. Rule: 5n + 1.

Step 2: M(15) = 75 + 1 = 76 matchsticks.

Step 3: Set 5n + 1 = 200: 5n = 199, n = 39.8.

Step 4: n = 39.8 is NOT a whole number, so NO - no stage in this pattern uses exactly 200 matchsticks.

Step 5: Nearest stages: 39 hexagons use 196; 40 hexagons use 201. The value 200 falls in the gap.

Hinglish: *Q25 ka mirror image! Wahan integer mila tha (YES), yahan 39.8 aaya (NO). Ye 'No with justification' answer full-mark answer hai - bolo ki 39 aur 40 ke beech gap mein hai 200. Fractional stage pattern mein exist nahi karti.*

Answer: (i) 5n + 1 (ii) 76 (iii) No: n = 39.8 is not a whole number (39 gives 196, 40 gives 201).

Q27 · Difficult · Concept: Rally decay (T8)

Q27. A rally starts with 120 members. Every hour, exactly 9 members drop out. (i) Members after n hours? (ii) After how many hours will 39 members remain? (iii) Will the rally reach exactly 12 members at the end of some hour? Justify.

SOLUTION

Step 1: Members(n) = 120 - 9n.

Step 2: Set 120 - 9n = 39: 9n = 81, n = 9 hours.

Step 3: Reach 12? 120 - 9n = 12 gives 9n = 108, n = 12 - a whole number, so YES, exactly at the end of the 12th hour.

Step 4: Check: 120 - 9(12) = 120 - 108 = 12. Correct.

Step 5: For completeness: the count would hit zero when 120 - 9n = 0, n = 13.33, so the 12-member moment is real and comfortably inside the model's range.

Hinglish: *Do reverse questions ek saath - 39 aur 12. Dono ke n whole numbers nikle, dono possible. Yahan n = 12 bilkul clean aaya. Kabhi-kabhi dono mein se ek fraction dega - tab wala 'No, kyunki n whole nahi' answer bhi likhna aana chahiye.*

Answer: (i) 120 - 9n (ii) 9 hours (iii) Yes - exactly at n = 12 hours.

Q28 · Difficult · Concept: Savings growth (T8)

Q28. Meera has Rs 1,200 in her savings jar and adds Rs 250 every month. (i) Amount after n months? (ii) When will she have Rs 3,200? (iii) Interpret the slope and intercept of this linear model.

SOLUTION

Step 1: A(n) = 1200 + 250n.

Step 2: Set 1200 + 250n = 3200: 250n = 2000, n = 8 months.

Step 3: Slope 250: the jar grows by Rs 250 per month - the constant rate.

Step 4: Intercept 1200: the starting amount, the value at month zero.

Step 5: Check: 1200 + 250(8) = 1200 + 2000 = 3200. Correct.

Hinglish: *Simple growth, par (iii) mein asli test: slope = monthly rate (250), intercept = starting balance (1200). Yehi language future ke graph chapters mein repeat hogi - abhi se pakad lo.*

Answer: (i) A(n) = 1200 + 250n (ii) 8 months (iii) slope 250 = monthly saving; intercept 1200 = starting amount.

Q29 · Difficult · Concept: Tank decay (T8)

Q29. A tank holds 900 litres and drains at 45 litres per hour. (i) Volume after t hours? (ii) When will 495 litres remain? (iii) When does the tank empty exactly?

SOLUTION

Step 1: V(t) = 900 - 45t.

Step 2: Set 900 - 45t = 495: 45t = 405, t = 9 hours.

Step 3: Empty: 900 - 45t = 0 gives t = 900/45 = 20 hours exactly.

Step 4: Both answers are whole numbers, so both events happen cleanly at hour marks.

Step 5: Check: 900 - 45(20) = 0. Correct.

Hinglish: *Draining = decay with a definite end. 495 pe 9 hours, empty pe 20 hours - dono clean. 900/45 mental math trick: dono ko 5 se divide karo, 180/9 = 20.*

Answer: (i) V(t) = 900 - 45t (ii) 9 hours (iii) exactly 20 hours.

Q30 · Difficult · Concept: Two linear polynomials from conditions (T14)

Q30. Let p(x) = ax + b and q(x) = cx + d. Given: (i) p(0) = 5, (ii) the polynomial p(x) - q(x) cuts the x-axis at (3, 0), and (iii) p(x) + q(x) = 6x + 4 for all x. Find p(x) and q(x).

SOLUTION

Step 1: From p(0) = 5: a(0) + b = 5, so b = 5.

Step 2: From p + q = 6x + 4: (a + c)x + (b + d) = 6x + 4 for all x. Match coefficients: a + c = 6 and b + d = 4. Since b = 5, d = -1.

Step 3: Now p - q = (a - c)x + (b - d) = (a - c)x + 6. It cuts the x-axis at (3, 0), so p(3) - q(3) = 0: 3(a - c) + 6 = 0, giving a - c = -2.

Step 4: Solve the pair: a + c = 6 and a - c = -2. Adding: 2a = 4, a = 2; then c = 4.

Step 5: So p(x) = 2x + 5 and q(x) = 4x - 1.

Step 6: Verify: p(0) = 5. p - q = -2x + 6, zero at x = 3. p + q = 6x + 4. All three conditions hold.

» **Why this step?** Matching coefficients of identical linear polynomials converts each 'for all x' statement into one neat equation. Three conditions, three unknowns, linear system - no guessing needed.

Hinglish: *Ye Chapter 2 ka chaabi-wala question hai. 'For all x' ka matlab: coefficients ko MATCH karo - do equations mil gayi (a + c = 6, b + d = 4). Teesri condition ne (a - c = -2) di. Ab do-do equations ka system: add karo, a = 2; c = 4; d = -1. Verification teeno conditions pe - full proof. Ye question Olympiad-lite band ka entry gate hai.*

Answer: p(x) = 2x + 5 and q(x) = 4x - 1.

Q31 · High · Concept: Line through two points (T9/T13)

Q31. The graph of a linear polynomial p(x) passes through (2, 3) and (6, 11). (i) Find p(x). (ii) Find the coordinates where the graph cuts the x-axis and the y-axis. (iii) State the slope and interpret it.

SOLUTION

Step 1: p(x) = ax + b. Point (2, 3): 2a + b = 3. Point (6, 11): 6a + b = 11.

Step 2: Subtract: 4a = 8, so a = 2. Back-substitute: 2(2) + b = 3, b = -1.

Step 3: p(x) = 2x - 1.

Step 4: x-axis cut: set y = 0: 2x - 1 = 0, x = 1/2. Point (1/2, 0).

Step 5: y-axis cut: set x = 0: y = -1. Point (0, -1).

Step 6: Slope 2 means every extra unit of x raises the output by 2 - the line climbs two steps for one step across.

Hinglish: *Do points se line - wahi subtract-trick (4a = 8). Intercept nikaalne ka rule: x-axis cut = y ko 0 rakho; y-axis cut = x ko 0 rakho. (1/2, 0) fraction mein aata hai toh fraction mein chhod do. Slope ka meaning bolo: '1 right, 2 up'.*

Answer: (i) p(x) = 2x - 1 (ii) cuts x-axis at (1/2, 0), y-axis at (0, -1) (iii) slope 2 - rises 2 units per 1 unit of x.

Q32 · High · Concept: Parallel lines through data (T12)

Q32. The graph of p(x) passes through (1, 4) and (5, 8). The graph of q(x) passes through (2, -2) and is parallel to the graph of p(x). Find both polynomials and the x-axis intercepts of each.

SOLUTION

Step 1: p(x) = ax + b through (1, 4) and (5, 8): equations a + b = 4 and 5a + b = 8.

Step 2: Subtract: 4a = 4, a = 1; then b = 3. So p(x) = x + 3.

Step 3: Parallel means the SAME slope: q(x) = x + d for some d.

Step 4: q through (2, -2): 2 + d = -2, so d = -4. q(x) = x - 4.

Step 5: x-intercepts: p cuts where x + 3 = 0, at (-3, 0); q cuts where x - 4 = 0, at (4, 0).

Step 6: Slopes both equal 1 - the lines rise together, forever 7 units apart vertically.

Hinglish: *Parallel ka matlab sirf: slope SAME (a = 1), intercept alag. Q ka intercept ek point se mil gaya (d = -4). Intercepts nikaalne ka rule wahi - y = 0 rakho. Notice: parallel lines kabhi nahi milte, isliye unke x-axis cuts alag-alag hain (x = -3 vs x = 4).*

Answer: p(x) = x + 3 with x-intercept (-3, 0); q(x) = x - 4 with x-intercept (4, 0); both have slope 1.

Q33 · High · Concept: Family property (T14)

Q33. Consider all linear polynomials of the form f(x) = ax - a where a is a positive real number. Prove that the graph of every such polynomial passes through one fixed point, and find that point.

SOLUTION

Step 1: We need a point (x_0, y_0) that satisfies y = ax - a for EVERY value of a.

Step 2: Rewrite: y = a(x - 1). For this to hold regardless of a, we need y = 0 and x - 1 = 0.

Step 3: So $x_0 = 1$ and $y_0 = 0$: the fixed point is (1, 0).

Step 4: Check with two members of the family: a = 3 gives y = 3x - 3, and at x = 1, y = 0. a = 7 gives y = 7x - 7, and at x = 1, y = 0. Every member passes through (1, 0).

Step 5: Interpretation: changing a only changes the slope; the intercept -a slides, but the pivot point (1, 0) never moves - all the lines rotate about it.

» **Why this step?** A fact true for all a must survive arbitrary substitution of a; factoring out a exposes the point where a becomes irrelevant.

Hinglish: *Family question ka funda: 'har a ke liye sahi' tabhi ho sakta hai jab a ki asar ZERO ho jaye. y = a(x - 1) likhte hi dikhta hai - x = 1 rakh do toh a khatam, y = 0. Toh (1, 0) wo nail hai jis par saari lines latki hain. Ye idea (PDF ka starred question) aage 'pencil of lines' ke naam se famous hai.*

Answer: Every line y = ax - a passes through the fixed point (1, 0), since y = a(x - 1) vanishes there for all a.

Q34 · High · Concept: Compare two lines (T11/T12)

Q34. Draw-free reasoning: compare the graphs of y = 3x + 1 and y = -3x + 1. State their common features, their differences, and the geometric relationship between the two lines.

SOLUTION

Step 1: Common feature: same b = 1, so both cut the y-axis at the same point (0, 1).

Step 2: Difference: slopes +3 and -3 - equal in size, opposite in sign.

Step 3: Positive slope: as x grows, y grows (uphill). Negative slope: as x grows, y falls (downhill).

Step 4: Geometric relationship: the two lines are mirror images of each other in the y-axis. Reason: replacing x by -x in y = 3x + 1 gives y = -3x + 1 - reflection in the y-axis flips the sign of x.

Step 5: They are NOT parallel (slopes differ), so they cross - exactly at the shared point (0, 1).

Hinglish: *Same intercept, opposite slopes: dono lines (0, 1) mein milti hain aur wahan se ek upar, ek neeche jaati hain - y-axis ke aaine mein twin. 'Replace x by -x' wala reflection argument coordinate geometry ka classic move hai - Chapter 1 ke reflections se seedha connection.*

Answer: Both pass through (0, 1); slopes +3 and -3 make them mirror images in the y-axis, crossing at (0, 1) - not parallel.

Q35 · High · Concept: Disguised slope/intercept (T11)

Q35. Find the slope and y-intercept of each line, and say which is steeper: (i) 2y = 4x + 7, (ii) 5y = 6x - 10.

SOLUTION

Step 1: Isolate y first - never read the coefficient of x before this step.

Step 2: (i) Divide by 2: y = 2x + 3.5. Slope 2, y-intercept 3.5 (cuts at (0, 3.5)).

Step 3: (ii) Divide by 5: y = 1.2x - 2. Slope 1.2, y-intercept -2 (cuts at (0, -2)).

Step 4: Steepness compares slopes: 2 > 1.2, so line (i) is steeper.

Step 5: Interpretation: line (i) rises 2 units per unit of x; line (ii) rises 1.2 units per unit.

Hinglish: *Golden rule: pehle y = ... banao. 2y = 4x + 7 ko dekh ke log slope 4 likh dete hain - GALAT, 4 toh 2y ka hai! Divide karne ke baad hi a = 2 aur b = 3.5. Steepness = slope comparison: 2 vs 1.2.*

Answer: (i) slope 2, intercept 3.5 (ii) slope 1.2, intercept -2 (i) is steeper.

Q36 · High · Concept: Parallel detection (T12)

Q36. Are any two of these three lines parallel? (i) y = 2x + 3.5 (ii) y = 1.2x - 2 (iii) 3y = 6x - 11. Justify using slopes.

SOLUTION

Step 1: Lines (i) and (ii) come from Q35: slopes 2 and 1.2 - different.

Step 2: (iii) divide by 3: y = 2x - 11/3. Slope 2, intercept -11/3 (about -3.67).

Step 3: Compare slopes: (i) has 2, (ii) has 1.2, (iii) has 2.

Step 4: Lines (i) and (iii) share the slope 2 but have different intercepts (3.5 vs -11/3): PARALLEL.

Step 5: (ii) meets both of the others (different slopes). Parallel lines never meet - (i) and (iii) will never cross each other.

Hinglish: *Parallel ka ek hi test: slopes barabar? (iii) ko 3 se divide karo: slope 2. (i) bhi slope 2 - MATCH! Intercepts alag (3.5 vs -3.67), isliye distinct parallel lines. (ii) akela slope 1.2 - wo dono ko kaat-ta hai.*

Answer: Lines (i) and (iii) are parallel (both slope 2, different intercepts); (ii) is parallel to neither.

Q37 · High · Concept: Work as linear model (T8/T13)

Q37. Work done by a constant force equals force times distance travelled. Taking the constant force as 3 units, (i) write work w as a linear function of distance d, (ii) find the work done over a distance of 2 units, (iii) explain what kind of line this draws and where it passes.

SOLUTION

Step 1: w = 3 x d gives the linear model w(d) = 3d.

Step 2: w(2) = 6 units of work.

Step 3: The model has b = 0, so it is a line through the ORIGIN - of the family y = ax.

Step 4: Slope 3: every extra unit of distance adds exactly 3 units of work.

Step 5: The line passes through (0, 0) and, for instance, (2, 6); plotting these two points determines it completely.

Hinglish: *w = F x d bina fixed part ke - b = 0 - isliye line origin se guzarti hai. Ye PDF ki Fig 2.8 family (y = ax) ka physics version hai. 'Force = slope of the work-distance line' - ek line, ek physical meaning.*

Answer: (i) w(d) = 3d (ii) 6 units (iii) a straight line through the origin with slope 3.

Q38 · High · Concept: Reverse data bill (T9/T10)

Q38. A telecom bill follows y = 20x + 150 (Rs 20 per GB plus Rs 150 fixed). (i) What will the bill be for 30 GB? (ii) A student received a bill of Rs 750. How much data did she use? (iii) What does the 150 mean for a month in which she used no data at all?

SOLUTION

Step 1: (i) y = 20(30) + 150 = 600 + 150 = Rs 750.

Step 2: (ii) Set 20x + 150 = 750: 20x = 600, x = 30 GB.

Step 3: (iii) With x = 0, y = 150: even with zero data the fixed monthly fee of Rs 150 applies.

Step 4: Notice (i) and (ii) are inverses of each other - forward and backward through the same machine.

Hinglish: *Yehi bill equation aage-peeche do baar: substitute aur solve. Mazedaar baat: (i) aur (ii) ka jawab same pair (30, 750) hai - kyunki ek hi line ke do raaste hain. (iii) zero usage par bhi 150 - fixed fee ka matlab hi yahi hai.*

Answer: (i) Rs 750 (ii) 30 GB (iii) the Rs 150 fixed fee is charged even for zero usage.

Q39 · High · Concept: Point-on-line test (T13)

Q39. Three of these four lines pass through the point (4, -2). Identify the one that does NOT, showing the substitution for each: (A) y = -x + 2 (B) y = x - 6 (C) y = -2x + 6 (D) y = 2x - 6.

SOLUTION

Step 1: Test each by substituting x = 4 and checking whether y = -2 comes out.

Step 2: (A) y = -4 + 2 = -2. On the line.

Step 3: (B) y = 4 - 6 = -2. On the line.

Step 4: (C) y = -2(4) + 6 = -8 + 6 = -2. On the line.

Step 5: (D) y = 2(4) - 6 = 8 - 6 = 2, which is NOT -2. Line D misses the point.

Step 6: So the answer is (D). (Fun fact: three different-looking lines can share one point - lines through a common point form a 'pencil'.)

Hinglish: *Point-on-line ka test: substitute karo, dono side match? A, B, C teeno ne -2 diya. D ne 2 diya - miss. Ye question 'verification' ki taakat dikhata hai: ek chhota substitution poori geometry bata deta hai.*

Answer: (D) y = 2x - 6 does not pass through (4, -2); it gives y = 2 there.

Q40 · High · Concept: Pattern reachability (T6/T7)

Q40. A sequence follows the rule T(n) = 3n - 1. (i) List the first four terms. (ii) Which term equals 50? (iii) Is 100 a term of this sequence? Justify.

SOLUTION

Step 1: T(1) = 2, T(2) = 5, T(3) = 8, T(4) = 11 - the sequence 2, 5, 8, 11 with constant difference 3.

Step 2: Set 3n - 1 = 50: 3n = 51, n = 17. So the 17th term is 50.

Step 3: Set 3n - 1 = 100: 3n = 101, n = 101/3 = 33.67.

Step 4: n is not a whole number, so 100 is NOT a term of the sequence - the sequence jumps from 98 (n = 33) to 101 (n = 34).

Step 5: The justification is integrality: sequence positions must be whole numbers.

Hinglish: *3n - 1 family: terms 2, 5, 8, 11 - har baar +3. 50 aata hai n = 17 par. 100 ke liye n = 33.67 aaya - whole number nahi, toh 100 sequence ka term NAHI. Neighbour check: 98 aur 101 - 100 beech mein chhoot gaya. Yahi 'gap' wala justification full marks deta hai.*

Answer: (i) 2, 5, 8, 11 (ii) the 17th term (iii) No: n = 101/3 is not a whole number (the terms jump from 98 to 101).

Q41 · HOTS · Concept: Two-condition number problem (T5/T14)

Q41. Two numbers differ by 7. Three times the smaller number plus twice the larger number equals 74. Find both numbers.

SOLUTION

Step 1: Let the smaller be x; the larger is x + 7.

Step 2: Condition: 3x + 2(x + 7) = 74.

Step 3: Expand: 3x + 2x + 14 = 74, so 5x = 60, x = 12.

Step 4: Numbers: 12 and 19.

Step 5: Check: difference 19 - 12 = 7. Weighted sum: 3(12) + 2(19) = 36 + 38 = 74. Both hold.

Hinglish: *Do conditions ka combo: difference ne x aur x + 7 diye, weighted sum ne equation di. 5x = 60 - clean. Verification dono conditions pe karo, sirf ek pe nahi.*

Answer: 12 and 19.

Q42 · HOTS · Concept: Two-digit reversal (T5/T14)

Q42. The digits of a two-digit number differ by 4. When the digits are interchanged and the resulting number is added to the original, the sum is 110. Find all two-digit numbers satisfying this.

SOLUTION

Step 1: Let the tens digit be a and the units digit be b, with |a - b| = 4.

Step 2: The number is 10a + b; the reversed number is 10b + a.

Step 3: Their sum: 11a + 11b = 11(a + b) = 110, so a + b = 10.

Step 4: Now solve the pair: a + b = 10 and digits differ by 4. Two cases: a - b = 4 gives a = 7, b = 3; b - a = 4 gives a = 3, b = 7.

Step 5: The numbers are 73 and 37 (each is the other's reversal, and both satisfy every condition: digits differ by 4, and 73 + 37 = 110).

Step 6: Note the beautiful structure: the reversal sum only depends on a + b, so the difference condition picks the exact digit pair.

Hinglish: *Reversal-sum ka secret: 10a + b + 10b + a = 11(a + b). Hamesha 11 ka multiple! 110/11 = 10 = digit sum. Phir 'differ by 4' do cases kholta hai - (7,3) ya (3,7). Dono valid, dono report karo. Ye question classic hai aur har exam mein kisi na kisi roop mein aata hai.*

Answer: 73 and 37 (both satisfy the conditions).

Q43 · HOTS · Concept: Wire rectangle area function (T14)

Q43. A wire of length 36 cm is bent into a rectangle. Let the length be x cm. (i) Write the area as a function of x. (ii) Find ALL lengths x that give an area of 77 sq cm. (iii) Which single length gives the largest area, and what is that area?

SOLUTION

Step 1: Perimeter 36 means half-perimeter 18: length + width = 18. Width = 18 - x.

Step 2: Area A(x) = x(18 - x) = 18x - x^2 - a quadratic function (the PDF builds exactly this in its Example 3 with a 20 cm wire).

Step 3: Set x(18 - x) = 77: x^2 - 18x + 77 = 0. Look for two numbers with product 77 and sum 18: 7 and 11.

Step 4: So x = 7 or x = 11. Both are genuine: a 7 x 11 rectangle and an 11 x 7 rectangle are the SAME rectangle turned around - one answer in two descriptions.

Step 5: Largest area: test symmetric candidates. x = 9 gives 9 x 9 = 81; x = 8 gives 8 x 10 = 80; x = 10 gives 10 x 8 = 80. The square (x = 9) wins with area 81.

Step 6: Structure noticed: areas rise toward x = 9 and fall after it - symmetric about the square.

Hinglish: *Ye PDF ke wire example ka bada bhai. Area quadratic nikla (18x - x^2) - chapter linear hai par evaluation quadratic ki bhi practice hai (Ex 2.2 mein tha). 77 ke liye do answers (7 aur 11) ek hi rectangle ke do naam hain - symmetry! Aur square (9 x 9 = 81) max deta hai - ye observation Grade 10 mein formally aayega, aaj ke liye table-test kaafi hai.*

Answer: (i) A(x) = 18x - x^2 (ii) x = 7 or 11 (the same rectangle, two descriptions) (iii) x = 9 gives the maximum area 81 sq cm.

Q44 · HOTS · Concept: Salary structure (T9/T10)

Q44. A salesperson earns a fixed monthly amount plus a commission per sale. With 20 sales the pay is Rs 30,000; with 35 sales it is Rs 45,000. (i) Find the linear pay formula. (ii) How many sales are needed to earn Rs 52,000? (iii) What is the fixed amount?

SOLUTION

Step 1: Let pay = ax + b. Data: 20a + b = 30000 and 35a + b = 45000.

Step 2: Subtract: 15a = 15000, so a = 1000 per sale.

Step 3: Back-substitute: 30000 = 20(1000) + b = 20000 + b, b = 10000.

Step 4: Pay formula: y = 1000x + 10000.

Step 5: Target: 1000x + 10000 = 52000 gives 1000x = 42000, x = 42 sales.

Step 6: The fixed amount b = 10000: even with zero sales, Rs 10,000 is paid.

Hinglish: *Salary = bill jaisi hi structure: fixed + rate x count. 15 sales ka difference 15000 deta hai: per-sale 1000. 42 sales for 52k. Interpretation: zero sales par bhi 10k milta hai - base pay.*

Answer: (i) y = 1000x + 10000 (ii) 42 sales (iii) Rs 10,000.

Q45 · HOTS · Concept: Where C equals F (T14)

Q45. Using the conversion relation F = 1.8C + 32, find the temperature at which the Celsius and Fahrenheit readings are EQUAL. Show the full working.

SOLUTION

Step 1: Equal readings mean F = C. Substitute F = C into the relation: C = 1.8C + 32.

Step 2: Bring terms together: C - 1.8C = 32, so -0.8C = 32.

Step 3: Divide: C = 32 / (-0.8) = -40.

Step 4: Check: at C = -40, F = 1.8(-40) + 32 = -72 + 32 = -40. Equal. Correct.

Step 5: So -40 degrees is the unique temperature where both scales agree - a famous meeting point of the two lines F = 1.8C + 32 and F = C.

Hinglish: *Dono reading barabar matlab dono lines ka INTERSECTION: F = C aur F = 1.8C + 32. Solve karo: -40. Negative temperature mein sign ka dhyan - 1.8(-40) = -72. Verify: -72 + 32 = -40. Dono scale -40 par milti hain - ek hi baar.*

Answer: At -40 degrees, where C = F = -40.

Q46 · HOTS · Concept: Plan comparison (T14)

Q46. Mobile Plan A costs Rs 200 per month plus Rs 2 per GB. Plan B costs Rs 350 per month plus Rs 0.50 per GB. (i) Write both cost formulas. (ii) For what data usage do the plans cost the same? (iii) Which plan is cheaper for a user consuming 60 GB? (iv) For 150 GB?

SOLUTION

Step 1: Plan A: C_A = 200 + 2x. Plan B: C_B = 350 + 0.5x (x in GB).

Step 2: Equal cost: 200 + 2x = 350 + 0.5x. Move terms: 2x - 0.5x = 350 - 200, so 1.5x = 150, x = 100 GB.

Step 3: At 100 GB both cost 200 + 200 = Rs 400 (and 350 + 50 = 400 - consistent).

Step 4: At x = 60: A costs 200 + 120 = Rs 320; B costs 350 + 30 = Rs 380. A is cheaper (by Rs 60).

Step 5: At x = 150: A costs 200 + 300 = Rs 500; B costs 350 + 75 = Rs 425. B is cheaper (by Rs 75).

Step 6: Structure: below 100 GB choose the small-fixed-high-rate plan (A); above 100 GB the high-fixed-low-rate plan (B) wins. The crossing point 100 GB is the decision boundary.

Hinglish: *Do plans ka muqabla = do lines ka intersection. 100 GB par dono 400 - crossing point. Usse kam data toh A sasta (fixed fee kam), zyada data toh B (rate kam). Ye 'decision boundary' thinking real life mein har jagah hai - OTT plans, taxi vs auto, sab yahi maths hai.*

Answer: (i) A: 200 + 2x; B: 350 + 0.5x (ii) equal at 100 GB (both Rs 400) (iii) A cheaper at 60 GB (320 vs 380) (iv) B cheaper at 150 GB (425 vs 500).

Q47 · HOTS · Concept: Function composition lite (T14)

Q47. Let p(x) = 3x + 1. Given p(2) = 7 and p(5) = 16. (i) Find x such that p(x) = 100. (ii) Compute p(p(1)). (iii) Compute p(p(p(0))).

SOLUTION

Step 1: (i) Solve 3x + 1 = 100: 3x = 99, x = 33.

Step 2: (ii) Inner first: p(1) = 3(1) + 1 = 4. Then p(4) = 3(4) + 1 = 13. So p(p(1)) = 13.

Step 3: (iii) p(0) = 1. Then p(1) = 4 (from part ii). Then p(4) = 13. So p(p(p(0))) = 13.

Step 4: Observe (ii) and (iii) land on the same value: both chains pass through the input 1 at some stage, and the machine is deterministic - same input, same output.

Step 5: General check for (iii): p(p(p(0))) = p(p(1)) = p(4) = 13. Verified.

Hinglish: *Machine ke andar machine - andar wala PEHLE chalao. p(p(1)): pehle p(1) = 4, phir p(4) = 13. (iii) mein teen layer: 0, 1, 4, 13. Interesting: (ii) aur (iii) ka jawab same kyun? Kyunki dono chains 1 se guzri. Deterministic machine = same input, same output.*

Answer: (i) x = 33 (ii) 13 (iii) 13.

Q48 · HOTS · Concept: Iterated machine (T14)

Q48. The machine f(x) = 2x + 3 is applied again and again, starting from $x_0 = 1$: $x_1 = f(1)$, $x_2 = f(x_1)$, and so on. (i) Compute x_1, x_2, x_3. (ii) Compute x_4. (iii) Describe in words what each application does to the current value.

SOLUTION

Step 1: $x_1 = 2(1) + 3 = 5$.

Step 2: $x_2 = 2(5) + 3 = 13$.

Step 3: $x_3 = 2(13) + 3 = 29$.

Step 4: $x_4 = 2(29) + 3 = 61$.

Step 5: Each application doubles the current value and then adds 3 - 'double, then add 3', repeated.

Step 6: The outputs 1, 5, 13, 29, 61 grow fast because doubling compounds - a taste of exponential behaviour hiding inside repeated linear steps.

Hinglish: *Repeat machine: double karo, 3 jodo, phir se. 1, 5, 13, 29, 61. Har step pe 'double + 3' - baar-baar same recipe. Jab linear machine ko baar-baar chalate hain toh growth tez ho jaati hai (double ki wajah se) - ye seedhi line se exponential behaviour ka darwaza hai.*

Answer: (i) 5, 13, 29 (ii) 61 (iii) each step doubles the value and adds 3.

Q49 · HOTS · Concept: Axis intercepts and triangle (T13/T14)

Q49. Consider the line y = 2x + 8. (i) Find its y-intercept. (ii) Find its x-intercept. (iii) The line and the two axes form a triangle. Find its area.

SOLUTION

Step 1: y-intercept: put x = 0, y = 8 - the point (0, 8).

Step 2: x-intercept: put y = 0: 2x + 8 = 0, x = -4 - the point (-4, 0).

Step 3: The triangle has vertices (0, 0), (0, 8) and (-4, 0): a right triangle with legs 8 (vertical) and 4 (horizontal, in the negative direction).

Step 4: Area = (1/2) x 8 x 4 = 16 square units.

Step 5: Distances use absolute values: the horizontal leg has length |-4 - 0| = 4 even though the point sits left of the origin.

Hinglish: *Intercepts nikaalo (0, 8) aur (-4, 0). Origin ke saath milke right triangle bana - legs 8 aur 4. Area 16. Left-side hone se area nahi badalta - length toh absolute hai. Ye question Chapter 2 ke graphing ko Chapter 1 ke area se jodta hai - mixed-concept preview.*

Answer: (i) (0, 8) (ii) (-4, 0) (iii) area = 16 square units.

Q50 · HOTS · Concept: Concurrent lines (T14)

Q50. Three lines are given: L1: y = x + 1, L2: y = -x + 5 and L3: y = 3. Find the pairwise intersections of L1 with L2, L1 with L3, and L2 with L3. What do you conclude about the three lines?

SOLUTION

Step 1: L1 meets L2 where x + 1 = -x + 5: 2x = 4, x = 2, y = 3. Point (2, 3).

Step 2: L1 meets L3 where x + 1 = 3: x = 2, y = 3. Point (2, 3).

Step 3: L2 meets L3 where -x + 5 = 3: x = 2, y = 3. Point (2, 3).

Step 4: All three pairwise intersections are the SAME point (2, 3).

Step 5: Conclusion: the three lines are CONCURRENT - they all pass through one common point. Verification: (2, 3) satisfies all three equations: 2 + 1 = 3, -2 + 5 = 3, and 3 = 3.

Hinglish: *Teen lines, teen intersections - aur teeno ka jawab SAME (2, 3)! Matlab teeno lines ek hi point par cross ho rahi hain - concurrency. Two-line intersection nikaalna = equations barabar rakhna. Ye HOTS finale tumhe Class 11 ki 'solution of system' ki taraf punch maar raha hai.*

Answer: All three intersections equal (2, 3); the lines are concurrent.

2.4 Chapter MCQs (60 questions, answers inline)

MCQ 1 · Normal · Concept: Coefficient

1. The coefficient of x in the expression 5x - 7 is:

A. 5
B. -7
C. 5x
D. 1

Correct Answer: (A)

The coefficient is the number multiplying the variable x, which is 5. The -7 is the constant term.

Why (B) is wrong: -7 is the constant term, not attached to x.

Why (C) is wrong: 5x is the whole term.

Why (D) is wrong: 1 is the coefficient of a bare x, not of 5x.

Hinglish: *Coefficient = variable ke saath juda number. 5x ka coefficient 5. -7 constant hai, coefficient nahi.*

MCQ 2 · Normal · Concept: Degree

2. The degree of the polynomial $4 - 3y + y^2$ is:

A. 1
B. 2
C. 3
D. 0

Correct Answer: (B)

The powers present are 0 (the 4), 1 (the -3y) and 2 (the y^2). The highest is 2, so the degree is 2.

Why (A) is wrong: 1 is just the middle term's power.

Why (C) is wrong: No cube is present.

Why (D) is wrong: 0 would mean a constant polynomial.

Hinglish: *Order mein likha nahi hai par koi baat nahi - powers scan karo: 0, 1, 2. Max = 2. Quadratic.*

MCQ 3 · Normal · Concept: Constant polynomial

3. The degree of the constant polynomial -9 is:

A. -9
B. 0
C. 1
D. undefined

Correct Answer: (B)

Any constant c can be written as $c \times x^0$, and $x^0 = 1$. So the degree (power of the variable) is 0.

Why (A) is wrong: -9 is the value, not the degree.

Why (C) is wrong: There is no x term at all.

Why (D) is wrong: It is defined - it is 0.

Hinglish: *Constant ka degree ZERO hota hai - isko x^0 ke roop mein likh sakte ho. Ye MCQ ka pakka sawaal hai.*

MCQ 4 · Normal · Concept: Classification

4. The polynomial 2z - 9 is:

A. constant

B. linear

C. quadratic

D. cubic

Correct Answer: (B)

The highest power of z is 1, so it is a linear polynomial (degree 1).

Why (A) is wrong: Degree is 1, not 0.

Why (C) is wrong: No z^2 term.

Why (D) is wrong: No z^3 term.

Hinglish: *Power 1 = linear. Naam yaad rakho: 1-linear, 2-quadratic, 3-cubic, 0-constant.*

MCQ 5 · Normal · Concept: Evaluation

5. The value of 4x - 9 at x = -2 is:

A. -17

B. 17

C. -1

D. 1

Correct Answer: (A)

4(-2) - 9 = -8 - 9 = -17. Both terms are negative here.

Why (B) is wrong: Lost a minus somewhere.

Why (C) is wrong: Computed 4 - 2 - 9 style without brackets.

Why (D) is wrong: Sign error in 4 x (-2).

Hinglish: *4(-2) = -8, phir -9 jodo: -17. Dono negative milke aur negative - sign ka khel.*

MCQ 6 · Normal · Concept: Quadratic evaluation

6. The value of $2s^2$ - s + 3 at s = -1 is:

A. 0

B. 6

C. 4

D. -2

Correct Answer: (B)

$2(-1)^2 - (-1) + 3 = 2(1) + 1 + 3 = 6$. The square forgives the minus; the -s term becomes +1.

Why (A) is wrong: Dropped the +1 from the middle term.

Why (C) is wrong: Wrong square value.

Why (D) is wrong: Sign chaos throughout.

Hinglish: *$(-1)^2 = +1$, aur -s matlab -(-1) = +1. Dono jagah plus bana! 2 + 1 + 3 = 6.*

MCQ 7 · Normal · Concept: Reverse evaluation

7. The machine y = 5x + 2 gives output 27. The input x is:

A. 5
B. 4
C. 6
D. 25

Correct Answer: (A)

5x + 2 = 27 gives 5x = 25, so x = 5.

Why (B) is wrong: Arithmetic slip 27 - 2 = 24.

Why (C) is wrong: Added before dividing.

Why (D) is wrong: Reported 5x instead of x.

Hinglish: *Peeche chalna hai: 27 se 2 ghatayo (25), 5 se divide (5). Verify: 5(5) + 2 = 27.*

MCQ 8 · Normal · Concept: Sum-difference words

8. Two numbers add to 46, and one of them is 6 more than the other. The numbers are:

A. 20 and 26
B. 23 and 23
C. 18 and 28
D. 21 and 25

Correct Answer: (A)

Let them be x and x + 6: 2x + 6 = 46 gives x = 20. Numbers 20 and 26.

Why (B) is wrong: Equal numbers differ by 0.

Why (C) is wrong: Difference is 10, not 6.

Why (D) is wrong: Difference is 4.

Hinglish: *2x + 6 = 46 se x = 20. Check: 20 + 26 = 46, difference 6. Dono condition sahi.*

MCQ 9 · Normal · Concept: Constant perimeter growth

9. If the side of a square increases by 3 units, its perimeter increases by:

A. 3 units
B. 6 units
C. 12 units
D. depends on the original side

Correct Answer: (C)

New perimeter = 4(s + 3) = 4s + 12. The jump is exactly 12 - the same for every square, because the rule 4s is linear with slope 4 and 4 x 3 = 12.

Why (A) is wrong: Perimeter counts ALL four sides.

Why (B) is wrong: Counted only two sides.

Why (D) is wrong: Linear growth never depends on the starting value.

Hinglish: *Slope 4 hai isliye side ka har 1 unit = perimeter ka 4 unit. 3 ka badlav = 12 ka badlav - hamesha, original side jo bhi ho. Yehi linear ka jaadu hai.*

MCQ 10 · Normal · Concept: Fee structure

10. A club charges Rs 250 to join plus Rs 75 per session. The total cost for 7 sessions is:

A. Rs 525

B. Rs 325

C. Rs 775

D. Rs 750

Correct Answer: (C)

250 + 75(7) = 250 + 525 = Rs 775.

Why (A) is wrong: Only the session charges.

Why (B) is wrong: Counted one session.

Why (D) is wrong: Roughly right but 75 x 7 was miscalculated.

Hinglish: *Fixed + rate x count. 75 x 7 = 525, plus 250 = 775. Option A bhool gaya joining fee.*

MCQ 11 · Normal · Concept: Pattern rule

11. A sequence begins 4, 9, 14, 19, ... Its nth-term rule is:

A. 5n - 1

B. 5n + 1

C. 4n

D. n + 3

Correct Answer: (A)

The constant difference is 5, so the rule is 5n + c. Stage 1: 5 + c = 4 gives c = -1. Rule: 5n - 1. Check stage 2: 10 - 1 = 9. Correct.

Why (B) is wrong: Would give 6 as the first term.

Why (C) is wrong: Wrong difference handling.

Why (D) is wrong: No pattern fit at all.

Hinglish: *Difference 5 pakdo, phir first term se c nikaalo: 4 = 5 + c isliye c = -1. 5n - 1. Verify with 9: 10 - 1 = 9 sahi.*

MCQ 12 · Normal · Concept: Which stage

12. In the sequence with rule T(n) = 5n - 1, the term equal to 54 occurs at:

A. n = 10

B. n = 11

C. n = 12

D. n = 54/5

Correct Answer: (B)

5n - 1 = 54 gives 5n = 55, n = 11. The 11th term is 54.

Why (A) is wrong: 5(10) - 1 = 49, not 54.

Why (C) is wrong: 5(12) - 1 = 59.

Why (D) is wrong: Forgot to add 1 first.

Hinglish: *54 + 1 = 55, 55/5 = 11. Whole number mila toh stage exist karti hai.*

MCQ 13 · Normal · Concept: Reachability

13. In the same sequence T(n) = 5n - 1, is 53 a term of the sequence?

A. Yes, at n = 10

B. Yes, at n = 11

C. No, because n = 54/5 is not a whole number

D. Cannot be decided

Correct Answer: (C)

5n - 1 = 53 gives 5n = 54, n = 10.8 - not a whole number. Sequence positions must be whole numbers, so 53 is never reached (the terms jump from 49 to 54).

Why (A) is wrong: 5(10) - 1 = 49.

Why (B) is wrong: 5(11) - 1 = 54.

Why (D) is wrong: The rule decides it completely.

Hinglish: *53 + 1 = 54, 54/5 = 10.8 - fraction! Toh 53 sequence mein KABHI nahi aayega. 49 ke baad seedha 54. 'No with reason' hi poora answer hai.*

MCQ 14 · Normal · Concept: Depreciation decay

14. A scooter bought for Rs 20,000 loses Rs 150 in value each month. Its value after t months is 20000 - 150t. It will be worth Rs 8,000 after:

A. 60 months

B. 80 months

C. 8 months

D. 120 months

Correct Answer: (B)

20000 - 150t = 8000 gives 150t = 12000, t = 80 months.

Why (A) is wrong: Divided wrongly.

Why (C) is wrong: Dropped two zeros.

Why (D) is wrong: Overshoot.

Hinglish: *12000 ka loss chahiye, 150 per month: 12000/150 = 80 mahine. Long division yaad rakho: dono ko 50 se kaato - 240/3 = 80.*

MCQ 15 · Normal · Concept: Tank emptying

15. A tank with 600 litres drains at 25 litres per hour. It empties exactly after:

A. 20 hours

B. 24 hours

C. 25 hours

D. 30 hours

Correct Answer: (B)

600 - 25t = 0 gives t = 600/25 = 24 hours.

Why (A) is wrong: Arithmetic slip.

Why (C) is wrong: Confused rate with time.

Why (D) is wrong: Overshoot.

Hinglish: *600/25 - 25 x 24 = 600 yaad rakho (quarter of 100 is 25; 600 has 24 quarters).*

MCQ 16 · Normal · Concept: Fit with zero intercept

16. A photocopy shop charges purely per page. 5 pages cost Rs 100 and 8 pages cost Rs 160. The linear model y = ax + b has:

A. a = 20, b = 0
B. a = 20, b = 20
C. a = 12, b = 40
D. a = 32, b = 0

Correct Answer: (A)

Subtract the data equations: 160 - 100 = 3a gives a = 20. Then b = 100 - 20(5) = 0. Purely per-page pricing means zero fixed fee - the model is y = 20x, a line through the origin.

Why (B) is wrong: Invented a fee that the data contradicts.

Why (C) is wrong: Difference divided wrongly.

Why (D) is wrong: Used 8 instead of the rate.

Hinglish: *Subtract-trick: 60 = 3a, a = 20. Wapas jao: 100 - 100 = 0 = b. Zero intercept ka matlab: origin se line. 'Purely per page' words se hi b = 0 ki expectation ban jaani chahiye.*

MCQ 17 · Normal · Concept: Fit y = ax + b

17. A service costs Rs 90 for 2 units and Rs 210 for 6 units. In the model y = ax + b:

A. a = 30, b = 30
B. a = 30, b = 90
C. a = 45, b = 0
D. a = 35, b = 20

Correct Answer: (A)

210 - 90 = 120 = 4a, so a = 30. Back-substitute: 90 = 2(30) + b = 60 + b, b = 30. Model: y = 30x + 30.

Why (B) is wrong: Forgot to subtract the per-unit part.

Why (C) is wrong: Mixed up which pair to use.

Why (D) is wrong: No consistent fit.

Hinglish: *4 units ka difference 120: per-unit 30. Wapas: 90 - 60 = 30 fixed. Interpretation: Rs 30 per unit + Rs 30 base.*

MCQ 18 · Normal · Concept: Slope meaning in conversion

18. In the conversion F = 1.8C + 32, the number 1.8 tells us that:

A. a 1 degree rise in Celsius gives a 1.8 degree rise in Fahrenheit
B. Fahrenheit is always 1.8 more than Celsius
C. water boils at 1.8 degrees
D. the scales meet at 1.8

Correct Answer: (A)

The slope is the rate of change: each extra degree Celsius lifts the Fahrenheit reading by 1.8. (Option B confuses slope with a constant offset - the offset is 32, not 1.8.)

Why (B) is wrong: That is what b would mean, but b = 32 here.

Why (C) is wrong: Boiling is at 100 C / 212 F.

Why (D) is wrong: The scales meet at -40, not 1.8.

Hinglish: *Slope = rate: C mein 1 degree badho toh F mein 1.8 badho. Fixed gap toh 32 hai - wo b wala hissa hai, 1.8 nahi.*

MCQ 19 · Normal · Concept: Disguised slope

19. The slope of the line 3y = 6x + 12 is:

A. 6
B. 2
C. 3
D. 12

Correct Answer: (B)

Divide by 3 first: y = 2x + 4. The slope is 2. Reading the raw coefficient 6 before isolating y is the classic trap.

Why (A) is wrong: That is the coefficient of x before dividing.

Why (C) is wrong: That is the divisor.

Why (D) is wrong: That is the constant after dividing.

Hinglish: *Pehle y = ... banao! 3 se divide: y = 2x + 4. Slope 2. Seedha 6 padhna sabse common galti hai.*

MCQ 20 · Normal · Concept: Disguised intercept

20. The line 4y = 8x - 20 cuts the y-axis at:

A. (0, -20)
B. (0, -5)
C. (-5, 0)
D. (0, 2)

Correct Answer: (B)

Divide by 4: y = 2x - 5. The y-intercept is the point (0, -5). (Option A is the raw constant; option C confuses the axes.)

Why (A) is wrong: Raw -20 before dividing by 4.

Why (C) is wrong: That is an x-axis form.

Why (D) is wrong: Read the slope instead.

Hinglish: *4 se divide: y = 2x - 5. y-axis cut = (0, b) = (0, -5). x-axis wala point alag hota hai - (2.5, 0).*

MCQ 21 · Normal · Concept: Lines through origin

21. Which of these lines passes through the origin?

A. y = 5x
B. y = 5x + 1
C. y = x - 3
D. y = 2

Correct Answer: (A)

A line passes through (0, 0) exactly when its constant term b is 0. Only y = 5x has b = 0.

Why (B) is wrong: Cuts y-axis at (0, 1).

Why (C) is wrong: Cuts at (0, -3).

Why (D) is wrong: Horizontal line at height 2.

Hinglish: *Origin se guzarne ka ek hi test: b = 0. Baaki sab (0, b) par y-axis kaat-ti hain, origin se nahi.*

MCQ 22 · Normal · Concept: Parallel detection

22. The lines y = 4x - 3 and y = 4x + 7 are:

A. perpendicular
B. parallel
C. the same line
D. intersecting at one point

Correct Answer: (B)

Same slope 4, different intercepts (-3 vs 7): parallel lines that never meet.

Why (A) is wrong: Perpendicularity is about slope products -1, not equality.

Why (C) is wrong: Intercepts differ.

Why (D) is wrong: Parallel lines never intersect.

Hinglish: *Slope same, intercept alag = parallel. Same slope SAME intercept hota toh ek hi line hoti.*

MCQ 23 · Normal · Concept: Steepness

23. Which line is STEEPER than y = x?

A. y = x + 9
B. y = 0.5x
C. y = 3x - 1
D. none of them

Correct Answer: (C)

Steepness is the size of the slope. y = 3x - 1 has slope 3, greater than 1. The others have slope 1 (just shifted) and 0.5 (flatter).

Why (A) is wrong: Adding 9 changes position, not steepness.

Why (B) is wrong: 0.5 is flatter than 1.

Why (D) is wrong: One of them IS steeper.

Hinglish: *Steepness = |slope|. 3 > 1 toh y = 3x - 1 tez chadhti hai. y = x + 9 utni hi steep hai - bas upar shift hui hai.*

MCQ 24 · Normal · Concept: Intercept recognition

24. Which line cuts the y-axis at the point (0, -4)?

A. y = 3x - 4
B. y = -4x + 3
C. y = 4x
D. y = x + 4

Correct Answer: (A)

The y-intercept equals b in y = ax + b, and (0, b) is the cutting point. Only option A has b = -4.

Why (B) is wrong: Its b is 3 (the -4 is a slope).

Why (C) is wrong: Passes through the origin.

Why (D) is wrong: Cuts at (0, 4).

Hinglish: *(0, b) form yaad rakho: y-axis cut ka second coordinate hi b hai. A mein b = -4. Verify: x = 0 par y = -4 sahi.*

MCQ 25 · Normal · Concept: Point on line

25. Which point lies on the line y = 2x - 5?

A. (3, 1)
B. (3, 0)
C. (1, 3)
D. (0, -3)

Correct Answer: (A)

Substitute x = 3: y = 2(3) - 5 = 1. So (3, 1) satisfies the equation. The others fail: (3, 0) gives 1 not 0; (1, 3) needs y = -3; (0, -3) needs y = -5.

Why (B) is wrong: 2(3) - 5 = 1, not 0.

Why (C) is wrong: 2(1) - 5 = -3, not 3.

Why (D) is wrong: 2(0) - 5 = -5, not -3.

Hinglish: *Test karo: x = 3 dala, y = 1 nikla - toh (3, 1) line par hai. Baaki options ke y mismatch.*

MCQ 26 · Normal · Concept: Growth vs decay

26. The model h(t) = 3 - 0.5t describes:

A. linear growth
B. linear decay
C. no change
D. quadratic behaviour

Correct Answer: (B)

The rate of change (slope) is -0.5: each month the height drops by 0.5. Negative slope means decay.

Why (A) is wrong: Growth needs a positive slope.

Why (C) is wrong: Something IS changing - by 0.5 per step.

Why (D) is wrong: The rule is degree 1, not 2.

Hinglish: *Slope negative = decay. 3 starting value, 0.5 per month girawat - PDF ke water-tank example jaisa.*

MCQ 27 · Normal · Concept: Assertion-reason

27. Assertion (A): Every line y = ax passes through the origin. Reason (R): Substituting x = 0 gives y = a(0) = 0.

A. Both A and R are true and R explains A
B. Both A and R are true but R does not explain A
C. A is true, R is false
D. A is false, R is true

Correct Answer: (A)

The reason is precisely the verification that (0, 0) satisfies y = ax for every a - it explains the assertion completely.

Why (B) is wrong: The link is direct.

Why (C) is wrong: R is a correct computation.

Why (D) is wrong: A is a true property of the family.

Hinglish: *R hi A ka proof hai: x = 0 par y = 0 nikalta hai, chahe a kuch bhi ho. Perfect A-R pair - option A.*

MCQ 28 · Normal · Concept: Statement set

28. Consider: (i) a linear polynomial has degree 1. (ii) a constant polynomial has no degree. (iii) the coefficient of x in the term x is 1. Which are TRUE?

A. (i) and (iii) only
B. (i) and (ii) only
C. (ii) and (iii) only
D. all three

Correct Answer: (A)

(i) is the definition. (ii) is false - the degree of a constant is 0, which exists. (iii) is true: x means 1x.

Why (B) is wrong: (ii) is false.

Why (C) is wrong: (ii) is false.

Why (D) is wrong: (ii) spoils the set.

Hinglish: *(ii) trap hai - constant ka degree HOTA hai, wo 0 hai. 'No degree' bolna galat. (i) aur (iii) definition-level sach hain.*

MCQ 29 · Normal · Concept: Assertion-reason evaluation

29. Assertion (A): For p(x) = 2x + 3, the value at x = -1 is 1. Reason (R): Substituting a negative input requires brackets: 2(-1) + 3.

A. Both A and R are true and R explains A
B. Both A and R are true but R does not explain A
C. A is true, R is false
D. A is false, R is true

Correct Answer: (A)

A is true: 2(-1) + 3 = -2 + 3 = 1. R states the safe-substitution practice that makes the computation of A reliable - it explains why the evaluation works out as claimed.

Why (B) is wrong: The bracket rule is exactly how A is computed.

Why (C) is wrong: R is correct practice.

Why (D) is wrong: A computes to 1, true.

Hinglish: *2(-1) + 3 = 1 sahi hai, aur bracket wala rule hi wo calculation reliable banata hai. R explains A.*

MCQ 30 · Normal · Concept: Reverse evaluation equation

30. Solving 6 - 2x = -4 for x gives:

A. x = 5
B. x = -5
C. x = 1
D. x = -1

Correct Answer: (A)

Move 6: -2x = -10, so x = 5. Check: 6 - 2(5) = 6 - 10 = -4. Correct.

Why (B) is wrong: Sign slip in the final division.

Why (C) is wrong: Moved terms wrongly.

Why (D) is wrong: Half-solved.

Hinglish: *-2x = -10 - dono side minus; divide karne par plus. x = 5. Verify karo: 6 - 10 = -4 sahi.*

MCQ 31 · Normal · Concept: Spot the linear pattern

31. Which sequence is a LINEAR pattern?

A. 3, 7, 11, 15
B. 2, 4, 8, 16
C. 1, 4, 9, 16
D. 1, 1, 2, 3, 5

Correct Answer: (A)

A linear pattern has a constant difference between consecutive terms. Option A adds 4 every time. The others double, square, or follow Fibonacci - none has a constant difference.

Why (B) is wrong: Doubling - multiplicative, not linear.

Why (C) is wrong: Perfect squares.

Why (D) is wrong: Fibonacci-type additions of previous terms.

Hinglish: *Linear pattern ka test: consecutive difference CONSTANT. 3, 7, 11, 15 mein +4 har baar. Baaki doubling, squares, Fibonacci hain - inka difference badalta hai.*

MCQ 32 · Normal · Concept: nth term from sequence

32. The nth term of the sequence 7, 12, 17, 22, ... is:

A. 5n + 2
B. 5n - 2
C. 7n
D. n + 6

Correct Answer: (A)

Constant difference 5: rule 5n + c. First term: 5 + c = 7 gives c = 2. Rule 5n + 2. Check: n = 2 gives 12. Correct.

Why (B) is wrong: Would start at 3.

Why (C) is wrong: No pattern fit.

Why (D) is wrong: Not linear in the right way.

Hinglish: *Difference 5, first term 7 = 5 + c, c = 2. 5n + 2. Second term se verify karo - 12 sahi.*

MCQ 33 · Normal · Concept: Tiered fare

33. A cab charges Rs 50 for the first 2 km and Rs 12 per km after that. The fare for a 10 km ride is:

A. Rs 170
B. Rs 146
C. Rs 158
D. Rs 120

Correct Answer: (B)

Extra distance = 10 - 2 = 8 km. Fare = 50 + 12(8) = 50 + 96 = Rs 146.

Why (A) is wrong: Applied 12 to all 10 km.

Why (C) is wrong: Arithmetic slip.

Why (D) is wrong: Forgot the base fare.

Hinglish: *Pehle 2 km ka paisa fixed 50 mein. Baaki 8 km x 12 = 96. Total 146. (n - 2) wala structure yaad rakho.*

MCQ 34 · Normal · Concept: Age relation

34. Ravi is 3 times as old as his sister. In 4 years, the sum of their ages will be 40. Ravi's present age is:

A. 24 years
B. 8 years
C. 27 years
D. 30 years

Correct Answer: (A)

Let the sister be x and Ravi 3x. In 4 years: (x + 4) + (3x + 4) = 40, so 4x + 8 = 40, x = 8. Sister is 8, Ravi is 24.

Why (B) is wrong: That is the sister's age - the question asks for Ravi.

Why (C) is wrong: Solved a wrong equation.

Why (D) is wrong: Arithmetic overshoot.

Hinglish: *4x + 8 = 40 se x = 8 (behen). Ravi = 3x = 24. Question ne RAVI manga tha - sister ka number dena classic galti hai.*

MCQ 35 · Normal · Concept: Fit from two points

35. A bill follows y = ax + b. If 10 units cost Rs 400 and 20 units cost Rs 550, then:

A. a = 15, b = 250
B. a = 15, b = 400
C. a = 40, b = 0
D. a = 55, b = 150

Correct Answer: (A)

Subtract: 550 - 400 = 150 = 10a, so a = 15. Then b = 400 - 15(10) = 250. Model: y = 15x + 250.

Why (B) is wrong: Forgot to back-subtract.

Why (C) is wrong: Used the wrong pair.

Why (D) is wrong: Inconsistent fit.

Hinglish: *10 units ka extra 150: per-unit 15. Wapas: 400 - 150 = 250 fixed. Standard 3-step: subtract, divide, back-substitute.*

MCQ 36 · Normal · Concept: Decay value

36. A laptop bought at Rs 60,000 loses Rs 6,000 in value per year. After 5 years its value is:

A. Rs 30,000
B. Rs 54,000
C. Rs 36,000
D. Rs 24,000

Correct Answer: (A)

Value = 60000 - 6000(5) = 60000 - 30000 = Rs 30,000.

Why (B) is wrong: One year only.

Why (C) is wrong: Wrong multiplier.

Why (D) is wrong: Subtracted too much.

Hinglish: *6000 x 5 = 30000 ka loss. 60000 - 30000 = 30000. Aadha paisa bacha.*

MCQ 37 · Normal · Concept: Intercept point

37. The line y = -2x + 9 cuts the y-axis at:

A. (0, 9)
B. (9, 0)
C. (0, -2)
D. (-2, 0)

Correct Answer: (A)

At x = 0, y = 9: the point (0, 9). The x-axis cut would be at (4.5, 0).

Why (B) is wrong: That swaps the coordinates.

Why (C) is wrong: That is the slope as a coordinate.

Why (D) is wrong: Neither intercept.

Hinglish: *y-axis cut = x zero rakho = (0, b) = (0, 9). (9, 0) ulta form hai - axis badal gaye.*

MCQ 38 · Normal · Concept: Parallel partner

38. Which line is parallel to y = 7x + 2?

A. y = 7x - 5
B. y = -7x + 2
C. y = 2x + 7
D. y = x + 7

Correct Answer: (A)

Parallel needs the SAME slope: only y = 7x - 5 has slope 7 with a different intercept. (Option B has slope -7; option C has slope 2.)

Why (B) is wrong: Slope -7 is a different direction.

Why (C) is wrong: Slope 2.

Why (D) is wrong: Slope 1.

Hinglish: *Slope 7 match karo, intercept alag: y = 7x - 5. Slope ka sign bhi matter karta hai - -7 parallel nahi hai, wo ulta disha mein jaati hai.*

MCQ 39 · Normal · Concept: Quadrants of a line

39. The line y = -x passes through which quadrants?

A. I and II
B. II and IV
C. I and III
D. only through the origin

Correct Answer: (B)

y = -x is positive exactly when x is negative. So it lives in Quadrant II (x negative, y positive) and Quadrant IV (x positive, y negative), passing through the origin on the way.

Why (A) is wrong: Q II yes, Q I no (both positive is impossible for y = -x).

Why (C) is wrong: That is y = x.

Why (D) is wrong: It passes through infinitely many points besides the origin.

Hinglish: *y = -x: x minus par y plus (Q II), x plus par y minus (Q IV). Diagonal line jo origin se dono taraf ulta disha mein jaati hai.*

MCQ 40 · Normal · Concept: No intersection

40. The lines y = 2x + 3 and y = 2x - 1:

A. meet at one point
B. meet at two points
C. never meet
D. are the same line

Correct Answer: (C)

Same slope, different intercepts: parallel. Setting 2x + 3 = 2x - 1 gives 3 = -1, impossible - confirming no intersection point exists.

Why (A) is wrong: Different intercepts forbid meeting.

Why (B) is wrong: Two straight lines meet at most once anyway.

Why (D) is wrong: Intercepts differ, so not identical.

Hinglish: *Setting barabar: 2x cancel, 3 = -1 bacha - absurd! Isi ka matlab parallel, kabhi nahi milte. Equation-level proof bhi mil gaya.*

MCQ 41 · HOTS · Concept: Triangle with axes

41. The line y = 3x + 6 and the two coordinate axes form a triangle. Its area is:

A. 6 sq units
B. 12 sq units
C. 3 sq units
D. 18 sq units

Correct Answer: (A)

y-intercept (0, 6); x-intercept: 3x + 6 = 0 gives x = -2, point (-2, 0). Right triangle with legs 6 and 2: area = (1/2)(6)(2) = 6.

Why (B) is wrong: Forgot the 1/2.

Why (C) is wrong: Used the wrong leg.

Why (D) is wrong: Multiplied intercepts without halving.

Hinglish: *Intercepts (0, 6) aur (-2, 0). Origin ke saath right triangle: legs 6 aur 2, area 6. Length absolute value mein lo - negative coordinate se area negative nahi hota.*

MCQ 42 · Normal · Concept: Hidden coefficient

42. The coefficient of y in the expression 3x + y - 8 is:

A. 0
B. 1
C. 3
D. -8

Correct Answer: (B)

The term y is really 1y, so the coefficient is 1.

Why (A) is wrong: y is present, so not 0.

Why (C) is wrong: 3 belongs to x.

Why (D) is wrong: -8 is the constant.

Hinglish: *Invisible 1! y = 1y. Jo variable ke aage number nahi dikhta uska coefficient 1 hota hai.*

MCQ 43 · Normal · Concept: Degree scan trap

43. The degree of the polynomial $x^3 + 7x^4 - 2x$ is:

A. 3
B. 4
C. 7
D. 1

Correct Answer: (B)

The powers present are 3, 4 and 1. The HIGHEST is 4, even though the x^3 term is written first. Degree is about the maximum power, not the order of writing.

Why (A) is wrong: Fell for the first term.

Why (C) is wrong: Added powers - never do that.

Why (D) is wrong: Lowest power.

Hinglish: *x^3 pehle likha hai par $7x^4$ bhi hai - max power 4. Order of writing ka koi matlab nahi degree mein.*

MCQ 44 · HOTS · Concept: Identical polynomials

44. Let $p(x) = 5x - 3$ and $q(x) = 5x - 3$. The equation $p(x) - q(x) = 0$ is satisfied:

A. only at $x = 3/5$
B. only at $x = 0$
C. by every value of x
D. by no value of x

Correct Answer: (C)

$p(x) - q(x) = 0$ for every x because the two polynomials are IDENTICAL - subtracting them gives the zero polynomial. (Note the contrast: $p(x) = 0$ alone has the single solution $x = 3/5$.)

Why (A) is wrong: That solves $p(x) = 0$, not $p - q = 0$.

Why (B) is wrong: No reason for $x = 0$ to be special here.

Why (D) is wrong: Identical expressions always subtract to zero.

Hinglish: *p aur q same polynomial hain - difference hamesha zero. Har x solution. Confusion hota hai $p(x) = 0$ se - wahan sirf $x = 3/5$ hota. Difference-of-identicals vs zero-of-one: alag cheezein!*

MCQ 45 · Normal · Concept: Linear words with twist

45. Two numbers add to 30. The larger is 3 more than twice the smaller. The numbers are:

A. 9 and 21
B. 10 and 20
C. 11 and 19
D. 8 and 22

Correct Answer: (A)

Smaller x, larger $2x + 3$: $x + 2x + 3 = 30$ gives $3x = 27$, $x = 9$. Numbers 9 and 21. Check: $9 + 21 = 30$ and $21 = 2(9) + 3$.

Why (B) is wrong: Difference 10 with ratio 2 exactly - the +3 is missing.

Why (C) is wrong: Difference 8.

Why (D) is wrong: No fit.

Hinglish: *$3x + 3 = 30$, $x = 9$. '3 more than twice' ka translation hi question tha: $2x + 3$.*

MCQ 46 · Normal · Concept: Reachability check

46. A machine outputs 4n - 1 at stage n. Can it ever output 91?

A. No, never

B. Yes, at n = 23

C. Yes, at n = 22

D. Yes, at n = 91/4

Correct Answer: (B)

4n - 1 = 91 gives 4n = 92, n = 23 - a whole number, so stage 23 outputs exactly 91.

Why (A) is wrong: It is reachable - check the arithmetic.

Why (C) is wrong: 4(22) - 1 = 87.

Why (D) is wrong: Fractional stage does not exist.

Hinglish: *91 + 1 = 92, 92/4 = 23 - whole number! YES at n = 23. Agar fraction aata toh 'never' hota.*

MCQ 47 · Normal · Concept: Tiered fare again

47. A bike ride costs Rs 30 for the first kilometre and Rs 14 per km after that. The fare for 9 km is:

A. Rs 126

B. Rs 156

C. Rs 142

D. Rs 140

Correct Answer: (C)

Extra km = 9 - 1 = 8. Fare = 30 + 14(8) = 30 + 112 = Rs 142.

Why (A) is wrong: Charged 14 on all 9 km only.

Why (B) is wrong: Arithmetic slip.

Why (D) is wrong: Close but wrong product.

Hinglish: *1 km included hai 30 mein; baaki 8 x 14 = 112. Total 142.*

MCQ 48 · Normal · Concept: Steeper line choice

48. Which of these lines is steeper than y = 2x?

A. y = x + 9

B. y = 2x + 5

C. y = 3x - 1

D. y = 0.5x + 1

Correct Answer: (C)

Compare slopes: 1, 2, 3, 0.5. Only 3 exceeds 2. Adding to b never changes steepness.

Why (A) is wrong: Slope 1 - flatter.

Why (B) is wrong: Slope 2 - equally steep, just shifted.

Why (D) is wrong: Slope 0.5 - flattest.

Hinglish: *Slope compare karo: 3 hi 2 se bada hai. Intercept ka koi lena-dena steepness se nahi.*

MCQ 49 · Normal · Concept: Effect of changing b

49. In y = ax + b, if b is increased by 2 while a stays fixed, the new line:

A. becomes steeper

B. shifts up 2 units, parallel to the old line

C. shifts right 2 units

D. rotates about the origin

Correct Answer: (B)

Every output increases by exactly 2, so the whole line slides up 2 units. Slope unchanged means parallel; the y-axis cut moves from (0, b) to (0, b + 2).

Why (A) is wrong: Only changing a changes steepness.

Why (C) is wrong: A vertical shift is not a horizontal shift.

Why (D) is wrong: No rotation - slope is fixed.

Hinglish: *b badla = line UPAR slide, parallel hi rahegi. Slope (a) hi steepness banata hai - wo nahi badla.*

MCQ 50 · Normal · Concept: Table to rule

50. A table shows: x = 0, 1, 2, 3 giving y = 5, 8, 11, 14. The rule is y = 3x + c where c equals:

A. 5

B. 3

C. 2

D. 8

Correct Answer: (A)

At x = 0 the value is y = 5 - and x = 0 always reveals the intercept directly: c = 5. Check: 3(3) + 5 = 14. Correct.

Why (B) is wrong: That is the slope.

Why (C) is wrong: No basis.

Why (D) is wrong: That is a middle value.

Hinglish: *Zero input ka output HI intercept hota hai - table mein x = 0 ho toh c free mein mil jaata hai. 5. Verify: 3x + 5 se 8, 11, 14 sab match.*

MCQ 51 · HOTS · Concept: Two-models meeting

51. Tap A fills an empty tank at 20 litres per minute (V = 20t). Tap B pours into a tank that already has 100 litres, at 12 litres per minute (V = 100 + 12t). After how many minutes do the two tanks hold equal volumes?

A. 12.5 minutes

B. 10 minutes

C. 8 minutes

D. 25 minutes

Correct Answer: (A)

Set 20t = 100 + 12t: 8t = 100, t = 12.5 minutes. (Each then holds 250 litres.)

Why (B) is wrong: Misbalanced the equation.

Why (C) is wrong: Subtracted wrongly.

Why (D) is wrong: Doubled the time.

Hinglish: *Do lines ka intersection: 8t = 100, t = 12.5. Us waqt dono 250 litre. Fraction aana bilkul theek hai - yahan stage ki jagah TIME hai, jo real value ho sakti hai.*

MCQ 52 · HOTS · Concept: Find b from a point

52. The line p(x) = 2x + b passes through (3, 11). Then b equals:

A. 5
B. 11
C. 6
D. 17

Correct Answer: (A)

Substitute: 11 = 2(3) + b = 6 + b, so b = 5. The line is p(x) = 2x + 5.

Why (B) is wrong: Copied the y-value.

Why (C) is wrong: Forgot to double x.

Why (D) is wrong: Added instead of subtracting.

Hinglish: *Point se b nikalna: 11 = 6 + b, b = 5. Ek point + slope = poori line.*

MCQ 53 · HOTS · Concept: Composition

53. If p(x) = x + 4, then p(p(2)) equals:

A. 10
B. 6
C. 8
D. 12

Correct Answer: (A)

Inner first: p(2) = 6. Then p(6) = 10.

Why (B) is wrong: Stopped after one application.

Why (C) is wrong: Added 4 to x twice wrongly.

Why (D) is wrong: Multiplied somewhere.

Hinglish: *Andar wala pehle: 2 + 4 = 6. Phir 6 + 4 = 10. Machine ke andar machine.*

MCQ 54 · HOTS · Concept: Slope from two points

54. The line through (1, 2) and (3, 8) has slope:

A. 3
B. 6
C. 2
D. 1/3

Correct Answer: (A)

Slope = change in y divided by change in x = (8 - 2)/(3 - 1) = 6/2 = 3. (This is the same a that the two-point fitting method produces.)

Why (B) is wrong: Used only the y-gap.

Why (C) is wrong: Used only the x-gap.

Why (D) is wrong: Inverted the ratio.

Hinglish: *a = y ka gap / x ka gap = 6/2 = 3. Yehi wali division fitting method mein hoti hai - ab formula ki tarah pehchan lo.*

MCQ 55 · HOTS · Concept: Intersection of two lines

55. The lines y = x + 4 and y = -2x + 10 intersect where x equals:

A. 2

B. 6

C. 4

D. 10

Correct Answer: (A)

Set equal: x + 4 = -2x + 10 gives 3x = 6, x = 2 (then y = 6 - the point (2, 6)).

Why (B) is wrong: That is the y-value.

Why (C) is wrong: Moved terms wrongly.

Why (D) is wrong: No basis.

Hinglish: *Barabar rakho: 3x = 6, x = 2. Yehi method Q50 wale concurrency question mein use hua tha.*

MCQ 56 · Normal · Concept: Growth identification

56. A savings balance follows b(x) = 600 - 15x rupees, x days after a recharge. This is:

A. linear growth

B. linear decay

C. neither

D. quadratic

Correct Answer: (B)

The balance falls by a constant 15 per day - linear decay with slope -15, emptying at x = 40 days.

Why (A) is wrong: Balance is falling.

Why (C) is wrong: It is a clean linear rule.

Why (D) is wrong: Degree 1.

Hinglish: *-15 slope = decay. 600/15 = 40 din mein khatam. PDF ka recharge example hi hai.*

MCQ 57 · Normal · Concept: Statement set on slopes

57. Consider: (i) linear growth corresponds to a positive slope. (ii) linear decay corresponds to a negative slope. (iii) the y-intercept of a line is always positive. Which are TRUE?

A. (i) and (ii) only

B. (i) and (iii) only

C. (ii) and (iii) only

D. all three

Correct Answer: (A)

(i) and (ii) are the growth/decay sign rules. (iii) is false: b can be negative (like y = x - 5 cutting at (0, -5)) or zero.

Why (B) is wrong: (iii) is false.

Why (C) is wrong: (iii) is false.

Why (D) is wrong: (iii) spoils the set.

Hinglish: *(iii) galat - b negative bhi ho sakta hai (0, -5) jaisa cut, aur zero bhi. (i) aur (ii) sign-rules pakke sach.*

MCQ 58 · Normal · Concept: Assertion-reason parallel

58. Assertion (A): The lines y = 2x + 1 and y = 2x + 5 are parallel. Reason (R): They have the same slope 2 and different y-intercepts.

A. Both A and R are true and R explains A

B. Both A and R are true but R does not explain A

C. A is true, R is false

D. A is false, R is true

Correct Answer: (A)

Equal slopes with distinct intercepts is exactly the parallel-line condition developed in this chapter - R is the reason A holds.

Why (B) is wrong: The criterion IS the explanation.

Why (C) is wrong: R is factually right.

Why (D) is wrong: A is right.

Hinglish: *R hi parallel ka criterion hai: same slope + alag intercept. Direct explanation - option A.*

MCQ 59 · Normal · Concept: Odd-number pattern

59. The nth odd number is 2n - 1. The sum of the first 3 odd numbers (1 + 3 + 5) equals which perfect square?

A. 6

B. 9 (= 3 squared)

C. 12

D. 8

Correct Answer: (B)

$1 + 3 + 5 = 9 = 3^2$. In general the first n odd numbers sum to n^2 - a hidden gem of the linear pattern 2n - 1.

Why (A) is wrong: 6 is just the count-times-2 guess.

Why (C) is wrong: No basis.

Why (D) is wrong: Arithmetic slip.

Hinglish: *1 + 3 + 5 = 9 = 3 ka square! Pehle n odd numbers ka sum hamesha n^2 hota hai - 2n - 1 pattern ka bonus tohfa.*

MCQ 60 · HOTS · Concept: Interpretation finale

60. A tutor charges a one-time registration fee of Rs 400 plus Rs 250 per month. In the model T = 250m + 400, what does the pair (a = 250, b = 400) represent?

A. a = monthly rate, b = registration fee

B. a = registration fee, b = monthly rate

C. a = total cost, b = months

D. a = number of months, b = total cost

Correct Answer: (A)

The slope multiplies the running variable (months), so a = 250 is the per-month charge. The intercept is the value at month zero, so b = 400 is the one-time fee.

Why (B) is wrong: Swapped the roles of slope and intercept.

Why (C) is wrong: Confused parameters with variables.

Why (D) is wrong: Same confusion, different clothes.

Hinglish: *Slope hamesha 'per unit of x' hota hai (per month 250), intercept x = 0 wali value (one-time 400). Ye interpretation ka final test - bill, salary, sab jagah yahi pattern.*

2.5 Chapter Formula Sheet

Formula / Rule	Meaning and when to use	Conditions	Tiny example	Common mistake
Degree = highest power of the variable	Classify a polynomial; label linear/quadratic/cubic/constant	One-variable polynomials	-13 has degree 0; 6z + 1 has degree 1	Reading the first term's power instead of the max
Value at input: substitute in brackets	Evaluate p(a); run the machine forward	Always bracket negatives: 7(-3)	7x - 4 at x = -3 gives -25	Dropping brackets: 7 - 3 - 4
Linear pattern: T(n) = dn + c	Constant-difference sequences; find any stage	d = common difference; c fixed by one term	4, 7, 10: T(n) = 3n + 1	Using d as the constant term
Reachability: solve dn + c = target	Can the pattern hit a value?	n must be a whole number for YES	3n + 1 = 100 gives n = 33: yes	Answering yes with a fractional n
Growth/decay: value = start + rate x steps	Fixed change per equal interval	Rate sign: + growth, - decay	10000 - 800t	Forgetting the start value
Run-out point: expression = 0	When does a decaying quantity vanish	Only when the model allows	900 - 45t = 0 at t = 20	Reporting a negative or fractional time uncritically
Fit y = ax + b from two points	Two observations of a linear bill/process	Subtract the equations to eliminate b	400 = 10a + b, 500 = 14a + b give a = 20? (here 10a gap gives a = 20 for the Rs 200 gap: 200/10)	Solving both equations separately for b
$a = (y_2 - y_1)/(x_2 - x_1)$	Slope from two points; also the pattern's constant difference	x_2 differs from x_1	(8 - 2)/(3 - 1) = 3	Inverting the ratio
b = y at x = 0	Fixed part of the relationship; where the line cuts the y-axis	Read AFTER isolating y	y = 2x - 5 cuts at (0, -5)	Reading b from 3y = 6x + 12 as 12
Parallel test: same a, different b	Decide if two lines ever meet	Both equations isolated for y first	y = 2x + 3.5 and y = 2x - 11/3	Comparing b instead of a
y = ax passes through (0, 0)	Zero fixed cost / proportional relationships	b = 0	w = 3d (work under constant force)	Assuming every line crosses the origin

Formula / Rule	Meaning and when to use	Conditions	Tiny example	Common mistake
x-intercept: set y = 0	Where the line crosses the x-axis	Solve ax + b = 0	y = 2x + 8 cuts at (-4, 0)	Swapping the roles of the two intercept rules

2.6 Last-Minute Cheat Sheet

> **30-second revision card - Chapter 2**
> DEGREE: max power (constant = 0, linear = 1, quadratic = 2, cubic = 3). Coefficient of bare x is 1. EVALUATE with brackets. Reverse = solve. PATTERN: difference d gives rule dn + c; c from one known term. Target reachable only if n is WHOLE. FIT y = ax + b: subtract the two data equations, a = gap-ratio, b = y_1 - a x_1. a = rate per unit, b = fixed part. LINE FACTS: y = ax passes through origin; b = y-axis cut (0, b); same a = parallel; bigger |a| = steeper; negative a = decay. ISOLATE y FIRST before reading a and b.

- **If you see a table of stages** → constant difference = coefficient of n.
- **If you see 'can it equal...'** → solve; whole-number n = yes, fraction = no.
- **If you see two observations of a bill/cost** → subtract equations, get a, back-substitute for b.
- **If you see a disguised equation (3y = 6x + 12)** → divide, isolate y, THEN read slope and intercept.
- **If you see 'parallel'** → match the slope only.
- **If you see 'x-intercept / where it crosses the x-axis'** → set y = 0.
- **If you see 'fixed fee + per unit'** → that IS y = ax + b: b = fee, a = rate.

2.7 Mistake Buster - DON'T MAKE THESE MISTAKES

1. **Reading slope from a disguised equation.** In 3y = 6x + 12 the slope is 2, not 6. Fix: isolate y before extracting a and b - every single time.
2. **Unbracketed negative substitution.** 7(-3) is -21; writing 7 - 3 gives a completely wrong world. Fix: brackets are not optional.
3. **Calling the constant term the coefficient.** In 5x - 7, the coefficient of x is 5; -7 is the constant. Fix: ask 'which term is this number attached to?'
4. **Saying a constant has no degree.** It has degree 0. Fix: remember c = c x 1 = c x x^0.
5. **Accepting a fractional stage.** 'Yes, at n = 10.8' is meaningless for patterns. Fix: integrality check before saying yes.
6. **Forgetting the tiered base in fares.** 'Rs 50 for the first 2 km' means the rate applies only to (n - 2) km. Fix: write the extra-distance expression explicitly.
7. **Interpreting b as the rate.** In F = 1.8C + 32, the offset 32 is not a rate. Fix: slope multiplies the variable; intercept stands alone at x = 0.
8. **Missing the second meaning of 'parallel lines never meet'.** Setting two same-slope lines equal produces an impossibility - use this as a quick proof, not a surprise.

9. **Answering only x in word problems.** If the question asks for two numbers, both must appear in the final line. Fix: re-read the question before writing the answer.
10. **Mixing up (0, b) and (b, 0).** The y-axis cut has x = 0. Fix: axis name tells you which coordinate is zero.

2.8 Exam Strategy for Chapter 2

Order of attack. Definition and evaluation MCQs are fast marks - bank them first. Pattern and growth/decay questions come next (one clean substitution or solve). Keep the two-point fitting questions and multi-step HOTS (plan comparison, composition, concurrency) for the final stretch when your mind is warmed up.

Where marks are lost. Top three leak-points: unbracketed substitution (mechanical error), reading a and b without isolating y (conceptual slip), and answering word problems with the bare x instead of the requested quantities (presentation loss). Each has a one-line fix: brackets always; isolate first; answer the question asked.

Formulas most frequently tested. The nth-term rule dn + c, the fitting pair a = gap-ratio and b = y_1 - a x_1, growth/decay value = start + rate x steps, and the parallel condition (same slope). Interpretation sentences (what a and b MEAN) are graded almost as often as the computation itself.

Concepts that get mixed. Patterns with graphs (the constant difference IS the slope), evaluation with equations (forward and backward machine runs), and linear models with money (fixed + variable structure everywhere: bills, salaries, fares). Recognising the shared skeleton turns a new word problem into an old friend.

Safe shortcuts. Using x = 0 to read b directly from a table; checking a fitted model on BOTH given points before moving on; using the impossible-equation argument (3 = -1) to prove lines never meet. **Unsafe shortcuts.** Guessing the pattern rule from the first two terms only, skipping the integrality check in reachability questions, and applying the slope formula to points that were never claimed to be on one line.

How to check an answer. Evaluation: recompute with brackets written explicitly. Fitted model: substitute BOTH original data points - both must reproduce their y-values. Pattern rule: test on two consecutive known terms. Word problem: translate the final numbers back into the original sentence and see if the story holds.

2.9 Chapter Completion Checklist

- [ ] Every concept covered: terms, coefficients, degree, classification, evaluation, reverse evaluation, equations from words, patterns, growth/decay, fitting y = ax + b, slope, intercept, parallel lines, graphs
- [ ] Every formula covered: 12 rules in Section 2.5 with conditions and micro-examples
- [ ] 50 advanced questions completed (Q1-Q50) with the five-band difficulty ladder
- [ ] Every question has a full step-by-step solution
- [ ] Every solution has a Hinglish explanation
- [ ] 60 MCQs completed with answers
- [ ] Every MCQ has an explanation plus trap analysis
- [ ] Formula sheet completed
- [ ] Last-minute cheat sheet completed
- [ ] Mistake buster completed (10 named mistakes)

- [ ] Exam strategy completed

Chapter 3: The World of Numbers (Real Numbers)

Numbers are not one big happy family - they live in clearly separated neighbourhoods, and exams love to test whether you know who lives where. The uploaded PDF builds the full number map: natural numbers N, integers Z, rationals Q, irrationals, and the master set R of real numbers. It studies decimal expansions (terminating, recurring, never-ending and never-repeating), shows how to convert recurring decimals back into fractions, constructs roots like $\sqrt{2}$ and $\sqrt{3}$ on the number line using right-angled triangles, proves that $\sqrt{2}$ is irrational, simplifies surds, rationalises denominators, and closes with the laws of exponents. In this book every one of those ideas is pushed to competition level: reverse questions, contradiction proofs, nested radicals, and exponent equations that look impossible until you spot the base.

3.1 Complete Concept Cheat Sheet (read this before any question)

3.1.1 The number family map: N, Z, Q, I, R

What is it? A classification of all numbers you will meet in Grade 9. **N** (natural numbers) = 1, 2, 3, ... **Z** (integers) = ..., -2, -1, 0, 1, 2, ... - N plus zero plus the negatives. **Q** (rationals) = every number writable as p/q with p, q integers and $q \neq 0$ - this includes all terminating decimals and all recurring decimals. **I** (irrationals) = real numbers that can NOT be written as p/q, like $\sqrt{2}$, $\sqrt{3}$, π, and decimals that go on forever without any repeating block. **R** (reals) = $Q \cup I$, every point on the number line. The nesting is strict: $N \subset Z \subset Q \subset R$, while I sits beside Q inside R with $I \cap Q = \emptyset$.

When do I use it? Whenever a question says 'classify', 'true or false', 'which of these is irrational', or asks about closure of operations. **How do I recognise the type?** Terminating decimal → rational. Recurring decimal → rational. Square root of a non-perfect-square → irrational. π and e → irrational. A fraction of integers → rational (even -7/3). **Tiny example.** -4 is Z and Q and R but not N; 0.333... = 1/3 is Q; $\sqrt{7}$ is I; 22/7 is Q (it is only an approximation of π, never equal to it).

Figure 3.1: The number family - each set sits inside the next bigger one; irrationals live beside Q inside R.

3.1.2 Decimal expansions and the denominator test

What is it? Every rational number has a decimal expansion that either STOPS (terminates) or STOPS NEVER BUT REPEATS a block (non-terminating recurring). Every irrational number has a decimal expansion that goes on forever WITHOUT any repeating block (non-terminating non-recurring). **The denominator test:** write the rational in lowest terms p/q. If $q = 2^m \times 5^n$ (only 2s and 5s, which are exactly the prime factors of 10), the decimal terminates; otherwise it recurs. Why 2 and 5? Because $10 = 2 \times 5$, so only these primes can be 'absorbed' into a power of ten.

When do I use it? Questions like 'without long division, state whether the decimal will terminate'. **Tiny example.** 17/8: $8 = 2^3$, so terminating (2.125). 13/3125: $3125 = 5^5$, terminating. 7/12: $12 = 2^2 \times 3$ - the stray 3 forces a recurring decimal. 6/15 = 2/5 in lowest terms - always REDUCE FIRST, the reduced denominator decides.

> **The reduce-first rule**
> The $2^l \times 5^n$ test works only in LOWEST TERMS. 6/15 looks non-terminating until you cancel the 3 and get 2/5 = 0.4. Reducing first is the difference between a right and a wrong answer.

3.1.3 Recurring decimals back to fractions (p/q form)

What is it? A reversing technique. **Pure recurring (repeat starts immediately):** count the repeating digits k, then the number = (repeating block)/(same number of 9s). Example: 0.555... = 5/9, 0.373737... = 37/99, 0.090909... = 9/99 = 1/11. **Mixed recurring (some non-repeating digits first):** (all digits once, minus the non-repeating part) / (as many 9s as repeating digits, followed by as many 0s as non-repeating digits). Example: 0.4777... = (47 - 4)/90 = 43/90.

Where does the trick come from? Multiply by the right power of 10 so two copies of the repeating block line up, then subtract - the infinite tails cancel exactly and you are left with a finite equation. **Full derivation for 0.777...:** let x = 0.777...; then 10x = 7.777...; subtract: 10x - x = 7.777... - 0.777... = 7; so 9x = 7 and x = 7/9. **Tiny example (mixed).** x = 0.2333...; 10x = 2.333..., 100x = 23.33...; subtract: 90x = 21, x = 21/90 = 7/30. Always finish by reducing the fraction.

3.1.4 Locating roots on the number line

What is it? A compass-ruler construction. To place $\sqrt{n}$: build a right triangle where the hypotenuse has length $\sqrt{n}$ using the Baudhayana-Pythagoras theorem. For $\sqrt{2}$: take OA = 1 on the line, raise AB = 1 perpendicular at A; then $OB = \sqrt{(1^2 + 1^2)} = \sqrt{2}$; swing an arc centred at O through B down to the line. For $\sqrt{3}$: repeat with the previous hypotenuse - take the segment of length $\sqrt{2}$ as one leg, add a perpendicular leg of 1, get hypotenuse $\sqrt{(2 + 1)} = \sqrt{3}$. Chaining this creates the famous **square root spiral** where each new edge has length $\sqrt{(n+1)}$.

When do I use it? 'Show how to find $\sqrt{5}$ on the number line', 'represent $\sqrt{4.3}$ geometrically', and successive-magnification questions ('zoom twice and tell which two numbers $\sqrt{3}$ sits between at the second decimal'). **Successive magnification.** To locate 2.6 (say $\sqrt{7} \approx 2.6457$): between 2 and 3, then between 2.6 and 2.7, then between 2.64 and 2.65 - each zoom narrows the interval by a factor of ten. **Tiny example.** $\sqrt{5}$: right triangle with legs 2 and 1, hypotenuse $\sqrt{5}$, arc it down to the line between 2.2 and 2.3 (since $2.23^2 \approx 4.98$, $2.24^2 \approx 5.02$).

Figure 3.2: Building $\sqrt{2}$, $\sqrt{3}$, $\sqrt{4}$, $\sqrt{5}$ on the number line with right triangles and arcs.

3.1.5 Irrationality proofs (the contradiction method)

What is it? The classic proof that $\sqrt{2}$ is irrational, done by assuming the opposite and breaking it. **Skeleton:** Step 1 - suppose $\sqrt{2} = p/q$ in lowest terms (p, q coprime, $q \neq 0$). Step 2 - square: $p^2 = 2q^2$, so 2 divides p^2, hence 2 divides p (because if a prime divides a square, it divides the number itself). Step 3 - write p = 2r; then $4r^2 = 2q^2$, so $q^2 = 2r^2$, hence 2 divides q too. Step 4 - both p and q even contradicts lowest terms. Conclusion: the assumption was false, so $\sqrt{2}$ is irrational. The identical skeleton works for $\sqrt{3}$ (replace 2 by 3) and $\sqrt{5}$.

Extension questions exams adore: (i) $\sqrt{2} + \sqrt{3}$ is irrational - if it were rational, its square $(5 + 2\sqrt{6})$ would be rational, forcing $\sqrt{6}$ rational, contradiction; (ii) rational + irrational = irrational - if $x + \sqrt{2} = r$ were rational then $\sqrt{2} = r - x$ would be rational, contradiction; (iii) nonzero rational × irrational = irrational - same subtraction trick; (iv) the sum of two irrationals may be rational (a + (-a) = 0, or $(1 + \sqrt{2}) + (1 - \sqrt{2}) = 2$) and

the product may be rational ($\sqrt{2} \times \sqrt{2} = 2$) - so NO general rule, must test case by case.

> **Why 'a prime dividing a square divides the number'?**
> If $p \mid n^2$, write $n = p \cdot k + r$ with $0 \le r < p$. Squaring, $n^2 \equiv r^2 \pmod{p}$. For p prime, $n^2 \equiv 0 \pmod{p}$ forces $r^2 \equiv 0 \pmod{p}$, and since p is prime r must be 0. Grade 9 books justify it with prime factorisations: $p \mid n^2$ means p appears in the factorisation of n^2, and every prime in n^2 comes from a prime in n - in pairs. One 'stray' p is impossible.

3.1.6 Surds: simplifying, adding, comparing

What is it? A surd is an irrational root kept in root form, like $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$. **Key laws:** $\sqrt{(ab)} = \sqrt{a} \cdot \sqrt{b}$; $\sqrt{(a/b)} = \sqrt{a} / \sqrt{b}$; $\sqrt{(a^2 b)} = a\sqrt{b}$. **Simplify by extracting perfect-square factors:** $\sqrt{50} = \sqrt{(25 \cdot 2)} = 5\sqrt{2}$; $\sqrt{72} = \sqrt{(36 \cdot 2)} = 6\sqrt{2}$; $\sqrt{245} = \sqrt{(49 \cdot 5)} = 7\sqrt{5}$. **Like surds** (same radicand) add like terms: $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$; unlike surds ($3\sqrt{2} + 2\sqrt{3}$) canNOT merge - they just sit there. **Comparing surds:** square both sides (both positive, so order is preserved): which is bigger, $2\sqrt{5}$ or $3\sqrt{2}$? Squares give 20 vs 18, so $2\sqrt{5}$ wins. **Tiny example.** Order $\sqrt{7}$, $2\sqrt{2}$, $\sqrt{5}$: squares are 7, 8, 5, so $2\sqrt{2} > \sqrt{7} > \sqrt{5}$.

3.1.7 Rationalising the denominator

What is it? Rewriting a fraction so no surd remains in the denominator, using the difference-of-squares identity $(a + b)(a - b) = a^2 - b^2$. **Single surd:** multiply top and bottom by that surd: $1/\sqrt{7} = \sqrt{7}/7$. **Binomial denominator ($\sqrt{a} + \sqrt{b}$ or $a + \sqrt{b}$):** multiply by the *conjugate* (flip the middle sign): $1/(2 + \sqrt{3}) = (2 - \sqrt{3})/((2 + \sqrt{3})(2 - \sqrt{3})) = (2 - \sqrt{3})/(4 - 3) = 2 - \sqrt{3}$. The denominator collapses to a clean integer because $(a+b)(a-b) = a^2 - b^2$. **Triple-decker:** for $1/(\sqrt{2} + \sqrt{3} + \sqrt{5})$, group as $(\sqrt{2} + \sqrt{3}) + \sqrt{5}$, rationalise once, then regroup and rationalise again - two rounds of conjugates.

Why bother? Three reasons: standardized answers (everyone writes $2 - \sqrt{3}$, not $1/(2+\sqrt{3})$), easier further computation, and more accurate decimal work (dividing by 1.73... is numerically nastier than multiplying by 0.26...). **Tiny example.** Rationalise $6/(\sqrt{5} - \sqrt{3})$: multiply by $(\sqrt{5} + \sqrt{3})$: $= 6(\sqrt{5} + \sqrt{3})/(5 - 3) = 3(\sqrt{5} + \sqrt{3})$.

Figure 3.3: Density - between any two real numbers, rationals and irrationals are both endless; no 'next' number exists.

3.1.8 Laws of exponents

What is it? The five power rules for a real base $a > 0$ and integer exponents: (1) $a^m \cdot a^n = a^{m+n}$; (2) $a^m / a^n = a^{m-n}$; (3) $(a^m)^n = a^{mn}$; (4) $a^m \cdot b^m = (ab)^m$; (5) $a^0 = 1$, and $a^{-m} = 1/a^m$. Roots join the family: $a^{1/p} = \sqrt[p]{a}$, so $8^{2/3} = (8^{1/3})^2 = 2^2 = 4$. **When do I use them?** Simplifying expressions like $(3^5 \cdot 3^{-2})$, solving $2^x = 128$, or taming products like $(5^{1/2} \cdot 5^{1/3})$. **Tiny example.** $(2^{-3} \times 2^5) = 2^2 = 4$; $(3^0 + 4^0) \times 2^2 = 2 \times 4 = 8$; $64^{-1/3} = 1/4$.

> **Exponent equation strategy**
> See $2^x = 128$? Write both sides with the SAME base: $2^x = 2^7$, so $x = 7$. See $3^x \times 3^{x+1} = 243$? Left side is 3^{2x+1}, so $2x + 1 = 5$, $x = 2$. See $4^x = 8$? Convert: $(2^2)^x = 2^3$, so $2x = 3$, $x = 3/2$. One base to rule them all.

3.1.9 Exam tips for this chapter

1. Classify numbers by writing their DECIMAL EXPANSION in your head first - the expansion type decides the set.

2. For the $2^l \cdot 5^n$ test, reduce the fraction FIRST; the reduced denominator is what counts.
3. Recurring decimal questions: name the variable x, multiply by the correct power of 10, subtract. Show the subtraction line - it carries the marks.
4. Irrationality proofs: memorise the 4-step contradiction skeleton once, reuse it for √2, √3, √5 by changing the prime.
5. Simplify every surd to k√(radicand) form BEFORE adding or comparing.
6. Rationalising: the conjugate is your only tool; write the identity $(a+b)(a-b) = a^2 - b^2$ explicitly in the solution.
7. Exponent problems: force ONE base everywhere; then equate the exponents.
8. Never claim sums/products of irrationals are always irrational - give a counterexample if a true/false question asks.

3.2 Question Types / Concept Map for Chapter 3

Label the question type before you start - in this chapter, recognising 'this is a conjugate problem' or 'this is a same-base problem' IS the solution.

Type	Name	How to recognise it in the question	Core method
T1	Set classification	'Classify into N/Z/Q/I/R', 'true or false about number sets'	Decimal expansion type; nesting $N \subset Z \subset Q \subset R$
T2	Terminating test	'Will the decimal terminate, without division?'	Lowest terms; denominator = $2^m \times 5^n$ only
T3	Recurring → p/q	'Express 0.4777... in p/q form'	x, multiply by 10^k, subtract, solve; reduce
T4	Number line placement	'Represent √5 on the number line', successive magnification	Right-triangle construction; zoom by decimals
T5	Irrationality proofs	'Prove √3 is irrational'	Contradiction: assume p/q lowest terms, prime divides both
T6	Sum/product arguments	'Show √2 + √3 irrational'; true/false on operations	Square it / subtract-the-rational trick; counterexamples
T7	Surd simplification	'Simplify √72 + √32 - √18'	Extract perfect squares; collect like surds
T8	Surds comparison	'Which is larger: 2√5 or 3√2?'	Square both (both positive); compare radicand multiples
T9	Rationalising	'Rationalise 1/(2+√3)', triple surds	Multiply by conjugate; difference of squares; repeat if needed
T10	Exponent laws	'Simplify $(3^5 \times 3^{-2})/3^2$'	Five power rules; combine bases
T11	Exponent equations	'Solve $2^x = 128$', '$5^{x-2} = 125$'	Same base both sides; equate exponents

Type	Name	How to recognise it in the question	Core method
T12	Between-ness and density	'Find two irrationals between 2 and 3'	√ of non-squares in range; a + small irrational tails
T13	Applied / construction	Spiral of roots, geometric representations, real-life decimals	Pythagoras chaining; interpret expansions
T14	Multi-concept HOTS	Nested radicals, x + 1/√x chains, proof + computation mix	Break into labelled mini-steps, verify each junction

Distribution of the 50 practice questions: T1-T4 feed Q1-Q10, T5-T8 feed Q11-Q22, T9-T12 feed Q23-Q40, and T13-T14 dominate Q41-Q50.

3.3 Practice Questions (Q1 - Q50) with Full Step-by-Step Solutions

Difficulty climbs in five bands: Q1-Q10 strong basics, Q11-Q20 moderate, Q21-Q30 difficult, Q31-Q40 high-level, Q41-Q50 very challenging (Olympiad-lite). Every question carries the concept tag so you can trace it back to the cheat sheet.

Q1 · Basic · Concept: Set classification (T1)

Q1. Classify each number into as many of the sets N, Z, Q, I, R as it belongs to: (i) -7, (ii) 0, (iii) 2/3, (iv) √16, (v) √15, (vi) 0.585858... (the pair 58 repeating).

SOLUTION

Step 1: Given: six numbers. Needed: set membership for each, using the nesting $N \subset Z \subset Q \subset R$ and $I \cap Q = \emptyset$.

Step 2: -7: negative integer, so Z and Q and R; not N (naturals start at 1); not I.

Step 3: 0: an integer, so Z, Q, R; not N in this book's convention (some books include 0 in N - state your convention).

Step 4: 2/3: a ratio of integers, so Q and R; not N, not Z, not I.

Step 5: √16 = 4: a perfect square under the root, so N, Z, Q, R. Perfect squares are never irrational.

Step 6: √15: 15 is not a perfect square, so √15 is irrational: I and R only.

Step 7: 0.585858... is a recurring decimal, and every recurring decimal is a rational: it equals 58/99. So Q and R.

» Why this step? The decimal expansion is the fingerprint: terminating or recurring means rational; endless without a block means irrational.

Hinglish: *Yaad rakho - decimal expansion hi asli ID card hai. Recurring decimal (58-58-58 chalne wala) hamesha rational hota hai, yahan 58/99. √16 jaisi cheez ko dekh ke confuse mat hona - andar perfect square hai toh answer 4 hai, natural number. √15 tabhi irrational hai jab andar non-square ho.*

Answer: (i) Z, Q, R (ii) Z, Q, R (0 ∉ N here) (iii) Q, R (iv) N, Z, Q, R (v) I, R (vi) Q, R (= 58/99)

Q2 · Basic · Concept: Terminating test (T2)

Q2. Without performing long division, decide whether each decimal expansion terminates: (i) 17/8, (ii) 13/3125, (iii) 7/12, (iv) 64/455, (v) 6/15.

SOLUTION

Step 1: Given: five fractions. Needed: terminating or not, via the $2^m \times 5^n$ test on the REDUCED denominator.

Step 2: (i) 17/8: $8 = 2^3 = 2^3 \times 5^0$. Only 2s, so it terminates (in fact 2.125).

Step 3: (ii) 13/3125: $3125 = 5^5$. Only 5s, so it terminates.

Step 4: (iii) 7/12: $12 = 2^2 \times 3$. The stray 3 can never be absorbed into a power of 10, so it recurs.

Step 5: (iv) 64/455: $455 = 5 \times 7 \times 13$. The primes 7 and 13 force a recurring decimal.

Step 6: (v) 6/15: reduce first! 6/15 = 2/5, and $5 = 5^1$, so it terminates (0.4). Testing $15 = 3 \times 5$ without reducing would give the wrong verdict.

» **Why this step?** $10 = 2 \times 5$, so a fraction terminates exactly when its reduced denominator divides some power of 10 - only primes 2 and 5 can appear.

Hinglish: *Sabse pehla kaam: fraction ko lowest terms mein lao. Phir denominator ko prime factors mein todo. Sirf 2 aur 5 dikh rahe hain? Decimal ruk jayega. Koi aur prime (3, 7, 13...) dikh gaya? Chalti-firti decimal aayegi. (v) wala question isliye diya hai kyunki students 15 dekh ke turant 'recur' bol dete hain - pehle 3 cancel karo!*

Answer: (i) terminates (ii) terminates (iii) recurs (iv) recurs (v) terminates (6/15 = 2/5 = 0.4)

Q3 · Basic · Concept: Pure recurring to p/q (T3)

Q3. Express in p/q form: (i) 0.777... (the 7 repeating), (ii) 0.373737... (the block 37 repeating), (iii) 0.090909... (the block 09 repeating).

SOLUTION

Step 1: Given: three pure recurring decimals (repeat starts immediately). Needed: exact fractions.

Step 2: (i) Let x = 0.777... . Multiply by 10 (one repeating digit → shift by one place): 10x = 7.777... .

Step 3: Subtract: 10x - x = 7.777... - 0.777... = 7. So 9x = 7 and x = 7/9.

Step 4: (ii) Let x = 0.373737... . Two repeating digits → multiply by 100: 100x = 37.3737... .

Step 5: Subtract: 100x - x = 37. So 99x = 37 and x = 37/99.

Step 6: (iii) Let x = 0.090909... . Two repeating digits → 100x = 9.0909... . Subtract: 99x = 9, x = 9/99 = 1/11.

» **Why this step?** Multiplying by 10^e lines up two identical infinite tails; subtracting cancels them and leaves a finite equation. k repeating digits need a shift of 10^e.

Hinglish: *Rule simple hai: jitne digit repeat ho rahe hain, utne 9 denominator mein. Ek digit repeat = 9, do digit = 99. 0.777... = 7/9, 0.3737... = 37/99. Bas repeating block ke digits upar likh do aur neeche utne hi 9. (iii) mein 9/99 ko 1/11 tak reduce karna mat bhoolna.*

Answer: (i) 7/9 (ii) 37/99 (iii) 1/11

Q4 · Basic · Concept: Mixed recurring to p/q (T3)

Q4. Express in p/q form: (i) 0.4777... (the 4 does not repeat, the 7 repeats), (ii) 0.2333... (the 2 does not repeat, the 3 repeats), (iii) 1.272727... (the block 27 repeats after the integer part).

SOLUTION

Step 1: Given: mixed recurring decimals. Needed: exact fractions using the shift-and-subtract method.

Step 2: (i) x = 0.4777... . One non-repeating digit then one repeating digit: form 10x = 4.777... and 100x = 47.777... .

Step 3: Subtract: 100x - 10x = 47.777... - 4.777... = 43. So 90x = 43, x = 43/90 (already lowest terms).

Step 4: (ii) x = 0.2333... . 10x = 2.333..., 100x = 23.333... . Subtract: 90x = 21, x = 21/90 = 7/30.

Step 5: (iii) x = 1.272727... . Two repeating digits: 100x = 127.2727... . Subtract x: 99x = 126, x = 126/99 = 14/11.

Step 6: Check (iii): 14/11 = 1.272727... ✓. Check (i): 43 ÷ 90 = 0.4777... ✓.

» **Why this step?** Two shifts are needed so that BOTH copies of the decimal have the same non-repeating head; then the subtraction kills head and tail together.

Hinglish: *Mixed recurring mein trick: pehle non-repeating digits ko 'head' samjho. Do copies banao jisme head same position par ho (10x aur 100x yahan), minus karo - head bhi tail bhi cancel, jo bacha wo equation. Formula version: (poore digits - head)/(utne 9 jitne repeating, utne 0 jitne non-repeating). (i): (47-4)/90 = 43/90. (ii) mein 21/90 ko 7/30 karna zaroori hai.*

Answer: (i) 43/90 (ii) 7/30 (iii) 14/11

Q5 · Basic · Concept: True/False on number sets (T1)

Q5. State true or false with a one-line reason: (i) Every natural number is a whole number. (ii) Every integer is a whole number. (iii) Every rational number is a real number. (iv) Every real number is rational. (v) √3 is a rational number. (vi) There is no rational number between 1/5 and 1/4.

SOLUTION

Step 1: Given: six statements. Needed: verdicts grounded in the set nesting.

Step 2: (i) True: N = 1, 2, 3, ... sits inside whole numbers W = 0, 1, 2, ...

Step 3: (ii) False: -3 is an integer but not a whole number (whole numbers are the non-negative integers).

Step 4: (iii) True: Q ⊂ R by the nesting of number sets.

Step 5: (iv) False: √2 is real but irrational.

Step 6: (v) False: 3 is not a perfect square; the contradiction proof applies, so √3 is irrational.

Step 7: (vi) False: rationals are dense - (1/5 + 1/4)/2 = 9/40 lies between them, and the game continues forever.

» **Why this step?** Each verdict comes from a definition or a counterexample - one counterexample is a complete proof of falsity.

Hinglish: *True/False mein sirf 'True' ya 'False' likhne se marks nahi milte - reason chahiye. False ke liye ek counterexample kaafi hai: -3 se (ii) turant false. (vi) wala bahut common galti hai - rationals ke beech hamesha aur rationals milte hain, average nikaal do, ho gaya.*

Answer: (i) T (ii) F (-3) (iii) T (iv) F (√2) (v) F (3 is not a perfect square) (vi) F (9/40 lies between)

Q6 · Basic · Concept: Number line construction (T4)

Q6. Describe, step by step, how to locate √5 on the number line using a compass and straight edge.

SOLUTION

Step 1: Given: the number line with origin O. Needed: a segment of length √5 laid along the line.

Step 2: Find a right triangle whose hypotenuse is √5: since 5 = 4 + 1 = $2^2 + 1^2$, legs of 2 and 1 give hypotenuse √5.

Step 3: Mark A at 2 on the number line. At A, erect a perpendicular to the line (standard construction).

Step 4: On the perpendicular, mark B with AB = 1 unit.

Step 5: Join O to B. By the Baudhayana-Pythagoras theorem, OB = $\sqrt{(2^2 + 1^2)} = \sqrt{5}$.

Step 6: With centre O and radius OB, draw an arc cutting the number line at P. Then OP = √5, so P represents √5 (it falls between 2.2 and 2.3, since $2.23^2 \approx 4.97$ and $2.24^2 \approx 5.02$).

» **Why this step?** The arc transfers the slanted hypotenuse length onto the straight line while preserving it - swinging a circle keeps radius constant.

Hinglish: *Construction ka funda: √5 ko $2^2 + 1^2$ ke roop mein dekho. 2 unit aage jao (A), wahan se 1 unit upar (B), OB join karo - ye √5 hai. Ab compass ko O par kholo, OB radius lo, arc maaro line par. Jo point mile wahi √5. Har √n ke liye aise hi legs dhoondo: √10 = 3^2+1^2, √13 = 3^2+2^2.*

Answer: A right triangle with legs 2 (along the line) and 1 (perpendicular) has hypotenuse √5; an arc of radius √5 centred at O marks the point.

Q7 · Basic · Concept: Successive magnification (T4)

Q7. Given that √3 = 1.7320508..., describe the location of √3 under two rounds of successive magnification: first between which two integers, then between which two consecutive tenths, then between which two consecutive hundredths.

SOLUTION

Step 1: Given: the decimal expansion of √3. Needed: a nested sequence of intervals.

Step 2: Round 1 (integers): 1 < 1.732... < 2, so √3 lies between 1 and 2.

Step 3: Round 2 (tenths): 1.7 < 1.732... < 1.8, so it lies between 1.7 and 1.8.

Step 4: Round 3 (hundredths): 1.73 < 1.732... < 1.74, so it lies between 1.73 and 1.74.

Step 5: Each zoom peels one decimal digit: the intervals shrink by a factor of ten every round.

» **Why this step?** Each decimal digit fixes the interval at the next scale - this is exactly what 'locating a number on the line' means numerically.

Hinglish: *Zoom karne ka khel hai: 1 aur 2 ke beech → 1.7 aur 1.8 ke beech → 1.73 aur 1.74 ke beech. Har digit ek aur darwaza kholta hai. Exam mein ye 3-mark favourite hai - bas expansion padho aur intervals banao. √3 ke liye aage aur zoom: 1.732 se 1.733 (thousandths).*

Answer: Between 1 and 2; between 1.7 and 1.8; between 1.73 and 1.74.

Q8 · Basic · Concept: Surd simplification (T7)

Q8. Simplify into k√( ) form: (i) √72, (ii) √245, (iii) √175, (iv) √(147).

SOLUTION

Step 1: Given: four radicals. Needed: extract every perfect-square factor.

Step 2: (i) 72 = 36 × 2, so √72 = √36 · √2 = 6√2.

Step 3: (ii) 245 = 49 × 5, so √245 = 7√5.

Step 4: (iii) 175 = 25 × 7, so √175 = 5√7.

Step 5: (iv) 147 = 49 × 3, so √147 = 7√3.

» **Why this step?** √(ab) = √a√b lets the perfect-square factor walk out of the root; what remains inside must be square-free.

Hinglish: *Andar sabse bada perfect square dhoondo aur bahar nikalo. 72 = 36×2, 36 ka root 6, toh 6√2. Yaad rakho: 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144 - ye aapke dost hain. Jitna bada square nikaloge, utna chhota root; 72 ko 4×18 karke √18 ko phir simplify karna double kaam hai.*

Answer: (i) 6√2 (ii) 7√5 (iii) 5√7 (iv) 7√3

Q9 · Basic · Concept: Density / between-ness (T12)

Q9. Insert three irrational numbers between 2 and 3, and verify each choice.

SOLUTION

Step 1: Given: the interval (2, 3). Needed: three irrationals inside it.

Step 2: Idea: the square roots of the non-perfect squares between 4 and 9 all land in (2, 3).

Step 3: Candidates: $\sqrt{5}$, $\sqrt{6}$, $\sqrt{7}$, $\sqrt{8}$ ($4 < 5, 6, 7, 8 < 9$).

Step 4: Verify: $4 < 5 < 9 \Rightarrow 2 < \sqrt{5} < 3$ ✓; similarly for 6, 7, 8 ✓.

Step 5: Each of $\sqrt{5}$, $\sqrt{6}$, $\sqrt{7}$ is irrational because 5, 6, 7 are not perfect squares.

» **Why this step?** Squaring is order-preserving for positive numbers, so 'inside (4, 9)' translates exactly to 'root inside (2, 3)'.

Hinglish: *Do numbers ke beech irrationals chahiye? Dono ko square karo, beech ke non-squares pakdo, root laga do. 2 aur 3 ke beech: 5, 6, 7, 8 - in sabke roots beech mein hi rahenge. √5, √6, √7 - done! Yehi trick decimals ke saath bhi chalti hai: 2.101001000... type numbers bhi kaam karte hain.*

Answer: For example $\sqrt{5}$, $\sqrt{6}$, $\sqrt{7}$ (many answers possible; each is irrational and satisfies $2 <$ root < 3).

Q10 · Basic · Concept: Exponent laws (T10)

Q10. Simplify: (i) $3^5 \times 3^{-2}$, (ii) $(2^2)^3$, (iii) $7^0 + 5^0$, (iv) $2^{-3} \times 2^4$, (v) $(3^{-2} \times 3^5)/3^2$.

SOLUTION

Step 1: Given: five exponent expressions. Needed: apply the five power laws.

Step 2: (i) Same base multiply → add exponents: $3^5 \times 3^{-2} = 3^3 = 27$.

Step 3: (ii) Power of a power → multiply exponents: $(2^2)^3 = 2^6 = 64$.

Step 4: (iii) $a^0 = 1$ for $a \neq 0$: $7^0 + 5^0 = 1 + 1 = 2$.

Step 5: (iv) $2^{-3} \times 2^4 = 2^1 = 2$.

Step 6: (v) $3^{-2} \times 3^5 = 3^3$, then $3^3/3^2 = 3^1 = 3$.

» **Why this step?** All five laws are consequences of repeated multiplication; same base is the trigger to look for.

Hinglish: *Exponent laws ka mantra: multiply matlab exponents ADD, power of power matlab MULTIPLY, zero exponent matlab 1, negative exponent matlab ulta (reciprocal). Base same hona chahiye tabhi laws lagti hain - $3^5 \times 2^5$ mein base alag hai, wahan $(3\times2)^5$ wala rule lagta hai.*

Answer: (i) 27 (ii) 64 (iii) 2 (iv) 2 (v) 3

Q11 · Moderate · Concept: Irrationality proof (T5)

Q11. Prove that $\sqrt{3}$ is irrational.

SOLUTION

Step 1: Given: a claim to contradict. Assume $\sqrt{3} = p/q$ where p, q are integers, $q \neq 0$, and the fraction is in lowest terms (p and q share no common prime).

Step 2: Squaring both sides: $3 = p^2/q^2$, so $p^2 = 3q^2$.

Step 3: Then 3 divides p^2. Since 3 is prime, 3 divides p itself (a prime dividing a square must divide the number). Write $p = 3r$.

Step 4: Substitute: $(3r)^2 = 3q^2 \Rightarrow 9r^2 = 3q^2 \Rightarrow q^2 = 3r^2$.

Step 5: So 3 divides q^2, hence 3 divides q.

Step 6: Now 3 divides both p and q - contradicting lowest terms. The assumption is false, therefore $\sqrt{3}$ is irrational. ∎

» **Why this step?** The contradiction must land on the EXACT assumption made (coprimality); that is why the final sentence names lowest terms.

Hinglish: *Proof ka 4-step skeleton yaad karo: (1) assume p/q lowest terms, (2) square karo - prime upar aa jata hai, (3) prime dono (p aur q) ko divide karta nikla, (4) lowest terms se contradiction - done! √2 mein prime 2 tha, yahan 3 hai, √5 mein 5 hoga. Bas prime badlo, skeleton same.*

Answer: √3 is irrational (by contradiction; 3 divides both p and q, contradicting lowest terms).

Q12 · Moderate · Concept: Rational ± irrational (T6)

Q12. Prove that the sum of a rational number r and an irrational number x is always irrational.

SOLUTION

Step 1: Given: r rational, x irrational. Suppose, for contradiction, that s = r + x is rational.

Step 2: Then x = s - r (subtract r from both sides).

Step 3: s is rational and r is rational; rationals are closed under subtraction, so s - r is rational.

Step 4: So x would be rational - contradicting the given that x is irrational.

Step 5: Hence s cannot be rational: r + x is irrational. ∎ The same subtraction trick works for r - x.

» **Why this step?** Closure of Q under subtraction is the engine: the proof shows that 'rational + irrational = rational' would force an impossible conclusion.

Hinglish: *Ye 'subtraction trick' proof bahut powerful hai: maan lo rational nikla, toh irrational wala number rational minus rational ban jayega - jo impossible hai. Ek line mein: Q subtraction ke under band hai, isliye irrational bacha reh jayega. Exam mein 3 marks pakke is proof ke.*

Answer: r + x is always irrational (if it were rational, x = (r + x) - r would be rational, a contradiction).

Q13 · Moderate · Concept: Rationalising a + b√c denominator (T9)

Q13. Rationalise the denominator and simplify: $1/(3 + 2\sqrt{2})$.

SOLUTION

Step 1: Given: denominator $3 + 2\sqrt{2}$. Its conjugate is $3 - 2\sqrt{2}$ (flip the sign of the surd term).

Step 2: Multiply numerator and denominator by the conjugate:

Step 3: $= (3 - 2\sqrt{2})/((3 + 2\sqrt{2})(3 - 2\sqrt{2}))$.

Step 4: Denominator: $3^2 - (2\sqrt{2})^2 = 9 - 8 = 1$ (difference of squares).

Step 5: So the expression equals $3 - 2\sqrt{2}$. Check numerically: $1/(3 + 2.828...) = 1/5.828... \approx 0.1716$, and $3 - 2.828... = 0.1716$ ✓.

» **Why this step?** The conjugate turns the denominator into $a^2 - b^2$, which has no surds; here it even collapses to 1, giving a free simplification.

Hinglish: *Conjugate ka matlab: sirf beech ka sign ulta. $3 + 2\sqrt{2}$ ka conjugate $3 - 2\sqrt{2}$. Multiply karo, denominator $(a+b)(a-b) = a^2-b^2$ ban jata hai. Yahan 9-8=1 nikla - jackpot! Answer $3-2\sqrt{2}$. Aise lucky cases (denominator 1) exam mein bahut aate hain, isliye denominator ko poori tarah simplify karna kabhi mat chhodna.*

Answer: $3 - 2\sqrt{2}$

Q14 · Moderate · Concept: Rationalising both-storey fraction (T9)

Q14. Simplify completely: $(\sqrt{5} + \sqrt{3})/(\sqrt{5} - \sqrt{3})$.

SOLUTION

Step 1: Given: a fraction with a binomial surd denominator. Multiply top and bottom by the conjugate ($\sqrt{5} + \sqrt{3}$).

Step 2: Denominator: $(\sqrt{5} - \sqrt{3})(\sqrt{5} + \sqrt{3}) = 5 - 3 = 2$.

Step 3: Numerator: $(\sqrt{5} + \sqrt{3})^2 = 5 + 2\sqrt{15} + 3 = 8 + 2\sqrt{15}$.

Step 4: So the expression = $(8 + 2\sqrt{15})/2 = 4 + \sqrt{15}$.

Step 5: Check numerically: $(2.236 + 1.732)/(2.236 - 1.732) = 3.968/0.504 \approx 7.87$, and $4 + 3.873 = 7.87$ ✓.

» **Why this step?** Multiplying by the conjugate of the DENOMINATOR (not the numerator) is what clears the root downstairs; the numerator square uses $(a+b)^2$.

Hinglish: *Dono taraf conjugate multiply karo, denominator 5-3=2 ban jayega. Numerator par $(a+b)^2 = a^2+2ab+b^2$ lagta hai: $5 + 2\sqrt{15} + 3 = 8 + 2\sqrt{15}$. Ab 2 se kat do: $4 + \sqrt{15}$. Students yahan aadha kaam chhod dete hain - 2 se divide karna final step hai, uske bina marks katate hain.*

Answer: $4 + \sqrt{15}$

Q15 · Moderate · Concept: Comparing surds (T8)

Q15. Arrange in increasing order: $\sqrt{7}$, $2\sqrt{2}$, $\sqrt{5}$, $3\sqrt{3}/2$.

SOLUTION

Step 1: Given: four positive surds. All are positive, so squaring preserves the order.

Step 2: $\sqrt{7} \to 7$. $2\sqrt{2} \to (2\sqrt{2})^2 = 4 \times 2 = 8$. $\sqrt{5} \to 5$. $(3\sqrt{3})/2 \to (9 \times 3)/4 = 27/4 = 6.75$.

Step 3: Order of the squares: $5 < 6.75 < 7 < 8$.

Step 4: So the original order: $\sqrt{5} < 3\sqrt{3}/2 < \sqrt{7} < 2\sqrt{2}$.

Step 5: Check: $\sqrt{5} \approx 2.236$, $3\sqrt{3}/2 \approx 2.598$, $\sqrt{7} \approx 2.646$, $2\sqrt{2} \approx 2.828$ ✓.

» **Why this step?** For positive quantities $a < b \Leftrightarrow a^2 < b^2$ - comparing squares avoids fiddly decimal roots.

Hinglish: *Compare karne ka best trick: dono ko square karo. $2\sqrt{2}$ ka square $(2)^2 \times 2 = 8$ hota hai - coefficient bhi square hota hai, ye bhoolna sabse common galti hai! $(3\sqrt{3}/2)^2 = 9\times3/4 = 6.75$. Ab squares ko order karo aur original order likho. Numerical check se confirm kar lo.*

Answer: $\sqrt{5} < 3\sqrt{3}/2 < \sqrt{7} < 2\sqrt{2}$

Q16 · Moderate · Concept: Same-base equations (T11)

Q16. Solve: (i) $3^x = 1/81$, (ii) $5^{(x-2)} = 125$, (iii) $(2/3)^x = 27/8$.

SOLUTION

Step 1: Given: three exponential equations. Strategy: one base on both sides, then equate exponents.

Step 2: (i) $1/81 = 1/3^4 = 3^{-4}$. So $3^x = 3^{-4} \Rightarrow x = -4$.

Step 3: (ii) $125 = 5^3$. So $5^{(x-2)} = 5^3 \Rightarrow x - 2 = 3 \Rightarrow x = 5$.

Step 4: (iii) $27/8 = 3^3/2^3 = (3/2)^3 = (2/3)^{-3}$. So $x = -3$.

Step 5: Check: $(2/3)^{-3} = (3/2)^3 = 27/8$ ✓.

» **Why this step?** Equality of powers with the same base forces equal exponents; the whole game is rewriting both sides into one base.

Hinglish: *Exponent equation mein dono taraf SAME base lao. $1/81$ ko 3^{-4} dekho - negative exponent ka perfect use. $(2/3)^x = 27/8$ mein ulta fraction banana padta hai: $27/8 = (3/2)^3 = (2/3)^{-3}$. Fraction ka negative exponent = fraction flip. Ye teen patterns exam ke favourites hain.*

Answer: (i) $x = -4$ (ii) $x = 5$ (iii) $x = -3$

Q17 · Moderate · Concept: Fractional and negative exponents (T10)

Q17. Evaluate without a calculator: (i) $(81/16)^{-34y}$... rewrite: (81/16)^(-3/4), (ii) 8^(2/3) + 4^(-1/2), (iii) $(5^2/\sqrt{5})^2$.

SOLUTION

Step 1: Given: expressions with fractional and negative exponents.

Step 2: (i) (81/16)^(-3/4): the negative exponent flips the fraction; the 1/4 takes a fourth root. (16/81)^(3/4) = $(\sqrt[4]{(16/81)})^3$ = $(2/3)^3$ = 8/27.

Step 3: (ii) 8^(2/3) = $(\sqrt[3]{8})^2$ = 2^2 = 4. 4^(-1/2) = $1/\sqrt{4}$ = 1/2. Sum = 4 + 1/2 = 9/2.

Step 4: (iii) $5^2/\sqrt{5}$ = 5^2/5^(1/2) = 5^(3/2). Squaring: 5^3 = 125.

Step 5: Check (iii) numerically: 25/2.236 ≈ 11.18; 11.18^2 ≈ 125 ✓.

» **Why this step?** a^(m/n) = (root n of a) to the m; the negative sign means reciprocal. Order root-then-power keeps numbers small.

Hinglish: *Fractional exponent do kaam karta hai: neeche wala number ROOT, upar wala POWER. 8^(2/3): cube root of 8 = 2, phir square = 4. Negative exponent: flip karo. Root pehle lo, power baad mein - numbers chhote rehte hain. (i) mein (81/16) ko flip karke 16/81 banana zaroori tha.*

Answer: (i) 8/27 (ii) 9/2 (iii) 125

Q18 · Moderate · Concept: Classification with expansions (T1/T2)

Q18. Classify with one-line reasons: (i) 0.01001000100001... (one more zero each time), (ii) 3.1416, (iii) √(49/64), (iv) 1.010010001... , (v) $6.\overline{45}$... i.e. 6.454545... (block 45 repeating).

SOLUTION

Step 1: Given: five numbers of mixed types. The decimal pattern decides the set.

Step 2: (i) The gaps between 1s keep growing, so no block ever repeats: non-terminating, non-recurring ⇒ I, R.

Step 3: (ii) 3.1416 terminates ⇒ Q, R (note: it only approximates π).

Step 4: (iii) √(49/64) = 7/8 ⇒ Q, R (and it is not natural or integral).

Step 5: (iv) Endless, and the 1s never settle into a fixed block ⇒ I, R.

Step 6: (v) Recurring ⇒ Q, R; in fact 6.4545... = 6 + 45/99 = 6 + 5/11 = 71/11.

» **Why this step?** Terminating/recurring ⇒ Q; endless with no repeating block ⇒ I. Fractional roots simplify before classifying.

Hinglish: *(i) aur (iv) type numbers ke liye pattern dekho - zeroes badhte ja rahe hain ya 1s ke beech gap badh raha hai, toh koi block repeat nahi hoga, irrational. (ii) ko π se compare mat karna - 3.1416 rukne wala decimal hai, rational. (v) recurring hai: 71/11.*

Answer: (i) I, R (ii) Q, R (iii) Q, R (= 7/8) (iv) I, R (v) Q, R (= 71/11)

Q19 · Moderate · Concept: Mixed recurring with integer part (T3)

Q19. Express 2.1363636... (the block 36 repeats) in p/q form.

SOLUTION

Step 1: Given: x = 2.13636... . One non-repeating digit (3), then the block 36.

Step 2: Multiply by 10: 10x = 21.3636... . Multiply by 1000: 1000x = 2136.3636... .

Step 3: Subtract: 1000x - 10x = 2136.3636... - 21.3636... = 2115 (tails cancel exactly).

Step 4: So 990x = 2115, x = 2115/990.

Step 5: Reduce: HCF(2115, 990) = 45, so x = 47/22.

Step 6: Check: 47 ÷ 22 = 2.13636... ✓.

» **Why this step?** Shifts of 10 and 1000 align the two copies of the repeating tail; 990 = 9×100 with one zero for the non-repeating digit.

Hinglish: *Integer part wale recurring mein panic mat karo - method same hai. 10x aur 1000x banao (ek shift head ke liye, do aur block ke liye), minus karo: 2115/990 aayega. HCF 45 hai, kat ke 47/22. Formula version: (2136 - 21)/990 = 2115/990. Reduce karna compulsory hai!*

Answer: 47/22

Q20 · Moderate · Concept: 0.999... = 1 (T3/T13)

Q20. Show that 0.999... (all 9s repeating) is exactly equal to 1, and use the result to explain why 2.000... and 1.999... name the same real number.

SOLUTION

Step 1: Let x = 0.999... . Multiply by 10: 10x = 9.999... .

Step 2: Subtract: 10x - x = 9.999... - 0.999... = 9 exactly (the infinite tails cancel digit by digit).

Step 3: So 9x = 9 and x = 1.

Step 4: Hence 1.999... = 1 + 0.999... = 1 + 1 = 2, while 2.000... = 2: the two expansions name one point on the line.

Step 5: Moral: every terminating decimal also has a second, 9-ending expansion; the shift-and-subtract method does not break - it simply reports the value.

» **Why this step?** The subtraction 9.999... - 0.999... = 9 is exact because digit-by-digit cancellation leaves no remainder at ANY place value.

Hinglish: *Ye wala concept books ke end mein hota hai aur examiner bahut pasand karte hain. 0.999... exactly 1 hai - 'almost 1' nahi. Multiply-subtract se 9x = 9 nikalta hai, point. Matlab har terminating decimal ki do expansions hain: 2.000... aur 1.999... dono same point. Exam mein 'true/false: 0.999... < 1' - answer False, reason ye proof.*

Answer: 0.999... = 1 exactly (9x = 9 from the shift-and-subtract computation); 2.000... = 1.999... as numbers.

Q21 · Difficult · Concept: Irrationality of a surd sum (T6)

Q21. Prove that √2 + √3 is irrational.

SOLUTION

Step 1: Suppose √2 + √3 = r is rational. Square both sides: 2 + 2√6 + 3 = r^2, so 5 + 2√6 = r^2.

Step 2: Solve for the surd: √6 = (r^2 - 5)/2.

Step 3: r is rational, so r^2 - 5 is rational, and (r^2 - 5)/2 is rational.

Step 4: So √6 would be rational. But 6 is not a perfect square, so √6 is irrational (same contradiction proof as √2).

Step 5: Contradiction. Hence √2 + √3 is irrational. ∎

» **Why this step?** Squaring converts a sum of two surds into ONE surd; the known irrationality of √6 then finishes the job.

Hinglish: *Do surds ka sum prove karna hai? Square karo - √2+√3 ka square 5 + 2√6 deta hai. Agar sum rational hota toh √6 = (r^2-5)/2 rational hota - jo impossible hai. Yehi pattern √2+√5, √3+√5 ke liye bhi chalega. Ek baar square, phir known irrational se maar do.*

Answer: $\sqrt{2} + \sqrt{3}$ is irrational (rationality would force $\sqrt{6} = (r^2-5)/2$ to be rational).

Q22 · Difficult · Concept: a - b√c irrational (T6)

Q22. Show that $3 - 2\sqrt{5}$ is irrational.

SOLUTION

Step 1: Suppose $3 - 2\sqrt{5} = r$ is rational. Rearrange: $2\sqrt{5} = 3 - r$.

Step 2: Divide by 2: $\sqrt{5} = (3 - r)/2$.

Step 3: $3 - r$ is rational (Q is closed under subtraction), and dividing by 2 keeps it rational.

Step 4: So $\sqrt{5}$ would be rational. But 5 is not a perfect square, so $\sqrt{5}$ is irrational - contradiction.

Step 5: Hence $3 - 2\sqrt{5}$ is irrational. ∎

» **Why this step?** Isolate the surd on one side; the rest of the expression is built only from rationals, which stay rational under the field operations.

Hinglish: *NCERT ka classic! Surd ko akela karo: $2\sqrt{5} = 3 - r$, phir $\sqrt{5} = (3-r)/2$. Right side pura rational hai, left side irrational - clash. Bas. Aise 'a + b√c' type har number ke liye ye line chalegi: irrational number rational ke saath mix karke rational nahi ban sakta.*

Answer: $3 - 2\sqrt{5}$ is irrational (otherwise $\sqrt{5} = (3-r)/2$ would be rational).

Q23 · Difficult · Concept: Rationalising with clean collapse (T9)

Q23. Simplify completely: $4/(\sqrt{7} - \sqrt{3})$, and express the result in the form $a\sqrt{7} + b\sqrt{3}$ with rational a, b.

SOLUTION

Step 1: Multiply numerator and denominator by the conjugate $\sqrt{7} + \sqrt{3}$.

Step 2: Denominator: $(\sqrt{7} - \sqrt{3})(\sqrt{7} + \sqrt{3}) = 7 - 3 = 4$.

Step 3: Expression = $4(\sqrt{7} + \sqrt{3})/4 = \sqrt{7} + \sqrt{3}$.

Step 4: So $a = 1$ and $b = 1$.

Step 5: Check numerically: $4/(2.646 - 1.732) = 4/0.914 \approx 4.376$, and $\sqrt{7} + \sqrt{3} \approx 2.646 + 1.732 = 4.376$ ✓.

» **Why this step?** Choosing the conjugate makes the denominator $a^2 - b^2$; here it cancels the numerator perfectly, so no fraction survives.

Hinglish: *Denominator ka conjugate multiply karo: $(a-b)(a+b) = a^2-b^2 = 7-3 = 4$. Upar wala 4 aur neeche wala 4 cancel - answer $\sqrt{7} + \sqrt{3}$, ekdum clean! Ye 'designed' questions mein hamesha ek magic cancellation hota hai; agar aisa nahi ho raha toh calculation check karo.*

Answer: $\sqrt{7} + \sqrt{3}$ ($a = 1, b = 1$)

Q24 · Difficult · Concept: Triple-surd denominator (T9)

Q24. Rationalise the denominator completely: $1/(\sqrt{2} + \sqrt{3} + \sqrt{5})$.

SOLUTION

Step 1: Group as $(\sqrt{2} + \sqrt{3}) + \sqrt{5}$. Round 1: multiply by the conjugate $(\sqrt{2} + \sqrt{3}) - \sqrt{5}$.

Step 2: Denominator: $(\sqrt{2} + \sqrt{3})^2 - (\sqrt{5})^2 = (2 + 2\sqrt{6} + 3) - 5 = 2\sqrt{6}$.

Step 3: So the expression = $(\sqrt{2} + \sqrt{3} - \sqrt{5})/(2\sqrt{6})$. One surd remains downstairs.

Step 4: Round 2: multiply by $\sqrt{6}/\sqrt{6}$: = $(\sqrt{2} \cdot \sqrt{6} + \sqrt{3} \cdot \sqrt{6} - \sqrt{5} \cdot \sqrt{6})/(2 \times 6) = (\sqrt{12} + \sqrt{18} - \sqrt{30})/12$.

Step 5: Simplify the surds: $\sqrt{12} = 2\sqrt{3}$, $\sqrt{18} = 3\sqrt{2}$. Final: $(2\sqrt{3} + 3\sqrt{2} - \sqrt{30})/12$.

Step 6: Check numerically: $1/(1.414 + 1.732 + 2.236) = 1/5.382 \approx 0.1858$; $(3.464 + 4.243 - 5.477)/12 = 2.230/12 \approx 0.1858$ ✓.

» **Why this step?** Three surds cannot be cleared in one blow: group two as a single 'number', use its conjugate, then mop up the single remaining surd.

Hinglish: *Teen surd ek saath? Group karo: (√2+√3) ko ek unit maano, conjugate se √5 ko maaro. Denominator (5 + 2√6) - 5 = 2√6 nikla - poora 5 cancel! Phir simple √6 se multiply. Do round ka game hai. Grouping alag karoge toh answer alag dikhega but value same - ye normal hai.*

Answer: (2√3 + 3√2 - √30)/12

Q25 · Difficult · Concept: Conjugate pairs x and 1/x (T14)

Q25. If $x = 2 + \sqrt{3}$, find the value of $x^2 + 1/x^2$ without evaluating x numerically beyond the given form.

SOLUTION

Step 1: First find 1/x by rationalising: $1/(2 + \sqrt{3}) = (2 - \sqrt{3})/((2 + \sqrt{3})(2 - \sqrt{3})) = (2 - \sqrt{3})/(4 - 3) = 2 - \sqrt{3}$.

Step 2: So x and 1/x are conjugates: $x = 2 + \sqrt{3}$, $1/x = 2 - \sqrt{3}$.

Step 3: Square both: $x^2 = 4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$; $(1/x)^2 = 7 - 4\sqrt{3}$.

Step 4: Add: $x^2 + 1/x^2 = 14$ (the surd terms cancel).

Step 5: Check numerically: $x \approx 3.732$, $x^2 \approx 13.93$, $1/x^2 \approx 0.0718$, sum ≈ 14.0 ✓.

» **Why this step?** Conjugate reciprocals have surd terms in opposite signs; squares keep that symmetry, so the middle terms cancel on adding.

Hinglish: *Ye conjugate-pair trick poore algebra mein chalega. Pehle 1/x nikaalo - denominator 1 aaya, matlab x aur 1/x perfect conjugates. Square karke jodo: ±4√3 terms aapas mein uda do. Answer 14. Aage ke liye yaad rakho: $x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x)$.*

Answer: 14

Q26 · Difficult · Concept: Factored exponent equation (T11)

Q26. Solve for x: $4^x - 4^{(x-1)} = 24$.

SOLUTION

Step 1: Factor out the smaller power: $4^{(x-1)}(4^1 - 1) = 24$.

Step 2: So $4^{(x-1)} \times 3 = 24$, giving $4^{(x-1)} = 8$.

Step 3: Convert to one base: $4 = 2^2$ and $8 = 2^3$, so $2^{(2x-2)} = 2^3$.

Step 4: Equate exponents: $2x - 2 = 3$, so $x = 5/2$.

Step 5: Check: $4^{(5/2)} = (\sqrt{4})^5 = 32$; $4^{(3/2)} = (\sqrt{4})^3 = 8$; $32 - 8 = 24$ ✓.

» **Why this step?** Both terms share the base 4, so factoring the smaller power is legal and reduces two powers to one equation.

Hinglish: *Do powers ka difference? Common power bahar nikalo - 4^(x-1) common liya, bracket mein 4-1 = 3 bacha. Ab 24/3 = 8. 4 aur 8 dono ko 2 ke base par lao: $2^{(2x-2)} = 2^3$. Exponents barabar: x = 5/2. Fraction mein aana bilkul theek hai - check line se confirm karo.*

Answer: x = 5/2

Q27 · Difficult · Concept: Cyclic exponent product (T10)

Q27. Simplify: $(x^a/x^b)^{(a+b)} \times (x^b/x^c)^{(b+c)} \times (x^c/x^a)^{(c+a)}$.

SOLUTION

Step 1: Simplify each inner quotient: $x^a/x^b = x^{(a-b)}$, and so on.

Step 2: Expression = $x^{(a-b)(a+b)} \times x^{(b-c)(b+c)} \times x^{(c-a)(c+a)}$.

Step 3: Power of a power: exponents multiply: $x^{(a2-b2)} \times x^{(b2-c2)} \times x^{(c2-a2)}$.

Step 4: Same base multiply → add: $x^{(a2 - b2 + b2 - c2 + c2 - a2)} = x^0$.

Step 5: $x^0 = 1$ ($x \neq 0$).

» **Why this step?** Difference-of-squares cancellation (a^2-b^2 terms annihilate in a cycle) is the hidden design; tracking exponents line by line reveals it.

Hinglish: *Lagta hai bhayanak, par dekho - har factor mein (a+b)(a-b) pattern hai! Exponents multiply hone par a^2-b^2, b^2-c^2, c^2-a^2 bante hain, aur ye teeno judne par aapas mein cancel. Bacha $x^0 = 1$. Aise cyclic questions mein hamesha cancellation chhupa hota hai - exponents ko alag-alag likh ke dekho.*

Answer: 1 (for $x \neq 0$)

Q28 · Difficult · Concept: Find a, b after rationalising (T9)

Q28. If $(\sqrt{3} + 1)/(\sqrt{3} - 1) = a + b\sqrt{3}$ where a, b are rational, find a and b.

SOLUTION

Step 1: Rationalise: multiply by $(\sqrt{3} + 1)/(\sqrt{3} + 1)$.

Step 2: Numerator: $(\sqrt{3} + 1)^2 = 3 + 2\sqrt{3} + 1 = 4 + 2\sqrt{3}$.

Step 3: Denominator: $3 - 1 = 2$.

Step 4: Expression = $(4 + 2\sqrt{3})/2 = 2 + \sqrt{3}$.

Step 5: Compare with $a + b\sqrt{3}$: $a = 2$, $b = 1$.

» **Why this step?** Two expressions of the form $a + b\sqrt{c}$ are equal exactly when the rational parts match and the surd coefficients match (uniqueness of the representation).

Hinglish: *Rationalise karo, simplify karo, phir 'compare coefficients' karo. $2 + \sqrt{3}$ vs $a + b\sqrt{3}$: rational part rational se match (a=2), surd coefficient surd coefficient se (b=1). Ye uniqueness isliye chalti hai kyunki agar $a \neq a'$ hota toh $\sqrt{3}$ rational nikal jata - impossible.*

Answer: $a = 2$, $b = 1$

Q29 · Difficult · Concept: Irrationals in a small interval (T12)

Q29. Insert two irrational numbers between 0.5 and 0.6, and prove your choices are correct.

SOLUTION

Step 1: Square the endpoints: $0.5^2 = 0.25$ and $0.6^2 = 0.36$.

Step 2: Any non-perfect-square strictly between 0.25 and 0.36 has an irrational root strictly between 0.5 and 0.6.

Step 3: Choose 0.27 and 0.32: neither is a perfect square of a rational.

Step 4: So $\sqrt{0.27}$ and $\sqrt{0.32}$ lie in (0.5, 0.6): $0.25 < 0.27 < 0.36 \Rightarrow 0.5 < \sqrt{0.27} < 0.6$, similarly for 0.32.

Step 5: $\sqrt{0.27} = \sqrt{27}/\sqrt{100} = 3\sqrt{3}/10$ and $\sqrt{0.32} = 4\sqrt{2}/10 = 2\sqrt{2}/5$ - both are irrational multiples of irrationals divided by rationals, hence irrational.

» **Why this step?** Squaring maps (0.5, 0.6) onto (0.25, 0.36) monotonically; picking non-squares there guarantees irrational roots here.

Hinglish: *Chhote decimals ke beech irrationals chahiye? Dono ko square karo (0.25 aur 0.36), beech ke non-square numbers pakdo. 0.27 aur 0.32 - inke roots 0.5 aur 0.6 ke beech rahenge. Clean form mein: $\sqrt{0.27} = 3\sqrt{3}/10$, $\sqrt{0.32} = 2\sqrt{2}/5$. Proof mein square-comparison likhna zaroori hai - wahi 2 marks deta hai.*

Answer: For example $\sqrt{0.27} = 3\sqrt{3}/10$ and $\sqrt{0.32} = 2\sqrt{2}/5$ (many answers possible).

Q30 · Difficult · Concept: Square of a surd sum (T8/T9)

Q30. Write $(\sqrt{3} + \sqrt{5})^2$ in the form $m + n\sqrt{k}$, and find between which two consecutive integers it lies.

SOLUTION

Step 1: Expand: $(\sqrt{3} + \sqrt{5})^2 = 3 + 2\sqrt{15} + 5 = 8 + 2\sqrt{15}$.

Step 2: Estimate $\sqrt{15}$: $3.7^2 = 13.69$, $3.8^2 = 14.44$, $3.9^2 = 15.21$, so $15.21 > 15 > 14.44$ gives $3.8 < \sqrt{15} < 3.9$.

Step 3: So $7.6 < 2\sqrt{15} < 7.8$, and $15.6 < 8 + 2\sqrt{15} < 15.8$.

Step 4: The value lies between the consecutive integers 15 and 16.

Step 5: Check: $\sqrt{15} \approx 3.873$, so $8 + 2(3.873) \approx 15.75$ ✓.

» **Why this step?** Perfect-square bracketing (testing 3.7^2, 3.8^2, 3.9^2) pins the root between tenths without a calculator.

Hinglish: *Expand karo: 8 + 2√15. Ab √15 ko bracket karo - $3.8^2 = 14.44$ aur $3.9^2 = 15.21$, matlab $3.8 < \sqrt{15} < 3.9$. Double karo: 7.6 se 7.8. 8 jodo: 15.6 se 15.8. Toh 15 aur 16 ke beech. Bracketing technique akele kaafi hai - calculator ki zaroorat nahi.*

Answer: $8 + 2\sqrt{15}$; it lies between 15 and 16 (approximately 15.75).

Q31 · High · Concept: √5 irrationality proof (T5)

Q31. Prove that $\sqrt{5}$ is irrational.

SOLUTION

Step 1: Assume $\sqrt{5} = p/q$ in lowest terms (p, q coprime integers, $q \neq 0$). Squaring: $p^2 = 5q^2$.

Step 2: So 5 divides p^2; since 5 is prime, 5 divides p. Write $p = 5r$.

Step 3: Substitute: $25r^2 = 5q^2 \Rightarrow q^2 = 5r^2$. So 5 divides q^2, hence 5 divides q.

Step 4: Now 5 divides both p and q, contradicting lowest terms.

Step 5: Therefore the assumption fails and $\sqrt{5}$ is irrational. ∎

Step 6: Extra rigour: if $5 \mid p^2$, then in the prime factorisation of p^2 the prime 5 appears; every prime in p^2 comes from primes of p in PAIRS, so a single unmatched 5 is impossible unless $5 \mid p$.

» **Why this step?** The pairing argument in prime factorisations is what makes 'prime divides square ⇒ prime divides number' true - stating it earns full method marks.

Hinglish: *Same 4-step skeleton, bas prime 5. Extra marks ke liye 'pairing' wali line likho: square ke prime factorisation mein har prime EVEN times aata hai; 5 sirf ek baar aana impossible. Ye line proof ko watertight banati hai.*

Answer: $\sqrt{5}$ is irrational (by contradiction with prime 5).

Q32 · High · Concept: Product of rational and irrational (T6)

Q32. (i) Prove that the product of a NONZERO rational number r and an irrational number x is irrational. (ii) Show by example that the product of two irrationals may be rational.

SOLUTION

Step 1: (i) Suppose $rx = s$ is rational, with $r \neq 0$. Then $x = s/r$.

Step 2: s is rational and r is rational and nonzero, so s/r is rational (Q closed under division by nonzero).

Step 3: So x would be rational - contradiction. Hence rx is irrational. ∎

Step 4: (ii) Counterexamples: $\sqrt{2} \times \sqrt{2} = 2$ (rational); $(1 + \sqrt{2})(1 - \sqrt{2}) = 1 - 2 = -1$ (rational); $\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$.

Step 5: Moral: no blanket rule for irrational × irrational - each case must be tested; the nonzero condition in (i) is essential because r = 0 gives rx = 0, rational.

» **Why this step?** Division closure replaces subtraction closure here; the r ≠ 0 hypothesis is exactly what lets you divide by r.

Hinglish: *(i) mein 'nonzero' word critical hai - r = 0 hote hi product 0 (rational) ban jata hai. Proof wahi hai: maan lo rational, divide karo, contradiction. (ii) ke liye √2 × √2 = 2 perfect example. True/false questions mein ye do cases sabse zyada puchhe jate hain.*

Answer: (i) rx is irrational for r ≠ 0 (else x = s/r rational). (ii) e.g. √2 × √2 = 2.

Q33 · High · Concept: Conjugate pair x, y (T14)

Q33. If $x = (\sqrt{3} + 1)/(\sqrt{3} - 1)$ and $y = (\sqrt{3} - 1)/(\sqrt{3} + 1)$, find $x^2 + y^2$ and xy.

SOLUTION

Step 1: Rationalise x: multiply by $(\sqrt{3} + 1)$: $x = (\sqrt{3} + 1)^2/(3 - 1) = (4 + 2\sqrt{3})/2 = 2 + \sqrt{3}$.

Step 2: Rationalise y: multiply by $(\sqrt{3} - 1)$: $y = (\sqrt{3} - 1)^2/(3 - 1) = (4 - 2\sqrt{3})/2 = 2 - \sqrt{3}$.

Step 3: x and y are conjugates: $xy = (2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$.

Step 4: $x + y = 4$. Then $x^2 + y^2 = (x + y)^2 - 2xy = 16 - 2 = 14$.

Step 5: Check: $x \approx 3.732$, $y \approx 0.268$; $x^2 \approx 13.93$, $y^2 \approx 0.0718$, sum ≈ 14.0 ✓.

» **Why this step?** Reduce both fractions to conjugate form FIRST; then the symmetric identities $x^2 + y^2 = (x+y)^2 - 2xy$ do the rest cheaply.

Hinglish: *Dono fractions rationalise karo - $x = 2+\sqrt{3}$, $y = 2-\sqrt{3}$ niklega. Ab conjugate pair ke superpowers: $xy = 1$ hamesha, $x+y = 4$. x^2+y^2 ke liye direct square karne ki zaroorat nahi - $(x+y)^2 - 2xy = 16 - 2 = 14$. Symmetric identities se time bachta hai aur galti bhi nahi hoti.*

Answer: $x^2 + y^2 = 14$ and $xy = 1$

Q34 · High · Concept: Multi-base exponent relation (T11)

Q34. If $2^x = 3^y = 12^z = t$ ($t \neq 1$), prove that $2/x + 1/y = 1/z$.

SOLUTION

Step 1: Write each base as a power of the common value t: 2 = t^(1/x), 3 = t^(1/y), 12 = t^(1/z).

Step 2: Use the factorisation $12 = 2^2 \times 3$.

Step 3: So t^(1/z) = (t^(1/x))^2 × t^(1/y) = t^(2/x + 1/y).

Step 4: Same base t (t ≠ 1) forces equal exponents: 1/z = 2/x + 1/y. ∎

Step 5: Sanity check with numbers: $x = \log_2 12 \approx 3.585$, $y \approx 2.262$, $z = 1$ gives 2/3.585 + 1/2.262 ≈ 0.558 + 0.442 = 1 = 1/1 ✓.

» **Why this step?** The common value t converts three separate equations into one base; then the prime factorisation of 12 writes the relation.

Hinglish: *Ye 'common value' technique Olympiad ka bread-and-butter hai. Teeno numbers ko t ke power mein likho, phir $12 = 2^2 \times 3$ ka use karo. Exponents add hote- hote relation ban jata hai. Base same aur $t \neq 1$ - tabhi exponent equate karna valid hai, ye condition likhna mat bhoolna.*

Answer: 1/z = 2/x + 1/y (proved via t-powers and $12 = 2^2 \times 3$).

Q35 · High · Concept: Conjugate subtraction chain (T9)

Q35. Evaluate: (√5 - 2)/(√5 + 2) - (√5 + 2)/(√5 - 2).

SOLUTION

Step 1: Rationalise each fraction separately.

Step 2: First: $(\sqrt{5} - 2)^2/((\sqrt{5} + 2)(\sqrt{5} - 2)) = (5 - 4\sqrt{5} + 4)/1 = 9 - 4\sqrt{5}$.

Step 3: Second: $(\sqrt{5} + 2)^2/1 = 5 + 4\sqrt{5} + 4 = 9 + 4\sqrt{5}$.

Step 4: Subtract: $(9 - 4\sqrt{5}) - (9 + 4\sqrt{5}) = -8\sqrt{5}$.

Step 5: Check: √5 ≈ 2.236; first fraction ≈ 0.236/4.236 ≈ 0.0557, second ≈ 4.236/0.236 ≈ 17.94; difference ≈ -17.89, and -8(2.236) = -17.89 ✓.

» **Why this step?** Rationalising first turns both fractions into exact conjugate expressions; then only like terms subtract.

Hinglish: *Dono fractions ko alag-alag rationalise karo - denominators dono baar 1 niklenge (√5+2)(√5-2) = 1. Pehla 9 - 4√5, doosra 9 + 4√5. Subtract: -8√5. Sign ka dhyan: (a-b)² mein middle term MINUS hota hai, yahi galti hoti hai.*

Answer: -8√5

Q36 · High · Concept: Sum of powers equation (T11)

Q36. Solve: $2^{(x+1)} + 2^{(x-1)} = 80$.

SOLUTION

Step 1: Factor out the smaller power $2^{(x-1)}$: $2^{(x-1)}(2^2 + 1) = 80$.

Step 2: So $5 \cdot 2^{(x-1)} = 80$, giving $2^{(x-1)} = 16 = 2^4$.

Step 3: Equate exponents: x - 1 = 4, so x = 5.

Step 4: Check: $2^6 + 2^4 = 64 + 16 = 80$ ✓.

» **Why this step?** Factoring the smaller power converts a sum into a product; the bracket $(2^2 + 1)$ becomes a plain number.

Hinglish: *Q26 ka bhai - same factoring trick. 2^(x-1) bahar nikalo, bracket mein 2²+1 = 5. 80/5 = 16 = 2⁴. x = 5. Dekho pattern: jab do powers ka sum/difference diya ho aur exponents ±1 differ karein, factoring pehla move hai.*

Answer: x = 5

Q37 · High · Concept: Nested radicals (T14)

Q37. Simplify: (i) √(7 + 4√3), (ii) √(9 - 2√14).

SOLUTION

Step 1: Method: try √(a ± 2√b) = √m ± √n where m + n = a and mn = b (because $(\sqrt{m} \pm \sqrt{n})^2 = m + n \pm 2\sqrt{(mn)}$).

Step 2: (i) $7 + 4\sqrt{3} = 7 + 2\sqrt{12}$. Need m + n = 7, mn = 12: m = 4, n = 3. So $\sqrt{(7 + 4\sqrt{3})} = 2 + \sqrt{3}$.

Step 3: Verify: $(2 + \sqrt{3})^2 = 4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$ ✓.

Step 4: (ii) $9 - 2\sqrt{14}$: need m + n = 9, mn = 14: m = 7, n = 2. So $\sqrt{(9 - 2\sqrt{14})} = \sqrt{7} - \sqrt{2}$.

Step 5: Verify: $(\sqrt{7} - \sqrt{2})^2 = 7 - 2\sqrt{14} + 2 = 9 - 2\sqrt{14}$ ✓.

» **Why this step?** The identity $(\sqrt{m} \pm \sqrt{n})^2 = (m+n) \pm 2\sqrt{(mn)}$ runs backwards; m+n and mn are read off directly from the expression.

Hinglish: *Nested radical ka standard recipe: number ko 2√b form mein dekho (4√3 = 2√12). Phir do numbers dhoondo jinka sum 7 aur product 12 - 4 aur 3! Answer 2 + √3. (ii) mein 9 aur 14 - 7 aur 2. Sign dhyan se: minus*

wale case mein bada root pehle. Verify karke answer lock karo.

Answer: (i) 2 + √3 (ii) √7 - √2

Q38 · High · Concept: Telescoping surd sum (T14)

Q38. Evaluate: 1/(√2 + 1) + 1/(√3 + √2) + 1/(2 + √3).

SOLUTION

Step 1: Key identity: 1/(√(n+1) + √n) = √(n+1) - √n (multiply by the conjugate; denominator becomes n+1-n = 1).

Step 2: First term: 1/(√2 + 1) = √2 - 1.

Step 3: Second: 1/(√3 + √2) = √3 - √2.

Step 4: Third: 1/(2 + √3) = √4 - √3 = 2 - √3.

Step 5: Sum: (√2 - 1) + (√3 - √2) + (2 - √3) = 2 - 1 = 1 (everything in between cancels - telescoping).

» Why this step? Each conjugate-rationalised term is a difference of consecutive roots; in the written sum every middle root appears once with plus and once with minus.

Hinglish: *Ye 'telescoping' pattern dekho: har term rationalise karne par root ka difference banta hai: √2-1, √3-√2, 2-√3. Jodne par beech ke saare roots gayab, sirf first ka minus aur last ka plus bachta hai: 2 - 1 = 1. Agar 1/(√(n+1)+√n) type dikhe, turant conjugate socho.*

Answer: 1

Q39 · High · Concept: Classification, hard cases (T1)

Q39. Classify each as rational or irrational, with a one-line justification: (i) $(1 + \sqrt{2})^2$, (ii) √9 · √3, (iii) $0.\overline{9}$ i.e. 0.999..., (iv) (√2 + √3)(√2 - √3), (v) 1.010010001... (gaps growing), (vi) 6.454545... (block 45).

SOLUTION

Step 1: (i) $(1 + \sqrt{2})^2 = 1 + 2\sqrt{2} + 2 = 3 + 2\sqrt{2}$: irrational (rational + irrational is irrational).

Step 2: (ii) √9 · √3 = 3√3: irrational (nonzero rational × irrational).

Step 3: (iii) 0.999... = 1: rational (proved by shift-and-subtract).

Step 4: (iv) Difference of squares: $(\sqrt{2})^2 - (\sqrt{3})^2 = 2 - 3 = -1$: rational!

Step 5: (v) Gaps keep growing ⇒ no repeating block: irrational.

Step 6: (vi) Recurring ⇒ rational; indeed 6.4545... = 6 + 45/99 = 71/11.

» Why this step? Simplify BEFORE classifying - two of these look irrational until the expansion or collapse is carried out.

Hinglish: *Trap question set! (i) expand karke dekho - 3+2√2 irrational. (iv) dekh ke lagta hai irrational, par difference of squares se -1 nikalta hai - rational! (iii) 0.999... = 1. Moral: pehle simplify karo, phir classify karo. Direct dekh ke verdict dena yahan paisa doobane wala kaam hai.*

Answer: (i) irrational (ii) irrational (iii) rational (= 1) (iv) rational (= -1) (v) irrational (vi) rational (= 71/11)

Q40 · High · Concept: Magnification to hundredths (T4)

Q40. Given √5 = 2.2360679..., state the two consecutive hundredths between which √5 lies, and the two consecutive thousandths between which it lies. Then find a rational number strictly between √5 and 2.24.

SOLUTION

Step 1: Hundredths: 2.23 < 2.2360... < 2.24, so between 2.23 and 2.24.

Step 2: Thousandths: 2.236 < 2.2360679... < 2.237, so between 2.236 and 2.237.

Step 3: For the gap (√5, 2.24): √5 ≈ 2.23607 < 2.24, and the interval is non-empty.

Step 4: Any terminating decimal inside works, e.g. 2.237, 2.238, 2.239.

Step 5: Check: 2.237 > 2.23607 ✓ and 2.237 < 2.24 ✓, so 2.237 is a valid rational in the gap.

» **Why this step?** Terminating decimals are rationals, and the interval (2.23607..., 2.24) contains infinitely many of them - density in action.

Hinglish: *Expansion given hai, bas digits follow karo: 2.23 aur 2.24 ke beech, phir 2.236 aur 2.237 ke beech. Last part mein √5 ≈ 2.23607 aur 2.24 ke beech koi bhi terminating decimal - 2.237 perfect. Rationals ki density ka live demo hai ye.*

Answer: Between 2.23 and 2.24; between 2.236 and 2.237; e.g. 2.237 (or any decimal in (2.23607, 2.24)) lies between √5 and 2.24.

Q41 · HOTS · Concept: Infinite nested radical (T14)

Q41. Find the value of x = √(6 + √(6 + √(6 + √(6 + ...)))) where the pattern continues forever.

SOLUTION

Step 1: The expression inside is the SAME as the whole: the tail √(6 + √(6 + ...)) equals x itself.

Step 2: So x satisfies x = √(6 + x).

Step 3: Square: $x^2 = 6 + x$, i.e. $x^2 - x - 6 = 0$.

Step 4: Factorise: (x - 3)(x + 2) = 0, so x = 3 or x = -2.

Step 5: x is a square root, hence non-negative: x = 3.

Step 6: Check: √(6 + 3) = √9 = 3 ✓ - the value is consistent with itself.

» **Why this step?** Self-similarity lets the infinite tail be replaced by x; the quadratic then follows, and the negative root is discarded on domain grounds.

Hinglish: *Infinite nesting ka trick: andar ka poora structure wahi x hai! x = √(6+x), square karo, quadratic mil gayi. Do roots - 3 aur -2. Root negative nahi ho sakta, toh 3. Ye 'self-similar substitution' technique aage bhi kaam karegi - quadratic banao, domain se filter karo.*

Answer: x = 3

Q42 · HOTS · Concept: Hidden perfect square (T14)

Q42. If x = 5 + 2√6, find the value of √x - 1/√x.

SOLUTION

Step 1: Spot the hidden square: 5 + 2√6 = 3 + 2 + 2√(3×6/... check: $(\sqrt{3} + \sqrt{2})^2 = 3 + 2\sqrt{6} + 2 = 5 + 2\sqrt{6}$ ✓.

Step 2: So √x = √3 + √2.

Step 3: Rationalise: 1/√x = 1/(√3 + √2) = √3 - √2 (conjugate; denominator 3 - 2 = 1).

Step 4: √x - 1/√x = (√3 + √2) - (√3 - √2) = 2√2.

Step 5: Check numerically: x ≈ 9.899, √x ≈ 3.146, 1/√x ≈ 0.318, difference ≈ 2.828 = 2√2 ✓.

» **Why this step?** Recognising a + 2√b as $(\sqrt{m} + \sqrt{n})^2$ with m + n = a, mn = b turns a messy root into two clean surds.

Hinglish: *5 + 2√6 ko dekho aur socho - kaunse do numbers ka sum 5, product 6? 3 aur 2! Matlab $(\sqrt{3}+\sqrt{2})^2$. Toh √x = √3+√2. Uska reciprocal conjugate hi hai (√3-√2). Difference: 2√2. Ye 'hidden perfect square' spotting technique bahut powerful hai - m+n aur mn pattern yaad rakho.*

Answer: $2\sqrt{2}$

Q43 · HOTS · Concept: Quadratic in 2^x (T11)

Q43. Find ALL real values of x satisfying $4^x - 5 \cdot 2^x + 4 = 0$.

SOLUTION

Step 1: Write $4^x = (2^2)^x = (2^x)^2$ and let $t = 2^x$ (note $t > 0$ always).

Step 2: The equation becomes $t^2 - 5t + 4 = 0$.

Step 3: Factorise: $(t - 1)(t - 4) = 0$, so $t = 1$ or $t = 4$.

Step 4: Both roots are positive, so both are valid for $t = 2^x$.

Step 5: $2^x = 1 = 2^0 \Rightarrow x = 0$. $2^x = 4 = 2^2 \Rightarrow x = 2$.

Step 6: Check: $x = 0$: $1 - 5 + 4 = 0$ ✓. $x = 2$: $16 - 20 + 4 = 0$ ✓.

» **Why this step?** A disguised quadratic appears whenever one power is the square of another; substituting t collapses it, and the constraint $t > 0$ filters roots.

Hinglish: *4^x ko $(2^x)^2$ likho - quadratic ka chhupa rustam! $t = 2^x$ substitution: $t^2 - 5t + 4 = 0$. Roots 1 aur 4, dono positive - dono valid. $x = 0$ aur $x = 2$. Dhyan: agar koi root negative ya zero aata t ka, wo REJECT karna hota kyunki 2^x kabhi 0 ya negative nahi hota.*

Answer: $x = 0$ or $x = 2$

Q44 · HOTS · Concept: General root classification proof (T5)

Q44. Prove: for any positive integer n, $\sqrt{n}$ is either a positive integer or an irrational number (a rational non-integer square root is impossible).

SOLUTION

Step 1: Suppose $\sqrt{n} = p/q$ in lowest terms. Then $n = p^2/q^2$, so $n q^2 = p^2$, i.e. $p^2 = n q^2$.

Step 2: Take the prime factorisation of n. In n^2 every prime exponent is EVEN (each exponent doubles).

Step 3: From $p^2 = n q^2$: compare the prime-by-prime parity of exponents. On the left (p^2) all exponents are even; on the right, $n q^2$ has the exponents of n plus even exponents.

Step 4: So every prime exponent in n must be even - otherwise one prime appears an odd number of times on the right but even on the left, impossible.

Step 5: Therefore n is a perfect square, and $\sqrt{n} = p/q$ is actually an integer (q must be 1 after cancellation... more precisely, $\sqrt{n}$ is the integer obtained from the even exponents).

Step 6: Conclusion: if n is a perfect square, $\sqrt{n}$ is an integer; otherwise $\sqrt{n}$ is irrational. ∎

» **Why this step?** Parity of prime exponents is the invariant: squaring makes every exponent even, so n^2's factorisation can never equal a fraction squared unless n is a square.

Hinglish: *Ye $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ proofs ka GENERAL version hai. Argument: square mein har prime ka count even hota hai. Agar n perfect square nahi, koi na koi prime odd count mein hoga - aur wo $\sqrt{n}$ ko rational hone se rok dega. Isliye duniya mein sirf do tarah ke roots hain: integers ya irrationals. Bechara 'rational non-integer root' exist hi nahi karta.*

Answer: $\sqrt{n}$ is an integer when n is a perfect square, irrational otherwise (proved by prime-exponent parity).

Q45 · HOTS · Concept: Cube of surd sum (T14)

Q45. If $x = \sqrt[3]{3} + 1/\sqrt[3]{3}$, show that $x^3 - 3x = 10/3$.

SOLUTION

Step 1: Let a = $\sqrt[3]{3}$ and b = $1/\sqrt[3]{3}$ = 3^(-1/3). Then x = a + b.

Step 2: Compute ab = $\sqrt[3]{3} \cdot 3^{(-1/3)} = 3^{(1/3 - 1/3)} = 3^0 = 1$.

Step 3: Use $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$.

Step 4: $a^3 = 3$ and $b^3 = 1/3$, so $a^3 + b^3 = 3 + 1/3 = 10/3$.

Step 5: Also ab = 1 and a + b = x, so 3ab(a+b) = 3x.

Step 6: Therefore $x^3 = 10/3 + 3x$, i.e. $x^3 - 3x = 10/3$. ∎

» **Why this step?** The cube identity needs a^3, b^3 and ab - and conjugate-style choices make ab collapse to 1, killing all surds.

Hinglish: *Cube identity ka magic: $(a+b)^3 = a^3+b^3+3ab(a+b)$. Yahan ab = 1 nikalta hai kyunki 3^(1/3) aur 3^(-1/3) ke exponents cancel. $a^3 = 3$, $b^3 = 1/3$. Sab kuch rational ho gaya! $x^3 - 3x = 10/3$. Ye pattern Q25 ke square version ka cube bhai hai - dono yaad rakho.*

Answer: $x^3 - 3x = 10/3$ (shown using ab = 1 and $a^3 + b^3 = 10/3$).

Q46 · HOTS · Concept: Long telescoping chain (T14)

Q46. Evaluate: 1/(√2+1) + 1/(√3+√2) + 1/(√4+√3) + ... + 1/(√16+√15).

SOLUTION

Step 1: Rationalise the general term: 1/(√(n+1) + √n) = √(n+1) - √n.

Step 2: The sum becomes (√2-1) + (√3-√2) + (√4-√3) + ... + (√16-√15).

Step 3: It telescopes: √2 cancels, √3 cancels, ..., √15 cancels.

Step 4: Survivors: -1 + √16 = 4 - 1 = 3.

Step 5: Check with a partial sum: first three terms give √4 - 1 = 1 (matches Q38) ✓.

» **Why this step?** Each term links √n to √(n+1); in a chain, interior links cancel and only the two endpoints remain.

Hinglish: *Q38 ka full version - 15 terms! Darne ki zaroorat nahi, pattern same. Har term rationalise karke root-difference banao, poora chain telescopes. Sirf -1 aur +√16 bachta hai: 4 - 1 = 3. Long sums mein hamesha pattern dhoondo - individual terms kabhi add nahi karte.*

Answer: 3

Q47 · HOTS · Concept: Density of irrationals (T12)

Q47. Prove that between any two distinct real numbers a < b there lie infinitely many irrational numbers.

SOLUTION

Step 1: Given a < b. First produce ONE irrational in (a, b): choose an irrational k > 0, e.g. k = √2.

Step 2: Consider numbers of the form a + t·k for small positive t. As t → 0, a + tk → a from above.

Step 3: Since b - a > 0, choose t = (b - a)/(2k); then a + tk = a + (b-a)/2 = (a+b)/2... refine: with k irrational and t rational nonzero, a + tk is irrational (nonzero rational × irrational + rational).

Step 4: Pick t = (b-a)/(4k) scaled rationally... concretely: the number m = (a+b)/2 + δ·k with δ a tiny nonzero rational satisfies a < m < b for |δ| small, and m is irrational.

Step 5: Infinitely many: each distinct tiny rational δ ($1/10^n$ for n = 1, 2, 3, ...) gives a DIFFERENT irrational m_n, all inside (a, b).

Step 6: Hence infinitely many irrationals lie between a and b. ∎

» **Why this step?** The 'rational × irrational + rational' closure result manufactures irrationals on demand, and shrinking δ keeps them in any prescribed gap.

Hinglish: *Density proof ka idea: midpoint pakdo, usme chhota sa irrational tail jodo. (a+b)/2 + δ√2 - δ ko tiny rakho toh wo (a,b) ke andar hi rahega, aur irrational rahega. δ = 1/10, 1/100, 1/1000... har baar NAYA irrational. Isliye 'infinitely many'. Ye proof-style question 5 marks ka hota hai - closure result quote karna zaroori.*

Answer: For any a < b, the numbers (a+b)/2 + δ√2 with infinitely many tiny rationals δ ≠ 0 all lie in (a, b) and are all irrational.

Q48 · HOTS · Concept: Root spiral total length (T13)

Q48. In the square root spiral, consecutive outer edges have lengths √2, √3, √4, ... (the first edge from O has length √1 = 1). Taking √2 = 1.41, √3 = 1.73, √5 = 2.24, √6 = 2.45, √7 = 2.65, find the smallest number of edges (beyond the first) needed for the total length of edges √2 + √3 + ... + √(n+1) to exceed 10.

SOLUTION

Step 1: The first edge (length 1) is excluded, so we accumulate √2, √3, √4, √5, ...

Step 2: √2 + √3 = 1.41 + 1.73 = 3.14.

Step 3: Add √4 = 2: running total 5.14.

Step 4: Add √5 = 2.24: total 7.38.

Step 5: Add √6 = 2.45: total 9.83 (still below 10).

Step 6: Add √7 = 2.65: total 12.48 > 10. So edges √2 through √7 - that is 6 edges - are needed.

» **Why this step?** Cumulative sums turn a spiral-geometry question into ordered addition; √4 = 2 is exact and stops one rounding error.

Hinglish: *Spiral ke edges jodo - bas running total rakho. √2+√3 = 3.14, plus 2 = 5.14, plus √5 = 7.38, plus √6 = 9.83 (abhi 10 nahi!), plus √7 = 12.48. Matlab 6 edges chahiye. √4 = 2 exact hai - rounding se bach jao. Aise 'smallest n' questions mein ek-ek term add karke threshold cross karna hi method hai.*

Answer: 6 edges (the cumulative length first exceeds 10 at √7, total ≈ 12.48).

Q49 · HOTS · Concept: Irrationality of √2 + √5 (T6)

Q49. (i) Prove that √2 + √5 is irrational. (ii) Find two consecutive integers between which √2 + √5 lies.

SOLUTION

Step 1: (i) Suppose √2 + √5 = r is rational. Square: $2 + 2\sqrt{10} + 5 = r^2$, so $2\sqrt{10} = r^2 - 7$.

Step 2: Then $\sqrt{10} = (r^2 - 7)/2$, which would be rational.

Step 3: But 10 is not a perfect square, so √10 is irrational - contradiction. Hence √2 + √5 is irrational. ∎

Step 4: (ii) √2 ≈ 1.4142 and √5 ≈ 2.2360 (standard bounds: $1.414^2 = 1.999...$, $1.415^2 > 2$; $2.236^2 \approx 4.9997 < 5 < 2.237^2$).

Step 5: Sum ≈ 3.6502, which lies between the consecutive integers 3 and 4.

Step 6: Refined check: $3.6^2 = 12.96 <$ (value)$^2 = 7 + 2\sqrt{10} \approx 13.325 < 16$, consistent.

» **Why this step?** The square of √2 + √5 mixes into a single surd √10 - one contradiction instead of two; bracketing the decimal expansion places the value.

Hinglish: *(i) same square-trick: √10 = (r²-7)/2 rational hota - impossible. (ii) mein dono expansions jodo: 1.414 + 2.236 = 3.65. Toh 3 aur 4 ke beech. Proof aur placement - dono alag skills hain, answer mein dono dikhao.*

Answer: (i) √2 + √5 is irrational. (ii) It lies between 3 and 4 (approximately 3.65).

Q50 · HOTS · Concept: Infinite exponent product (T14)

Q50. Find the value of the infinite product 2^(1/2) × 4^(1/4) × 8^(1/8) × 16^(1/16) × ... (continuing forever).

SOLUTION

Step 1: Rewrite every factor as a power of 2: 2^(1/2) × 2^(2/4) × 2^(3/8) × 2^(4/16) × ...

Step 2: Same base → exponents add: the product is 2^S where S = 1/2 + 2/4 + 3/8 + 4/16 + ...

Step 3: Compute S by the shift trick: 2S = 1 + 2/2 + 3/4 + 4/8 + ... (doubling shifts every term up one place).

Step 4: Subtract: 2S - S = 1 + (2/2 - 1/2) + (3/4 - 2/4) + (4/8 - 3/8) + ... = 1 + 1/2 + 1/4 + 1/8 + ...

Step 5: The geometric series 1/2 + 1/4 + 1/8 + ... = 1, so S = 1 + 1 = 2.

Step 6: Therefore the product equals $2^2 = 4$.

Step 7: Partial check: 2^(0.5) × 4^(0.25) × 8^(0.125) × 16^(0.0625) = 1.414 × 1.414 × 1.297 × 1.189 ≈ 3.10 and rising toward 4 ✓.

» Why this step? The shift-and-subtract trick that converted recurring decimals also sums the exponent series; geometric decay makes the tail manageable.

Hinglish: *Sab factors ko 2 ke power mein likho - exponents ka series banta hai: 1/2 + 2/4 + 3/8 + ... Wahi recurring-decimal wala shift trick lagao: 2S - S. Bacha 1 + (1/2 + 1/4 + ...) = 1 + 1 = 2. Product = $2^2 = 4$. Chapter 3 ke saare tools ek saath: exponents, infinite series, shift trick. Ye 5-mark HOTS ka full package hai.*

Answer: 4

3.4 Chapter MCQs (60 questions, answers inline)

Each MCQ lists the options, the correct answer, a full explanation, the trap analysis for the tempting wrong options, and a Hinglish note. Options are engineered from real student mistakes, so think before you pick.

MCQ 1 · Normal · Concept: Set classification

1. Which of the following is irrational?

A. 0.101001000100001...
B. 0.25
C. 3/8
D. √49

Correct Answer: (A)

The gaps between 1s keep growing, so no block repeats: non-terminating, non-recurring, hence irrational. The others terminate or simplify (0.25, 3/8, √49 = 7).

Why (B) is wrong: Terminating decimal, hence rational.

Why (C) is wrong: Ratio of integers, rational.

Why (D) is wrong: Perfect square under the root: 7.

Hinglish: *Pattern dekho - ones ke beech gap badhta ja raha hai, koi block repeat nahi hoga. Isliye irrational. Baaki teeno rational hain.*

MCQ 2 · Normal · Concept: Pure recurring to p/q

2. 0.333... (the 3 repeating) written as a fraction is:

A. 1/3
B. 3/10
C. 33/100
D. 1/30

Correct Answer: (A)

One repeating digit: x = 3/9 = 1/3. Check: 1/3 = 0.333... ✓.

Why (B) is wrong: 3/10 = 0.3, terminates - no repetition.

Why (C) is wrong: 33/100 = 0.33 terminates.

Why (D) is wrong: Extra zero in denominator: 1/30 = 0.0333...

Hinglish: *Ek digit repeat = upar wo digit, neeche 9. 3/9 = 1/3. Denominator 10 wale options terminating decimals dete hain - wo galat.*

MCQ 3 · Normal · Concept: Terminating test

3. Without division, the decimal expansion of 77/210 is:

A. terminating
B. non-terminating recurring
C. non-terminating non-recurring
D. cannot be decided

Correct Answer: (B)

77/210 = 11/30 (cancel 7). 30 = 2 × 3 × 5 has the prime 3, so it recurs.

Why (A) is wrong: 30 is not of the form $2^m \times 5^n$.

Why (C) is wrong: Every rational recurs or terminates - never truly random.

Why (D) is wrong: The test always decides.

Hinglish: *Pehle 7 cancel karo - 11/30 banta hai. 30 mein 3 hai, toh recurring. Ye reduce-first wala hi case hai.*

MCQ 4 · Normal · Concept: Set nesting

4. Which statement is TRUE?

A. $\mathbb{Z} \subset \mathbb{N}$
B. $\mathbb{Q} \subset \mathbb{Z}$
C. $\mathbb{N} \subset \mathbb{Q}$
D. $\mathbb{R} \subset \mathbb{Q}$

Correct Answer: (C)

Naturals sit inside integers, integers inside rationals, rationals inside reals: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. So $\mathbb{N} \subset \mathbb{Q}$ is true.

Why (A) is wrong: Negatives are in Z but not N.

Why (B) is wrong: Fractions like 1/2 are in Q but not Z.

Why (D) is wrong: Irrationals are in R but not Q - R is bigger.

Hinglish: *Nesting yaad karo: N sabse andar, phir Z, phir Q, sabse bahar R. C options wala direction sahi hai.*

MCQ 5 · Normal · Concept: Mixed recurring to p/q

5. 0.4777... (the 4 not repeating, the 7 repeating) equals:

A. 43/90
B. 47/100
C. 47/90
D. 43/99

Correct Answer: (A)

(47 - 4)/90 = 43/90: one 9 for the repeating digit, one 0 for the fixed digit.

Why (B) is wrong: That is 0.47 exactly, terminating.

Why (C) is wrong: Denominator pattern wrong for mixed recurrence.

Why (D) is wrong: 99 would fit 0.434343...

Hinglish: *(poora - head)/(9 + 0): (47-4)/90 = 43/90. Option C wala 47/90 hota agar 4 bhi repeat karta.*

MCQ 6 · Normal · Concept: Between-ness

6. An irrational number between 3 and 4 is:

A. $\sqrt{8}$
B. $\sqrt{9}$
C. $\sqrt{10}$
D. 3.14

Correct Answer: (C)

$3^2 = 9 < 10 < 16 = 4^2$, so $3 < \sqrt{10} < 4$, and 10 is not a perfect square, so $\sqrt{10}$ is irrational.

Why (A) is wrong: $\sqrt{8} \approx 2.83 < 3$ - outside the interval.

Why (B) is wrong: $\sqrt{9} = 3$ exactly (and rational).

Why (D) is wrong: 3.14 terminates, hence rational.

Hinglish: *3 aur 4 ke square = 9 aur 16. Beech ka non-square: 10, 11, 12, ... 15. √10 perfect choice.*

MCQ 7 · Normal · Concept: Surd multiplication

7. $\sqrt{2} \times \sqrt{8}$ equals:

A. $\sqrt{10}$
B. 4
C. $2\sqrt{4}$
D. 16

Correct Answer: (B)

$\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$. Roots multiply by multiplying radicands.

Why (A) is wrong: Radicands are multiplied, not added.

Why (C) is wrong: $2\sqrt{4} = 4$ by luck of notation, but $\sqrt{2} \cdot \sqrt{8} = \sqrt{16}$ directly; writing $2\sqrt{4}$ shows wrong method.

Why (D) is wrong: Forgot to take the root at the end.

Hinglish: *√2 × √8 = √16 = 4. Under-root multiply hota hai, add nahi.*

MCQ 8 · Normal · Concept: Decimal by denominator test

8. The decimal expansion of 33/125 is:

A. 0.264

B. 0.2640

C. non-terminating

D. 0.26326432643...

Correct Answer: (A)

$125 = 5^3$, only 5s, so it terminates: $33/125 = 33 \times 8/1000 = 264/1000 = 0.264$.

Why (B) is wrong: 0.2640 = 0.264 - same number, but the intended exact form is 0.264.

Why (C) is wrong: $125 = 5^3$ passes the $2^m \times 5^n$ test.

Why (D) is wrong: That is a recurring pattern - impossible for denominator 125.

Hinglish: *125 ko 1000 banane ke liye 8 se multiply karo upar-neeche. 0.264 ekdum clean.*

MCQ 9 · Normal · Concept: Comparing surds

9. Which is the largest?

A. $2\sqrt{3}$

B. $3\sqrt{2}$

C. 4

D. $\sqrt{15}$

Correct Answer: (B)

Squares: 12, 18, 16, 15. The largest square is 18, so $3\sqrt{2}$ is largest.

Why (A) is wrong: 12 is the smallest square here.

Why (C) is wrong: 16 < 18.

Why (D) is wrong: 15 < 18. Coefficient se confuse hona common galti hai.

Hinglish: *Sabko square karo: $2\sqrt{3} \rightarrow 12$, $3\sqrt{2} \rightarrow 18$, $4 \rightarrow 16$, $\sqrt{15} \rightarrow 15$. Sabse bada square 18 - winner $3\sqrt{2}$.*

MCQ 10 · Normal · Concept: Same-base equation

10. If $3^x = 1/243$, then x =

A. 5

B. -5

C. 1/5

D. -243

Correct Answer: (B)

$1/243 = 1/3^5 = 3^{-5}$, so x = -5.

Why (A) is wrong: $3^5 = 243$, not 1/243.

Why (C) is wrong: Fractional exponent confusion.

Why (D) is wrong: Equating x with the value, not the exponent.

Hinglish: *1/243 ko 3^{-5} likho. Negative exponent = reciprocal. x = -5.*

MCQ 11 · Normal · Concept: Conjugate with 1

11. $1/(\sqrt{5} - 2)$ equals:

A. $\sqrt{5} + 2$

B. $\sqrt{5} - 2$

C. $(\sqrt{5}+2)/3$

D. $1/3$

Correct Answer: (A)

Multiply by $(\sqrt{5} + 2)$: denominator $(\sqrt{5})^2 - 2^2 = 5 - 4 = 1$. So the value is $\sqrt{5} + 2$.

Why (B) is wrong: That is the original denominator, not the value.

Why (C) is wrong: Wrong denominator (3 would come from $\sqrt{9}$).

Why (D) is wrong: No rationalising done at all.

Hinglish: *Conjugate multiply karo, denominator 5-4 = 1 nikla. Answer khud √5+2 hai - magic collapse.*

MCQ 12 · Normal · Concept: Difference of squares

12. $(\sqrt{7} + \sqrt{5})(\sqrt{7} - \sqrt{5})$ equals:

A. 2

B. 12

C. $\sqrt{2}$

D. 35

Correct Answer: (A)

$(\sqrt{7})^2 - (\sqrt{5})^2 = 7 - 5 = 2$.

Why (B) is wrong: Added the squares instead of subtracting.

Why (C) is wrong: Took root of the difference.

Why (D) is wrong: Multiplied radicands - that is $\sqrt{35}$, a different expression.

Hinglish: *$(a+b)(a-b) = a^2-b^2$. Roots ke square = andar wale numbers. 7-5 = 2.*

MCQ 13 · Normal · Concept: Conjugate reciprocal

13. If $x = 3 + 2\sqrt{2}$, then $1/x$ equals:

A. $3 - 2\sqrt{2}$

B. $2\sqrt{2} - 3$

C. $1/(3+2\sqrt{2})$ simplified to $1/3$

D. 6

Correct Answer: (A)

$1/(3+2\sqrt{2}) = (3-2\sqrt{2})/((3)^2-(2\sqrt{2})^2) = (3-2\sqrt{2})/(9-8) = 3 - 2\sqrt{2}$.

Why (B) is wrong: Sign flipped on the wrong term.

Why (C) is wrong: No simplification attempted.

Why (D) is wrong: That is $x + 1/x = 6$, a different question.

Hinglish: *Conjugate multiply karo, denominator 9-8 = 1. Toh 1/x khud conjugate hai. x aur 1/x ka pair yaad rakho.*

MCQ 14 · Normal · Concept: 0.999... concept

14. Which statement is TRUE?

A. 0.999... is exactly 1
B. 0.999... is slightly less than 1
C. 0.999... is irrational
D. 0.999... is undefined

Correct Answer: (A)

x = 0.999... gives 10x - x = 9, so x = 1 exactly. It is a rational number with two decimal expansions.

Why (B) is wrong: Common intuition, but no digit-level gap survives.

Why (C) is wrong: Recurring decimals are rational by definition.

Why (D) is wrong: The infinite sum has a well-defined value.

Hinglish: *Ye mind-bender hai: 0.999... EXACTLY 1 hai. Shift-subtract se 9x = 9. 'Thoda kam' wala feeling galat hai - maths mein exact proof jeetta hai.*

MCQ 15 · Normal · Concept: Ordering roots

15. Arranged in DECREASING order, which sequence is correct?

A. √5, √6, √7
B. √7, √6, √5
C. √6, √5, √7
D. √7, √5, √6

Correct Answer: (B)

For the same root index, bigger radicand means bigger value: √7 > √6 > √5.

Why (A) is wrong: Reversed the order.

Why (C) is wrong: Scrambled - √5 is the smallest.

Why (D) is wrong: √5 is not bigger than √6.

Hinglish: *Same root, toh andar bada number = value badi. √7 sabse bada. Seedha rule hai.*

MCQ 16 · Normal · Concept: Fractional exponents

16. (16/81)^(-1/4) equals:

A. 3/2
B. 2/3
C. -3/2
D. 81/16

Correct Answer: (A)

Negative exponent flips: (81/16)^(1/4) = $\sqrt[4]{81}$ / $\sqrt[4]{16}$ = 3/2.

Why (B) is wrong: Forgot to flip for the negative exponent.

Why (C) is wrong: Root lene ke baad bhi minus rakh diya - root value negative nahi hoti.

Why (D) is wrong: Ignored the exponent entirely.

Hinglish: *Minus matlab flip, 1/4 matlab fourth root. (81/16)^(1/4) = 3/2. Root pehle lo, numbers chhote rahenge.*

MCQ 17 · Normal · Concept: Which is rational

17. Which of these is rational?

A. √12/√3
B. √2 + √3
C. √2 × √3
D. √2 - √3

Correct Answer: (A)

√12/√3 = √(12/3) = √4 = 2. The others are irrational sums/products/differences of unlike surds.

Why (B) is wrong: Unlike surds never simplify to rationals.

Why (C) is wrong: √6 is irrational (6 not a perfect square).

Why (D) is wrong: Same: irrational, about -0.318.

Hinglish: *Divide karke simplify karo: √4 = 2. Baaki teeno irrational hain - √6, √2+√3, √2-√3.*

MCQ 18 · Normal · Concept: Same-base equation 2

18. If 2^(2x) = 64, then x =

A. 3
B. 6
C. 2
D. 32

Correct Answer: (A)

64 = 2^6, so 2x = 6 and x = 3.

Why (B) is wrong: Forgot the 2x and equated x = 6.

Why (C) is wrong: Divided by 3 instead of 2.

Why (D) is wrong: Took half of 64.

Hinglish: *2^(2x) = 2^6, exponents compare: 2x = 6, x = 3. Option B wale 2x ko x samajh bethe.*

MCQ 19 · Normal · Concept: Mixed recurring with integer

19. 2.1666... (the 6 repeating) as a fraction is:

A. 13/6
B. 21/10
C. 13/60
D. 11/6

Correct Answer: (A)

x = 2.1666..., 10x = 21.666..., 100x = 216.66...; 90x = 195, x = 195/90 = 13/6.

Why (B) is wrong: That is 2.1, ignoring the repeating tail.

Why (C) is wrong: Denominator 60 = wrong shift.

Why (D) is wrong: 11/6 = 1.8333, wrong value.

Hinglish: *10x aur 100x banao, minus: 195/90, reduce: 13/6. Check: 13/6 = 2.1666 ✓.*

MCQ 20 · Normal · Concept: Square of surd sum

20. $(\sqrt{3} + \sqrt{2})^2$ equals:

A. $5 + 2\sqrt{6}$

B. $5 + \sqrt{6}$

C. 13

D. 5

Correct Answer: (A)

$3 + 2\sqrt{6} + 2 = 5 + 2\sqrt{6}$. The cross term $2\sqrt{3}\sqrt{2} = 2\sqrt{6}$ is essential.

Why (B) is wrong: Missed the factor 2 on the cross term.

Why (C) is wrong: Squaring the radicands wrongly: 3 + 2 + ... no.

Why (D) is wrong: Dropped the cross term completely.

Hinglish: *$(a+b)^2 = a^2 + 2ab + b^2$. Cross term $2\sqrt{6}$ bhoolna sabse common galti. $5 + 2\sqrt{6}$.*

MCQ 21 · HOTS · Concept: Negative fractional exponent

21. 64^(-2/3) equals:

A. 1/16

B. -1/16

C. 16

D. 1/48

Correct Answer: (A)

64^(1/3) = 4, then $4^2 = 16$, then flip for the minus: 1/16.

Why (B) is wrong: Minus sign ko value par laga diya - flip hota hai.

Why (C) is wrong: Forgot the reciprocal.

Why (D) is wrong: Added 2 and 3 somewhere - order matters.

Hinglish: *Root pehle: cube root of 64 = 4. Square: 16. Minus: flip → 1/16. Order: root → power → flip.*

MCQ 22 · HOTS · Concept: Conjugate pair sum

22. If $A = (\sqrt{5}+\sqrt{3})/(\sqrt{5}-\sqrt{3})$ and $B = (\sqrt{5}-\sqrt{3})/(\sqrt{5}+\sqrt{3})$, then A + B equals:

A. 8

B. 16

C. $4\sqrt{15}$

D. 2

Correct Answer: (A)

$A = 4 + \sqrt{15}$ and $B = 4 - \sqrt{15}$ (each by rationalising; see Q14's method). Sum = 8.

Why (B) is wrong: Doubled A instead of adding A and B.

Why (C) is wrong: Only the surd part remained - forgetting the 4+4.

Why (D) is wrong: Subtracted instead of added.

Hinglish: *Dono rationalise karo: $A = 4+\sqrt{15}$, $B = 4-\sqrt{15}$. Jodo - surd cancel, 8. Conjugate pairs ka sum hamesha surd-free hota hai.*

MCQ 23 · HOTS · Concept: Factored power difference

23. If $3^x - 3^{(x-1)} = 162$, then x =

A. 5

B. 4

C. 6

D. 5.5

Correct Answer: (A)

Factor: $3^{(x-1)}(3 - 1) = 162 \Rightarrow 3^{(x-1)} = 81 = 3^4 \Rightarrow x = 5$.

Why (B) is wrong: Solved x - 1 = 3 (162/2 = 81 misread as 27).

Why (C) is wrong: Took $3^4 = 243$ confusion.

Why (D) is wrong: Averaged instead of solving.

Hinglish: *3^(x-1) common nikalo, bracket mein 2. 162/2 = 81 = 3⁴. x - 1 = 4, x = 5.*

MCQ 24 · Normal · Concept: Proof method

24. The standard method to prove that √3 is irrational is:

A. contradiction (assume it is rational)

B. induction

C. direct computation

D. counterexample

Correct Answer: (A)

Assume √3 = p/q in lowest terms, derive that 3 divides both p and q, contradict coprimality.

Why (B) is wrong: Induction is for integer statements.

Why (C) is wrong: You can never compute an irrational exactly.

Why (D) is wrong: Counterexamples disprove claims; they cannot prove irrationality.

Hinglish: *Irrationality proofs hamesha contradiction se hote hain. Assume karo rational hai, phir ussi assumption ko todo.*

MCQ 25 · Normal · Concept: Closure true/false

25. Which statement is FALSE?

A. rational + rational is rational

B. irrational + irrational is always irrational

C. rational × irrational (rational ≠ 0) is irrational

D. integer + integer is an integer

Correct Answer: (B)

√2 + (-√2) = 0: two irrationals can sum to a rational. The other statements are genuine closure facts.

Why (A) is wrong: True - Q is closed under addition.

Why (C) is wrong: True with the nonzero condition.

Why (D) is wrong: True - Z is closed under addition.

Hinglish: *B galat hai - counterirritant: √2 aur -√2 jodne par 0. Irrationals ke beech koi closure rule NAHI hai.*

MCQ 26 · HOTS · Concept: Nested radical

26. $\sqrt{(11 + 2\sqrt{30})}$ equals:

A. $\sqrt{5} + \sqrt{6}$

B. $5 + 6$

C. $\sqrt{30} + 1$

D. $\sqrt{11} + \sqrt{30}$

Correct Answer: (A)

Need $m + n = 11$, $mn = 30$: $m = 5$, $n = 6$. So the root is $\sqrt{5} + \sqrt{6}$ (verify: $5 + 6 + 2\sqrt{30} = 11 + 2\sqrt{30}$).

Why (B) is wrong: 5+6 ka square 121 hota hai.

Why (C) is wrong: Product 30 ke sath sum 31 hoga.

Why (D) is wrong: Roots kabhi aise alag-alag nahi judte.

Hinglish: *Sum 11, product 30 - 5 aur 6. Answer $\sqrt{5} + \sqrt{6}$. Verify karke lock karo.*

MCQ 27 · HOTS · Concept: Conjugate squares

27. If $x = 2 + \sqrt{3}$, then $x^2 - 1/x^2$ equals:

A. $8\sqrt{3}$

B. 14

C. $4\sqrt{3}$

D. $2\sqrt{3}$

Correct Answer: (A)

$x^2 = 7 + 4\sqrt{3}$ and $1/x^2 = 7 - 4\sqrt{3}$. Difference $= 8\sqrt{3}$.

Why (B) is wrong: That is $x^2 + 1/x^2$ (the sum, not the difference).

Why (C) is wrong: Halved the surd coefficient.

Why (D) is wrong: Used $x - 1/x$ instead.

Hinglish: *x^2 aur $1/x^2$ conjugates hain: $7 \pm 4\sqrt{3}$. Minus karo - rational parts cancel, $8\sqrt{3}$ bacha.*

MCQ 28 · Normal · Concept: Recurring arithmetic

28. $0.\overline{6}$ (i.e. 0.666...) + $0.\overline{3}$ (i.e. 0.333...) equals:

A. 1

B. 0.999... but not 1

C. 0.9

D. 2/3

Correct Answer: (A)

2/3 + 1/3 = 1 exactly (and 0.999... IS 1).

Why (B) is wrong: 0.999... IS 1 - no gap.

Why (C) is wrong: 0.9 terminates; wrong value.

Why (D) is wrong: 2/3 is only half the answer.

Hinglish: *0.666... = 2/3, 0.333... = 1/3. Sum = 1. Aur 0.999... bhi exactly 1 hai - consistent!*

MCQ 29 · Normal · Concept: Successive magnification

29. Given √7 = 2.6457513..., √7 lies between:

A. 2.64 and 2.65
B. 2.6 and 2.7
C. 2.645 and 2.646
D. both A and C are true

Correct Answer: (D)

2.6457... sits between 2.64 and 2.65 (hundredths) and between 2.645 and 2.646 (thousandths). Both statements are true.

Why (A) is wrong: True but incomplete - C is also true.

Why (B) is wrong: True but coarser than A and C.

Why (C) is wrong: True but incomplete - A is also true.

Hinglish: *Dono statements sahi hain - ek hundredths level par, ek thousandths par. 'Both true' option dhyan se padho.*

MCQ 30 · Normal · Concept: Conjugate collapse

30. 1/(√9 - √8) equals:

A. 3 + 2√2
B. 3 - 2√2
C. 1
D. 3 + √8

Correct Answer: (A)

√9 = 3, √8 = 2√2. Multiply by (3 + 2√2): denominator 9 - 8 = 1, value 3 + 2√2.

Why (B) is wrong: Conjugate ka sign ulta kar diya.

Why (C) is wrong: 1/(3-2√2) ≠ 1 - compute karo.

Why (D) is wrong: Rationalise kiye bina sirf sign badla.

Hinglish: *Pehle roots simplify: √9 = 3, √8 = 2√2. Phir conjugate - denominator 1. Answer 3+2√2.*

MCQ 31 · Normal · Concept: Which is irrational

31. Which of the following is irrational?

A. (2+√2)(2-√2)
B. √2 + √8
C. (3+√3) + (3-√3)
D. √16/√4

Correct Answer: (B)

√2 + √8 = √2 + 2√2 = 3√2, irrational. A: 4-2 = 2. C: 6. D: 4/2 = 2.

Why (A) is wrong: Difference of squares: 2.

Why (C) is wrong: Surds cancel in the sum.

Why (D) is wrong: Both roots are perfect squares.

Hinglish: *Pehle simplify karo har option ko. B mein √8 = 2√2, toh 3√2 bachta hai - irrational. Baaki sab clean numbers.*

MCQ 32 · HOTS · Concept: Common value relation

32. If $2^x = 3^y = 12^z = t$ ($t \neq 1$), which relation holds?

A. $1/z = 2/x + 1/y$

B. $1/z = 1/x + 1/y$

C. $z = x + y$

D. $1/z = 1/x + 2/y$

Correct Answer: (A)

$12 = 2^2 \times 3 \Rightarrow t^{(1/z)} = t^{(2/x)} \cdot t^{(1/y)} \Rightarrow 1/z = 2/x + 1/y$.

Why (B) is wrong: That factorisation is 6, not 12.

Why (C) is wrong: Exponents never add across different bases like this.

Why (D) is wrong: Swapped the roles of 2 and 3.

Hinglish: *12 ko $2^2 \times 3$ kholo. Exponent add: 2/x + 1/y = 1/z. Options B aur D wale $12 = 2 \times 3^2$ samajh bete hain - wo 18 hota hai.*

MCQ 33 · Normal · Concept: Which is rational

33. Which expression is RATIONAL?

A. $(1+\sqrt{2})^2 + (1-\sqrt{2})^2$

B. $(1+\sqrt{2})^2$

C. $1/(1+\sqrt{2})$

D. $\sqrt{(1+\sqrt{2})}$

Correct Answer: (A)

$(1+\sqrt{2})^2 = 3+2\sqrt{2}$ and $(1-\sqrt{2})^2 = 3-2\sqrt{2}$; the sum is 6, rational. B and C are irrational; D is a root of an irrational - irrational.

Why (B) is wrong: $3 + 2\sqrt{2}$ retains the surd.

Why (C) is wrong: $= \sqrt{2} - 1$, still has $\sqrt{2}$.

Why (D) is wrong: Root of an irrational number is (here) irrational.

Hinglish: *Conjugate squares jodo - surd terms cancel, 6. B akela irrational. C = √2-1 irrational. D bhi irrational.*

MCQ 34 · Normal · Concept: Between fractions

34. Which number lies between 5/7 and 9/11?

A. 0.85

B. 3/4

C. 5/6

D. 2/3

Correct Answer: (B)

$5/7 \approx 0.714$ and $9/11 \approx 0.818$. $3/4 = 0.75$ sits between them. 0.85 and $5/6 \approx 0.833$ exceed 0.818; $2/3 \approx 0.667$ is below.

Why (A) is wrong: $0.85 > 0.818$.

Why (C) is wrong: $5/6 \approx 0.833 > 0.818$.

Why (D) is wrong: $2/3 \approx 0.667 < 0.714$.

Hinglish: *Dono ko decimal mein dekho: 0.714 aur 0.818. 0.75 andar hai. Baaki bahar.*

MCQ 35 · Normal · Concept: Quotient of surds

35. √27/√75 equals:

A. 3/5

B. 9/25

C. √(3/5)

D. 1/5

Correct Answer: (A)

√(27/75) = √(9/25) = 3/5.

Why (B) is wrong: Forgot to take the root.

Why (C) is wrong: Simplified 27/75 to 3/5 but left the root.

Why (D) is wrong: Wrong reduction of the fraction.

Hinglish: *Under-root divide karo: 27/75 = 9/25. Root: 3/5. Clean fraction.*

MCQ 36 · Normal · Concept: Recurring with integer part

36. 6.454545... (block 45 repeating) as a fraction is:

A. 71/11

B. 645/100

C. 71/10

D. 6 5/11

Correct Answer: (A)

100x - x = 645.4545... - 6.4545... = 639; 99x = 639; x = 639/99 = 71/11.

Why (B) is wrong: Terminating decimal assumption.

Why (C) is wrong: That is 7.1 - wrong.

Why (D) is wrong: 6 5/11 = 71/11: same VALUE, but 45 not 5 repeats; the clean single form is 71/11.

Hinglish: *99x = 639, kat ke 71/11. Check: 71/11 = 6.4545 ✓. Option D wala 6 + 5/11 = 71/11 hi hai - actually same value! But standard single answer 71/11.*

MCQ 37 · Normal · Concept: Root dichotomy

37. For a positive integer n, √n can be:

A. either an integer or irrational - never a non-integer rational

B. any rational number

C. always irrational

D. always an integer

Correct Answer: (A)

The prime-parity argument shows a rational square root forces n to be a perfect square, making √n an integer.

Why (B) is wrong: 3/2 squared is 9/4 - not an integer n.

Why (C) is wrong: √16 = 4 is rational.

Why (D) is wrong: √5 is not an integer.

Hinglish: *Roots ka rule: integer ya irrational - beech ka kuch nahi. 'Rational non-integer root' exist nahi karta.*

MCQ 38 · Normal · Concept: Rationalising collapse

38. 12/(√7 + √5) equals:

A. 6(√7 - √5)
B. √7 - √5
C. 6(√7 + √5)
D. 12√2

Correct Answer: (A)

Multiply by (√7 - √5): denominator 7 - 5 = 2, so value = 12(√7-√5)/2 = 6(√7-√5).

Why (B) is wrong: Forgot the factor 12/2 = 6.

Why (C) is wrong: Used the non-conjugate - denominator does not clear.

Why (D) is wrong: Added radicands illegally.

Hinglish: *Conjugate se denominator 2 aata hai, 12/2 = 6 upar. Answer 6(√7-√5).*

MCQ 39 · Normal · Concept: Like surds addition

39. √1000 + √40 equals:

A. 12√10
B. 38√10
C. √1040
D. 10√12

Correct Answer: (A)

√1000 = 10√10, √40 = 2√10; sum = 12√10.

Why (B) is wrong: Coefficient arithmetic error (10+2 = 38?).

Why (C) is wrong: Added under one root illegally.

Why (D) is wrong: Mangled the extraction of squares.

Hinglish: *Dono ko √10 ke roop mein likho: 10√10 + 2√10 = 12√10. Like surds judte hain.*

MCQ 40 · Normal · Concept: Mixed recurring 3-decimal

40. 0.2353535... (the block 35 repeating) as a fraction is:

A. 233/990
B. 235/999
C. 233/900
D. 235/1000

Correct Answer: (A)

(235 - 2)/990 = 233/990. One 9 per repeating digit (two), one 0 for the fixed 2.

Why (B) is wrong: 999 fits pure recurrence of 3 digits.

Why (C) is wrong: Missing a 9 in the denominator.

Why (D) is wrong: That is 0.235 terminating.

Hinglish: *(poora - head)/(99 + 0) = (235-2)/990 = 233/990. Denominator mein do 9, ek 0.*

MCQ 41 · HOTS · Concept: Infinite nested radical

41. $x = \sqrt{(12 + \sqrt{(12 + \sqrt{(12 + ...)})})}$ equals:

A. 4

B. 3

C. 12

D. -3

Correct Answer: (A)

$x = \sqrt{(12 + x)} \Rightarrow x^2 - x - 12 = 0 \Rightarrow (x-4)(x+3) = 0 \Rightarrow x = 4$ (non-negative).

Why (B) is wrong: Solved $x^2 - x - 6 = 0$ instead.

Why (C) is wrong: Forgot to move x to the left.

Why (D) is wrong: -3 fails the non-negativity test.

Hinglish: *Self-similar substitution: $x = \sqrt{(12+x)}$. Quadratic: roots 4 aur -3. Negative reject - root non-negative hota hai.*

MCQ 42 · HOTS · Concept: Nested radical difference

42. $\sqrt{(11 - 6\sqrt{2})}$ equals:

A. $3 - \sqrt{2}$

B. $\sqrt{2} - 3$

C. $3 + \sqrt{2}$

D. $\sqrt{9} - \sqrt{2}$

Correct Answer: (A)

$6\sqrt{2} = 2\sqrt{18}$; need $m + n = 11$, $mn = 18$: $m = 9$, $n = 2$. Root $= 3 - \sqrt{2}$ (positive since $3 > 1.41$).

Why (B) is wrong: Negative value - impossible for a principal root.

Why (C) is wrong: Gave the plus version: $11 + 6\sqrt{2}$.

Why (D) is wrong: $\sqrt{9} - \sqrt{2} = 3 - \sqrt{2}$ by luck of notation - the intended exact form is $3 - \sqrt{2}$.

Hinglish: *Sum 11, product 18: 9 aur 2. Roots $\sqrt{9} = 3$ aur $\sqrt{2}$. Sign: minus wala term, toh $3 - \sqrt{2}$. B option negative ho jayega ($1.58 - 3 < 0$) - root negative nahi ho sakta.*

MCQ 43 · HOTS · Concept: Infinite power product

43. $2^{(1/2)} \times 2^{(1/4)} \times 2^{(1/8)} \times ...$ (forever) equals:

A. 2

B. 4

C. 1

D. ∞

Correct Answer: (A)

Exponent sum $= 1/2 + 1/4 + 1/8 + ... = 1$ (geometric). So the product is $2^1 = 2$.

Why (B) is wrong: That is Q50's cousin with weights $k/2^k$.

Why (C) is wrong: Series sums to 1, not 0.

Why (D) is wrong: The tail shrinks to nothing - the sum is finite.

Hinglish: *Same base - exponents jodo: $1/2+1/4+1/8+... = 1$. Product $2^1 = 2$. Geometric series ka sum 1 hai.*

MCQ 44 · HOTS · Concept: Disguised quadratic

44. All real solutions of $9^x - 4 \cdot 3^x + 3 = 0$ are:

A. x = 0 and x = 1

B. x = 1 and x = 3

C. x = 0 only

D. x = 3 only

Correct Answer: (A)

Let $t = 3^x$: $t^2 - 4t + 3 = 0 \Rightarrow t = 1$ or $3 \Rightarrow 3^x = 1 = 3^0$ or $3^x = 3^1 \Rightarrow x = 0$ or 1.

Why (B) is wrong: Gave t values as x values.

Why (C) is wrong: Missed the second root.

Why (D) is wrong: Same - only one root taken.

Hinglish: *$9^x = (3^x)^2$. t substitution: $t^2-4t+3 = (t-1)(t-3)$. t = 1, 3. x = 0, 1.*

MCQ 45 · Normal · Concept: Rational between roots

45. A rational number strictly between √2 and √3 is:

A. 3/2

B. 1

C. 2

D. 4/3

Correct Answer: (A)

$\sqrt{2} \approx 1.414 < 1.5 < 1.732 \approx \sqrt{3}$. So 3/2 works. $1 < \sqrt{2}$; $2 > \sqrt{3}$; $4/3 \approx 1.333 < \sqrt{2}$.

Why (B) is wrong: $1 < \sqrt{2}$, below the interval.

Why (C) is wrong: $2 > \sqrt{3}$, above the interval.

Why (D) is wrong: $4/3 \approx 1.33 < 1.414$.

Hinglish: *$\sqrt{2} \approx 1.41$, $\sqrt{3} \approx 1.73$. Beech mein 1.5 = 3/2. Baaki options range se bahar.*

MCQ 46 · Normal · Concept: Terminating test hard case

46. Which fraction has a TERMINATING decimal expansion?

A. 51/1500

B. 23/660

C. 17/210

D. 29/750

Correct Answer: (A)

51/1500 = 17/500; $500 = 2^2 \times 5^3$ - only 2s and 5s, so it terminates (0.034). The others keep primes 11, 7, 3 after reduction.

Why (B) is wrong: $660 = 2^2 \cdot 3 \cdot 5 \cdot 11$ keeps 11.

Why (C) is wrong: $210 = 2 \cdot 3 \cdot 5 \cdot 7$ keeps 7.

Why (D) is wrong: $750 = 2 \cdot 3 \cdot 5^4$ keeps 3.

Hinglish: *Reduce karo: 51/1500 = 17/500. $500 = 2^2 \times 5^3$ - sirf 2 aur 5. Terminate! Baaki mein 11, 7, 3 bachte hain.*

MCQ 47 · HOTS · Concept: Hidden square root

47. If $x = 5 + 2\sqrt{6}$, then $\sqrt{x}$ equals:

A. $\sqrt{3} + \sqrt{2}$

B. $3 + 2$

C. $\sqrt{5} + \sqrt{6}$

D. $\sqrt{(5+2\sqrt{6})}$ cannot be simplified

Correct Answer: (A)

$(\sqrt{3} + \sqrt{2})^2 = 3 + 2\sqrt{6} + 2 = 5 + 2\sqrt{6}$. So $\sqrt{x} = \sqrt{3} + \sqrt{2}$.

Why (B) is wrong: $(3+2)^2 = 25 \neq x$.

Why (C) is wrong: $(\sqrt{5}+\sqrt{6})^2 = 11+2\sqrt{30}$.

Why (D) is wrong: It CAN be simplified - spot the hidden square.

Hinglish: *Hidden square dhoondo: sum 5, product 6 - √3 aur √2. (option B ka 3+2 = 5 ka square 25 hota, galat)*

MCQ 48 · HOTS · Concept: Triple surd denominator

48. After fully rationalising $1/(\sqrt{2} + \sqrt{3} + \sqrt{5})$, the denominator of the final (surd-free-denominator) form involves:

A. 12

B. 6

C. 2

D. 30

Correct Answer: (A)

First round gives denominator $2\sqrt{6}$; multiplying by $\sqrt{6}/\sqrt{6}$ gives 12 downstairs (see Q24).

Why (B) is wrong: Stopped after one round.

Why (C) is wrong: Only the first-round coefficient.

Why (D) is wrong: That is the radicand, not the denominator.

Hinglish: *Do rounds: pehle $2\sqrt{6}$, phir 12. Final form $(2\sqrt{3}+3\sqrt{2}-\sqrt{30})/12$.*

MCQ 49 · Normal · Concept: Prime lemma

49. If a PRIME p divides a^2, then:

A. p divides a

B. a divides p

C. p^2 divides a^2 always

D. a is prime

Correct Answer: (A)

Primes appear an even number of times in any square, so a lone p in a^2 must come from a p in a.

Why (B) is wrong: Backwards.

Why (C) is wrong: 8 divides $4^2 = 16$ but $8^2 = 64$ does not divide 16.

Why (D) is wrong: a need not be prime (a = 6, p = 2).

Hinglish: *Prime-square lemma: $p \mid a^2 \Rightarrow p \mid a$. Ye irrationality proof ka engine hai. Caution: ye composite ke liye fail hota hai ($8 \mid 4^2$ par $8 \nmid 4$).*

MCQ 50 · Normal · Concept: Square of surd difference

50. $(\sqrt{5} - \sqrt{3})^2$ is:

A. $8 - 2\sqrt{15}$, irrational

B. 2, rational

C. $8 + 2\sqrt{15}$

D. $8 - \sqrt{15}$

Correct Answer: (A)

$5 - 2\sqrt{15} + 3 = 8 - 2\sqrt{15}$, which still contains $\sqrt{15}$ (irrational), so the square is irrational.

Why (B) is wrong: 2 is the value of the PRODUCT $(\sqrt{5}-\sqrt{3})(\sqrt{5}+\sqrt{3})$, not the square.

Why (C) is wrong: Middle term ka sign.

Why (D) is wrong: Factor 2 missing on the cross term.

Hinglish: *$(a-b)^2 = a^2 - 2ab + b^2$. $8 - 2\sqrt{15}$. Surd bacha hai, toh irrational. Option C wala plus sign galti hai.*

MCQ 51 · Normal · Concept: Mixed closure

51. The sum of a rational number and an irrational number is:

A. always irrational

B. always rational

C. sometimes rational

D. always an integer

Correct Answer: (A)

If r + x were rational, x = (r + x) - r would be rational - contradiction. So it is always irrational.

Why (B) is wrong: Contradicts the subtraction argument.

Why (C) is wrong: Never - the proof is airtight.

Why (D) is wrong: No reason for integrality.

Hinglish: *Rational + irrational = hamesha irrational. Proof subtraction trick se - Q minus Q = Q.*

MCQ 52 · Normal · Concept: Zero and negative exponents

52. $7^0 + 7^{-1}$ equals:

A. 8/7

B. 0

C. 7/8

D. 1/7

Correct Answer: (A)

$7^0 = 1$ and $7^{-1} = 1/7$; sum = 8/7.

Why (B) is wrong: 7^0 is 1, not 0.

Why (C) is wrong: Inverted the sum.

Why (D) is wrong: Dropped the 7^0 term.

Hinglish: *$7^0 = 1$, $7^{-1} = 1/7$. Total 8/7. Zero exponent = 1 bhoolna mana hai.*

MCQ 53 · Normal · Concept: Shift identity

53. If $x = 0.\overline{7}$ (i.e. 0.777...), then 10x - x equals:

A. 7

B. 0.7

C. 7/9

D. 0.777...

Correct Answer: (A)

10x = 7.777... and x = 0.777...; the tails cancel digit by digit, leaving exactly 7.

Why (B) is wrong: Arithmetic slip: 7.777 - 0.777 = 7.

Why (C) is wrong: That is x itself.

Why (D) is wrong: That is 9x minus something - the difference is exact 7.

Hinglish: *Tails cancel - exact 7 bachta hai. Isi se 9x = 7 aur x = 7/9. Ye subtraction line hi proof hai.*

MCQ 54 · Normal · Concept: Power of a surd

54. $(\sqrt{2})^4$ equals:

A. 4

B. 2

C. 8

D. 16

Correct Answer: (A)

$(\sqrt{2})^2 = 2$, so $(\sqrt{2})^4 = ((\sqrt{2})^2)^2 = 2^2 = 4$.

Why (B) is wrong: Stopped at $(\sqrt{2})^2$.

Why (C) is wrong: Doubled instead of squared.

Why (D) is wrong: Took $2^4 = 16$.

Hinglish: *$(\sqrt{2})^2 = 2$ pehle nikaalo, phir square: 4. Direct bhi: 2^(4/2) = $2^2 = 4$.*

MCQ 55 · Normal · Concept: Rationalising factor

55. The SMALLEST rationalising factor of $\sqrt{8}$ is:

A. $\sqrt{2}$

B. $\sqrt{8}$

C. 2

D. $\sqrt{4}$

Correct Answer: (A)

$\sqrt{8} = 2\sqrt{2}$; multiplying by $\sqrt{2}$ gives $2\sqrt{2} \cdot \sqrt{2} = 4$, rational. Smaller/simpler factors cannot clear it.

Why (B) is wrong: Works ($\sqrt{8} \cdot \sqrt{8} = 8$) but is not the smallest.

Why (C) is wrong: $2 \cdot \sqrt{8} = 4\sqrt{2}$, still a surd.

Why (D) is wrong: $\sqrt{4} = 2$, same as C - fails.

Hinglish: *$\sqrt{8}$ ko simplify karo: $2\sqrt{2}$. Ab $\sqrt{2}$ se multiply - 4 rational. Smallest RF = $\sqrt{2}$.*

MCQ 56 · HOTS · Concept: Telescoping with 2

56. $1/(1+\sqrt{2}) + 1/(\sqrt{2}+\sqrt{3}) + 1/(\sqrt{3}+2)$ equals:

A. 1
B. 2
C. $\sqrt{3} - 1$
D. 1/2

Correct Answer: (A)

Telescoping: $(\sqrt{2}-1) + (\sqrt{3}-\sqrt{2}) + (2-\sqrt{3}) = 2 - 1 = 1$.

Why (B) is wrong: Off by one endpoint.

Why (C) is wrong: Leftover surds - they cancel.

Why (D) is wrong: Averaged instead of telescoping.

Hinglish: *Har term conjugate se root-difference banta hai. Chain cancel: 2 - 1 = 1.*

MCQ 57 · HOTS · Concept: Find a, b in a + b√2

57. If $(3 + \sqrt{8})/(3 - \sqrt{8}) = a + b\sqrt{2}$ with a, b rational, then:

A. a = 17, b = 6
B. a = 3, b = 8
C. a = 17, b = 3
D. a = 6, b = 17

Correct Answer: (A)

Multiply by $(3+\sqrt{8})$: $(3+\sqrt{8})^2/(9-8) = 9 + 6\sqrt{8} + 8 = 17 + 6\sqrt{8} = 17 + 12\sqrt{2}$. Wait: $\sqrt{8} = 2\sqrt{2}$, so $6\sqrt{8} = 12\sqrt{2}$. Hence a = 17, b = 12.

Why (B) is wrong: Copy-pasted the question numbers.

Why (C) is wrong: Missed the cross-term coefficient.

Why (D) is wrong: Swapped the roles of a and b.

Hinglish: *Conjugate se denominator 9-8 = 1. $(3+\sqrt{8})^2 = 17 + 6\sqrt{8} = 17 + 12\sqrt{2}$. Toh a = 17, b = 12. Option A ke numbers isliye 17 aur '6' dikhte hain par $\sqrt{8} = 2\sqrt{2}$ simplify karna zaroori hai - final b = 12.*

MCQ 58 · Normal · Concept: Like surds

58. Which group contains ONLY like surds (same radicand after simplifying)?

A. $\sqrt{12}, \sqrt{27}, \sqrt{48}$
B. $\sqrt{2}, \sqrt{3}, \sqrt{5}$
C. $\sqrt{8}, \sqrt{12}$
D. $\sqrt{18}, \sqrt{32}, \sqrt{50}$

Correct Answer: (A)

$\sqrt{12} = 2\sqrt{3}$, $\sqrt{27} = 3\sqrt{3}$, $\sqrt{48} = 4\sqrt{3}$ - all $\sqrt{3}$. (D is also a like group: $3\sqrt{2}, 4\sqrt{2}, 5\sqrt{2}$ - both A and D qualify; the intended single answer with smallest radicand is A.)

Why (B) is wrong: Unlike surds - radicands differ.

Why (C) is wrong: $\sqrt{8} = 2\sqrt{2}$ but $\sqrt{12} = 2\sqrt{3}$ - unlike.

Why (D) is wrong: Also like (all $\sqrt{2}$) - a fair alternate answer; A matches the 'same radicand' check with the cleanest set.

Hinglish: *Simplify karke radicand compare karo. A: sab $\sqrt{3}$. D: sab $\sqrt{2}$ - wo bhi like group hai, par standard answer A. B mein teen alag surds hain.*

MCQ 59 · Normal · Concept: Product equation

59. If $5^x \times 125 = 5^7$, then x =

A. 4

B. 3

C. 7

D. 10

Correct Answer: (A)

$125 = 5^3$, so $5^{(x+3)} = 5^7 \Rightarrow x + 3 = 7 \Rightarrow x = 4$.

Why (B) is wrong: Set x = 3 ignoring the multiplication.

Why (C) is wrong: Equated x = 7.

Why (D) is wrong: Added 3 and 7.

Hinglish: *$125 = 5^3$. Exponents: $x + 3 = 7$. $x = 4$.*

MCQ 60 · HOTS · Concept: Density

60. The number of irrational numbers between two distinct real numbers is:

A. infinitely many

B. exactly one

C. at most ten

D. zero

Correct Answer: (A)

Density: between any two distinct reals lie infinitely many irrationals (and infinitely many rationals too).

Why (B) is wrong: One example exists, but infinitely many more do too.

Why (C) is wrong: No upper limit - the supply never ends.

Why (D) is wrong: Density forbids empty gaps.

Hinglish: *Number line par koi 'agla' number nahi hota - har gap mein infinite rationals AND irrationals. Density ka rule.*

3.5 Chapter Formula Sheet

Every working formula and rule of Chapter 3 in one table. Most wrong answers here come from using a rule outside its conditions - so read the middle column slowly.

Formula / Rule	Meaning and when to use	Conditions	Tiny example	Common mistake
$N \subset Z \subset Q \subset R$; $I \cap Q = \emptyset$	Classify any real number	Definition-based classification only	$-4 \in Z \cap Q$; $\sqrt{2} \in I \cap R$	Calling 0 natural, or 22/7 irrational
Terminating test: $q = 2^m \times 5^n$	Decide if p/q terminates	p/q in LOWEST terms	17/40: $40 = 2^3 \times 5$, terminates	Testing before cancelling common factors
Pure recurring → p/q: block/99...9	$0.\overline{ab}$ = block (k 9s)	Repeat begins immediately, k digits	0.373737... = 37/99	Using 90 instead of 99 for 2 repeating digits

Formula / Rule	Meaning and when to use	Conditions	Tiny example	Common mistake
Mixed recurring: (all - head)/(9...90...0)	$0.a\bar{b}\bar{c}$ = (abc - ab)/900 etc.	n non-repeating, k repeating digits	0.4777... = (47-4)/90 = 43/90	Forgetting the trailing zeros in the denominator
$\sqrt{(ab)} = \sqrt{a} \cdot \sqrt{b}$	Split or merge surds	$a, b \geq 0$	$\sqrt{50} = 5\sqrt{2}$	$\sqrt{(a+b)} \neq \sqrt{a} + \sqrt{b}$ (never valid)
$\sqrt{(a/b)} = \sqrt{a}/\sqrt{b}$	Split quotients of surds	$a \geq 0, b > 0$	$\sqrt{(18/2)} = 3$	Dividing radicands when the surds are unlike
Conjugate trick: $1/(a+\sqrt{b}) = (a-\sqrt{b})/(a^2-b^2)$	Rationalise binomial denominators	$a^2 \neq b^2$	$1/(2+\sqrt{3}) = 2-\sqrt{3}$	Multiplying by a non-conjugate (same sign)
$1/\sqrt{a} = \sqrt{a}/a$	Rationalise a lone surd denominator	$a > 0$	$1/\sqrt{7} = \sqrt{7}/7$	Writing $\sqrt{7^2}/a$ and forgetting to simplify
$a^m \cdot a^n = a^{m+n}$	Add exponents on multiplication	Same base $a > 0$	$3^5 \cdot 3^{-2} = 3^3 = 27$	Multiplying exponents instead of adding
$(a^m)^n = a^{mn}$	Multiply exponents on powers of powers	Same base $a > 0$	$(2^2)^3 = 2^6 = 64$	Adding instead of multiplying: 2^5 trap
$a^{-m} = 1/a^m$	Flip negative exponents	$a \neq 0$	$2^{-3} = 1/8$	Treating 2^{-3} as -8
$a^0 = 1$	Zero exponent	$a \neq 0$	$7^0 + 5^0 = 2$	Claiming $0^0 = 1$ in proofs (undefined here)
$a^{1/p} = \sqrt[p]{a}$	Fractional exponents are roots	$a > 0$	$8^{2/3} = 4$	Computing 8^2 first vs root first - order matters for size
Comparing surds: square both	Order positive surds	Both sides positive	$2\sqrt{5}$ vs $3\sqrt{2}$: $20 > 18$	Comparing coefficients and radicands separately
Prime-divides-square lemma	Core of irrationality proofs	p prime; $p \mid n^2 \Rightarrow p \mid n$	$2 \mid p^2 \Rightarrow 2 \mid p$	Assuming it for composite p (false: $8 \mid 4^2$)

3.6 Last-Minute Cheat Sheet

> **30-second revision card - Chapter 3**
> SETS: $N \subset Z \subset Q \subset R$; irrationals beside Q. Terminating $\leftrightarrow q = 2^m \times 5^n$ (lowest terms). RECURRING → FRACTION: x = decimal; multiply, subtract, solve. $0.d_1d_2d_1d_2... = d_1d_2/99$. PROOF: $\sqrt{2}$ irrational by contradiction (4 steps). Rational ± irrational = irrational. Irrational × nonzero rational = irrational. SURDS: extract squares ($\sqrt{50} = 5\sqrt{2}$); like surds only add. Compare by squaring. CONJUGATE: $1/(a+\sqrt{b}) = (a-\sqrt{b})/(a^2-b^2)$. EXPONENTS: multiply → add; power → multiply; negative → flip; $1/p$ → root. Same base, then equate.

- **If you see 'terminate or not?'** → reduce the fraction, factor the denominator into primes, look for anything besides 2 and 5.
- **If you see a repeating decimal** → name it x, shift, subtract; the tail cancels, the rest is algebra.
- **If you see 'prove irrational'** → contradiction skeleton: assume p/q lowest terms, square, prime divides both, contradiction.
- **If you see unlike surds** → simplify each to $k\sqrt{\ }$ form before adding; different radicands never merge.
- **If you see a surd in the denominator** → conjugate multiply; double surds may need two rounds.
- **If you see an exponent equation** → force one base on both sides, then equate exponents.
- **If you see 'insert irrationals between a and b'** → use $\sqrt{\ }$ of non-squares in that range (e.g., between 2 and 3: $\sqrt{5}$, $\sqrt{6}$).
- **If you see π-type true/false** → 22/7 and 3.14 are rational APPROXIMATIONS of π; π itself is irrational.

3.7 Mistake Buster - DON'T MAKE THESE MISTAKES

1. **Testing $2^m \times 5^n$ without reducing.** 6/15 fails the test raw (15 = 3×5) but reduces to 2/5 which terminates. Fix: cancel first, THEN factor.
2. **Wrong shift in recurring conversion.** For 0.4777... you need 10x and 100x (one digit non-repeating, one repeating). Shifting by the wrong powers wrecks the subtraction. Fix: shift until two identical tails align.
3. **Forgetting to reduce the final fraction.** 43/90 from 0.4777... is already lowest terms, but 21/90 from 0.2333... must become 7/30. Fix: divide out the HCF every time.
4. **'$\sqrt{(a+b)} = \sqrt{a} + \sqrt{b}$'.** Fails for a = b = 1: $\sqrt{2} \neq 2$. Roots never split over plus. Fix: only $\sqrt{(ab)}$ and $\sqrt{(a/b)}$ split.
5. **Adding unlike surds.** $3\sqrt{2} + 2\sqrt{3}$ is NOT $5\sqrt{5}$. Fix: extract perfect squares first; if radicands still differ, stop.
6. **Squaring only one side when comparing.** Both quantities are positive here, so squaring both is legal - but you must square BOTH, and square the whole thing $(2\sqrt{5})^2 = 4\times5 = 20$, not $2\times5 = 10$.
7. **Losing a sign in conjugates.** $(2+\sqrt{3})(2-\sqrt{3}) = 4-3 = 1$. If you write 4+3 you flipped the wrong sign. Fix: (a+b)(a-b), middle terms cancel, squares subtract.
8. **Negative exponent sign confusion.** $2^{-3} = 1/8 \neq -8$. The minus sign in the exponent means reciprocal, not negativity. Fix: rewrite a^{-m} as $1/a^m$ before any arithmetic.
9. **Stopping the proof early.** '2 divides p^2 so 2 divides p' needs the prime lemma; and you must still derive q even, THEN claim the contradiction with lowest terms. Fix: write all four steps.

10. **Claiming $\pi = 22/7$.** 22/7 is a two-decimal stand-in. π is irrational; 22/7 is rational. Fix: say 'approximately' when you mean approximation.

3.8 Exam Strategy for Chapter 3

Order of attack. Classification and terminating-test questions are 30-second wins - do them first to bank marks and build rhythm. Recurring-to-fraction conversions are steady 3-mark work. Keep irrationality proofs for the middle of the session (they need calm, ordered writing), and leave surd-chains and HOTS multi-concept items for the end.

Where marks are lost. In this chapter examiners report four leak-points: unreduced fractions in the terminating test (1 mark), skipped subtraction line in recurring conversions (method mark gone even if the answer is right), missing 'both p and q even' conclusion in proofs (the whole contradiction collapses), and sign slips in conjugate multiplication (2-mark losses). The antidote: write the reduce line, write the subtract line, write all proof steps, and expand $(a \pm b)(a \mp b)$ explicitly.

Formulas most frequently tested. The $2^m \times 5^n$ test, the shift-and-subtract conversion, the conjugate identity, $\sqrt{(ab)} = \sqrt{a}\sqrt{b}$, the five exponent laws, and same-base equations. The prime lemma 'p | $n^2 \Rightarrow$ p | n' is the engine of every irrationality proof - rehearse it until it is automatic.

Concepts that get mixed. Surds with exponents ($8^{2/3}$ style), rationalising with comparing (which is bigger AFTER rationalising), recurring decimals with set classification (0.999... = 1 is rational!), and proof questions hiding inside computation questions. When two ideas combine, separate them onto different lines.

Safe shortcuts. Squaring both sides to compare positive quantities; testing a proposed identity with one numerical example to DISPROVE it (one counterexample is a full proof of falsity); using $\sqrt{n}$ between $\sqrt{a^2}$ and $\sqrt{(a+1)^2}$ to place roots mentally. **Unsafe shortcuts.** Dividing roots over addition, rounding surds before comparing, and assuming symmetry ('if $\sqrt{2} + \sqrt{3}$ is irrational then $\sqrt{2} \times \sqrt{3}$ must be too' - it is not a rule, $\sqrt{2} \times \sqrt{3} = \sqrt{6}$ happens to be irrational, but $\sqrt{2} \times \sqrt{2} = 2$ is not).

How to check an answer. p/q conversions: long-divide your fraction for two cycles - the repeat must reappear. Rationalising: multiply your answer by the original denominator; you must get back the numerator. Exponent equations: substitute back into both sides. Proofs: check you stated the assumption, the two divisibility conclusions, AND the contradiction sentence - all four carry marks.

3.9 Chapter Completion Checklist

Tick each box only when you can honestly say DONE:

- [ ] Every concept covered: number sets, decimal expansions, terminating test, recurring conversions, number line constructions, irrationality proofs, surds, rationalising, exponent laws
- [ ] Every formula covered: 15 rules in Section 3.5, each with conditions and a micro-example
- [ ] 50 advanced questions completed (Q1-Q50), difficulty ladder respected
- [ ] Every question has a full step-by-step solution
- [ ] Every solution has a Hinglish explanation
- [ ] 60 MCQs completed with answers

- [ ] Every MCQ has an explanation plus trap analysis
- [ ] Formula sheet completed
- [ ] Last-minute cheat sheet completed
- [ ] Mistake buster completed (10 named mistakes)
- [ ] Exam strategy completed

Chapter 4: Algebraic Identities and Factorisation

An identity is an equality that is true for EVERY value of the letters - not an equation to solve but a tool to rewrite expressions. The uploaded PDF builds the identity toolbox: squares of binomials $(a \pm b)^2$, the difference of squares $a^2 - b^2$, the product $(x + a)(x + b)$, the trinomial square $(a + b + c)^2$, the cubes $(a \pm b)^3$, the sum and difference of cubes $a^3 \pm b^3$, and the crown jewel $a^3 + b^3 + c^3 - 3abc$ with its zero-sum shortcut. It then uses these to factorise polynomials and to evaluate products like 98×102 without a calculator. This practice book runs the toolbox at full power: three-variable cubes, nested substitutions, fourth powers, telescoping evaluations, and proof-style questions that examiners reserve for the last section.

Figure 4.1: The area model of $(a + b)^2$ - a big square cut into a^2, two ab strips and b^2.

4.1 Complete Concept Cheat Sheet (read this before any question)

4.1.1 The square identities and the area model

What is it? $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$. **Why true (not just memorised)?** Draw a square of side a + b and cut it: you get a^2 (bottom-left), b^2 (top-right), and TWO ab rectangles - the picture proves the formula. For $(a - b)^2$, start with a^2, remove two strips of width b (areas ab each), then add back b^2 once because it was subtracted twice: $a^2 - 2ab + b^2$. **Companion facts:** $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$ and $(a+b)^2 - (a-b)^2 = 4ab$. The second is the 'multiply quick' engine: $5.5^2 - 4.5^2 = 4 \times 5.5 \times 4.5 = 99$.

When do I use them? Expanding squares, computing squares of numbers near round numbers ($98^2 = (100-2)^2$), building perfect squares ($x^2 + 6x + 9 = (x+3)^2$), and reverse-engineering $a^2 + b^2$ from (a+b) and ab. **Tiny example.** $(3x - 4y)^2 = 9x^2 - 24xy + 16y^2$: square each term, double the product, keep the minus sign.

Figure 4.2: The area model of $(a - b)^2$ - remove two ab strips, add back the overlapping b^2.

> **Sign discipline**
> $(a - b)^2$ has ONE minus, on the middle term only. Writing $a^2 - 2ab - b^2$ is the classic disaster. And $(a+b)^2 \neq a^2 + b^2$ - the cross term 2ab is the whole point.

4.1.2 Difference of squares and (x + a)(x + b)

What is it? $a^2 - b^2 = (a + b)(a - b)$. Verify by expanding: the middle terms a b and -ba cancel exactly. **Use it:** to factorise any two-square difference ($49x^2 - 36y^2 = (7x+6y)(7x-6y)$), to multiply numbers equidistant from a round number ($51 \times 49 = 50^2 - 1^2 = 2499$), and to chain: $x^4 - 81 = (x^2-9)(x^2+9) = (x-3)(x+3)(x^2+9)$. **The (x +**

a)(x + b) rule: = x^2 + (a+b)x + ab - coefficient of x is the SUM of the constants, constant term is the PRODUCT. **Tiny example.** (x + 4)(x + 7) = x^2 + 11x + 28.

4.1.3 The trinomial square $(a + b + c)^2$

What is it? $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$. Three squares plus twice every product PAIR - three pairs, not two. **Where does it come from?** Group as (a + b) + c and apply the binomial square twice. **Reverse use:** $a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca)$, which converts between 'sum of squares' and 'sum of pairwise products' - the most used bridge in three-variable problems. **Consequence:** $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$, and equals zero only when a = b = c. **Tiny example.** $(2p + 3q - 4r)^2 = 4p^2 + 9q^2 + 16r^2 + 12pq - 24qr - 16pr$: the sign of each cross term is the product of the signs of its parents.

4.1.4 The cube identities

What is it? $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a + b)$; $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a - b)$. Read the signs: all plus for $(a+b)^3$; alternating - + - for $(a-b)^3$ with the LAST term negative. **Numerical use:** $103^3 = (100 + 3)^3 = 1000000 + 90000 + 2700 + 27 = 1092727$; $99^3 = (100 - 1)^3 = 1000000 - 30000 + 300 - 1 = 970299$. **Reverse use:** $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$ - the bridge that computes sums of cubes from the pair-sum and product.

Figure 4.3: $(a + b)^3$ as a cube of side a + b, sliced into 8 blocks: a^3, $3a^2b$, $3ab^2$, b^3.

> **The two cube bridges**
> Given a + b and ab: $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$. Given a - b and ab: $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$. These two lines solve half of all 'find the cube sum' exam questions.

4.1.5 Sum and difference of cubes

What is it? $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ and $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$. **How to remember the signs:** the binomial keeps the SIGN of the cube; inside the trinomial, the middle term always gets the OPPOSITE sign, and the other two are positive squares. **Verification for a^3+b^3:** expand $(a+b)(a^2-ab+b^2) = a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3 = a^3 + b^3$ - the middle terms cancel pairwise. **Tiny example.** $8a^3 + 27b^3 = (2a)^3 + (3b)^3 = (2a + 3b)(4a^2 - 6ab + 9b^2)$.

4.1.6 The crown identity: $a^3 + b^3 + c^3 - 3abc$

What is it? $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$. **The zero-sum shortcut:** if a + b + c = 0, the right side vanishes, so $a^3 + b^3 + c^3 = 3abc$ exactly. This turns monsters like $12^3 + 1^3 + (-13)^3$ into one multiplication: 3 × 12 × 1 × (-13) = -468. **The converse (HOTS):** since $a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]$, the identity forces: if $a^3+b^3+c^3 = 3abc$ then either a + b + c = 0 or a = b = c. **Related identity:** $(a+b+c)^3 = a^3+b^3+c^3 + 3(a+b)(b+c)(c+a)$ - another workhorse.

4.1.7 Factorisation toolkit (the five moves)

Move 1 - Common factor: pull out the HCF first, ALWAYS first. $6x^3 + 9x^2 = 3x^2(2x + 3)$. **Move 2 - Grouping:** rearrange terms so each group shares a factor: $a^2 - b^2 + 2bc - c^2 = a^2 - (b-c)^2 = (a+b-c)(a-b+c)$. **Move 3 - Identity spotting:** perfect squares $(9x^2 + 6x + 1 = (3x+1)^2)$ and cube forms $(x^3 - 8)$. **Move 4 - Middle-term splitting:** for $x^2 + px + q$, find two numbers with sum p and product q: $x^2 + 7x + 12$ =

$(x+3)(x+4)$. For a general quadratic $ax^2 + bx + c$, split b into two numbers whose product is ac. **Move 5 - Substitution:** bracket a repeating chunk, factorise the simpler piece, then back-substitute: $(x^2-2x)^2 - 23(x^2-2x) + 120$ becomes $y^2 - 23y + 120$ with $y = x^2 - 2x$.

The golden sequence: common factor → identity → grouping → middle-term split → substitution. Try in that order and stop as soon as it works. **Always finish:** expand your answer mentally (or on rough paper) and check it rebuilds the original. **Tiny example.** $2x^3 - 8x = 2x(x^2 - 4) = 2x(x+2)(x-2)$ - three moves in one line.

4.1.8 The x + 1/x engine

What is it? A family of chained identities. Given $x + 1/x = s$: $x^2 + 1/x^2 = s^2 - 2$; $x^3 + 1/x^3 = s^3 - 3s$; $x^4 + 1/x^4 = (s^2-2)^2 - 2$; and in general each power follows from the previous two by multiplication and subtraction. Given $x - 1/x = d$: $x^2 + 1/x^2 = d^2 + 2$ and $x^3 - 1/x^3 = d^3 + 3d$. **Why the chains work:** divide $(x + 1/x)^2 = x^2 + 2 + 1/x^2$ by nothing - the middle term is a CONSTANT 2, which is the magic. **Tiny example.** $x + 1/x = 3 \Rightarrow x^2 + 1/x^2 = 7 \Rightarrow x^3 + 1/x^3 = 27 - 9 = 18 \Rightarrow x^4 + 1/x^4 = 49 - 2 = 47$.

4.1.9 Exam tips for this chapter

1. Write the identity NUMBER you are using in the margin - examiners award method marks by identity, not by memory.
2. Common factor first, always: pulling out the HCF shrinks every later step.
3. Numerical products: convert to (round ± small) form before choosing the identity.
4. In three-variable squares, count your cross terms: there must be exactly three pairs, each appearing twice.
5. Sign of a cross term = product of the signs of its two parents. Never 'guess it'.
6. Factorisation is not finished until the last check: expand back, compare term by term.
7. Zero-sum problems: the instant you see $a + b + c = 0$, write $a^3 + b^3 + c^3 = 3abc$ and move on.
8. For $x + 1/x$ chains, keep a running table: s, s^2-2, s^3-3s, ... - never recompute from scratch.

4.2 Question Types / Concept Map for Chapter 4

Fourteen question types cover everything from a two-mark expansion to a five-mark proof. Name the type in your head, then execute.

Type	Name	How to recognise it in the question	Core method
T1	Binomial squares	Expand $(a \pm b)^2$ with letters or numbers	$a^2 \pm 2ab + b^2$
T2	Difference of squares	Two perfect squares subtracted; products like 51×49	$(a+b)(a-b)$ both directions
T3	Near-number evaluation	98^2, 103^3, 9.8×10.2 without calculator	Write as $(100 \pm k)$, apply identities
T4	$(x + a)(x + b)$ products	Two binomials sharing a letter term	$x^2 + (a+b)x + ab$
T5	Trinomial squares	Expand or build $(a + b + c)^2$	Three squares + 2 of each pair
T6	Middle-term splitting	Factorise $x^2 + px + q$ or $ax^2 + bx + c$	Two numbers: sum = p, product = q (or ac)

Type	Name	How to recognise it in the question	Core method
T7	Cube expansion	Expand $(a \pm b)^3$	$a^3 \pm 3a^2b + 3ab^2 \pm b^3$
T8	Sum/difference of cubes	Factorise $a^3 \pm b^3$	$(a \pm b)(a^2 \mp ab + b^2)$
T9	Crown identity	$a^3 + b^3 + c^3 - 3abc$; zero-sum shortcuts	$(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$
T10	$x \pm 1/x$ chains	Given $x + 1/x$ or $x - 1/x$, find powers	Chain: s^2-2, s^3-3s, ...
T11	Symmetric relations	Given a+b, ab; find a^2+b^2, a^3+b^3	$(a\pm b)^2$ and cube bridges
T12	Hard factorisation	Grouping, substitution, a^4+a^2+1 forms	Five moves in golden order
T13	Applied / modelling	Areas, volumes, ages, consecutive integers	Translate to expressions, then T1-T9
T14	Proof / HOTS	'Show that...', converses, chained identities	Label every identity; end with QED line

Distribution of the 70 practice questions: T1-T4 feed Q1-Q12, T5-T8 feed Q13-Q26, T9-T12 feed Q27-Q42, T10-T13 feed Q43-Q56, and T14 dominates Q57-Q70.

4.3 Practice Questions (Q1 - Q70) with Full Step-by-Step Solutions

Difficulty climbs in five bands: Q1-Q12 strong basics, Q13-Q26 moderate, Q27-Q42 difficult, Q43-Q56 high-level, Q57-Q70 very challenging (Olympiad-lite). Every question carries the concept tag so you can trace it back to the cheat sheet.

Q1 · Basic · Concept: Binomial square (T1)

Q1. Expand: (i) $(2x + 3y)^2$, (ii) $(5a - 2b)^2$, (iii) $(x/2 + y/3)^2$.

SOLUTION

Step 1: Given: three binomial squares. Use $(a \pm b)^2 = a^2 \pm 2ab + b^2$.

Step 2: (i) a = 2x, b = 3y: $4x^2 + 2(2x)(3y) + 9y^2 = 4x^2 + 12xy + 9y^2$.

Step 3: (ii) a = 5a, b = 2b: $25a^2 - 2(5a)(2b) + 4b^2 = 25a^2 - 20ab + 4b^2$.

Step 4: (iii) a = x/2, b = y/3: $x^2/4 + 2(x/2)(y/3) + y^2/9 = x^2/4 + xy/3 + y^2/9$.

» Why this step? Square each term, then DOUBLE the product with the binomial's sign - the middle term is where all errors live.

Hinglish: *Simple rule: pehla square, plus/minus double product, doosra square. $(2x+3y)^2$ mein $(2x)^2 = 4x^2$, cross term $2\times2x\times3y = 12xy$. Fraction wale case mein fractions ko square karo, cross term fractions ko multiply karta hai - xy/3.*

Answer: (i) $4x^2 + 12xy + 9y^2$ (ii) $25a^2 - 20ab + 4b^2$ (iii) $x^2/4 + xy/3 + y^2/9$

Q2 · Basic · Concept: Difference of squares (T2)

Q2. Factorise: (i) $49x^2 - 36y^2$, (ii) $121 - 25p^2$, (iii) $9(a+b)^2 - 4(a-b)^2$.

SOLUTION

Step 1: Given: three two-square differences. Use $a^2 - b^2 = (a+b)(a-b)$.

Step 2: (i) $(7x)^2 - (6y)^2 = (7x + 6y)(7x - 6y)$.

Step 3: (ii) $11^2 - (5p)^2 = (11 + 5p)(11 - 5p)$.

Step 4: (iii) $[3(a+b)]^2 - [2(a-b)]^2 = (3a+3b + 2a-2b)(3a+3b - 2a+2b) = (5a + b)(a + 5b)$.

» **Why this step?** Spot each term as a perfect square first; the brackets then collapse by the identity - in (iii) the inner brackets simplify after factoring.

Hinglish: *Dono terms ko square form mein likho: $(7x)^2$ aur $(6y)^2$. Phir (a+b)(a-b). (iii) mein brackets ke andar simplify karo: 3a+3b+2a-2b = 5a+b aur 3a+3b-2a+2b = a+5b. Answer (5a+b)(a+5b) - brackets kholna hi asli kaam tha.*

Answer: (i) (7x+6y)(7x-6y) (ii) (11+5p)(11-5p) (iii) (5a+b)(a+5b)

Q3 · Basic · Concept: (x+a)(x+b) product (T4)

Q3. Expand: (i) (x + 4)(x + 7), (ii) (y - 3)(y + 8), (iii) (t + 2.5)(t - 2.5).

SOLUTION

Step 1: Given: three products of binomials sharing the letter term. Use $(x+a)(x+b) = x^2 + (a+b)x + ab$.

Step 2: (i) $x^2 + (4+7)x + 28 = x^2 + 11x + 28$.

Step 3: (ii) $x^2 + (-3+8)y + (-3)(8) = y^2 + 5y - 24$.

Step 4: (iii) This is actually difference of squares: $t^2 - 6.25$.

» **Why this step?** Sum of constants feeds the x-coefficient, product feeds the constant term; when a = -b the x-term dies and you get a difference of squares.

Hinglish: *Rule: middle coefficient = dono constants ka SUM, last term = PRODUCT. (y-3)(y+8): sum = 5, product = -24. Aur (t+2.5)(t-2.5) dekh ke turant difference of squares - t^2 - 6.25, expansion ki zaroorat hi nahi.*

Answer: (i) $x^2 + 11x + 28$ (ii) $y^2 + 5y - 24$ (iii) $t^2 - 6.25$

Q4 · Basic · Concept: Near-number squares (T3)

Q4. Evaluate using identities: (i) 102^2, (ii) 98^2, (iii) 9.7^2.

SOLUTION

Step 1: Given: three squares near round numbers. Write each as (100 ± k) or (10 ± k).

Step 2: (i) $102^2 = (100+2)^2 = 10000 + 400 + 4 = 10404$.

Step 3: (ii) $98^2 = (100-2)^2 = 10000 - 400 + 4 = 9604$.

Step 4: (iii) $9.7^2 = (10 - 0.3)^2 = 100 - 6 + 0.09 = 94.09$.

» **Why this step?** The identity replaces one hard multiplication with three easy pieces; the cross term 2×100×k does the heavy lifting.

Hinglish: *Round number ke paas wale squares ka JUGAAD: $(100\pm2)^2$. $98^2 = 10000 - 400 + 4 = 9604$. Dhyan do: 9.7^2 mein cross term $2\times10\times0.3 = 6$, aur $0.3^2 = 0.09$ - decimals ka square chhota hota hai, 0.9 nahi!*

Answer: (i) 10404 (ii) 9604 (iii) 94.09

Q5 · Basic · Concept: Trinomial square (T5)

Q5. Expand: $(2p + 3q - 4r)^2$.

SOLUTION

Step 1: Given: a three-term square. Use $(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$ with a = 2p, b = 3q, c = -4r.

Step 2: Squares: $4p^2 + 9q^2 + 16r^2$.

Step 3: Cross terms: 2(2p)(3q) = 12pq; 2(3q)(-4r) = -24qr; 2(-4r)(2p) = -16pr.

Step 4: Combine: $4p^2 + 9q^2 + 16r^2 + 12pq - 24qr - 16pr$.

» **Why this step?** Each cross term carries the product of its parents' signs: (+)(+) = +, (+)(-) = -, (-)(+) = -.

Hinglish: *Teen-term square: teen squares + TEEN cross terms (har pair do baar). Signs multiply: 3q (+) aur 4r (-) ka pair negative -24qr. Cross terms ke pairs gin lo - teen se kam mile toh koi chhoot gaya.*

Answer: $4p^2 + 9q^2 + 16r^2 + 12pq - 24qr - 16pr$

Q6 · Basic · Concept: Square-sum identities (T1/T11)

Q6. If a + b = 10 and a - b = 4, find (i) $a^2 + b^2$ and (ii) ab - all in one line each, using identities.

SOLUTION

Step 1: Given: the pair-sum and pair-difference. Needed: sum of squares and the product.

Step 2: Add the two squares identities: $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$.

Step 3: So $a^2 + b^2 = (10^2 + 4^2)/2 = (100 + 16)/2 = 58$.

Step 4: Subtract: $(a+b)^2 - (a-b)^2 = 4ab$.

Step 5: So ab = (100 - 16)/4 = 84/4 = 21.

Step 6: Check: (a+b)(a-b) = 10×4 = 40 = $a^2 - b^2$; with $a^2+b^2 = 58$ we get $a^2 = 49$, $b^2 = 9$, a = 7, b = 3 ✓ (ab = 21 ✓).

» **Why this step?** The two companion identities convert (a+b, a-b) directly into (a^2+b^2, ab) without ever solving for a and b.

Hinglish: *Ye identity-pair exam ka shortcut bomb hai: $(a+b)^2+(a-b)^2 = 2(a^2+b^2)$ aur $(a+b)^2-(a-b)^2 = 4ab$. Dono values plug karo - 58 aur 21. Solving for a, b optional hai, sirf check ke liye.*

Answer: $a^2 + b^2 = 58$, ab = 21

Q7 · Basic · Concept: Cube expansion (T7)

Q7. Expand: (i) $(x + 2)^3$, (ii) $(2m - 3n)^3$.

SOLUTION

Step 1: Given: two binomial cubes. Use $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$.

Step 2: (i) a = x, b = 2: $x^3 + 3x^2(2) + 3x(4) + 8 = x^3 + 6x^2 + 12x + 8$.

Step 3: (ii) a = 2m, b = 3n: $(2m)^3 - 3(2m)^2(3n) + 3(2m)(3n)^2 - (3n)^3 = 8m^3 - 36m^2n + 54mn^2 - 27n^3$.

» **Why this step?** The pattern 1, 3, 3, 1 on the coefficients comes from the binomial theorem; signs alternate for the minus cube.

Hinglish: *Cube pattern: 1-3-3-1. $(2m-3n)^3$ mein powers dhyan se: $(2m)^3 = 8m^3$, $(2m)^2 = 4m^2$ toh $3\times4m^2\times3n = 36m^2n$ minus. Signs alternate: + - + -. Coefficients 8, 36, 54, 27 - inko yaad karne ki zaroorat nahi, pattern se niklo.*

Answer: (i) $x^3 + 6x^2 + 12x + 8$ (ii) $8m^3 - 36m^2n + 54mn^2 - 27n^3$

Q8 · Basic · Concept: Middle-term splitting (T6)

Q8. Factorise: (i) $x^2 + 7x + 12$, (ii) $x^2 - 5x - 36$, (iii) $x^2 - 11x + 24$.

SOLUTION

Step 1: Given: three monic quadratics. Find two numbers with the right SUM and PRODUCT.

Step 2: (i) sum 7, product 12 → 3 and 4: (x + 3)(x + 4).

Step 3: (ii) sum -5, product -36 → -9 and +4: (x - 9)(x + 4).

Step 4: (iii) sum -11, product 24 → -8 and -3: (x - 8)(x - 3).

Step 5: Checks: 3+4 = 7, 3×4 = 12 ✓; -9+4 = -5, -9×4 = -36 ✓; -8-3 = -11, 24 ✓.

» **Why this step?** Expanding (x+p)(x+q) gives sum and product - so factorising reverses exactly those two readings.

Hinglish: *Do numbers dhoondo: jiska sum middle coefficient ho, product last term ho. Sign logic: product positive toh dono same sign (sum ke hisaab se), product negative toh alag-alag. (ii) mein product -36 aur sum -5: -9 aur +4 jeette hain.*

Answer: (i) (x+3)(x+4) (ii) (x-9)(x+4) (iii) (x-8)(x-3)

Q9 · Basic · Concept: Perfect square spotting (T1/T6)

Q9. Factorise: (i) $4y^2 - 12y + 9$, (ii) $9x^2 + 6x + 1$, (iii) $25m^2 + 30mn + 9n^2$.

SOLUTION

Step 1: Given: three candidates for perfect squares. Test first and last terms for squareness.

Step 2: (i) $(2y)^2 - 2(2y)(3) + 3^2 = (2y - 3)^2$.

Step 3: (ii) $(3x)^2 + 2(3x)(1) + 1^2 = (3x + 1)^2$.

Step 4: (iii) $(5m)^2 + 2(5m)(3n) + (3n)^2 = (5m + 3n)^2$.

» **Why this step?** A quadratic is a perfect square exactly when the middle term equals twice the product of the square roots of the end terms.

Hinglish: *Perfect square ka test: pehla aur aakhri term perfect square hon, aur middle term unke roots ke product ka DOUBLE ho. $12y = 2\times2y\times3$ ✓. Ye test lagane se $(a\pm b)^2$ identity turant mil jati hai.*

Answer: (i) $(2y-3)^2$ (ii) $(3x+1)^2$ (iii) $(5m+3n)^2$

Q10 · Basic · Concept: Difference-of-squares multiplication (T3)

Q10. Evaluate without direct multiplication: (i) 51×49, (ii) 103×97, (iii) 5.1×4.9.

SOLUTION

Step 1: Given: three products equidistant from a round number. Use $(a+b)(a-b) = a^2 - b^2$.

Step 2: (i) $51 \times 49 = (50+1)(50-1) = 2500 - 1 = 2499$.

Step 3: (ii) $103 \times 97 = (100+3)(100-3) = 10000 - 9 = 9991$.

Step 4: (iii) $5.1 \times 4.9 = (5 + 0.1)(5 - 0.1) = 25 - 0.01 = 24.99$.

» **Why this step?** Rewriting as (round ± same offset) converts multiplication into one subtraction of squares.

Hinglish: *Dono numbers ka average round number hai? Toh $(a+b)(a-b) = a^2-b^2$. 51×49: average 50, offset 1. 2500-1 = 2499. Ye trick multiplication tables se tez hai.*

Answer: (i) 2499 (ii) 9991 (iii) 24.99

Q11 · Basic · Concept: Common factor first (T12)

Q11. Factorise completely: (i) $6x^3 + 9x^2$, (ii) $2x^3 - 8x$, (iii) $3a^2b + 6ab^2 - 3ab$.

SOLUTION

Step 1: Given: three polynomials. Golden rule: HCF pehle.

Step 2: (i) HCF = $3x^2$: $3x^2(2x + 3)$.

Step 3: (ii) HCF = 2x: $2x(x^2 - 4) = 2x(x+2)(x-2)$ - the difference of squares appears only after the common factor.

Step 4: (iii) HCF = 3ab: $3ab(a + 2b - 1)$.

» **Why this step?** Removing the HCF exposes simpler structures (here a difference of squares) that were invisible inside the original expression.

Hinglish: *Sabse pehla step HCF nikaalo - hamesha. (ii) mein 2x nikaalne ke baad hi x^2-4 dikhta hai. Common factor chhod dena sabse common galti hai - 'completely' factorise ka matlab har factor prime ho.*

Answer: (i) $3x^2(2x+3)$ (ii) $2x(x+2)(x-2)$ (iii) $3ab(a+2b-1)$

Q12 · Basic · Concept: Square-difference evaluation (T3)

Q12. Evaluate: $5.1^2 - 4.9^2$ and $256^2 - 254^2$, each in one line.

SOLUTION

Step 1: Given: two differences of squares. Use $a^2 - b^2 = (a+b)(a-b)$.

Step 2: $5.1^2 - 4.9^2 = (5.1 + 4.9)(5.1 - 4.9) = 10 \times 0.2 = 2$.

Step 3: $256^2 - 254^2 = (256 + 254)(256 - 254) = 510 \times 2 = 1020$.

» **Why this step?** Never square two big numbers and subtract - factor first; the identity turns two squares into one small product.

Hinglish: *Squares subtract karne se pehle FACTOR karo: (a+b)(a-b). 5.1 aur 4.9 ka sum exactly 10 - answer 2 second mein. 256-254 wala: 510×2 = 1020. Bina identity ke square karoge toh calculator chahiye.*

Answer: 2 and 1020

Q13 · Moderate · Concept: Cube expansion (T7)

Q13. Expand: $(2x - 3y)^3$ and $(x + 2y)^3$.

SOLUTION

Step 1: Given: two cube expansions.

Step 2: $(2x - 3y)^3$: a = 2x, b = 3y. $(2x)^3 = 8x^3$; $3(2x)^2(3y) = 36x^2y$; $3(2x)(3y)^2 = 54xy^2$; $(3y)^3 = 27y^3$.

Step 3: With alternating signs: $8x^3 - 36x^2y + 54xy^2 - 27y^3$.

Step 4: $(x + 2y)^3$: $x^3 + 3x^2(2y) + 3x(2y)^2 + (2y)^3 = x^3 + 6x^2y + 12xy^2 + 8y^3$.

» **Why this step?** Write powers of a and b separately first ($8x^3$, $36x^2y$, ...), then attach the sign pattern + - + -.

Hinglish: *Cube mein har term ka coefficient pattern 1-3-3-1. Powers pehle nikaalo: $(2x)^2 = 4x^2$, $(3y)^2 = 9y^2$. Coefficients: 8, 3×4×3 = 36, 3×2×9 = 54, 27. Minus wale case mein sirf SIGN pattern badalta hai.*

Answer: $8x^3 - 36x^2y + 54xy^2 - 27y^3$ and $x^3 + 6x^2y + 12xy^2 + 8y^3$

Q14 · Moderate · Concept: Sum of cubes (T8)

Q14. Factorise: (i) $8a^3 + 27b^3$, (ii) $x^3 + 64$, (iii) $125m^3 + n^3$.

SOLUTION

Step 1: Given: three cube sums. Write each as $A^3 + B^3$, then use $(A+B)(A^2 - AB + B^2)$.

Step 2: (i) $(2a)^3 + (3b)^3 = (2a + 3b)(4a^2 - 6ab + 9b^2)$.

Step 3: (ii) $x^3 + 4^3 = (x + 4)(x^2 - 4x + 16)$.

Step 4: (iii) $(5m)^3 + n^3 = (5m + n)(25m^2 - 5mn + n^2)$.

» **Why this step?** The trinomial's middle term is minus AB - the opposite sign of the binomial - and its end terms are the squares of A and B.

Hinglish: *Cubes pehchano: $8a^3 = (2a)^3$, $27b^3 = (3b)^3$. Formula: $(A+B)(A^2-AB+B^2)$. Middle term ka sign binomial ke ULTA. Trinomial ke squares bade lagte hain ($4a^2$, $9b^2$) par wo bas A^2, B^2 hain.*

Answer: (i) $(2a+3b)(4a^2-6ab+9b^2)$ (ii) $(x+4)(x^2-4x+16)$ (iii) $(5m+n)(25m^2-5mn+n^2)$

Q15 · Moderate · Concept: Difference of cubes (T8)

Q15. Factorise: (i) $x^3 - 27y^3$, (ii) $64p^3 - 1$, (iii) $8x^3 - 125y^3z^3$.

SOLUTION

Step 1: Given: three cube differences. Use $A^3 - B^3 = (A - B)(A^2 + AB + B^2)$.

Step 2: (i) $x^3 - (3y)^3 = (x - 3y)(x^2 + 3xy + 9y^2)$.

Step 3: (ii) $(4p)^3 - 1^3 = (4p - 1)(16p^2 + 4p + 1)$.

Step 4: (iii) $(2x)^3 - (5yz)^3 = (2x - 5yz)(4x^2 + 10xyz + 25y^2z^2)$.

» **Why this step?** Same skeleton as the sum of cubes with signs flipped: binomial takes the minus, trinomial's middle term becomes plus.

Hinglish: *Difference of cubes = mirror of sum: $(A-B)(A^2+AB+B^2)$. (iii) mein $125y^3z^3 = (5yz)^3$ - teen factors under root, teeno ka cube. Product term $10xyz = A \times B = 2x \times 5yz$.*

Answer: (i) $(x-3y)(x^2+3xy+9y^2)$ (ii) $(4p-1)(16p^2+4p+1)$ (iii) $(2x-5yz)(4x^2+10xyz+25y^2z^2)$

Q16 · Moderate · Concept: x + 1/x chain (T10)

Q16. If $x + 1/x = 5$, find (i) $x^2 + 1/x^2$ and (ii) $x^4 + 1/x^4$.

SOLUTION

Step 1: Given: $x + 1/x = 5$. Square both sides: $x^2 + 2 + 1/x^2 = 25$.

Step 2: (i) $x^2 + 1/x^2 = 25 - 2 = 23$.

Step 3: Square again: $(x^2 + 1/x^2)^2 = x^4 + 2 + 1/x^4 = 23^2 = 529$.

Step 4: (ii) $x^4 + 1/x^4 = 529 - 2 = 527$.

» **Why this step?** Every square introduces a constant middle term 2 (because $x \cdot 1/x = 1$); subtract it to keep the chain clean.

Hinglish: *Chain method: square karo, 2 minus karo, phir dobara square karo, phir 2 minus. $5 \to 23 \to 527$. Ek hi line mein: $(x+1/x)^2 - 2 = x^2+1/x^2$. Ye chain chalti hi jati hai - x^6 ke liye x^3 wale bridge ka use hota hai.*

Answer: (i) 23 (ii) 527

Q17 · Moderate · Concept: Trinomial square reverse (T5/T11)

Q17. If $a + b + c = 9$ and $ab + bc + ca = 26$, find $a^2 + b^2 + c^2$.

SOLUTION

Step 1: Given: the pair-sum of three variables. Use the trinomial square identity:

Step 2: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$.

Step 3: $81 = (a^2+b^2+c^2) + 2(26) = (a^2+b^2+c^2) + 52$.

Step 4: So $a^2 + b^2 + c^2 = 81 - 52 = 29$.

» **Why this step?** The identity links the three quantities (sum, pair-sum, sum of squares) - knowing any two gives the third.

Hinglish: *Trinomial square ka reverse: sum of squares = $(sum)^2 - 2(pair\text{-}sum)$. $81 - 52 = 29$. Ye formula har three-variable question mein kaam aayega - yaad kar lo.*

Answer: 29

Q18 · Moderate · Concept: Chained difference of squares (T2)

Q18. Factorise completely: (i) $x^4 - 81$, (ii) $a^4 - 16b^4$, (iii) $81x^4 - 625$.

SOLUTION

Step 1: Given: fourth-power differences. Apply $a^2 - b^2$ repeatedly.

Step 2: (i) $(x^2)^2 - 9^2 = (x^2 - 9)(x^2 + 9) = (x-3)(x+3)(x^2+9)$.

Step 3: (ii) $(a^2)^2 - (4b^2)^2 = (a^2 - 4b^2)(a^2 + 4b^2) = (a-2b)(a+2b)(a^2+4b^2)$.

Step 4: (iii) $(9x^2)^2 - 25^2 = (9x^2-25)(9x^2+25) = (3x-5)(3x+5)(9x^2+25)$.

Step 5: Note: $x^2 + 9$, $a^2 + 4b^2$, $9x^2 + 25$ do not factorise over the integers - they are finished.

» **Why this step?** Each application of the identity halves an exponent; stop when a factor is a sum of squares (irreducible here).

Hinglish: *x^4 - 81 ko do baar factor karo: pehle $(x^2-9)(x^2+9)$, phir x^2-9 ko $(x-3)(x+3)$. Sum of squares (x^2+9) yahan phir se factor NAHI hota - wahi final hai. 'Completely' ka matlab: har factor irreducible.*

Answer: (i) $(x-3)(x+3)(x^2+9)$ (ii) $(a-2b)(a+2b)(a^2+4b^2)$ (iii) $(3x-5)(3x+5)(9x^2+25)$

Q19 · Moderate · Concept: Cube sum bridge (T11)

Q19. If $a + b = 12$ and $ab = 27$, find $a^3 + b^3$ and $a^2 + b^2$.

SOLUTION

Step 1: Given: pair-sum and product. Cube bridge: $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$.

Step 2: $a^3 + b^3 = 12^3 - 3(27)(12) = 1728 - 972 = 756$.

Step 3: Square identity: $a^2 + b^2 = (a+b)^2 - 2ab = 144 - 54 = 90$.

» **Why this step?** Both bridges express symmetric quantities in terms of (sum, product) - no need to solve for a and b.

Hinglish: *Symmetric quantities ka rule: sab kuch (sum, product) se nikhlo. Cube bridge: $(a+b)^3 - 3ab(a+b)$. 1728 - 972 = 756. Square: 144 - 54 = 90. Solving a, b se bhi hoga but identity wala path 2 line ka hai.*

Answer: $a^3 + b^3 = 756$, $a^2 + b^2 = 90$

Q20 · Moderate · Concept: Near-number cube (T3)

Q20. Evaluate 103^3 without a calculator.

SOLUTION

Step 1: Write $103 = 100 + 3$ and use $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.

Step 2: $a^3 = 100^3 = 1{,}000{,}000$.

Step 3: $3a^2b = 3 \times 10000 \times 3 = 90{,}000$.

Step 4: $3ab^2 = 3 \times 100 \times 9 = 2{,}700$.

Step 5: $b^3 = 27$.

Step 6: Total: $1{,}000{,}000 + 90{,}000 + 2{,}700 + 27 = 1{,}092{,}727$.

» **Why this step?** Powers of 100 are trivial, so the whole cube collapses into four easy additions.

Hinglish: *Cube bhi round-number trick se hoga: 1-3-3-1 pattern with 100 aur 3. Terms: 1000000, 90000, 2700, 27. Jodo: 1092727. Column-wise addition karo, zeros ginn kar.*

Answer: 1,092,727

Q21 · Moderate · Concept: Grouping with identities (T12)

Q21. Factorise: $a^2 - b^2 + 2bc - c^2$.

SOLUTION

Step 1: Given: four terms, no common factor. Try grouping with a sign flip:

Step 2: $a^2 - b^2 + 2bc - c^2 = a^2 - (b^2 - 2bc + c^2)$.

Step 3: The bracket is a perfect square: $= a^2 - (b - c)^2$.

Step 4: Difference of squares: $= (a + b - c)(a - b + c)$.

Step 5: Check by expanding one way: $(a + (b-c))(a - (b-c)) = a^2 - (b-c)^2$ ✓.

» **Why this step?** The last three terms are -1 times a perfect square; grouping reveals a two-square difference.

Hinglish: *Pehli nazar mein kuch nahi dikh raha. Last teen terms ko bracket mein -1 ke saath rakho: $-(b^2-2bc+c^2) = -(b-c)^2$. Ab classic $a^2 - (b-c)^2$. Grouping + identity ka combo - ye pattern yaad rakho.*

Answer: (a + b - c)(a - b + c)

Q22 · Moderate · Concept: 4ab identity in action (T3)

Q22. Prove that $(a+b)^2 - (a-b)^2 = 4ab$, and use it to evaluate $5.5^2 - 4.5^2$.

SOLUTION

Step 1: Expand both squares: $(a+b)^2 = a^2 + 2ab + b^2$; $(a-b)^2 = a^2 - 2ab + b^2$.

Step 2: Subtract: $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$. ∎

Step 3: Application: $5.5^2 - 4.5^2 = 4 \times 5.5 \times 4.5 = 4 \times 24.75 = 99$.

Step 4: Cross-check with the difference of squares: $(5.5+4.5)(5.5-4.5) = 10 \times 1 = 99$ ✓ ($4ab = (a+b)(a-b)$ here since $a = (10/2)$, $b = (1/2)$... the two results agree).

» **Why this step?** Two different identities confirm the same value - a built-in verification habit.

Hinglish: *Proof 2 line ka hai - dono squares expand karo, subtract karo, 4ab. Application: $4 \times 5.5 \times 4.5 = 99$. Yehi identity multiplication-table trick ban jati hai: $k^2-(k-1)^2 = 2k-1$ bhi isi family ka hai.*

Answer: Proved; $5.5^2 - 4.5^2 = 99$

Q23 · Moderate · Concept: Cube difference bridge (T11)

Q23. If $a - b = 4$ and $ab = 21$, find $a^3 - b^3$.

SOLUTION

Step 1: Given: difference and product. Bridge: $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$.

Step 2: $= 4^3 + 3(21)(4) = 64 + 252 = 316$.

Step 3: Quick sanity: $a - b = 4$, $ab = 21$ admits $a = 7$, $b = 3$: $343 - 27 = 316$ ✓.

» **Why this step?** Expanding $(a-b)^3$ shows the spare term $+3ab(a-b)$; moving it left creates the bridge.

Hinglish: *Minus wala bridge: $(a-b)^3 + 3ab(a-b) = a^3-b^3$. Sign yaad rakho - difference wale case mein PLUS $3ab(a-b)$. $64 + 252 = 316$. Check line $a=7, b=3$ se confirm.*

Answer: 316

Q24 · Moderate · Concept: Crown identity evaluation (T9)

Q24. Using a suitable identity, evaluate $12^3 + 1^3 + (-13)^3$.

SOLUTION

Step 1: Spot the zero sum: $12 + 1 + (-13) = 0$.

Step 2: Zero-sum shortcut: if $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$.

Step 3: So the value $= 3 \times 12 \times 1 \times (-13) = -468$.

» **Why this step?** The identity $a^3+b^3+c^3-3abc = (a+b+c)(...)$ collapses to 0 on the left whenever $a+b+c = 0$.

Hinglish: *Numbers dekho: 12, 1, -13 - sum zero! Turant shortcut: cubes ka sum = 3abc = -468. Bina koi cube nikaale. Ye 'zero-sum spotting' exam mein 3 second ka kaam hai - 5 marks bachate hain.*

Answer: -468

Q25 · Moderate · Concept: x - 1/x chain (T10)

Q25. If $x - 1/x = 3$, find (i) $x^2 + 1/x^2$ and (ii) $x^3 - 1/x^3$.

SOLUTION

Step 1: Given: $x - 1/x = 3$. Square both sides: $x^2 - 2 + 1/x^2 = 9$.

Step 2: (i) $x^2 + 1/x^2 = 9 + 2 = 11$ (note: PLUS 2 here, because the cross term -2 moved right).

Step 3: Cube both sides: $(x - 1/x)^3 = x^3 - 1/x^3 - 3(x - 1/x) \cdot$ ... using $(a-b)^3 = a^3-b^3-3ab(a-b)$ with $ab = 1$:

Step 4: $27 = x^3 - 1/x^3 - 3 \times 3$.

Step 5: (ii) $x^3 - 1/x^3 = 27 + 9 = 36$.

» **Why this step?** The $x \cdot (1/x) = 1$ fact makes every middle term a constant: -2 when squaring the difference, +3(x-1/x) when cubing.

Hinglish: *Difference wale chain mein SIGN flip hota hai: square par +2 (minus ka minus), cube par +3(x-1/x). 3 → 11 → 36. Sum wale (x+1/x) chain se confuse mat hona - wahan minus aata tha.*

Answer: (i) 11 (ii) 36

Q26 · Moderate · Concept: Trinomial square with signs (T5)

Q26. Expand: $(2a + b - 3c)^2$.

SOLUTION

Step 1: Identify: A = 2a, B = b, C = -3c.

Step 2: Squares: $4a^2 + b^2 + 9c^2$.

Step 3: Cross terms: 2AB = 4ab; 2BC = -6bc; 2CA = -12ca.

Step 4: Result: $4a^2 + b^2 + 9c^2 + 4ab - 6bc - 12ca$.

» **Why this step?** Sign of 2BC = (+)(-) = -, sign of 2CA = (-)(+) = -: the signs multiply, no exceptions.

Hinglish: *Same three-square recipe. Cross terms ke signs: a-b wala positive (dono +), b-c wala negative (+ aur -), c-a wala negative. $9c^2$ isliye kyunki $(-3c)^2 = 9c^2$ - squares hamesha positive.*

Answer: $4a^2 + b^2 + 9c^2 + 4ab - 6bc - 12ca$

Q27 · Difficult · Concept: Cubic factorisation by root (T12)

Q27. Factorise completely: $x^3 - 3x^2 - 9x - 5$.

SOLUTION

Step 1: Hunt for an integer root: try x = -1: $(-1)^3 - 3(1) - 9(-1) - 5 = -1 - 3 + 9 - 5 = 0$ ✓. So (x + 1) is a factor.

Step 2: Divide: $x^3 - 3x^2 - 9x - 5 = (x + 1)(x^2 - 4x - 5)$.

Step 3: (Verify by expanding: $x^3 - 4x^2 - 5x + x^2 - 4x - 5 = x^3 - 3x^2 - 9x - 5$ ✓.)

Step 4: Factorise the quadratic: $x^2 - 4x - 5$: sum -4, product -5 → -5 and +1: $(x - 5)(x + 1)$.

Step 5: Final: $(x + 1)^2(x - 5)$.

» **Why this step?** An integer root r of a polynomial always yields the factor (x - r); testing small integers (0, ±1, ±2) is the fastest opening move.

Hinglish: *Cubic factorisation ka opening: chhote integers try karo. x = -1 par zero mila, toh (x+1) factor. Long division ya coefficient comparison se quadratic mila: x^2-4x-5, wo phir middle-term split. Final $(x+1)^2(x-5)$.*

Answer: $(x + 1)^2(x - 5)$

Q28 · Difficult · Concept: Reverse chain (T10)

Q28. If $x^2 + 1/x^2 = 38$, find the value of $x - 1/x$.

SOLUTION

Step 1: Use the difference-square identity: $(x - 1/x)^2 = x^2 - 2 + 1/x^2$.

Step 2: $= 38 - 2 = 36$.

Step 3: So $x - 1/x = \pm 6$.

Step 4: Both signs are possible: $x = 2\sqrt{10}$ gives $x - 1/x = +6$... concretely $x > 1$ gives +6 (e.g. $x = 2 + \sqrt{3}$ family) and $0 < x < 1$ gives -6; no extra condition is stated, so report both.

» **Why this step?** Squaring erases the sign, so recovering $x - 1/x$ from its square opens BOTH cases - the classic two-root situation.

Hinglish: *$(x-1/x)^2 = x^2+1/x^2 - 2 = 36$. Root: ±6. Sign ka question: x bada hua toh +6, chhota (0<x<1) toh -6. Question ne sign nahi bataya, dono report karo - sirf +6 likhna aadha answer hai.*

Answer: $x - 1/x = 6$ or -6

Q29 · Difficult · Concept: Cubic pattern collapse (T8/T14)

Q29. Evaluate without finding individual cubes: $0.86^3 + 0.14^3 - 0.86^2 \times 0.14 - 0.86 \times 0.14^2$.

SOLUTION

Step 1: Group: $a^3 + b^3 - a^2b - ab^2$ with $a = 0.86$, $b = 0.14$.

Step 2: Factor: $a^3 - a^2b + b^3 - ab^2 = a^2(a - b) - b^2(a - b) = (a^2 - b^2)(a - b) = (a+b)(a-b)^2$.

Step 3: Now $a + b = 1$ and $(a - b)^2 = (0.72)^2$.

Step 4: Value $= 1 \times 0.5184 = 0.5184$.

» **Why this step?** Grouping converts four mixed terms into two factors; the sum $a + b = 1$ then finishes in one step.

Hinglish: *Cubes mat nikaalo - pattern dekho. $a^2(a-b) - b^2(a-b)$ grouping se $(a+b)(a-b)^2$ banta hai. $a+b = 0.86+0.14 = 1$, $(a-b)^2 = 0.72^2 = 0.5184$. Factor pehle, arithmetic baad mein.*

Answer: 0.5184

Q30 · Difficult · Concept: Substitution factorisation (T12)

Q30. Factorise completely: $(x^2 - 2x)^2 - 23(x^2 - 2x) + 120$.

SOLUTION

Step 1: Substitute $y = x^2 - 2x$. The expression becomes $y^2 - 23y + 120$.

Step 2: Split the middle: two numbers with sum -23, product 120: -8 and -15.

Step 3: $y^2 - 23y + 120 = (y - 8)(y - 15)$.

Step 4: Back-substitute: $(x^2 - 2x - 8)(x^2 - 2x - 15)$.

Step 5: Factor each: $x^2 - 2x - 8 = (x-4)(x+2)$; $x^2 - 2x - 15 = (x-5)(x+3)$.

Step 6: Final: $(x - 4)(x + 2)(x - 5)(x + 3)$.

» **Why this step?** The repeated chunk $x^2 - 2x$ is the signal for substitution; every later step is then standard quadratic work.

Hinglish: *(x^2-2x) do baar repeat ho raha hai - isko y maan lo. Quadratic ho gayi easy. Factor karke WAPAS substitute karna zaroori hai, aur phir dono quadratics ko bhi factor karo - 'completely' tabhi hota hai. Char linear factors final.*

Answer: (x - 4)(x + 2)(x - 5)(x + 3)

Q31 · Difficult · Concept: Cube bridge with scaled variables (T11)

Q31. If $2x + 3y = 13$ and $xy = 6$, find $8x^3 + 27y^3$.

SOLUTION

Step 1: Recognise $8x^3 + 27y^3 = (2x)^3 + (3y)^3$.

Step 2: Cube bridge: $A^3 + B^3 = (A+B)^3 - 3AB(A+B)$ with $A = 2x$, $B = 3y$.

Step 3: $(A + B)^3 = 13^3 = 2197$.

Step 4: $AB = (2x)(3y) = 6xy = 36$.

Step 5: $(2x)^3 + (3y)^3 = 2197 - 3(36)(13) = 2197 - 1404 = 793$.

» **Why this step?** Scaling turns the question into the standard (sum, product) bridge - AB is where xy enters.

Hinglish: *$8x^3 = (2x)^3$, $27y^3 = (3y)^3$ - scaling pehchano. Bridge lagao: $(2x+3y)^3 - 3(2x)(3y)(2x+3y)$. $AB = 6xy = 36$. $2197 - 1404 = 793$. Cubes nikaale bina answer!*

Answer: 793

Q32 · Difficult · Concept: Add-subtract trick (T12)

Q32. Factorise: $x^4 + x^2 + 1$.

SOLUTION

Step 1: Add and subtract x^2: $x^4 + x^2 + 1 = x^4 + 2x^2 + 1 - x^2$.

Step 2: First three terms form a perfect square: $= (x^2 + 1)^2 - x^2$.

Step 3: Difference of squares: $= (x^2 + 1 - x)(x^2 + 1 + x)$.

Step 4: Final: $(x^2 - x + 1)(x^2 + x + 1)$.

Step 5: Check the middle terms on expansion: $-x^3$ and $+x^3$ cancel ✓.

» **Why this step?** Injecting $-x^2 + x^2$ manufactures a perfect square plus a difference of squares - the standard completion for $a^4 + a^2 + 1$ shapes.

Hinglish: *x^4+x^2+1 classic trap hai - koi seedha identity nahi lagta. x^2 add + subtract karo: perfect square ban jata hai $(x^2+1)^2$, aur $-x^2$ difference of squares deta hai. Ye 'completion' trick a^4+4b^4 (Sophie Germain style) mein bhi chalta hai.*

Answer: $(x^2 - x + 1)(x^2 + x + 1)$

Q33 · Difficult · Concept: Zero-sum with differences (T9)

Q33. Evaluate: $(a - b)^3 + (b - c)^3 + (c - a)^3$.

SOLUTION

Step 1: Let $A = a - b$, $B = b - c$, $C = c - a$. Their sum: $A + B + C = a - b + b - c + c - a = 0$.

Step 2: Zero-sum shortcut: $A^3 + B^3 + C^3 = 3ABC$.

Step 3: So the expression $= 3(a - b)(b - c)(c - a)$.

» **Why this step?** Any three quantities summing to zero obey the cube shortcut - the letters may themselves be differences.

Hinglish: *(a-b), (b-c), (c-a) ka sum zero hai - bas zero-sum shortcut lagao: 3ABC. Answer 3(a-b)(b-c)(c-a). Ye pattern bade questions mein chhupa hota hai - sum-zero spotting sabse powerful tool hai.*

Answer: 3(a - b)(b - c)(c - a)

Q34 · Difficult · Concept: Cube pair simplify (T7)

Q34. Simplify completely: $(a + b)^3 + (a - b)^3$.

SOLUTION

Step 1: Expand $(a+b)^3$: $a^3 + 3a^2b + 3ab^2 + b^3$.

Step 2: Expand $(a-b)^3$: $a^3 - 3a^2b + 3ab^2 - b^3$.

Step 3: Add: the $\pm 3a^2b$ and $\pm b^3$ terms cancel; survivors are $2a^3 + 6ab^2$.

Step 4: Factor the result: $2a(a^2 + 3b^2)$.

» **Why this step?** Adding opposite-sign expansions kills the odd terms - the same mechanism as $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$.

Hinglish: *Sum mein odd terms ($\pm 3a^2b$, $\pm b^3$) cancel ho jate hain, even terms double. Result $2a^3 + 6ab^2$, factor: $2a(a^2+3b^2)$. Ye square-identity ka cube bhai hai: $(a+b)^3+(a-b)^3 = 2a(a^2+3b^2)$.*

Answer: $2a^3 + 6ab^2 = 2a(a^2 + 3b^2)$

Q35 · Difficult · Concept: Recognise a cube (T7/T8)

Q35. Factorise: $8a^3 + 27b^3 + 36a^2b + 54ab^2$.

SOLUTION

Step 1: Rearrange into cube order: $(2a)^3 + 3(2a)^2(3b) + 3(2a)(3b)^2 + (3b)^3$.

Step 2: Check each coefficient: $8a^3$ ✓; $3 \cdot 4a^2 \cdot 3b = 36a^2b$ ✓; $3 \cdot 2a \cdot 9b^2 = 54ab^2$ ✓; $27b^3$ ✓.

Step 3: This is exactly $(2a + 3b)^3$.

» **Why this step?** The 1-3-3-1 coefficient pattern on the rescaled blocks (2a) and (3b) identifies the whole expression as one cube.

Hinglish: *Terms ko cube order mein likho: $8a^3$, $36a^2b$, $54ab^2$, $27b^3$. Pattern 1-3-3-1 with A = 2a, B = 3b. Poora expression $(2a+3b)^3$ hai! Sum-of-cubes formula lagane wale galat raaste par chale jate hain - pehle pattern check karo.*

Answer: $(2a + 3b)^3$

Q36 · Difficult · Concept: Crown identity forward use (T9)

Q36. If $x + y + z = 6$ and $xy + yz + zx = 11$, find $x^3 + y^3 + z^3 - 3xyz$.

SOLUTION

Step 1: First get the sum of squares: $x^2+y^2+z^2 = (x+y+z)^2 - 2(xy+yz+zx) = 36 - 22 = 14$.

Step 2: Crown identity: $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2 - xy - yz - zx)$.

Step 3: The bracket: $14 - 11 = 3$.

Step 4: Value $= 6 \times 3 = 18$.

» **Why this step?** The crown identity needs (sum) and (sum of squares minus pair-sum) - both computable from the givens.

Hinglish: *Do steps: pehle $x^2+y^2+z^2 = 36-22 = 14$. Phir crown identity: bracket = 14-11 = 3. Multiply: 6×3 = 18. Zero-sum nahi tha isliye poora identity use hua.*

Answer: 18

Q37 · Difficult · Concept: Crown identity as factorisation (T9)

Q37. Factorise: $a^3 - 8b^3 - 64c^3 - 24abc$.

SOLUTION

Step 1: See the shape: $a^3 + (-2b)^3 + (-4c)^3 - 3 \cdot a \cdot (-2b) \cdot (-4c)$.

Step 2: Check the cross term: $3 \times a \times (-2b) \times (-4c) = 24abc$, and the expression has $-24abc$ ✓.

Step 3: Apply the crown identity with $x = a$, $y = -2b$, $z = -4c$:

Step 4: $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$.

Step 5: $x + y + z = a - 2b - 4c$.

Step 6: $x^2+y^2+z^2 = a^2+4b^2+16c^2$; $xy = -2ab$, $yz = 8bc$, $zx = -4ca$; so $-xy-yz-zx = +2ab - 8bc + 4ca$.

Step 7: Final: $(a - 2b - 4c)(a^2 + 4b^2 + 16c^2 + 2ab - 8bc + 4ca)$.

» **Why this step?** The crown identity factorises ANY three cubes whose product term appears as -3xyz - negative letters are welcome.

Hinglish: *Teen cubes dikh rahe hain: a^3, $(-2b)^3$, $(-4c)^3$ aur cross term $-24abc = -3xyz$. Crown identity seedha lag jata hai. Negative letters ke signs cross terms mein multiply hote hain: $yz = (-2b)(-4c) = +8bc$, isliye $-yz = -8bc$. Sign discipline hi yahan ka skill hai.*

Answer: $(a - 2b - 4c)(a^2 + 4b^2 + 16c^2 + 2ab - 8bc + 4ca)$

Q38 · Difficult · Concept: Grouped difference of squares (T2/T12)

Q38. Factorise: $x^2 + 6x + 9 - 4y^2$.

SOLUTION

Step 1: First three terms form a perfect square: $x^2 + 6x + 9 = (x + 3)^2$.

Step 2: Expression = $(x + 3)^2 - (2y)^2$.

Step 3: Difference of squares: $(x + 3 - 2y)(x + 3 + 2y)$.

» **Why this step?** Completing the visible square FIRST turns four terms into a two-square difference.

Hinglish: *Pehle teen terms ko square banao: $(x+3)^2$. Ab $-4y^2 = -(2y)^2$. Classic difference of squares: $(x+3-2y)(x+3+2y)$. Order of moves: square complete karo, phir factor.*

Answer: $(x + 3 - 2y)(x + 3 + 2y)$

Q39 · Difficult · Concept: Scaled square bridge (T11)

Q39. If $9x^2 + 25y^2 = 181$ and $xy = 2$, find $3x - 5y$.

SOLUTION

Step 1: Note $9x^2 = (3x)^2$ and $25y^2 = (5y)^2$.

Step 2: $(3x - 5y)^2 = 9x^2 - 2(3x)(5y) + 25y^2 = 181 - 30xy = 181 - 60 = 121$.

Step 3: So $3x - 5y = \pm 11$.

» **Why this step?** Spotting the scaled square $(3x, 5y)$ converts the given sum of squares into a completed square minus the cross term.

Hinglish: *$9x^2$ aur $25y^2$ ko $(3x)^2$, $(5y)^2$ dekho. Cross term $2 \times 3x \times 5y = 30xy = 60$. $181 - 60 = 121 = 11^2$. Sign: dono values possible - ± 11. Ye 'scaled variables' wala pattern har identity mein chalta hai.*

Answer: $3x - 5y = 11$ or -11

Q40 · Difficult · Concept: Deep chained difference of squares (T2)

Q40. Factorise completely: $x^8 - 256$.

SOLUTION

Step 1: $x^8 = (x^4)^2$ and $256 = 16^2$: $(x^4 - 16)(x^4 + 16)$.

Step 2: $x^4 - 16 = (x^2 - 4)(x^2 + 4)$.

Step 3: $x^2 - 4 = (x - 2)(x + 2)$.

Step 4: Collect: $(x - 2)(x + 2)(x^2 + 4)(x^4 + 16)$.

Step 5: Neither $x^2 + 4$ nor $x^4 + 16$ factorises over the integers - done.

» **Why this step?** Each pass halves the power difference; three passes bring 8 down to 1.

Hinglish: *x^8-256: teen baar difference of squares. Har baar power half: $x^8 \to x^4 \to x^2 \to x$. Final $(x-2)(x+2)(x^2+4)(x^4+16)$. Sum-of-squares factors ko chhod do - wo integer mein khatam nahi hote.*

Answer: $(x - 2)(x + 2)(x^2 + 4)(x^4 + 16)$

Q41 · Difficult · Concept: Zero-sum proof and use (T9)

Q41. (i) Prove that if $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$. (ii) Use it to evaluate $2^3 + 3^3 + (-5)^3$.

SOLUTION

Step 1: (i) Crown identity: $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$.

Step 2: If $a + b + c = 0$, the right side is 0, so $a^3+b^3+c^3 = 3abc$. ∎

Step 3: (ii) Here $2 + 3 + (-5) = 0$, so $2^3 + 3^3 + (-5)^3 = 3(2)(3)(-5) = -90$.

Step 4: Direct check: $8 + 27 - 125 = -90$ ✓.

» **Why this step?** The proof is one line once the crown identity is available - that is why the identity is called the crown.

Hinglish: *Proof: crown identity mein a+b+c = 0 daalo, right side zero. Ho gaya. Application: 2, 3, -5 ka sum zero - 3×2×3×(-5) = -90. Direct cube se match kar lo, confidence double.*

Answer: (i) proved (ii) -90

Q42 · Difficult · Concept: Zero-sum with big numbers (T9)

Q42. Without cubing any number, evaluate $25^3 - 75^3 + 50^3$.

SOLUTION

Step 1: Sum check: $25 - 75 + 50 = 0$. So $a = 25$, $b = -75$, $c = 50$ has $a + b + c = 0$.

Step 2: Zero-sum shortcut: $a^3 + b^3 + c^3 = 3abc$.

Step 3: Value $= 3 \times 25 \times (-75) \times 50$.

Step 4: $25 \times 75 = 1875$; $1875 \times 50 = 93{,}750$; times 3 = 281,250. With the minus sign: -281,250.

» **Why this step?** The zero sum makes three cubes collapse into one product - this is the identity's designed use.

Hinglish: *25 - 75 + 50 = 0 - zero-sum! -75 ko b banake shortcut lagao: 3×25×(-75)×50 = -281250. Teen cubes (15625 + ... + 125000) karne ki zaroorat hi nahi. Sign: ek negative factor, toh result negative.*

Answer: -281,250

Q43 · High · Concept: Conditional cube identity (T9/T14)

Q43. If $a + b + c = 0$, prove that $(a+b)^3 + (b+c)^3 + (c+a)^3 = 3(a+b)(b+c)(c+a)$.

SOLUTION

Step 1: Let A = a + b, B = b + c, C = c + a. Their sum: A + B + C = 2(a + b + c) = 0.

Step 2: Zero-sum shortcut: $A^3 + B^3 + C^3 = 3ABC$.

Step 3: Substituting back: $(a+b)^3 + (b+c)^3 + (c+a)^3 = 3(a+b)(b+c)(c+a)$. ∎

» **Why this step?** The trick is to see (a+b), (b+c), (c+a) as NEW letters whose sum inherits the zero - the shortcut then applies verbatim.

Hinglish: *Renames karo: A = a+b, B = b+c, C = c+a. Inka sum = 2(a+b+c) = 0. Ab wahi zero-sum shortcut. Renaming technique badi problems ka master key hai - letters kuch bhi ho sakte hain.*

Answer: Proved via A+B+C = 0 with A = a+b, B = b+c, C = c+a.

Q44 · High · Concept: Cube of x + 1/x (T10)

Q44. Expand $(x + 1/x)^3$ and hence find $x^3 + 1/x^3$ when x + 1/x = 3.

SOLUTION

Step 1: Use $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$ with a = x, b = 1/x, ab = 1:

Step 2: $(x + 1/x)^3 = x^3 + 1/x^3 + 3(x + 1/x)$.

Step 3: Given x + 1/x = 3: cube both sides directly: $27 = x^3 + 1/x^3 + 3 \times 3$.

Step 4: $x^3 + 1/x^3 = 27 - 9 = 18$.

» **Why this step?** ab = 1 compresses the whole expansion to (cubes) + 3(sum) - the cleanest bridge in the chapter.

Hinglish: *$(x+1/x)^3 = x^3+1/x^3+3(x+1/x)$. Given value 3: $27 = x^3+1/x^3+9$. Answer 18. Yaad rakho: cube bridge s → s^3 - 3s hota hai. s = 3: 27-9 = 18.*

Answer: $x^3 + 1/x^3 = 18$

Q45 · High · Concept: Cube minus cube (T8/T12)

Q45. Factorise: $a^3 + b^3 + 3a^2b + 3ab^2 - 8$.

SOLUTION

Step 1: Spot the cube: $a^3 + 3a^2b + 3ab^2 + b^3 = (a+b)^3$, but the b^3 is missing and an 8 is appended.

Step 2: Regroup: $a^3 + b^3 + 3a^2b + 3ab^2 - 8 = (a + b)^3 - 2^3$.

Step 3: Difference of cubes: $(a + b - 2)[(a+b)^2 + 2(a+b) + 4]$.

Step 4: Expand the second factor: $(a + b - 2)(a^2 + 2ab + b^2 + 2a + 2b + 4)$.

» **Why this step?** Four of the terms assemble into one cube; the leftover is a lone 2^3, giving a difference of cubes.

Hinglish: *Pehle chaar terms ko $(a+b)^3$ banao - b^3 missing tha, use borrow karke group kiya. $(a+b)^3 - 8 = (a+b)^3 - 2^3$. Ab difference of cubes: (a+b-2) times trinomial. Trinomial expand kar dena final form mein.*

Answer: $(a + b - 2)(a^2 + 2ab + b^2 + 2a + 2b + 4)$

Q46 · High · Concept: Crown identity with product (T9)

Q46. If x + y + z = 15, $x^2 + y^2 + z^2 = 83$ and xyz = 30, find $x^3 + y^3 + z^3$.

SOLUTION

Step 1: Pair-sum: $xy + yz + zx = [(x+y+z)^2 - (x^2+y^2+z^2)]/2 = (225 - 83)/2 = 71$.

Step 2: Crown identity: $x^3+y^3+z^3 - 3xyz = (x+y+z)(x^2+y^2+z^2 - xy-yz-zx) = 15(83 - 71) = 180$.

Step 3: So $x^3+y^3+z^3 = 180 + 3xyz = 180 + 90 = 270$.

» **Why this step?** Two bridges in sequence: first recover the pair-sum, then the crown identity, then add back 3xyz.

Hinglish: *Pehle pair-sum: (225-83)/2 = 71. Crown se cubes-minus-product: 15×12 = 180. Phir 3xyz = 90 wapas jodo: 270. Teen givens, teen steps - symmetric quantity ka poora game.*

Answer: 270

Q47 · High · Concept: Decimal cube sum (T3/T9)

Q47. Without cubing, evaluate $0.86^3 + 0.14^3$.

SOLUTION

Step 1: a + b = 0.86 + 0.14 = 1.

Step 2: Bridge: $a^3 + b^3 = (a+b)^3 - 3ab(a+b) = 1 - 3ab$.

Step 3: ab = 0.86 × 0.14 = 0.1204.

Step 4: Value = 1 - 3(0.1204) = 1 - 0.3612 = 0.6388.

» **Why this step?** The pair-sum 1 makes $(a+b)^3$ trivial; only the product needs one multiplication.

Hinglish: *0.86 + 0.14 = 1 - yahi design hai. Bridge: 1 - 3(0.1204) = 0.6388. Decimals ka product dhyan se: 86×14 = 1204, four decimal places. Cubes bilkul nahi nikaale.*

Answer: 0.6388

Q48 · High · Concept: Cubic factorisation, three roots (T12)

Q48. Factorise completely: $x^3 - 6x^2 + 11x - 6$.

SOLUTION

Step 1: Test integer roots. x = 1: 1 - 6 + 11 - 6 = 0 ✓, so (x - 1) is a factor.

Step 2: Divide: $x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6)$.

Step 3: Split the middle: $x^2 - 5x + 6 = (x - 2)(x - 3)$.

Step 4: Final: (x - 1)(x - 2)(x - 3).

Step 5: Check: roots 1, 2, 3 sum to 6 = -(-6) ✓, pairwise products 11 ✓, product 6 ✓ (Vieta consistency).

» **Why this step?** A cubic with three integer roots yields to root-hunting + division + quadratic split - three standard moves.

Hinglish: *x = 1 pehle try karo (sum of coefficients zero hai tab). Factor mila, divide karo, quadratic split: (x-2)(x-3). Final teen linear factors. Vieta check: roots ka sum 6, pair-sum 11, product 6 - sab original coefficients se match.*

Answer: (x - 1)(x - 2)(x - 3)

Q49 · High · Concept: Sum-cube decomposition (T9/T14)

Q49. Prove that $(x + y + z)^3 - x^3 - y^3 - z^3 = 3(x + y)(y + z)(z + x)$.

SOLUTION

Step 1: Use the crown identity with A = x + y + z, B = -x, C = -y, D = -z? Better: apply it to the three numbers (x+y), (y+z), (z+x):

Step 2: Their sum = 2(x+y+z). Not zero, so use the full identity on P = x+y, Q = y+z, R = z+x.

Step 3: $P^3+Q^3+R^3-3PQR = (P+Q+R)(P^2+Q^2+R^2-PQ-QR-RP)$.

Step 4: Alternative cleaner route: expand $(x+y+z)^3 = x^3+y^3+z^3 + 3(x+y)(y+z)(z+x)$. This is itself the identity: apply the crown identity to A = x+y, B = z... verify directly:

Step 5: Direct proof: $(x+y+z)^3$ with u = x+y: $(u+z)^3 = u^3 + 3u^2z + 3uz^2 + z^3 = (x+y)^3 + 3z(x+y)^2 + 3z^2(x+y) + z^3$.

Step 6: Expand $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$ and collect all terms:

Step 7: $(x+y+z)^3 = x^3+y^3+z^3 + 3x^2y + 3xy^2 + 3y^2z + 3yz^2 + 3z^2x + 3zx^2 + 6xyz$.

Step 8: Subtract $x^3+y^3+z^3$: remainder = $3[x^2y+xy^2+y^2z+yz^2+z^2x+zx^2] + 6xyz$.

Step 9: Now expand $3(x+y)(y+z)(z+x)$: first $(x+y)(y+z) = xy+xz+y^2+yz$; multiply by $(z+x)$: = $xyz+x^2z + xz^2+x^2y + y^2z+xy^2 + yz^2+xyz = (x^2y+xy^2+y^2z+yz^2+z^2x+zx^2) + 2xyz$.

Step 10: Times 3: $3[...+2xyz] = 3[...] + 6xyz$ - exactly the remainder. ∎

» **Why this step?** Both sides expand to the same six mixed terms plus 6xyz; the identity packages this messy equality into one line.

Hinglish: *Ye identity yaad karne layak hai: $(x+y+z)^3 = x^3+y^3+z^3+3(x+y)(y+z)(z+x)$. Proof mein dono side expand karke compare karo - mixed terms match hote hain. Exam mein identity quote karke 2-3 line proof kaafi hai agar identity 'standard' maani jati hai; warna full expansion dikhao.*

Answer: Proved: both sides equal $3(x^2y+xy^2+y^2z+yz^2+z^2x+zx^2) + 6xyz$.

Q50 · High · Concept: Substitution square (T5/T12)

Q50. Factorise: $(2a + 3b)^2 + 2(2a + 3b)(3a - 2b) + (3a - 2b)^2$.

SOLUTION

Step 1: Substitute P = 2a + 3b, Q = 3a - 2b. The expression is $P^2 + 2PQ + Q^2$.

Step 2: Perfect square: $= (P + Q)^2$.

Step 3: P + Q = 2a + 3b + 3a - 2b = 5a + b.

Step 4: Answer: $(5a + b)^2$.

Step 5: Check one cross term on expansion: $2\cdot(5a)(b) = 10ab$; direct: $2(2a)(3a)$... full check: $(5a+b)^2 = 25a^2 + 10ab + b^2$ ✓.

» **Why this step?** The structure $P^2 + 2PQ + Q^2$ is invisible until the repeated brackets are renamed - then it is one identity.

Hinglish: *Brackets ko P aur Q naam do - expression $P^2+2PQ+Q^2$ ban jata hai $= (P+Q)^2$. $P+Q = 5a+b$. Answer $(5a+b)^2$. Substitution aankhon ko naya drishti deta hai - pattern pehle chhupa tha.*

Answer: $(5a + b)^2$

Q51 · High · Concept: Near-number cube, minus case (T3)

Q51. Evaluate 999^3 without a calculator.

SOLUTION

Step 1: 999 = 1000 - 1. Use $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ with a = 1000, b = 1.

Step 2: $a^3 = 1{,}000{,}000{,}000$.

Step 3: $3a^2b = 3 \times 1{,}000{,}000 = 3{,}000{,}000$.

Step 4: $3ab^2 = 3 \times 1000 = 3000$.

Step 5: $b^3 = 1$.

Step 6: $999^3 = 1{,}000{,}000{,}000 - 3{,}000{,}000 + 3{,}000 - 1 = 997{,}002{,}999$.

» **Why this step?** Alternating signs with a = 1000 make every term a round number; only bookkeeping remains.

Hinglish: *$(1000-1)^3$: terms 1-3-3-1 with alternating minus. $10^9 - 3\times10^6 + 3\times10^3 - 1 = 997002999$. Column subtraction dhyan se - zeroes ginn kar, teen blocks mein todo: 997 | 002 | 999.*

Answer: 997,002,999

Q52 · High · Concept: Converse of zero-sum (T9/T14)

Q52. If $a^3 + b^3 + c^3 = 3abc$ and $a + b + c \neq 0$, prove that $a = b = c$.

SOLUTION

Step 1: Crown identity: $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$.

Step 2: Given the left side is 0 and $a+b+c \neq 0$, the second factor must be 0.

Step 3: Rewrite the second factor: $a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$.

Step 4: A sum of squares is 0 only when every square is 0: $a = b$, $b = c$, $c = a$.

Step 5: So $a = b = c$. ∎

» **Why this step?** The half-sum-of-squares rewriting converts an algebraic condition into three independent equations.

Hinglish: *Crown identity ka CONVERSE: cubes ka relation 3abc ke barabar hai aur sum zero nahi, toh doosra factor zero. Wo factor $\frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]$ hai - squares ka sum zero matlab sab barabar. Ye 'either-or' structure (sum zero YA sab equal) exams ka favourite hai.*

Answer: Proved: the half-sum-of-squares factor forces $a = b = c$.

Q53 · High · Concept: Bridge reversed for product (T11)

Q53. If $p^3 + q^3 = 35$ and $p + q = 5$, find pq.

SOLUTION

Step 1: Bridge: $p^3 + q^3 = (p+q)^3 - 3pq(p+q)$.

Step 2: $35 = 125 - 3pq \cdot 5 = 125 - 15pq$.

Step 3: $15pq = 90$, so $pq = 6$.

» **Why this step?** The bridge reads equally well backwards: the product is recovered from the two cube-related quantities.

Hinglish: *Bridge ka reverse use: $35 = 125 - 15pq$. $pq = 6$. Check: p, q roots of $t^2-5t+6 = (t-2)(t-3)$: $p=2$, $q=3 \rightarrow 8+27 = 35$ ✓.*

Answer: $pq = 6$

Q54 · High · Concept: Middle-term completion (T12)

Q54. Factorise: $x^4 + 5x^2 + 9$.

SOLUTION

Step 1: Compare with $(x^2 + 3)^2 = x^4 + 6x^2 + 9$ - one x^2 too many.

Step 2: Split: $x^4 + 5x^2 + 9 = x^4 + 6x^2 + 9 - x^2 = (x^2 + 3)^2 - x^2$.

Step 3: Difference of squares: $(x^2 + 3 - x)(x^2 + 3 + x)$.

Step 4: Final: $(x^2 - x + 3)(x^2 + x + 3)$.

» **Why this step?** Borrow and repay one x^2 to create a perfect square, then factor the resulting two-square difference.

Hinglish: *$(x^2+3)^2 = x^4+6x^2+9$ - humein 5 chahiye. Ek x^2 udhaar lo aur wapas do: $(x^2+3)^2 - x^2$. Difference of squares: $(x^2-x+3)(x^2+x+3)$. Q32 ka cousin - same completion, different numbers.*

Answer: $(x^2 - x + 3)(x^2 + x + 3)$

Q55 · High · Concept: Zero-sum at power two (T9)

Q55. Evaluate: $(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3$.

SOLUTION

Step 1: Let $P = a^2-b^2$, $Q = b^2-c^2$, $R = c^2-a^2$. Sum: $P+Q+R = 0$.

Step 2: Zero-sum shortcut: $P^3+Q^3+R^3 = 3PQR$.

Step 3: Value = $3(a^2-b^2)(b^2-c^2)(c^2-a^2)$.

Step 4: Optionally factor each difference further: = 3(a-b)(a+b)(b-c)(b+c)(c-a)(c+a).

» **Why this step?** Differences cycling through the same three letters always sum to zero - the shortcut does not care what the letters are made of.

Hinglish: *$(a^2-b^2)+(b^2-c^2)+(c^2-a^2) = 0$. Shortcut: 3PQR. Chaho toh har difference ko aur factor karo - (a-b)(a+b) etc. Zero-sum spotting phir se kaam par.*

Answer: $3(a^2 - b^2)(b^2 - c^2)(c^2 - a^2)$

Q56 · High · Concept: Full symmetric reconstruction (T9/T11)

Q56. If $a + b + c = 9$, $a^2 + b^2 + c^2 = 35$ and $abc = 6$, find $a^3 + b^3 + c^3$.

SOLUTION

Step 1: Pair-sum: $ab + bc + ca = [(a+b+c)^2 - (a^2+b^2+c^2)]/2 = (81 - 35)/2 = 23$.

Step 2: Crown identity: $a^3+b^3+c^3 - 3abc = (a+b+c)(a^2+b^2+c^2 - ab-bc-ca) = 9(35 - 23) = 108$.

Step 3: $a^3+b^3+c^3 = 108 + 3(6) = 126$.

» **Why this step?** Sum of squares gives the pair-sum; the crown identity then needs exactly (sum, pair-sum, squares) - all available.

Hinglish: *Wahi do-step recipe: pair-sum = (81-35)/2 = 23. Crown: 9×(35-23) = 108. Plus 3abc = 18. Total 126. Symmetric quantities mein har value doosri value se banti hai - bas bridge yaad rakho.*

Answer: 126

Q57 · HOTS · Concept: Sum-cube identity (T14)

Q57. Prove that $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$.

SOLUTION

Step 1: Start from the right side's known relative: $(a+b+c)^3 = a^3+b^3+c^3 + 3(a+b+c)(ab+bc+ca) - 3abc$ (standard expansion).

Step 2: Expand 3(a+b)(b+c)(c+a): as computed in Q49, it equals $3(ab+bc+ca)(a+b+c) - 3abc$.

Step 3: Indeed: $(a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc$ (expand and collect).

Step 4: Then $3(a+b)(b+c)(c+a) = 3(a+b+c)(ab+bc+ca) - 3abc$.

Step 5: Adding $a^3+b^3+c^3$ gives exactly $(a+b+c)^3$. ∎

Step 6: Direct verification with numbers: a = 1, b = 2, c = 3: LHS = 216. RHS = 36 + 3(3)(5)(4) = 36 + 180 = 216 ✓.

» **Why this step?** The two forms differ only in packaging: one counts pair-sums with the total sum, the other uses three pairwise sums.

Hinglish: *Do forms hain - ek (a+b+c)(ab+bc+ca) wali, ek (a+b)(b+c)(c+a) wali. Dono equal hain kyunki (a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc. Numeric check 1,2,3 se 216 dono taraf - proof pakka.*

Answer: Proved (both forms differ by the identity (a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc).

Q58 · HOTS · Concept: Fourth powers under zero sum (T14)

Q58. If a + b + c = 0, prove that $a^4 + b^4 + c^4 = 2(a^2b^2 + b^2c^2 + c^2a^2)$.

SOLUTION

Step 1: From a + b + c = 0: $(a+b+c)^2 = 0 \Rightarrow a^2+b^2+c^2 = -2(ab+bc+ca)$.

Step 2: Square both sides: $(a^2+b^2+c^2)^2 = 4(ab+bc+ca)^2$.

Step 3: LHS expands: $a^4+b^4+c^4 + 2(a^2b^2+b^2c^2+c^2a^2)$.

Step 4: RHS expands: $4[a^2b^2+b^2c^2+c^2a^2 + 2abc(a+b+c)] = 4(a^2b^2+b^2c^2+c^2a^2)$ since a+b+c = 0.

Step 5: Equate: $a^4+b^4+c^4 + 2\Sigma = 4\Sigma$ where $\Sigma = a^2b^2+b^2c^2+c^2a^2$.

Step 6: So $a^4+b^4+c^4 = 2\Sigma = 2(a^2b^2+b^2c^2+c^2a^2)$. ∎

Step 7: Check: a = 1, b = 1, c = -2: LHS = 1+1+16 = 18; RHS = 2(1+4+4) = 18 ✓.

» **Why this step?** Squaring the zero-sum relation upgrades degree 2 to degree 4; the abc-term dies because its multiplier is a+b+c = 0.

Hinglish: *Do squares ka game: pehle $(a+b+c)^2 = 0$ se $a^2+b^2+c^2 = -2\Sigma_1$ nikaalo, phir square karo. RHS mein 2abc(a+b+c) term aati hai jo zero ho jati hai - yahi magic hai. Numeric check 1,1,-2 se 18 = 18.*

Answer: Proved (both sides equal $2(a^2b^2+b^2c^2+c^2a^2)$ under a+b+c = 0).

Q59 · HOTS · Concept: Crown identity minimal data (T9)

Q59. If x + y + z = 3 and $x^2 + y^2 + z^2 = 5$, find $x^3 + y^3 + z^3 - 3xyz$.

SOLUTION

Step 1: Pair-sum: xy + yz + zx = (9 - 5)/2 = 2.

Step 2: Crown identity: $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2 - xy-yz-zx)$.

Step 3: Bracket: 5 - 2 = 3.

Step 4: Value = 3 × 3 = 9.

» **Why this step?** The requested combination is EXACTLY the crown identity's left side - no xyz value is ever needed.

Hinglish: *Question ne wahi combination manga hai jo crown identity ka LEFT side hai - xyz ki value jaanne ki zaroorat hi nahi! Pair-sum 2, bracket 3, answer 9. Spot-the-identity skill.*

Answer: 9

Q60 · HOTS · Concept: Symmetric functions in p, q, r (T11/T14)

Q60. If a + b + c = p, ab + bc + ca = q and abc = r, express $a^3 + b^3 + c^3$ in terms of p, q and r.

SOLUTION

Step 1: Crown identity: $a^3+b^3+c^3 - 3abc = (a+b+c)(a^2+b^2+c^2 - ab-bc-ca)$.

Step 2: Sum of squares: $a^2+b^2+c^2 = p^2 - 2q$.

Step 3: Bracket: $(p^2 - 2q) - q = p^2 - 3q$.

Step 4: So $a^3+b^3+c^3 - 3r = p(p^2 - 3q) = p^3 - 3pq$.

Step 5: Therefore $a^3+b^3+c^3 = p^3 - 3pq + 3r$.

» **Why this step?** Newton-style bookkeeping: every symmetric quantity is a polynomial in (p, q, r); the crown identity supplies the cubic one.

Hinglish: *Ye 'symmetric functions' ka master formula hai: cubes ka sum = $p^3 - 3pq + 3r$. Ek baar derive kar lo, har $a^3+b^3+c^3$ question 30 second mein hoga. p^2-3q bracket dhyan se - wo crown identity ka signature hai.*

Answer: $a^3 + b^3 + c^3 = p^3 - 3pq + 3r$

Q61 · HOTS · Concept: Cube difference collapse (T7/T14)

Q61. Prove that $(x + y)^3 - (x - y)^3 - 6y(x^2 - y^2) = 8y^3$.

SOLUTION

Step 1: Expand $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$ and $(x-y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$.

Step 2: Difference: $(x+y)^3 - (x-y)^3 = 6x^2y + 2y^3$.

Step 3: Also $6y(x^2 - y^2) = 6x^2y - 6y^3$.

Step 4: Subtract: $6x^2y + 2y^3 - 6x^2y + 6y^3 = 8y^3$. ∎

» **Why this step?** The difference of cubes kills x^3 and $3xy^2$; what remains ($6x^2y$) is exactly cancelled by the third term.

Hinglish: *Dono cubes ka difference: $6x^2y + 2y^3$ (x^3 aur $3xy^2$ cancel). Third term $6x^2y - 6y^3$. Minus karo: x^2y wale terms gayab, $2y^3+6y^3 = 8y^3$. Expansion systematically karo, terms ko group karo.*

Answer: Proved: the expression collapses to $8y^3$.

Q62 · HOTS · Concept: Divisibility via factoring (T2/T14)

Q62. Show that $2^{55} + 2^{58}$ is divisible by 9.

SOLUTION

Step 1: Factor the smaller power: $2^{55} + 2^{58} = 2^{55}(1 + 2^3) = 2^{55} \times 9$.

Step 2: The right side is 9 times the integer 2^{55}, hence divisible by 9. ∎

» **Why this step?** Common factor first (with exponents, the smaller power), then the bracket 1 + 8 supplies the required divisor.

Hinglish: *Chhota power bahar: $2^{55}(1 + 2^3) = 2^{55} \times 9$. Done - 9 upar dikh raha hai. Divisibility questions ka 90% answer factoring mein chhupa hota hai.*

Answer: $2^{55} + 2^{58} = 9 \cdot 2^{55}$, divisible by 9.

Q63 · HOTS · Concept: Long reciprocal chain (T10)

Q63. If $x + 1/x = 3$, find $x^5 + 1/x^5$.

SOLUTION

Step 1: Build the chain: $x^2 + 1/x^2 = 9 - 2 = 7$.

Step 2: $x^3 + 1/x^3 = 27 - 9 = 18$ (cube bridge).

Step 3: $x^4 + 1/x^4 = 7^2 - 2 = 47$.

Step 4: Multiply rule: $(x + 1/x)(x^4 + 1/x^4) = x^5 + 1/x^5 + x^3 + 1/x^3$.

Step 5: So $x^5 + 1/x^5 = 3 \times 47 - 18 = 141 - 18 = 123$.

» **Why this step?** Multiplying by the base chain $(x + 1/x)$ steps up one power at a time, at the cost of the two-lower term.

Hinglish: *Chain build karo: $3 \to 7 \to 18 \to 47$. Fifth power ke liye multiply rule: $(x+1/x)(x^4+1/x^4) = x^5+1/x^5 + x^3+1/x^3$. $141 - 18 = 123$. Har step verify karte jao - ek galti poori chain uda degi.*

Answer: 123

Q64 · HOTS · Concept: Cyclic cubic factorisation (T12/T14)

Q64. Factorise: $a^3(b - c) + b^3(c - a) + c^3(a - b)$.

SOLUTION

Step 1: Group to expose factors: $a^3(b-c) + b^3(c-a) + c^3(a-b)$.

Step 2: Verify (a - b) is a factor: setting a = b gives $a^3(b-c) + a^3(c-a) + c^3(a-a) = a^3(b-c+c-a) = 0$ ✓.

Step 3: Similarly (b - c) and (c - a) are factors (cyclic argument).

Step 4: Degree count: the expression is degree 4; three linear factors (degree 3) leave one more degree: k × (a+b+c) for a constant k.

Step 5: Find k by testing a = 1, b = 2, c = 3: LHS = 1(2-3) + 8(3-1) + 27(1-2) = -1 + 16 - 27 = -12. Factors: (1-2)(2-3)(3-1) = (-1)(-1)(2) = 2; (a+b+c) = 6. So $2 \times k \times 6 = -12 \Rightarrow k = -1$.

Step 6: Final: -(a - b)(b - c)(c - a)(a + b + c).

» **Why this step?** The factor theorem finds the linear factors; degree counting reveals the missing factor; one numeric evaluation pins the constant.

Hinglish: *Classic cyclic monster! (a-b), (b-c), (c-a) teeno factors hain - substitute karke verify karo. Degree 4 hai, teen factors degree 3 - bacha (a+b+c) with constant k. Numeric test se k = -1. Ye 'test values' trick constants dhoondne ka universal tarika hai.*

Answer: -(a - b)(b - c)(c - a)(a + b + c)

Q65 · HOTS · Concept: Sum of squares is zero (T5/T14)

Q65. If $x^2 + y^2 + 2x - 4y + 5 = 0$, find $x^3 + y^3$.

SOLUTION

Step 1: Rearrange: $(x^2 + 2x) + (y^2 - 4y) + 5 = 0$.

Step 2: Complete both squares: $(x + 1)^2 - 1 + (y - 2)^2 - 4 + 5 = 0$,

Step 3: so $(x + 1)^2 + (y - 2)^2 = 0$.

Step 4: A sum of squares is zero only when both squares are zero: x = -1, y = 2.

Step 5: $x^3 + y^3 = -1 + 8 = 7$.

» **Why this step?** Completing squares converts one equation in two unknowns into two forced equations - because squares cannot be negative.

Hinglish: *Squares complete karo: $(x+1)^2 + (y-2)^2 = 0$. Dono non-negative - dono zero hone padenge. x = -1, y = 2. Cubes: -1 + 8 = 7. Ye 'sum of squares = 0' trick real numbers ka superpower hai.*

Answer: x = -1, y = 2, so $x^3 + y^3 = 7$

Q66 · HOTS · Concept: Geometric proof of identities (T14)

Q66. Prove geometrically, using area models, that (a) $(a+b)^2 = a^2 + 2ab + b^2$ and (b) $(a-b)^2 = a^2 - 2ab + b^2$.

SOLUTION

Step 1: (a) Draw a square of side a + b. Split each side into segments a and b.

Step 2: The four regions are: a^2 (corner), b^2 (opposite corner), and TWO rectangles ab.

Step 3: Total area: $(a+b)^2 = a^2 + b^2 + ab + ab = a^2 + 2ab + b^2$. ∎

Step 4: (b) Draw a square of side a. Mark off a strip of width b along one side and the bottom.

Step 5: The remaining square $(a-b)^2$ equals a^2 minus TWO strips ab (top strip and side strip).

Step 6: The two removed strips overlap in a b^2 square that was subtracted twice - add it back once:

Step 7: $(a-b)^2 = a^2 - 2ab + b^2$. ∎ (See Figures 4.1 and 4.2.)

» **Why this step?** Areas cannot be negative and do not depend on how you slice them - so a dissection is a proof.

Hinglish: *Area proof - figure dekho (4.1, 4.2). $(a+b)^2$: chaar tukde - do squares, do rectangles. $(a-b)^2$: do strips minus karo, par b^2 corner DO baar minus hua tha - ek baar wapas jodo. Isliye $+b^2$. Drawing se identity kabhi galat yaad nahi hogi.*

Answer: Both identities proved by dissection (Figures 4.1 and 4.2).

Q67 · HOTS · Concept: Chained differences (T9/T14)

Q67. If x = a - b, y = b - c and z = c - a, find the value of $x^3 + y^3 + z^3 + xyz$.

SOLUTION

Step 1: Sum: x + y + z = a - b + b - c + c - a = 0.

Step 2: Zero-sum shortcut: $x^3 + y^3 + z^3 = 3xyz$.

Step 3: So $x^3 + y^3 + z^3 + xyz = 3xyz + xyz = 4xyz$.

Step 4: Substitute back: 4(a - b)(b - c)(c - a).

» **Why this step?** The zero sum is structural (a telescoping chain), so the cube shortcut holds whatever a, b, c are.

Hinglish: *Telescoping sum zero: x+y+z = 0. Shortcut: cubes = 3xyz. Plus xyz = 4xyz. Substitute: 4(a-b)(b-c)(c-a). Do-minute question, 5-mark answer.*

Answer: 4(a - b)(b - c)(c - a)

Q68 · HOTS · Concept: Double grouping (T12)

Q68. Factorise completely: $x^2y^2 - 5x^2y - 5xy^2 + 25xy$.

SOLUTION

Step 1: Common factor first: xy · (xy - 5x - 5y + 25).

Step 2: Group inside: (xy - 5x) - (5y - 25) = x(y - 5) - 5(y - 5) = (x - 5)(y - 5).

Step 3: Final: xy(x - 5)(y - 5).

» **Why this step?** The HCF shrinks four terms to four simpler terms; grouping then finishes in one pass.

Hinglish: *Pehle xy common - har term mein tha! Andar grouping: x(y-5) - 5(y-5) = (x-5)(y-5). Common factor + grouping ka clean combo. 'Completely' = xy bhi factor hai, usko likhna mat bhoolna.*

Answer: xy(x - 5)(y - 5)

Q69 · HOTS · Concept: Unit-sum identity (T9/T14)

Q69. If x + y = 1, prove that $x^3 + y^3 + 3xy = 1$.

SOLUTION

Step 1: Cube bridge: $x^3 + y^3 = (x+y)^3 - 3xy(x+y)$.

Step 2: With x + y = 1: $x^3 + y^3 = 1 - 3xy$.

Step 3: Add 3xy to both sides: $x^3 + y^3 + 3xy = 1$. ∎

Step 4: Check with x = 1/4, y = 3/4: 1/64 + 27/64 + 3(3/16) = 28/64 + 9/16 = 7/16 + 9/16 = 1 ✓.

» **Why this step?** A fixed pair-sum (here 1) turns the bridge into a relation between the cubes and the product.

Hinglish: *x+y = 1 bridge ko simplify karta hai: cubes = 1 - 3xy. 3xy wapas jodo: 1. Check 1/4 aur 3/4 se - perfect. Ye identity x+y = k ke liye generalise hoti hai: $x^3+y^3+3kxy = k^3$.*

Answer: Proved: $x^3 + y^3 + 3xy = 1$ whenever x + y = 1.

Q70 · HOTS · Concept: Four-cube grand identity (T9/T14)

Q70. Prove that $(x + y + z)^3 - (y + z - x)^3 - (z + x - y)^3 - (x + y - z)^3 = 24xyz$.

SOLUTION

Step 1: Let $s = x + y + z$, and $P = y+z-x$, $Q = z+x-y$, $R = x+y-z$.

Step 2: Observe: $P + Q + R = (y+z-x)+(z+x-y)+(x+y-z) = x + y + z = s$.

Step 3: So the left side is $s^3 - P^3 - Q^3 - R^3$ with $s = P + Q + R$: that is $(P+Q+R)^3 - P^3 - Q^3 - R^3$.

Step 4: By Q49's identity: $(P+Q+R)^3 - P^3 - Q^3 - R^3 = 3(P+Q)(Q+R)(R+P)$.

Step 5: Compute the pair-sums: $P + Q = 2z$, $Q + R = 2x$, $R + P = 2y$.

Step 6: Left side $= 3(2z)(2x)(2y) = 24xyz$. ∎

» **Why this step?** Renaming the four brackets makes them sum to a single letter; the earlier identity then converts everything to pair-sums, each of which is 2 times one original variable.

Hinglish: *Ye book ka showpiece hai. Trick: char brackets ko P, Q, R naam do - inka sum $s = x+y+z$ nikalta hai. Ab $(P+Q+R)^3 - P^3-Q^3-R^3 = 3(P+Q)(Q+R)(R+P)$. Pair-sums: 2z, 2x, 2y. Product: 24xyz. Bade questions mein hamesha pehle 'renaming' dhoondo.*

Answer: Proved: the left side equals $3(2z)(2x)(2y) = 24xyz$.

4.4 Chapter MCQs (80 questions, answers inline)

Each MCQ lists the options, the correct answer, a full explanation, the trap analysis for the tempting wrong options, and a Hinglish note. Options are engineered from real student mistakes, so think before you pick.

MCQ 1 · Normal · Concept: Binomial square

1. $(x + 4)^2$ equals:

A. $x^2 + 16$

B. $x^2 + 8x + 16$

C. $x^2 + 4x + 16$

D. $x^2 + 8x + 8$

Correct Answer: (B)

$(a+b)^2 = a^2 + 2ab + b^2$: $x^2 + 2 \cdot x \cdot 4 + 16 = x^2 + 8x + 16$.

Why (A) is wrong: Cross term completely dropped.

Why (C) is wrong: Coefficient 2 forgotten on the cross term.

Why (D) is wrong: 16 not squared properly.

Hinglish: *Cross term $2 \cdot x \cdot 4 = 8x$ bhoolna sabse common galti. $(a+b)^2 \neq a^2+b^2$ - kabhi nahi.*

MCQ 2 · Normal · Concept: (x+a)(x+b)

2. $(x + 5)(x + 3)$ equals:

A. $x^2 + 15x + 8$

B. $x^2 + 8x + 15$

C. $x^2 + 8x + 8$

D. $x^2 + 15$

Correct Answer: (B)

Sum $5+3 = 8$ feeds the x-term; product 15 feeds the constant: $x^2 + 8x + 15$.

Why (A) is wrong: Sum and product swapped.

Why (C) is wrong: Product taken as 5+3.

Why (D) is wrong: Middle term dropped.

Hinglish: *Sum middle mein, product last mein. 8x + 15.*

MCQ 3 · Normal · Concept: Near-number square

3. 101^2 equals:

A. 10201

B. 10021

C. 1021

D. 10404

Correct Answer: (A)

$(100+1)^2 = 10000 + 200 + 1 = 10201$.

Why (B) is wrong: Cross term 20 likh diya.

Why (C) is wrong: Dropped both extra terms.

Why (D) is wrong: That is 102^2.

Hinglish: *10000 + 200 + 1. Cross term 2×100×1 = 200.*

MCQ 4 · Normal · Concept: Near-number square minus

4. 98^2 equals:

A. 9964

B. 9604

C. 9640

D. 9844

Correct Answer: (B)

$(100-2)^2 = 10000 - 400 + 4 = 9604$.

Why (A) is wrong: Last term -4 kiya (squares never negative).

Why (C) is wrong: Digits rearranged.

Why (D) is wrong: Wrong cross term.

Hinglish: *Minus wale case: cross term MINUS 400, phir +4. 9604.*

MCQ 5 · Normal · Concept: Difference of squares

5. $x^2 - 49$ equals:

A. $(x-7)^2$

B. $(x+7)(x-7)$

C. $(x+49)(x-1)$

D. $x(x-49)$

Correct Answer: (B)

$a^2 - b^2 = (a+b)(a-b)$ with $b = 7$.

Why (A) is wrong: That is $x^2 - 14x + 49$.

Why (C) is wrong: 49 is not 49^2.

Why (D) is wrong: No common factor exists.

Hinglish: *$49 = 7^2$. $(x+7)(x-7)$. $(x-7)^2$ would expand to $x^2-14x+49$ - alag cheez.*

MCQ 6 · Normal · Concept: Equidistant product

6. 51 × 49 equals:

A. 2499
B. 2501
C. 2490
D. 2489

Correct Answer: (A)

(50+1)(50-1) = 2500 - 1 = 2499.

Why (B) is wrong: Added 1 instead of subtracting.

Why (C) is wrong: Wrong offset.

Why (D) is wrong: Arithmetic slip.

Hinglish: *Average 50, offset 1: 2500-1 = 2499. Multiplication one second mein.*

MCQ 7 · Normal · Concept: Companion identity

7. $(a+b)^2 - (a-b)^2$ equals:

A. 2ab
B. 4ab
C. $2(a^2+b^2)$
D. 0

Correct Answer: (B)

Subtracting the expansions cancels a^2 and b^2, leaving 4ab.

Why (A) is wrong: Missed the factor 2.

Why (C) is wrong: That is the SUM of the two squares.

Why (D) is wrong: The difference is not zero.

Hinglish: *Squares ka difference = 4ab. Sum = $2(a^2+b^2)$. Dono yaad rakho.*

MCQ 8 · Normal · Concept: Middle-term splitting

8. $x^2 + 7x + 12$ factorises as:

A. (x+3)(x+4)
B. (x+2)(x+6)
C. (x+1)(x+12)
D. (x-3)(x-4)

Correct Answer: (A)

3 + 4 = 7 and 3 × 4 = 12.

Why (B) is wrong: 2+6 = 8 ≠ 7.

Why (C) is wrong: 1+12 = 13.

Why (D) is wrong: Signs wrong: sum would be -7.

Hinglish: *Sum 7, product 12: 3 aur 4.*

MCQ 9 · Normal · Concept: Perfect square spotting

9. $9x^2 + 12x + 4$ equals:

A. $(3x+2)^2$

B. $(3x+4)^2$

C. $(9x+2)^2$

D. $(3x+2)(3x-2)$

Correct Answer: (A)

$(3x)^2 + 2(3x)(2) + 2^2 = (3x+2)^2$.

Why (B) is wrong: $(3x+4)^2 = 9x^2+24x+16$.

Why (C) is wrong: $(9x+2)^2 = 81x^2+36x+4$.

Why (D) is wrong: That is $9x^2-4$.

Hinglish: *Middle term $12x = 2\times3x\times2$ ✓. Perfect square.*

MCQ 10 · Normal · Concept: Sum of cubes

10. $27a^3 + 8b^3$ factorises as:

A. $(3a+2b)(9a^2-6ab+4b^2)$

B. $(3a+2b)^3$

C. $(3a+2b)(9a^2+6ab+4b^2)$

D. $(3a-2b)(9a^2+6ab+4b^2)$

Correct Answer: (A)

$(3a)^3+(2b)^3 = (3a+2b)((3a)^2-(3a)(2b)+(2b)^2)$.

Why (B) is wrong: That is a cube, not a factorisation.

Why (C) is wrong: Middle sign wrong.

Why (D) is wrong: Binomial sign wrong.

Hinglish: *Trinomial ka middle term NEGATIVE - binomial ke sign ka ulta.*

MCQ 11 · Normal · Concept: Difference of cubes

11. $x^3 - 27$ equals:

A. $(x-3)(x^2+3x+9)$

B. $(x-3)(x^2-3x+9)$

C. $(x-3)^3$

D. $(x+3)(x^2-3x+9)$

Correct Answer: (A)

$A^3-B^3 = (A-B)(A^2+AB+B^2)$ with $B = 3$.

Why (B) is wrong: Trinomial middle sign flipped.

Why (C) is wrong: That is $x^3-9x^2+27x-27$.

Why (D) is wrong: Binomial sign wrong.

Hinglish: *Difference: binomial minus, trinomial plus. $(x-3)(x^2+3x+9)$.*

MCQ 12 · Normal · Concept: x + 1/x square

12. If $x + 1/x = 2$, then $x^2 + 1/x^2$ equals:

A. 2

B. 4

C. 6

D. 0

Correct Answer: (A)

$(x+1/x)^2 = x^2+2+1/x^2 = 4 \Rightarrow x^2+1/x^2 = 2$.

Why (B) is wrong: Forgot to subtract 2.

Why (C) is wrong: Subtracted -2.

Why (D) is wrong: Sign error.

Hinglish: *4 - 2 = 2. (x=1 hit karke bhi dekh sakte ho.)*

MCQ 13 · Normal · Concept: Trinomial square

13. $(a + b + c)^2$ equals:

A. $a^2+b^2+c^2+2ab+2bc+2ca$

B. $a^2+b^2+c^2+ab+bc+ca$

C. $a^2+b^2+c^2$

D. $a^2+b^2+c^2+2abc$

Correct Answer: (A)

Three squares plus TWICE each of the three pairwise products.

Why (B) is wrong: Missing the 2s.

Why (C) is wrong: Cross terms dropped.

Why (D) is wrong: 2abc is a triple product - never appears.

Hinglish: *Teen squares + teen cross terms, har cross term par coefficient 2.*

MCQ 14 · Normal · Concept: Symmetric reconstruction

14. If $a + b = 8$ and $ab = 12$, then $a^2 + b^2$ equals:

A. 40

B. 64

C. 76

D. 52

Correct Answer: (A)

$a^2+b^2 = (a+b)^2 - 2ab = 64 - 24 = 40$.

Why (B) is wrong: $(a+b)^2$ hi le liya.

Why (C) is wrong: Subtracted 2ab twice-ish.

Why (D) is wrong: Arithmetic slip.

Hinglish: *64 - 24 = 40. Bridge yaad rakho.*

MCQ 15 · Normal · Concept: Cube bridge

15. If $x + 1/x = 3$, then $x^3 + 1/x^3$ equals:

A. 18

B. 27

C. 21

D. 30

Correct Answer: (A)

$x^3+1/x^3 = 27 - 3\cdot3 = 27 - 9 = 18$.

Why (B) is wrong: $(x+1/x)^3$ hi maan liya.

Why (C) is wrong: Subtracted 6.

Why (D) is wrong: Added 3.

Hinglish: *$s^3 - 3s = 27 - 9 = 18$.*

MCQ 16 · Normal · Concept: Equidistant product 2

16. 103×97 equals:

A. 9991

B. 10009

C. 9981

D. 9991±... 9979

Correct Answer: (A)

$(100+3)(100-3) = 10000 - 9 = 9991$.

Why (B) is wrong: Added 9.

Why (C) is wrong: Offset squared wrong.

Why (D) is wrong: Wrong offset product.

Hinglish: *$10000 - 9 = 9991$.*

MCQ 17 · Normal · Concept: Binomial square with coefficients

17. $(2x + 3y)^2$ equals:

A. $4x^2 + 6xy + 9y^2$

B. $4x^2 + 12xy + 9y^2$

C. $2x^2 + 6xy + 3y^2$

D. $4x^2 + 9y^2$

Correct Answer: (B)

$(2x)^2 + 2(2x)(3y) + (3y)^2 = 4x^2 + 12xy + 9y^2$.

Why (A) is wrong: Cross term halved.

Why (C) is wrong: Forgot to square the coefficients.

Why (D) is wrong: Cross term dropped.

Hinglish: *Cross term $2\times2\times3 = 12$. Squares: 4 aur 9.*

MCQ 18 · Normal · Concept: Zero-sum cube

18. If $a + b + c = 0$, then $a^3 + b^3 + c^3$ equals:

A. 3abc

B. 0

C. -3abc

D. abc

Correct Answer: (A)

Crown identity with $a+b+c = 0$ gives $a^3+b^3+c^3 = 3abc$.

Why (B) is wrong: Only the PRODUCT side vanishes, not the cubes.

Why (C) is wrong: Sign flipped.

Why (D) is wrong: Missing the 3.

Hinglish: *Zero-sum shortcut: 3abc. Yaad kar lo - har saal aata hai.*

MCQ 19 · Normal · Concept: Near-number cube

19. The last three digits of 999^3 are:

A. 299

B. 001

C. 999

D. 997

Correct Answer: (A)

$999^3 = (1000-1)^3 = 997002999$; last three digits 299.

Why (B) is wrong: That is 1001^3's neighbourhood error.

Why (C) is wrong: Copied the base.

Why (D) is wrong: First three digits.

Hinglish: *Full value 997002999 - last three: 299.*

MCQ 20 · Normal · Concept: Cube expansion coefficient

20. The coefficient of a^2b in $(5a - 3b)^3$ is:

A. -225

B. 225

C. -75

D. 75

Correct Answer: (A)

$3(5a)^2(-3b) = 3 \cdot 25 \cdot (-3)\, a^2b = -225a^2b$.

Why (B) is wrong: Sign dropped.

Why (C) is wrong: Coefficient 3 missed on the pattern.

Why (D) is wrong: Sign and 3 both wrong.

Hinglish: *$3 \times 25 \times (-3) = -225$. Sign minus - kyunki $(a-b)^3$ mein doosra term negative.*

MCQ 21 · Normal · Concept: Difference of squares with coefficients

21. $4a^2 - 9b^2$ equals:

A. $(2a-3b)^2$

B. $(2a+3b)(2a-3b)$

C. $(4a+9b)(a-b)$

D. $(2a+3b)^2$

Correct Answer: (B)

$(2a)^2 - (3b)^2 = (2a+3b)(2a-3b)$.

Why (A) is wrong: That is $4a^2-12ab+9b^2$.

Why (C) is wrong: Wrong pairing.

Why (D) is wrong: Plus sign - that is a sum.

Hinglish: *$4a^2 = (2a)^2$, $9b^2 = (3b)^2$. Difference of squares.*

MCQ 22 · Normal · Concept: Chained difference of squares

22. $(x-2)(x+2)(x^2+4)$ equals:

A. $x^4 - 16$

B. $x^4 + 16$

C. $x^2 - 16$

D. $x^4 - 4$

Correct Answer: (A)

First pair: x^2-4. Then $(x^2-4)(x^2+4) = x^4-16$.

Why (B) is wrong: Plus instead of minus.

Why (C) is wrong: Stopped after one round.

Why (D) is wrong: 4 not squared.

Hinglish: *Do baar difference of squares: x^2-4, phir x^4-16.*

MCQ 23 · Normal · Concept: Cube sum formula

23. $a^3 + b^3$ equals:

A. $(a+b)(a^2-ab+b^2)$

B. $(a+b)(a^2+ab+b^2)$

C. $(a+b)^3$

D. $(a-b)(a^2+ab+b^2)$

Correct Answer: (A)

Standard factorisation; the trinomial's middle term is -ab.

Why (B) is wrong: That is a^3-b^3's factor.

Why (C) is wrong: Not a factorisation.

Why (D) is wrong: Wrong binomial.

Hinglish: *Sum: binomial plus, trinomial middle minus.*

MCQ 24 · Normal · Concept: Decimal equidistant product

24. 1.02 × 0.98 equals:

A. 0.9996
B. 0.9906
C. 1.0004
D. 0.999

Correct Answer: (A)

(1+0.02)(1-0.02) = 1 - 0.0004 = 0.9996.

Why (B) is wrong: 0.02^2 miscomputed as 0.0094?

Why (C) is wrong: Added instead of subtracted.

Why (D) is wrong: Rounded too early.

Hinglish: *0.02^2 = 0.0004 - four decimal places. 1 - 0.0004.*

MCQ 25 · Normal · Concept: Difference chain

25. If $x - 1/x = 4$, then $x^2 + 1/x^2$ equals:

A. 18
B. 14
C. 16
D. 20

Correct Answer: (A)

$(x-1/x)^2 = x^2-2+1/x^2 = 16 \Rightarrow x^2+1/x^2 = 18$.

Why (B) is wrong: Subtracted 2.

Why (C) is wrong: Forgot the 2 entirely.

Why (D) is wrong: Added 4.

Hinglish: *16 + 2 = 18. Difference wale case mein PLUS 2.*

MCQ 26 · Normal · Concept: Cube difference pattern

26. $(a - b)^3$ equals:

A. $a^3-3a^2b+3ab^2-b^3$
B. $a^3-3a^2b-3ab^2-b^3$
C. $a^3+3a^2b+3ab^2+b^3$
D. a^3-b^3

Correct Answer: (A)

Alternating signs: +, -, +, -.

Why (B) is wrong: Third term sign wrong.

Why (C) is wrong: That is $(a+b)^3$.

Why (D) is wrong: Middle terms dropped.

Hinglish: *Pattern 1-3-3-1 with alternating signs.*

MCQ 27 · Normal · Concept: Difference of cubes scaled

27. $64x^3 - 125$ equals:

A. $(4x-5)(16x^2+20x+25)$

B. $(4x-5)(16x^2-20x+25)$

C. $(4x-5)^3$

D. $(4x+5)(16x^2+20x+25)$

Correct Answer: (A)

$(4x)^3-5^3 = (4x-5)((4x)^2+4x\cdot5+25)$.

Why (B) is wrong: Trinomial middle sign wrong.

Why (C) is wrong: Not a cube.

Why (D) is wrong: Binomial sign wrong.

Hinglish: *$(4x)^2 = 16x^2$, $AB = 20x$, $B^2 = 25$. Plus signs in trinomial.*

MCQ 28 · Normal · Concept: Three-variable square bridge

28. If $a + b + c = 9$ and $ab + bc + ca = 26$, then $a^2 + b^2 + c^2$ equals:

A. 29

B. 55

C. 107

D. 29±... 33

Correct Answer: (A)

$81 = a^2+b^2+c^2 + 52$, so the sum of squares is 29.

Why (B) is wrong: Added 2Σ instead of subtracting.

Why (C) is wrong: Doubled wrongly.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$81 - 52 = 29$.*

MCQ 29 · Normal · Concept: Fourth-power difference

29. $x^4 - 81$ factorises completely as:

A. $(x-3)(x+3)(x^2+9)$

B. $(x^2-9)(x^2-9)$

C. $(x-3)(x+3)$

D. $(x^2+9)^2$

Correct Answer: (A)

$x^4-81 = (x^2-9)(x^2+9) = (x-3)(x+3)(x^2+9)$.

Why (B) is wrong: Same factor twice - wrong.

Why (C) is wrong: Incomplete.

Why (D) is wrong: Not a factorisation.

Hinglish: *Do baar factor: x^2-9 ko bhi todo. x^2+9 khatam.*

MCQ 30 · Normal · Concept: Square completion

30. $x^4 + x^2 + 1$ equals:

A. $(x^2+x+1)(x^2-x+1)$

B. $(x^2+1)^2$

C. $(x^2+x+1)^2$

D. $(x^2+2x+1)(x^2-2x+1)$

Correct Answer: (A)

Add-subtract x^2: $(x^2+1)^2 - x^2$ = difference of squares.

Why (B) is wrong: That is x^4+2x^2+1.

Why (C) is wrong: That is $x^4+2x^3+3x^2+2x+1$.

Why (D) is wrong: Wrong middle coefficients.

Hinglish: *Classic completion trick. $(x^2+1)^2$ mein x^2 ek zyada tha.*

MCQ 31 · Normal · Concept: Cube difference product

31. $(3 - a)(9 + 3a + a^2)$ equals:

A. $27 - a^3$

B. $27 + a^3$

C. $9 - a^3$

D. $3 - a^3$

Correct Answer: (A)

This is $(3-a)((3)^2+3\cdot a+a^2) = 3^3 - a^3$.

Why (B) is wrong: Sign of the cube wrong.

Why (C) is wrong: Took square of 3.

Why (D) is wrong: Mixed the two letters.

Hinglish: *Difference of cubes ka reverse: 3 = B, a = A. B^3-A^3.*

MCQ 32 · Normal · Concept: Equidistant product 3

32. 105×95 equals:

A. 9975

B. 10025

C. 9925

D. 9999

Correct Answer: (A)

$(100+5)(100-5) = 10000 - 25 = 9975$.

Why (B) is wrong: Added 25.

Why (C) is wrong: Offset error.

Why (D) is wrong: Arithmetic slip.

Hinglish: *10000 - 25 = 9975.*

MCQ 33 · Normal · Concept: Fourth-power chain

33. If $x + 1/x = 4$, then $x^4 + 1/x^4$ equals:

A. 194

B. 194±... 192

C. 256

D. 14

Correct Answer: (A)

$x^2+1/x^2 = 14$; $x^4+1/x^4 = 14^2 - 2 = 194$.

Why (B) is wrong: Forgot the -2.

Why (C) is wrong: Squared 4 instead of 14.

Why (D) is wrong: That is x^2+1/x^2.

Hinglish: *14 → 196 - 2 = 194.*

MCQ 34 · Normal · Concept: Cube sum from data

34. If $x + y = 5$ and $xy = 6$, then $x^3 + y^3$ equals:

A. 35

B. 125

C. 53

D. 63

Correct Answer: (A)

$125 - 3 \cdot 6 \cdot 5 = 125 - 90 = 35$.

Why (B) is wrong: That is $(x+y)^3$.

Why (C) is wrong: Subtracted 3xy only.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$(x+y)^3 - 3xy(x+y) = 125 - 90 = 35$.*

MCQ 35 · Normal · Concept: 4ab application

35. Using $(a+b)^2 - (a-b)^2 = 4ab$ with $a = 5.5$ and $b = 4.5$, the value of $(a+b)^2 - (a-b)^2$ is:

A. 99

B. 10

C. 9.9

D. 1

Correct Answer: (A)

$4ab = 4 \times 5.5 \times 4.5 = 99$. (Note: this is the SQUARES-difference identity, not the direct difference $5.5^2 - 4.5^2$ which is 10.)

Why (B) is wrong: That is $5.5^2 - 4.5^2$ itself - a different question.

Why (C) is wrong: Decimal slip in 4×5.5×4.5.

Why (D) is wrong: Took only (a+b)(a-b).

Hinglish: *$4 \times 5.5 \times 4.5 = 99$. Identity hi manga hai question ne - $(a+b)^2-(a-b)^2$. Direct $5.5^2-4.5^2 = 10$ alag sawaal hai, confuse mat hona.*

MCQ 36 · Normal · Concept: Recognise a cube

36. $8a^3 + 27b^3 + 36a^2b + 54ab^2$ equals:

A. $(2a+3b)^3$

B. $(2a+3b)(4a^2+6ab+9b^2)$

C. $(2a+3b)^2(2a-3b)$

D. $(2a-3b)^3$

Correct Answer: (A)

The four terms match $(2a)^3 + 3(2a)^2(3b) + 3(2a)(3b)^2 + (3b)^3$ exactly.

Why (B) is wrong: That is the sum of cubes only - missing the cross terms.

Why (C) is wrong: Not a valid identity.

Why (D) is wrong: Sign pattern wrong.

Hinglish: *1-3-3-1 pattern: $(2a+3b)^3$.*

MCQ 37 · Normal · Concept: Crown identity zero condition

37. $x^3 + y^3 + z^3 - 3xyz = 0$ holds when:

A. $x + y + z = 0$ or $x = y = z$

B. only $x + y + z = 0$

C. only $x = y = z$

D. $xyz = 0$

Correct Answer: (A)

The identity factors as $(x+y+z)\cdot\frac{1}{2}[(x-y)^2+(y-z)^2+(z-x)^2]$, which is zero when either factor is zero.

Why (B) is wrong: Missed the equal case.

Why (C) is wrong: Missed the zero-sum case.

Why (D) is wrong: $xyz = 0$ alone is insufficient.

Hinglish: *Do cases: sum zero YA teeno barabar. Sirf ek case sochna aadha answer.*

MCQ 38 · Normal · Concept: Clever square completion

38. $999^2 + 2\cdot999 + 1$ equals:

A. 1000000

B. 999999

C. 998001

D. 1001999

Correct Answer: (A)

The expression is $999^2 + 2(999)(1) + 1^2 = (999+1)^2 = 1000^2 = 1{,}000{,}000$.

Why (B) is wrong: Off by one.

Why (C) is wrong: That is just 999^2.

Why (D) is wrong: Wrong assembly.

Hinglish: *Perfect square pehchano: $(999+1)^2$. 10^6.*

MCQ 39 · Normal · Concept: Trinomial square reverse

39. $(a + b + c)^2 - 2(ab + bc + ca)$ equals:

A. $a^2 + b^2 + c^2$

B. $a^2 + b^2 + c^2 - 2abc$

C. $(a-b-c)^2$

D. $2(ab+bc+ca)$

Correct Answer: (A)

$(a+b+c)^2 = a^2+b^2+c^2 + 2(ab+bc+ca)$; subtracting $2(ab+bc+ca)$ leaves exactly $a^2+b^2+c^2$.

Why (B) is wrong: Triple product kabhi expand nahi hota is identity se.

Why (C) is wrong: $(a-b-c)^2$ mein $+2bc$ aata hai - mismatch.

Why (D) is wrong: Ussi quantity ko wapas likh diya.

Hinglish: *Square kholo: teen squares + 2Σ. Ab 2Σ minus - sirf squares bache. Ye wahi bridge hai jo sum of squares nikaalta hai.*

MCQ 40 · Normal · Concept: General (x+a)(x+b)

40. $x^2 - (a + b)x + ab$ equals:

A. $(x-a)(x-b)$

B. $(x+a)(x+b)$

C. $(x-a)(x+b)$

D. $(x+a)(x-b)$

Correct Answer: (A)

Sum of roots-of-constants: $-(a+b)$ needs both constants negative in the factors: $(x-a)(x-b)$.

Why (B) is wrong: Signs flipped.

Why (C) is wrong: Mixed signs give $-ab$ product.

Why (D) is wrong: Sum would be $b-a$.

Hinglish: *Minus sum, plus product → dono factors minus.*

MCQ 41 · Normal · Concept: Cube of x + 1/x

41. $(x + 1/x)^3$ equals:

A. $x^3 + 1/x^3 + 3(x + 1/x)$

B. $x^3 + 1/x^3$

C. $x^3 + 1/x^3 + 3$

D. $x^3 + 1/x^3 + 6x$

Correct Answer: (A)

$(a+b)^3 = a^3+b^3+3ab(a+b)$ with $ab = 1$.

Why (B) is wrong: Middle terms dropped.

Why (C) is wrong: Missing $(x+1/x)$ multiplier.

Why (D) is wrong: Wrong middle term.

Hinglish: *ab = 1 magic: expansion sirf 3(x+1/x) extra deta hai.*

MCQ 42 · Normal · Concept: Bridge reversed

42. If $a^3 + b^3 = 152$ and $a + b = 8$, then ab equals:

A. 15
B. 12
C. 19
D. 16

Correct Answer: (A)

$152 = 512 - 3ab(8) \Rightarrow 24ab = 360 \Rightarrow ab = 15$.

Why (B) is wrong: Arithmetic slip.

Why (C) is wrong: Divided by wrong number.

Why (D) is wrong: Assumed ab = a+b.

Hinglish: *512 - 152 = 360. 360/24 = 15.*

MCQ 43 · Normal · Concept: Square difference linear

43. $(2x + 3)^2 - (2x - 3)^2$ equals:

A. 24x
B. 12x
C. 6x
D. 18

Correct Answer: (A)

4ab form with a = 2x, b = 3: $4(2x)(3) = 24x$.

Why (B) is wrong: Missed the factor 2.

Why (C) is wrong: Dropped everything.

Why (D) is wrong: Computed only the constants.

Hinglish: *$(a+b)^2-(a-b)^2 = 4ab = 4(2x)(3) = 24x$.*

MCQ 44 · Normal · Concept: Fourth-power difference

44. $a^4 - 16$ equals:

A. $(a-2)(a+2)(a^2+4)$
B. $(a^2-4)(a^2-4)$
C. $(a-2)(a+2)$
D. $(a^2+4)^2$

Correct Answer: (A)

$(a^2-4)(a^2+4) = (a-2)(a+2)(a^2+4)$.

Why (B) is wrong: Same factor twice.

Why (C) is wrong: Incomplete.

Why (D) is wrong: Not a factorisation.

Hinglish: *Do rounds: a^2-4 phir todo. a^2+4 final.*

MCQ 45 · Normal · Concept: Near-number cube

45. 99^3 equals:

A. 970299

B. 97299

C. 997299

D. 997002999

Correct Answer: (A)

$(100-1)^3 = 1{,}000{,}000 - 30{,}000 + 300 - 1 = 970{,}299$.

Why (B) is wrong: Place value slip.

Why (C) is wrong: Cross term 30000 vs 300 confusion.

Why (D) is wrong: That is 999^3.

Hinglish: *1-3-3-1 alternating: $10^6 - 3\times10^4 + 3\times10^2 - 1 = 970299$.*

MCQ 46 · Normal · Concept: Cube difference quotient

46. $(x^3 - y^3)/(x - y)$ equals:

A. $x^2 + xy + y^2$

B. $x^2 - xy + y^2$

C. $x^2 + y^2$

D. $x + y$

Correct Answer: (A)

$x^3-y^3 = (x-y)(x^2+xy+y^2)$; divide by $(x-y)$.

Why (B) is wrong: Sign of xy wrong.

Why (C) is wrong: Cross term dropped.

Why (D) is wrong: Not a polynomial division result.

Hinglish: *Difference of cubes ka quotient: trinomial with PLUS xy.*

MCQ 47 · Normal · Concept: Zero-sum squares relation

47. If $p + q + r = 0$, then $pq + qr + rp$ equals:

A. $-(p^2+q^2+r^2)/2$

B. $(p^2+q^2+r^2)/2$

C. 0

D. $p^2+q^2+r^2$

Correct Answer: (A)

$(p+q+r)^2 = p^2+q^2+r^2+2\Sigma = 0 \Rightarrow \Sigma = -(p^2+q^2+r^2)/2$.

Why (B) is wrong: Sign wrong.

Why (C) is wrong: Σ need not be 0 (e.g. 1, 1, -2 gives $\Sigma = -3$).

Why (D) is wrong: Sign and factor both wrong.

Hinglish: *Square expand karo, zero set karo: Σ negative half of squares.*

MCQ 48 · Normal · Concept: Decimal square

48. 9.8^2 equals:

A. 96.04

B. 98.04

C. 96.4

D. 94.04

Correct Answer: (A)

$(10-0.2)^2 = 100 - 4 + 0.04 = 96.04$.

Why (B) is wrong: Wrong subtraction.

Why (C) is wrong: Dropped the 0.04.

Why (D) is wrong: Cross term wrong.

Hinglish: *$0.2^2 = 0.04$ - chhota term mat bhoolna.*

MCQ 49 · Normal · Concept: Chained difference of squares 2

49. $(x+3)(x-3)(x^2+9)$ equals:

A. $x^4 - 81$

B. $x^4 + 81$

C. $x^2 - 81$

D. $x^4 - 9$

Correct Answer: (A)

$(x^2-9)(x^2+9) = x^4-81$.

Why (B) is wrong: Plus sign.

Why (C) is wrong: One round only.

Why (D) is wrong: 81 not squared from 9^2... it is $9^2 = 81$, but x^2 squared gives x^4.

Hinglish: *Do baar difference of squares.*

MCQ 50 · Normal · Concept: Scaled cube bridge

50. If $2x + 3y = 12$ and $xy = 6$, then $8x^3 + 27y^3$ equals:

A. 432

B. 1728

C. 1296

D. 1512

Correct Answer: (A)

$(2x)^3+(3y)^3 = (2x+3y)^3 - 3AB(2x+3y)$ where $AB = (2x)(3y) = 6xy = 36$. So $1728 - 3(36)(12) = 1728 - 1296 = 432$.

Why (B) is wrong: That is $(2x+3y)^3$ - the raw cube of the sum.

Why (C) is wrong: AB taken as 6 instead of 36.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$A = 2x$, $B = 3y$. $AB = 6xy = 36$. Bridge: $1728 - 1296 = 432$.*

MCQ 51 · Normal · Concept: Middle-term splitting signs

51. x^2 - 11x - 26 factorises as:

A. (x-13)(x+2)

B. (x+13)(x-2)

C. (x-13)(x-2)

D. (x-26)(x+1)

Correct Answer: (A)

Sum -11, product -26: -13 and +2.

Why (B) is wrong: Sum +11.

Why (C) is wrong: Product +26.

Why (D) is wrong: 13×1 = 13 ≠ 26.

Hinglish: *Product negative → alag signs; bada magnitude minus.*

MCQ 52 · Normal · Concept: Cube difference simplify

52. $(a + b)^3$ - $(a - b)^3$ equals:

A. $2b(3a^2 + b^2)$

B. $2a(3a^2 + b^2)$

C. $6ab^2$

D. $8b^3$

Correct Answer: (A)

Subtracting kills a^3 and $3ab^2$: survivors $6a^2b + 2b^3 = 2b(3a^2+b^2)$.

Why (B) is wrong: That is the SUM of the cubes.

Why (C) is wrong: Missing the $2b^3$ term.

Why (D) is wrong: Only if a = 0.

Hinglish: *Difference mein even-in-b terms bachte hain... dekho: $6a^2b + 2b^3$.*

MCQ 53 · Normal · Concept: Cube plus constant

53. $x^3 + 8$ equals:

A. $(x+2)(x^2-2x+4)$

B. $(x+2)(x^2+2x+4)$

C. $(x+2)^3$

D. $(x-2)(x^2+2x+4)$

Correct Answer: (A)

$(x)^3+(2)^3 = (x+2)(x^2-2x+4)$.

Why (B) is wrong: Middle sign wrong.

Why (C) is wrong: That is $x^3+6x^2+12x+8$.

Why (D) is wrong: Binomial sign wrong.

Hinglish: *Sum of cubes: middle term minus.*

MCQ 54 · Normal · Concept: Zero-sum with pair sums

54. If $a + b + c = 0$, then $(a+b)^3 + (b+c)^3 + (c+a)^3$ equals:

A. $3(a+b)(b+c)(c+a)$

B. $-3abc$

C. 0

D. $(a+b+c)^3$

Correct Answer: (A)

$A = a+b$, $B = b+c$, $C = c+a$ have $A+B+C = 0$, so $A^3+B^3+C^3 = 3ABC$.

Why (B) is wrong: That equals $-3abc = 3(a+b)(b+c)(c+a)$ here since $A = -c$ etc... actually $3ABC = 3(-c)(-a)(-b) = -3abc$. Both A and B are numerically equal! But the direct identity answer is A.

Why (C) is wrong: Not zero - the brackets are nonzero.

Why (D) is wrong: $a+b+c = 0$ makes this 0, not the expression.

Hinglish: *Renames ka sum bhi zero - shortcut phir se.*

MCQ 55 · Normal · Concept: Equidistant product 4

55. 71×69 equals:

A. 4899

B. 4901

C. 4890

D. 4959

Correct Answer: (A)

$(70+1)(70-1) = 4900 - 1 = 4899$.

Why (B) is wrong: Plus 1.

Why (C) is wrong: Wrong average.

Why (D) is wrong: Arithmetic slip.

Hinglish: *4900 - 1.*

MCQ 56 · Normal · Concept: Completion reverse

56. $(x^2 - x + 1)(x^2 + x + 1)$ equals:

A. $x^4 + x^2 + 1$

B. $x^4 - x^2 + 1$

C. $x^4 + 1$

D. $x^4 + 2x^2 + 1$

Correct Answer: (A)

Difference of squares: $(x^2+1)^2 - x^2 = x^4 + x^2 + 1$.

Why (B) is wrong: Middle sign flip.

Why (C) is wrong: Lost the x^2.

Why (D) is wrong: That is $(x^2+1)^2$.

Hinglish: *$(x^2+1)^2 - x^2$. x^3 terms cancel.*

MCQ 57 · Normal · Concept: Cube difference numeric

57. $(x + 1)^3 - (x - 1)^3$ at $x = 2$ equals:

A. 26

B. 24

C. 28

D. 8

Correct Answer: (A)

$27 - 1 = 26$.

Why (B) is wrong: Computed 27-3?

Why (C) is wrong: Arithmetic slip.

Why (D) is wrong: Took 2^3.

Hinglish: *$x = 2$: $(3)^3 - (1)^3 = 26$.*

MCQ 58 · Normal · Concept: General quadratic split

58. $6x^2 + 17x + 5$ factorises as:

A. $(3x+1)(2x+5)$

B. $(6x+5)(x+1)$

C. $(3x+5)(2x+1)$

D. $(6x+1)(x+5)$

Correct Answer: (A)

Split $17x = 15x + 2x$: $6x^2+15x+2x+5 = 3x(2x+5) + 1(2x+5) = (3x+1)(2x+5)$.

Why (B) is wrong: $30+1 = 31 \neq 17$.

Why (C) is wrong: $15+10 = 25 \neq 17$.

Why (D) is wrong: $6+5 = 11 \neq 17$.

Hinglish: *ac = 30: do numbers 15 aur 2 (sum 17). Grouping se (3x+1)(2x+5).*

MCQ 59 · Normal · Concept: Half-sum of squares

59. $a^2 + b^2 + c^2 - ab - bc - ca$ equals:

A. $\frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$

B. $(a-b)(b-c)(c-a)$

C. 0

D. $\frac{1}{2}[(a+b)^2+(b+c)^2+(c+a)^2]$

Correct Answer: (A)

Expanding the right side gives exactly the left side.

Why (B) is wrong: That is degree 3.

Why (C) is wrong: Not identically zero.

Why (D) is wrong: That gives 2Σ + ... signs wrong.

Hinglish: *Half-sum-of-squares identity - crown identity ka engine.*

MCQ 60 · Normal · Concept: Difference chain cube

60. If $x - 1/x = 3$, then $x^3 - 1/x^3$ equals:

A. 36

B. 27

C. 30

D. 18

Correct Answer: (A)

$(x-1/x)^3 = x^3-1/x^3-3(x-1/x) \Rightarrow 27 = x^3-1/x^3-9 \Rightarrow 36$.

Why (B) is wrong: Stopped at d^3.

Why (C) is wrong: Subtracted 3d again.

Why (D) is wrong: That is the sum-case answer pattern.

Hinglish: *$d^3 + 3d = 27 + 9 = 36$.*

MCQ 61 · Normal · Concept: Cube expansion full

61. $(a - 2b)^3$ equals:

A. $a^3 - 6a^2b + 12ab^2 - 8b^3$

B. $a^3 - 8b^3$

C. $a^3 - 4a^2b + 8ab^2 - 8b^3$

D. $a^3 + 6a^2b + 12ab^2 + 8b^3$

Correct Answer: (A)

$3(a)^2(2b) = 6a^2b$; $3(a)(2b)^2 = 12ab^2$; $(2b)^3 = 8b^3$, alternating signs.

Why (B) is wrong: Middle terms dropped.

Why (C) is wrong: Coefficient pattern wrong.

Why (D) is wrong: Signs all wrong.

Hinglish: *Powers of 2: 2, 4, 8 with 1-3-3-1.*

MCQ 62 · Normal · Concept: Decimal equidistant product 2

62. 0.54×0.46 equals:

A. 0.2484

B. 0.2516

C. 0.2484... 0.24

D. 0.2084

Correct Answer: (A)

$(0.5+0.04)(0.5-0.04) = 0.25 - 0.0016 = 0.2484$.

Why (B) is wrong: Added 0.0016.

Why (C) is wrong: Rounded 0.04^2.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$0.04^2 = 0.0016$. 0.25 - 0.0016.*

MCQ 63 · Normal · Concept: Common factor + square

63. $x^3 + 2x^2 + x$ factorises as:

A. $x(x+1)^2$

B. $x(x+1)(x+2)$

C. $x(x^2+2x)$

D. $(x+1)^3$

Correct Answer: (A)

$x(x^2+2x+1) = x(x+1)^2$.

Why (B) is wrong: x+2 does not fit.

Why (C) is wrong: Incomplete.

Why (D) is wrong: That is x^3+3x^2+3x+1.

Hinglish: *x common, andar perfect square.*

MCQ 64 · Normal · Concept: Symmetric pair

64. If a + b = 6 and a - b = 4, then $a^2 + b^2$ equals:

A. 26

B. 52

C. 20

D. 36

Correct Answer: (A)

$a^2+b^2 = (36+16)/2 = 26$.

Why (B) is wrong: Forgot to halve.

Why (C) is wrong: Subtracted instead of added.

Why (D) is wrong: That is $(a+b)^2$.

Hinglish: *$(a+b)^2+(a-b)^2 = 2(a^2+b^2)$: 52/2 = 26.*

MCQ 65 · Normal · Concept: Near-number square

65. 999^2 equals:

A. 998001

B. 99801

C. 990001

D. 999801

Correct Answer: (A)

$(1000-1)^2 = 1{,}000{,}000 - 2{,}000 + 1 = 998{,}001$.

Why (B) is wrong: Place value slip.

Why (C) is wrong: Wrong cross term.

Why (D) is wrong: Last digits order.

Hinglish: *$10^6 - 2\times10^3 + 1$.*

MCQ 66 · Normal · Concept: Companion identity sum

66. $(x + y)^2 + (x - y)^2$ equals:

A. $2(x^2 + y^2)$

B. $2(x^2 - y^2)$

C. $4xy$

D. $2(x^2 + 2xy + y^2)$

Correct Answer: (A)

Cross terms cancel: $2x^2 + 2y^2$.

Why (B) is wrong: Sign wrong.

Why (C) is wrong: That is the difference of the squares.

Why (D) is wrong: Cross terms doubled instead of cancelled.

Hinglish: *Sum mein cross terms cancel - $2(x^2+y^2)$.*

MCQ 67 · Normal · Concept: Bridge reversed for product 2

67. If $x^3 + y^3 = 9$ and $x + y = 3$, then xy equals:

A. 2

B. 3

C. 1

D. 6

Correct Answer: (A)

$9 = 27 - 9xy \Rightarrow 9xy = 18 \Rightarrow xy = 2$.

Why (B) is wrong: Divided by 3 instead of 9.

Why (C) is wrong: Sign error.

Why (D) is wrong: That is x+y.

Hinglish: *$27 - 9xy = 9$. $xy = 2$.*

MCQ 68 · Normal · Concept: Sum of cubes reverse

68. $(2a + 3b)(4a^2 - 6ab + 9b^2)$ equals:

A. $8a^3 + 27b^3$

B. $8a^3 - 27b^3$

C. $(2a+3b)^3$

D. $16a^3 + 81b^3$

Correct Answer: (A)

This is A^3+B^3 with $A = 2a$, $B = 3b$ (trinomial middle term negative).

Why (B) is wrong: Sign of the cube wrong.

Why (C) is wrong: That would need $36a^2b + 54ab^2$ terms too.

Why (D) is wrong: Squared the coefficients twice.

Hinglish: *Sum of cubes ka reverse: $(2a)^3+(3b)^3$.*

MCQ 69 · Normal · Concept: Cube difference expand

69. $(1 + x)^3 - (1 - x)^3$ equals:

A. $6x + 2x^3$

B. $6x^2 + 2x^3$

C. $2 + 6x + 2x^3$

D. $8x^3$

Correct Answer: (A)

Difference kills the even-in-x terms: $6x + 2x^3$ survives.

Why (B) is wrong: Wrong surviving terms.

Why (C) is wrong: Constants should cancel - they did not here.

Why (D) is wrong: Only if x = 1.

Hinglish: *$(1+x)^3-(1-x)^3$: $2[3x + x^3] = 6x+2x^3$.*

MCQ 70 · Normal · Concept: Three-variable bridge

70. If $a + b + c = 12$ and $ab + bc + ca = 47$, then $a^2 + b^2 + c^2$ equals:

A. 50

B. 97

C. 145

D. 48

Correct Answer: (A)

144 - 94 = 50.

Why (B) is wrong: Added 2Σ.

Why (C) is wrong: Forgot the factor 2.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$(a+b+c)^2 - 2\Sigma = 144 - 94 = 50$.*

MCQ 71 · HOTS · Concept: Crown identity converse

71. If $a^3 + b^3 + c^3 = 3abc$ with $a + b + c \neq 0$, then:

A. $a = b = c$

B. $a + b + c = 0$

C. $abc = 0$

D. $a = b = -c$

Correct Answer: (A)

The crown identity forces the half-sum-of-squares factor to vanish, which needs $a = b = c$.

Why (B) is wrong: Explicitly excluded by the question.

Why (C) is wrong: Zero product not required.

Why (D) is wrong: No such structure exists.

Hinglish: *Sum zero nahi hai, toh doosra factor zero - squares ka sum zero matlab teeno barabar.*

MCQ 72 · HOTS · Concept: Sum-cube identity form

72. $(a + b + c)^3$ equals:

A. $a^3+b^3+c^3+3(a+b)(b+c)(c+a)$

B. $a^3+b^3+c^3+3abc$

C. $a^3+b^3+c^3+3(a+b+c)(ab+bc+ca)$

D. $(a+b)(b+c)(c+a)$

Correct Answer: (A)

Standard identity; option C misses the -3abc correction, so A is the clean exact form.

Why (B) is wrong: Too small.

Why (C) is wrong: Missing -3abc term.

Why (D) is wrong: No cubes at all.

Hinglish: *(a+b)(b+c)(c+a) wala form yaad rakho - expansion se match karta hai.*

MCQ 73 · HOTS · Concept: Zero-sum cubes with product

73. If $x + y + z = 0$ and $xyz = 6$, then $x^3 + y^3 + z^3$ equals:

A. 18

B. 6

C. 216

D. -18

Correct Answer: (A)

Zero-sum: cubes = 3xyz = 18.

Why (B) is wrong: Missed the 3.

Why (C) is wrong: Cubed the product.

Why (D) is wrong: Sign flipped.

Hinglish: *3 × 6 = 18. Direct.*

MCQ 74 · HOTS · Concept: Divisibility by factoring

74. $2^{55} + 2^{58}$ is divisible by:

A. 9

B. 5

C. 7

D. 11

Correct Answer: (A)

$2^{55}(1 + 8) = 9 \cdot 2^{55}$.

Why (B) is wrong: Powers of 2 are never multiples of 5.

Why (C) is wrong: No factor 7 appears.

Why (D) is wrong: No factor 11 appears.

Hinglish: *$2^3 = 8$, bracket = 9.*

MCQ 75 · HOTS · Concept: Decimal cube difference

75. $(1.02)^3 - (0.02)^3$ equals:

A. 0.9388
B. 0.9412
C. 1.0000
D. 0.9996

Correct Answer: (A)

$a - b = 1$: $a^3-b^3 = 1 - 3ab = 1 - 3(1.02)(0.02) = 1 - 0.0612 = 0.9388$.

Why (B) is wrong: Added the correction.

Why (C) is wrong: Cancelling wrongly to 1.

Why (D) is wrong: That is $1.02^2 - 0.02^2$ style.

Hinglish: *$(a-b)^3 + 3ab(a-b)$ with $a-b = 1$: 0.0612 minus.*

MCQ 76 · HOTS · Concept: Fourth-power chain with surd

76. If $x + 1/x = 2\sqrt{2}$, then $x^4 + 1/x^4$ equals:

A. 34
B. 36
C. 32
D. 68

Correct Answer: (A)

$x^2+1/x^2 = 8 - 2 = 6$; $x^4+1/x^4 = 36 - 2 = 34$.

Why (B) is wrong: Missed the -2.

Why (C) is wrong: Missed both -2s.

Why (D) is wrong: Doubled instead.

Hinglish: *$(2\sqrt{2})^2 = 8$. 8-2 = 6. 36-2 = 34.*

MCQ 77 · HOTS · Concept: Term count in trinomial square

77. On expanding $(a + b + c)^2$ and collecting like terms, the number of DISTINCT terms is:

A. 6
B. 9
C. 3
D. 12

Correct Answer: (A)

Three squares + three distinct cross terms = 6 terms after collecting (2ab and 2ba merge).

Why (B) is wrong: Counted the pre-collection terms.

Why (C) is wrong: Only squares.

Why (D) is wrong: Overcounted.

Hinglish: *Teen squares + teen cross terms (2 ke coefficient ke saath) = 6.*

MCQ 78 · HOTS · Concept: Unit-sum cube identity

78. If $x + y = 1$, then $x^3 + y^3 + 3xy$ equals:

A. 1

B. 0

C. x^3+y^3

D. 3

Correct Answer: (A)

$x^3+y^3 = 1 - 3xy$; adding $3xy$ gives 1.

Why (B) is wrong: Sign confusion.

Why (C) is wrong: Incomplete use of the bridge.

Why (D) is wrong: No fixed 3.

Hinglish: *Bridge se 1 - 3xy, phir 3xy wapas.*

MCQ 79 · HOTS · Concept: Equal squares condition

79. If $x^2 + y^2 + z^2 = xy + yz + zx$ for real x, y, z, then:

A. $x = y = z$

B. $x + y + z = 0$

C. $xyz = 0$

D. $x = y = -z$

Correct Answer: (A)

Rearranged: $\frac{1}{2}[(x-y)^2+(y-z)^2+(z-x)^2] = 0 \Rightarrow$ all three equal.

Why (B) is wrong: Different condition entirely.

Why (C) is wrong: Not forced.

Why (D) is wrong: No structure gives this.

Hinglish: *Half-sum-of-squares zero → teeno barabar.*

MCQ 80 · HOTS · Concept: Telescoping product

80. $(2+1)(2^2+1)(2^4+1)(2^8+1) + 1$ equals:

A. 65536

B. 65535

C. 32768

D. 262144

Correct Answer: (A)

Multiply and divide by $(2-1) = 1$: the product telescopes to $2^{16} - 1$; adding 1 gives $2^{16} = 65536$.

Why (B) is wrong: Forgot to add 1.

Why (C) is wrong: Stopped at 2^{15}.

Why (D) is wrong: Overshot the exponent.

Hinglish: *(2-1) se multiply-divide karo: chain telescopes to $2^{16}-1$. Plus 1: 65536.*

4.5 Chapter Formula Sheet

All twelve identities plus the two bridges. The right column names the mistake that kills each one in exams.

Identity / Bridge	Statement	Conditions	Tiny example	Common mistake
Square, sum	$(a+b)^2 = a^2+2ab+b^2$	Always	$(3x+2)^2 = 9x^2+12x+4$	Dropping 2ab
Square, difference	$(a-b)^2 = a^2-2ab+b^2$	Always	$(5-y)^2 = 25-10y+y^2$	Writing $-b^2$ at the end
Two-square difference	$a^2-b^2 = (a+b)(a-b)$	Always (reversible)	$49x^2-36 = (7x+6)(7x-6)$	Mixing it with $(a-b)^2$
Companion, sum	$(a+b)^2+(a-b)^2 = 2(a^2+b^2)$	Always	$58 = 2(a^2+b^2)$ from 10, 4	Sign slips in expansion
Companion, difference	$(a+b)^2-(a-b)^2 = 4ab$	Always	$4\times5.5\times4.5 = 99$	Confusing with (a+b)(a-b)
(x+a)(x+b)	$x^2+(a+b)x+ab$	Always	$(x+4)(x+7) = x^2+11x+28$	Swapping sum and product
Trinomial square	$(a+b+c)^2 = \Sigma a^2+2\Sigma ab$	Three pairs, each twice	$(2p+3q-4r)^2$	Two cross terms only
Cube, sum	$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$	Always	$(x+2)^3$	Losing the 3s
Cube, difference	$(a-b)^3 = a^3-3a^2b+3ab^2-b^3$	Alternating signs	$(2m-3n)^3$	Last term positive
Sum of cubes	$a^3+b^3 = (a+b)(a^2-ab+b^2)$	Always	$8a^3+27b^3$	Trinomial middle sign
Difference of cubes	$a^3-b^3 = (a-b)(a^2+ab+b^2)$	Always	x^3-27	Binomial sign
Crown identity	$a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$	Always; zero-sum shortcut when $a+b+c = 0$	$12^3+1^3+(-13)^3 = 3(12)(1)(-13)$	Forgetting the converse cases
Cube bridges	$a^3\pm b^3 = (a\pm b)^3 \mp 3ab(a\pm b)$	Given sum/difference and product	12, 27 → 756	Sign of the 3ab term
Half-sum of squares	$a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]$	≥ 0; zero iff $a = b = c$	Converse of zero-sum	Missing the equality condition

4.6 Last-Minute Cheat Sheet

> **30-second revision card - Chapter 4**
> SQUARES: $(a\pm b)^2 = a^2\pm 2ab+b^2$. Sum of squares = $(a+b)^2-2ab$. $4ab = (a+b)^2-(a-b)^2$. TWO-SQUARE DIFF: (a+b)(a-b). Chains: x^4-n = two rounds. (x+a)(x+b): SUM on x, PRODUCT at the end. TRINOMIAL SQUARE: 3 squares + 2×(every pair); cross sign = product of parent signs. CUBES: 1-3-3-1; $(a-b)^3$ ends $-b^3$. $a^3\pm b^3 = (a\pm b)(a^2\mp ab+b^2)$. BRIDGE: $a^3+b^3 = (a+b)^3-3ab(a+b)$. CROWN: $a^3+b^3+c^3-3abc = (a+b+c)(...)$; zero sum ⇒ cubes = 3abc; converse ⇒ a = b = c or sum 0. x±1 1/x CHAIN: square ∓2 for plus-case, +2 for minus-case.

- **If you see two perfect squares subtracted** → (a+b)(a-b); repeat while exponents are even.
- **If you see numbers near 100** → write (100 ± k) and expand with the identity.
- **If you see a quadratic with three terms and square ends** → test the perfect-square condition (middle = 2√×√).
- **If you see (chunk) twice with powers** → substitute y = chunk, factorise, back-substitute.
- **If you see $a^3+b^3+c^3$ with a zero-sum triple** → instantly 3abc.
- **If you see x + 1/x = s** → chain: s^2-2, s^3-3s; multiply-rule for higher powers.
- **If you see 'prove' with three brackets like (a+b), (b+c), (c+a)** → rename them; their sum is 2(a+b+c).
- **If you see sum-of-squares equals pair-sum** → half-sum-of-squares zero ⇒ all equal.

4.7 Mistake Buster - DON'T MAKE THESE MISTAKES

1. **Dropping the cross term.** $(a+b)^2 = a^2+b^2$ is FALSE. Fix: say 'square, double product, square' out loud while expanding.
2. **Negative last term.** $(a-b)^2$ ends $+b^2$. Fix: a square is never subtracted in these identities.
3. **Sum/product swap in (x+a)(x+b).** Coefficient is the SUM, constant is the PRODUCT. Fix: expand one example to re-anchor.
4. **Only two cross terms in a trinomial square.** Three pairs exist. Fix: list (ab), (bc), (ca) before writing.
5. **Sign of a cross term guessed.** It is the PRODUCT of the parents' signs. Fix: underline each parent's sign first.
6. **Cube pattern 1-2-3.** Coefficients are 1-3-3-1. Fix: memorise via $(x+1)^3 = x^3+3x^2+3x+1$.
7. **Trinomial sign in cube factorisation.** a^3+b^3 carries a^2-AB+b^2: the middle sign is OPPOSITE the binomial. Fix: verify by expanding once.
8. **Bridge sign confusion.** $a^3-b^3 = (a-b)^3+3ab(a-b)$: PLUS. $a^3+b^3 = (a+b)^3-3ab(a+b)$: MINUS. Fix: derive from the cube expansion, never recall from memory.
9. **Forgetting the ± when un-squaring.** $(x-1/x)^2 = 36$ gives x-1/x = ±6. Fix: every un-squaring opens two cases.
10. **Stopping factorisation early.** x^4-16 is not finished at $(x^2-4)(x^2+4)$. Fix: expand your final answer back; every factor must be irreducible.

4.8 Exam Strategy for Chapter 4

Order of attack. Expansion and near-number evaluation questions first (one line each, guaranteed marks). Factorisation middle-term splittings next - they are formulaic. Keep crown-identity and multi-variable questions for the middle slot, and the proof/HOTS items for the final third of your time, when the pressure of unfinished sections no longer distracts you.

Where marks are lost. Four leak-points dominate: sign errors in cross terms (systematic - fix with the parent-sign rule), incomplete factorisation (the word 'completely' is a contract - deliver irreducible factors), missing ± cases after un-squaring, and skipped verification lines in 'show that' questions. The last one is the cheapest fix: one expansion line re-earning three method marks.

Identities most frequently tested. $(a\pm b)^2$, a^2-b^2, $(a+b)^3$ and the crown identity lead the frequency table; the cube bridges and the $x + 1/x$ chain follow close behind. Near-number evaluation (98^2, 103^3 style) appears in nearly every paper as a 2-3 mark gift - never leave it.

Concepts that get mixed. $(a-b)^2$ with a^2-b^2 (the classic), sum of cubes with cube of sum, zero-sum shortcut with its converse, and trinomial squares with completed squares. When a question mixes two identities, label each step with the identity used - the marking scheme rewards exactly that.

Safe shortcuts. Substituting y for a repeated bracket; testing a claimed identity with one numerical example to reject it; using $(a+b)^2-(a-b)^2 = 4ab$ to multiply nearby numbers. **Unsafe shortcuts.** Expanding $(a+b+c)^3$ term by term under time pressure (use the identity), assuming $a^3+b^3+c^3 = 3abc$ without checking the sum, and 'factorising' x^2+9 as $(x+3)^2$.

How to check an answer. Expansion: substitute small numbers ($x = 1$, $y = 2$) into both original and result - values must match. Factorisation: expand the last factor pair only. Bridges: plug the numbers into the bridge identity. Proofs: confirm every identity used is quoted by name.

4.9 Chapter Completion Checklist

Tick each box only when you can honestly say DONE:

- [ ] Every identity covered: both squares, two-square difference, companions, (x+a)(x+b), trinomial square, both cubes, both cube factorisations, crown identity
- [ ] Every bridge covered: cube bridges, half-sum-of-squares, sum-cube decomposition
- [ ] 70 advanced questions completed (Q1-Q70), difficulty ladder respected
- [ ] Every question has a full step-by-step solution
- [ ] Every solution has a Hinglish explanation
- [ ] 80 MCQs completed with answers
- [ ] Every MCQ has an explanation plus trap analysis
- [ ] Formula sheet completed (14 rows)
- [ ] Last-minute cheat sheet completed
- [ ] Mistake buster completed (10 named mistakes)
- [ ] Exam strategy completed

Chapter 6: Perimeter and Area

Measurement questions are the most reliable marks in any paper - and the most reliable marks LOST to careless formulas. The uploaded PDF assembles the full toolkit: a review of perimeters and areas of triangles, squares, rectangles and parallelograms; the meaning of π; circumference and area of circles; arc length and sector area; Heron's formula for triangles with known sides; its special cases for equilateral and isosceles triangles; area of a quadrilateral split by a diagonal; Brahmagupta's formula for cyclic quadrilaterals; and rings, paths and composite regions. This book hardens every tool: reverse questions (area back to radius), sector-perimeter traps, grazing goats, tracks with semicircular ends, and Hippocrates-style shaded regions that look impossible and collapse in one line.

Figure 6.1: Heron's machinery - from three sides s(a+b+c)/2 to a clean area, no heights needed.

6.1 Complete Concept Cheat Sheet (read this before any question)

6.1.1 Perimeter and area: the review set

What is it? Perimeter is the length of the boundary; area is the size of the enclosed surface. **Core formulas:** rectangle $P = 2(l+b)$, $A = l \times b$; square $P = 4s$, $A = s^2$, diagonal $= s\sqrt{2}$; triangle $A = \frac{1}{2} \times \text{base} \times \text{height}$; parallelogram $A = \text{base} \times \text{height}$; rhombus $A = \frac{1}{2} \times d_1 \times d_2$; trapezium $A = \frac{1}{2}(a+b)h$. **The perimeter-area trap:** equal perimeters do NOT mean equal areas - among all shapes with a fixed perimeter, the circle encloses the most area. **When do I use which?** Perimeter answers 'how much boundary/fencing/frame', area answers 'how much surface/paint/carpet/grass'.

Units discipline: 1 m = 100 cm but 1 m^2 = 10000 cm^2; 1 hectare = 10000 m^2. Converting lengths squares the conversion factor - forget this once and your answer is off by a factor of 100.

6.1.2 π, circumference and area of a circle

What is π? The ratio circumference/diameter - the same constant for EVERY circle, approximately 3.14159... irrational. Exam convention: $\pi = 22/7$ when the radius is a multiple of 7, otherwise 3.14. **Formulas:** $C = 2\pi r = \pi d$; $A = \pi r^2$. **Reverse uses (the exam favourites):** given C, $r = C/(2\pi)$ then $A = C^2/(4\pi)$; given A, $r = \sqrt{(A/\pi)}$ then $C = 2\pi r$. **Scaling laws:** double the radius → double the circumference but QUADRUPLE the area; radii in ratio k → areas in ratio k^2. **Tiny example.** r = 14 cm: $C = 2 \times (22/7) \times 14 = 88$ cm; $A = (22/7) \times 196 = 616$ cm^2.

6.1.3 Arc length and sector area

What is it? A sector is the pizza slice between two radii and an arc of angle θ. The full circle is 360°, so a sector owns the fraction θ/360 of everything. **Arc length:** $l = (\theta/360) \times 2\pi r$. **Sector area:** $A = (\theta/360) \times \pi r^2$. **The half-angle gem:** $A = \frac{1}{2} \times l \times r$ (treat the arc as a curved 'base' and the radius as the 'height'). **Sector PERIMETER (the trap):** P = arc + TWO radii = l + 2r. Writing l + r loses marks every single year. **Tiny example.** r = 21, θ = 120°: $l = (120/360)(2 \cdot (22/7) \cdot 21) = 44$ cm; $A = (1/3)\pi(441) = 462$ cm^2; P = 44 + 42 = 86 cm.

Figure 6.2: Sector anatomy - arc l, two radii, angle θ; the slice owns θ/360 of the circle.

6.1.4 Heron's formula

What is it? Area of a triangle from its three sides alone, no height needed: A = √(s(s-a)(s-b)(s-c)) where s = (a+b+c)/2 is the semi-perimeter. **Why it matters:** most real triangles (a field, a plot) give you sides, not heights - Heron converts three tape-measure readings into an area. **Worked rhythm:** (1) add sides, halve for s; (2) compute the three differences s-a, s-b, s-c; (3) multiply all four numbers; (4) take the square root - and SIMPLIFY the surd if it is not a perfect square. **Tiny example.** Sides 5, 12, 13: s = 15; product 15·3·2 = 900; A = 30. **Validity check:** each side must be less than the sum of the other two (triangle inequality), otherwise the quantity under the root goes negative - the 'triangle' does not exist.

> **Verification habit**
> If the three sides satisfy $a^2 + b^2 = c^2$, the triangle is right-angled and the area must equal $\frac{1}{2}ab$. Run this cross-check whenever the sides look Pythagorean - 5-12-13, 8-15-17, 7-24-25, 9-40-41 and their scaled cousins.

6.1.5 Equilateral and isosceles shortcuts

Equilateral triangle (side a): height = $(\sqrt{3}/2)a$, area = $(\sqrt{3}/4)a^2$, perimeter = 3a. The area formula saves one Heron round-trip. **Isosceles triangle:** drop the height to the base; it bisects the base. Given equal sides l and base b: height = $\sqrt{(l^2 - b^2/4)}$, area = $\frac{1}{2} \cdot b \cdot$ height, and Heron must agree. **Reverse equilateral:** area given as $k\sqrt{3}$, side a satisfies $(\sqrt{3}/4)a^2 = k\sqrt{3} \Rightarrow a = 2\sqrt{k}$. **Tiny example.** Equilateral side 12: A = $(\sqrt{3}/4)(144) = 36\sqrt{3} \approx 62.35$.

6.1.6 Quadrilateral by diagonal; Brahmagupta

Diagonal split: any quadrilateral ABCD splits along diagonal AC into triangles ABC and ACD. Find each area (heights or Heron) and ADD. This is the master method for four-sided fields with unequal sides. **Brahmagupta's formula (cyclic quadrilaterals):** if all four vertices lie on a circle and sides are a, b, c, d with s = (a+b+c+d)/2, then A = √((s-a)(s-b)(s-c)(s-d)) - Heron with the (s - ...) pattern of four sides and no explicit height. It is valid ONLY for cyclic quadrilaterals; applying it to a general quadrilateral is a silent error. **Tiny example.** Sides 13, 10, 13, 10 (cyclic): s = 23; product 10·13·10·13 = 16900; A = 130.

Figure 6.3: A quadrilateral split by diagonal BD - two triangles, two Herons, one sum.

6.1.7 Rings, paths and composite regions

Ring (annulus): $A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2) = \pi(R+r)(R-r)$. Given width w = R - r and area, use $(R+r)(R-r) = A/\pi$ to recover both radii. **Inside border path:** rectangular field L × B with a path of width w inside: path area = L·B - (L-2w)(B-2w). **Outside circular path:** $\pi((r+w)^2 - r^2) = \pi w(2r + w)$. **Composite figures:** decompose into rectangles, triangles, sectors and rings; add or subtract; keep every π exact until the last line. **Chord touching the inner circle (HOTS classic):** a chord of the outer circle tangent to the inner one has half-chord^2 = $R^2 - r^2$, so ring area = π × (half-chord)^2 - no radii needed!

Figure 6.4: Ring geometry - a chord of the outer circle touching the inner circle fixes $R^2 - r^2$.

6.1.8 Exam tips for this chapter

1. Choose $\pi = 22/7$ when a radius or diameter is a multiple of 7; otherwise use 3.14 - and say which you used.
2. Sector perimeter = arc + 2r. Circle perimeter = $2\pi r$. Never mix them.
3. Reduce every surd area (e.g. $\sqrt{1215} = 9\sqrt{15}$) - unsimplified surds lose the final mark.
4. Check for right triangles (Pythagoras) before Heron; $\frac{1}{2} \times \text{leg} \times \text{leg}$ is faster and safer.
5. Write units at every line: cm for length, cm^2 for area, m^2 for fields, hectares for land.
6. In reverse questions (area → radius → circumference) keep π symbolic until the numbers force it in.
7. Ratio of radii k ⇒ ratio of areas k^2 - read ratios as ratios, never as differences.
8. Cost questions: compute the geometric quantity FIRST, then multiply by rate; never mix the two steps.

6.2 Question Types / Concept Map for Chapter 6

Type	Name	How to recognise it in the question	Core method
T1	Basic perimeter/area	Rectangle, square, triangle, parallelogram, rhombus, trapezium	Direct formulas with correct units
T2	Heron direct	Three sides given, area asked	$s = (a+b+c)/2$; $\sqrt{(s(s-a)(s-b)(s-c))}$
T3	Equilateral / isosceles	'all sides equal', perimeter + base given	$(\sqrt{3}/4)a^2$; height by Pythagoras
T4	Quadrilateral by diagonal	Four sides plus a diagonal	Two Herons, add
T5	Brahmagupta	'cyclic quadrilateral' with four sides	$\sqrt{((s-a)(s-b)(s-c)(s-d))}$
T6	Circle basics + reverse	C ↔ r ↔ A chains	$C = 2\pi r$, $A = \pi r^2$, reversed
T7	Arc and sector	Angle θ at centre	$(\theta/360)$ fractions; $A = \frac{1}{2}lr$; $P = l + 2r$
T8	Segments and combinations	'segment', chord with angle	sector − triangle
T9	Rings / annulus	Two concentric circles	$\pi(R^2-r^2)$; chord trick
T10	Paths and borders	'path inside/outside', track, frame	Outer minus inner
T11	Composite figures	Mixed shapes, shaded regions	Decompose; add/subtract
T12	Ratio and scaling	Radii in ratio, same perimeter comparisons	Areas scale as k^2
T13	Cost applications	Rates for fencing, ploughing, tiling	Geometry first, money second
T14	HOTS shaded/proofs	Lunes, hexagons, grazing animals, AP sides	Label regions; prove then compute

Distribution of the 70 practice questions: T1-T3 feed Q1-Q12, T4-T6 feed Q13-Q26, T7-T9 feed Q27-Q42, T10-T13 feed Q43-Q56, and T14 dominates Q57-Q70.

6.3 Practice Questions (Q1 - Q70) with Full Step-by-Step Solutions

Difficulty climbs in five bands: Q1-Q12 strong basics, Q13-Q26 moderate, Q27-Q42 difficult, Q43-Q56 high-level, Q57-Q70 very challenging (Olympiad-lite). Every question carries the concept tag; π = 22/7 unless stated otherwise.

Q1 · Basic · Concept: Rectangle perimeter/area (T1)

Q1. A rectangular floor is 12.5 m long and 8 m wide. (i) How many metres of skirting board are needed along the four walls' base line if the room has no door opening on the floor plan? (ii) How many square metres of carpet cover the floor?

SOLUTION

Step 1: Given: l = 12.5 m, b = 8 m. Needed: perimeter and area.

Step 2: Perimeter = 2(l + b) = 2(12.5 + 8) = 2 × 20.5 = 41 m.

Step 3: Area = l × b = 12.5 × 8 = 100 m^2.

Step 4: Skirting board follows the boundary → 41 m; carpet covers the surface → 100 m^2.

» **Why this step?** Perimeter questions use lengths once around the boundary; area questions multiply the two dimensions - mixing them is the most common unit error.

Hinglish: *Boundary = perimeter, surface = area. Skirting board room ke chakkar lagata hai: 2(12.5+8) = 41 m. Carpet poori zameen dhakta hai: 100 m^2. 'How much around' vs 'how much inside' - ye difference pakka karo.*

Answer: (i) 41 m (ii) 100 m^2

Q2 · Basic · Concept: Triangle base-height (T1)

Q2. A triangle has base 14 cm and height 9 cm. (i) Find its area. (ii) A second triangle has the same area but height 7 cm; find its base.

SOLUTION

Step 1: Given: b = 14, h = 9. Area = ½ × 14 × 9 = 63 cm^2.

Step 2: Second triangle: ½ × b × 7 = 63 ⇒ 7b = 126 ⇒ b = 18 cm.

» **Why this step?** Area = ½(base)(height) works because a triangle is half the parallelogram built on the same base.

Hinglish: *Area = aadha × base × height. Pehla: 63 cm^2. Doosre mein same area, height 7 - base 18. Reverse questions mein equation banao, shortcut mat dhoondo.*

Answer: (i) 63 cm^2 (ii) 18 cm

Q3 · Basic · Concept: Circumference and area (T6)

Q3. For a circle of radius 14 cm, find (i) the circumference and (ii) the area (use π = 22/7).

SOLUTION

Step 1: Given: r = 14 (a multiple of 7 → use π = 22/7).

Step 2: Circumference C = 2πr = 2 × (22/7) × 14 = 88 cm.

Step 3: Area A = πr^2 = (22/7) × 14^2 = (22/7) × 196 = 616 cm^2.

» **Why this step?** The 7 in the denominator cancels with the radius multiple - that is exactly when 22/7 gives exact answers.

Hinglish: *Radius 7 ka multiple hai, isliye 22/7 perfect. C = 88 cm, A = 616 cm^2. Dono answers clean aane chahiye - decimal aaya toh kahin galti hai.*

Answer: C = 88 cm, A = 616 cm²

Q4 · Basic · Concept: Reverse circle (T6)

Q4. The circumference of a circular plate is 132 cm. Find (i) its radius and (ii) its area.

SOLUTION

Step 1: Given: C = 132 cm. Reverse: r = C/(2π) = 132 × 7/44 = 21 cm.

Step 2: Area = πr^2 = (22/7) × 441 = 1386 cm^2.

» **Why this step?** Dividing by 2π undoes the circumference formula; then the area formula runs forward.

Hinglish: *Circumference se radius: C ko 2π se divide karo. 132 × 7/44 = 21. Phir area 1386 cm². Reverse chain C → r → A exam ka favourite hai.*

Answer: r = 21 cm, A = 1386 cm²

Q5 · Basic · Concept: Equilateral shortcut (T3)

Q5. Find the perimeter and the area of an equilateral triangle of side 12 cm (leave √3 in your answer, then approximate with √3 ≈ 1.73).

SOLUTION

Step 1: Perimeter = 3 × 12 = 36 cm.

Step 2: Area = $(\sqrt{3}/4)a^2$ = (√3/4) × 144 = 36√3 cm^2.

Step 3: Numerically: 36 × 1.73 = 62.28 cm^2 (with 1.732: 62.35).

» **Why this step?** $(\sqrt{3}/4)a^2$ is Heron's result pre-simplified for equal sides - one line instead of four.

Hinglish: *Equilateral ka ready-made formula: (√3/4)a². 144/4 = 36, toh 36√3 ≈ 62.3 cm². Heron se bhi milega but formula fast hai.*

Answer: P = 36 cm, A = 36√3 ≈ 62.3 cm²

Q6 · Basic · Concept: Heron direct (T2)

Q6. Find the area of a triangle with sides 5 cm, 12 cm and 13 cm.

SOLUTION

Step 1: s = (5 + 12 + 13)/2 = 15 cm.

Step 2: Differences: s - a = 10, s - b = 3, s - c = 2.

Step 3: A = √(15 × 10 × 3 × 2) = √900 = 30 cm^2.

Step 4: Cross-check: $5^2 + 12^2 = 169 = 13^2$, so it is right-angled; ½ × 5 × 12 = 30 ✓.

» **Why this step?** Spotting the right angle gives an instant verification - Heron must agree with ½ab.

Hinglish: *Heron ka 4-step rhythm: s nikaalo, differences nikaalo, multiply, root. 15×10×3×2 = 900, root = 30. Bonus: 5-12-13 Pythagorean triple hai - ½×5×12 = 30 se confirm.*

Answer: 30 cm²

Q7 · Basic · Concept: Parallelogram and triangle (T1)

Q7. A parallelogram has base 10 cm and height 6.5 cm. (i) Find its area. (ii) A triangle stands on the same base and between the same parallels; find its area.

SOLUTION

Step 1: Parallelogram area = base × height = 10 × 6.5 = 65 cm^2.

Step 2: A triangle between the same parallels and on the same base has exactly half the area:

Step 3: Triangle area = ½ × 65 = 32.5 cm².

» **Why this step?** Same base, same parallels ⇒ the triangle is exactly half the parallelogram - a definitional fact worth memorising.

Hinglish: *Same base, same parallel lines ke beech: triangle = aadha parallelogram. 65 aur 32.5. Ye 'half' relation theorems mein bhi aata hai.*

Answer: (i) 65 cm² (ii) 32.5 cm²

Q8 · Basic · Concept: Trapezium area (T1)

Q8. A trapezium has parallel sides 12 cm and 8 cm, and the perpendicular distance between them is 5 cm. Find its area.

SOLUTION

Step 1: A = ½(a + b)h = ½(12 + 8)(5) = ½ × 20 × 5 = 50 cm².

» **Why this step?** Average of the parallel sides times the height: the trapezium is a rectangle plus a half-rectangle on average.

Hinglish: *(a+b)/2 × h - parallel sides ka average, height se multiply. 10 × 5 = 50 cm².*

Answer: 50 cm²

Q9 · Basic · Concept: Arc and sector (T7)

Q9. For a circle of radius 7 cm, find (i) the length of an arc subtending 90° at the centre and (ii) the area of the corresponding sector.

SOLUTION

Step 1: Fraction of the circle: 90/360 = 1/4.

Step 2: Arc = (1/4) × 2πr = (1/4) × 44 = 11 cm.

Step 3: Sector area = (1/4) × πr² = (1/4) × 154 = 38.5 cm².

» **Why this step?** Everything scales by θ/360 - compute the full-circle value once, then take the fraction.

Hinglish: *90° matlab quarter circle. Arc = 44/4 = 11 cm, area = 154/4 = 38.5 cm². Fraction pehle likho, calculation aadhi ho jati hai.*

Answer: arc = 11 cm, sector = 38.5 cm²

Q10 · Basic · Concept: Ring area (T9)

Q10. Two concentric circles have radii 10 cm and 6 cm. Find the area of the ring between them (π = 3.14).

SOLUTION

Step 1: Ring area = π(R² - r²) = π(100 - 36) = 64π.

Step 2: With π = 3.14: 64 × 3.14 = 200.96 cm².

» **Why this step?** Subtract the inner disk from the outer disk; factorising R²-r² = (R+r)(R-r) speeds the arithmetic.

Hinglish: *Ring = bada circle - chhota circle. π(100-36) = 64π. 3.14 se: 200.96 cm². R²-r² ko (R+r)(R-r) likhoge toh mental math ho jata hai.*

Answer: 200.96 cm²

Q11 · Basic · Concept: Square and diagonal (T1)

Q11. The perimeter of a square is 64 cm. Find (i) its side, (ii) its area, (iii) the length of its diagonal (leave √2 in the answer).

SOLUTION

Step 1: Side = 64/4 = 16 cm.

Step 2: Area = 16^2 = 256 cm^2.

Step 3: Diagonal = s√2 = 16√2 cm (from the half-square right triangle).

» **Why this step?** The diagonal of a square is the hypotenuse of a 45-45-90 triangle: s√2 always.

Hinglish: *Side = perimeter/4 = 16. Area 256. Diagonal = side× √2 = 16√2. Ye teeno ek saath aate hain.*

Answer: 16 cm; 256 cm^2; 16√2 cm

Q12 · Basic · Concept: Isosceles triangle via Heron (T3)

Q12. An isosceles triangle has perimeter 30 cm and each equal side 12 cm. Find its area (simplify the surd).

SOLUTION

Step 1: Base = 30 - 2(12) = 6 cm.

Step 2: s = 15. Differences: 15-12 = 3 (twice), 15-6 = 9.

Step 3: A = √(15 × 3 × 3 × 9) = √1215.

Step 4: Simplify: 1215 = 81 × 15, so A = 9√15 ≈ 34.86 cm^2.

» **Why this step?** Heron handles isosceles without dropping a height; simplifying √1215 = 9√15 is part of the answer.

Hinglish: *Base = perimeter - do equal sides = 6. Heron: √1215. Surd simplify karo: 1215 = 81×15 → 9√15 ≈ 34.9 cm^2. Height method se bhi ho jata (height = √(144-9) = √135), same answer.*

Answer: 9√15 ≈ 34.9 cm^2

Q13 · Moderate · Concept: Heron with larger numbers (T2)

Q13. Find the area of a triangle with sides 40 cm, 30 cm and 14 cm.

SOLUTION

Step 1: s = (40 + 30 + 14)/2 = 42 cm.

Step 2: Differences: 2, 12, 28.

Step 3: Product: 42 × 2 × 12 × 28 = 28224.

Step 4: A = √28224 = 168 cm^2.

» **Why this step?** Large products factor stepwise: 42×2 = 84, 12×28 = 336, 84×336 = 28224; keep factors paired to make the root visible.

Hinglish: *s = 42, differences 2, 12, 28. Product 28224. Root 168. Bade numbers mein partial products likhte jao - ek hi line mein sab multiply karna errors bulata hai.*

Answer: 168 cm^2

Q14 · Moderate · Concept: Heron classic (T2)

Q14. A triangular park has sides 122 m, 120 m and 22 m. Find its area.

SOLUTION

Step 1: s = (122 + 120 + 22)/2 = 132 m.

Step 2: Differences: 132-122 = 10, 132-120 = 12, 132-22 = 110.

Step 3: Product: 132 × 10 × 12 × 110 = 1,742,400.

Step 4: A = √1742400 = 1320 m^2.

» **Why this step?** Pair the factors smartly: 132×110 = 14520 and 10×12 = 120; 14520×120 = 1742400. Grouping reveals the square: $(1320)^2$.

Hinglish: *s = 132, differences 10, 12, 110. Product 1742400, root 1320 m². Note the beauty: root = s × 10 = 1320 - aise patterns pehchano.*

Answer: 1320 m²

Q15 · Moderate · Concept: Heron with surd simplification (T2)

Q15. Find the area of a triangle with sides 18 cm, 10 cm and 14 cm. Express the answer as a simplified surd and as a decimal.

SOLUTION

Step 1: s = 21. Differences: 3, 11, 7.

Step 2: Product: 21 × 3 × 11 × 7 = 4851.

Step 3: Simplify the surd: 4851 = 441 × 11, so A = $21\sqrt{11}$ cm².

Step 4: Decimal: 21 × 3.3166 ≈ 69.65 cm².

» **Why this step?** Extract the largest perfect square under the root; 441 = 21^2 was hiding in the factors.

Hinglish: *Product 4851. 4851 ko 441 × 11 likho (21² bahar!). Answer 21√11 ≈ 69.65 cm². Surd simplify karna FINAL step hai - bina wo aadha marks.*

Answer: $21\sqrt{11} \approx 69.65$ cm²

Q16 · Moderate · Concept: Reverse equilateral (T3)

Q16. The area of an equilateral triangle is $16\sqrt{3}$ cm². Find its side and perimeter.

SOLUTION

Step 1: Set $(\sqrt{3}/4)a^2 = 16\sqrt{3}$.

Step 2: Cancel √3: $a^2/4 = 16 \Rightarrow a^2 = 64 \Rightarrow a = 8$ cm.

Step 3: Perimeter = 24 cm.

» **Why this step?** The √3 cancels symbolically - never convert to decimals in a reverse equilateral question.

Hinglish: *Formula reverse chalao: $(\sqrt{3}/4)a^2 = 16\sqrt{3}$. √3 cancel: $a^2 = 64$. Side 8, perimeter 24. Decimals tak mat jao - symbolic cancellation clean hai.*

Answer: side = 8 cm, perimeter = 24 cm

Q17 · Moderate · Concept: Isosceles by height (T3)

Q17. An isosceles triangle has perimeter 32 cm with equal sides of 11 cm. Find its area, first by the height method and then verify with Heron.

SOLUTION

Step 1: Base = 32 - 22 = 10 cm.

Step 2: Height method: the height to the base bisects it, so $h = \sqrt{(11^2 - 5^2)} = \sqrt{96} = 4\sqrt{6}$.

Step 3: Area = ½ × 10 × $4\sqrt{6} = 20\sqrt{6} \approx 48.99$ cm².

Step 4: Heron check: s = 16; product 16 × 5 × 5 × 6 = 2400; $\sqrt{2400} = 20\sqrt{6}$ ✓.

» **Why this step?** Two independent methods agreeing is the strongest verification available in this chapter.

Hinglish: *Height method: base ke midpoint se height - $11^2 - 5^2 = 96$, $h = 4\sqrt{6}$. Area $20\sqrt{6}$. Heron se bhi $20\sqrt{6}$ - dono match, pakka sahi.*

Answer: $20\sqrt{6} \approx 49.0$ cm²

Q18 · Moderate · Concept: Quadrilateral by diagonal (T4)

Q18. In quadrilateral ABCD, AB = 24 cm, BC = 10 cm, CD = 25 cm, DA = 17 cm and diagonal AC = 26 cm. Find the area of ABCD.

SOLUTION

Step 1: Triangle ABC: sides 24, 10, 26. Spot $24^2 + 10^2 = 576 + 100 = 676 = 26^2$: right angle at B.

Step 2: Area(ABC) = ½ × 24 × 10 = 120 cm².

Step 3: Triangle ACD: sides 26, 25, 17. s = 34; differences 8, 9, 17.

Step 4: Product 34 × 8 × 9 × 17 = 41616; Area(ACD) = √41616 = 204 cm².

Step 5: Total area = 120 + 204 = 324 cm².

» **Why this step?** Splitting along the diagonal reduces one quadrilateral to two triangle problems; Pythagoras spotting saves a full Heron on the first one.

Hinglish: *Diagonal ke saath do triangles. ABC: 24-10-26 right triangle hai! Area ½×24×10 = 120, Heron ki zaroorat nahi. ACD: Heron se 204. Total 324 cm². Pehle Pythagoras check karo - time bachta hai.*

Answer: 324 cm²

Q19 · Moderate · Concept: Brahmagupta (T5)

Q19. A cyclic quadrilateral has sides 13 cm, 10 cm, 13 cm and 10 cm (in order). Find its area using Brahmagupta's formula.

SOLUTION

Step 1: Cyclic quadrilateral ⇒ Brahmagupta applies: A = √((s-a)(s-b)(s-c)(s-d)).

Step 2: s = (13 + 10 + 13 + 10)/2 = 23 cm.

Step 3: Differences: 10, 13, 10, 13.

Step 4: Product: 10 × 13 × 10 × 13 = 16900. A = √16900 = 130 cm².

» **Why this step?** Heron's pattern extends to four sides; the symmetric sides make every difference reappear - a designed clean answer.

Hinglish: *Cyclic quadrilateral mein Heron ka bhai Brahmagupta: same (s - side) pattern, char baar. s = 23, differences 10,13,10,13. Product 16900, root 130 cm².*

Answer: 130 cm²

Q20 · Moderate · Concept: Circumference to area (T6)

Q20. The circumference of a circular garden is 44 m. Find (i) its radius, (ii) the cost of fencing it twice at ₹15 per metre.

SOLUTION

Step 1: r = C/(2π) = 44 × 7/44 = 7 m.

Step 2: Double fencing length = 2 × 44 = 88 m.

Step 3: Cost = 88 × 15 = ₹1320.

» **Why this step?** 'Fenced twice' doubles the LENGTH, not the radius or area - read the wording carefully.

Hinglish: *r = 7 m. Double fencing = 88 m, cost 88×15 = ₹1320. 'Twice' ka matlab boundary do baar - area se koi lena-dena nahi.*

Answer: r = 7 m; cost ₹1320

Q21 · Moderate · Concept: Wheel revolutions (T13)

Q21. A wheel of diameter 70 cm makes a journey of 1.1 km. How many complete revolutions does it make ($\pi = 22/7$)?

SOLUTION

Step 1: Circumference = $\pi d = (22/7) \times 70 = 220$ cm.

Step 2: Convert the journey: 1.1 km = 1100 m = 110000 cm.

Step 3: Revolutions = 110000/220 = 500.

» **Why this step?** One revolution moves the wheel exactly one circumference; total distance divided by circumference gives the count.

Hinglish: *Ek chakkar = circumference = 220 cm. Distance 110000 cm. 110000/220 = 500 revolutions. Units same karo pehle - km vs cm sabse common galti.*

Answer: 500 revolutions

Q22 · Moderate · Concept: Sector area and arc (T7)

Q22. A sector of a circle of radius 21 cm has central angle 120°. Find (i) the arc length, (ii) the sector area, (iii) the sector's perimeter.

SOLUTION

Step 1: Fraction: 120/360 = 1/3.

Step 2: Full circle: $C = 2\pi(21) = 132$ cm; $A = \pi(441) = 1386$ cm^2.

Step 3: Arc = 132/3 = 44 cm.

Step 4: Sector area = 1386/3 = 462 cm^2.

Step 5: Perimeter = arc + 2 radii = 44 + 42 = 86 cm.

» **Why this step?** The sector owns one third of the circumference AND one third of the area; its perimeter needs the two straight edges too.

Hinglish: *120° = one-third. Arc 44, area 462. Perimeter mein DO radii: 44 + 42 = 86 cm. Arc + ek radius likhna classic mistake hai.*

Answer: arc 44 cm; area 462 cm^2; perimeter 86 cm

Q23 · Moderate · Concept: Sector perimeter (T7)

Q23. Find the perimeter of a sector of radius 14 cm and angle 45° ($\pi = 22/7$).

SOLUTION

Step 1: Fraction: 45/360 = 1/8.

Step 2: Arc = $(1/8) \times 2\pi r = (1/8) \times 88 = 11$ cm.

Step 3: Perimeter = arc + 2r = 11 + 28 = 39 cm.

» **Why this step?** Perimeter of a curved region = every boundary piece: one arc plus two straight radii.

Hinglish: *45° = one-eighth. Arc 11. Perimeter 11 + 14 + 14 = 39 cm. Dono radii ginn kar.*

Answer: 39 cm

Q24 · Moderate · Concept: Ratio scaling (T12)

Q24. The radii of two circles are in the ratio 3 : 4. Find (i) the ratio of their circumferences, (ii) the ratio of their areas. If the smaller circle's area is 154 cm^2, find the larger's.

SOLUTION

Step 1: Let radii be 3k and 4k.

Step 2: Circumferences: $2\pi(3k) : 2\pi(4k) = 3 : 4$ (linear scaling).

Step 3: Areas: $\pi(3k)^2 : \pi(4k)^2 = 9 : 16$ (square scaling).

Step 4: Larger area = $154 \times 16/9 = 2464/9 \approx 273.78$ cm^2.

» **Why this step?** Lengths scale linearly with k; areas carry k^2 - every ratio question here hinges on this difference.

Hinglish: *Circumference ratio = radius ratio (3:4). Area ratio = SQUARE (9:16). 154 ko 16/9 se scale karo: 273.78 cm². 'k² rule' yaad rakho.*

Answer: 3 : 4 and 9 : 16; larger area = 2464/9 ≈ 273.8 cm²

Q25 · Moderate · Concept: Heron moderate (T2)

Q25. Find the area of a triangle with sides 51 cm, 37 cm and 20 cm.

SOLUTION

Step 1: s = (51 + 37 + 20)/2 = 54 cm.

Step 2: Differences: 3, 17, 34.

Step 3: Product: $54 \times 3 \times 17 \times 34 = 93636$.

Step 4: A = $\sqrt{93636} = 306$ cm^2.

» **Why this step?** Differences 3 and 34 pair to 102; 17 and 54 pair to 918; regrouping makes the root visible: $(306)^2$.

Hinglish: *s = 54, differences 3, 17, 34. Product 93636, root 306 cm². Bade products ko chhote-chhote multiply karke likho.*

Answer: 306 cm²

Q26 · Moderate · Concept: Wire re-bending (T12)

Q26. A wire of length 132 cm is first bent into an equilateral triangle and then re-bent into a circle. Which shape encloses more area, and by how much ($\sqrt{3} \approx 1.73$, $\pi = 22/7$)?

SOLUTION

Step 1: Equilateral triangle: side = 132/3 = 44 cm.

Step 2: Area = $(\sqrt{3}/4)(44)^2 = (\sqrt{3}/4)(1936) = 484\sqrt{3} \approx 837.3$ cm^2.

Step 3: Circle: circumference 132 $\Rightarrow$ r = $132 \times 7/44 = 21$ cm.

Step 4: Area = $\pi(441) = 1386$ cm^2.

Step 5: Difference = 1386 - 837.3 = 548.7 cm^2. The circle wins.

» **Why this step?** The same perimeter always encloses more area as a circle than as any polygon - this question quantifies it.

Hinglish: *Same wire, do shapes. Triangle: 484√3 ≈ 837 cm². Circle: 1386 cm². Circle 549 cm² zyada. Ye universal truth hai - fixed perimeter par circle sabse bada area deta hai.*

Answer: Circle: 1386 vs 484√3 ≈ 837.3; the circle encloses about 548.7 cm² more.

Q27 · Difficult · Concept: Heron large (T2)

Q27. Find the area of a triangle whose sides are 51 cm, 37 cm and 20 cm... verify your answer by checking whether the triangle is right-angled or not.

SOLUTION

Step 1: Q25 computed this area: s = 54, differences 3, 17, 34, A = 306 cm^2.

Step 2: Pythagoras check on the largest side: $51^2 = 2601$; $37^2 + 20^2 = 1369 + 400 = 1769$.

Step 3: $2601 \neq 1769$, so the triangle is NOT right-angled - Heron was genuinely needed.

Step 4: No shortcut formula applies; the answer stands at 306 cm^2.

» **Why this step?** The Pythagoras test tells you whether the ½-leg-leg shortcut is legal; here it is not, and saying so is part of the verification.

Hinglish: *Same numbers as Q25 - par ab CHECK karna hai. $51^2 = 2601$ vs $37^2+20^2 = 1769$. Barabar nahi, toh right angle nahi - Heron compulsory tha. Verification habit banao.*

Answer: 306 cm^2; the triangle is not right-angled ($51^2 \neq 37^2 + 20^2$).

Q28 · Difficult · Concept: Quadrilateral via diagonal (T4)

Q28. In quadrilateral ABCD, AB = 7 cm, BC = 24 cm, CD = 26 cm, DA = 17 cm and AC = 25 cm. Find the area of ABCD.

SOLUTION

Step 1: Triangle ABC: sides 7, 24, 25. Check $7^2 + 24^2 = 49 + 576 = 625 = 25^2$: right angle at B.

Step 2: Area(ABC) = $\frac{1}{2} \times 7 \times 24 = 84$ cm^2.

Step 3: Triangle ACD: sides 25, 26, 17. s = 34; differences 9, 8, 17.

Step 4: Product: $34 \times 9 \times 8 \times 17 = 41616$; Area(ACD) = 204 cm^2.

Step 5: Total = 84 + 204 = 288 cm^2.

» **Why this step?** One diagonal creates two independent triangle problems; a hidden right triangle in one of them halves the work.

Hinglish: *ABC: 7-24-25 triple! ½×7×24 = 84. ACD: Heron se 204. Total 288 cm^2. Diagonal questions mein pehle Pythagoras triple hunt karo.*

Answer: 288 cm^2

Q29 · Difficult · Concept: Brahmagupta with surd (T5)

Q29. A cyclic quadrilateral has sides 9 cm, 12 cm, 15 cm and 12 cm. Find its area, expressing the answer as a simplified surd.

SOLUTION

Step 1: s = (9 + 12 + 15 + 12)/2 = 24 cm.

Step 2: Differences: 15, 12, 9, 12.

Step 3: Product: $15 \times 12 \times 9 \times 12 = 19440$.

Step 4: Simplify: $19440 = 1296 \times 15$, so $A = 36\sqrt{15}$ cm^2.

Step 5: Decimal: $36 \times 3.873 \approx 139.4$ cm^2.

» **Why this step?** Brahmagupta's product often carries a large square factor (here 36^2) - extract it before decimalising.

Hinglish: *s = 24, differences 15, 12, 9, 12. Product 19440 = 1296×15. Root: $36\sqrt{15} \approx 139.4$ cm^2. Brahmagupta mein bhi surd simplify hota hai.*

Answer: $36\sqrt{15} \approx 139.4$ cm^2

Q30 · Difficult · Concept: Area to circumference (T6)

Q30. The area of a circular ring-shaped garden bed is 616 m^2 and it is actually a full circle. Find its radius and circumference ($\pi = 22/7$).

SOLUTION

Step 1: $\pi r^2 = 616 \Rightarrow r^2 = 616 \times 7/22 = 196 \Rightarrow r = 14$ m.

Step 2: $C = 2\pi r = 2 \times (22/7) \times 14 = 88$ m.

» **Why this step?** Area → square root → radius → circumference: the standard reverse chain; keep π as 22/7 to stay exact.

Hinglish: *Area se radius: $r^2 = 616 \times 7/22 = 196$. $r = 14$. $C = 88$ m. Reverse chain mein π ko decimal mat karo - 22/7 se sab clean.*

Answer: r = 14 m, C = 88 m

Q31 · Difficult · Concept: Sector reverse (T7)

Q31. The area of a sector of a circle with central angle 60° is 231 cm^2. Find the radius of the circle and the arc length of the sector ($\pi = 22/7$).

SOLUTION

Step 1: Sector area = $(\theta/360)\pi r^2 = (1/6)\pi r^2 = 231$.

Step 2: $\pi r^2 = 1386 \Rightarrow r^2 = 1386 \times 7/22 = 441 \Rightarrow r = 21$ cm.

Step 3: Arc = $(1/6) \times 2\pi r = (1/6)(132) = 22$ cm.

» **Why this step?** Work the sector formula backwards: undo the fraction first, then π, then the square.

Hinglish: *60° = one-sixth. $(1/6)\pi r^2 = 231 \rightarrow \pi r^2 = 1386 \rightarrow r^2 = 441$. $r = 21$. Arc 22 cm. Reverse sector questions mein fraction pehle undo karo.*

Answer: r = 21 cm, arc = 22 cm

Q32 · Difficult · Concept: Segment area (T8)

Q32. Find the area of the minor segment of a circle of radius 14 cm cut off by a chord subtending 90° at the centre ($\pi = 22/7$).

SOLUTION

Step 1: Segment = sector - triangle.

Step 2: Sector (90°) = $(1/4)\pi(196) = 154$ cm^2.

Step 3: The chord and two radii form a right isosceles triangle with legs 14: area = $\frac{1}{2} \times 14 \times 14 = 98$ cm^2.

Step 4: Segment = 154 - 98 = 56 cm^2.

» **Why this step?** Every segment problem is 'pizza minus triangle'; the triangle's area uses the two radii as legs when the angle is 90°.

Hinglish: *Segment = sector - triangle. Sector 154, triangle $\frac{1}{2} \times 14 \times 14 = 98$. Answer 56 cm^2. 90° wale case mein triangle right-angled hota hai - radii legs.*

Answer: 56 cm^2

Q33 · Difficult · Concept: Ring from inner circumference (T9)

Q33. A circular ring has width 7 cm. The circumference of its inner circle is 44 cm. Find the area of the ring ($\pi = 22/7$).

SOLUTION

Step 1: Inner radius: $44 = 2\pi r \Rightarrow r = 7$ cm.

Step 2: Width 7 ⇒ outer radius $R = 7 + 7 = 14$ cm.

Step 3: Ring area $= \pi(R^2 - r^2) = (22/7)(196 - 49) = (22/7)(147) = 462$ cm^2.

» **Why this step?** The width adds to the radius (not the diameter); $R^2-r^2 = (R+r)(R-r) = 21\times7$ makes the arithmetic one line.

Hinglish: *Width radius mein JUDTI hai: r = 7, R = 14. Ring = π(147) = 462 cm². Width → diameter confusion sabse common galti hai.*

Answer: 462 cm^2

Q34 · Difficult · Concept: Inside rectangular path (T10)

Q34. A rectangular garden is 30 m by 20 m. A path 2 m wide runs inside it along all four sides. Find the area of the path.

SOLUTION

Step 1: Outer area $= 30 \times 20 = 600$ m^2.

Step 2: The inner rectangle loses 2 m on EACH side: inner length $= 30 - 4 = 26$ m; inner width $= 20 - 4 = 16$ m.

Step 3: Inner area $= 26 \times 16 = 416$ m^2.

Step 4: Path area $= 600 - 416 = 184$ m^2.

» **Why this step?** An inside border subtracts the width TWICE per dimension - once for each of the two parallel sides.

Hinglish: *Path andar hai: dono dimensions se 2w = 4 minus. Inner 26×16 = 416. Path = 600-416 = 184 m². '2w' rule yaad rakho.*

Answer: 184 m^2

Q35 · Difficult · Concept: Outside circular path (T10)

Q35. A circular park of radius 17 m is surrounded by a path 4 m wide running outside it. Find the area of the path ($\pi = 22/7$).

SOLUTION

Step 1: Outer radius $R = 17 + 4 = 21$ m.

Step 2: Path area $= \pi(R^2 - r^2) = (22/7)(441 - 289) = (22/7)(152)$.

Step 3: $= 3344/7 \approx 477.7$ m^2.

» **Why this step?** An outside path adds to the radius once; the ring formula $\pi w(2r + w)$ is the fast version: $(22/7)(4)(38)$.

Hinglish: *Bahar wala path: R = 21. π(152) ≈ 477.7 m². Shortcut: πw(2r+w) = (22/7)×4×38 - same answer, direct.*

Answer: $152\pi \approx 477.7$ m^2

Q36 · Difficult · Concept: Area-sum circle (T6)

Q36. Two circles have radii 5 cm and 12 cm. Find the radius of a third circle whose area equals the SUM of their areas.

SOLUTION

Step 1: $\pi R^2 = \pi(25) + \pi(144) = \pi(169)$.

Step 2: $R^2 = 169 \Rightarrow R = 13$ cm.

» **Why this step?** π cancels; the question is secretly 25 + 144 = 169 - the 5-12-13 triple wearing a circle costume.

Hinglish: *Areas add: $\pi R^2 = \pi(25+144) = \pi(169)$. R = 13. Ye 5-12-13 triple hai - areas jodne par hypotenuse ban jata hai!*

Answer: 13 cm

Q37 · Difficult · Concept: Composite figure (T11)

Q37. A rectangle 20 m × 16 m has a semicircular grass bed built on one of its 16 m sides (outside the rectangle). Find (i) the total area and (ii) the perimeter of the combined figure ($\pi = 22/7$).

SOLUTION

Step 1: Semicircle diameter = 16 ⇒ radius = 8 m.

Step 2: Semicircle area = $\frac{1}{2} \times \pi \times 64 = 32\pi = 704/7 \approx 100.57$ m^2.

Step 3: Total area = 320 + 100.57 = 420.57 m^2.

Step 4: Perimeter: the 16 m side is now internal - it disappears from the boundary. Boundary = 20 + 16 + 20 + semicircular arc.

Step 5: Arc = $\pi \times 8 = 176/7 \approx 25.14$ m. Total perimeter ≈ 76 + 25.14 = 101.14 m.

» **Why this step?** When shapes join, the shared edge vanishes from the perimeter but stays inside the area sum.

Hinglish: *Rectangle + semicircle: area jodo (320 + 100.57). Perimeter mein shared 16 m side GAYAB ho jati hai - uski jagah arc aata hai: 76 + 25.14 = 101.14 m. Composite figures ka pehla rule: shared edge boundary se hatao.*

Answer: (i) ≈ 420.6 m² (ii) ≈ 101.1 m

Q38 · Difficult · Concept: Segment with 120° (T8)

Q38. Find the area of the minor segment of a circle of radius 21 cm whose chord subtends 120° at the centre (use $\sqrt{3} \approx 1.73$, $\pi = 22/7$).

SOLUTION

Step 1: Sector (120°) = $(1/3)\pi(441) = 462$ cm^2.

Step 2: The triangle formed by the two radii and the chord is isosceles with included angle 120°.

Step 3: Its area = $\frac{1}{2} \cdot r^2 \cdot \sin 120°$; equivalently $\frac{1}{2} \cdot r^2 \cdot \sin 60°$ per half. Directly: area = $\frac{1}{2}(21)(21)\cdot\sin 120° = \frac{1}{2}(441)(\sqrt{3}/2) = 441\sqrt{3}/4 = 110.25\sqrt{3}$.

Step 4: Numerically: $110.25 \times 1.73 \approx 190.73$ cm^2.

Step 5: Segment = 462 - 190.73 ≈ 271.3 cm^2.

» **Why this step?** For non-90° sectors the 'triangle' needs the sine form $\frac{1}{2}r^2\sin\theta$ (here $\theta = 120°$) - still just $\frac{1}{2}$ × base × height in disguise.

Hinglish: *120° wale segment mein triangle formula: $\frac{1}{2}r^2 \sin 120° = (441\sqrt{3})/4 \approx 190.7$. Sector 462. Segment ≈ 271.3 cm². sin 120° = sin 60° = √3/2 - ye value yaad rakho.*

Answer: ≈ 271.3 cm² (462 - 110.25√3)

Q39 · Difficult · Concept: Minute hand sweep (T13)

Q39. The minute hand of a clock is 14 cm long. Find the area swept by it in 40 minutes ($\pi = 22/7$).

SOLUTION

Step 1: In 60 minutes the hand sweeps 360°; in 40 minutes it sweeps 240°.

Step 2: Fraction: 240/360 = 2/3.

Step 3: Full circle area = $\pi(196) = 616$ cm^2.

Step 4: Swept area = $(2/3) \times 616 = 1232/3 \approx 410.67$ cm^2.

» **Why this step?** The hand is a rotating radius: the swept region is a sector whose angle is proportional to time.

Hinglish: *40 min = 240° = two-thirds. 616 × 2/3 = 410.67 cm². Time ko angle mein convert karo, phir sector.*

Answer: 1232/3 ≈ 410.7 cm²

Q40 · Difficult · Concept: Circle in square (T11)

Q40. A circle is inscribed in a square of side 14 cm. Find the area of the region inside the square but outside the circle ($\pi = 22/7$).

SOLUTION

Step 1: Inscribed circle: diameter = side ⇒ r = 7 cm.

Step 2: Circle area = $\pi(49) = 154$ cm^2.

Step 3: Square area = 196 cm^2.

Step 4: Gap = 196 - 154 = 42 cm^2.

» **Why this step?** Inscribed means the circle touches all four sides, so the diameter equals the side - the only geometric input needed.

Hinglish: *Inscribed circle: diameter = side. r = 7. Gap = 196 - 154 = 42 cm². Four corner pieces ka total.*

Answer: 42 cm²

Q41 · Difficult · Concept: Sector from perimeter (T7)

Q41. The perimeter of a sector of a circle is 68 cm and its radius is 21 cm. Find the area of the sector.

SOLUTION

Step 1: Perimeter = arc + 2r ⇒ arc = 68 - 42 = 26 cm.

Step 2: Sector area = ½ × arc × r = ½ × 26 × 21 = 273 cm^2.

» **Why this step?** The A = ½l·r form avoids computing the angle entirely - fastest route when perimeter is given.

Hinglish: *Perimeter se arc nikaalo: 68 - 42 = 26. Area = ½ × l × r = 273 cm². Angle ki zaroorat hi nahi - ye half-arc-radius shortcut strong hai.*

Answer: 273 cm²

Q42 · Difficult · Concept: Ratio sides with perimeter (T2)

Q42. The sides of a triangular plot are in the ratio 13 : 14 : 15 and its perimeter is 84 m. Find the area.

SOLUTION

Step 1: Let sides be 13k, 14k, 15k: 42k = 84 ⇒ k = 2.

Step 2: Sides: 26 m, 28 m, 30 m.

Step 3: s = 42; differences 16, 14, 12.

Step 4: Product: 42 × 16 × 14 × 12 = 112896. A = $\sqrt{112896} = 336$ m^2.

» **Why this step?** Convert ratios to actual sides FIRST; Heron only works on real lengths.

Hinglish: *Ratio + perimeter → pehle actual sides: 26, 28, 30. Heron: 336 m². Ratio mein Heron lagana mana hai.*

Answer: 336 m²

Q43 · High · Concept: Pythagorean ratio (T2/T3)

Q43. The sides of a triangle are in the ratio 3 : 4 : 5 and its perimeter is 72 cm. Find the area by TWO methods and confirm they agree.

SOLUTION

Step 1: Sides: 18k... let 3k + 4k + 5k = 72 ⇒ k = 6: sides 18, 24, 30 cm.

Step 2: Method 1 (right triangle): $18^2 + 24^2 = 324 + 576 = 900 = 30^2$, so right angle between the legs 18 and 24.

Step 3: Area = ½ × 18 × 24 = 216 cm^2.

Step 4: Method 2 (Heron): s = 36; differences 18, 12, 6; product 36×18×12×6 = 46656; √46656 = 216 ✓.

» **Why this step?** A 3-4-5 ratio triangle is always right-angled (the ratio is scale-invariant), so both methods must match.

Hinglish: *3:4:5 hamesha right triangle hota hai. Sides 18, 24, 30. ½×18×24 = 216. Heron bhi 216 deta hai - dono match. Ratio dekh ke triple pehchano.*

Answer: 216 cm^2 (both methods)

Q44 · High · Concept: Reverse equilateral with height (T3)

Q44. The area of an equilateral triangle is $81\sqrt{3}$ cm^2. Find its side, its height, and its perimeter.

SOLUTION

Step 1: $(\sqrt{3}/4)a^2 = 81\sqrt{3} \Rightarrow a^2 = 324 \Rightarrow a = 18$ cm.

Step 2: Height = $(\sqrt{3}/2)a = 9\sqrt{3} \approx 15.59$ cm.

Step 3: Perimeter = 54 cm.

» **Why this step?** All three quantities come from a; find it first by cancelling √3 exactly.

Hinglish: *$a^2 = 324 \rightarrow a = 18$. Height $9\sqrt{3}$, perimeter 54. Equilateral ke teeno answers ek hi variable se nikalte hain.*

Answer: side 18 cm, height $9\sqrt{3}$ cm, perimeter 54 cm

Q45 · High · Concept: Isosceles reverse (T3)

Q45. An isosceles triangle has area 48 cm^2 and base 16 cm. Find the equal sides and the perimeter.

SOLUTION

Step 1: ½ × 16 × h = 48 ⇒ h = 6 cm.

Step 2: The height bisects the base: half-base = 8 cm.

Step 3: Equal side = $\sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$ cm.

Step 4: Perimeter = 10 + 10 + 16 = 36 cm.

» **Why this step?** Recover the height from the area, then the side from the right triangle (half-base, height, side).

Hinglish: *Area se height: 48 = ½×16×h → h = 6. Half-base 8. Side √(64+36) = 10. Perimeter 36. 6-8-10 triple hidden inside!*

Answer: equal sides 10 cm each, perimeter 36 cm

Q46 · High · Concept: Trapezium via Heron (T4)

Q46. A field is in the shape of a trapezium whose parallel sides are 77 m and 60 m, and the non-parallel sides are 25 m and 26 m. Find its area.

SOLUTION

Step 1: Draw the height through the shorter parallel side's endpoint; it creates a helper triangle whose base is 77 - 60 = 17 m (after transferring the full 60 to the other side).

Step 2: Helper triangle sides: 25, 26, 17. s = 34; differences 9, 8, 17.

Step 3: Area of helper triangle = $\sqrt{(34\times9\times8\times17)} = \sqrt{41616} = 204$ m^2.

Step 4: Height: $204 = \frac{1}{2} \times 17 \times h \Rightarrow h = 24$ m.

Step 5: Trapezium area = $\frac{1}{2}(77 + 60)(24) = \frac{1}{2} \times 137 \times 24 = 1644$ m^2.

» **Why this step?** A general trapezium hides a triangle built from the slanted sides and the difference of the parallel sides; Heron extracts the height.

Hinglish: *Trapezium ka height chhupa hai helper triangle mein: base = 77-60 = 17, sides 25, 26. Heron se area 204. h = 2×204/17 = 24. Phir ½(a+b)h = 1644 m². Ye 4-step method ratta maar lo.*

Answer: 1644 m²

Q47 · High · Concept: Rhombus from perimeter (T1)

Q47. The perimeter of a rhombus is 40 cm and one diagonal is 12 cm. Find the other diagonal and the area.

SOLUTION

Step 1: Side = 40/4 = 10 cm.

Step 2: Diagonals bisect at right angles: half of the given diagonal = 6 cm.

Step 3: Half of the other diagonal = $\sqrt{(10^2 - 6^2)} = \sqrt{64} = 8$ cm, so $d_2 = 16$ cm.

Step 4: Area = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 12 \times 16 = 96$ cm^2.

» **Why this step?** The half-diagonals and a side form a right triangle - the whole rhombus question lives inside it.

Hinglish: *Side 10. Half-diagonal 6. Other half √(100-36) = 8 → d_2 = 16. Area ½×12×16 = 96 cm². Rhombus = 4 right triangles.*

Answer: d_2 = 16 cm, area = 96 cm²

Q48 · High · Concept: Circle in equilateral triangle (T3/T11)

Q48. Find the area of the circle inscribed in an equilateral triangle of side 42 cm ($\pi = 22/7$).

SOLUTION

Step 1: Inradius of an equilateral triangle: $r = a/(2\sqrt{3}) = 42/(2\sqrt{3}) = 21/\sqrt{3} = 7\sqrt{3}$ cm.

Step 2: $r^2 = 147$ cm^2.

Step 3: Circle area = $\pi r^2 = (22/7)(147) = 462$ cm^2.

» **Why this step?** The inradius formula $r = a/(2\sqrt{3})$ (or equivalently r = height/3) comes from the height's trisection by the incentre.

Hinglish: *$r = a/(2\sqrt{3}) = 7\sqrt{3}$. $r^2 = 147$. Area 462 cm². Equilateral mein inradius = height/3 hota hai - height $21\sqrt{3}$, uska 1/3 = $7\sqrt{3}$. Same baat.*

Answer: 462 cm²

Q49 · High · Concept: Circle in square gap (T11)

Q49. A circle is inscribed in a square of side 28 cm. Find the total area of the four corner regions ($\pi = 22/7$).

SOLUTION

Step 1: r = 14 cm; circle area = $\pi(196) = 616$ cm^2.

Step 2: Square area = 784 cm^2.

Step 3: Corners total = 784 - 616 = 168 cm^2.

Step 4: Bonus: each corner alone is 42 cm^2 - the four are congruent.

» **Why this step?** Subtraction gives the total; symmetry then divides it into four equal corners.

Hinglish: *Square 784, circle 616. Gap 168 cm^2 - chaar corners, har ek 42. Symmetry se divide karo.*

Answer: 168 cm^2 (42 cm^2 per corner)

Q50 · High · Concept: Circumference sum (T6)

Q50. Two circles have circumferences 88 cm and 132 cm. A third circle has circumference equal to the SUM of theirs. Find the radius and area of the third circle ($\pi = 22/7$).

SOLUTION

Step 1: Third circumference = 88 + 132 = 220 cm.

Step 2: $r = C/(2\pi) = 220 \times 7/44 = 35$ cm.

Step 3: Area = $\pi(35^2) = (22/7)(1225) = 3850$ cm^2.

Step 4: Note: radii add, NOT areas - 14 + 21 = 35 is the additive quantity here.

» **Why this step?** Circumference is linear in r, so circumferences add exactly like radii - the reverse of the area-sum situation (Q36).

Hinglish: *Circumferences JUDTE hain: 220 → r = 35, area 3850 cm^2. Q36 mein areas judte the; yahan circumferences. Linear quantity add hota hai, quadratic nahi - ye contrast yaad rakho.*

Answer: r = 35 cm, area = 3850 cm^2

Q51 · High · Concept: Ring via chord (T9)

Q51. In two concentric circles, a chord of the OUTER circle of length 28 cm touches the inner circle. Find the area of the ring ($\pi = 22/7$).

SOLUTION

Step 1: Let O be the common centre, M the touch point. OM is perpendicular to the chord (radius to a tangent point... here the chord is tangent to the inner circle), so OM = inner radius r.

Step 2: The perpendicular from the centre bisects the chord: AM = 14 cm, where A is a chord endpoint.

Step 3: In right triangle OMA: $R^2 = r^2 + 14^2$, so $R^2 - r^2 = 196$.

Step 4: Ring area = $\pi(R^2 - r^2) = (22/7)(196) = 616$ cm^2.

» **Why this step?** The tangent-radius is perpendicular to the chord at the touch point, and the half-chord theorem gives $AM^2 = R^2 - r^2$ directly - individual radii are never needed.

Hinglish: *Magic question! Chord inner circle ko touch karti hai → half-chord² = R^2-r^2 = 14^2 = 196. Ring area = $\pi \times 196 = 616$ cm^2. Na R chahiye na r - sirf chord.*

Answer: 616 cm^2

Q52 · High · Concept: Running track (T11)

Q52. A running track is 90 m long on each straight and has two semicircular ends of diameter 56 m. Find (i) the perimeter of the track and (ii) the area enclosed ($\pi = 22/7$).

SOLUTION

Step 1: Perimeter = two straights + two semicircular arcs = 90 + 90 + one full circle of diameter 56.

Step 2: Circle circumference = $\pi d = (22/7)(56) = 176$ m.

Step 3: Perimeter = 180 + 176 = 356 m.

Step 4: Area = rectangle 90×56 + full circle $\pi(28^2) = 5040 + (22/7)(784) = 5040 + 2464 = 7504$ m^2.

» **Why this step?** Two semicircles of equal diameter combine into one full circle - for both length and area.

Hinglish: *Do semicircles = ek full circle. Perimeter 180 + 176 = 356 m. Area 5040 + 2464 = 7504 m². Track problems mein halves ko jod kar full banao.*

Answer: (i) 356 m (ii) 7504 m^2

Q53 · High · Concept: Circle in equilateral gap (T11)

Q53. A circle is inscribed in an equilateral triangle of side 12 cm. Find the area of the triangle not covered by the circle ($\sqrt{3} \approx 1.73$, $\pi = 3.14$).

SOLUTION

Step 1: Triangle area = $(\sqrt{3}/4)(144) = 36\sqrt{3} \approx 62.28$ cm^2.

Step 2: Inradius $r = a/(2\sqrt{3}) = 12/(2\sqrt{3}) = 2\sqrt{3}$ cm; $r^2 = 12$.

Step 3: Circle area = $\pi r^2 = 3.14 \times 12 = 37.68$ cm^2.

Step 4: Gap = 62.28 - 37.68 = 24.6 cm^2.

» **Why this step?** Mixing 22/7-style exactness with 3.14 is fine as long as both approximations are stated; the gap inherits the rounding.

Hinglish: *Triangle $36\sqrt{3} \approx 62.28$, circle $r = 2\sqrt{3} \rightarrow 12\pi \approx 37.68$. Gap 24.6 cm². Teen gap regions hain - inka total ye hai.*

Answer: ≈ 24.6 cm^2

Q54 · High · Concept: Costs: fence and plough (T13)

Q54. A circular garden has radius 35 m. Fencing costs ₹20 per metre and ploughing costs ₹0.50 per m^2. Find the total cost of fencing the boundary once and ploughing the whole garden ($\pi = 22/7$).

SOLUTION

Step 1: Circumference = $2\pi r = 2 \times (22/7) \times 35 = 220$ m.

Step 2: Fencing cost = $220 \times 20 =$ ₹4400.

Step 3: Area = $\pi r^2 = (22/7)(1225) = 3850$ m^2.

Step 4: Ploughing cost = $3850 \times 0.50 =$ ₹1925.

Step 5: Total = 4400 + 1925 = ₹6325.

» **Why this step?** Perimeter multiplies per-metre rates; area multiplies per-square-metre rates - never interchange them.

Hinglish: *Fencing = boundary × rate/m: 220×20 = 4400. Ploughing = area × rate/m²: 3850×0.5 = 1925. Total ₹6325. Rate ke saath unit dekho - /m ya /m².*

Answer: ₹6325

Q55 · High · Concept: Same perimeter comparison (T12)

Q55. A wire of length 44 cm is bent into (i) a square and (ii) a circle. Find each area and state the percentage increase when the square is replaced by the circle ($\pi = 22/7$).

SOLUTION

Step 1: Square: side = 44/4 = 11 cm; area = 121 cm^2.

Step 2: Circle: $r = 44/(2\pi) = 7$ cm; area = 154 cm^2.

Step 3: Increase = 154 - 121 = 33 cm^2.

Step 4: Percentage increase = (33/121) × 100 = 27.27% (exactly 100/3.675...; commonly stated as 27.3%).

» **Why this step?** Same perimeter always favours the circle; the gain is measured relative to the SMALLER area (the square's).

Hinglish: *Square 121, circle 154. Increase 33. Percentage 33/121 = 27.3%. Base hamesha PEHLE wala (chhota) area hota hai.*

Answer: 121 cm² and 154 cm²; increase ≈ 27.3%

Q56 · High · Concept: Ring reverse with width (T9)

Q56. The area of a circular ring is 1386 cm² and its width is 3 cm. Find the two radii (π = 22/7).

SOLUTION

Step 1: $\pi(R^2 - r^2) = 1386 \Rightarrow R^2 - r^2 = 441$.

Step 2: Factorise: (R + r)(R - r) = 441.

Step 3: Width: R - r = 3 ⇒ R + r = 441/3 = 147.

Step 4: Solve: R = (147 + 3)/2 = 75, r = (147 - 3)/2 = 72.

Step 5: Check: $\pi(75^2 - 72^2) = (22/7)(5625 - 5184) = (22/7)(441) = 1386$ ✓.

» **Why this step?** The difference-of-squares factorisation splits one equation into the sum and the difference - both trivially solvable.

Hinglish: *R^2-r^2 = 441 ko (R+r)(R-r) mein todo. Width = R-r = 3 → R+r = 147. R = 75, r = 72. Factorise-and-solve technique rings ke saare reverse questions mein chalti hai.*

Answer: R = 75 cm, r = 72 cm

Q57 · HOTS · Concept: Ratio sides large (T2)

Q57. The sides of a triangular park are in the ratio 12 : 17 : 25 and its perimeter is 540 m. Find the area and the cost of levelling it at ₹3 per m².

SOLUTION

Step 1: 12k + 17k + 25k = 54k = 540 ⇒ k = 10: sides 120, 170, 250 m.

Step 2: s = 270; differences 150, 100, 20.

Step 3: Product: 270 × 150 × 100 × 20 = 81,000,000.

Step 4: A = √81000000 = 9000 m².

Step 5: Cost = 9000 × 3 = ₹27000.

» **Why this step?** Scale first, compute second; the giant product collapses because every factor carries a factor of 10.

Hinglish: *Sides 120, 170, 250. s = 270. Product 8.1 crore, root 9000 m². Cost ₹27000. Zeros ginte waqt dhyan se - pairs of 10s root ke andar square banate hain.*

Answer: 9000 m²; ₹27000

Q58 · HOTS · Concept: Heron surd area (T2)

Q58. A triangle has sides 35 cm, 54 cm and 61 cm. Find its area as a simplified surd and as a decimal (to 2 dp).

SOLUTION

Step 1: s = (35 + 54 + 61)/2 = 75 cm.

Step 2: Differences: 40, 21, 14.

Step 3: Product: $75 \times 40 \times 21 \times 14 = 882000$.

Step 4: Simplify: $882000 = 176400 \times 5$ and $176400 = 420^2$.

Step 5: $A = 420\sqrt{5} \approx 939.15$ cm^2.

» **Why this step?** Extract the square part of the product systematically: pair off factors of 2, 3, 5, 7 until only primes with odd exponents remain.

Hinglish: *s = 75, differences 40, 21, 14. Product 882000. $420^2 = 176400$ bahar nikala: $420\sqrt{5} \approx 939.15$ cm². Factor pairing hi surd ka tool hai.*

Answer: $420\sqrt{5} \approx 939.15$ cm^2

Q59 · HOTS · Concept: Quadrilateral field (T4)

Q59. A quadrilateral field ABCD has AB = 42 m, BC = 21 m, CD = 29 m, DA = 34 m and diagonal BD = 20 m. Find its area.

SOLUTION

Step 1: Triangle ABD: sides 42, 34, 20. s = 48; differences 6, 14, 28.

Step 2: Product: $48 \times 6 \times 14 \times 28 = 112896$; area = 336 m^2.

Step 3: Triangle BCD: sides 21, 29, 20. s = 35; differences 14, 6, 15.

Step 4: Product: $35 \times 14 \times 6 \times 15 = 44100$; area = 210 m^2.

Step 5: Total = 336 + 210 = 546 m^2.

» **Why this step?** Both Heron products land on perfect squares - a sign the question was designed for clean addition at the end.

Hinglish: *Do Herons: 336 + 210 = 546 m². Diagonal BD ke around dono triangles organize karo, answers ko JODO.*

Answer: 546 m^2

Q60 · HOTS · Concept: Brahmagupta symmetric proof (T5/T14)

Q60. For a cyclic quadrilateral with sides a, a, b, b (in order), show that Brahmagupta's formula reduces to A = (s-a)(s-b), and evaluate it for a = 25, b = 39.

SOLUTION

Step 1: $s = (a + a + b + b)/2 = a + b$.

Step 2: Differences: $s - a = b$ (twice), $s - b = a$ (twice).

Step 3: $A = \sqrt{(b \cdot b \cdot a \cdot a)} = ab$... careful: $(s-a)(s-a)(s-b)(s-b) = b^2a^2$, so $A = ab$ with the direct reading.

Step 4: Wait - recompute: $s - a = (a+b) - a = b$ and $s - b = (a+b) - b = a$. Product $= b^2a^2 \Rightarrow A = ab$.

Step 5: But the intended general form: $A = (s-a)(s-b)$ holds when the product of the four differences is $(s-a)^2(s-b)^2$ - true here since two sides equal a and two equal b. For a = 25, b = 39: s = 64, $A = 25 \times 39 = 975$.

» **Why this step?** Symmetric side pairs make the four differences repeat in pairs, so the root becomes a clean product.

Hinglish: *Sides a,a,b,b: s = a+b. Differences: b, b, a, a. Area = ab. a = 25, b = 39 → 975. Brahmagupta symmetric case mein direct product deta hai.*

Answer: A = (s-a)(s-b) = ab here; for 25, 25, 39, 39 the area is 975.

Q61 · HOTS · Concept: Hexagon segment (T8/T14)

Q61. A regular hexagon is inscribed in a circle of radius 14 cm. Find the area of one of the six segments between the hexagon and the circle ($\sqrt{3} \approx 1.73$, $\pi = 22/7$).

SOLUTION

Step 1: Each side of the inscribed hexagon equals the radius: chord = 14 cm, subtending 60°.

Step 2: Sector (60°) = $(1/6)\pi(196) = 308/3 \approx 102.67$ cm^2.

Step 3: The triangle in that sector is equilateral with side 14: area = $(\sqrt{3}/4)(196) = 49\sqrt{3} \approx 84.77$ cm^2.

Step 4: One segment = 102.67 - 84.77 ≈ 17.9 cm^2.

» **Why this step?** An inscribed regular hexagon turns each 60° slice into sector minus equilateral triangle - the hexagon's defining property (side = radius) does the setup.

Hinglish: *Hexagon ki side = radius hoti hai. 60° sector 102.67, equilateral triangle $49\sqrt{3} \approx 84.77$. Ek segment ≈ 17.9 cm². Six segments total ≈ 107.4.*

Answer: ≈ 17.9 cm² per segment

Q62 · HOTS · Concept: Revolutions to distance (T13)

Q62. A wheel of radius 28 cm makes 3000 revolutions. How far does it travel, in kilometres ($\pi = 22/7$)?

SOLUTION

Step 1: Circumference = $2\pi r = 2 \times (22/7) \times 28 = 176$ cm.

Step 2: Distance = 3000 × 176 = 528000 cm.

Step 3: Convert: 528000 cm = 5280 m = 5.28 km.

» **Why this step?** Revolutions × circumference gives arc length travelled; unit conversion is the only other job.

Hinglish: *3000 × 176 = 528000 cm = 5.28 km. Revolutions multiply circumference; phir cm → km (5 zeros hatane hain).*

Answer: 5.28 km

Q63 · HOTS · Concept: Grazing goat (T13/T14)

Q63. A goat is tied to one corner of a rectangular field 40 m by 30 m with a rope 21 m long. Find the area it can graze, assuming it cannot leave the field ($\pi = 22/7$).

SOLUTION

Step 1: At a corner of a rectangle, only 90° of the circle's sweep lies inside the field: a quarter circle.

Step 2: Grazing area = $(1/4)\pi r^2 = (1/4) \times (22/7) \times 441 = 346.5$ m^2.

» **Why this step?** The two fences of the corner cut the reachable disk to a quarter; the rope length is the radius.

Hinglish: *Corner par sirf 90° ka sector available - quarter circle. ¼ × π × 441 = 346.5 m². Rope = radius, corner = center.*

Answer: 346.5 m²

Q64 · HOTS · Concept: Nested circle-square ratio (T12/T14)

Q64. A circle is inscribed in a square, and the square is inscribed in a larger circle. Prove that the areas of the larger and smaller circles are in the ratio 2 : 1.

SOLUTION

Step 1: Let the inner circle have radius r. Then the square has side 2r.

Step 2: The square's diagonal = $2r\sqrt{2}$; this is the diameter of the outer circle.

Step 3: Outer radius $R = r\sqrt{2}$.

Step 4: Area ratio = $\pi R^2 : \pi r^2 = 2r^2 : r^2 = 2 : 1$. ∎

» **Why this step?** The diagonal of the inscribed square is the diameter of the circumscribed circle - one picture, one chain of equalities.

Hinglish: *Chain banao: inner circle $r \to$ square side $2r \to$ diagonal $2r\sqrt{2} \to$ outer radius $r\sqrt{2}$. Area ratio = $(r\sqrt{2})^2 : r^2 = 2:1$. Nested figures mein chain diagram banao.*

Answer: Proved: the ratio is 2 : 1.

Q65 · HOTS · Concept: Hippocrates lunes (T14)

Q65. On the legs of a right triangle with legs 12 cm and 16 cm, semicircles are drawn OUTWARD. On the hypotenuse, the semicircle is drawn inward (toward the triangle). Show that the total area of the two lunes (the crescent-shaped regions between the outward semicircles and the hypotenuse semicircle) equals the area of the triangle.

SOLUTION

Step 1: Right triangle: hypotenuse = $\sqrt{(144 + 256)} = 20$ cm; triangle area = $\frac{1}{2} \times 12 \times 16 = 96$ cm^2.

Step 2: Semicircle on leg 12: area $\frac{1}{2}\pi(6^2) = 18\pi$; on leg 16: $\frac{1}{2}\pi(8^2) = 32\pi$.

Step 3: Semicircle on hypotenuse 20: $\frac{1}{2}\pi(10^2) = 50\pi$.

Step 4: Key identity: $18\pi + 32\pi = 50\pi$ - the two leg semicircles exactly fill the hypotenuse semicircle (because $6^2 + 8^2 = 10^2$).

Step 5: The two lunes together = (leg semicircles) + (triangle) - (hypotenuse semicircle) = $18\pi + 32\pi + 96 - 50\pi = 96$ cm^2. ∎

» **Why this step?** Pythagoras on the RADII makes the curved areas cancel, leaving the flat triangle - Hippocrates' classic squaring of lunes.

Hinglish: *Lunes ka magic: leg semicircles ka sum = hypotenuse semicircle ($18\pi+32\pi = 50\pi$). Lunes = 96 cm^2 = triangle ka area exactly. Curved areas cancel ho gaye - 2500 saal purana result!*

Answer: The lunes total exactly 96 cm^2, equal to the triangle's area.

Q66 · HOTS · Concept: Perpendicular paths in circle (T10/T11)

Q66. A circular park of radius 42 m has two straight paths, each 7 m wide, crossing at the centre - one north-south, one east-west. Find the total area covered by the paths ($\pi = 22/7$).

SOLUTION

Step 1: Each path is a rectangle of length = diameter 84 m and width 7 m: area = 588 m^2.

Step 2: The two paths overlap in a 7×7 square at the centre: area 49 m^2.

Step 3: Total path area = 588 + 588 - 49 = 1127 m^2.

» **Why this step?** Overlapping regions are counted twice in the raw sum; subtract the intersection once.

Hinglish: *Do rectangles 588 + 588, beech ka overlap $7\times7 = 49$ minus. 1127 m^2. Inclusion-exclusion ka basic form - jod ke overlap hatao.*

Answer: 1127 m^2

Q67 · HOTS · Concept: Three-shape comparison (T12)

Q67. A wire of 66 cm is bent into (i) an equilateral triangle, (ii) a square, (iii) a circle. Arrange the three areas in increasing order ($\sqrt{3} \approx 1.73$, $\pi = 22/7$).

SOLUTION

Step 1: Triangle: side 22; area = $(\sqrt{3}/4)(484) = 121\sqrt{3} \approx 209.3$ cm^2.

Step 2: Square: side 16.5; area = 272.25 cm^2.

Step 3: Circle: r = 66 × 7/44 = 10.5; area = (22/7)(110.25) = 346.5 cm^2.

Step 4: Order: triangle < square < circle.

» **Why this step?** For a fixed perimeter, area grows as the shape gains symmetry: triangle < square < circle - a general theorem, illustrated numerically.

Hinglish: *Same wire: triangle 209.3 < square 272.25 < circle 346.5. Circle jeet gaya - fixed perimeter par circle maximum area deta hai. Ye ordering har comparison question mein same rehti hai.*

Answer: $121\sqrt{3} \approx 209.3 < 272.25 < 346.5$ (triangle < square < circle)

Q68 · HOTS · Concept: Wheel journey km (T13)

Q68. A bicycle wheel has radius 28 cm. How many revolutions will it make in a journey of 4.4 km ($\pi = 22/7$)?

SOLUTION

Step 1: Circumference = $2\pi r$ = 176 cm.

Step 2: Journey = 4.4 km = 440000 cm.

Step 3: Revolutions = 440000/176 = 2500.

» **Why this step?** Distance per revolution is fixed at the circumference; the count is a division.

Hinglish: *440000/176 = 2500 revolutions. Unit conversion pehle karo - km ko cm mein.*

Answer: 2500 revolutions

Q69 · HOTS · Concept: AP sides triangle (T2/T14)

Q69. The sides of a triangle are in arithmetic progression, its perimeter is 60 cm and the shortest side is 15 cm. (i) Find all sides. (ii) Show the triangle is right-angled and find its area.

SOLUTION

Step 1: Let sides be 15, 15 + d, 15 + 2d: sum = 45 + 3d = 60 ⇒ d = 5.

Step 2: Sides: 15, 20, 25 cm.

Step 3: Check: $15^2 + 20^2 = 225 + 400 = 625 = 25^2$: right angle opposite the 25 side. ∎

Step 4: Area = ½ × 15 × 20 = 150 cm^2.

Step 5: Heron confirms: s = 30; product 30 × 15 × 10 × 5 = 22500; $\sqrt{22500} = 150$ ✓.

» **Why this step?** AP condition turns three unknowns into one (d); the Pythagoras test then classifies the triangle for free.

Hinglish: *AP + perimeter: sides 15, 20, 25. 15-20-25 = 3-4-5 ka 5×! Right triangle, area 150. Heron se confirm. AP wale side questions mein shortest side + d method use karo.*

Answer: 15, 20, 25 cm; right-angled; area 150 cm^2

Q70 · HOTS · Concept: Hexagon vs inscribed circle (T14)

Q70. A regular hexagon of side 14 cm is inscribed in a circle. (i) Find the hexagon's area. (ii) Find the area of the inscribed circle of the hexagon. (iii) Express the ratio of the two areas in the form $2\sqrt{3} : \pi$ and verify it numerically ($\sqrt{3} \approx 1.73$, $\pi = 22/7$).

SOLUTION

Step 1: (i) Hexagon = 6 equilateral triangles of side 14: area = $6 \times (\sqrt{3}/4)(196) = 294\sqrt{3} \approx 508.6$ cm^2.

Step 2: (ii) The inscribed circle has radius equal to the apothem = $(\sqrt{3}/2)(14) = 7\sqrt{3}$; $r^2 = 147$.

Step 3: Circle area = $\pi(147) = (22/7)(147) = 462$ cm^2.

Step 4: (iii) Ratio = $294\sqrt{3} : 147\pi = 2\sqrt{3} : \pi$ (dividing both by 147).

Step 5: Numerically: $508.6 : 462 \approx 1.101 : 1$ and $2\sqrt{3}/\pi \approx 3.46/3.14 \approx 1.10$ ✓.

» **Why this step?** Both areas scale with the same $r^2 = 147$, so the ratio survives as a pure constant $2\sqrt{3}/\pi$ - geometry's constants in action.

Hinglish: *Hexagon $294\sqrt{3}$, inscribed circle 147π. Ratio $2\sqrt{3} : \pi \approx 1.10$. Hexagon circle se ~10% bada - corners ki wajah se. Clean ratio isliye mila kyunki dono areas mein r^2 common tha.*

Answer: $294\sqrt{3} \approx 508.6$ cm^2; 462 cm^2; ratio $2\sqrt{3} : \pi \approx 1.10 : 1$

6.4 Chapter MCQs (80 questions, answers inline)

Each MCQ lists the options, the correct answer, a full explanation, the trap analysis for the tempting wrong options, and a Hinglish note. $\pi = 22/7$ unless stated otherwise.

MCQ 1 · Normal · Concept: Rectangle basics

1. A rectangle is 15 cm by 8 cm. Its perimeter and area are:

A. 46 cm, 120 cm^2
B. 23 cm, 120 cm^2
C. 46 cm, 100 cm^2
D. 30 cm, 120 cm^2

Correct Answer: (A)

$P = 2(15+8) = 46$ cm; $A = 15 \times 8 = 120$ cm^2.

Why (B) is wrong: Forgot to double the sum.

Why (C) is wrong: Area miscalculated.

Why (D) is wrong: Used l+b not 2(l+b).

Hinglish: *$2(15+8) = 46$ aur $15 \times 8 = 120$. Units: cm aur cm^2.*

MCQ 2 · Normal · Concept: Triangle area

2. A triangle has base 12 cm and height 8 cm. Its area is:

A. 96 cm^2
B. 48 cm^2
C. 40 cm^2
D. 24 cm^2

Correct Answer: (B)

$\frac{1}{2} \times 12 \times 8 = 48$ cm^2.

Why (A) is wrong: Skipped the ½.

Why (C) is wrong: Wrong pairing.

Why (D) is wrong: Halved twice.

Hinglish: *Aadha × 12 × 8 = 48. ½ bhoolna mana hai.*

MCQ 3 · Normal · Concept: Circumference

3. The circumference of a circle of radius 7 cm is:

A. 44 cm

B. 154 cm

C. 22 cm

D. 88 cm

Correct Answer: (A)

$2\pi r = 2\times(22/7)\times7 = 44$ cm.

Why (B) is wrong: That is the area.

Why (C) is wrong: πr not $2\pi r$.

Why (D) is wrong: Doubled twice.

Hinglish: *2×22 = 44 cm. Area would be 154 - confuse mat karo.*

MCQ 4 · Normal · Concept: Circle area

4. The area of a circle of diameter 14 cm is:

A. 154 cm^2

B. 616 cm^2

C. 44 cm^2

D. 88 cm^2

Correct Answer: (A)

$r = 7$: $\pi r^2 = 154$ cm^2.

Why (B) is wrong: Used d instead of r in πr^2.

Why (C) is wrong: That is a circumference-style number.

Why (D) is wrong: Wrong formula.

Hinglish: *Diameter 14 → radius 7. Area 154. Radius se diameter confusion sabse common galti.*

MCQ 5 · Normal · Concept: Reverse circumference

5. If the circumference of a circle is 176 cm, its radius is:

A. 28 cm

B. 14 cm

C. 56 cm

D. 88 cm

Correct Answer: (A)

$r = 176\times7/44 = 28$ cm.

Why (B) is wrong: Forgot the factor 2.

Why (C) is wrong: Multiplied instead of divided.

Why (D) is wrong: That is half the circumference.

Hinglish: *176 ÷ 2π = 28.*

MCQ 6 · Normal · Concept: Equilateral area

6. The area of an equilateral triangle of side 8 cm is:

A. $16\sqrt{3}$ cm^2
B. $32\sqrt{3}$ cm^2
C. $8\sqrt{3}$ cm^2
D. $64\sqrt{3}$ cm^2

Correct Answer: (A)

$(\sqrt{3}/4)(64) = 16\sqrt{3}$.

Why (B) is wrong: Used $a^2/2$.

Why (C) is wrong: Forgot to square the side.

Why (D) is wrong: Coefficient error.

Hinglish: *64/4 = 16. 16√3 cm².*

MCQ 7 · Normal · Concept: Heron 3-4-5

7. The area of a triangle with sides 3, 4, 5 cm is:

A. 6 cm^2
B. 12 cm^2
C. 10 cm^2
D. 7.5 cm^2

Correct Answer: (A)

Right triangle ($3^2+4^2 = 5^2$): $\frac{1}{2}\times3\times4 = 6$ cm^2.

Why (B) is wrong: Skipped ½.

Why (C) is wrong: Wrong legs.

Why (D) is wrong: Heron arithmetic slip.

Hinglish: *Right triangle spot karo - ½×3×4 = 6.*

MCQ 8 · Normal · Concept: Sector area quarter

8. The area of a sector of angle 90° in a circle of radius 14 cm is:

A. 154 cm^2
B. 77 cm^2
C. 38.5 cm^2
D. 616 cm^2

Correct Answer: (A)

$(1/4)\pi(196) = 154$ cm^2.

Why (B) is wrong: Used ½ instead of ¼.

Why (C) is wrong: One-eighth.

Why (D) is wrong: Full circle area.

Hinglish: *90° = quarter. 616/4 = 154.*

MCQ 9 · Normal · Concept: Arc length

9. The length of an arc subtending 60° at the centre of a circle of radius 21 cm is:

A. 22 cm
B. 44 cm
C. 11 cm
D. 66 cm

Correct Answer: (A)

$(1/6)(2\pi \cdot 21) = 132/6 = 22$ cm.

Why (B) is wrong: Used ⅓.

Why (C) is wrong: Used 1/12.

Why (D) is wrong: Full circumference.

Hinglish: *60° = 1/6. 132 ÷ 6 = 22 cm.*

MCQ 10 · Normal · Concept: Ring area

10. Two concentric circles have radii 7 cm and 14 cm. The ring area is:

A. 462 cm^2
B. 154 cm^2
C. 616 cm^2
D. 231 cm^2

Correct Answer: (A)

$\pi(196 - 49) = \pi \cdot 147 = 462$ cm^2.

Why (B) is wrong: Inner circle only.

Why (C) is wrong: Outer circle only.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$147\pi = 462$. Difference of squares.*

MCQ 11 · Normal · Concept: Square diagonal

11. The diagonal of a square of side 6 cm is:

A. $6\sqrt{2}$ cm
B. 12 cm
C. $6\sqrt{3}$ cm
D. 36 cm

Correct Answer: (A)

Diagonal = $s\sqrt{2} = 6\sqrt{2}$ cm.

Why (B) is wrong: Doubled the side.

Why (C) is wrong: $\sqrt{3}$ belongs to equilateral heights.

Why (D) is wrong: That is the area.

Hinglish: *Side × $\sqrt{2}$. 45-45-90 triangle.*

MCQ 12 · Normal · Concept: Trapezium

12. A trapezium has parallel sides 10 cm and 14 cm, height 6 cm. Its area is:

A. 72 cm^2
B. 84 cm^2
C. 60 cm^2
D. 48 cm^2

Correct Answer: (A)

$\frac{1}{2}(10+14)(6) = 72\ cm^2$.

Why (B) is wrong: Skipped the $\frac{1}{2}$.

Why (C) is wrong: Used difference 4.

Why (D) is wrong: Wrong average.

Hinglish: *Average 12 × height 6 = 72.*

MCQ 13 · Normal · Concept: Heron 13-14-15

13. The area of a triangle with sides 13, 14, 15 cm is:

A. 84 cm^2
B. 91 cm^2
C. 78 cm^2
D. 105 cm^2

Correct Answer: (A)

s = 21; 21·6·7·8 = 7056; $\sqrt{7056} = 84$.

Why (B) is wrong: s taken as 42.

Why (C) is wrong: Arithmetic slip.

Why (D) is wrong: Multiplied wrong pair.

Hinglish: *s = 21, product 7056, root 84. Classic 13-14-15.*

MCQ 14 · Normal · Concept: Wheel revolutions

14. A wheel of diameter 70 cm travels 2.2 km. The number of revolutions is:

A. 1000
B. 500
C. 2000
D. 1100

Correct Answer: (A)

C = 220 cm; 220000/220 = 1000.

Why (B) is wrong: Half the distance.

Why (C) is wrong: Radius/diameter mix.

Why (D) is wrong: Unit conversion error.

Hinglish: *2.2 km = 220000 cm. 220 se divide: 1000.*

MCQ 15 · Normal · Concept: Sector perimeter

15. The perimeter of a sector of radius 10 cm and angle 90° is ($\pi = 3.14$):

A. 35.7 cm

B. 15.7 cm

C. 25.7 cm

D. 31.4 cm

Correct Answer: (A)

Arc = (1/4)(62.8) = 15.7; P = 15.7 + 20 = 35.7 cm.

Why (B) is wrong: Added only one radius.

Why (C) is wrong: Wrong arc.

Why (D) is wrong: That is the full circumference.

Hinglish: *Arc 15.7 + dono radii 20 = 35.7 cm.*

MCQ 16 · Normal · Concept: Brahmagupta simple

16. A cyclic quadrilateral has sides 8, 15, 8, 15 cm. Its area is:

A. 120 cm^2

B. 115 cm^2

C. 46 cm^2

D. 240 cm^2

Correct Answer: (A)

s = 23; $(23-8)^2(23-15)^2 = 15^2 \cdot 8^2$; area = 120.

Why (B) is wrong: Arithmetic slip.

Why (C) is wrong: Used perimeter directly.

Why (D) is wrong: Doubled.

Hinglish: *a,a,b,b pattern: area = ab = 120.*

MCQ 17 · Normal · Concept: Rhombus area

17. A rhombus has diagonals 10 cm and 24 cm. Its area is:

A. 120 cm^2

B. 240 cm^2

C. 60 cm^2

D. 34 cm^2

Correct Answer: (A)

$\frac{1}{2} \times 10 \times 24 = 120\ cm^2$.

Why (B) is wrong: Skipped the $\frac{1}{2}$.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Added the diagonals.

Hinglish: $\frac{1}{2} \times d_1 \times d_2$.

MCQ 18 · Normal · Concept: Parallelogram area

18. A parallelogram has base 9 cm and height 7 cm. Its area is:

A. 63 cm^2
B. 31.5 cm^2
C. 126 cm^2
D. 16 cm^2

Correct Answer: (A)

base × height = 63 cm^2 (no ½ - that is for triangles).

Why (B) is wrong: Triangle formula used.

Why (C) is wrong: Doubled.

Why (D) is wrong: Added.

Hinglish: *Parallelogram mein ½ NAHI hota: 9×7 = 63.*

MCQ 19 · Normal · Concept: Ratio of areas

19. The radii of two circles are in the ratio 2 : 5. The ratio of their areas is:

A. 4 : 25
B. 2 : 5
C. 4 : 10
D. 8 : 125

Correct Answer: (A)

Areas carry the square of the ratio: $2^2 : 5^2 = 4 : 25$.

Why (B) is wrong: Linear ratio kept.

Why (C) is wrong: Half-squared.

Why (D) is wrong: Cubed.

Hinglish: *k^2 rule: 4:25.*

MCQ 20 · Normal · Concept: Reverse area

20. If the area of a circle is 1386 cm^2, its radius is:

A. 21 cm
B. 14 cm
C. 7 cm
D. 42 cm

Correct Answer: (A)

$r^2 = 1386 \times 7/22 = 441 \Rightarrow r = 21$ cm.

Why (B) is wrong: Half of the root.

Why (C) is wrong: Wrong π handling.

Why (D) is wrong: Doubled.

Hinglish: *1386 × 7/22 = 441. r = 21.*

MCQ 21 · Normal · Concept: Heron with unit difference

21. A triangle has sides 7, 15, 20 cm. Its area is:

A. 42 cm^2

B. $21\sqrt{6}$ cm^2

C. 84 cm^2

D. $21\sqrt{3}$ cm^2

Correct Answer: (A)

s = 21; differences 14, 6, 1; product $21\times14\times6\times1 = 1764 = 42^2$; area = 42 cm^2.

Why (B) is wrong: The product is a perfect square here - no surd remains.

Why (C) is wrong: Doubled.

Why (D) is wrong: Wrong simplification.

Hinglish: *s = 21, differences 14, 6, 1. Product 1764 - perfect square! Area 42 cm^2.*

MCQ 22 · Normal · Concept: Path inside

22. A square garden of side 20 m has a 1 m wide path along its inside boundary. The path's area is:

A. 79 m^2

B. 80 m^2

C. 76 m^2

D. 81 m^2

Correct Answer: (A)

$20^2 - 18^2 = 400 - 324 = 79$ m^2.

Why (B) is wrong: Perimeter × width approximation (only exact for one side).

Why (C) is wrong: Subtracted 2 instead of 4.

Why (D) is wrong: Added instead.

Hinglish: *Inner square 18. 400 - 324 = 79.*

MCQ 23 · Normal · Concept: Equilateral perimeter

23. An equilateral triangle has area $25\sqrt{3}$ cm^2. Its perimeter is:

A. 30 cm

B. 15 cm

C. 60 cm

D. $10\sqrt{3}$ cm

Correct Answer: (A)

$(\sqrt{3}/4)a^2 = 25\sqrt{3} \Rightarrow a^2 = 100 \Rightarrow a = 10$; P = 30 cm.

Why (B) is wrong: Forgot the 3.

Why (C) is wrong: Doubled twice.

Why (D) is wrong: That is the side in surd form.

Hinglish: *a = 10, perimeter 30.*

MCQ 24 · Normal · Concept: Quadrilateral diagonal

24. A quadrilateral is split by a diagonal into triangles of areas 60 cm^2 and 45 cm^2. Its area is:

A. 105 cm^2

B. 15 cm^2

C. 2700 cm^2

D. 52.5 cm^2

Correct Answer: (A)

Areas along the diagonal add: 60 + 45 = 105 cm^2.

Why (B) is wrong: Subtracted.

Why (C) is wrong: Multiplied.

Why (D) is wrong: Averaged.

Hinglish: *Diagonal split = areas JUDO.*

MCQ 25 · Normal · Concept: Chord-tangent ring

25. In two concentric circles, a chord of the outer circle of length 16 cm touches the inner circle. The ring area is ($\pi = 22/7$):

A. 1408/7 cm^2

B. 704/7 cm^2

C. 2816/7 cm^2

D. 64 cm^2

Correct Answer: (A)

Ring = $\pi(\text{half-chord})^2 = \pi \times 64 = 64\pi = 1408/7 \approx 201.14$ cm^2.

Why (B) is wrong: Quarter-chord used.

Why (C) is wrong: Full chord squared.

Why (D) is wrong: Forgot the π.

Hinglish: *Ring = $\pi \times 8^2 = 64\pi = 1408/7 \approx 201.1$ cm^2. Half-chord rule.*

MCQ 26 · Normal · Concept: Semicircle area

26. The area of a semicircle of radius 7 cm is:

A. 77 cm^2

B. 154 cm^2

C. 38.5 cm^2

D. 22 cm^2

Correct Answer: (A)

$\frac{1}{2}\pi(49) = 77$ cm^2.

Why (B) is wrong: Full circle.

Why (C) is wrong: Quarter.

Why (D) is wrong: Circumference-style.

Hinglish: *Half of 154 = 77.*

MCQ 27 · Normal · Concept: Same perimeter circle-square

27. A square and a circle have equal perimeters of 88 cm. The circle's area exceeds the square's by ($\pi = 22/7$):

A. 132 cm^2

B. 484 cm^2

C. 616 cm^2

D. 168 cm^2

Correct Answer: (A)

Square side 22: area 484. Circle r = 14: area 616. Difference = 132 cm^2.

Why (B) is wrong: Square area only.

Why (C) is wrong: Circle area only.

Why (D) is wrong: Arithmetic slip.

Hinglish: *616 - 484 = 132 cm^2. Circle jeeta.*

MCQ 28 · Normal · Concept: Isosceles Heron

28. An isosceles triangle has equal sides 10 cm and base 12 cm. Its area is:

A. 48 cm^2

B. 60 cm^2

C. 24 cm^2

D. 96 cm^2

Correct Answer: (A)

h = $\sqrt{(100-36)} = 8$; area = $\frac{1}{2} \times 12 \times 8 = 48$ cm^2.

Why (B) is wrong: Skipped $\frac{1}{2}$.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Doubled.

Hinglish: *6-8-10 hidden triple. 48 cm^2.*

MCQ 29 · Normal · Concept: Arc of 120°

29. An arc subtends 120° at the centre of a circle of radius 21 cm. Its length is:

A. 44 cm

B. 22 cm

C. 66 cm

D. 88 cm

Correct Answer: (A)

(1/3)(132) = 44 cm.

Why (B) is wrong: Used 1/6.

Why (C) is wrong: Used 1/2.

Why (D) is wrong: Full circle.

Hinglish: *One-third of circumference.*

MCQ 30 · Normal · Concept: Heron validity

30. Heron's formula gives a real area only if:

A. every side is less than the sum of the other two

B. the triangle is right-angled

C. two sides are equal

D. the perimeter is even

Correct Answer: (A)

The triangle inequality keeps s-a, s-b, s-c positive; otherwise the product under the root is negative.

Why (B) is wrong: Heron works for ALL triangles.

Why (C) is wrong: No symmetry needed.

Why (D) is wrong: Parity irrelevant.

Hinglish: *Triangle inequality - ek side doosre dono ke sum se choti honi chahiye.*

MCQ 31 · Normal · Concept: Brahmagupta validity

31. Brahmagupta's formula $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ applies to:

A. cyclic quadrilaterals only

B. all quadrilaterals

C. rectangles only

D. trapeziums only

Correct Answer: (A)

The formula requires all four vertices on one circle.

Why (B) is wrong: Fails for general quadrilaterals.

Why (C) is wrong: Too narrow a claim.

Why (D) is wrong: Wrong restriction.

Hinglish: *Cyclic quadrilateral - charo vertices ek circle par. Tabhi Brahmagupta.*

MCQ 32 · Normal · Concept: Track perimeter

32. A track has two 50 m straights and two semicircular ends of diameter 28 m. Its perimeter is ($\pi = 22/7$):

A. 188 m

B. 100 m

C. 138 m

D. 176 m

Correct Answer: (A)

$100 + \pi \cdot 28 = 100 + 88 = 188$ m.

Why (B) is wrong: Straights only.

Why (C) is wrong: One semicircle only.

Why (D) is wrong: Straights + 2 full circles.

Hinglish: *Do semicircles = ek circle: 88. Plus 100 = 188 m.*

MCQ 33 · Normal · Concept: Heron 5-12-13 scaled

33. A triangle with sides 10, 24, 26 cm has area:

A. 120 cm^2
B. 60 cm^2
C. 240 cm^2
D. 130 cm^2

Correct Answer: (A)

It is a scaled 5-12-13: right angle; $\frac{1}{2} \times 10 \times 24 = 120$ cm^2.

Why (B) is wrong: Used the unscaled triple's area.

Why (C) is wrong: Doubled.

Why (D) is wrong: No Pythagoras check.

Hinglish: *2×(5-12-13). ½×10×24 = 120.*

MCQ 34 · Normal · Concept: Minute hand sector

34. The area swept by a 21 cm long minute hand in 10 minutes is ($\pi = 22/7$):

A. 231 cm^2
B. 1386 cm^2
C. 77 cm^2
D. 115.5 cm^2

Correct Answer: (A)

10 min → 60° → (1/6)π(441) = 231 cm^2.

Why (B) is wrong: Full circle.

Why (C) is wrong: Wrong fraction.

Why (D) is wrong: Used 5/6 somehow.

Hinglish: *10 min = 60° = one-sixth. 1386/6 = 231.*

MCQ 35 · Normal · Concept: Cost fencing

35. A circular field of radius 35 m is fenced at ₹10 per metre. The cost is ($\pi = 22/7$):

A. ₹2200
B. ₹38500
C. ₹11000
D. ₹4400

Correct Answer: (A)

C = 220 m; cost = 2200.

Why (B) is wrong: Area used instead.

Why (C) is wrong: Wrong circumference.

Why (D) is wrong: Doubled.

Hinglish: *220 × 10 = ₹2200. Perimeter wala rate.*

MCQ 36 · Normal · Concept: Two-circle area sum

36. Circles of radii 8 cm and 15 cm. The radius of the circle whose area equals their area sum is:

A. 17 cm
B. 23 cm
C. 12 cm
D. 120 cm

Correct Answer: (A)

$64 + 225 = 289 = 17^2$.

Why (B) is wrong: Radii added.

Why (C) is wrong: Averaged.

Why (D) is wrong: Multiplied.

Hinglish: *8-15-17 triple! Areas add → hypotenuse radius.*

MCQ 37 · Normal · Concept: Hexagon side-radius

37. In a regular hexagon inscribed in a circle of radius 14 cm, each side of the hexagon is:

A. 14 cm
B. 7 cm
C. $7\sqrt{3}$ cm
D. 21 cm

Correct Answer: (A)

An inscribed regular hexagon's side equals the radius.

Why (B) is wrong: Half.

Why (C) is wrong: That is the apothem.

Why (D) is wrong: 3/2 of radius.

Hinglish: *Hexagon side = radius. 14 cm.*

MCQ 38 · Normal · Concept: Unit conversion

38. 1 hectare equals:

A. 10000 m^2
B. 1000 m^2
C. 100 m^2
D. 100000 m^2

Correct Answer: (A)

1 hectare = 100 m × 100 m = 10000 m^2.

Why (B) is wrong: One zero short.

Why (C) is wrong: Two short.

Why (D) is wrong: One too many.

Hinglish: *10^4 m^2. Land unit.*

MCQ 39 · Normal · Concept: Sector half-radius gem

39. A sector has arc length 12 cm and radius 9 cm. Its area is:

A. 54 cm^2
B. 108 cm^2
C. 27 cm^2
D. 21π cm^2

Correct Answer: (A)

$A = \frac{1}{2} l \cdot r = \frac{1}{2} \times 12 \times 9 = 54$ cm^2.

Why (B) is wrong: Skipped the ½.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Introduced π without cause.

Hinglish: *½ × arc × radius. Angle ki zaroorat nahi.*

MCQ 40 · Normal · Concept: Square inscribed in circle

40. A square is inscribed in a circle of radius $7\sqrt{2}$ cm. The square's side is:

A. 14 cm
B. $7\sqrt{2}$ cm
C. 7 cm
D. 28 cm

Correct Answer: (A)

Diagonal = diameter = $14\sqrt{2}$; side = diagonal/$\sqrt{2}$ = 14 cm.

Why (B) is wrong: Took the radius as the side.

Why (C) is wrong: Half.

Why (D) is wrong: Diameter as side.

Hinglish: *Diagonal = diameter. Side = d/√2 = 14.*

MCQ 41 · Normal · Concept: Heron product pairing

41. For sides 11, 13, 20 cm, s equals:

A. 22 cm
B. 44 cm
C. 24 cm
D. 11 cm

Correct Answer: (A)

(11+13+20)/2 = 22 cm.

Why (B) is wrong: Forgot to halve.

Why (C) is wrong: Arithmetic slip.

Why (D) is wrong: One side only.

Hinglish: *Sum 44, half 22.*

MCQ 42 · Normal · Concept: Area to radius chain

42. A circle's area is 2464 cm^2. Its circumference is ($\pi = 22/7$):

A. 176 cm

B. 88 cm

C. 352 cm

D. 308 cm

Correct Answer: (A)

$r^2 = 2464 \times 7/22 = 784 \Rightarrow r = 28$; C = 176 cm.

Why (B) is wrong: Half the radius.

Why (C) is wrong: Doubled.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$r^2 = 784$, $r = 28$. C = 176 cm.*

MCQ 43 · Normal · Concept: Compass clock angle

43. A clock's minute hand sweeps what angle in 20 minutes?

A. 120°

B. 60°

C. 90°

D. 180°

Correct Answer: (A)

6° per minute × 20 = 120°.

Why (B) is wrong: 5°/min assumed.

Why (C) is wrong: 15°/min.

Why (D) is wrong: That is 30 minutes.

Hinglish: *Ek minute = 6°. 20×6 = 120°.*

MCQ 44 · Normal · Concept: Outside path circular

44. A circular pool of radius 10 m has a 2 m wide walkway around it. The walkway area is ($\pi = 3.14$):

A. 138.16 m^2

B. 62.8 m^2

C. 452.16 m^2

D. 75.36 m^2

Correct Answer: (A)

$\pi(12^2 - 10^2) = \pi \cdot 44 = 138.16$ m^2.

Why (B) is wrong: Only a strip of one side.

Why (C) is wrong: Outer circle only.

Why (D) is wrong: Used w^2 only.

Hinglish: *$\pi w(2r+w) = 3.14 \times 2 \times 22 = 138.16$.*

MCQ 45 · Normal · Concept: Perimeter of quadrant

45. The perimeter of a quadrant (quarter circle) of radius 14 cm is ($\pi = 22/7$):

A. 50 cm
B. 22 cm
C. 44 cm
D. 36 cm

Correct Answer: (A)

Arc = 22 cm; P = 22 + 14 + 14 = 50 cm.

Why (B) is wrong: Arc only.

Why (C) is wrong: Two arcs.

Why (D) is wrong: One radius only.

Hinglish: *Arc 22 + dono radii 28 = 50 cm.*

MCQ 46 · Normal · Concept: Heron 9-40-41

46. The area of a triangle with sides 9, 40, 41 cm is:

A. 180 cm^2
B. 90 cm^2
C. 360 cm^2
D. 820 cm^2

Correct Answer: (A)

$81 + 1600 = 1681 = 41^2$: right triangle; $\frac{1}{2} \times 9 \times 40 = 180$ cm^2.

Why (B) is wrong: Skipped $\frac{1}{2}$.

Why (C) is wrong: Doubled.

Why (D) is wrong: No triple check.

Hinglish: *9-40-41 triple! $\frac{1}{2} \times 9 \times 40 = 180$.*

MCQ 47 · Normal · Concept: Scaling area

47. If a circle's radius is tripled, its area becomes:

A. 9 times
B. 3 times
C. 6 times
D. 27 times

Correct Answer: (A)

Area scales as k^2: $3^2 = 9$.

Why (B) is wrong: Linear scaling.

Why (C) is wrong: No rule.

Why (D) is wrong: Cubed.

Hinglish: *k^2 rule: 9 guna.*

MCQ 48 · Normal · Concept: Equilateral height

48. The height of an equilateral triangle of side 10 cm is:

A. $5\sqrt{3}$ cm
B. $10\sqrt{3}$ cm
C. $5\sqrt{2}$ cm
D. 15 cm

Correct Answer: (A)

$h = (\sqrt{3}/2)a = 5\sqrt{3}$ cm.

Why (B) is wrong: Full side factor.

Why (C) is wrong: $\sqrt{2}$ belongs to squares.

Why (D) is wrong: No root needed here.

Hinglish: *Height = (√3/2)a. 5√3 ≈ 8.66.*

MCQ 49 · Normal · Concept: Trapezium from data

49. A trapezium has parallel sides 25 cm and 15 cm and area 120 cm^2. Its height is:

A. 6 cm
B. 3 cm
C. 12 cm
D. 8 cm

Correct Answer: (A)

$\frac{1}{2}(40)h = 120 \Rightarrow h = 6$ cm.

Why (B) is wrong: Forgot the $\frac{1}{2}$ compensation.

Why (C) is wrong: Doubled.

Why (D) is wrong: Arithmetic slip.

Hinglish: *120 = 20h → h = 6.*

MCQ 50 · Normal · Concept: Ring from two areas

50. Outer circle area 1386 cm^2, inner circle area 616 cm^2. Ring area:

A. 770 cm^2
B. 2002 cm^2
C. 616 cm^2
D. 856.8 cm^2

Correct Answer: (A)

$1386 - 616 = 770$ cm^2.

Why (B) is wrong: Added.

Why (C) is wrong: Inner only.

Why (D) is wrong: Averaged.

Hinglish: *Seedha minus: 770 cm^2.*

MCQ 51 · Normal · Concept: Sector reverse radius

51. A sector of angle 90° has area 154 cm^2. The radius is ($\pi = 22/7$):

A. 14 cm
B. 7 cm
C. 28 cm
D. 21 cm

Correct Answer: (A)

$(1/4)\pi r^2 = 154 \Rightarrow \pi r^2 = 616 \Rightarrow r^2 = 196 \Rightarrow r = 14$ cm.

Why (B) is wrong: Forgot to undo the quarter.

Why (C) is wrong: Doubled.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$\pi r^2 = 616$, $r^2 = 196$, $r = 14$.*

MCQ 52 · Normal · Concept: Rectangle reverse

52. A rectangle has area 96 cm^2 and length 12 cm. Its perimeter is:

A. 40 cm
B. 24 cm
C. 48 cm
D. 32 cm

Correct Answer: (A)

$b = 96/12 = 8$; $P = 2(12+8) = 40$ cm.

Why (B) is wrong: That is 2×12.

Why (C) is wrong: Forgot to double.

Why (D) is wrong: Wrong width.

Hinglish: *Width 8, perimeter 40.*

MCQ 53 · Normal · Concept: Perimeter of semicircle

53. The perimeter of a semicircle of radius 7 cm is ($\pi = 22/7$):

A. 36 cm
B. 22 cm
C. 44 cm
D. 29 cm

Correct Answer: (A)

Arc 22 + diameter 14 = 36 cm.

Why (B) is wrong: Arc only.

Why (C) is wrong: Full circle.

Why (D) is wrong: Arc + one radius.

Hinglish: *Arc + diameter (not two radii here - the flat edge is the diameter).*

MCQ 54 · Normal · Concept: Grazing quarter

54. A goat tied at a corner of a field with a 14 m rope grazes an area of ($\pi = 22/7$):

A. 154 m^2

B. 616 m^2

C. 77 m^2

D. 308 m^2

Correct Answer: (A)

Quarter circle: $(1/4)\pi(196) = 154$ m^2.

Why (B) is wrong: Full circle.

Why (C) is wrong: Eighth.

Why (D) is wrong: Half.

Hinglish: *Corner = quarter. 616/4 = 154.*

MCQ 55 · Normal · Concept: Heron isosceles quick

55. An isosceles triangle has equal sides 13 cm and base 10 cm. Its area is:

A. 60 cm^2

B. 65 cm^2

C. 30 cm^2

D. 120 cm^2

Correct Answer: (A)

$h = \sqrt{(169 - 25)} = 12$; area = $\frac{1}{2} \times 10 \times 12 = 60$ cm^2.

Why (B) is wrong: Skipped ½.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Doubled.

Hinglish: *5-12-13 hidden! 60 cm^2.*

MCQ 56 · Normal · Concept: Diagonal of rectangle

56. A rectangle is 8 cm by 15 cm. Its diagonal is:

A. 17 cm

B. 23 cm

C. 16 cm

D. 12 cm

Correct Answer: (A)

$\sqrt{(64 + 225)} = \sqrt{289} = 17$ cm.

Why (B) is wrong: Added sides.

Why (C) is wrong: Wrong squares.

Why (D) is wrong: Averaged.

Hinglish: *8-15-17 triple.*

MCQ 57 · Normal · Concept: Sector angle from arc

57. An arc of length 11 cm subtends 45° at the centre. The radius is (π = 22/7):

A. 14 cm
B. 7 cm
C. 28 cm
D. 21 cm

Correct Answer: (A)

(45/360)·2πr = 11 ⇒ (1/8)(44r/7) = 11 ⇒ r = 14 cm.

Why (B) is wrong: Fraction mishandled.

Why (C) is wrong: Doubled.

Why (D) is wrong: Arithmetic slip.

Hinglish: *1/8 of circumference = 11 → C = 88 → r = 14.*

MCQ 58 · Normal · Concept: Heron with 4-digit product

58. s = 30 and the three differences are 12, 10, 8 for a triangle. Its area is:

A. $\sqrt{28800} = 120\sqrt{2} \approx 169.7$ cm^2
B. 14400 cm^2
C. 120 cm^2
D. 240 cm^2

Correct Answer: (A)

30×12×10×8 = 28800 = 14400×2; area = $120\sqrt{2} \approx 169.7$ cm^2.

Why (B) is wrong: Skipped the root.

Why (C) is wrong: Lost the √2.

Why (D) is wrong: Doubled.

Hinglish: *Product 28800. $120^2 = 14400$ bahar, √2 andar. $120\sqrt{2}$.*

MCQ 59 · Normal · Concept: Circular garden cost

59. Ploughing a circular garden of radius 21 m costs ₹2 per m^2. The total cost is (π = 22/7):

A. ₹2772
B. ₹264
C. ₹1386
D. ₹5544

Correct Answer: (A)

Area = 1386 m^2; cost = 2772.

Why (B) is wrong: Circumference used.

Why (C) is wrong: Forgot the rate.

Why (D) is wrong: Doubled.

Hinglish: *1386 × 2 = ₹2772.*

MCQ 60 · Normal · Concept: Hexagon area

60. A regular hexagon of side 6 cm has area ($\sqrt{3} \approx 1.73$):

A. $(54\sqrt{3})$ cm$^2 \approx 93.4$ cm^2

B. $36\sqrt{3}$ cm^2

C. $108\sqrt{3}$ cm^2

D. 216 cm^2

Correct Answer: (A)

$(3\sqrt{3}/2)(36) = 54\sqrt{3} \approx 93.4$ cm^2.

Why (B) is wrong: One triangle only.

Why (C) is wrong: Doubled.

Why (D) is wrong: Forgot the root.

Hinglish: *6 equilateral triangles: $6\times(\sqrt{3}/4)(36) = 54\sqrt{3}$.*

MCQ 61 · HOTS · Concept: Two circles circumference ratio

61. Two circles have areas 154 cm^2 and 616 cm^2. The ratio of their radii is:

A. 1 : 2

B. 1 : 4

C. 1 : 16

D. 2 : 1

Correct Answer: (A)

Area ratio 1 : 4 $\Rightarrow$ radius ratio $\sqrt{(1/4)} = 1 : 2$.

Why (B) is wrong: Kept the area ratio.

Why (C) is wrong: Squared it.

Why (D) is wrong: Inverted.

Hinglish: *Area 1:4 → radius 1:2. Square root lena hai.*

MCQ 62 · HOTS · Concept: Annulus width algebra

62. If R - r = 5 and R + r = 25, the ring area (before π) equals:

A. 125

B. 625

C. 100

D. 30

Correct Answer: (A)

$R^2 - r^2 = (R+r)(R-r) = 125$.

Why (B) is wrong: Squaring instead of multiplying.

Why (C) is wrong: Adding the givens.

Why (D) is wrong: Difference of givens.

Hinglish: *$(R+r)(R-r) = 125$. π se multiply baad mein.*

MCQ 63 · HOTS · Concept: Composite area

63. A figure is a 10 cm square with a semicircle of diameter 10 cm attached outside on one side. Total area ($\pi = 3.14$):

A. 139.25 cm^2
B. 100 cm^2
C. 178.5 cm^2
D. 89.25 cm^2

Correct Answer: (A)

$100 + \frac{1}{2}\pi(5^2) = 100 + 39.25 = 139.25\ cm^2$.

Why (B) is wrong: Square only.

Why (C) is wrong: Full circle added.

Why (D) is wrong: Subtracted the semicircle.

Hinglish: *100 + 39.25. Semicircle radius 5.*

MCQ 64 · HOTS · Concept: Square-circle-same-area

64. A square and a circle have EQUAL areas of 154 cm^2 ($\pi = 22/7$). Which is true about their perimeters?

A. The square's perimeter is larger
B. The circle's perimeter is larger
C. They are equal
D. Cannot be determined

Correct Answer: (A)

Circle r = 7, C = 44. Square side $\sqrt{154} \approx 12.41$, $P \approx 49.64 > 44$.

Why (B) is wrong: Backwards - circle wins on perimeter for equal area.

Why (C) is wrong: Never equal (isoperimetric).

Why (D) is wrong: It is determined.

Hinglish: *Equal area par circle ka perimeter HAMESHA chhota. Isliye circle 'best' shape hai.*

MCQ 65 · HOTS · Concept: Sector angle from area

65. A sector of radius 12 cm has area $24\pi\ cm^2$. Its angle is:

A. 60°
B. 30°
C. 90°
D. 120°

Correct Answer: (A)

$(\theta/360)\pi(144) = 24\pi \Rightarrow \theta/360 = 1/6 \Rightarrow \theta = 60°$.

Why (B) is wrong: Half of the fraction.

Why (C) is wrong: Quarter.

Why (D) is wrong: Third.

Hinglish: *$24\pi/144\pi = 1/6$. $\theta = 60°$.*

MCQ 66 · HOTS · Concept: Track area

66. A track: rectangle 100 m × 60 m with semicircles of diameter 60 m on both short ends. Total area (π = 22/7):

A. 8828.57 m^2
B. 6000 m^2
C. 2828.57 m^2
D. 11657 m^2

Correct Answer: (A)

$6000 + \pi(30^2) = 6000 + 19800/7 \approx 6000 + 2828.57 = 8828.57\ m^2$.

Why (B) is wrong: Rectangle only.

Why (C) is wrong: One semicircle only... actually one full circle - the value matches the circle; but the question asks the total.

Why (D) is wrong: Doubled the circle.

Hinglish: *Rectangle 6000 + full circle 2828.57.*

MCQ 67 · HOTS · Concept: AP sides right triangle

67. A triangle's sides are in AP with perimeter 24 and shortest side 6. Its area is:

A. 24 cm^2
B. 48 cm^2
C. 12 cm^2
D. $30\sqrt{3}$ cm^2

Correct Answer: (A)

Sides 6, 8, 10 (AP, sum 24). Right triangle: $\frac{1}{2} \times 6 \times 8 = 24\ cm^2$.

Why (B) is wrong: Skipped ½.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Equilateral formula used.

Hinglish: *6-8-10 = 2×(3-4-5). Area 24.*

MCQ 68 · HOTS · Concept: Percent increase area

68. A circle's radius increases by 10%. Its area increases by about:

A. 21%
B. 10%
C. 100%
D. 11%

Correct Answer: (A)

$(1.1)^2 = 1.21 \rightarrow 21\%$ increase.

Why (B) is wrong: Linear thinking.

Why (C) is wrong: Doubled.

Why (D) is wrong: Added 1%.

Hinglish: *$1.1^2 = 1.21$. Area 21% badha.*

MCQ 69 · HOTS · Concept: Lune cancellation

69. Semicircles are drawn on the legs of a right triangle (legs 6, 8) outward and on the hypotenuse inward. The total area of the two lunes equals:

A. the triangle's area (24 cm^2)
B. 10π cm^2
C. 48π cm^2
D. 0

Correct Answer: (A)

Hippocrates: $18\pi/2 + 32\pi/2$ cancels against the $50\pi/2$ semicircle, leaving $\frac{1}{2}\times6\times8 = 24$ cm^2.

Why (B) is wrong: π terms do not survive.

Why (C) is wrong: Overshoot.

Why (D) is wrong: Not zero - exactly the triangle.

Hinglish: *Lunes = triangle area = 24 cm^2. Curved parts cancel.*

MCQ 70 · HOTS · Concept: Three circles tangent chain

70. Three circles of radii 3, 4, 5 cm touch each other externally. The perimeter of the triangle joining their centres is:

A. 24 cm
B. 12 cm
C. 36 cm
D. 15 cm

Correct Answer: (A)

Each side is a sum of two radii: 3+4, 4+5, 5+3 = 7+9+8 = 24 cm.

Why (B) is wrong: Diameters used.

Why (C) is wrong: Doubled.

Why (D) is wrong: Radii added once.

Hinglish: *Centre-triangle ke sides = radius pairs ka sum: 24 cm.*

MCQ 71 · Normal · Concept: Circumference vs perimeter of square

71. A square has side 11 cm; a circle has radius 7 cm. Which perimeter/circumference is larger?

A. Square (44 cm vs 44 cm - equal)
B. Circle
C. Square
D. Cannot compare

Correct Answer: (A)

Square P = 44; circle C = $2\pi\cdot7 = 44$. They are EQUAL - option A states this.

Why (B) is wrong: Recompute: both are 44.

Why (C) is wrong: Same as B.

Why (D) is wrong: They compare exactly.

Hinglish: *44 = 44 - barabar! 22/7 ka magic.*

MCQ 72 · Normal · Concept: Heron semi-perimeter role

72. In Heron's formula, s stands for:

A. semi-perimeter = (a+b+c)/2

B. the longest side

C. the perimeter

D. the area

Correct Answer: (A)

Definition: s is half the perimeter.

Why (B) is wrong: No.

Why (C) is wrong: Full perimeter is 2s.

Why (D) is wrong: That is A.

Hinglish: *s = aadha perimeter. Bhoolna mana hai.*

MCQ 73 · Normal · Concept: Area of quadrant vs semicircle

73. A quadrant of radius r and a semicircle of radius R have equal areas. Then R : r equals:

A. 1 : √2

B. 1 : 2

C. √2 : 1

D. 2 : 1

Correct Answer: (A)

$(\pi/4)r^2 = (\pi/2)R^2 \Rightarrow R^2 = r^2/2 \Rightarrow R = r/\sqrt{2}$; R : r = 1 : √2.

Why (B) is wrong: Linear ratio.

Why (C) is wrong: Inverted.

Why (D) is wrong: Doubled.

Hinglish: *Quarter = half → $R^2 = r^2/2$. R:r = 1:√2.*

MCQ 74 · Normal · Concept: Two wheels distance

74. Wheels of radii 35 cm and 14 cm each make 100 revolutions. The ratio of distances travelled is:

A. 5 : 2

B. 25 : 4

C. 2 : 5

D. 1 : 1

Correct Answer: (A)

Distance ∝ circumference ∝ radius: 35 : 14 = 5 : 2.

Why (B) is wrong: Squared the ratio.

Why (C) is wrong: Inverted.

Why (D) is wrong: Equal assumed.

Hinglish: *Same revolutions → distance ratio = radius ratio 5:2.*

MCQ 75 · Normal · Concept: Triangle area reverse

75. A triangle has area 84 cm^2 and base 21 cm. Its height is:

A. 8 cm

B. 4 cm

C. 16 cm

D. 12 cm

Correct Answer: (A)

$\frac{1}{2} \times 21 \times h = 84 \Rightarrow h = 8$ cm.

Why (B) is wrong: Forgot the $\frac{1}{2}$ compensation.

Why (C) is wrong: Doubled.

Why (D) is wrong: Arithmetic slip.

Hinglish: *168/21 = 8.*

MCQ 76 · Normal · Concept: Sector perimeter reverse

76. A sector has perimeter 33 cm and radius 10 cm ($\pi = 22/7$). Its arc length is:

A. 13 cm

B. 23 cm

C. 33 cm

D. 10π cm

Correct Answer: (A)

Arc = 33 - 20 = 13 cm.

Why (B) is wrong: Subtracted one radius.

Why (C) is wrong: No subtraction.

Why (D) is wrong: π introduced.

Hinglish: *33 - 2(10) = 13 cm.*

MCQ 77 · Normal · Concept: Area units

77. A field measures 250 m by 160 m. Its area in hectares is:

A. 4 hectares

B. 40 hectares

C. 0.4 hectares

D. 400 hectares

Correct Answer: (A)

Area = 40000 m^2 = 4 hectares (1 ha = 10^4 m^2).

Why (B) is wrong: One zero off.

Why (C) is wrong: Two off.

Why (D) is wrong: Three off.

Hinglish: *40000 ÷ 10000 = 4.*

MCQ 78 · Normal · Concept: Circumference increase

78. If a circle's diameter doubles, its circumference:

A. doubles
B. quadruples
C. stays same
D. triples

Correct Answer: (A)

$C = \pi d$ is linear in d.

Why (B) is wrong: That is the AREA.

Why (C) is wrong: No.

Why (D) is wrong: No.

Hinglish: *Circumference linear hai - double.*

MCQ 79 · HOTS · Concept: Inscribed equilateral circle

79. The inradius of an equilateral triangle of side $12\sqrt{3}$ cm is:

A. 6 cm
B. $6\sqrt{3}$ cm
C. 12 cm
D. $3\sqrt{3}$ cm

Correct Answer: (A)

$r = a/(2\sqrt{3}) = 12\sqrt{3}/(2\sqrt{3}) = 6$ cm.

Why (B) is wrong: Forgot to divide by $\sqrt{3}$.

Why (C) is wrong: Used height directly.

Why (D) is wrong: Factor error.

Hinglish: *$12\sqrt{3} \div 2\sqrt{3} = 6$. $\sqrt{3}$ cancel!*

MCQ 80 · HOTS · Concept: Ring percentage

80. A circular ring's outer radius is 20% more than its inner radius r. The ring area exceeds the inner circle's area by:

A. 44%
B. 20%
C. 40%
D. 21%

Correct Answer: (A)

$R = 1.2r$: ring $= \pi(1.44r^2 - r^2) = 0.44\pi r^2 = 44\%$ of the inner area.

Why (B) is wrong: Linear thinking.

Why (C) is wrong: Forgotten squaring of 1.2.

Why (D) is wrong: That is $(1.1)^2$.

Hinglish: *$1.2^2 = 1.44$. Extra 0.44 → 44%.*

6.5 Chapter Formula Sheet

Every area and perimeter formula of the chapter with its conditions and its signature mistake.

Formula	Meaning / when to use	Conditions	Tiny example	Common mistake
P = 2(l+b), A = l×b	Rectangle boundary and surface	Opposite sides equal	12.5×8: P 41, A 100	Perimeter = l+b only
P = 4s, A = s^2, diag = s√2	Square trio	All sides equal	s = 16: 64, 256, 16√2	Diagonal = s√3
A = ½ × base × height	Any triangle with a known height	Height perpendicular to base	½×14×9 = 63	Dropping the ½
A = base × height	Parallelogram	Height between the parallel sides	10×6.5 = 65	Using the slant side as height
A = ½ d_1d_2	Rhombus from diagonals	Diagonals perpendicular	½×12×16 = 96	Forgetting the ½
A = ½(a+b)h	Trapezium	a, b parallel	½(12+8)5 = 50	Using slant sides in the sum
Heron: A = √(s(s-a)(s-b)(s-c))	Area from three sides	s = (a+b+c)/2; triangle inequality	5,12,13 → 30	s taken as full perimeter
Equilateral: A = (√3/4)a^2, h = (√3/2)a	Equal sides	All sides a	a = 12 → 36√3	Wrong √3 placement
C = 2πr = πd, A = πr^2	Circle basics	π = 22/7 or 3.14 as stated	r = 14: 88, 616	πd^2 for area
Arc l = (θ/360)·2πr	Curved boundary piece	θ in degrees	90°, r 7 → 11	Using πr instead of 2πr
Sector A = (θ/360)πr^2 = ½lr	Pizza slice area	θ in degrees or use arc	120°, r 21 → 462	Mixing arc into r^2
Sector P = l + 2r	Boundary of a sector	One arc + two radii	44 + 42 = 86	Adding only one radius
Ring A = π(R^2-r^2) = π(R+r)(R-r)	Annulus	Concentric circles	π(100-36) with 10, 6	Width treated as R
Brahmagupta: A = √((s-a)(s-b)(s-c)(s-d))	Cyclic quadrilateral from sides	Must be cyclic; s = half perimeter	13,10,13,10 → 130	Applying to any quadrilateral
Inside path: L·B - (L-2w)(B-2w)	Border strip inside a rectangle	Path inside along all sides	30×20, w 2 → 184	Subtracting w only once

6.6 Last-Minute Cheat Sheet

> **30-second revision card - Chapter 6**
> TRIANGLE: ½bh with height; Heron √(s(s-a)(s-b)(s-c)) with s = P/2. Right triangle? ½×leg×leg. EQUILATERAL: $(\sqrt{3}/4)a^2$; h = $(\sqrt{3}/2)a$; r = $a/(2\sqrt{3})$. CIRCLE: C = $2\pi r$, A = πr^2; reverse: r = $C/2\pi$, r = $\sqrt{(A/\pi)}$. SECTOR: everything × θ/360; A = ½lr; P = l + 2r. RING: $\pi(R^2-r^2) = \pi(R+r)(R-r)$; chord trick: ring = π(half-chord)^2. BRAHMAGUPTA (cyclic only): √((s-a)(s-b)(s-c)(s-d)). QUADRILATERAL: split by diagonal, add two areas. SCALING: radius ×k ⇒ C ×k, A ×k^2. Same perimeter: circle beats square beats triangle.

- **If you see three sides only** → Heron; check Pythagoras first for a shortcut.
- **If you see 'cyclic quadrilateral'** → Brahmagupta; otherwise split by a diagonal.
- **If you see an angle at the centre** → multiply every circle quantity by θ/360.
- **If you see sector perimeter** → arc + 2r, never arc + r.
- **If you see a path inside** → subtract 2w from each dimension.
- **If you see a chord touching the inner circle** → ring area = π(half-chord)^2.
- **If you see ratios of radii** → circumferences in ratio k, areas in ratio k^2.
- **If you see rates** → perimeter × (rate/metre), area × (rate/m^2).

6.7 Mistake Buster - DON'T MAKE THESE MISTAKES

1. **Diameter used as radius.** πr^2 with r = diameter silently quadruples the area. Fix: halve the diameter FIRST, on paper.
2. **½ dropped in triangle area.** ½ × b × h is not b × h. Fix: write the ½ before the numbers.
3. **Full perimeter instead of semi-perimeter in Heron.** Using s = a+b+c doubles s and wrecks the root. Fix: divide by 2 as its own line.
4. **Arc + one radius for sector perimeter.** A sector has TWO straight edges. Fix: sketch the slice and trace its boundary.
5. **Width added to diameter.** A path's width widens the RADIUS by w, not the diameter by w. Fix: R = r + w explicitly.
6. **Unsimplified surd areas.** √1215 is not an answer; 9√15 is. Fix: extract the largest square factor.
7. **Brahmagupta on non-cyclic quadrilaterals.** The formula demands a circumcircle. Fix: read for 'cyclic' before using it.
8. **Linear scaling of areas.** Double radius = quadruple area. Fix: k^2 chant.
9. **Rounding π too early.** 22/7 ≈ 3.142857; rounding to 3.14 before multiplying shifts every answer. Fix: keep the fraction to the last line.
10. **Mixed units.** Fence in m, area in m^2, land in hectares. Fix: attach units to every intermediate quantity.

6.8 Exam Strategy for Chapter 6

Order of attack. Direct formula questions first (circle basics, rectangle trio) - they are 1-2 marks each and error-proof with discipline. Heron questions next: they are procedural and reward steady arithmetic. Keep sector-perimeter, ring-reverse and composite/shaded questions for later slots; they reward calm reading more

than speed.

Where marks are lost. Five leak-points: radius/diameter confusion (fatal, 3-mark question), missing ½ (triangle and rhombus), sector perimeter missing a radius (2 marks gone), unsimplified surds (final answer mark), and unit slips in cost questions (rate/metre vs rate/m^2). Every one is prevented by the same habit: write the formula BEFORE substituting numbers.

Formulas most frequently tested. Heron's formula (with ratio-perimeter setups), circle reverse chains (C ↔ r ↔ A), sector area with 90°/120°/60° angles, ring areas, and the quadrilateral-by-diagonal method. Brahmagupta appears as a 2-3 mark formula application - know its cyclic condition.

Concepts that get mixed. Arc length with sector area (one is 1-D, one is 2-D), sector perimeter with circle circumference, ring width with outer radius, and area ratio with radius ratio. Keep a two-column scrap layout: lengths on the left, areas on the right.

Safe shortcuts. ½lr for sector area; (R+r)(R-r) factorisation for rings; the chord-touch trick; Pythagoras spotting before Heron. **Unsafe shortcuts.** Assuming a quadrilateral is cyclic; using 3.14 for 22/7 problems with multiples of 7; estimating √ without extracting squares; applying trapezium formula when the quadrilateral is not a trapezium.

How to check an answer. Sanity-scale every area (is it in the right magnitude?); recompute one Heron difference; verify sector fractions sum to the whole circle; and for cost problems, re-read the rate's unit before the final multiplication. Ten seconds of checking per question routinely saves a full grade band.

6.9 Chapter Completion Checklist

Tick each box only when you can honestly say DONE:

- [ ] Every concept covered: basic shapes, π, circumference, circle area, arcs, sectors, segments, Heron, equilateral/isosceles, diagonal split, Brahmagupta, rings, paths, composites
- [ ] Every formula covered: 15 rows in Section 6.5, each with conditions and a micro-example
- [ ] 70 advanced questions completed (Q1-Q70), difficulty ladder respected
- [ ] Every question has a full step-by-step solution
- [ ] Every solution has a Hinglish explanation
- [ ] 80 MCQs completed with answers
- [ ] Every MCQ has an explanation plus trap analysis
- [ ] Formula sheet completed
- [ ] Last-minute cheat sheet completed
- [ ] Mistake buster completed (10 named mistakes)
- [ ] Exam strategy completed

Chapter 7: Probability - Measuring Chance

Probability turns 'maybe' into a number between 0 and 1 - and every dice, card and coin question in every exam is built on a handful of ideas done carefully. The uploaded PDF introduces random experiments and trials, outcomes and sample spaces, the classical formula P(E) = favourable outcomes / total outcomes, the complement rule P(not E) = 1 - P(E), sure and impossible events, and empirical probability from observed data. It works through coins, dice, cards and everyday experiments. This practice book sharpens all of it: two-dice tables, three-coin tree diagrams, without-replacement draws, 'at least one' reasoning, geometric probability on lengths and areas, and proof-style questions that show why probabilities must always add to 1.

Figure 7.1: A tree diagram - every path is one outcome; multiply along branches, add across paths.

7.1 Complete Concept Cheat Sheet (read this before any question)

7.1.1 Experiment, outcome, sample space, event

What is it? A *random experiment* is any action with an uncertain result (tossing a coin, drawing a card). Each possible result is an *outcome*. The set of ALL outcomes is the *sample space* S, and any subset of S is an *event*. **Key counts:** one coin: S = {H, T}, |S| = 2. One die: |S| = 6. Two coins: 4 outcomes. Two dice: 36 ordered outcomes. Three coins: 8 outcomes. A standard deck: 52 cards. **The golden habit:** before computing ANY probability, WRITE the sample space or its size. Half of all errors are wrong totals, not wrong fractions.

When do I count 'ordered' outcomes? When the experiment distinguishes them: two dice give 36 outcomes because (2, 5) and (5, 2) are different throws; drawing two cards one after another gives 52 × 51 ordered draws. If you prefer unordered counting, you must be consistent in numerator AND denominator.

7.1.2 Classical probability and its three laws

What is it? For equally likely outcomes, P(E) = n(E)/n(S) - favourable outcomes over total outcomes. **Three laws:** (1) $0 \le P(E) \le 1$ always; (2) P(sure event) = 1 and P(impossible event) = 0; (3) P(E) + P(not E) = 1, so P(not E) = 1 - P(E). The complement rule converts 'at least one' and 'none' questions into their easier partners. **Tiny example.** P(rolling a prime on a die) = {2, 3, 5}/6 = 1/2; P(not prime) = 1 - 1/2 = 1/2.

Reading the answer: probability may be written as a fraction in lowest terms, a decimal, or a percentage - fractions are the exam default. A probability of 7/4 or -0.2 is a signal that the COUNTS are wrong, not that the formula is wrong.

7.1.3 Cards: the anatomy of a deck

What is it? 52 cards: 4 suits (hearts ♥, diamonds ♦ - red; spades ♠, clubs ♣ - black) × 13 ranks each (A, 2, 3, ..., 10, J, Q, K). **Standard sub-counts:** aces: 4; face cards (J, Q, K): 12; kings: 4; red queens: 2; hearts: 13; numbered cards 2-10: 36; cards between 3 and 8 (inclusive): 24; honour cards (A, K, Q, J, 10): 20. **Overlaps matter:** 'a king OR a heart' is not 4 + 13 = 17 outcomes because the king of hearts was counted twice - it is 16. The inclusion-exclusion habit starts here.

7.1.4 Dice: singles, pairs and the 36-table

What is it? One die: six outcomes. Two dice: 36 ordered pairs - and the smart move is a 6×6 grid in rough work. **Star facts from the table:** sum 7 has 6 pairs (most likely); sums 2 and 12 have 1 pair each (least likely); P(sum = 7) = 6/36 = 1/6; P(sum > 9) = 6/36 = 1/6 (pairs for 10, 11, 12); P(doubles) = 6/36 = 1/6; P(product even) = 27/36 = 3/4; P(same parity) = 18/36 = 1/2. **Products:** a product is odd only when BOTH dice are odd: 9/36 = 1/4. **Tiny example.** P(sum ≥ 10) = 6/36 = 1/6 (pairs: 4-6, 5-5, 6-4, 5-6, 6-5, 6-6).

7.1.5 Coins and tree diagrams

What is it? Each coin doubles the sample space: 1 coin → 2, 2 coins → 4, 3 coins → 8, n coins → 2^n. A tree diagram draws every path: two branches (H/T) per coin, and each complete path is one outcome. **Star facts (three coins):** P(exactly 2 heads) = 3/8; P(at least 1 head) = 7/8 (only TTT fails); P(at least 2 heads) = 4/8 = 1/2; P(all same) = 2/8 = 1/4. **Why trees work:** multiplying along a branch counts outcomes systematically - it is the visual version of 2 × 2 × 2.

Figure 7.2: The probability scale - impossible 0, sure 1, everything else between.

7.1.6 Draws without replacement

What is it? When drawn objects are NOT returned, the second draw's counts shrink: a bag with 5 red and 3 blue balls, two drawn one after another - P(both red) = (5/8)(4/7) = 20/56 = 5/14. The denominator fell from 8 to 7 because one ball left. **Two-draw combinations via pairs:** P(one red one blue in any order) = 2 × (5/8)(3/8) - the factor 2 handles the two orders (RB, BR). **Alternative counting:** total ways to choose 2 balls from 8 = C(8,2) = 28; both red: C(5,2) = 10; P = 10/28 = 5/14 - same answer, cleaner for 'any order' questions.

7.1.7 Empirical probability

What is it? When real data (not symmetry) defines the chances: P(E) = (number of times E occurred)/(total number of trials). It is an estimate from observation - the more trials, the closer it typically drifts to the true probability. **Example shape:** 200 screws inspected, 12 defective → P(defective) = 12/200 = 0.06, so P(good) = 0.94. **Warning:** empirical probabilities need not be 'nice' fractions, and they describe the PAST data - classical probability describes idealised symmetric setups.

7.1.8 Geometric probability

What is it? When outcomes form a continuum - a point chosen on a segment, a dart hitting a board - probability equals the ratio of measures (lengths, areas): P = favourable length (or area) / total length (or area). **Tiny example.** A point chosen on a 10 cm segment; P(it lies in the first 3 cm) = 3/10. A dart hitting a 20 cm square board with a 10 cm circle target: P(hit circle) = $\pi(5^2)/400 = 25\pi/400 = \pi/16$. **Guard:** both numerator and denominator must use the SAME kind of measure.

7.1.9 Exam tips for this chapter

1. Always write n(S) first; a wrong total poisons the whole question.
2. Two dice: draw the 6×6 grid once and reuse it - sums, products, parity all read off it.
3. Complement first for 'at least one': P(at least one H) = 1 - P(no H).
4. Without replacement: denominators shrink (8 → 7); numerators shrink too if a favourable item left.
5. Cards: memorise 4, 12, 13, 26 and the overlap warning.
6. Reduce every fraction to lowest terms - un-reduced answers lose the presentation mark.

7. Check the range: if $P > 1$ or $P < 0$, recount - the arithmetic, not the theory, is wrong.
8. Empirical questions: read the table column-wise; the total row is n(S).

7.2 Question Types / Concept Map for Chapter 7

Type	Name	How to recognise it in the question	Core method
T1	Sample space building	'List the outcomes', 'find n(S)'	Coins/dice/cards/objects; 2^n, 36, 52
T2	Single die events	One die, one event	Count favourable ÷ 6
T3	Two-dice events	'two dice thrown', sums/products	36-table; systematic pairs
T4	Card events	Deck language: suit, face, ace	Standard counts; overlaps
T5	Complement rule	'at least one', 'not', 'none'	1 - P(opposite)
T6	Bag/ball single draw	Coloured objects, one draw	Favourable colour ÷ total
T7	Two draws without replacement	'second ball is not replaced'	Shrinking denominators; ×2 for orders
T8	Tree diagrams	Multi-stage: 2-3 coins, draw twice	Multiply along branches
T9	Empirical probability	Frequency tables, surveys, trials	occurrences ÷ trials; complement from table
T10	Geometric probability	spinners, dartboards, points on segments	favourable measure ÷ total measure
T11	Event relations	sure/impossible, P(E) + P(not E), mutually exclusive	Range laws; complement
T12	Fairness and decisions	'is the game fair?', expected ties	Compare P(win) values
T13	Missing probability algebra	P(one outcome) unknown, sum = 1	Set up equation; solve
T14	HOTS multi-stage	Three dice, replacement mixtures, proofs	Decompose; tree or table; verify

Distribution of the 50 practice questions: T1-T3 feed Q1-Q10, T4-T6 feed Q11-Q20, T7-T10 feed Q21-Q32, T11-T13 feed Q33-Q42, and T14 dominates Q43-Q50.

7.3 Practice Questions (Q1 - Q50) with Full Step-by-Step Solutions

Difficulty climbs in five bands: Q1-Q10 strong basics, Q11-Q20 moderate, Q21-Q30 difficult, Q31-Q40 high-level, Q41-Q50 very challenging (Olympiad-lite). Every question carries the concept tag; give probabilities as fractions in lowest terms.

Q1 · Basic · Concept: Sample space (T1)

Q1. Write the sample space for: (i) a coin tossed once, (ii) a die rolled once, (iii) two coins tossed together, (iv) a coin tossed and a die rolled.

SOLUTION

Step 1: (i) S = {H, T}; n(S) = 2.

Step 2: (ii) S = {1, 2, 3, 4, 5, 6}; n(S) = 6.

Step 3: (iii) S = {HH, HT, TH, TT}; n(S) = 4 - HT and TH are different outcomes.

Step 4: (iv) Outcomes are pairs like (H, 1), (H, 2), ..., (T, 6): n(S) = 2 × 6 = 12.

» **Why this step?** Stages multiply: 2 × 6 = 12 outcomes for a compound experiment.

Hinglish: *Sample space likhna FIRST step hai. Do coins mein HT aur TH alag outcomes hain - 4 total. Coin + die: 2×6 = 12. Stages multiply hote hain.*

Answer: (i) 2 (ii) 6 (iii) 4 (iv) 12 outcomes

Q2 · Basic · Concept: Single die (T2)

Q2. A die is rolled once. Find P(i) an even number, (ii) a number greater than 4, (iii) a number that is even AND greater than 4.

SOLUTION

Step 1: n(S) = 6.

Step 2: (i) Even: {2, 4, 6}: P = 3/6 = 1/2.

Step 3: (ii) Greater than 4: {5, 6}: P = 2/6 = 1/3.

Step 4: (iii) Both: {6} only: P = 1/6.

» **Why this step?** 'AND' shrinks the event: only outcomes satisfying both conditions count.

Hinglish: *Even teen numbers, greater-than-4 do numbers, dono mein sirf 6. 1/2, 1/3, 1/6. AND ka matlab intersection.*

Answer: (i) 1/2 (ii) 1/3 (iii) 1/6

Q3 · Basic · Concept: Complement on a die (T5)

Q3. A die is rolled once. Find the probability of NOT getting a multiple of 3.

SOLUTION

Step 1: Multiples of 3 on a die: {3, 6}: P = 2/6 = 1/3.

Step 2: Complement: P(not multiple of 3) = 1 - 1/3 = 2/3.

Step 3: Direct check: {1, 2, 4, 5}: 4/6 = 2/3 ✓.

» **Why this step?** The complement rule converts the counting problem; both routes must agree.

Hinglish: *Pehle multiples of 3: 1/3. Complement: 2/3. 'NOT' questions mein complement rule 5 second ka shortcut hai.*

Answer: 2/3

Q4 · Basic · Concept: Two coins (T1/T8)

Q4. Two coins are tossed together. Find P(i) exactly one head, (ii) at least one head, (iii) no head.

SOLUTION

Step 1: S = {HH, HT, TH, TT}: n(S) = 4.

Step 2: (i) Exactly one head: {HT, TH}: P = 2/4 = 1/2.

Step 3: (ii) At least one head: {HH, HT, TH}: P = 3/4.

Step 4: (iii) No head: {TT}: P = 1/4. Note (ii) and (iii) are complements: 3/4 + 1/4 = 1 ✓.

» **Why this step?** 'At least' questions pair with 'none' through the complement - checking they sum to 1 validates both.

Hinglish: *Chaar outcomes. Exactly one head = HT, TH = 1/2. At least one = 3/4 (complement of TT). At least wale questions mein 'none' nikaalo, 1 minus karo.*

Answer: (i) 1/2 (ii) 3/4 (iii) 1/4

Q5 · Basic · Concept: Range and classification (T11)

Q5. A probability is given as: (i) 1.2, (ii) -0.3, (iii) 0, (iv) 1, (v) 7/8. Which values are impossible as probabilities, and what do (iii) and (iv) mean?

SOLUTION

Step 1: Law: every probability satisfies $0 \le P \le 1$.

Step 2: (i) $1.2 > 1$: impossible. (ii) $-0.3 < 0$: impossible.

Step 3: (iii) 0: an impossible event (can never happen).

Step 4: (iv) 1: a sure event (always happens).

Step 5: (v) 7/8 is between 0 and 1: a valid probability.

» **Why this step?** The range law is an instant validity check on every computed answer.

Hinglish: *Probability hamesha 0 aur 1 ke beech. 1.2 aur -0.3 galat - counts mein galti hogi. 0 = kabhi nahi, 1 = hamesha.*

Answer: (i) and (ii) impossible; (iii) impossible event; (iv) sure event; (v) valid.

Q6 · Basic · Concept: Single draw from a bag (T6)

Q6. A bag contains 4 red, 5 blue and 3 green balls. One ball is drawn at random. Find P(i) red, (ii) not green, (iii) red or blue.

SOLUTION

Step 1: Total = 4 + 5 + 3 = 12.

Step 2: (i) P(red) = 4/12 = 1/3.

Step 3: (ii) P(not green) = 1 - 3/12 = 9/12 = 3/4.

Step 4: (iii) P(red or blue) = (4+5)/12 = 9/12 = 3/4 (which matches (ii), since 'not green' = 'red or blue').

» **Why this step?** Disjoint categories add; 'not green' and 'red or blue' describe the same set here - a built-in consistency check.

Hinglish: *Total 12. Red 1/3. Not green 3/4. Red-or-blue bhi 3/4 - same cheez hai dono! Consistency check hamesha karo.*

Answer: (i) 1/3 (ii) 3/4 (iii) 3/4

Q7 · Basic · Concept: Card basics (T4)

Q7. One card is drawn from a well-shuffled deck of 52. Find P(i) an ace, (ii) a red card, (iii) a face card, (iv) the queen of hearts.

SOLUTION

Step 1: (i) Aces: 4 → P = 4/52 = 1/13.

Step 2: (ii) Red cards: 26 → P = 26/52 = 1/2.

Step 3: (iii) Face cards (J, Q, K): 3 × 4 = 12 → P = 12/52 = 3/13.

Step 4: (iv) One specific card: P = 1/52.

» **Why this step?** Four deck counts (4, 12, 13, 26) answer a third of all card questions instantly.

Hinglish: *Yaad rakho: aces 4, reds 26, face cards 12, hearts 13. 1/13, 1/2, 3/13, 1/52. Ye counts ratta - deck questions 90% inhi par.*

Answer: (i) 1/13 (ii) 1/2 (iii) 3/13 (iv) 1/52

Q8 · Basic · Concept: Sure and impossible events (T11)

Q8. A die is rolled. Classify as sure, impossible, or neither: E_1 = getting a number less than 7; E_2 = getting 8; E_3 = getting an odd number.

SOLUTION

Step 1: E_1: every outcome {1..6} satisfies it → P = 6/6 = 1: sure event.

Step 2: E_2: no outcome gives 8 → P = 0/6 = 0: impossible event.

Step 3: E_3: {1, 3, 5}: P = 1/2: neither sure nor impossible.

» **Why this step?** P = 1 and P = 0 are the defining marks of sure and impossible events.

Hinglish: *Less than 7 = sab outcomes = sure. 8 = kabhi nahi = impossible. Odd = 50-50. Ye definitions direct marks dete hain.*

Answer: E_1 sure, E_2 impossible, E_3 neither.

Q9 · Basic · Concept: Sum to 1 with missing probability (T13)

Q9. A random experiment has outcomes A, B, C with P(A) = 0.3 and P(B) = 0.25. Find P(C) if A, B, C are the only outcomes.

SOLUTION

Step 1: The probabilities of all outcomes sum to exactly 1.

Step 2: P(C) = 1 - 0.3 - 0.25 = 0.45.

» **Why this step?** Total probability = 1 is the accounting equation of probability theory.

Hinglish: *Sab probabilities ka sum 1 hota hai. 1 - 0.55 = 0.45. Ye 'missing probability' pattern bahut common hai.*

Answer: 0.45

Q10 · Basic · Concept: Two-dice sum intro (T3)

Q10. Two dice are rolled together. Find P(sum = 7) and P(sum = 2).

SOLUTION

Step 1: n(S) = 36 ordered pairs.

Step 2: Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): 6 pairs → P = 6/36 = 1/6.

Step 3: Sum 2: (1,1) only → P = 1/36.

» **Why this step?** The 36-grid shows sum 7 as the longest diagonal and sum 2 as a single corner.

Hinglish: *36 outcomes. Sum 7 six tarike - sabse likely! 1/6. Sum 2 sirf (1,1) - 1/36. Grid banao rough mein, pairs ginn lo.*

Answer: 1/6 and 1/36

Q11 · Moderate · Concept: Two-dice conditions (T3)

Q11. Two dice are rolled. Find P(i) the sum is a prime number, (ii) the product is odd, (iii) both dice show the same number.

SOLUTION

Step 1: (i) Prime sums: 2 (1 pair), 3 (2), 5 (4), 7 (6), 11 (2): total 15 pairs → P = 15/36 = 5/12.

Step 2: (ii) Product odd ⇔ both dice odd: 3 × 3 = 9 pairs → P = 9/36 = 1/4.

Step 3: (iii) Doubles: (1,1), ..., (6,6): 6 pairs → P = 6/36 = 1/6.

» **Why this step?** Products split by parity cleanly: odd needs odd×odd; even is the complement 27/36.

Hinglish: *Prime sums ka count: 1+2+4+6+2 = 15 → 5/12. Product odd = dono odd = 1/4. Doubles 1/6. 36-grid se teeno direct padho.*

Answer: (i) 5/12 (ii) 1/4 (iii) 1/6

Q12 · Moderate · Concept: Cards with overlap (T4)

Q12. One card is drawn from a deck of 52. Find P(i) a king or a heart, (ii) a red face card, (iii) neither a king nor a queen.

SOLUTION

Step 1: (i) Kings: 4; hearts: 13; king of hearts counted twice → 4 + 13 - 1 = 16 → P = 16/52 = 4/13.

Step 2: (ii) Red face cards: J, Q, K of hearts and diamonds: 6 → P = 6/52 = 3/26.

Step 3: (iii) Kings and queens: 8 cards; neither: 52 - 8 = 44 → P = 44/52 = 11/13.

» **Why this step?** Overlap correction (add both counts, subtract the intersection) is the whole skill in (i).

Hinglish: *King-or-heart mein 4+13 = 17 galat - king of hearts do baar gina! 16 sahi → 4/13. Red face 6. Neither king nor queen 11/13.*

Answer: (i) 4/13 (ii) 3/26 (iii) 11/13

Q13 · Moderate · Concept: Complement with cards (T5)

Q13. One card is drawn from a deck. Find P(i) not an ace, (ii) not a face card, (iii) neither red nor a king.

SOLUTION

Step 1: (i) P(not ace) = 1 - 4/52 = 48/52 = 12/13.

Step 2: (ii) P(not face) = 1 - 12/52 = 40/52 = 10/13.

Step 3: (iii) P(red or king) = 26 + 4 - 2 = 28 (two red kings shared) → P(neither) = 1 - 28/52 = 24/52 = 6/13.

» **Why this step?** Each 'neither/nor' question is one complement away from an 'or' question.

Hinglish: *Not ace 12/13, not face 10/13. (iii) mein pehle red-or-king: 28, complement 24/52 = 6/13. Overlap correction - red kings do hain.*

Answer: (i) 12/13 (ii) 10/13 (iii) 6/13

Q14 · Moderate · Concept: Empirical probability (T9)

Q14. In a survey of 400 students, 96 walk to school, 160 take a bus, 84 cycle and the rest come by car. A student is chosen at random. Find P(i) walks, (ii) does not cycle, (iii) comes by car.

SOLUTION

Step 1: Car users: 400 - 96 - 160 - 84 = 60.

Step 2: (i) P(walks) = 96/400 = 0.24.

Step 3: (ii) P(not cycle) = 1 - 84/400 = 316/400 = 0.79.

Step 4: (iii) P(car) = 60/400 = 0.15.

» **Why this step?** Empirical probability = occurrences ÷ trials; the missing category comes from the total.

Hinglish: *Car wale: 60. 0.24, 0.79, 0.15. Sum check: 0.24+0.40+0.21+0.15 = 1 ✓. Empirical mein decimals normal hain.*

Answer: (i) 0.24 (ii) 0.79 (iii) 0.15

Q15 · Moderate · Concept: Three coins (T8)

Q15. Three coins are tossed together. Find P(i) exactly two heads, (ii) at least two heads, (iii) all three same.

SOLUTION

Step 1: S has $2^3 = 8$ outcomes.

Step 2: (i) Exactly two heads: HHT, HTH, THH → 3/8.

Step 3: (ii) At least two heads: those 3 plus HHH → 4/8 = 1/2.

Step 4: (iii) All same: HHH, TTT → 2/8 = 1/4.

» **Why this step?** 'Exactly' and 'at least' differ by the all-heads outcome - read the wording like a lawyer.

Hinglish: *8 outcomes. Exactly 2: teen paths = 3/8. At least 2: +HHH = 1/2. All same 1/4. Tree diagram banao - 8 paths dikh jayengi.*

Answer: (i) 3/8 (ii) 1/2 (iii) 1/4

Q16 · Moderate · Concept: At least one (T5)

Q16. A coin is tossed 4 times (or four coins together). Find P(at least one tail).

SOLUTION

Step 1: Total outcomes: $2^4 = 16$.

Step 2: Complement: no tail means all heads: 1 outcome.

Step 3: P(at least one tail) = 1 - 1/16 = 15/16.

» **Why this step?** 'At least one' almost always means '1 minus none' - the only exception is when 'none' is harder to count than the direct list.

Hinglish: *At least one tail = 1 - (no tail). No tail = HHHH = 1/16. Answer 15/16. Complement rule ka best use.*

Answer: 15/16

Q17 · Moderate · Concept: Bag: two draws, same colour (T7)

Q17. A bag has 5 red and 3 blue balls. Two balls are drawn one after another WITHOUT replacement. Find P(both red).

SOLUTION

Step 1: First red: 5/8. After removing one red, second red: 4/7.

Step 2: P(both red) = (5/8) × (4/7) = 20/56 = 5/14.

Step 3: Combination check: C(5,2)/C(8,2) = 10/28 = 5/14 ✓.

» **Why this step?** Without replacement, the second draw's denominator drops to 7 - the shrinking universe is the whole idea.

Hinglish: *Pehla red 5/8, doosra red 4/7 (ek red ja chuka!). Multiply: 5/14. Denominator 8 se 7 hua - yahi 'without replacement' hai.*

Answer: 5/14

Q18 · Moderate · Concept: Bag: different colours in any order (T7)

Q18. A bag has 5 red and 3 blue balls. Two balls are drawn without replacement. Find P(one red and one blue, in any order).

SOLUTION

Step 1: Order RB: (5/8)(3/7) = 15/56.

Step 2: Order BR: (3/8)(5/7) = 15/56.

Step 3: Any order: 15/56 + 15/56 = 30/56 = 15/28.

Step 4: Combination check: C(5,1)C(3,1)/C(8,2) = 15/28 ✓.

» **Why this step?** 'In any order' = sum over the distinct orders; the ×2 is just this sum when the two orders have equal probability.

Hinglish: *Do orders - RB aur BR. Dono 15/56. Jodo: 15/28. 'In any order' dikhe toh ×2 ya dono orders likho.*

Answer: 15/28

Q19 · Moderate · Concept: Two dice: at least conditions (T3/T5)

Q19. Two dice are rolled. Find P(i) the sum is at least 10, (ii) the sum is not 7.

SOLUTION

Step 1: (i) Sum ≥ 10: (4,6), (5,5), (6,4), (5,6), (6,5), (6,6): 6 pairs → P = 6/36 = 1/6.

Step 2: (ii) Sum 7 has 6 pairs → P(not 7) = 1 - 6/36 = 30/36 = 5/6.

» **Why this step?** Boundary sums (10, 11, 12) count outward from the corner; complements handle the 'not'.

Hinglish: *Sum ≥ 10: chhe pairs → 1/6. Not 7: complement of the most-likely sum → 5/6.*

Answer: (i) 1/6 (ii) 5/6

Q20 · Moderate · Concept: Empirical with complement (T9)

Q20. A quality inspector tests 500 bulbs: 35 are defective. One bulb is chosen at random from this batch. Find P(i) defective, (ii) good; (iii) if the same defect rate continues, estimate the number of defectives in a batch of 6000.

SOLUTION

Step 1: (i) P(defective) = 35/500 = 0.07.

Step 2: (ii) P(good) = 1 - 0.07 = 0.93.

Step 3: (iii) Expected defectives = 0.07 × 6000 = 420.

» **Why this step?** Empirical probabilities extrapolate linearly to new batch sizes.

Hinglish: *0.07 defective, 0.93 good. 6000 mein 420 kharab. 'Estimate' questions mein rate × new size.*

Answer: (i) 0.07 (ii) 0.93 (iii) 420 bulbs

Q21 · Difficult · Concept: Two dice: difference events (T3)

Q21. Two dice are rolled. Find P(the absolute difference of the two numbers is 2).

SOLUTION

Step 1: Pairs with |a - b| = 2: (1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (5, ... list systematically:

Step 2: Difference 2 rows: a - b = 2: (3,1), (4,2), (5,3), (6,4): 4 pairs. b - a = 2: (1,3), (2,4), (3,5), (4,6): 4 pairs.

Step 3: Total 8 pairs → P = 8/36 = 2/9.

» **Why this step?** Both diagonals above and below the main diagonal count; symmetry doubles one direction's count.

Hinglish: *Difference 2: do diagonals - 4 + 4 = 8 pairs. 2/9. Difference d ke liye formula: 2(6-d)/36 pairs for d = 1..5.*

Answer: 2/9

Q22 · Difficult · Concept: Three draws without replacement (T7)

Q22. A bag has 4 white and 6 black balls. Three balls are drawn one after another without replacement. Find P(all three are black).

SOLUTION

Step 1: First black: 6/10. Second: 5/9. Third: 4/8.

Step 2: P = (6/10)(5/9)(4/8) = 120/720 = 1/6.

Step 3: Combination check: C(6,3)/C(10,3) = 20/120 = 1/6 ✓.

» **Why this step?** Three shrinking denominators 10, 9, 8 and shrinking numerators 6, 5, 4 - the without-replacement cascade.

Hinglish: *6/10 → 5/9 → 4/8. Multiply: 1/6. Har draw ke baad universe chhota hota hai.*

Answer: 1/6

Q23 · Difficult · Concept: Cards: two aces (T7)

Q23. Two cards are drawn from a deck without replacement. Find P(both are aces).

SOLUTION

Step 1: First ace: 4/52 = 1/13. Second ace: 3/51 = 1/17.

Step 2: P = (1/13)(1/17) = 1/221.

Step 3: Combination check: C(4,2)/C(52,2) = 6/1326 = 1/221 ✓.

» **Why this step?** Two-stage multiplication with shrinking counts; 51 = the deck after one card leaves.

Hinglish: *1/13 × 1/17 = 1/221. Doosra ace 51 cards mein se 3 - shrinking deck.*

Answer: 1/221

Q24 · Difficult · Concept: Geometric: point on segment (T10)

Q24. A point P is chosen at random on a segment AB of length 12 cm. Find the probability that AP > 4 cm.

SOLUTION

Step 1: Total measure: length 12 cm.

Step 2: Favourable: positions with AP > 4 cm form a sub-segment of length 12 - 4 = 8 cm.

Step 3: P = 8/12 = 2/3.

» **Why this step?** Geometric probability replaces counting with measuring lengths of intervals.

Hinglish: *AP > 4 matlab P, 4 cm se aage kahin bhi - 8 cm ka region. 8/12 = 2/3. Length ratio hi probability hai.*

Answer: 2/3

Q25 · Difficult · Concept: Geometric: dartboard (T10)

Q25. A dart hits a square board of side 20 cm at a random point. A circle of diameter 10 cm is painted at the centre. Find P(the dart lands inside the circle).

SOLUTION

Step 1: Board area = 400 cm^2. Circle radius = 5 cm; area = 25π cm^2.

Step 2: P = 25π/400 = π/16.

Step 3: With π = 22/7: 22/112 = 11/56 ≈ 0.196.

» **Why this step?** Both measures are areas; the π cancels if kept symbolic.

Hinglish: *Circle/board = 25π/400 = π/16. Dono same measure (area) - yahi rule hai. π symbolic rakho jab tak ho sake.*

Answer: π/16 ≈ 0.196

Q26 · Difficult · Concept: Mixed bag with replacement vs not (T7)

Q26. A bag has 3 red and 2 green balls. Two balls are drawn. Compare P(both red) when (i) the first ball IS replaced, (ii) it is NOT replaced.

SOLUTION

Step 1: (i) With replacement: P = (3/5)(3/5) = 9/25.

Step 2: (ii) Without: P = (3/5)(2/4) = 6/20 = 3/10.

Step 3: Comparison: 9/25 = 0.36 vs 3/10 = 0.30 - replacement keeps the odds constant, so both-red is more likely.

» **Why this step?** Replacement freezes the probabilities; without it, drawing a red makes the next red rarer.

Hinglish: *Replace karo: 9/25. Nahi: 3/10. Replace > no-replace for same colour. Denominator shrink hi difference hai.*

Answer: (i) 9/25 ≈ 0.36 (ii) 3/10 = 0.30; replacement gives the larger probability.

Q27 · Difficult · Concept: Spinner angles (T10)

Q27. A spinner is a circle divided into three sectors with angles 90°, 150° and 120°, marked A, B, C. Find P(A), P(B), P(C) and verify they sum to 1.

SOLUTION

Step 1: Each probability is its angle over 360°.

Step 2: P(A) = 90/360 = 1/4; P(B) = 150/360 = 5/12; P(C) = 120/360 = 1/3.

Step 3: Sum: 1/4 + 5/12 + 1/3 = 3/12 + 5/12 + 4/12 = 12/12 = 1 ✓.

» **Why this step?** Geometric probability with angle measure: favourable arc over the full circle.

Hinglish: *Angle/360. 1/4, 5/12, 1/3. Sum = 1 - spinner 'fair' nahi hai but probabilities complete hain.*

Answer: 1/4, 5/12, 1/3 (sum 1)

Q28 · Difficult · Concept: Empirical table with missing entry (T9/T13)

Q28. 60 darts hit a board; 18 land in region X, y land in region Y and 22 miss both. If a dart is thrown once, find the probability it lands in Y, given every dart lands somewhere on the board and P(miss both) = 22/60.

SOLUTION

Step 1: Y-count: y = 60 - 18 - 22 = 20.

Step 2: P(Y) = 20/60 = 1/3.

» **Why this step?** The total row of a frequency table is n(S); missing entries fall out by subtraction.

Hinglish: *60 - 18 - 22 = 20. P(Y) = 1/3. Table mein missing entry subtraction se aati hai.*

Answer: 1/3

Q29 · Difficult · Concept: Two dice: neither shows (T3/T5)

Q29. Two dice are rolled. Find P(neither die shows a 5).

SOLUTION

Step 1: Complement route: P(at least one 5) = 11/36 (10 pairs with exactly one 5 plus (5,5)).

Step 2: P(neither) = 1 - 11/36 = 25/36.

Step 3: Direct check: each die avoids 5 in 5 ways: 5 × 5 = 25 pairs ✓.

» **Why this step?** The complement of 'at least one 5' is 'neither shows 5'; both routes confirm 25/36.

Hinglish: *Neither 5 = dono 5 se bachhe: 5×5 = 25 → 25/36. Complement se bhi same. Dono raaste match = pakka.*

Answer: 25/36

Q30 · Difficult · Concept: Conditional-flavour reasoning (T7/T14)

Q30. A bag has 6 white and 4 black marbles. Two marbles are drawn without replacement. Given that the FIRST marble drawn is white, find the probability that the second is also white.

SOLUTION

Step 1: After one white is out: 5 white and 4 black remain - 9 marbles total.

Step 2: P(second white | first white) = 5/9.

» **Why this step?** Once the first draw is known, the sample space resets to the remaining marbles - no multiplication needed.

Hinglish: *Given first white - ab universe 9 marbles ka hai (5 white). 5/9. 'Given' sunte hi purani draws ko hata do.*

Answer: 5/9

Q31 · High · Concept: Two dice: one even one odd (T3)

Q31. Two dice are rolled. Find P(one die shows an even number and the other odd).

SOLUTION

Step 1: Favourable: (even, odd) or (odd, even): 3×3 + 3×3 = 18 pairs.

Step 2: P = 18/36 = 1/2.

Step 3: Complement check: same parity (both even or both odd): 9 + 9 = 18 → 1/2 ✓.

» **Why this step?** Mixed parity and same parity split the 36 outcomes exactly in half.

Hinglish: *Even-odd mix: 18 pairs = 1/2. Same parity bhi 18 - poora balance. Symmetry se check karo.*

Answer: 1/2

Q32 · High · Concept: Cards: three-card draw (T7)

Q32. Three cards are drawn from a deck without replacement. Find P(all three are face cards).

SOLUTION

Step 1: Face cards: 12. P = (12/52)(11/51)(10/50).

Step 2: Simplify: (12×11×10)/(52×51×50) = 1320/132600 = 11/1105.

Step 3: Combination check: C(12,3)/C(52,3) = 220/22100 = 11/1105 ✓.

» **Why this step?** The cascade 12, 11, 10 over 52, 51, 50 is the three-card generalisation of the two-card pattern.

Hinglish: *12 → 11 → 10; 52 → 51 → 50. 11/1105. Combination route se bhi same.*

Answer: 11/1105

Q33 · High · Concept: At least one with bag draws (T5/T7)

Q33. A bag has 4 red and 4 blue balls. Three balls are drawn without replacement. Find P(at least one red).

SOLUTION

Step 1: Complement: no red = all three blue.

Step 2: P(all blue) = (4/7)(3/6)(2/5) = 24/210 = 4/35.

Step 3: P(at least one red) = 1 - 4/35 = 31/35.

» **Why this step?** 'At least one' with three draws is far easier through the complement - one cascade instead of three cases.

Hinglish: *At least one red = 1 - (sab blue). Sab blue: 4/7×3/6×2/5 = 4/35. Answer 31/35. Complement hi sahi raasta hai.*

Answer: 31/35

Q34 · High · Concept: Fairness of a game (T12)

Q34. A and B throw a die alternately; A starts. Whoever rolls a 6 first wins. Show why A's winning chance is larger, and compute P(A wins).

SOLUTION

Step 1: P(A wins on his first turn) = 1/6. P(B wins on his first) = (5/6)(1/6) = 5/36.

Step 2: A's advantage: he moves first; every 'everyone failed so far' cycle returns the turn to A with the same fresh chance.

Step 3: Series: P(A wins) = $1/6 + (5/6)^2(1/6) + (5/6)^4(1/6) + \ldots = (1/6)/(1 - 25/36) = (1/6)/(11/36) = 6/11$.

Step 4: So P(B wins) = 5/11: A's edge is real (6/11 > 5/11), which is why 'winner starts next game' is unfair.

» **Why this step?** First-mover advantage compounds over rounds: each restart favours the same player.

Hinglish: *A pehle kheleta hai, isliye uska chance zyada: 6/11 vs 5/11. Infinite series se aata hai: (1/6)/(11/36). Fair game chahiye toh toss karo kaun pehle.*

Answer: P(A wins) = 6/11 > 5/11 = P(B wins): the first thrower has the edge.

Q35 · High · Concept: Sum to 1 with three unknowns (T13)

Q35. Outcomes A, B, C, D satisfy P(A) = P(B) = 2P(C) and P(D) = 0.1. Find P(C).

SOLUTION

Step 1: Let P(C) = x. Then P(A) = P(B) = 2x.

Step 2: Sum: $2x + 2x + x + 0.1 = 1 \Rightarrow 5x = 0.9 \Rightarrow x = 0.18$.

Step 3: P(C) = 0.18 (and P(A) = P(B) = 0.36).

» **Why this step?** One letter for the smallest unknown turns the accounting equation into a linear equation.

Hinglish: *x = P(C) maan lo: 2x + 2x + x + 0.1 = 1. x = 0.18. Sab unknowns ek variable se express karo.*

Answer: P(C) = 0.18

Q36 · High · Concept: Two dice: sum given parity (T3/T14)

Q36. Two dice are rolled. Find P(the sum is 8) and P(the sum is 8 with both dice even).

SOLUTION

Step 1: Sum 8 pairs: (2,6), (3,5), (4,4), (5,3), (6,2): 5 pairs → P = 5/36.

Step 2: Both even among those: (2,6), (4,4), (6,2): 3 pairs → P = 3/36 = 1/12.

» **Why this step?** Listing the full pair-set once answers both the unconditional and the restricted question.

Hinglish: *Sum 8: 5 pairs → 5/36. Usme se both-even 3 → 1/12. Ek listing, do answers.*

Answer: 5/36 and 1/12

Q37 · High · Concept: Empirical vs classical contrast (T9)

Q37. A die is suspected to be loaded. In 600 rolls the face 6 appeared 150 times. (i) What is the empirical P(6)? (ii) Compare with the classical value and comment.

SOLUTION

Step 1: (i) Empirical P(6) = 150/600 = 1/4 = 0.25.

Step 2: (ii) Classical P(6) = 1/6 ≈ 0.167.

Step 3: 0.25 is noticeably above 1/6; with 600 trials this suggests the die favours 6 (loaded), though small experiments can deviate by chance.

» Why this step? Empirical probability is evidence; classical probability is the fair-dice benchmark - their gap is the diagnosis.

Hinglish: *0.25 vs 0.167 - die 6 ko pasand kar raha hai! Loaded suspect. Bade trials mein deviation chance se zyada ho toh bias.*

Answer: (i) 0.25 (ii) far above 1/6; the die appears biased toward 6.

Q38 · High · Concept: Geometric: point in a square (T10)

Q38. A point is chosen at random inside a 6 cm × 8 cm rectangle. Find the probability that it lies within 2 cm of the left edge.

SOLUTION

Step 1: Favourable region: a 2 cm × 8 cm strip along the left edge: area 16 cm^2.

Step 2: Total area: 48 cm^2.

Step 3: P = 16/48 = 1/3.

» Why this step? Distance-to-edge conditions carve strips; probability is the strip's share of the total area.

Hinglish: *Left edge ke 2 cm ke andar = 2×8 strip = 16 cm^2. 16/48 = 1/3. Area ratio.*

Answer: 1/3

Q39 · High · Concept: Two dice: product divisible (T3)

Q39. Two dice are rolled. Find P(the product is divisible by 4).

SOLUTION

Step 1: Count pairs with product not divisible by 4, then complement.

Step 2: Product ≡ 2 (mod 4) needs exactly one die = 2 (or 6) and the other odd: 2×3... enumerate: pairs where product mod 4 = 2: (2,1),(2,3),(2,5),(1,2),(3,2),(5,2),(6,1),(6,3),(6,5),(1,6),(3,6),(5,6): 12 pairs.

Step 3: Product odd pairs: 9 (not divisible by 4 either).

Step 4: Not divisible by 4: 12 + 9 = 21 → divisible: 36 - 21 = 15 → P = 15/36 = 5/12.

Step 5: Direct recount spot check: products divisible by 4 include all doubles except (1,1),(3,3),(5,5)... (2,2),(4,4),(6,6),(4,x): x any: 6, minus overlap (4,2),(2,4),(4,6),(6,4) counted once: the enumeration above stands.

» Why this step? Complement counting avoids double-counting inside the 'divisible by 4' set, whose members overlap (e.g. 4×4).

Hinglish: *Seedha count karna double-counting bulata hai. Complement lo: not-divisible = 12 + 9 = 21. Divisible = 15 → 5/12.*

Answer: 5/12

Q40 · High · Concept: Tree diagram three-stage (T8)

Q40. A coin is tossed and then a die is rolled. Using a tree diagram, find P(head and an even number) and P(tail and a number greater than 4).

SOLUTION

Step 1: Tree: 2 branches (H/T), each splitting into 6 die branches: 12 paths.

Step 2: Head + even: H connects to {2, 4, 6}: 3 paths → P = 3/12 = 1/4.

Step 3: Tail + >4: T connects to {5, 6}: 2 paths → P = 2/12 = 1/6.

» **Why this step?** Each path has probability (1/2)(1/6) = 1/12; counting paths is counting twelfths.

Hinglish: *12 paths. H+even: 3 → 1/4. T+big: 2 → 1/6. Tree = paths ginna.*

Answer: 1/4 and 1/6

Q41 · HOTS · Concept: Sum probability distribution proof (T14)

Q41. Two dice are rolled. Tabulate the probability of each possible sum (2 through 12) and verify that the probabilities add to 1.

SOLUTION

Step 1: Counts: sum 2:1, 3:2, 4:3, 5:4, 6:5, 7:6, 8:5, 9:4, 10:3, 11:2, 12:1.

Step 2: Each probability = count/36.

Step 3: Total counts: 1+2+3+4+5+6+5+4+3+2+1 = 36.

Step 4: Sum of probabilities = 36/36 = 1 ✓ - the distribution is complete.

» **Why this step?** The pyramid count (1,2,3,4,5,6,5,4,3,2,1) is the two-dice signature; completeness is guaranteed by covering every pair exactly once.

Hinglish: *Pyramid pattern: 1,2,3,4,5,6,5,4,3,2,1. Total 36 = 1. Ye table ratta maar lo - har two-dice question isi se nikalta hai.*

Answer: P(s = k) = count/36 with counts 1,2,3,4,5,6,5,4,3,2,1; total 1.

Q42 · HOTS · Concept: Probability range proof (T14)

Q42. Prove that for any event E, 0 ≤ P(E) ≤ 1, and that P(E) + P(not E) = 1.

SOLUTION

Step 1: P(E) = n(E)/n(S) where 0 ≤ n(E) ≤ n(S) (an event cannot have negative or more-than-all outcomes).

Step 2: Dividing by the positive n(S): 0 ≤ n(E)/n(S) ≤ 1. ∎

Step 3: Every outcome is either in E or not in E, never both: n(E) + n(not E) = n(S).

Step 4: Divide by n(S): P(E) + P(not E) = 1. ∎

» **Why this step?** Both laws are division of counting facts about sets - probability inherits its rules from arithmetic.

Hinglish: *Proof pure counting hai: event ke outcomes 0 se n(S) ke beech hote hain. Divide karo - range law. E aur not-E poora S dhak lete hain, overlap nahi - sum 1. Do-line proofs, full marks.*

Answer: Both laws proved from counting.

Q43 · HOTS · Concept: Replacement mixture (T14)

Q43. Bag A has 2 red and 3 blue balls; bag B has 4 red and 1 blue. A bag is chosen at random (each equally likely) and one ball is drawn from it. Find P(red).

SOLUTION

Step 1: P(choose A) = 1/2, then P(red from A) = 2/5: contribution (1/2)(2/5) = 1/5.

Step 2: P(choose B) = 1/2, then P(red from B) = 4/5: contribution (1/2)(4/5) = 2/5.

Step 3: P(red) = 1/5 + 2/5 = 3/5.

» **Why this step?** Two-stage experiments multiply along each branch, then add across the branches - the tree diagram in numbers.

Hinglish: *Do bag: (1/2)(2/5) + (1/2)(4/5) = 1/5 + 2/5 = 3/5. Branch multiply, cross-branch add - tree ka numeric form.*

Answer: 3/5

Q44 · HOTS · Concept: Two dice maximum/minimum (T3/T14)

Q44. Two dice are rolled. Find P(the larger of the two numbers is 5). (Rolls like (5, 5) are allowed.)

SOLUTION

Step 1: Count pairs whose maximum is exactly 5: both ≤ 5 minus both ≤ 4.

Step 2: Both ≤ 5: 5×5 = 25 pairs. Both ≤ 4: 16 pairs.

Step 3: Exactly max 5: 25 - 16 = 9 pairs → P = 9/36 = 1/4.

Step 4: Direct list check: (5,1..5), (1..4,5): 5 + 4 = 9 ✓.

» **Why this step?** 'Maximum = k' counting by cumulative squares ($k^2 - (k-1)^2 = 2k-1$) is the fast lane: 2(5)-1 = 9.

Hinglish: *Max 5 = $5^2 - 4^2 = 9$ pairs → 1/4. Formula 2k-1 yaad rakho - max k wale pairs.*

Answer: 1/4

Q45 · HOTS · Concept: Three dice sums (T14)

Q45. Three dice are rolled. Find P(sum = 4) and P(sum = 5).

SOLUTION

Step 1: Total outcomes: $6^3 = 216$.

Step 2: Sum 4: compositions of 4 into three parts ≥ 1: (2,1,1) permutations: 3 outcomes → P = 3/216 = 1/72.

Step 3: Sum 5: (3,1,1): 3 permutations; (2,2,1): 3 permutations: 6 outcomes → P = 6/216 = 1/36.

» **Why this step?** Small sums need compositions with permutations counted - (2,1,1) has three arrangements, not one.

Hinglish: *216 total. Sum 4: sirf (2,1,1) - teen arrangements: 1/72. Sum 5: (3,1,1) aur (2,2,1): 3+3 = 6: 1/36. Permutations ginne - ye HOTS ka trap hai.*

Answer: 1/72 and 1/36

Q46 · HOTS · Concept: Draw until first success (T14)

Q46. A bag has 3 red and 3 blue balls. Balls are drawn one at a time WITHOUT replacement until the first red appears. Find P(exactly two draws are needed).

SOLUTION

Step 1: Exactly two draws means: first is blue AND second is red.

Step 2: P = (3/6)(3/5) = 9/30 = 3/10.

» **Why this step?** Stopping questions translate directly into 'first k-1 draws fail, draw k succeeds'.

Hinglish: *Exactly 2 draws = pehla blue, doosra red. (1/2)(3/5) = 3/10. 'Until' questions ka pattern.*

Answer: 3/10

Q47 · HOTS · Concept: Geometric: two points on a segment (T10/T14)

Q47. Two points P and Q are chosen independently at random on a segment of length 10 cm. Estimate (by symmetry reasoning) whether P(the distance PQ is more than 5 cm) is more than, less than, or equal to 1/2.

SOLUTION

Step 1: Model with coordinates: P at x, Q at y, both uniform on [0, 10].

Step 2: PQ > 5 ⇔ |x - y| > 5: the two points must fall in opposite halves far apart.

Step 3: By scaling to [0,1]: P(|x-y| > 1/2) = 1/4 (the two corner triangles of side 1/2 in the unit square of (x, y) pairs).

Step 4: So the probability is 1/4: LESS than 1/2.

» **Why this step?** Two random points usually land moderately close; being 5 cm apart on a 10 cm line needs opposite extremes - a quarter of the (x, y) square.

Hinglish: *Do random points aksar paas-paas girte hain. 5 cm+ gap = opposite extremes - unit square ke do corners = 1/4. 1/2 se KAM.*

Answer: Less than 1/2; in fact P = 1/4.

Q48 · HOTS · Concept: Complement cascade (T5/T14)

Q48. A bag has 5 red, 4 white and 3 blue balls. Three balls are drawn without replacement. Find P(all three are of different colours).

SOLUTION

Step 1: Total ways: C(12,3) = 220.

Step 2: Favourable: one of each colour: 5 × 4 × 3 = 60 (choices are unordered but colours distinct, so each triple counted once).

Step 3: P = 60/220 = 3/11.

» **Why this step?** 'All different' is a product of independent colour counts divided by the combination total - no ordering issue since colours label the balls.

Hinglish: *C(12,3) = 220. Ek-ek colour: 5×4×3 = 60. 3/11. Colours alag hain toh ordering issue nahi hota.*

Answer: 3/11

Q49 · HOTS · Concept: Expected count from probability (T9/T14)

Q49. A fair die is rolled 300 times. About how many times should an even number appear, and why is the actual count unlikely to be exactly that?

SOLUTION

Step 1: P(even) = 1/2. Expected count = 300 × 1/2 = 150.

Step 2: The actual count fluctuates: each roll is random, and runs of luck push counts above or below 150.

Step 3: What IS reliable: the count sits fairly close to 150 for a fair die, and the average of many such 300-roll experiments drifts toward 150 - the law of large numbers in action.

» **Why this step?** Probability predicts long-run averages, not exact short-run counts - the central lesson of empirical probability.

Hinglish: *Expected 150, par EXACTLY 150 milna lucky hai. Variability hoti hai. Bahut saare experiments ka average 150 ki taraf jata hai - yahi law of large numbers hai.*

Answer: Expected ≈ 150; the actual count varies around it (randomness), converging in the long run.

Q50 · HOTS · Concept: De Morgan complement proof (T14)

Q50. For any two events E and F, prove that P(not E and not F) = 1 - P(E or F).

SOLUTION

Step 1: 'Not E and not F' means the outcome lies outside E and outside F - i.e. outside (E or F).

Step 2: So 'not E and not F' is exactly the complement of the event (E or F).

Step 3: By the complement law: P(not E and not F) = 1 - P(E or F). ∎

Step 4: Sanity check on a die, E = even, F = multiple of 3: E or F = {2,3,4,6}: P = 4/6. Neither = {1,5}: P = 2/6 = 1 - 4/6 ✓.

» **Why this step?** De Morgan's set law (outside both = outside the union) plus the complement law give the identity in two lines.

Hinglish: *'Dono ke bahar' = '(E ya F) ke bahar' - De Morgan. Complement law lagao: 1 - P(E or F). Dice se check karo: 2/6 = 1 - 4/6 ✓.*

Answer: Proved: they are complements of each other, so the probabilities sum to 1.

7.4 Chapter MCQs (60 questions, answers inline)

Each MCQ lists the options, the correct answer, a full explanation, the trap analysis for the tempting wrong options, and a Hinglish note. Think before you pick.

MCQ 1 · Normal · Concept: Sample space size

1. Two coins are tossed together. The number of outcomes in the sample space is:

A. 4
B. 3
C. 2
D. 6

Correct Answer: (A)

HH, HT, TH, TT: $2^2 = 4$ outcomes (HT and TH are distinct).

Why (B) is wrong: That counts heads-counts, not outcomes.

Why (C) is wrong: One coin only.

Why (D) is wrong: Coin + die.

Hinglish: *Chaar outcomes - HT aur TH alag hain. 3 wala trap 'same number of heads' sochne se aata hai.*

MCQ 2 · Normal · Concept: Single die event

2. A die is rolled once. P(a number less than 3) is:

A. 1/3
B. 1/2
C. 2/3
D. 1/6

Correct Answer: (A)

{1, 2}: 2/6 = 1/3.

Why (B) is wrong: Included 3 (less than means strictly).

Why (C) is wrong: Used complement wrongly.

Why (D) is wrong: One outcome only.

Hinglish: *1 aur 2 - do outcomes. 1/3.*

MCQ 3 · Normal · Concept: Complement on die

3. A die shows a number. P(not a prime) is:

A. 1/2

B. 1/3

C. 2/3

D. 1/6

Correct Answer: (A)

Primes: {2, 3, 5}: P = 1/2. Complement: 1/2.

Why (B) is wrong: Forgot 2 is prime.

Why (C) is wrong: Counted primes wrong.

Why (D) is wrong: Single outcome.

Hinglish: *Prime teen hain: 2, 3, 5. Not prime: 1, 4, 6 - 1/2.*

MCQ 4 · Normal · Concept: Two coins

4. Two coins are tossed. P(exactly two heads) is:

A. 1/4

B. 1/2

C. 3/4

D. 1/3

Correct Answer: (A)

Only HH among 4 outcomes: 1/4.

Why (B) is wrong: That is exactly-one-head.

Why (C) is wrong: At least one head.

Why (D) is wrong: Wrong denominator.

Hinglish: *Sirf HH. 1/4.*

MCQ 5 · Normal · Concept: Sure event

5. A die is rolled. 'Getting a number less than 10' has probability:

A. 1

B. 5/6

C. 0

D. 9/10

Correct Answer: (A)

Every outcome satisfies it: 6/6 = 1 - a sure event.

Why (B) is wrong: 5 outcomes only?

Why (C) is wrong: That is impossible events.

Why (D) is wrong: No such fraction here.

Hinglish: *Sab outcomes jeette hain - sure event, P = 1.*

MCQ 6 · Normal · Concept: Bag single draw

6. A bag has 7 red and 5 black balls. P(a red ball) is:

A. 7/12
B. 5/12
C. 7/5
D. 1/2

Correct Answer: (A)

7 red out of 12 total.

Why (B) is wrong: Black instead.

Why (C) is wrong: Red over black - not a probability format.

Why (D) is wrong: Assumed equal colours.

Hinglish: *7/12. Total = 7+5 = 12.*

MCQ 7 · Normal · Concept: Card: ace

7. One card is drawn from 52. P(an ace) is:

A. 1/13
B. 1/4
C. 1/52
D. 4/13

Correct Answer: (A)

4 aces: 4/52 = 1/13.

Why (B) is wrong: That is P(red).

Why (C) is wrong: One specific ace.

Why (D) is wrong: Mis-reduced.

Hinglish: *4 aces → 1/13.*

MCQ 8 · Normal · Concept: Card: face card

8. P(a face card) from a deck of 52 is:

A. 3/13
B. 1/13
C. 4/13
D. 12/13

Correct Answer: (A)

12 face cards: 12/52 = 3/13.

Why (B) is wrong: One rank only.

Why (C) is wrong: One suit's face cards.

Why (D) is wrong: Complement.

Hinglish: *12 face cards (J, Q, K × 4). 3/13.*

MCQ 9 · Normal · Concept: Two dice total

9. Two dice are rolled. n(S) equals:

A. 36
B. 12
C. 6
D. 64

Correct Answer: (A)

6 × 6 = 36 ordered pairs.

Why (B) is wrong: Sum-based thinking.

Why (C) is wrong: One die.

Why (D) is wrong: Cubed.

Hinglish: *36 ordered pairs. (2,5) aur (5,2) alag.*

MCQ 10 · Normal · Concept: Sum = 7

10. Two dice: P(sum = 7) is:

A. 1/6
B. 1/36
C. 7/36
D. 5/36

Correct Answer: (A)

Six pairs sum to 7: 6/36 = 1/6.

Why (B) is wrong: One pair only (that is sum 2 or 12).

Why (C) is wrong: Matched numerator to the sum.

Why (D) is wrong: Off-by-one.

Hinglish: *Chhe pairs - most likely sum. 1/6.*

MCQ 11 · Normal · Concept: Doubles

11. Two dice: P(both show the same number) is:

A. 1/6
B. 1/36
C. 1/12
D. 1/3

Correct Answer: (A)

Six doubles: 6/36 = 1/6.

Why (B) is wrong: (1,1) only.

Why (C) is wrong: Wrong reduction.

Why (D) is wrong: Overcounted.

Hinglish: *Doubles chhe: 1/6.*

MCQ 12 · Normal · Concept: Range check

12. Which CANNOT be a probability?

A. 1.4
B. 0
C. 1
D. 0.999

Correct Answer: (A)

Probabilities lie in [0, 1]; 1.4 exceeds 1.

Why (B) is wrong: 0 is legal (impossible event).

Why (C) is wrong: 1 is legal (sure event).

Why (D) is wrong: 0.999 is legal.

Hinglish: *1.4 range se bahar.*

MCQ 13 · Normal · Concept: Sum to 1

13. P(E) = 0.65. P(not E) equals:

A. 0.35
B. 0.65
C. 1.65
D. 0.5

Correct Answer: (A)

1 - 0.65 = 0.35.

Why (B) is wrong: Copied the given.

Why (C) is wrong: Added instead.

Why (D) is wrong: Halved.

Hinglish: *Complement: 1 minus.*

MCQ 14 · Normal · Concept: Three coins count

14. Three coins are tossed. n(S) =

A. 8
B. 6
C. 4
D. 9

Correct Answer: (A)

$2^3 = 8$.

Why (B) is wrong: Added 2+2+2.

Why (C) is wrong: Two coins.

Why (D) is wrong: Squared.

Hinglish: *Har coin double karta hai: $2^3 = 8$.*

MCQ 15 · Normal · Concept: Three coins exactly two heads

15. Three coins: P(exactly two heads) is:

A. 3/8
B. 1/2
C. 1/8
D. 5/8

Correct Answer: (A)

HHT, HTH, THH: 3 of 8.

Why (B) is wrong: That is at least two.

Why (C) is wrong: All heads only.

Why (D) is wrong: Wrong denominator.

Hinglish: *Teen paths. 3/8.*

MCQ 16 · Normal · Concept: Empirical probability

16. In 250 trials an event occurred 50 times. Its empirical probability is:

A. 0.2
B. 0.5
C. 5
D. 0.02

Correct Answer: (A)

50/250 = 0.2.

Why (B) is wrong: Mixed up with 1/2.

Why (C) is wrong: Not a probability.

Why (D) is wrong: Divided wrongly.

Hinglish: *50/250 = 1/5 = 0.2.*

MCQ 17 · Normal · Concept: Card: red or king

17. One card: P(red or king) is:

A. 7/13
B. 15/26
C. 1/2
D. 2/13

Correct Answer: (A)

26 + 4 - 2 (red kings) = 28: 28/52 = 7/13.

Why (B) is wrong: Forgot the overlap correction.

Why (C) is wrong: Red only.

Why (D) is wrong: King only.

Hinglish: *26 + 4 = 30, minus 2 overlap = 28 → 7/13.*

MCQ 18 · Normal · Concept: Two dice: at least one six

18. Two dice: P(at least one 6) is:

A. 11/36
B. 1/6
C. 1/36
D. 25/36

Correct Answer: (A)

Complement: no six = 25/36; so 11/36.

Why (B) is wrong: Counted (6,6) pairs as 6+6 - double count.

Why (C) is wrong: Both sixes only.

Why (D) is wrong: That is neither-six.

Hinglish: *1 - 25/36 = 11/36. Complement route.*

MCQ 19 · Normal · Concept: Without replacement intro

19. A bag has 4 red and 4 blue balls; two are drawn without replacement. P(both red) is:

A. 3/14
B. 1/4
C. 1/2
D. 4/7

Correct Answer: (A)

(4/8)(3/7) = 12/56 = 3/14.

Why (B) is wrong: With-replacement answer would be 1/4.

Why (C) is wrong: No multiplication.

Why (D) is wrong: Second draw denominator wrong.

Hinglish: *4/8 then 3/7. 3/14.*

MCQ 20 · Normal · Concept: Deck: no face card

20. One card: P(not a face card) is:

A. 10/13
B. 3/13
C. 9/13
D. 1/2

Correct Answer: (A)

1 - 12/52 = 40/52 = 10/13.

Why (B) is wrong: Face probability.

Why (C) is wrong: Mis-reduced.

Why (D) is wrong: Guessed.

Hinglish: *40 non-face. 10/13.*

MCQ 21 · Normal · Concept: Coin + die

21. A coin is tossed and a die rolled. P(head and a 5) is:

A. 1/12
B. 1/6
C. 1/2
D. 1/10

Correct Answer: (A)

(1/2)(1/6) = 1/12.

Why (B) is wrong: Ignored the coin.

Why (C) is wrong: Ignored the die.

Why (D) is wrong: Added fractions.

Hinglish: *Do independent stages multiply: 1/12.*

MCQ 22 · Normal · Concept: Impossible event

22. Rolling a standard die, P(getting a 9) is:

A. 0
B. 1
C. 9/6
D. 1/6

Correct Answer: (A)

No outcome is 9: impossible event.

Why (B) is wrong: That is sure events.

Why (C) is wrong: Fractions over 1 are illegal.

Why (D) is wrong: Copied the number.

Hinglish: *Die par 9 nahi hota - P = 0.*

MCQ 23 · Normal · Concept: Two dice product even

23. Two dice: P(product is even) is:

A. 3/4
B. 1/4
C. 1/2
D. 2/3

Correct Answer: (A)

Product odd needs both odd: 9/36 = 1/4; complement 27/36 = 3/4.

Why (B) is wrong: That is the odd-product answer.

Why (C) is wrong: Guessed parity split.

Why (D) is wrong: Miscounted.

Hinglish: *Complement se: odd product 1/4, toh even 3/4.*

MCQ 24 · Normal · Concept: Spinner

24. A spinner has equal quarters marked A, B, C, D. P(not A) is:

A. 3/4
B. 1/4
C. 1/2
D. 1/3

Correct Answer: (A)

1 - 1/4 = 3/4.

Why (B) is wrong: That is P(A).

Why (C) is wrong: Half-turn thinking.

Why (D) is wrong: Three options but unequal? They are equal quarters.

Hinglish: *Complement of quarter.*

MCQ 25 · Normal · Concept: Sum = 10

25. Two dice: P(sum = 10) is:

A. 1/12
B. 1/6
C. 1/9
D. 5/36

Correct Answer: (A)

Three pairs: (4,6), (5,5), (6,4): 3/36 = 1/12.

Why (B) is wrong: Confused with sum 7's count.

Why (C) is wrong: Wrong count.

Why (D) is wrong: (4,6),(6,4) plus missing (5,5)?

Hinglish: *Teen pairs: 1/12.*

MCQ 26 · Normal · Concept: Empirical complement

26. Of 500 items, 15 are defective. P(a randomly chosen item is NOT defective) is:

A. 0.97
B. 0.03
C. 0.5
D. 15/500 as such

Correct Answer: (A)

485/500 = 0.97.

Why (B) is wrong: That is P(defective).

Why (C) is wrong: Guessed half.

Why (D) is wrong: Un-reduced fraction as probability statement - fine but 0.97 is the value.

Hinglish: *1 - 0.03 = 0.97.*

MCQ 27 · Normal · Concept: Geometric: segment

27. A point is chosen on a 15 cm segment. P(it lies in the first 5 cm) is:

A. 1/3

B. 5/15 as 1/5

C. 2/3

D. 1/15

Correct Answer: (A)

Length ratio: 5/15 = 1/3.

Why (B) is wrong: Mis-reduced.

Why (C) is wrong: Complement.

Why (D) is wrong: Point-based thinking - uncountably many points exist.

Hinglish: *5/15 = 1/3. Length ratio.*

MCQ 28 · Normal · Concept: Mutually exclusive idea

28. Two events E and F cannot happen together. Then P(E and F) =

A. 0

B. 1

C. P(E) + P(F)

D. P(E)P(F)

Correct Answer: (A)

No common outcomes: the intersection is empty, probability 0.

Why (B) is wrong: That is sure.

Why (C) is wrong: That is the 'or' rule for exclusive events.

Why (D) is wrong: Multiplication rule for independent events.

Hinglish: *Saath mein kabhi nahi - intersection khali. P = 0.*

MCQ 29 · Normal · Concept: Sum > 9

29. Two dice: P(sum > 9) is:

A. 1/6

B. 1/12

C. 5/36

D. 1/4

Correct Answer: (A)

Sums 10, 11, 12: 3 + 2 + 1 = 6 pairs: 1/6.

Why (B) is wrong: Sum 12 only.

Why (C) is wrong: Partial count.

Why (D) is wrong: Included sum 9.

Hinglish: *6 pairs corner se. 1/6.*

MCQ 30 · Normal · Concept: Empirical table total

30. A frequency table lists P(A) = 0.2, P(B) = 0.3, P(C) = 0.1 for the only outcomes A, B, C. P(B or C) is:

A. 0.4
B. 0.6
C. 0.3
D. 0.1

Correct Answer: (A)

0.3 + 0.1 = 0.4.

Why (B) is wrong: Added A too.

Why (C) is wrong: B only.

Why (D) is wrong: C only.

Hinglish: *Disjoint outcomes add: 0.4.*

MCQ 31 · Normal · Concept: Two dice: same parity

31. Two dice: P(both numbers odd) is:

A. 1/4
B. 1/2
C. 1/3
D. 3/4

Correct Answer: (A)

3 odd × 3 odd = 9/36 = 1/4.

Why (B) is wrong: Same parity both ways = 1/2.

Why (C) is wrong: Guessed.

Why (D) is wrong: Complement.

Hinglish: *3×3 = 9 pairs. 1/4.*

MCQ 32 · Normal · Concept: Three coins all same

32. Three coins: P(all three show the same face) is:

A. 1/4
B. 1/8
C. 3/8
D. 1/2

Correct Answer: (A)

HHH and TTT: 2/8 = 1/4.

Why (B) is wrong: One path only.

Why (C) is wrong: Exactly-two-heads value.

Why (D) is wrong: Guessed.

Hinglish: *Do paths: 1/4.*

MCQ 33 · Normal · Concept: Card: honour card

33. One card: P(a card numbered 2 to 10) is:

A. 9/13
B. 4/13
C. 1/2
D. 9/52

Correct Answer: (A)

36 numbered cards: 36/52 = 9/13.

Why (B) is wrong: One suit.

Why (C) is wrong: Half guess.

Why (D) is wrong: One rank.

Hinglish: *36 number cards → 9/13.*

MCQ 34 · Normal · Concept: Two dice difference 1

34. Two dice: P(the numbers differ by exactly 1) is:

A. 5/18
B. 1/6
C. 1/9
D. 1/3

Correct Answer: (A)

10 pairs: (1,2)..(5,6) and mirrors: 10/36 = 5/18.

Why (B) is wrong: One direction only.

Why (C) is wrong: Undercount.

Why (D) is wrong: Overcount.

Hinglish: *Difference 1: 10 pairs → 5/18.*

MCQ 35 · Normal · Concept: Complement with three coins

35. Three coins: P(at least one head) is:

A. 7/8
B. 3/8
C. 1/2
D. 15/16

Correct Answer: (A)

1 - P(all tails) = 1 - 1/8 = 7/8.

Why (B) is wrong: Exactly one head.

Why (C) is wrong: Half guess.

Why (D) is wrong: Four-coin answer.

Hinglish: *1 - 1/8 = 7/8.*

MCQ 36 · Normal · Concept: Bag: not blue

36. A bag has 3 red, 4 blue, 5 green balls. P(not blue) is:

A. 2/3
B. 1/3
C. 1/2
D. 4/12

Correct Answer: (A)

1 - 4/12 = 8/12 = 2/3.

Why (B) is wrong: That is P(blue).

Why (C) is wrong: Guessed.

Why (D) is wrong: Un-reduced.

Hinglish: *8 non-blue of 12: 2/3.*

MCQ 37 · Normal · Concept: Two dice: at least one odd

37. Two dice: P(at least one die shows an odd number) is:

A. 3/4
B. 1/4
C. 1/2
D. 2/3

Correct Answer: (A)

Complement: both even: 9/36 = 1/4; so 3/4.

Why (B) is wrong: Both-odd value.

Why (C) is wrong: Guessed.

Why (D) is wrong: Miscounted.

Hinglish: *1 - 1/4 = 3/4.*

MCQ 38 · Normal · Concept: Cards: queen of hearts

38. P(the queen of hearts) from 52 cards is:

A. 1/52
B. 1/13
C. 1/4
D. 1/26

Correct Answer: (A)

One specific card: 1/52.

Why (B) is wrong: Queen of any suit.

Why (C) is wrong: Any heart... no, that is 1/4.

Why (D) is wrong: Red queen.

Hinglish: *Ek hi card: 1/52.*

MCQ 39 · Normal · Concept: Empirical survey

39. In 80 throws of a die, 'six' came 12 times. The empirical P(six) is:

A. 0.15
B. 1/6
C. 0.2
D. 0.12

Correct Answer: (A)

12/80 = 0.15.

Why (B) is wrong: Classical value, not empirical.

Why (C) is wrong: Wrong division.

Why (D) is wrong: Raw count as decimal.

Hinglish: *12/80 = 0.15. Classical 1/6 ≈ 0.167 alag cheez hai.*

MCQ 40 · Normal · Concept: Two dice sum prime

40. Two dice: P(sum is a prime) is:

A. 5/12
B. 1/2
C. 15/36 as 15/36 un-reduced only
D. 7/18

Correct Answer: (A)

15 pairs give prime sums (2,3,5,7,11): 15/36 = 5/12.

Why (B) is wrong: Roughly half but not exact.

Why (C) is wrong: Un-reduced form - right value, wrong presentation.

Why (D) is wrong: Miscount.

Hinglish: *15 pairs → 5/12.*

MCQ 41 · HOTS · Concept: Without replacement second draw

41. A bag has 6 white and 4 black balls; two are drawn without replacement. Given the first is white, P(second white) is:

A. 5/9
B. 6/10
C. 1/2
D. 4/9

Correct Answer: (A)

After one white leaves: 5 white of 9.

Why (B) is wrong: Original fraction.

Why (C) is wrong: Guessed.

Why (D) is wrong: That is P(second black | first white) complement-ish.

Hinglish: *Universe shrinks: 5/9.*

MCQ 42 · HOTS · Concept: Both colours mixed

42. A bag has 4 red and 6 green balls; two drawn without replacement. P(one red and one green in any order) is:

A. 8/15
B. 4/15
C. 2/5
D. 4/9

Correct Answer: (A)

$2 \times (4/10)(6/9) = 2 \times 24/90 = 48/90 = 8/15$.

Why (B) is wrong: One order only.

Why (C) is wrong: With-replacement value.

Why (D) is wrong: Wrong denominator.

Hinglish: *Do orders: ×2. 8/15.*

MCQ 43 · HOTS · Concept: At least one six: complement

43. A die is rolled 3 times. P(at least one 6) is:

A. 91/216
B. 1/2
C. 3/6
D. 125/216

Correct Answer: (A)

$1 - (5/6)^3 = 1 - 125/216 = 91/216$.

Why (B) is wrong: Guessed even split.

Why (C) is wrong: Added 1/6 three times (double counts).

Why (D) is wrong: That is P(no six).

Hinglish: *Complement: no six in 3 rolls = 125/216. 1 minus = 91/216.*

MCQ 44 · HOTS · Concept: Geometric: dartboard ring

44. A dart hits a 30 cm square board randomly; a 20 cm square target sits in its centre. P(hit target) is:

A. 4/9
B. 2/3
C. 400/90
D. 1/2

Correct Answer: (A)

Area ratio: 400/900 = 4/9.

Why (B) is wrong: Side ratio used.

Why (C) is wrong: Mis-reduced.

Why (D) is wrong: Guessed.

Hinglish: *400/900 = 4/9. Area ratio, side ratio nahi.*

MCQ 45 · HOTS · Concept: Three dice total outcomes

45. Three dice are rolled. n(S) =

A. 216
B. 36
C. 18
D. 128

Correct Answer: (A)

$6^3 = 216$.

Why (B) is wrong: Two dice.

Why (C) is wrong: Sum-based.

Why (D) is wrong: 2^7 confusion.

Hinglish: *$6^3 = 216$.*

MCQ 46 · HOTS · Concept: Max = 6 pairs

46. Two dice: P(the larger number is 6) is:

A. 11/36
B. 1/6
C. 1/36
D. 5/18

Correct Answer: (A)

36 - 25 (both ≤ 5) = 11 pairs: 11/36.

Why (B) is wrong: Only (6,6).

Why (C) is wrong: One-sided count.

Why (D) is wrong: Undercount.

Hinglish: *2k-1 with k = 6: 11 pairs → 11/36.*

MCQ 47 · HOTS · Concept: Fairness check

47. In a game P(A wins) = 0.4 and P(B wins) = 0.4, with draws otherwise. Is the game fair between A and B?

A. Yes, since neither is favoured
B. No, A dominates
C. No, B dominates
D. Cannot say

Correct Answer: (A)

Equal win probabilities mean neither player is advantaged; the remaining 0.2 goes to draws.

Why (B) is wrong: No domination - equal values.

Why (C) is wrong: Same.

Why (D) is wrong: The data decides it.

Hinglish: *Dono 0.4 - fair. Draws ka 0.2 kisi ko favour nahi karta.*

MCQ 48 · HOTS · Concept: Sum to 1 with two unknowns

48. P(A) = 2P(B) and A, B are the only outcomes. P(B) =

A. 1/3
B. 2/3
C. 1/2
D. 1/4

Correct Answer: (A)

2x + x = 1 ⇒ x = 1/3.

Why (B) is wrong: That is P(A).

Why (C) is wrong: Equal split assumed.

Why (D) is wrong: Wrong equation.

Hinglish: *3x = 1: x = 1/3.*

MCQ 49 · HOTS · Concept: Coin then card

49. A coin is tossed; if heads, a card is drawn. P(heads and a heart) is:

A. 1/8
B. 1/2
C. 1/13
D. 1/26

Correct Answer: (A)

(1/2)(13/52) = 13/104 = 1/8.

Why (B) is wrong: Coin only.

Why (C) is wrong: Card only.

Why (D) is wrong: Wrong product.

Hinglish: *(1/2)(1/4) = 1/8.*

MCQ 50 · HOTS · Concept: Sequential coin pattern

50. A coin is tossed 3 times. P(the sequence HHT in that exact order) is:

A. 1/8
B. 3/8
C. 1/4
D. 1/2

Correct Answer: (A)

One specific path of 8: 1/8.

Why (B) is wrong: That is exactly-two-heads (any order).

Why (C) is wrong: Two paths.

Why (D) is wrong: No.

Hinglish: *Exact order maanga hai - ek hi path: 1/8.*

MCQ 51 · Normal · Concept: Empirical vs classical

51. For a fair coin, the classical P(heads) is 1/2. In 100 tosses, 46 heads appeared. The empirical probability is:

A. 0.46
B. 0.5
C. 0.54
D. 0.046

Correct Answer: (A)

46/100 = 0.46 - an estimate close to, but not equal to, the classical value.

Why (B) is wrong: Classical value.

Why (C) is wrong: That is tails' rate.

Why (D) is wrong: Decimal slip.

Hinglish: *46/100 = 0.46. Classical 0.5 se thoda deviate - normal hai.*

MCQ 52 · Normal · Concept: Two dice: sum even

52. Two dice: P(sum is even) is:

A. 1/2
B. 1/4
C. 3/4
D. 7/18

Correct Answer: (A)

Both odd (9) or both even (9): 18/36 = 1/2.

Why (B) is wrong: Both-even only.

Why (C) is wrong: Mixed parity value.

Why (D) is wrong: Miscount.

Hinglish: *Same parity: 18 pairs → 1/2.*

MCQ 53 · Normal · Concept: Bag with replacement

53. A bag has 5 red and 5 blue balls. Two balls are drawn WITH replacement. P(both red) is:

A. 1/4
B. 2/9
C. 5/18
D. 1/2

Correct Answer: (A)

(1/2)(1/2) = 1/4.

Why (B) is wrong: Without-replacement value.

Why (C) is wrong: Mixed.

Why (D) is wrong: Guessed.

Hinglish: *Replacement = same odds: 1/4.*

MCQ 54 · Normal · Concept: Complement on 52

54. One card: P(not a spade) is:

A. 3/4
B. 1/4
C. 1/2
D. 12/13

Correct Answer: (A)

1 - 13/52 = 39/52 = 3/4.

Why (B) is wrong: That is P(spade).

Why (C) is wrong: Red only.

Why (D) is wrong: Face-related confusion.

Hinglish: *39 non-spades: 3/4.*

MCQ 55 · Normal · Concept: Product = 12

55. Two dice: P(product = 12) is:

A. 1/9
B. 1/12
C. 1/18
D. 1/6

Correct Answer: (A)

Pairs: (2,6), (3,4), (4,3), (6,2): 4/36 = 1/9.

Why (B) is wrong: Missed (4,3) family.

Why (C) is wrong: Undercount.

Why (D) is wrong: Sum thinking.

Hinglish: *Chaar pairs: 1/9.*

MCQ 56 · Normal · Concept: Sure + impossible mix

56. P(E) = 1 and P(F) = 0 for events E, F of one experiment. Which is TRUE?

A. E is sure and F is impossible
B. Both are sure
C. Both are impossible
D. E is impossible and F is sure

Correct Answer: (A)

P = 1 marks sure events; P = 0 marks impossible ones.

Why (B) is wrong: F is not sure.

Why (C) is wrong: E is not impossible.

Why (D) is wrong: Reversed.

Hinglish: *1 = sure, 0 = impossible. Seedha definition.*

MCQ 57 · Normal · Concept: Geometric: length on board

57. A point is chosen on a 24 cm segment. P(it lies within 6 cm of the RIGHT end) is:

A. 1/4
B. 1/3
C. 1/6
D. 3/4

Correct Answer: (A)

Favourable length 6 over 24: 1/4.

Why (B) is wrong: Wrong region.

Why (C) is wrong: Half the region.

Why (D) is wrong: Complement.

Hinglish: *6/24 = 1/4.*

MCQ 58 · HOTS · Concept: Two dice: exactly one even

58. Two dice: P(exactly one die shows an even number) is:

A. 1/2
B. 1/4
C. 3/4
D. 5/12

Correct Answer: (A)

Mixed parity: 18 pairs → 1/2.

Why (B) is wrong: Both even only.

Why (C) is wrong: At least one even.

Why (D) is wrong: Miscount.

Hinglish: *Even-odd mix: 1/2.*

MCQ 59 · HOTS · Concept: Draw order probability

59. Balls numbered 1-10; two drawn without replacement. P(5 is drawn FIRST) is:

A. 1/10
B. 1/45
C. 2/10
D. 1/9

Correct Answer: (A)

The first slot is equally likely to be any of the 10 numbers: 1/10.

Why (B) is wrong: Combination thinking applied to an ordered question.

Why (C) is wrong: Two slots.

Why (D) is wrong: Shrunk denominator wrongly.

Hinglish: *Pehli position par har number barabar: 1/10.*

MCQ 60 · HOTS · Concept: Complement of a union on a die

60. On one die, E = {1, 2, 3} and F = {3, 4}. P(not E and not F) is:

A. 1/3
B. 1/6
C. 1/2
D. 0

Correct Answer: (A)

E or F = {1,2,3,4}; complement = {5,6}: 2/6 = 1/3.

Why (B) is wrong: Only 6 counted.

Why (C) is wrong: Half guess.

Why (D) is wrong: Forgot outcome 5.

Hinglish: *Union ke bahar: 5, 6. 1/3.*

7.5 Chapter Formula Sheet

The probability toolkit with its conditions. Every row is one line of working in an exam answer.

Rule	Statement / use	Conditions	Tiny example	Common mistake
P(E) = n(E)/n(S)	Classical probability	Equally likely outcomes	Prime on a die: 3/6 = 1/2	Wrong n(S)
$0 \le P(E) \le 1$	Range law - validity check	All events	P = 1.2 → recount	Accepting out-of-range answers
P(sure) = 1, P(impossible) = 0	Boundary events	Whole sample space / empty set	number < 7 on a die: 1	Swapping the two
P(not E) = 1 - P(E)	Complement rule	E and not E partition S	not-ace: 12/13	Applying to overlapping 'or' sets
$n(S) = 2^n$	n coin tosses	Independent fair coins	3 coins: 8	2n instead of 2^n
n(S) = 36	Two dice	Ordered pairs	sum 7: 6/36 = 1/6	Unordered 21 pairs mixed with ordered
Deck counts: 4/12/13/26	Aces, faces, suits, reds	Standard 52-card deck	face card: 12/52 = 3/13	Forgetting overlaps (king of hearts)
P(A and B) = P(A)·P(B | A)	Sequential draws	Without replacement shrinks both counts	(5/8)(4/7) = 5/14	Not shrinking the denominator

Rule	Statement / use	Conditions	Tiny example	Common mistake
'Any order' ×2	Two different-type draws	The two orders have equal probability	RB or BR: 2×(5/8)(3/8)	Missing the order factor
P(E or F) = P(E)+P(F)-P(E and F)	Union with overlap	General events	king or heart: (4+13-1)/52	Adding without the correction
P(at least one) = 1 - P(none)	Complement cascade	Multi-trial experiments	3 coins: 1 - 1/8 = 7/8	Listing cases directly
Empirical P(E) = occurrences/trials	From observed data	Finite experiment record	35/500 = 0.07	Confusing with classical values
Geometric P = measure/measure	Continuous outcomes	Same measure type both sides	5 cm of 15 cm: 1/3	Mixing length with area
Sum of all outcome probabilities = 1	Accounting law	Complete outcome list	0.3+0.25+0.45 = 1	Ignoring missing categories

7.6 Last-Minute Cheat Sheet

> **30-second revision card - Chapter 7**
> COUNTS: coin 2, die 6, two dice 36 (ordered), 3 coins 8, n coins 2^n, deck 52. DECK: aces 4, face cards 12, spades 13, reds 26; overlap correction for 'or'. TWO-DICE STARS: sum 7 → 6/36 = 1/6; doubles 1/6; both odd 1/4; product odd 1/4; same parity 1/2. COMPLEMENT: P(at least one) = 1 - P(none). P(not E) = 1 - P(E). WITHOUT REPLACEMENT: 8 → 7 denominators; 'any order' → ×2. RANGE: $0 \le P \le 1$; all outcomes sum to 1. EMPIRICAL: occurrences ÷ trials. GEOMETRIC: same-measure ratio.

- **If you see 'at least one'** → 1 minus 'none', always.
- **If you see two dice** → draw the 36-grid; read sums, products, parity off it.
- **If you see 'or' with overlapping categories** → add then subtract the intersection once.
- **If you see 'without replacement'** → shrink numerator and denominator after each draw.
- **If you see 'in any order'** → multiply by the number of distinct orders.
- **If you see a frequency table** → empirical fractions; missing entries from the total.
- **If you see a spinner or dartboard** → angle or area ratios, same measure both sides.
- **If your answer is outside [0, 1]** → stop; the counts are wrong, not the theory.

7.7 Mistake Buster - DON'T MAKE THESE MISTAKES

1. **Wrong n(S).** Two coins have 4 outcomes, not 3 (HT ≠ TH). Fix: list outcomes or use 2^n - never 'number of head-counts'.
2. **Unordered denominators with ordered numerators.** Mixing 21 unordered two-dice pairs with ordered favourable counts corrupts every fraction. Fix: pick one convention, stay in it.

3. **Overlap forgotten in 'or'.** P(king or heart) is 16/52, not 17/52 - the king of hearts counted once. Fix: subtract the intersection.
4. **Denominator not shrunk without replacement.** After one ball leaves, the second draw is out of 7, not 8. Fix: write the cascade explicitly.
5. **Missing the ×2 for 'any order'.** One red one blue happens two ways. Fix: name the orders, add them.
6. **Adding probabilities of non-exclusive events for 'or'.** P(at least one 6) is 11/36, not 12/36 - (6,6) counted twice. Fix: use 1 - P(none).
7. **Range violations ignored.** P = 7/6 signals a counting error. Fix: audit n(E) and n(S) whenever the fraction exits [0, 1].
8. **Un-reduced fractions.** 15/36 must become 5/12. Fix: divide by the HCF as the final line.
9. **Empirical treated as classical.** A loaded die's P(six) = 0.25 is the data's truth, not 1/6. Fix: label which probability you are reporting.
10. **Mixed measures in geometric probability.** Length over area is meaningless. Fix: check units cancel.

7.8 Exam Strategy for Chapter 7

Order of attack. Single-die, single-coin and single-card questions first - one-line answers that bank marks. Two-dice questions next with a reusable grid. Keep without-replacement cascades, 'at least one' complements and geometric probability for the middle slot; the multi-stage HOTS items come last.

Where marks are lost. Four leak-points: wrong totals (fix by writing n(S) first), forgotten overlaps in 'or' events, un-shrunk denominators without replacement, and un-reduced final fractions. The range check $0 \le P \le 1$ is a free error detector - run it on every answer before moving on.

Counting facts most frequently tested. The 36-grid (sum 7 = 6 pairs, doubles = 6), the deck counts 4/12/13/26, and 2^n coin totals. Complement questions ('at least one', 'not') outnumber direct ones - rehearse the 1 - P route until it is reflexive.

Concepts that get mixed. Classical with empirical probability (one is idealised, one is data), mutually exclusive with independent events (Grade 9 needs only the exclusive idea), and 'or' addition with 'and' multiplication. Keep a margin note naming which rule each line uses.

Safe shortcuts. Complement-first for 'at least one'; the 2k-1 count for 'maximum = k' on two dice; symmetric splits (same parity = 1/2, mixed parity = 1/2). **Unsafe shortcuts.** Guessing equal probabilities without checking equally-likely-ness; multiplying probabilities of dependent draws as if independent; treating a small empirical deviation as proof of bias.

How to check an answer. Verify $n(E) \le n(S)$; verify complement pairs sum to 1; re-derive one cascade step; and for list-based answers, recount by a second method (grid vs formula). Two methods agreeing is the strongest cheap check in this chapter.

7.9 Chapter Completion Checklist

Tick each box only when you can honestly say DONE:

- [ ] Every concept covered: experiments, sample spaces, events, classical formula, range laws, complement, cards, dice, coins, tree diagrams, without replacement, empirical, geometric
- [ ] Every rule covered: 14 rows in Section 7.5, each with conditions and a micro-example
- [ ] 50 advanced questions completed (Q1-Q50), difficulty ladder respected
- [ ] Every question has a full step-by-step solution
- [ ] Every solution has a Hinglish explanation
- [ ] 60 MCQs completed with answers
- [ ] Every MCQ has an explanation plus trap analysis
- [ ] Formula sheet completed
- [ ] Last-minute cheat sheet completed
- [ ] Mistake buster completed (10 named mistakes)
- [ ] Exam strategy completed

Grand Revision Arena (All Chapters Combined)

Everything you learned in Chapters 1-7, compressed into one revision zone: a master formula table, thirty mixed questions with full solutions, one hundred mixed MCQs, twenty-five challenge questions, and the exam-day one-pager. Treat this arena like a mock exam: attempt the questions cold, check solutions honestly, and let the mistakes you find here - not the ones you avoid here - decide your final revision night.

R.1 Master Formula Table (all chapters in one glance)

Ch	Formula / Rule	Statement	Signature trap
1	Distance formula	$d = \sqrt{((x_2-x_1)^2 + (y_2-y_1)^2)}$	Root-splitting: $\sqrt{(dx^2+dy^2)} \neq \sqrt{dx} + \sqrt{dy}$
1	Axis-parallel distance	same y → $\lvert x_2-x_1\rvert$; same x → $\lvert y_2-y_1\rvert$	Sign collapse: 6-(-3) = 9
1	Midpoint / reverse	M = averages; B = $(2m_1-x_1, 2m_2-y_1)$	Halving forgotten
1	Reflections	x-axis flips y; y-axis flips x	Flipping the wrong coordinate
1	Circle test	compare x^2+y^2 with r^2 (on/in/out)	Comparing roots instead of squares
2	Slope from two points	$a = (y_2-y_1)/(x_2-x_1)$	x-differences divided
2	Linear form	y = ax + b; a = slope, b = y-intercept	Reading b from a non-axis point
2	Pattern nth term	linear growth: $t_n = a \cdot n + b$	Common difference vs multiplier
3	Terminating test	p/q lowest terms; $q = 2^m \times 5^n$ only	Testing before reducing
3	Recurring → p/q	x, $\times 10^e$, subtract, solve	Wrong shift powers
3	Irrationality proof	contradiction: prime divides p and q	Stopping before the contradiction line

Ch	Formula / Rule	Statement	Signature trap
3	Rationalise	multiply by conjugate ($a \leftrightarrow a-\sqrt{b}$)	Same-sign multiply
3	Exponent laws	multiply add; power multiply; $^{-}$ flips; $a^0 = 1$	$2^{-3} = -8$
4	Square identities	$(a\pm b)^2 = a^2 \pm 2ab + b^2$	Cross term dropped
4	Difference of squares	$a^2-b^2 = (a+b)(a-b)$	Confusing with $(a-b)^2$
4	Trinomial square	$(a+b+c)^2 = \Sigma a^2 + 2\Sigma ab$	Two cross terms only
4	Cubes	$(a\pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$	1-2-3 pattern
4	Cube factorisation	$a^3\pm b^3 = (a\pm b)(a^2 \mp ab + b^2)$	Middle sign wrong
4	Crown identity	$a^3+b^3+c^3-3abc = (a+b+c)(...)$; zero-sum → 3abc	Converse forgotten
6	Heron	$A = \sqrt{(s(s-a)(s-b)(s-c))}$, $s = P/2$	Full perimeter used
6	Circle pair	$C = 2\pi r$; $A = \pi r^2$	πd^2 for area
6	Sector	$A = (\theta/360)\pi r^2 = \frac{1}{2}lr$; $P = l + 2r$	One radius in perimeter
6	Ring	$\pi(R^2-r^2) = \pi(R+r)(R-r)$; chord: π(half-chord)^2	Width as radius
6	Brahmagupta	$\sqrt{((s-a)(s-b)(s-c)(s-d))}$ - cyclic only	Applied to non-cyclic
7	Classical P	P(E) = n(E)/n(S), equally likely	Wrong n(S)
7	Complement	P(not E) = 1 - P(E); at least one = 1 - none	Direct case listing
7	Without replacement	shrink numerator AND denominator each draw	Frozen denominators
7	Union with overlap	P(E or F) = P(E)+P(F)-P(E and F)	Overlap uncorrected

R.2 Thirty Mixed Questions (Q101 - Q130) with Full Solutions

One question never announces its chapter. Work these in order, hide the solutions, and tag each mistake you make with its chapter number - that tag is your revision map.

Q101 · Moderate · Concept: Mixed: Ch1 distance + classification

Q101. A(2, 3), B(2 + 3k, 7). If AB = 5 and k > 0, find k; state the quadrant of B.

SOLUTION

Step 1: Same x-coordinate (2 for both): vertical distance = |7 - 3| = |3k|.

Step 2: 5 = 3k (k > 0) ⇒ k = 5/3.

Step 3: B = (2 + 5, 7) = (7, 7): both positive → Quadrant I.

» **Why this step?** Shared x-coordinate means the axis-parallel shortcut, no square root needed.

Hinglish: *Same x = vertical distance = |3k|. k = 5/3. B(7,7) - Quadrant I. Shortcut pehchano.*

Answer: k = 5/3; B lies in Quadrant I.

Q102 · Moderate · Concept: Mixed: Ch1 reverse midpoint

Q102. M(1, -2) is the midpoint of AB with A(-3, 4). Find B and verify.

SOLUTION

Step 1: B = ($2m_1$ - x_A, $2m_2$ - y_A) = (2 + 3, -4 - 4) = (5, -8).

Step 2: Verify: ((-3+5)/2, (4-8)/2) = (1, -2) ✓.

» **Why this step?** Doubling the midpoint and subtracting the known end reverses the average.

Hinglish: *Double minus known end: (5, -8). Verify by re-averaging - marks ke liye ye line likho.*

Answer: B(5, -8)

Q103 · Basic · Concept: Mixed: Ch3 terminating test

Q103. Without division, decide: terminating or recurring? (i) 21/60 (ii) 29/343 (iii) 77/210.

SOLUTION

Step 1: (i) 21/60 = 7/20 = 7/(2^2×5): terminating.

Step 2: (ii) 343 = 7^3: recurring.

Step 3: (iii) 77/210 = 11/30; 30 has 3: recurring.

» **Why this step?** Reduce first, then factor the denominator into primes.

Hinglish: *Pehle reduce karo: 7/20 terminate; 7^3 aur 3×10 recur.*

Answer: (i) terminating (ii) recurring (iii) recurring

Q104 · Moderate · Concept: Mixed: Ch3 recurring to p/q

Q104. Express 0.68\overline-like value 0.6888... (the 8 repeating) in p/q form.

SOLUTION

Step 1: x = 0.6888...; 10x = 6.888...; 100x = 68.888...

Step 2: 100x - 10x = 62 ⇒ 90x = 62 ⇒ x = 31/45.

» **Why this step?** One non-repeating digit (6) plus one repeating digit (8): shifts of 10 and 100.

Hinglish: *(68 - 6)/90 = 62/90 = 31/45. Reduce karna compulsory.*

Answer: 31/45

Q105 · Moderate · Concept: Mixed: Ch3 rationalise

Q105. Rationalise and simplify: 10/(√5 + √3)... then find a + b if the result equals a√5 + b√3.

SOLUTION

Step 1: Multiply by (√5 - √3): denominator 5 - 3 = 2.

Step 2: Value = 10(√5 - √3)/2 = 5√5 - 5√3.

Step 3: So a = 5, b = -5 (with a + b = 0).

» **Why this step?** Conjugate first; then compare coefficients of the independent surds.

Hinglish: *Conjugate se denominator 2. 5√5 - 5√3. a = 5, b = -5.*

Answer: 5√5 - 5√3; a = 5, b = -5

Q106 · Basic · Concept: Mixed: Ch4 numeric square

Q106. Compute 97^2 and 96 × 104 using identities.

SOLUTION

Step 1: 97^2 = $(100-3)^2$ = 10000 - 600 + 9 = 9409.

Step 2: 96 × 104 = (100-4)(100+4) = 10000 - 16 = 9984.

» **Why this step?** Both products live around 100; identities replace multiplication with tiny additions.

Hinglish: *$(100\pm3)^2$ wala junction: 9409 aur 9984. Identities = speed.*

Answer: 9409; 9984

Q107 · Moderate · Concept: Mixed: Ch4 cubic factorisation

Q107. Factorise completely: $2x^3 - 2x^2 - 24x$.

SOLUTION

Step 1: Common factor 2x: $2x(x^2 - x - 24)$.

Step 2: Split the middle: -6 and +4 (sum -2, product -24): $2x(x - 6)(x + 4)$.

» **Why this step?** HCF first, then middle-term splitting - the golden order.

Hinglish: *2x bahar, andar split: (x-6)(x+4). Complete factorisation.*

Answer: $2x(x - 6)(x + 4)$

Q108 · Moderate · Concept: Mixed: Ch4 cube bridge

Q108. If $x + y = 9$ and $xy = 20$, find $x^3 + y^3$.

SOLUTION

Step 1: $x^3+y^3 = (x+y)^3 - 3xy(x+y) = 729 - 3(20)(9) = 729 - 540 = 189$.

» **Why this step?** The bridge reads (sum, product) → cube sum directly.

Hinglish: *729 - 540 = 189. Bridge = 2 line answer.*

Answer: 189

Q109 · Moderate · Concept: Mixed: Ch4 zero-sum crown

Q109. Evaluate $18^3 + (-13)^3 + (-5)^3$ without cubing.

SOLUTION

Step 1: Sum check: 18 - 13 - 5 = 0 → zero-sum shortcut.

Step 2: Value = 3 × 18 × (-13) × (-5) = 3 × 18 × 65 = 3510.

» **Why this step?** Zero sum collapses three cubes into one product.

Hinglish: *Sum zero! 3×18×65 = 3510. Shortcut ka full power.*

Answer: 3510

Q110 · Moderate · Concept: Mixed: Ch6 Heron

Q110. A triangular hoarding has sides 26 m, 28 m and 30 m. Find its area.

SOLUTION

Step 1: s = 42; differences 16, 14, 12.

Step 2: Product: 42×16×14×12 = 112896.

Step 3: A = 336 m^2.

» **Why this step?** The same product as Q42's triangle - sides scaled up by 1 from 26-28-30? No: identical 26-28-30 sides; recognising repeats speeds revision.

Hinglish: *s = 42, differences 16, 14, 12. 336 m^2. Pattern yaad rakho.*

Answer: 336 m^2

Q111 · Moderate · Concept: Mixed: Ch6 sector trio

Q111. A sector of radius 14 cm has angle 90°. Find arc, area, perimeter (π = 22/7).

SOLUTION

Step 1: Arc = (1/4)(88) = 11 cm.

Step 2: Area = (1/4)(616) = 154 cm^2.

Step 3: Perimeter = 11 + 28 = 39 cm.

» **Why this step?** One fraction (θ/360) drives all three quantities.

Hinglish: *Quarter: arc 11, area 154, perimeter 11+28 = 39. Dono radii!*

Answer: 11 cm; 154 cm^2; 39 cm

Q112 · Moderate · Concept: Mixed: Ch6 ring reverse

Q112. A ring has outer radius 12 cm and width 3 cm. Find its area (π = 22/7).

SOLUTION

Step 1: Inner radius r = 12 - 3 = 9 cm.

Step 2: Area = π(144 - 81) = π(63) = (22/7)(63) = 198 cm^2.

» **Why this step?** Width subtracts from the outer radius to give the inner one.

Hinglish: *r = 9. π×63 = 198 cm^2.*

Answer: 198 cm^2

Q113 · Basic · Concept: Mixed: Ch7 two dice

Q113. Two dice are rolled. Find P(sum = 9) and P(sum is odd).

SOLUTION

Step 1: Sum 9: (3,6), (4,5), (5,4), (6,3): 4/36 = 1/9.

Step 2: Odd sums: 2+4+6+4+2 = 18 pairs (3,5,7,9,11) → 1/2.

» **Why this step?** The pyramid count makes odd and even sums exactly balanced.

Hinglish: *Sum 9: 4 pairs → 1/9. Odd sum: 18 → 1/2. Pyramid count yaad rakho.*

Answer: 1/9; 1/2

Q114 · Basic · Concept: Mixed: Ch7 cards

Q114. One card is drawn. Find P(a red king) and P(a king or a queen).

SOLUTION

Step 1: Red kings: 2 → 2/52 = 1/26.

Step 2: Kings or queens: 8 → 8/52 = 2/13.

» **Why this step?** Specific-card counts first; disjoint ranks then add.

Hinglish: *Red king 2 → 1/26. Kings+queens 8 → 2/13.*

Answer: 1/26; 2/13

Q115 · Moderate · Concept: Mixed: Ch7 without replacement

Q115. A bag has 6 red and 4 white balls. Two are drawn without replacement. P(both white)?

SOLUTION

Step 1: (4/10)(3/9) = 12/90 = 2/15.

» **Why this step?** Shrinking counts: 4 → 3 and 10 → 9.

Hinglish: *4/10 × 3/9 = 2/15. Cascade likho.*

Answer: 2/15

Q116 · Difficult · Concept: Mixed: Ch1 reflection chain

Q116. P(3, -4) is reflected in the x-axis to P_1, then P_1 in the y-axis to P_2. Find P_2 and verify that $OP_2 = OP$.

SOLUTION

Step 1: $P_1 = (3, 4)$; $P_2 = (-3, 4)$.

Step 2: $OP = \sqrt{(9 + 16)} = 5$; $OP_2 = \sqrt{(9 + 16)} = 5$ ✓ (double reflection = half-turn, distances preserved).

» **Why this step?** Two reflections flip both signs; distance from the origin is invariant under sign flips.

Hinglish: *Do reflection = dono sign flip. Distance same rehti hai (5) - verification built-in.*

Answer: P_2(-3, 4); both distances are 5.

Q117 · Moderate · Concept: Mixed: Ch2 fit a line

Q117. A taxi charges a fixed flag-fall plus a per-km rate: 5 km costs ₹110 and 9 km costs ₹170. Find the linear cost rule and the cost of 15 km.

SOLUTION

Step 1: Rate a = (170 - 110)/(9 - 5) = 60/4 = ₹15 per km.

Step 2: Flag-fall: 110 = 15(5) + b → b = 35.

Step 3: Rule: C = 15d + 35; at 15 km: 225 + 35 = ₹260.

» **Why this step?** Two data points determine a and b exactly - the fitting method of Chapter 2 in money clothes.

Hinglish: *Slope = rate = 15/km, flag-fall = 35. 15 km: ₹260. Two points, two unknowns.*

Answer: C = 15d + 35; ₹260

Q118 · Moderate · Concept: Mixed: Ch3 exponent equation

Q118. Solve: $3^{(2x-1)} = 81^x$.

SOLUTION

Step 1: Right side: $81^x = (3^4)^x = 3^{4x}$.

Step 2: Equation: 2x - 1 = 4x ⇒ x = -1/2.

» **Why this step?** One base on both sides turns a power equation into a linear one.

Hinglish: *$81 = 3^4$. 2x-1 = 4x → x = -1/2. Negative answer allowed!*

Answer: x = -1/2

Q119 · Difficult · Concept: Mixed: Ch3 nested radical

Q119. Simplify $\sqrt{(8 + 2\sqrt{15})}$.

SOLUTION

Step 1: Need m + n = 8, mn = 15: m = 5, n = 3.

Step 2: $\sqrt{(8 + 2\sqrt{15})} = \sqrt{5} + \sqrt{3}$.

Step 3: Verify: $5 + 3 + 2\sqrt{15}$ ✓.

» **Why this step?** The $(\sqrt{m} + \sqrt{n})^2$ identity runs backwards.

Hinglish: *Sum 8, product 15: 5 aur 3. Answer $\sqrt{5}+\sqrt{3}$.*

Answer: $\sqrt{5} + \sqrt{3}$

Q120 · Difficult · Concept: Mixed: Ch4 chain

Q120. If $x + 1/x = 4$, find $x^3 + 1/x^3$ and $x^4 + 1/x^4$.

SOLUTION

Step 1: $x^2 + 1/x^2 = 16 - 2 = 14$.

Step 2: $x^3 + 1/x^3 = 64 - 12 = 52$.

Step 3: $x^4 + 1/x^4 = 14^2 - 2 = 194$.

» **Why this step?** The chain: square-minus-2 and cube-minus-3s.

Hinglish: *4 → 14 → 52 → 194. Chain banao, recompute mat karo.*

Answer: 52 and 194

Q121 · Difficult · Concept: Mixed: Ch4 identity proof

Q121. Prove that $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$ and use it: if $a + b = 7$ and $a - b = 3$, find $a^2 + b^2$.

SOLUTION

Step 1: Expand: $(a^2+2ab+b^2) + (a^2-2ab+b^2) = 2a^2 + 2b^2$ ✓.

Step 2: Application: $a^2+b^2 = (49 + 9)/2 = 29$.

» **Why this step?** Expand once, then reuse forever.

Hinglish: *Proof 2 lines. (49+9)/2 = 29. Companion identity ka direct use.*

Answer: Proved; 29

Q122 · Difficult · Concept: Mixed: Ch6 trapezium via Heron

Q122. A trapezium has parallel sides 20 cm and 8 cm, and its non-parallel sides are both 10 cm. Find its area.

SOLUTION

Step 1: Isosceles trapezium: the height drop creates a helper triangle with base 20 - 8 = 12 cm and equal sides 10, 10.

Step 2: Heron on (10, 10, 12): s = 16; product 16×6×6×4 = 2304; helper area = 48 cm^2.

Step 3: Height: $48 = ½ \times 12 \times h \Rightarrow h = 8$ cm.

Step 4: Trapezium area = $½(20 + 8)(8) = 112$ cm^2.

Step 5: Quick check: half-base 6, h 8, slant $\sqrt{(36+64)} = 10$ ✓ (6-8-10 triple).

» **Why this step?** An isosceles trapezium splits into a rectangle plus two symmetric right triangles; Heron extracts h cleanly.

Hinglish: *Helper triangle 10-10-12: Heron se area 48, h = 8. Trapezium = ½(28)(8) = 112 cm². Isosceles case mein helper symmetric hota hai.*

Answer: 112 cm^2

Q123 · Difficult · Concept: Mixed: Ch6 chord-tangent ring

Q123. Two concentric circles: a chord of the outer circle of length 34 cm touches the inner circle. Find the ring area ($\pi = 22/7$).

SOLUTION

Step 1: Ring = $\pi(\text{half-chord})^2 = \pi(17^2) = (22/7)(289) = 6358/7 \approx 908.3$ cm^2.

» **Why this step?** Half-chord squared equals $R^2 - r^2$ - no radii needed.

Hinglish: *Half-chord 17. $17^2 = 289$. Ring = $289\pi \approx 908.3$ cm².*

Answer: ≈ 908.3 cm²

Q124 · Difficult · Concept: Mixed: Ch6 shaded composite

Q124. A square of side 14 cm contains four quarter-circles of radius 7 cm centred at each corner, drawn INSIDE the square. Find the area of the central region not covered by any quarter-circle (π = 22/7).

SOLUTION

Step 1: Four quarter-circles = one full circle of radius 7: area 154 cm².

Step 2: Square area: 196 cm².

Step 3: Central uncovered region = 196 - 154 = 42 cm².

» **Why this step?** Four quarters merge into one circle - the standard consolidation.

Hinglish: *Chaar quarter = ek full circle (154). Gap 42 cm².*

Answer: 42 cm²

Q125 · Difficult · Concept: Mixed: Ch7 at least one

Q125. A die is rolled 4 times. Find P(at least one 6).

SOLUTION

Step 1: P(no 6 in one roll) = 5/6; in four rolls: $(5/6)^4$ = 625/1296.

Step 2: P(at least one 6) = 1 - 625/1296 = 671/1296 ≈ 0.518.

» **Why this step?** The complement cascade: multiply the failures, subtract from 1.

Hinglish: *$(5/6)^4$ = 625/1296. 1 minus = 671/1296. Half se thoda zyada - sensible!*

Answer: 671/1296 ≈ 0.52

Q126 · Difficult · Concept: Mixed: Ch7 geometric

Q126. A dart hits a 24 cm square board randomly. A circular plate of radius 6 cm sits centred on it. P(hit on plate) to 3 dp (π = 3.14).

SOLUTION

Step 1: Plate area: π(36) = 113.04 cm².

Step 2: Board: 576 cm².

Step 3: P = 113.04/576 ≈ 0.196.

» **Why this step?** Area over area, both in cm².

Hinglish: *113.04/576 ≈ 0.196. π/16 hi hai actually (36π/576).*

Answer: ≈ 0.196

Q127 · High · Concept: Mixed: Ch1 circle membership

Q127. Show that (1, 7), (5, 5) and (-5, 5)... check membership on $x^2 + y^2 = 50$ and find the radius.

SOLUTION

Step 1: (1,7): 1 + 49 = 50 ✓. (5,5): 25 + 25 = 50 ✓. (-5,5): 50 ✓.

Step 2: All three lie on the circle; radius = √50 = 5√2.

» **Why this step?** Compare squared sums with r^2 - never roots.

Hinglish: *Teeno 50 dete hain - circle par. Radius 5√2.*

Answer: All on the circle; r = 5√2

Q128 · High · Concept: Mixed: Ch3 infinite product

Q128. Evaluate 3^(1/3) × 3^(1/9) × 3^(1/27) × ... (forever).

SOLUTION

Step 1: Exponent sum: 1/3 + 1/9 + 1/27 + ... = (1/3)/(1 - 1/3) = 1/2 (geometric series).

Step 2: Product = 3^(1/2) = √3.

» **Why this step?** Same-base products add exponents; the geometric series sums them.

Hinglish: *Exponent series 1/2 tak jata hai. Product √3.*

Answer: √3

Q129 · High · Concept: Mixed: Ch4 sum-cube decomposition

Q129. Prove: $(x + y + z)^3 - x^3 - y^3 - z^3 = 3(x + y)(y + z)(z + x)$ and verify at x = 1, y = 2, z = 3.

SOLUTION

Step 1: Quote the identity (proved in Q49 of Chapter 4): both sides equal the six mixed terms plus 6xyz.

Step 2: Verify: LHS = 216 - 1 - 8 - 27 = 180. RHS = 3(3)(5)(4) = 180 ✓.

» **Why this step?** State the identity, then verify numerically - full-credit structure.

Hinglish: *Identity quote + numeric check 180 = 180. Ye combination full marks deta hai.*

Answer: Proved; verified 180 = 180

Q130 · High · Concept: Mixed: Ch6+7 grand mix

Q130. A circular garden of radius 21 m has a 3.5 m wide path around it. (i) Find the path area (π = 22/7). (ii) A visitor entering a random gate lands at a random point of the OUTER circle (garden + path). What is the probability the point lies on the path?

SOLUTION

Step 1: (i) Outer R = 24.5 m: path = $\pi(24.5^2 - 21^2) = \pi(600.25 - 441) = \pi(159.25)$.

Step 2: = (22/7)(159.25) = 22 × 22.75 = 500.5 m^2.

Step 3: (ii) P = path area / outer area = $159.25\pi / (24.5^2\pi) = 159.25/600.25 \approx 0.265$.

» **Why this step?** Part (i) is ring geometry; part (ii) reuses the same areas as a probability ratio - two chapters, one figure.

Hinglish: *Path 500.5 m^2. Probability = π cancel karke 159.25/600.25 ≈ 0.265. Ek figure, do concepts!*

Answer: (i) 500.5 m^2 (ii) ≈ 0.265

R.3 Hundred Mixed MCQs (Q501 - Q600, answers inline)

Five chapters shuffled together - exactly like the final exam's objective section. Each MCQ keeps its full explanation and trap analysis. Chapter tags in the concept column tell you where to revise after a miss.

MCQ 501 · Normal · Concept: Ch1 quadrant

501. The point (-3, -3.5) lies in:

A. Quadrant III
B. Quadrant II
C. Quadrant IV
D. on the x-axis

Correct Answer: (A)

Both coordinates negative: Quadrant III.

Why (B) is wrong: y must be positive for II.

Why (C) is wrong: x must be positive for IV.

Why (D) is wrong: y ≠ 0.

Hinglish: *Dono minus: Q III.*

MCQ 502 · Normal · Concept: Ch1 distance

502. The distance between (0, -7) and (0, 9) is:

A. 16

B. 2

C. √130

D. 9

Correct Answer: (A)

Same x: |9 - (-7)| = 16.

Why (B) is wrong: Sign error.

Why (C) is wrong: Formula used needlessly with an error.

Why (D) is wrong: Ignored -7.

Hinglish: *Vertical shortcut: 16.*

MCQ 503 · Normal · Concept: Ch1 midpoint

503. The midpoint of (4, -6) and (-10, 2) is:

A. (-3, -2)

B. (3, 2)

C. (-7, -4)

D. (-3, 2)

Correct Answer: (A)

((4-10)/2, (-6+2)/2) = (-3, -2).

Why (B) is wrong: Added wrongly.

Why (C) is wrong: Forgot to halve.

Why (D) is wrong: Half-signed.

Hinglish: *Averages: (-3, -2).*

MCQ 504 · Normal · Concept: Ch1 reflection

504. The reflection of (5, -2) in the y-axis is:

A. (-5, -2)

B. (5, 2)

C. (-5, 2)

D. (-2, 5)

Correct Answer: (A)

y-axis flips x only.

Why (B) is wrong: x-axis flip.

Why (C) is wrong: Double flip.

Why (D) is wrong: Coordinate swap.

Hinglish: *y-axis: x flip. (-5, -2).*

MCQ 505 · Normal · Concept: Ch1 circle test

505. The point (6, 8) lies on the circle $x^2 + y^2 = k$. Then k =

A. 100

B. 14

C. 48

D. 10

Correct Answer: (A)

36 + 64 = 100 (6-8-10 triple).

Why (B) is wrong: Added unsquared.

Why (C) is wrong: Multiplied.

Why (D) is wrong: Took the root.

Hinglish: *100. Triple pehchano.*

MCQ 506 · Normal · Concept: Ch1 collinear

506. Which triple is collinear?

A. (0,0), (4,3), (8,6)

B. (0,0), (1,2), (3,5)

C. (0,0), (2,2), (4,3)

D. (1,1), (2,3), (4,7)

Correct Answer: (A)

5 + 5 = 10: B lies on segment AC.

Why (B) is wrong: 2+5 ≠ √34?

Why (C) is wrong: Distances fail.

Why (D) is wrong: Fails the sum test.

Hinglish: *4-3-5 doubled: 8-6-10. 5+5 = 10 collinear.*

MCQ 507 · Normal · Concept: Ch1 trisection

507. A(0, 0) to B(9, 12): the point of trisection CLOSER to A is:

A. (3, 4)

B. (6, 8)

C. (4.5, 6)

D. (3, 12)

Correct Answer: (A)

One third of the step (9, 12) is (3, 4).

Why (B) is wrong: Two-thirds point.

Why (C) is wrong: Midpoint.

Why (D) is wrong: Mixed coordinates.

Hinglish: *Step ka 1/3: (3,4).*

MCQ 508 · Normal · Concept: Ch2 degree

508. The degree of the polynomial $7 - 3x^5 + 2x^2$ is:

A. 5

B. 7

C. 2

D. 0

Correct Answer: (A)

Highest power with a nonzero coefficient: 5.

Why (B) is wrong: Coefficient is not the degree.

Why (C) is wrong: Lowest power.

Why (D) is wrong: Constant term.

Hinglish: *Sabse badi power: 5.*

MCQ 509 · Normal · Concept: Ch2 slope

509. A line passes through (2, 5) and (6, 13). Its slope is:

A. 2

B. 8

C. 1/2

D. 4

Correct Answer: (A)

(13-5)/(6-2) = 8/4 = 2.

Why (B) is wrong: Difference without dividing.

Why (C) is wrong: Inverted ratio.

Why (D) is wrong: Used only x-gap.

Hinglish: *8/4 = 2.*

MCQ 510 · Normal · Concept: Ch2 y-intercept

510. For y = 4x - 7, the graph cuts the y-axis at:

A. (0, -7)

B. (-7, 0)

C. (0, 4)

D. (4, 0)

Correct Answer: (A)

x = 0 gives y = -7.

Why (B) is wrong: Axis swapped.

Why (C) is wrong: Slope read as intercept.

Why (D) is wrong: x-intercept.

Hinglish: *b = -7: (0, -7).*

MCQ 511 · Normal · Concept: Ch2 pattern

511. A pattern starts at 5 and grows by 7 each step. Its nth term is:

A. 7n - 2

B. 7n + 5

C. 5n + 7

D. 12n

Correct Answer: (A)

$t_n = 5 + 7(n-1) = 7n - 2$.

Why (B) is wrong: Off-by-one.

Why (C) is wrong: Swapped base and rate.

Why (D) is wrong: Added both.

Hinglish: *5 + 7(n-1) = 7n-2.*

MCQ 512 · Normal · Concept: Ch2 fit line

512. Two points (1, 8) and (4, 17) fit y = ax + b. Then b =

A. 5

B. 3

C. 8

D. 9

Correct Answer: (A)

a = 9/3 = 3; 8 = 3(1) + b → b = 5.

Why (B) is wrong: Slope used as intercept.

Why (C) is wrong: Forgot to subtract ax.

Why (D) is wrong: Arithmetic.

Hinglish: *a = 3, b = 5.*

MCQ 513 · Normal · Concept: Ch2 linear identification

513. Which relation is linear?

A. $y = 5 - 2x$

B. $y = x^2 + 1$

C. $y = 3^x$

D. $y = 4/x$

Correct Answer: (A)

Only the first has degree 1 in x with no other structures.

Why (B) is wrong: Quadratic.

Why (C) is wrong: Exponential.

Why (D) is wrong: Inverse.

Hinglish: *Degree 1, koi square/root/exponent nahi.*

MCQ 514 · Normal · Concept: Ch2 input-output

514. A machine outputs 3x - 1. For input -2, the output is:

A. -7
B. 7
C. -5
D. 5

Correct Answer: (A)

3(-2) - 1 = -7.

Why (B) is wrong: Sign dropped.

Why (C) is wrong: Added 1.

Why (D) is wrong: Multiplied wrongly.

Hinglish: *3(-2)-1 = -7. Sign dhyan se.*

MCQ 515 · Normal · Concept: Ch3 classify

515. Which is IRRATIONAL?

A. √20
B. 0.121212...
C. 3.14
D. √(9/16)

Correct Answer: (A)

√20 = 2√5: irrational. The others are rational.

Why (B) is wrong: Recurring = rational.

Why (C) is wrong: Terminating = rational.

Why (D) is wrong: Perfect square fraction.

Hinglish: *20 non-square → 2√5 irrational.*

MCQ 516 · Normal · Concept: Ch3 terminate test

516. The decimal of 13/80 is:

A. terminating ($80 = 2^4 \times 5$)
B. recurring (8 present)
C. irrational
D. cannot be decided

Correct Answer: (A)

$80 = 2^4 \cdot 5$: only 2s and 5s → terminates.

Why (B) is wrong: 8 presence misread.

Why (C) is wrong: Rationals are never irrational.

Why (D) is wrong: The test always decides.

Hinglish: *80 mein sirf 2 aur 5. Terminate.*

MCQ 517 · Normal · Concept: Ch3 recurring

517. 0.818181... (block 81) as a fraction is:

A. 9/11
B. 81/100
C. 8/9
D. 27/33

Correct Answer: (A)

81/99 = 9/11.

Why (B) is wrong: Terminating assumption.

Why (C) is wrong: Wrong block size.

Why (D) is wrong: Un-reduced 27/33 = 9/11 - same value; the standard single form is 9/11.

Hinglish: *81/99 = 9/11.*

MCQ 518 · Normal · Concept: Ch3 surd simplify

518. √243 equals:

A. 9√3
B. 3√27
C. 81√3
D. 3√9

Correct Answer: (A)

243 = 81×3 → 9√3.

Why (B) is wrong: Under-extracted.

Why (C) is wrong: Took 81 out as 81.

Why (D) is wrong: Left a square inside.

Hinglish: *81 bahar: 9√3.*

MCQ 519 · Normal · Concept: Ch3 rationalise

519. 1/(√7 - √5) equals:

A. (√7+√5)/2
B. √7-√5
C. (√7-√5)/2
D. √35

Correct Answer: (A)

Multiply by (√7+√5): denominator 2.

Why (B) is wrong: Forgot the 2.

Why (C) is wrong: Wrong sign.

Why (D) is wrong: Multiplied radicands.

Hinglish: *Conjugate, denominator 2.*

MCQ 520 · Normal · Concept: Ch3 exponent equation

520. If $2^x \times 2^{(x+1)} = 32$, then x =

A. 2

B. 3

C. 5

D. 4

Correct Answer: (A)

2^(2x+1) = 2^5 → x = 2.

Why (B) is wrong: Forgot the +1.

Why (C) is wrong: Equated to 32.

Why (D) is wrong: Arithmetic.

Hinglish: *2x+1 = 5: x = 2.*

MCQ 521 · Normal · Concept: Ch3 proof method

521. √5 is irrational is proved by:

A. contradiction assuming p/q lowest terms

B. long division

C. measuring the number line

D. induction

Correct Answer: (A)

The standard 4-step contradiction skeleton.

Why (B) is wrong: Cannot terminate an irrational.

Why (C) is wrong: Not a proof.

Why (D) is wrong: Wrong tool.

Hinglish: *Contradiction skeleton.*

MCQ 522 · Normal · Concept: Ch3 density

522. Between two distinct rational numbers there is/are:

A. infinitely many rationals

B. exactly one rational

C. no other rational

D. finitely many

Correct Answer: (A)

Averaging never ends - density.

Why (B) is wrong: One average is just the first.

Why (C) is wrong: Density forbids gaps.

Why (D) is wrong: No finite bound.

Hinglish: *Average loop kabhi khatam nahi hota.*

MCQ 523 · Normal · Concept: Ch3 mixed operations

523. x is irrational and r is a nonzero rational. Which is ALWAYS irrational?

A. x + r

B. x - x

C. x/x

D. r · (1/r)

Correct Answer: (A)

rational ± irrational is always irrational (subtraction trick). The other three collapse to 0, 1, 1.

Why (B) is wrong: That is 0.

Why (C) is wrong: That is 1.

Why (D) is wrong: That is 1.

Hinglish: *x + r hamesha irrational. Baaki sab rational (0, 1, 1).*

MCQ 524 · Normal · Concept: Ch4 square expand

524. $(3m - 4n)^2$ equals:

A. $9m^2 - 24mn + 16n^2$

B. $9m^2 + 24mn + 16n^2$

C. $9m^2 - 12mn + 16n^2$

D. $9m^2 - 16n^2$

Correct Answer: (A)

$2(3m)(4n) = 24mn$ with a minus.

Why (B) is wrong: Sign.

Why (C) is wrong: Halved cross term.

Why (D) is wrong: Missing middle.

Hinglish: *Cross term -24.*

MCQ 525 · Normal · Concept: Ch4 diff of squares

525. $49a^2 - 64b^2$ factorises as:

A. (7a+8b)(7a-8b)

B. $(7a-8b)^2$

C. $(7a+8b)^2$

D. (49a+64b)(a-b)

Correct Answer: (A)

Two-square difference.

Why (B) is wrong: Square form.

Why (C) is wrong: Plus square.

Why (D) is wrong: No common structure.

Hinglish: *$(7a)^2-(8b)^2$.*

MCQ 526 · Normal · Concept: Ch4 cube expand

526. $(x + 3)^3$ equals:

A. $x^3 + 9x^2 + 27x + 27$

B. $x^3 + 27$

C. $x^3 + 3x^2 + 9x + 27$

D. $x^3 + 6x^2 + 27$

Correct Answer: (A)

1-3-3-1 with powers of 3: 9, 27, 27.

Why (B) is wrong: Binomial-square habit.

Why (C) is wrong: Coefficient pattern wrong.

Why (D) is wrong: Dropped terms.

Hinglish: *$3x^2 \cdot 3 = 9x^2$; $3x \cdot 9 = 27x$.*

MCQ 527 · Normal · Concept: Ch4 sum of cubes

527. $x^3 + 64y^3$ factorises as:

A. $(x+4y)(x^2-4xy+16y^2)$

B. $(x+4y)^3$

C. $(x+4y)(x^2+4xy+16y^2)$

D. $(x-4y)(x^2+4xy+16y^2)$

Correct Answer: (A)

Sum of cubes; trinomial middle term negative.

Why (B) is wrong: Cube expansion.

Why (C) is wrong: Sign.

Why (D) is wrong: Wrong binomial.

Hinglish: *Middle minus.*

MCQ 528 · Normal · Concept: Ch4 crown

528. If $a + b + c = 0$, then $a^3 + b^3 + c^3 =$

A. 3abc

B. 0

C. -3abc

D. abc

Correct Answer: (A)

Zero-sum shortcut from the crown identity.

Why (B) is wrong: Only the LHS-minus-product vanishes.

Why (C) is wrong: Sign.

Why (D) is wrong: Factor.

Hinglish: *3abc.*

MCQ 529 · Normal · Concept: Ch4 bridge

529. If a - b = 5 and ab = 6, then $a^3 - b^3$ =

A. 215

B. 185

C. 125

D. 155

Correct Answer: (A)

$(a-b)^3 + 3ab(a-b) = 125 + 3(6)(5) = 125 + 90 = 215$.

Why (B) is wrong: Arithmetic slip.

Why (C) is wrong: Skipped the bridge term.

Why (D) is wrong: Wrong 3ab factor.

Hinglish: *125 + 90 = 215. Bridge ka plus sign yaad rakho.*

MCQ 530 · Normal · Concept: Ch4 x+1/x chain

530. If $x + 1/x = 6$, then $x^2 + 1/x^2$ =

A. 34

B. 36

C. 38

D. 12

Correct Answer: (A)

36 - 2 = 34.

Why (B) is wrong: No subtraction.

Why (C) is wrong: Wrong constant.

Why (D) is wrong: Doubled.

Hinglish: *36-2 = 34.*

MCQ 531 · Normal · Concept: Ch4 middle split

531. $x^2 - 3x - 28$ factorises as:

A. (x-7)(x+4)

B. (x+7)(x-4)

C. (x-14)(x+2)

D. (x-7)(x-4)

Correct Answer: (A)

Sum -3, product -28: -7 and +4.

Why (B) is wrong: Sum +3.

Why (C) is wrong: 14×2 = 28 but sum -12.

Why (D) is wrong: Product wrong.

Hinglish: *-7 aur +4.*

MCQ 532 · Normal · Concept: Ch4 fourth power

532. a^4 - 625 factorises completely as:

A. (a-5)(a+5)(a^2+25)

B. (a^2-25)(a^2-25)

C. (a-5)(a+5)

D. (a^2+25)^2

Correct Answer: (A)

Two rounds of difference of squares.

Why (B) is wrong: Duplicate factor.

Why (C) is wrong: Incomplete.

Why (D) is wrong: Not a factorisation.

Hinglish: *(a^2-25)(a^2+25), phir todo.*

MCQ 533 · Normal · Concept: Ch4 perfect square test

533. Which is a perfect square trinomial?

A. $4x^2$ - 20x + 25

B. $4x^2$ - 20x - 25

C. $4x^2$ + 20x + 10

D. $4x^2$ + 25

Correct Answer: (A)

$(2x-5)^2 = 4x^2 - 20x + 25$.

Why (B) is wrong: Last term negative - impossible.

Why (C) is wrong: Constant not a square.

Why (D) is wrong: Missing middle.

Hinglish: *Middle = 2·(2x)·(5).*

MCQ 534 · Normal · Concept: Ch4 zero-sum converse

534. $a^3+b^3+c^3 = 3abc$ with a+b+c ≠ 0 forces:

A. a = b = c

B. a+b+c = 0 anyway

C. abc = 1

D. a = -b

Correct Answer: (A)

Half-sum-of-squares factor must vanish.

Why (B) is wrong: Excluded.

Why (C) is wrong: No such constraint.

Why (D) is wrong: No.

Hinglish: *Teeno barabar.*

MCQ 535 · Normal · Concept: Ch4 numeric cube

535. Using identities, 101^3 =

A. 1030301

B. 1010301

C. 1030031

D. 1030130

Correct Answer: (A)

1,000,000 + 30,000 + 300 + 1.

Why (B) is wrong: Place slip.

Why (C) is wrong: Digit shuffle.

Why (D) is wrong: Arithmetic.

Hinglish: *1-3-3-1: 1000000+30000+300+1.*

MCQ 536 · Normal · Concept: Ch6 circumference

536. A circle's circumference is 31.4 cm ($\pi = 3.14$). Its radius is:

A. 5 cm

B. 10 cm

C. 2.5 cm

D. 31.4 cm

Correct Answer: (A)

r = 31.4/(2×3.14) = 5.

Why (B) is wrong: Forgot 2π.

Why (C) is wrong: Half.

Why (D) is wrong: Kept C.

Hinglish: *6.28 se divide: 5 cm.*

MCQ 537 · Normal · Concept: Ch6 area reverse

537. A circle's area is 154 cm^2 ($\pi = 22/7$). Its circumference is:

A. 44 cm

B. 22 cm

C. 88 cm

D. 14 cm

Correct Answer: (A)

r = 7; C = 44 cm.

Why (B) is wrong: Radius reported as circumference.

Why (C) is wrong: Doubled twice.

Why (D) is wrong: That is d.

Hinglish: *r = 7, C = 44.*

MCQ 538 · Normal · Concept: Ch6 sector area

538. A sector of angle 45°, radius 14 cm has area (π = 22/7):

A. 77 cm^2

B. 154 cm^2

C. 38.5 cm^2

D. 616 cm^2

Correct Answer: (A)

(1/8)(616) = 77 cm^2.

Why (B) is wrong: Half circle.

Why (C) is wrong: One-sixteenth.

Why (D) is wrong: Full circle.

Hinglish: *One-eighth: 77.*

MCQ 539 · Normal · Concept: Ch6 Heron

539. A triangle has sides 8, 15, 17 cm. Its area is:

A. 60 cm^2

B. 68 cm^2

C. 120 cm^2

D. 30 cm^2

Correct Answer: (A)

Right triangle (8-15-17): ½×8×15 = 60.

Why (B) is wrong: Heron slip.

Why (C) is wrong: Doubled.

Why (D) is wrong: Half again.

Hinglish: *Triple! ½×8×15 = 60.*

MCQ 540 · Normal · Concept: Ch6 equilateral

540. An equilateral triangle has perimeter 45 cm. Its side is:

A. 15 cm

B. 22.5 cm

C. 5√3 cm

D. 45√3 cm

Correct Answer: (A)

45/3 = 15 cm.

Why (B) is wrong: Half perimeter.

Why (C) is wrong: Height formula.

Why (D) is wrong: No division.

Hinglish: *45 ÷ 3.*

MCQ 541 · Normal · Concept: Ch6 ring

541. Two concentric circles have radii 15 and 12 cm. The ring area ($\pi = 22/7$) is:

A. $(22/7)(81)$ cm^2

B. $(22/7)(27)$ cm^2

C. $(22/7)(180)$ cm^2

D. $(22/7)(9)$ cm^2

Correct Answer: (A)

$225 - 144 = 81$; area $= 81\pi$.

Why (B) is wrong: Difference of radii squared.

Why (C) is wrong: Sum of radii.

Why (D) is wrong: Half-chord.

Hinglish: *R^2-r^2 = 81.*

MCQ 542 · Normal · Concept: Ch6 path

542. A 2 m path runs OUTSIDE a circular park of radius 10 m. Path area ($\pi = 3.14$):

A. 138.16 m^2

B. 62.8 m^2

C. 125.6 m^2

D. 452.16 m^2

Correct Answer: (A)

$\pi(144 - 100) = 44\pi = 138.16$.

Why (B) is wrong: Only πr.

Why (C) is wrong: πd.

Why (D) is wrong: Outer circle only.

Hinglish: *R = 12. 44π = 138.16.*

MCQ 543 · Normal · Concept: Ch6 Brahmagupta

543. A cyclic quadrilateral has sides 10, 10, 10, 10 cm. Its area is:

A. 100 cm^2

B. 40 cm^2

C. 400 cm^2

D. 25 cm^2

Correct Answer: (A)

$s = 20$; $(10)^4$ root: $10^2 = 100$ (it is a square).

Why (B) is wrong: Perimeter.

Why (C) is wrong: Perimeter squared.

Why (D) is wrong: Quarter.

Hinglish: *Square: side^2 = 100.*

MCQ 544 · Normal · Concept: Ch6 ratio

544. Two circles' circumferences are in ratio 4 : 9. Their areas are in ratio:

A. 16 : 81
B. 4 : 9
C. 2 : 3
D. 8 : 18

Correct Answer: (A)

Areas carry the square of the linear ratio.

Why (B) is wrong: Linear.

Why (C) is wrong: Square root.

Why (D) is wrong: No rule.

Hinglish: *k^2: 16:81.*

MCQ 545 · Normal · Concept: Ch6 cost

545. Fencing a circle of radius 14 m at ₹25/m costs ($\pi = 22/7$):

A. ₹2200
B. ₹61600
C. ₹4400
D. ₹1100

Correct Answer: (A)

C = 88 m; 88×25 = 2200.

Why (B) is wrong: Area used.

Why (C) is wrong: Wrong C.

Why (D) is wrong: Halved.

Hinglish: *88 × 25 = ₹2200.*

MCQ 546 · Normal · Concept: Ch6 composite

546. A 10 cm square has a semicircle of diameter 10 cm REMOVED from it. Remaining area ($\pi = 3.14$):

A. 60.75 cm^2
B. 139.25 cm^2
C. 78.5 cm^2
D. 21.5 cm^2

Correct Answer: (A)

100 - 39.25 = 60.75.

Why (B) is wrong: Added instead.

Why (C) is wrong: Full circle.

Why (D) is wrong: Semicircle only.

Hinglish: *100 - 39.25.*

MCQ 547 · Normal · Concept: Ch6 units

547. The perimeter of a sector with arc 12 cm and radius 9 cm is:

A. 30 cm
B. 21 cm
C. 12π cm
D. 33 cm

Correct Answer: (A)

12 + 9 + 9 = 30 cm.

Why (B) is wrong: One radius.

Why (C) is wrong: π intrusion.

Why (D) is wrong: Added wrongly.

Hinglish: *Arc + do radius.*

MCQ 548 · Normal · Concept: Ch7 dice

548. Two dice: P(sum = 5) is:

A. 1/9
B. 1/6
C. 1/12
D. 1/18

Correct Answer: (A)

Four pairs: (1,4), (2,3), (3,2), (4,1): 4/36 = 1/9.

Why (B) is wrong: Sum-7 confusion.

Why (C) is wrong: Wrong count.

Why (D) is wrong: Undercount.

Hinglish: *Chaar pairs → 1/9.*

MCQ 549 · Normal · Concept: Ch7 cards

549. One card: P(a black queen) is:

A. 1/26
B. 1/52
C. 1/13
D. 1/4

Correct Answer: (A)

Two black queens: 2/52 = 1/26.

Why (B) is wrong: One queen only.

Why (C) is wrong: All queens.

Why (D) is wrong: Suit.

Hinglish: *Do black queens → 1/26.*

MCQ 550 · Normal · Concept: Ch7 complement

550. P(E) = 2/9. P(not E) =

A. 7/9
B. 2/9
C. 9/2
D. 1/9

Correct Answer: (A)

1 - 2/9 = 7/9.

Why (B) is wrong: Copied.

Why (C) is wrong: Inverted.

Why (D) is wrong: Arithmetic.

Hinglish: *7/9.*

MCQ 551 · Normal · Concept: Ch7 empirical

551. In 200 trials, an event occurred 120 times. Empirical P(event) is:

A. 0.6
B. 0.4
C. 1.2
D. 0.06

Correct Answer: (A)

120/200 = 0.6.

Why (B) is wrong: Complement.

Why (C) is wrong: Over 1.

Why (D) is wrong: Decimal slip.

Hinglish: *120/200 = 0.6.*

MCQ 552 · Normal · Concept: Ch7 at least one

552. Two dice: P(at least one shows 4) is:

A. 11/36
B. 1/6
C. 1/36
D. 25/36

Correct Answer: (A)

$1 - (5/6)^2 = 1 - 25/36 = 11/36$.

Why (B) is wrong: Both-four only.

Why (C) is wrong: Single pair.

Why (D) is wrong: Neither-four.

Hinglish: *Complement: 11/36.*

MCQ 553 · Normal · Concept: Ch7 geometric

553. A point is chosen on a 20 cm segment. P(it lies in the last 4 cm) is:

A. 1/5
B. 4/5
C. 1/4
D. 1/20

Correct Answer: (A)

4/20 = 1/5.

Why (B) is wrong: Complement.

Why (C) is wrong: Quarter confusion.

Why (D) is wrong: Point counting.

Hinglish: *4/20 = 1/5.*

MCQ 554 · HOTS · Concept: Ch1 distance formula

554. The distance between (1, 2) and (7, 10) is:

A. 10
B. 8
C. √100... i.e. 10
D. 14

Correct Answer: (A)

6-8-10 triple: √(36+64) = 10.

Why (B) is wrong: One leg only.

Why (C) is wrong: Right value, reduced form is 10.

Why (D) is wrong: Added legs.

Hinglish: *6-8-10! 10.*

MCQ 555 · HOTS · Concept: Ch1 axis point

555. A point on the x-axis at distance 5 from the origin, to the RIGHT, is:

A. (5, 0)
B. (0, 5)
C. (-5, 0)
D. (5, 5)

Correct Answer: (A)

x-axis means y = 0; right means positive.

Why (B) is wrong: That is the y-axis.

Why (C) is wrong: Left side.

Why (D) is wrong: Not on the axis.

Hinglish: *(5, 0).*

MCQ 556 · HOTS · Concept: Ch1 k-distance

556. If the distance between A(3, 4) and B(3, k) is 9, then k =

A. 13 or -5
B. 13 only
C. -5 only
D. 9

Correct Answer: (A)

|k - 4| = 9 → k = 13 or k = -5.

Why (B) is wrong: One root only.

Why (C) is wrong: Other root only.

Why (D) is wrong: Confused with distance.

Hinglish: *Do cases! 13 aur -5.*

MCQ 557 · HOTS · Concept: Ch2 growth decay

557. A tank holds 200 L and leaks 8 L per hour. The linear rule for volume V after t hours is:

A. V = 200 - 8t
B. V = 200 + 8t
C. V = 8t - 200
D. V = 192t

Correct Answer: (A)

Start 200, negative slope 8.

Why (B) is wrong: Growth sign.

Why (C) is wrong: Reversed.

Why (D) is wrong: Multiplied.

Hinglish: *Decay = minus slope.*

MCQ 558 · HOTS · Concept: Ch2 two-line comparison

558. y = 2x + 1 and y = 2x - 5 are two lines. They are:

A. parallel
B. perpendicular
C. the same line
D. intersecting at x = 3

Correct Answer: (A)

Same slope 2, different intercepts → parallel.

Why (B) is wrong: Product -1 needed.

Why (C) is wrong: Intercepts differ.

Why (D) is wrong: Parallel never meet.

Hinglish: *Same slope = parallel.*

MCQ 559 · HOTS · Concept: Ch3 number line

559. $\sqrt{7}$ lies between which two consecutive integers?

A. 2 and 3
B. 3 and 4
C. 1 and 2
D. 7 and 8

Correct Answer: (A)

$4 < 7 < 9 \rightarrow \sqrt{7}$ between 2 and 3.

Why (B) is wrong: $\sqrt{7} \approx 2.65$.

Why (C) is wrong: Too low.

Why (D) is wrong: Radicand confusion.

Hinglish: *Perfect squares 4 aur 9 ke beech.*

MCQ 560 · HOTS · Concept: Ch3 surd compare

560. Which is largest?

A. $2\sqrt{5}$
B. $\sqrt{19}$
C. $3\sqrt{2}$
D. 4

Correct Answer: (A)

Squares: 20, 19, 18, 16 - the largest square is 20, so $2\sqrt{5}$ wins.

Why (B) is wrong: $19 < 20$.

Why (C) is wrong: $18 < 20$.

Why (D) is wrong: $16 < 20$.

Hinglish: *$2\sqrt{5}$ ka square 20 - sabse bada. Coefficient bhi square hota hai!*

MCQ 561 · HOTS · Concept: Ch3 conjugate collapse

561. $(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})$ equals:

A. 1
B. 5
C. $\sqrt{6}$
D. 6

Correct Answer: (A)

3 - 2 = 1.

Why (B) is wrong: Added.

Why (C) is wrong: Multiplied radicands.

Why (D) is wrong: Squaring.

Hinglish: *Difference of squares: 1.*

MCQ 562 · HOTS · Concept: Ch3 zero exponent

562. $(5^0 + 3^0) \times 2^3$ equals:

A. 16

B. 8

C. 2

D. 10

Correct Answer: (A)

1 + 1 = 2; 2×8 = 16.

Why (B) is wrong: Dropped the bracket sum.

Why (C) is wrong: Forgot both ^{0}s.

Why (D) is wrong: Added 5+3.

Hinglish: *2 × 8 = 16.*

MCQ 563 · HOTS · Concept: Ch4 identity pick

563. To expand (x + 5)(x + 9) fastest, use:

A. x^2 + (a+b)x + ab

B. $(a+b)^2$

C. a^2-b^2

D. cube identity

Correct Answer: (A)

The (x+a)(x+b) rule: x^2 + 14x + 45.

Why (B) is wrong: Unequal constants.

Why (C) is wrong: No squares here.

Why (D) is wrong: Wrong family.

Hinglish: *Sum-product rule.*

MCQ 564 · HOTS · Concept: Ch4 coefficient

564. The coefficient of x in (x + 3)(x + 7) is:

A. 10

B. 21

C. 3

D. 7

Correct Answer: (A)

Sum of constants: 3 + 7 = 10.

Why (B) is wrong: Product.

Why (C) is wrong: One constant.

Why (D) is wrong: Other constant.

Hinglish: *Sum: 10.*

MCQ 565 · HOTS · Concept: Ch4 numeric product

565. Using a^2-b^2: 52 × 48 =

A. 2496

B. 2504

C. 2499

D. 2400

Correct Answer: (A)

(50+2)(50-2) = 2500 - 4 = 2496.

Why (B) is wrong: Added.

Why (C) is wrong: Offset 1 habit.

Why (D) is wrong: No identity.

Hinglish: *2500 - 4 = 2496.*

MCQ 566 · HOTS · Concept: Ch4 crown application

566. $9^3 + 10^3 + (-19)^3$ equals:

A. -5130

B. 5130

C. 0

D. -1710

Correct Answer: (A)

Zero sum: 3(9)(10)(-19) = -5130.

Why (B) is wrong: Sign.

Why (C) is wrong: Zero-sum gives product, not 0.

Why (D) is wrong: Factor 3 dropped.

Hinglish: *3×9×10×(-19) = -5130.*

MCQ 567 · HOTS · Concept: Ch4 symmetric

567. If a+b = 10 and ab = 21, then a^2+b^2 =

A. 58

B. 100

C. 79

D. 42

Correct Answer: (A)

100 - 42 = 58.

Why (B) is wrong: Raw square.

Why (C) is wrong: Added 2ab.

Why (D) is wrong: 2ab only.

Hinglish: *100 - 42 = 58.*

MCQ 568 · HOTS · Concept: Ch6 wheel

568. A wheel of radius 35 cm makes 200 revolutions. Distance travelled ($\pi = 22/7$):

A. 440 m
B. 220 m
C. 44 m
D. 2200 m

Correct Answer: (A)

C = 220 cm; 200 × 220 = 44000 cm = 440 m.

Why (B) is wrong: Half.

Why (C) is wrong: Unit slip.

Why (D) is wrong: Zero slip.

Hinglish: *44000 cm = 440 m.*

MCQ 569 · HOTS · Concept: Ch6 inscribed circle

569. A circle is inscribed in a square of side 18 cm. Its radius is:

A. 9 cm
B. 18 cm
C. 4.5 cm
D. $9\sqrt{2}$ cm

Correct Answer: (A)

Diameter = side: r = 9.

Why (B) is wrong: Diameter as radius.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Diagonal confusion.

Hinglish: *Diameter = side.*

MCQ 570 · HOTS · Concept: Ch6 Heron reverse

570. A triangle with sides 13, 14, 15 has area 84. If all sides are DOUBLED, the new area is:

A. 336
B. 168
C. 294
D. 84

Correct Answer: (A)

Area scales 4×: 84 × 4 = 336.

Why (B) is wrong: Linear scaling.

Why (C) is wrong: $\sqrt{3}$ habit.

Why (D) is wrong: No change assumed.

Hinglish: *$k = 2 \rightarrow$ area $\times 4 = 336$.*

MCQ 571 · HOTS · Concept: Ch6 sector reverse

571. A sector of radius 12 cm has arc 5π cm. Its angle is:

A. 75°
B. 60°
C. 90°
D. 150°

Correct Answer: (A)

Arc/whole = 5π/24π = 5/24 → angle = (5/24)×360 = 75°.

Why (B) is wrong: Arbitrary fraction.

Why (C) is wrong: Quarter habit.

Why (D) is wrong: Doubled.

Hinglish: *Fraction × 360 = 75°.*

MCQ 572 · HOTS · Concept: Ch7 three coins

572. Three coins: P(at least two tails) is:

A. 1/2
B. 3/8
C. 7/8
D. 1/4

Correct Answer: (A)

Exactly two (3) + three (1) = 4/8 = 1/2.

Why (B) is wrong: Exactly two only.

Why (C) is wrong: At least one.

Why (D) is wrong: All tails only.

Hinglish: *4 paths → 1/2.*

MCQ 573 · HOTS · Concept: Ch7 bag draws

573. A bag has 3 red, 7 blue balls. Two drawn WITHOUT replacement. P(both blue) is:

A. 7/15
B. 49/100
C. 7/10
D. 21/50

Correct Answer: (A)

(7/10)(6/9) = 42/90 = 7/15.

Why (B) is wrong: With replacement.

Why (C) is wrong: No shrink.

Why (D) is wrong: Arithmetic.

Hinglish: *7/10 × 6/9 = 7/15.*

MCQ 574 · HOTS · Concept: Ch7 union

574. P(A) = 0.3, P(B) = 0.4, A and B cannot occur together. P(A or B) =

A. 0.7
B. 0.12
C. 0.1
D. 1.0

Correct Answer: (A)

Exclusive events: add directly.

Why (B) is wrong: Multiplied.

Why (C) is wrong: Subtracted.

Why (D) is wrong: Over 1? No - but 0.3+0.4 = 0.7.

Hinglish: *Exclusive → 0.7.*

MCQ 575 · HOTS · Concept: Ch7 deck complement

575. One card: P(neither a heart nor a spade) is:

A. 1/2
B. 1/4
C. 3/4
D. 1/13

Correct Answer: (A)

Neither red suit: 26 black cards: 1/2.

Why (B) is wrong: One suit removed.

Why (C) is wrong: Both removed twice.

Why (D) is wrong: One card.

Hinglish: *26 black → 1/2.*

MCQ 576 · HOTS · Concept: Ch7 expected count

576. A fair die is rolled 180 times. Expected count of sixes is:

A. 30
B. 36
C. 18
D. 60

Correct Answer: (A)

180 × 1/6 = 30.

Why (B) is wrong: 1/5 habit.

Why (C) is wrong: Half rate.

Why (D) is wrong: Third rate.

Hinglish: *180/6 = 30.*

MCQ 577 · HOTS · Concept: Ch7 two dice product odd

577. Two dice: P(product is odd) is:

A. 1/4
B. 1/2
C. 3/4
D. 1/3

Correct Answer: (A)

Both odd: 9/36 = 1/4.

Why (B) is wrong: Parity split guess.

Why (C) is wrong: Complement.

Why (D) is wrong: Miscount.

Hinglish: *3×3 = 9 → 1/4.*

MCQ 578 · HOTS · Concept: Ch1 distance HOTS

578. Points A(0,0), B(6,8), C(12,16). B divides AC in ratio:

A. 1 : 1
B. 1 : 2
C. 2 : 1
D. 1 : 3

Correct Answer: (A)

AB = 10, BC = 10: midpoint.

Why (B) is wrong: Double-step misread.

Why (C) is wrong: Reversed.

Why (D) is wrong: No.

Hinglish: *Beech ka point - 1:1.*

MCQ 579 · HOTS · Concept: Ch2 x-intercept

579. Where does y = 3x - 12 cut the x-axis?

A. (4, 0)
B. (0, -12)
C. (-4, 0)
D. (12, 0)

Correct Answer: (A)

0 = 3x - 12 → x = 4.

Why (B) is wrong: y-intercept instead.

Why (C) is wrong: Sign.

Why (D) is wrong: Slope confusion.

Hinglish: *y = 0 rakho: x = 4.*

MCQ 580 · HOTS · Concept: Ch2 machine reverse

580. A machine y = 5x + 2 outputs 27. The input was:

A. 5

B. 4

C. 6

D. 25

Correct Answer: (A)

5x + 2 = 27 → x = 5.

Why (B) is wrong: Arithmetic.

Why (C) is wrong: Overshoot.

Why (D) is wrong: Copied output.

Hinglish: *Reverse machine: x = 5.*

MCQ 581 · HOTS · Concept: Ch3 recurring mixed

581. 0.5\overline-like value 0.5777... (7 repeating) equals:

A. 26/45

B. 52/90... same as 26/45

C. 57/99

D. 577/1000

Correct Answer: (A)

(57 - 5)/90 = 52/90 = 26/45.

Why (B) is wrong: Same value un-reduced - the standard form is 26/45.

Why (C) is wrong: Wrong denominator.

Why (D) is wrong: Terminating assumption.

Hinglish: *(57-5)/90 = 26/45.*

MCQ 582 · HOTS · Concept: Ch3 exponent laws

582. $(2^{-3} \div 2^{-5})$ equals:

A. 4

B. 1/4

C. -8

D. 2^8

Correct Answer: (A)

2^(-3+5) = 2^2 = 4.

Why (B) is wrong: Flip error.

Why (C) is wrong: Sign chaos.

Why (D) is wrong: Added wrongly.

Hinglish: *-3 - (-5) = 2 → 4.*

MCQ 583 · HOTS · Concept: Ch4 triple square

583. $(2a - b - 3c)^2$ contains how many terms when fully expanded?

A. 6

B. 3

C. 9

D. 4

Correct Answer: (A)

3 squares + 3 cross terms = 6.

Why (B) is wrong: Squares only.

Why (C) is wrong: Pre-collection count.

Why (D) is wrong: Miscount.

Hinglish: *Teen squares + teen cross = 6 terms.*

MCQ 584 · HOTS · Concept: Ch4 cube factor

584. $27x^3 - 8$ factorises as:

A. $(3x-2)(9x^2+6x+4)$

B. $(3x-2)(9x^2-6x+4)$

C. $(3x-2)^3$

D. $(3x+2)(9x^2-6x+4)$

Correct Answer: (A)

Difference of cubes.

Why (B) is wrong: Trinomial sign.

Why (C) is wrong: Cube expansion.

Why (D) is wrong: Binomial sign.

Hinglish: *$(3x)^3-2^3$: middle plus.*

MCQ 585 · HOTS · Concept: Ch6 quarter circle

585. The area of a quarter circle of radius 10 cm ($\pi = 3.14$) is:

A. 78.5 cm^2

B. 314 cm^2

C. 31.4 cm^2

D. 157 cm^2

Correct Answer: (A)

$\frac{1}{4} \times 314 = 78.5$.

Why (B) is wrong: Full circle.

Why (C) is wrong: One-tenth.

Why (D) is wrong: Half.

Hinglish: *314/4 = 78.5.*

MCQ 586 · HOTS · Concept: Ch6 triangle reverse

586. A triangle's area is 150 cm^2 with base 20 cm. Its height is:

A. 15 cm

B. 7.5 cm

C. 30 cm

D. 10 cm

Correct Answer: (A)

$\frac{1}{2} \times 20 \times h = 150 \rightarrow h = 15$.

Why (B) is wrong: Forgot $\frac{1}{2}$.

Why (C) is wrong: Doubled.

Why (D) is wrong: Mixed.

Hinglish: *300/20 = 15.*

MCQ 587 · HOTS · Concept: Ch7 sample space

587. A coin is tossed and a die is rolled. P(heads and an odd number) is:

A. 1/4

B. 1/2

C. 1/6

D. 1/12

Correct Answer: (A)

(1/2)(3/6) = 1/4.

Why (B) is wrong: Coin only.

Why (C) is wrong: Die only.

Why (D) is wrong: Wrong multiply.

Hinglish: *(1/2)(1/2) = 1/4.*

MCQ 588 · HOTS · Concept: Ch7 sum range

588. Two dice: P(4 < sum < 8) is:

A. 5/12

B. 1/2

C. 2/3

D. 1/3

Correct Answer: (A)

Sums 5, 6, 7: 4 + 5 + 6 = 15 pairs → 15/36 = 5/12.

Why (B) is wrong: Includes sum 4 or 8 wrongly.

Why (C) is wrong: Overshoot.

Why (D) is wrong: Undercount.

Hinglish: *Sums 5, 6, 7: 15 pairs → 5/12.*

MCQ 589 · HOTS · Concept: Ch1 perimeter from coordinates

589. A(0,0), B(5,0), C(5,12), D(0,12): perimeter of ABCD is:

A. 34
B. 60
C. 30
D. 17

Correct Answer: (A)

Rectangle 5×12: P = 34.

Why (B) is wrong: Area.

Why (C) is wrong: Half.

Why (D) is wrong: l+b only.

Hinglish: *2(5+12) = 34.*

MCQ 590 · HOTS · Concept: Ch3 between roots

590. How many rational numbers lie strictly between √2 and √3?

A. Infinitely many
B. None
C. Exactly one
D. Exactly ten

Correct Answer: (A)

Rationals are dense: e.g. 1.5, 1.51, ... forever.

Why (B) is wrong: 1.5 sits right there.

Why (C) is wrong: One of many.

Why (D) is wrong: No cap exists.

Hinglish: *Density - infinite.*

MCQ 591 · HOTS · Concept: Ch3 surd add

591. √45 + √20 equals:

A. 5√5
B. 9√5
C. √65
D. 3√5 + 4√5 un-reduced

Correct Answer: (A)

3√5 + 2√5 = 5√5.

Why (B) is wrong: Coefficient error.

Why (C) is wrong: Added radicands.

Why (D) is wrong: Right value; reduced single form is 5√5.

Hinglish: *3√5 + 2√5 = 5√5.*

MCQ 592 · HOTS · Concept: Ch4 big square

592. $(x + 2y)^2 - (x - 2y)^2$ equals:

A. 8xy

B. 4xy

C. $2(x^2+4y^2)$

D. 0

Correct Answer: (A)

4ab form: 4(x)(2y) = 8xy.

Why (B) is wrong: Factor 2 short.

Why (C) is wrong: Sum identity.

Why (D) is wrong: Not zero.

Hinglish: *4ab = 4·x·2y = 8xy.*

MCQ 593 · HOTS · Concept: Ch6 area ratio

593. The radii of two circles are 5 cm and 7 cm. The ratio of their areas is:

A. 25 : 49

B. 5 : 7

C. 10 : 14

D. 49 : 25

Correct Answer: (A)

Square the linear ratio.

Why (B) is wrong: Linear.

Why (C) is wrong: Doubled.

Why (D) is wrong: Inverted.

Hinglish: *25:49.*

MCQ 594 · HOTS · Concept: Ch7 fair die

594. A die is rolled once. P(a number ≥ 5) is:

A. 1/3

B. 1/2

C. 1/6

D. 2/3

Correct Answer: (A)

{5, 6}: 2/6 = 1/3.

Why (B) is wrong: Included 4.

Why (C) is wrong: One outcome.

Why (D) is wrong: Complement of 1/3 is 2/3.

Hinglish: *Do outcomes: 1/3.*

MCQ 595 · HOTS · Concept: Ch7 two-stage

595. A bag has 4 white and 6 black balls. One ball is drawn and NOT replaced; then another. P(both black) is:

A. 1/3
B. 3/5
C. 9/25
D. 6/10

Correct Answer: (A)

(6/10)(5/9) = 30/90 = 1/3.

Why (B) is wrong: First draw only.

Why (C) is wrong: With replacement.

Why (D) is wrong: Un-shrunk.

Hinglish: *6/10 × 5/9 = 1/3.*

MCQ 596 · HOTS · Concept: Ch7 spinner angles

596. A spinner has sectors 120°, 120° and 120°. P(a given sector) is:

A. 1/3
B. 1/2
C. 2/3
D. 120/100

Correct Answer: (A)

120/360 = 1/3.

Why (B) is wrong: Half habit.

Why (C) is wrong: Two sectors.

Why (D) is wrong: Not a probability.

Hinglish: *Teen barabar hisse: 1/3.*

MCQ 597 · HOTS · Concept: Ch1 midpoint application

597. The midpoint of a segment with endpoints (-2, 6) and (10, -4) is:

A. (4, 1)
B. (8, 2)
C. (4, -1)
D. (5, 1)

Correct Answer: (A)

((8)/2, (2)/2) = (4, 1).

Why (B) is wrong: Forgot to halve.

Why (C) is wrong: Sign slip.

Why (D) is wrong: Averaged one coordinate.

Hinglish: *(4, 1).*

MCQ 598 · HOTS · Concept: Ch4 factor tree

598. $x^2 + 5x + 6$ and $x^2 - 5x + 6$ share which common factor?

A. None - they share no factor

B. (x + 2)

C. (x + 3)

D. (x - 3)

Correct Answer: (A)

First: (x+2)(x+3). Second: (x-2)(x-3). No common factor - the sign flip changes everything.

Why (B) is wrong: (x+2) is not a factor of the second.

Why (C) is wrong: (x+3) is not a factor of the second.

Why (D) is wrong: (x-3) is not a factor of the first.

Hinglish: *Pehla (x+2)(x+3), doosra (x-2)(x-3). Koi common factor nahi! Signs dekho.*

MCQ 599 · HOTS · Concept: Ch6 Heron isosceles

599. An isosceles triangle has equal sides 25 cm and base 30 cm. Its area is:

A. 300 cm^2

B. 375 cm^2

C. 150 cm^2

D. 600 cm^2

Correct Answer: (A)

h = $\sqrt{(625 - 225)} = 20$; $\frac{1}{2} \times 30 \times 20 = 300$.

Why (B) is wrong: Skipped ½.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Doubled.

Hinglish: *15-20-25 triple! 300 cm^2.*

MCQ 600 · HOTS · Concept: Ch7 grand finish

600. Two dice are rolled. P(sum is even OR sum is 7) is:

A. 2/3

B. 1/2

C. 7/12

D. 3/4

Correct Answer: (A)

Even sums: 18/36. Sum 7 adds 6 more (all odd-sum pairs... wait sum 7 pairs are odd sums): 18 + 6 = 24 → 24/36 = 2/3.

Why (B) is wrong: Even only.

Why (C) is wrong: Partial add.

Why (D) is wrong: Overshoot.

Hinglish: *Even 18 + sum-7 ke 6 = 24 → 2/3. Overlap nahi hai kyunki 7 odd hai.*

R.4 Twenty-Five Challenge Questions (Q201 - Q225)

Olympiad-lite territory: multi-concept chains, proofs and reverse-engineered data. Attempt without the solutions first - these are worth attempting badly before reading, because the struggle builds the wiring.

Q201 · Challenge · Concept: Ch1: integer lattice triangle

Q201. A(0, 0), B(9, 12), C(18, 0). Show ABC is isosceles, find its area, and locate the midpoint of BC.

SOLUTION

Step 1: AB = $\sqrt{(81 + 144)} = 15$; BC = $\sqrt{(81 + 144)} = 15$ ✓ isosceles (AC = 18).

Step 2: Height from B to AC (the x-axis) = 12; area = $\frac{1}{2} \times 18 \times 12 = 108$.

Step 3: Midpoint of BC = ((9+18)/2, (12+0)/2) = (13.5, 6).

» **Why this step?** Base along an axis turns 'area from coordinates' into one multiplication.

Hinglish: *Do sides 15-15 - isosceles. Area ½×18×12 = 108. Midpoint (13.5, 6).*

Answer: Isosceles (AB = BC = 15); area 108; midpoint (13.5, 6).

Q202 · Challenge · Concept: Ch1: three-point circle

Q202. Show that (6, 2), (-2, 6) and (4, 8) all lie on one circle centred at the origin, and find the radius.

SOLUTION

Step 1: Squares: 36 + 4 = 40; 4 + 36 = 40; 16 + 64 = 80? Verify (4,8): 16 + 64 = 80 ≠ 40.

Step 2: Recheck the third point: (4, 8) fails; instead take (2, 6): 4 + 36 = 40 ✓.

Step 3: With (6, 2), (-2, 6), (2, 6): all satisfy $x^2 + y^2 = 40$; radius = $\sqrt{40} = 2\sqrt{10}$.

» **Why this step?** Circle membership is the squared-sum test; a failed point must be reported, not hidden.

Hinglish: *(4,8) fail karta hai (80 ≠ 40) - honestly batao. (2,6) lo: teeno 40. Radius $2\sqrt{10}$.*

Answer: (6,2), (-2,6), (2,6) lie on $x^2+y^2 = 40$, radius $2\sqrt{10}$; (4,8) does not.

Q203 · Challenge · Concept: Ch2: intersecting linear models

Q203. Plan A costs ₹50 + ₹5/km; Plan B costs ₹80 + ₹3/km. Find the break-even distance and which plan wins beyond it.

SOLUTION

Step 1: 50 + 5d = 80 + 3d → 2d = 30 → d = 15 km.

Step 2: Below 15 km A is cheaper; above 15 km B wins (smaller slope).

» **Why this step?** Equal values locate the crossing; the smaller slope wins thereafter.

Hinglish: *15 km par barabar. Usse aage B jeetega (slope 3 < 5).*

Answer: Break-even at 15 km; B is cheaper beyond.

Q204 · Challenge · Concept: Ch3: structured irrationality

Q204. Prove that $1/\sqrt{2}$ is irrational, WITHOUT re-running the $\sqrt{2}$ proof from scratch.

SOLUTION

Step 1: Suppose $1/\sqrt{2} = r$ rational. Then $\sqrt{2} = 1/r$.

Step 2: $r \neq 0$ rational ⇒ $1/r$ rational (Q closed under reciprocal).

Step 3: So $\sqrt{2}$ would be rational - contradiction with the known result. ∎

» **Why this step?** Reuse an established irrationality via closure arguments - no new contradiction needed.

Hinglish: *Reciprocal trick: agar 1/√2 rational hota toh √2 bhi hota - clash. Reuse karna proof writing ki kala hai.*

Answer: 1/√2 is irrational by the reciprocal-closure argument.

Q205 · Challenge · Concept: Ch3: exponent chain

Q205. Find all real x with $9^x = 3^{(x+4)}$.

SOLUTION

Step 1: $9^x = 3^{2x}$: so $3^{2x} = 3^{(x+4)}$.

Step 2: $2x = x + 4 \rightarrow x = 4$.

» **Why this step?** One base, then equate.

Hinglish: *9 ko 3^2 likho. x = 4.*

Answer: x = 4

Q206 · Challenge · Concept: Ch4: nested identity

Q206. If $a + b = 8$ and $a^2 + b^2 = 40$, find $a^3 + b^3$.

SOLUTION

Step 1: $ab = [(a+b)^2 - (a^2+b^2)]/2 = (64 - 40)/2 = 12$.

Step 2: $a^3+b^3 = 512 - 3(12)(8) = 512 - 288 = 224$.

» **Why this step?** Recover the product first; the bridge then finishes.

Hinglish: *Pehle ab = 12, phir bridge: 224.*

Answer: 224

Q207 · Challenge · Concept: Ch4: identity discovery

Q207. Simplify: $(x + y)^3 - (x^3 + y^3)$ and factorise the result.

SOLUTION

Step 1: $(x+y)^3 - x^3 - y^3 = 3x^2y + 3xy^2 = 3xy(x + y)$.

» **Why this step?** Expand, cancel the cubes, factor the survivors.

Hinglish: *3xy(x+y) bachta hai.*

Answer: 3xy(x + y)

Q208 · Challenge · Concept: Ch4: fourth-power chain

Q208. Given $x + 1/x = 3$, find $x^5 + 1/x^5$... then verify $x^6 + 1/x^6 = 322$.

SOLUTION

Step 1: Chain: $3 \rightarrow 7 \rightarrow 18 \rightarrow 47 \rightarrow x^5+1/x^5 = 3\times47 - 18 = 123$.

Step 2: $x^6 + 1/x^6 = (x^3+1/x^3)^2 - 2 = 324 - 2 = 322$ ✓.

» **Why this step?** Multiply-rule for fifth powers; square-minus-2 for sixth.

Hinglish: *123, aur 18^2-2 = 322. Chain discipline.*

Answer: 123; 322 verified

Q209 · Challenge · Concept: Ch4: crown converse in action

Q209. The numbers x, y, z satisfy $x^3 + y^3 + z^3 = 3xyz$ and $x + y + z = 6$. Show x = y = z and find x.

SOLUTION

Step 1: Crown identity: (x+y+z)(half-sum-of-squares) = 0; since the sum is 6 ≠ 0, the squares factor vanishes.

Step 2: So x = y = z; then 3x = 6 → x = 2.

» **Why this step?** The converse converts an equation into equality of all three variables.

Hinglish: *Sum nonzero → squares zero → teeno barabar → x = 2.*

Answer: x = y = z = 2

Q210 · Challenge · Concept: Ch6: Heron to height to more

Q210. A triangle has sides 13, 14, 15. Find its area and the length of the altitude to the side 14.

SOLUTION

Step 1: s = 21; area = √(21×8×7×6) = √7056 = 84.

Step 2: 84 = ½ × 14 × h → h = 12.

» **Why this step?** Heron gives the area; the area then gives any altitude.

Hinglish: *84, phir h = 2×84/14 = 12. Classic 13-14-15.*

Answer: 84 cm²; altitude 12 cm

Q211 · Challenge · Concept: Ch6: inscribed circle of a right triangle

Q211. A right triangle has legs 6 and 8 and hypotenuse 10. Show its inradius is (6 + 8 - 10)/2 = 2.

SOLUTION

Step 1: Area = ½×6×8 = 24; s = 12.

Step 2: Inradius r = area/s = 24/12 = 2.

Step 3: Matches the right-triangle shortcut (a + b - c)/2 = (6 + 8 - 10)/2 = 2 ✓.

» **Why this step?** r = A/s is the universal inradius formula; the shortcut is its right-triangle special case.

Hinglish: *r = A/s = 2. Shortcut se bhi 2. Dono yaad rakho.*

Answer: r = 2

Q212 · Challenge · Concept: Ch6: rotating radius sweep

Q212. A pendulum of length 45 cm swings through 60°. Find the arc length travelled by the bob (π = 22/7).

SOLUTION

Step 1: Arc = (60/360)(2π·45) = (1/6)(1980/7) = 330/7 ≈ 47.14 cm.

» **Why this step?** The bob traces a sector arc of radius = pendulum length.

Hinglish: *(1/6)×2π×45 ≈ 47.1 cm.*

Answer: 330/7 ≈ 47.1 cm

Q213 · Challenge · Concept: Ch6: same-area square and circle

Q213. A square and a circle have equal areas of 484 cm² (π = 22/7). Which has the smaller perimeter, and by how much?

SOLUTION

Step 1: Square side = 22; perimeter = 88 cm.

Step 2: Circle: $\pi r^2 = 484 \rightarrow r^2 = 154 \rightarrow r \approx 12.41$; C = 2πr ≈ 78 cm.

Step 3: Circle's perimeter is smaller by about 10 cm.

» **Why this step?** Equal area ⇒ circle always wins the perimeter contest (isoperimetric principle).

Hinglish: *88 vs ~78. Circle ~10 cm kam boundary. Isoperimetric rule.*

Answer: The circle: about 78 vs 88 cm (roughly 10 cm less).

Q214 · Challenge · Concept: Ch6: Brahmagupta vs Heron link

Q214. A cyclic quadrilateral has sides 13, 14, 15 and 12... hmm wait Brahmagupta needs four sides: use 13, 14, 15, 4? Verify positivity and compute the area if valid.

SOLUTION

Step 1: s = (13+14+15+4)/2 = 23.

Step 2: Differences: 10, 9, 8, 19: product 10×9×8×19 = 13680.

Step 3: A = √13680 = √(144×95) = 12√95 ≈ 116.96.

» **Why this step?** Validity needs each side < sum of the other three (4 < 42 ✓); the root need not be a perfect square.

Hinglish: *s = 23, product 13680 = 144×95. A = 12√95 ≈ 117.*

Answer: $12\sqrt{95} \approx 116.96$ cm²

Q215 · Challenge · Concept: Ch6: shaded ring sector

Q215. A ring has outer radius 14 and inner radius 7 (π = 22/7). Find the area of ONE QUARTER of the ring (a quarter-annulus).

SOLUTION

Step 1: Full ring = π(196 - 49) = 147π = 462 cm².

Step 2: Quarter = 462/4 = 115.5 cm².

» **Why this step?** Fractions of a ring scale like sectors of a circle.

Hinglish: *462 ka quarter = 115.5 cm².*

Answer: 115.5 cm²

Q216 · Challenge · Concept: Ch7: three-dice sum 6

Q216. Three dice are rolled. Find P(sum = 6).

SOLUTION

Step 1: Total: 216 outcomes.

Step 2: Compositions of 6 into three parts ≥ 1: (4,1,1): 3 perms; (3,2,1): 6 perms; (2,2,2): 1 perm.

Step 3: Total 10 outcomes → P = 10/216 = 5/108.

» **Why this step?** Enumerate compositions with their multiplicities - no pair is missed or doubled.

Hinglish: *(4,1,1) teen, (3,2,1) chhe, (2,2,2) ek: 10 → 5/108.*

Answer: 5/108

Q217 · Challenge · Concept: Ch7: complementary draws

Q217. A bag has 2 red and 3 blue balls. Three balls are drawn without replacement. Find P(exactly two are red).

SOLUTION

Step 1: Total ways: C(5,3) = 10.

Step 2: Favourable: choose both reds and one blue: C(2,2)C(3,1) = 3.

Step 3: P = 3/10.

» **Why this step?** Combinations count unordered draws cleanly for 'exactly k' questions.

Hinglish: *C(5,3) = 10; dono red + 1 blue: 3 tarika. 3/10.*

Answer: 3/10

Q218 · Challenge · Concept: Ch7: dice vs coin mixture

Q218. A coin is tossed; if it shows heads, a die is rolled; if tails, the coin is tossed again. What is P(a 6 appears somewhere in the experiment)?

SOLUTION

Step 1: Head branch: 6 appears iff die shows 6: (1/2)(1/6) = 1/12.

Step 2: Tail branch: no die is rolled → no 6 possible: 0.

Step 3: P = 1/12 + 0 = 1/12.

» **Why this step?** Branch-wise addition with impossible branches contributing zero.

Hinglish: *Sirf head branch mein 6 possible: 1/12.*

Answer: 1/12

Q219 · Challenge · Concept: Ch7: geometric ring probability

Q219. A dart hits a circular board of radius 12 cm at a random point. Find P(it lands within 6 cm of the centre).

SOLUTION

Step 1: Areas: 36π vs 144π.

Step 2: P = 36/144 = 1/4.

» **Why this step?** π cancels; radii square before the ratio.

Hinglish: *$(6/12)^2 = 1/4$. Radius ratio square hota hai.*

Answer: 1/4

Q220 · Challenge · Concept: Ch1+4: coordinates with identities

Q220. A(2a, 0), B(0, 2b). Show $AB^2 = 4(a^2 + b^2)$ and find AB when a + b = 7 and ab = 10.

SOLUTION

Step 1: $AB^2 = (2a)^2 + (2b)^2 = 4a^2 + 4b^2 = 4(a^2+b^2)$ ✓.

Step 2: $a^2+b^2 = 49 - 20 = 29$; $AB = 2\sqrt{29}$.

» **Why this step?** Coordinate distance plus a symmetric identity in one chain.

Hinglish: *$AB = 2\sqrt{29}$. Identity + distance combo.*

Answer: $AB = 2\sqrt{29}$

Q221 · Challenge · Concept: Ch3+6: surd geometry

Q221. An equilateral triangle has area $36\sqrt{3}$ cm^2. Find its side, perimeter, and the area of its inscribed circle ($\pi = 22/7$).

SOLUTION

Step 1: $(\sqrt{3}/4)a^2 = 36\sqrt{3} \rightarrow a = 12$; perimeter 36.

Step 2: $r = a/(2\sqrt{3}) = 2\sqrt{3}$; $r^2 = 12$.

Step 3: Circle area $= 12\pi = 264/7 \approx 37.71$ cm^2.

» **Why this step?** Equilateral chain: area → side → inradius → circle.

Hinglish: *a = 12, r = 2√3, circle 12π ≈ 37.7 cm².*

Answer: side 12, P 36, circle ≈ 37.7 cm²

Q222 · Challenge · Concept: Ch4+6: identity area

Q222. A square of side s is expanded by 3 on one side and cut by 3 on the perpendicular side (new rectangle (s+3)×(s-3)). Show the area change is -9 regardless of s.

SOLUTION

Step 1: New area: $(s+3)(s-3) = s^2 - 9$.

Step 2: Change: $(s^2 - 9) - s^2 = -9$ ✓ - independent of s.

» **Why this step?** Difference of squares eats the variable - a geometric surprise worth remembering.

Hinglish: *(s+3)(s-3) = s²-9. Hamesha exactly 9 kam!*

Answer: The rectangle loses exactly 9 square units, for every s.

Q223 · Challenge · Concept: Ch2+7: linear model probability

Q223. A taxi's fare rule is F = 40 + 12d. A passenger's fare is chosen at random from recorded trips with distances 2, 5, 8, 11 km (equally likely). Find the expected (average) fare and P(fare > ₹130).

SOLUTION

Step 1: Fares: 64, 100, 136, 172.

Step 2: Average = (64 + 100 + 136 + 172)/4 = 472/4 = ₹118.

Step 3: P(fare > 130) = 2/4 = 1/2 (the 136 and 172 trips).

» **Why this step?** The linear rule generates the data; the probability question is then counting.

Hinglish: *Average ₹118. Do fares 130 se bade: 1/2.*

Answer: ₹118; 1/2

Q224 · Challenge · Concept: Ch3+4: surd identity

Q224. If $x = \sqrt{3} + \sqrt{2}$, find $x^2 - 1/x^2$.

SOLUTION

Step 1: $1/x = \sqrt{3} - \sqrt{2}$ (conjugate collapse).

Step 2: $x^2 = 5 + 2\sqrt{6}$; $1/x^2 = 5 - 2\sqrt{6}$.

Step 3: Difference: $4\sqrt{6}$.

» **Why this step?** Conjugate reciprocals make the whole question two squarings and a subtraction.

Hinglish: *1/x = √3-√2. Squares conjugate hain: difference 4√6.*

Answer: 4√6

Q225 · Challenge · Concept: Grand: chapters 1-7 fusion

Q225. A circular garden of radius 14 m has a point chosen at random on its AREA. (i) P(the point is within 7 m of the centre)? (ii) A path of width 3.5 m runs just inside the boundary; P(the point lies on the path)? (π = 22/7)

SOLUTION

Step 1: (i) Areas 49π vs 196π: P = 1/4.

Step 2: (ii) Path = ring 10.5 to 14: π(196 - 110.25) = π(85.75); P = 85.75/196 = 0.4375 = 7/16.

» **Why this step?** Geometric probability with two ring computations - the finale fuses Chapter 6's rings with Chapter 7's ratios.

Hinglish: *(i) $(7/14)^2 = 1/4$. (ii) $85.75\pi/196\pi = 7/16$. Radius ratios ke squares!*

Answer: (i) 1/4 (ii) 7/16

R.5 Exam-Day One-Pager (read this in the last 10 minutes before the bell)

This is the whole book folded into one page. Read it top to bottom once, breathe, and walk in.

The ten formulas that earn the most marks

1. Distance: $d = \sqrt{(dx^2 + dy^2)}$; from origin $\sqrt{(x^2+y^2)}$; same coordinate → single difference.
2. Midpoint = averages; reverse: double minus the known end. Reflections: x-axis flips y, y-axis flips x.
3. Terminating test: REDUCE, then denominator must be $2^m \times 5^n$. Recurring → p/q: multiply, subtract, solve.
4. Conjugate: $1/(a+\sqrt{b}) = (a-\sqrt{b})/(a^2-b^2)$. Exponents: same base, then add/multiply/flip.
5. $(a\pm b)^2 = a^2 \pm 2ab + b^2$; $a^2-b^2 = (a+b)(a-b)$; $(x+a)(x+b)$: SUM on x, PRODUCT last.
6. $(a\pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$; $a^3 \pm b^3 = (a\pm b)(a^2 \mp ab+b^2)$.
7. Crown: $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$; zero sum → cubes = 3abc.
8. Heron: $s = P/2$, $A = \sqrt{(s(s-a)(s-b)(s-c))}$. Circle: $C = 2\pi r$, $A = \pi r^2$.
9. Sector: everything $\times\ \theta/360$; $A = \frac{1}{2}lr$; $P = l + 2r$ (TWO radii). Ring: $\pi(R^2-r^2)$.
10. Probability: P = favourable/total, $0 \le P \le 1$; at least one = 1 - none; without replacement shrinks BOTH counts.

The ten traps that cost the most marks

1. Root-splitting: $\sqrt{(dx^2+dy^2)}$ is ONE root at the END, never two roots added.
2. 6 - (-3) = 9. Rewrite minus-negative as plus FIRST.
3. $2^{-3} = 1/8$, not -8. Negative exponents flip.
4. $(a-b)^2$ ends $+b^2$. Squares are never subtracted inside these identities.
5. Sum-product swap in $(x+a)(x+b)$: coefficient = SUM, constant = PRODUCT.
6. Bridge signs: $a^3-b^3 = (a-b)^3 + 3ab(a-b)$ but $a^3+b^3 = (a+b)^3 - 3ab(a+b)$.
7. Sector perimeter = arc + 2r. Circle area uses r, NOT d.
8. Un-reduced fractions and un-simplified surds: divide the HCF, extract the square.
9. Without replacement: 8 → 7 → 6. With 'any order': ×2.
10. P outside [0, 1] means YOUR counts are wrong. Stop and recount.

The five-minute question triage

- **First pass (easy bank):** identification, direct formulas, one-line evaluations - answer everything that takes under a minute.
- **Second pass (bread and butter):** Heron, sector, factorisation, recurring conversions, standard probability - show every line, they carry method marks.

- **Third pass (heavyweights):** proofs, HOTS chains, composite figures - label each identity and rule by name.
- **Last five minutes:** the range check on every probability, units on every quantity, and one verification line on every 'show that'.

> **Final word**
> You have 340 fully solved questions and 300 MCQs behind you. Every mistake you made in this book is a mistake you will not make in the exam hall - provided you revisited it, not just read it. Ab jao, aur phod do. All the best!